# Mathematical Olympiad Training

Levels 1–4 — The Saudi Arabian National Program

Compiled from the training materials of the Saudi Arabian
Mathematical Olympiad Program, 2016–2026

# Preface

This book collects, in one place, ten years of training material written for the Saudi Arabian Mathematical Olympiad Program: camp handouts, online-training sheets, and weekly e-training sets prepared between 2016 and 2026 by the program's trainers. The material is organized by student level (Levels 1–4, matching the program's own tracks) and, within each level, by subject: Algebra, Combinatorics, Geometry, and Number Theory.

By design this volume contains *theory and problems only* — theorem statements, definitions, techniques, and problem statements, without solutions. It is meant to be used the way the original handouts were used: as working material for training sessions. Where a classical theorem was posed by a trainer as an exercise, it remains an exercise here.

Recurring ideas are deliberately revisited: a tool introduced at one level returns at higher levels with harder problems, and such sections carry an explicit pointer to the earlier material.

# Acknowledgments

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# How to Use This Book

- **Levels.** Part I (Level 1) assumes only school mathematics. Part II (Level 2) corresponds roughly to the JBMO track, Part III (Level 3) to national-olympiad level, and Part IV (Level 4) to the IMO track.
- **Numbering.** Problems are numbered by chapter: Problem 7.23 is the 23rd problem of Chapter 7. Theory items (definitions, theorems, lemmas, techniques) share a single sequence per chapter.
- **Sources.** A gray bracket such as [IMO 2006] after a problem reproduces the attribution printed in the original handout; a box at the start of each chapter lists the handouts the chapter was built from.
- **Hard problems.** A star (★) marks problems the source material itself flagged as hard for the level.
- **Revisits.** Italic notes such as "*Revisits...*" point back to the earlier level where a tool was first introduced.
- **Verification marks.** A dagger (†) marks the few statements whose constants could not be fully verified against the (damaged) source files; Appendix A lists them.

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# Part I

# Level 1

# Chapter 1

# Numbers First: Divisibility, Digits, Parity

**Sources for this chapter:** E-training 2022–2023, Level 1 weekly sheets (weeks 1–14, September 2022 – January 2023); Winter Camp 2021 Level 1 algebra handout by Asaad Mohammedsaleh (the three compositeness problems of its factoring section).

Almost every number-theoretic problem at this level is solved by one of four moves: divide with remainder, look at the digits, look at the parity, or recognise a square. The boxes below collect the handful of facts the training sheets actually use; everything else is practice.

## 1.1 Divisibility and remainders

**Technique 1.1 (Reduce a divisibility to a constant).** To decide for which integers $n$ a fixed expression $f(n)$ is divisible by $n + a$ (or by another expression), rewrite $f(n)$ as a multiple of the divisor plus a constant remainder $r$. Then the divisor must divide $r$, and only finitely many $n$ have to be tested.

**Theorem 1.2 (Sums of odd powers).** *If $n$ is odd, then $a+b$ divides $a^n+b^n$ for all integers $a$ and $b$.*

**Theorem 1.3 (Bezout's identity).** *For integers $m$ and $n$, not both zero, there exist integers $u$ and $v$ with*

$$um + vn = \gcd(m, n).$$

**Problem 1.1.** Prove that $1^9 + 2^9 + 3^9 + \ldots + 16^9$ is divisible by 17.

**Problem 1.2.** Prove that $(x^2 + xy + y^2)^7 + (x^2 - xy + y^2)^7$ is divisible by $x^2 + y^2$.

**Problem 1.3.** Find the largest integer $n$ such that $n^3 + 100$ is divisible by $n + 10$.

**Problem 1.4.** Find all positive integers $n$ such that

$$3^{n-1} + 5^{n-1} \mid 3^n + 5^n.$$

**Problem 1.5.** Prove that if all three of

$$\frac{x^2-3x+4}{49}, \qquad \frac{x^2+5x-9}{169}, \qquad \frac{x^2+3x+15}{121}$$

are integers, then $x$ is not an integer.

**Problem 1.6.** Ali computed the sum of 10 consecutive powers of 2 (for example $2^7+2^8+2^9+\ldots+2^{16}$) and Baba computed the sum of 9 consecutive integers (for example $45+46+47+\ldots+53$). Is there a chance that they get the same answer?

**Problem 1.7.** Find the number of positive integers $n$ less than 10000 for which $2^n - n^2$ is divisible by 7.

**Problem 1.8.** The numbers in the sequence 101, 104, 109, 116, ... are of the form $a_n = 100+n^2$, where $n = 1, 2, 3, \ldots$. For each $n$, let $d_n$ be the greatest common divisor of $a_n$ and $a_{n+1}$. Find the maximum value of $d_n$ as $n$ ranges through the positive integers.

**Problem 1.9.** Find all positive integers $k$ such that $\gcd(kn+1,\, 4n+1) = 1$ for every positive integer $n$.

**Problem 1.10.** Prove that for any integers $1 \le m \le n$ the value of

$$\frac{\gcd(m,n)}{n}\binom{n}{m}$$

is an integer.

**Problem 1.11.** Let $a$ be an odd integer and let $m$ be such that $2^m \mid a+1$ and $2^{m+1} \nmid a+1$. Prove that for any positive integer $k$ one has

$$2^{k+m+1} \mid (2a+1)^{2^k} - 1 \qquad \text{and} \qquad 2^{k+m+2} \nmid (2a+1)^{2^k} - 1.$$

## 1.2 Divisors, primes and factorisation

**Theorem 1.4 (Number of divisors).** *If $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ is the prime factorisation of a positive integer $n$, then $n$ has exactly*

$$(\alpha_1+1)(\alpha_2+1)\cdots(\alpha_k+1)$$

*positive divisors.*

**Problem 1.12.** Find the smallest positive integer having

a) 9,

b) 12,

c) 8

divisors.

**Problem 1.13.** Let $S(n)$ be the sum of the divisors of $n$ (for example $S(6) = 1+2+3+6 = 12$). Find all $n$ for which $S(2n) = 3S(n)$.

**Problem 1.14.** Find all pairs of prime numbers $(p, q)$ such that $p^2 + q^2 - 2$ is also prime.

**Problem 1.15.** Prove that for any positive integer $n$ the number $(n+1)^5 + n$ is not prime.

**Problem 1.16.** Prove that for any positive integers $m$ and $n$ the number

$$8m^6 + 27m^3n^3 + 27n^6$$

is composite.

**Problem 1.17.** Let $a, b, c > 1$ be integers such that $\gcd(a-1, b-1, c-1) > 1$. Prove that $abc - 1$ is not a prime.

**Problem 1.18.** The prime numbers $p$, $q$, $r$, $s$ satisfy $p > q > r > s$ and $p - q = q - r = r - s$. Prove that $18 \mid p - s$.

## 1.3 Digits and their sums

**Theorem 1.5 (Digit sums modulo 9 and 3).** *Let $S(x)$ denote the sum of the decimal digits of the positive integer $x$. Then*

$$x \equiv S(x) \pmod 9, \qquad \text{and in particular} \qquad x \equiv S(x) \pmod 3.$$

*Consequently $x$ is divisible by 9 (respectively by 3) exactly when $S(x)$ is.*

**Remark.** Throughout, $\overline{a_1 a_2 \ldots a_k}$ denotes the integer whose decimal digits are $a_1, a_2, \ldots, a_k$ in that order.

**Problem 1.19.** Let

$$35! = 10333147966386144929 * 66651337523200000000,$$

where one digit has been replaced by $*$. Find that missing digit.

**Problem 1.20.** Let $S(x)$ be the sum of digits of $x$. Solve the equation

$$x + S(x) + S(S(x)) = 2023.$$

**Problem 1.21.** Find the smallest positive integer $\overline{abcd}$ such that

$$\overline{abcd} = \overline{ab} \cdot \overline{cd} + \overline{ab} + \overline{cd}.$$

**Problem 1.22.** Find all three-digit integers $\overline{abc}$ that satisfy the equation

$$\overline{abc} = 11(a^2 + b^2 + c^2).$$

## 1.4 Parity

**Technique 1.6 (Parity).** Before computing anything, ask whether each side of an equation is forced to be odd or even, and whether a quantity that is supposed to change keeps its parity. An equality between a number that must be odd and one that must be even is impossible, and this alone settles many problems.

**Problem 1.23.** Solve for integers $x$ and $y$:

$$x^2 - x + 1 = 2^y.$$

**Problem 1.24.** Do there exist integers $a$, $b$ such that $a^2 + b$ and $a + b^2$ are consecutive integers?

**Problem 1.25.** Let the numbers $x_1, x_2, \ldots, x_n$ be given, each of them equal to either $+1$ or $-1$. Prove that if

$$x_1x_2 + x_2x_3 + \ldots + x_nx_1 = 0,$$

then $n$ is divisible by 4.

## 1.5 Squares, cubes and factorials

**Lemma 1.7 (Cubes modulo 9).** *The cube of an integer leaves remainder* 0, 1 *or* 8 *upon division by* 9.

**Technique 1.8 (Trapping a square).** If an integer $N$ satisfies $k^2 < N < (k+1)^2$ for some integer $k$, then $N$ is not a perfect square. Writing the expression under consideration between two consecutive squares (or between two consecutive cubes) is the standard way to rule out perfect powers.

**Problem 1.26.** Find all positive integers $n$ for which $n^2 + 3n$ is a perfect square.

**Problem 1.27.** Solve the equation in integers:

$$x! + 13 = y^2.$$

**Problem 1.28.** Does there exist a positive integer $n$ such that $n! + 223$ is a perfect cube?

**Problem 1.29.** Let positive integers $a$, $b$, $c$ be given. Prove that among the expressions $a^2+b+c$, $b^2 + c + a$, $c^2 + a + b$ at least one is not a perfect square.

## 1.6 Equations in integers

**Problem 1.30.** Find all integers $x$ and $y$ for which $(2x + y)(5x + 3y) = 7$.

**Problem 1.31.** There are 30 students in a class. Each of them received a grade 2, 3, 4 or 5, and the total sum of the grades is equal to 90. More students received 3 than 4, and more students received 4 than 5. The number of students that received 4 is divisible by 5 and the

number of students that received 3 is divisible by 7. Determine which grades the students of the class received.†

**Problem 1.32.** Solve for positive integers $x$, $y$ and $z$:

$$x + \frac{1}{y + \frac{1}{z}} = \frac{10}{7}.$$

**Problem 1.33.** Find all pairs $(x, y)$ of integers satisfying

$$\begin{cases} x^2 + 11 = xy + y^4, \\ y^2 - 30 = xy. \end{cases}$$

**Problem 1.34.** Find all integer solutions to the equation

$$x^2 - 6xy + 13y^2 = 100.$$

**Problem 1.35.** Let $n \geq 2$ be an integer. Consider the solutions of the system

$$\begin{cases} n = a + b - c, \\ n = a^2 + b^2 - c^2, \end{cases}$$

where $a$, $b$, $c$ are integers. Show that there is at least one solution and that the solutions are finitely many.

**Problem 1.36.** Is it true that for any nonzero rational numbers $a$ and $b$ one can find integers $m$ and $n$ such that the number $(am + b)^2 + (a + nb)^2$ is an integer? [Belarus 2019]

# Chapter 2

# Algebra First: Equations, Radicals, Inequalities

**Sources for this chapter:** E-training 2022–2023, Level 1 weekly sheets (weeks 1–14, September 2022 – January 2023); Winter Camp 2021 Level 1 algebra handout by Asaad Mohammedsaleh (systems and completing the square, inequalities, factoring).

The algebra of the first level is short on theory and long on manoeuvres: a symmetric substitution, a completed square, a factorisation recognised in time. The three boxes below are the only named results the training sheets lean on; the rest of the chapter is the manoeuvres.

## 2.1 Identities, factoring and evaluation

**Technique 2.1 (Symmetric substitution).** If a system or an expression is symmetric in $x$ and $y$, put $s = x + y$ and $p = xy$ and rewrite everything in terms of $s$ and $p$; useful identities are

$$x^2 + y^2 = s^2 - 2p, \qquad x^3 + y^3 = s^3 - 3ps.$$

Once $s$ and $p$ are known, $x$ and $y$ are exactly the two roots of $z^2 - sz + p = 0$.

**Problem 2.1.** Factorize

$$a^3 + a - 2.$$

**Problem 2.2.** Factor the expressions:

a) $a^3 + b^3 + c^3 - 3abc$,

b) $a^4 + 4b^4$,

c) $(x^2y + y^2z + z^2x) + (xy^2 + yz^2 + zx^2) + 2xyz$.

**Problem 2.3.** Factor the expression $n^5 + n^4 + 1$.

**Problem 2.4.** Simplify

$$\frac{1^4 + a^4 + (a+1)^4}{1^2 + a^2 + (a+1)^2}.$$

**Problem 2.5.** Given that $x + \frac{1}{x} = 6$, compute the value of $x^3 + \frac{1}{x^3}$.

**Problem 2.6.** Find the value of

$$\frac{(a+b)(b+c)(c+a)}{abc}$$

when it is known that

$$\frac{a+b-c}{c} = \frac{b+c-a}{a} = \frac{c+a-b}{b}.$$

**Problem 2.7.** Find the maximum possible value of $x^6 + y^6$ if it is known that $x^2 + y^2 = 1$.

**Problem 2.8.** Let $a$, $b$, $c$ be distinct real numbers such that

$$a^2(1-b+c) + b^2(1-c+a) + c^2(1-a+b) = ab + bc + ca.$$

Prove that

$$\frac{1}{(a-b)^2} + \frac{1}{(b-c)^2} + \frac{1}{(c-a)^2} = 1.$$

**Problem 2.9.** ★ Let $x$, $y$ be real numbers such that

$$x^3 + y^3 + (x+y)^3 + 30xy = 2000.$$

Prove that $x + y = 10$.

## 2.2 Equations and systems

**Problem 2.10.** Solve the equation

$$(x^2+x)^2 + \left|x^2+x\right| - 2 = 0.$$

**Problem 2.11.** Solve the inequality

$$\frac{2x^2-5x-2}{3x-x^2-7} \le 1.$$

**Problem 2.12.** How many integer solutions does the following inequality have?

$$\left(x-\frac{1}{2}\right)^1 \cdot \left(x-\frac{3}{2}\right)^3 \cdot \ldots \cdot \left(x-\frac{2017}{2}\right)^{2017} < 0.$$

**Problem 2.13.** Solve the system of equations

$$\begin{cases} \dfrac{1}{x} + \dfrac{1}{y} = \dfrac{5}{6}, \\ \dfrac{1}{x^3} + \dfrac{1}{y^3} = \dfrac{35}{216}. \end{cases}$$

**Problem 2.14.** Solve in real numbers the system of equations

$$\begin{cases} x^2 + xy + xz = 20, \\ y^2 + yx + yz = 30, \\ z^2 + zx + zy = 50. \end{cases}$$

**Problem 2.15.** Let $a$, $b$, $c$ be nonzero real numbers such that $a^2+b^2=c^2$. Solve the system of equations

$$\begin{cases} x^2+y^2=z^2, \\ (x+a)^2+(y+b)^2=(z+c)^2. \end{cases}$$

**Problem 2.16.** Solve in real numbers the system of equations

$$\begin{cases} (x^2+1)(y^2+1)=2(xy+1)(x+y), \\ (y^2+1)(z^2+1)=3(yz+1)(y+z), \\ (z^2+1)(x^2+1)=6(zx+1)(z+x). \end{cases}$$

**Problem 2.17.** Solve in real numbers the system of equations

$$\begin{cases} \left(x^2+x+\frac{1}{2}\right)\left(y^2-y+\frac{1}{2}\right)=z^2, \\ \left(y^2+y+\frac{1}{2}\right)\left(z^2-z+\frac{1}{2}\right)=x^2, \\ \left(z^2+z+\frac{1}{2}\right)\left(x^2-x+\frac{1}{2}\right)=y^2. \end{cases}$$

†

## 2.3 Radicals and surds

**Problem 2.18.** Simplify

$$\frac{\sqrt{2}+\sqrt{6}}{\sqrt{2+\sqrt{3}}}.$$

**Problem 2.19.** Construct a polynomial with integer coefficients having $\sqrt{2}+\sqrt{3}$ as a root.

**Problem 2.20.** Solve in real numbers the system of equations

$$\begin{cases} x-y=2016, \\ x+y-2\sqrt{xy}=72. \end{cases}$$

**Problem 2.21.** Find all real numbers $x_1, x_2, \ldots, x_{10}$ for which

$$\sqrt{x_1-1^2}+2\sqrt{x_2-2^2}+\ldots+10\sqrt{x_{10}-10^2}=\frac{x_1+x_2+\ldots+x_{10}}{2}.$$

**Problem 2.22.** Find all real numbers $x, y, z \geq \frac{1}{2}$ for which

$$x\sqrt{2y-1}+y\sqrt{2z-1}+z\sqrt{2x-1}=xy+yz+zx.$$

**Problem 2.23.** Find all positive real numbers $x$, $y$ such that

$$(x+y)\sqrt{1+xy}+4=2\left(\sqrt{2x+1}+\sqrt{2y+1}\right).$$

†

## 2.4 Quadratics and Vieta's formulas

**Theorem 2.2 (Vieta's formulas).** *If $x_1$ and $x_2$ are the roots of $ax^2+bx+c=0$ with $a \neq 0$, then*

$$x_1 + x_2 = -\frac{b}{a}, \qquad x_1 x_2 = \frac{c}{a}.$$

*Conversely, the numbers $s$ and $p$ are the sum and the product of the roots of $x^2 - sx + p = 0$.*

**Problem 2.24.** Let $a$, $b$ be the solutions of the equation $x^2+2x-13=0$. Construct an equation whose roots are $a+1$ and $b+1$.

**Problem 2.25.** Let $x_1$ and $x_2$ be the roots of the equation $x^2 + 5x - 11 = 0$. Find a quadratic polynomial whose roots are $x_1x_2$ and $x_1^2x_2^2$.

**Problem 2.26.** Find $m$ for which the difference of the solutions of the equation $x^2+3x+m=0$ is equal to 6.

**Problem 2.27.** ★ Find all positive integers $a$, $b$, $c$ such that the equations

$$x^2 - ax + b = 0, \qquad x^2 - bx + c = 0, \qquad x^2 - cx + a = 0$$

all have integer roots.

## 2.5 Graphs and regions

**Problem 2.28.** Draw the graph of the solutions of the equation $|x| + |y| = 2x$.

**Problem 2.29.** Find the maximum value of the expression $\sqrt{x^2+y^2}$ if it is known that

$$-4 \leq y - 2x \leq 2, \qquad 1 \leq y - x \leq 2.$$

## 2.6 Inequalities

**Theorem 2.3 (AM-GM for two numbers).** *For all nonnegative reals $x$ and $y$,*

$$\frac{x+y}{2} \geq \sqrt{xy},$$

*equivalently $(x+y)^2 \geq 4xy$ and $x^2+y^2 \geq 2xy$, with equality exactly when $x=y$.*

### 2.6.1 Getting started

**Problem 2.30.** The product of two positive integers is equal to 120. Find the minimal possible value of their sum.

**Problem 2.31.** Positive real numbers $x$, $y$ are such that their product is bigger than their sum. Prove that their sum is bigger than 4.

**Problem 2.32.** Let $a^2+b^2 > a+b$ with $a>0$ and $b>0$. Prove that

$$a^3 + b^3 > a^2 + b^2.$$

**Problem 2.33.** Prove that for any numbers $a$, $b$, $c > 0$ the following inequality holds:

$$\frac{a}{bc} + \frac{b}{ca} + \frac{c}{ab} \geq \frac{2}{a} + \frac{2}{b} - \frac{2}{c}.$$

**Problem 2.34.** Prove that for positive $a$, $b$, $c$ one has

$$\frac{a}{\sqrt{a+b}} + \frac{b}{\sqrt{b+c}} + \frac{c}{\sqrt{c+a}} > \sqrt{a+b+c}.$$

### 2.6.2 Sums and estimates

**Problem 2.35.** Prove the inequality

$$\frac{1}{2} - \frac{1}{3} + \frac{1}{4} - \frac{1}{5} + \frac{1}{6} - \ldots - \frac{1}{99} + \frac{1}{100} > \frac{1}{5}.$$

**Problem 2.36.** Let $n > 1$. Prove that

$$\frac{1}{n} + \frac{1}{n+1} + \frac{1}{n+2} + \ldots + \frac{1}{n^2-1} + \frac{1}{n^2} > 1.$$

[Hungary 1934]

### 2.6.3 Classical inequalities

**Problem 2.37.** For $a$, $b$, $c > 0$, show that

$$\frac{a}{\sqrt{(a+b)(a+c)}} + \frac{b}{\sqrt{(b+a)(b+c)}} + \frac{c}{\sqrt{(c+a)(c+b)}} \leq \frac{3}{2}.$$

**Problem 2.38.** For $a$, $b$, $c > 0$, show that

$$\frac{a}{a+2b+c} + \frac{b}{b+2c+a} + \frac{c}{c+2a+b} \geq \frac{3}{4}.$$

**Problem 2.39.** Let $x$, $y$, $z$ be positive reals with $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 2$. Prove that

$$\sqrt{x+y+z} \geq \sqrt{x-1} + \sqrt{y-1} + \sqrt{z-1}.$$

**Problem 2.40.** Let $a$, $b$, $c > 0$ with $abc = 1$. Prove that

$$\frac{a}{1+b+c} + \frac{b}{1+c+a} + \frac{c}{1+a+b} \geq 1.$$

**Problem 2.41.** Let $x$, $y$, $z > 0$. Show that

$$\frac{x}{x+\sqrt{(x+y)(x+z)}} + \frac{y}{y+\sqrt{(y+z)(y+x)}} + \frac{z}{z+\sqrt{(z+x)(z+y)}} \leq 1.$$

**Problem 2.42.** If $a$, $b$, $c \geq 0$, then

$$\frac{a}{1+a} + \frac{b}{1+b} + \frac{c}{1+c} + \frac{1}{1+a+b+c} \geq 1.$$

**Problem 2.43.** Let $a$, $b$, $c$ be positive real numbers. Prove that

$$\frac{a}{2a+b+c} + \frac{b}{a+2b+c} + \frac{c}{a+b+2c} \leq \frac{3}{4}.$$

**Problem 2.44.** Let $a$, $b$, $c$ be positive real numbers. Prove that

$$\frac{ab}{a+b+2c} + \frac{ca}{a+2b+c} + \frac{bc}{2a+b+c} \leq \frac{a+b+c}{4}.$$

**Problem 2.45.** Let $a$, $b$, $c > 0$. Prove that

$$\sqrt{\frac{1+a^2}{a+b}} + \sqrt{\frac{1+b^2}{b+c}} + \sqrt{\frac{1+c^2}{c+a}} \geq 3.$$

**Problem 2.46.** Let $a$, $b$, $c > 0$. Prove that

$$(a^2+bc)(b^2+ca)(c^2+ab) \geq abc(a+b)(b+c)(c+a).$$

**Problem 2.47.** Let $a$, $b$, $c > 0$. Prove that

$$\frac{a^2}{b^2} + \frac{b^2}{c^2} + \frac{c^2}{a^2} \geq \frac{a}{b} + \frac{b}{c} + \frac{c}{a}.$$

**Problem 2.48.** Let $x$, $y > 0$. Prove that

$$\frac{1}{(1+x)^2} + \frac{1}{(1+y)^2} \geq \frac{1}{1+xy}.$$

**Problem 2.49.** Let $a$, $b$, $c > 0$ with $abc = 1$. Prove that

$$\frac{1}{a^2+2b^2+3} + \frac{1}{b^2+2c^2+3} + \frac{1}{c^2+2a^2+3} \leq \frac{1}{2}.$$

**Problem 2.50.** Let $a$, $b$, $c > 0$. Prove that

$$\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{a} \geq a+b+c+\frac{4(a-b)^2}{a+b+c}.$$

[BMO 2005]

**Problem 2.51.** If $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 1$ for $x$, $y$, $z > 0$, prove that

$$(x-1)(y-1)(z-1) \geq 8.$$

**Problem 2.52.** Let $a$, $b$, $c$ be the side lengths of a triangle. Prove that

$$\frac{a}{a-b+c}+\frac{b}{b-c+a}+\frac{c}{c-a+b}\geq 3.$$

**Problem 2.53.** Let $a$, $b$, $c>0$. Prove that

$$\frac{a^3}{a^2+ab+b^2}+\frac{b^3}{b^2+bc+c^2}+\frac{c^3}{c^2+ca+a^2}\geq\frac{a+b+c}{3}.$$

**Problem 2.54.** Let $a$, $b$, $c>0$. Prove that

$$\frac{1+a(b+c)}{(1+b+c)^2}+\frac{1+b(c+a)}{(1+c+a)^2}+\frac{1+c(a+b)}{(1+a+b)^2}\geq 1.$$

**Problem 2.55.** Let $x$, $y$, $z>0$. Prove that

$$(x+y)\sqrt{(z+x)(z+y)}+(y+z)\sqrt{(x+y)(x+z)}+(z+x)\sqrt{(y+z)(y+x)}\geq 4(xy+yz+zx).$$

[BMO 2012]

**Problem 2.56.** Let $a$, $b$, $c>0$ be such that $a+b+c=3$. Find the minimum value of the expression

$$A=\frac{2-a^3}{a}+\frac{2-b^3}{b}+\frac{2-c^3}{c}.$$

[JBMO 2015]

# Chapter 3

# How to Count and How to Argue

**Sources for this chapter:** E-training 2022–2023, Level 1 weekly sheets (weeks 1–14, September 2022 – January 2023).

Combinatorics at this level asks two kinds of question: how many, and is it possible. The first is answered by counting carefully in two different ways; the second almost always by exhibiting a quantity that the allowed moves cannot change.

## 3.1 Counting

**Theorem 3.1 (Inclusion and exclusion for two sets).** *For any finite sets $A$ and $B$,*

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

*In particular, the objects having at least one of two properties are counted by adding the two counts and subtracting the objects having both.*

**Problem 3.1.** In how many ways can one form a basketball team of 5 players such that one of them is the captain, if there are 30 candidates for team membership?

**Problem 3.2.** Find the number of 7-digit positive integers such that the second digit is odd and the product of the third and the fifth digits is even.

**Problem 3.3.** Find the number of 7-digit positive integers whose digits are ordered in

a) strictly increasing order,

b) strictly decreasing order.

**Problem 3.4.** Find the number of positive integers that are less than 1 000 000 and contain both the digit 1 and the digit 2.

**Problem 3.5.** Find the coefficient of $x^{100}$ after opening the brackets and grouping terms of $(1 + x + x^2 + \ldots + x^{100})^3$.

## 3.2 The pigeonhole principle

**Technique 3.2 (Pigeonhole principle).** If $N$ objects are distributed among $k$ boxes and $N > k$, then some box contains at least two objects; more generally, if $N > km$, then some box contains at least $m + 1$ objects. Using the principle means choosing what the objects are and what the boxes are, so that two objects landing in one box is exactly the conclusion wanted.

**Problem 3.6.** There are 85 cubes in a shop, each of them painted in one colour. Prove that either there exist 10 cubes all of which have the same colour, or there exist 10 cubes all of which have different colours.

**Problem 3.7.** 50 humans and 51 aliens played chess against each other, in such a way that any pair played at most one game. It turned out that all the humans played the same number of games against aliens. Prove that there exist 2 aliens that played the same number of games against humans.

**Problem 3.8.** 10 players participate in a chess tournament. Each day they play 5 games – one game each. After 9 days all the players have played against each other. It turns out that most of the games are played between players from Jeddah. Prove that every day at least two players from Jeddah played against each other.

**Problem 3.9.** We are given an infinite set of rectangles in the plane, each with vertices of the form $(0, 0)$, $(0, m)$, $(n, 0)$ and $(n, m)$, where $m$ and $n$ are positive integers. Prove that there exist two rectangles in the set such that one contains the other.

**Problem 3.10.** Each integer is given one of three colours. Prove that there exist two distinct integers of the same colour whose difference is a square of an integer.

**Problem 3.11.** Let 2022 numbers be written on a blackboard. It is known that no matter how one splits them into 1011 pairs, there are two pairs whose sums are equal. Find the maximal possible number of different values written on the blackboard.

**Problem 3.12.** Let the positive integers from 1 to $22^2$ be distributed in the cells of a $22 \times 22$ square. Prove that one can find 2 cells having a common side or a common vertex such that the sum of the numbers written in these cells is divisible by 4.

## 3.3 Invariants and processes

**Technique 3.3 (Look for an invariant).** To show that a position cannot be reached from a starting position by the allowed moves, find a quantity attached to a position – a sum, an alternating sum, a residue modulo a fixed number, a parity – which every allowed move leaves unchanged, or changes only in a controlled way. If that quantity differs at the start and at the target, the target is unreachable. The same quantity, when it does not obstruct, often tells you how many moves are needed.

**Problem 3.13.** A triple $(1, 1, 1)$ is given. On each step one chooses 2 of the numbers and increases each of them by 1. Is it possible to obtain the numbers $(2016, 2016, 2016)$ after some steps?

**Problem 3.14.** Let the numbers $(1, 2, 3, 4)$ be given. On each step one chooses 2 neighbouring numbers (the first and the fourth numbers are also considered neighbouring) and increases each of them by 1. Is it possible to obtain the numbers $(2021, 2022, 2023, 2022)$ after some steps?

**Problem 3.15.** A circle is divided into 6 sectors and each sector contains exactly 1 coin. At each step Ali is allowed to move two coins, each into a neighbouring sector (two sectors are called neighbours if they have a common side). Is it possible to achieve a configuration where all the coins are in one sector?

**Problem 3.16.** There are three piles with $n$ tokens each. In every step we are allowed to choose two piles, take one token from each of those two piles and add a token to the third pile. Using these moves, is it possible to end up having only one token?

**Problem 3.17.** Is it possible to write 7 integers around a table with total sum equal to 1000, such that the difference of any two neighbouring numbers is equal in absolute value to

a) 1,

b) 2?

**Problem 3.18.** The numbers 1, 1 are written on a board. At each step, between every two neighbouring numbers Aziz writes their sum. Below are the results after the first three steps:

$$1,\ 2,\ 1$$

$$1,\ 3,\ 2,\ 3,\ 1$$

$$1,\ 4,\ 3,\ 5,\ 2,\ 5,\ 3,\ 4,\ 1$$

Find the total sum of the numbers written after the 100th step.

## 3.4 Colourings, boards and labellings

**Problem 3.19.** A chess king started from some cell and, passing over each cell exactly once, came back to its original position. Prove that the king made an even number of diagonal moves.

**Problem 3.20.** One puts stones on the squares of a chessboard such that each row and each column contains exactly 1 stone. The chessboard is divided into 4 squares of size $4 \times 4$. Prove that the bottom-left square of size $4 \times 4$ contains as many stones as the top-right square of size $4 \times 4$.

**Problem 3.21.** Is it possible to write the numbers from 1 to 10 (each once) on the edges and the vertices of a triangular pyramid in such a way that the number on any edge is the arithmetic mean of the numbers written at the endpoints of that edge?

## 3.5 Constructions, games and strategies

**Problem 3.22.** Let $k$ be given and let the numbers from 1 to 100 be written on the board. Ali erases from the board $k$ arbitrary numbers. Is it true that Bob may then choose $k$ numbers written on the board whose sum is equal to 100? Consider the cases

a) $k = 8$,

b) $k = 9$.

**Problem 3.23.** Let $3n$ different points be given in the plane such that no 3 of them lie on the same line. Prove that one can draw $n$ triangles all of whose vertices are among the given $3n$ points, so that no 2 of the triangles have a common point (not necessarily a common vertex).

**Problem 3.24.** Let $n$ be a positive integer. Two players, Alice and Bob, are playing the following game.

- Alice chooses $n$ real numbers, not necessarily distinct.
- Alice writes all pairwise sums on a sheet of paper and gives it to Bob. (There are $\frac{n(n-1)}{2}$ such sums, not necessarily distinct.)
- Bob wins if he correctly finds the initial $n$ numbers chosen by Alice with only one guess.

For which $n$ can Bob be sure to win?[†] [JBMO 2013]

# Chapter 4

# Triangles, Circles, and Their Centers

**Sources for this chapter:** Level-1 camp handout *Centers of Triangle* by Dren Neziri (January 2025 camp, KAUST); Level-1 e-training weekly sheets (2022–2023).

Every triangle carries a handful of remarkable points — the center of the circle through its vertices, the meeting point of its altitudes, the meeting point of its angle bisectors, the meeting point of its medians. This chapter introduces those four centers, the excenters that come with them, and the two famous configurations they generate: the Euler line and the nine-point circle. Almost everything here is proved by angle chasing together with one ratio criterion, Ceva's theorem.

**Remark.** Notation used throughout the chapter. For three points $X$, $Y$, $Z$ we write $(XYZ)$ for the circle passing through them. In a triangle $ABC$ we always write $O$ for its circumcenter, $H$ for its orthocenter, $I$ for its incenter and $G$ for its centroid.

## 4.1 Angles in triangles and circles

We begin with four problems that need nothing beyond angle chasing, the inscribed angle theorem and the criterion for four points to be concyclic.

**Problem 4.1.** Prove that if all three sides of a triangle are shorter than 1, then the area of the triangle is less than $\sqrt{3}/4$.

**Problem 4.2.** Draw a triangle together with its three altitudes. Find six cyclic quadrilaterals in the resulting figure.

**Problem 4.3.** Let $X$, $Y$, $Z$ be points on the sides $BC$, $CA$, $AB$ of a triangle $ABC$, respectively. Prove that the circles $(BXZ)$, $(CXY)$ and $(AZY)$ have a common point.

**Problem 4.4.** Let $ABCD$ be a quadrilateral. The circumcircle of triangle $ABC$ meets the lines $CD$ and $AD$ again at $E$ and $F$ respectively, and the circumcircle of triangle $ACD$ meets the lines $AB$ and $BC$ again at $P$ and $Q$ respectively. The lines $BE$ and $BF$ meet the line $PQ$ at $X$ and $Y$ respectively. Show that $E$, $F$, $Y$, $X$ are concyclic.

## 4.2 Concurrency: the four classical centers

Three lines drawn in a triangle rarely pass through one point; the four constructions below are the classical exceptions. Each definition records one concurrency together with the name of the resulting point — proving the concurrencies is the business of the problems that follow.

**Definition 4.1 (Circumcenter and circumcircle).** The three perpendicular bisectors of the sides of a triangle $ABC$ meet at one point, called the *circumcenter* of the triangle and denoted by $O$. It is equidistant from $A$, $B$ and $C$, so the circle centered at $O$ through the three vertices — the *circumcircle* $(ABC)$ — is well defined.

**Definition 4.2 (Orthocenter).** The three altitudes of a triangle $ABC$ meet at one point, called the *orthocenter* of the triangle and denoted by $H$.

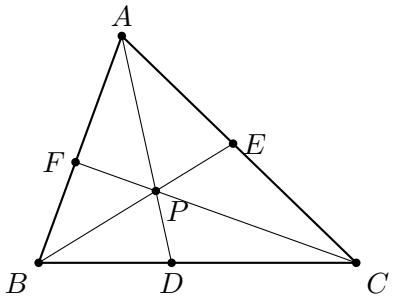
**Definition 4.3 (Incenter and incircle).** The three internal angle bisectors of a triangle $ABC$ meet at one point, called the *incenter* of the triangle and denoted by $I$. It is equidistant from the three sides, so there is a circle centered at $I$ tangent to all three sides: the *incircle*.

**Definition 4.4 (Centroid).** The three medians of a triangle $ABC$ meet at one point, called the *centroid* of the triangle and denoted by $G$.

The single tool behind most concurrency statements at this level is the following ratio criterion. A line joining a vertex of a triangle to a point of the opposite side is called a *cevian*.

**Theorem 4.5 (Ceva's theorem).** *Let $ABC$ be a triangle, and let $D$, $E$, $F$ be points on the lines $BC$, $CA$, $AB$, respectively. Then the lines $AD$, $BE$, $CF$ are concurrent if and only if*

$$\frac{AF}{FB}\cdot\frac{BD}{DC}\cdot\frac{CE}{EA}=1.$$




Two auxiliary triangles appear again and again, so it is worth naming them once and for all.

**Definition 4.6 (The medial and orthic triangles).** The triangle whose vertices are the midpoints of the three sides of $ABC$ is called the *medial triangle* of $ABC$. The triangle whose vertices are the feet of the three altitudes of $ABC$ is called the *orthic triangle* of $ABC$.

**Problem 4.5.** Prove that the three angle bisectors of a triangle are concurrent.

**Problem 4.6.** Prove that the three medians of a triangle are concurrent.

**Problem 4.7.** Let $I$ be the incenter of a triangle $ABC$. Show that $\angle BIC = 90^\circ + \frac{1}{2}\angle A$.

**Problem 4.8.** Let $M$, $N$, $P$ be the midpoints of the sides $BC$, $CA$, $AB$ of a triangle $ABC$ and let $G$ be its centroid. Prove that the six triangles into which the three medians cut $ABC$ all have

the same area. Prove also that

$$\frac{AG}{GM} = \frac{BG}{GN} = \frac{CG}{GP} = 2.$$

**Problem 4.9.** Prove that the circumcenter of a triangle $ABC$ is the orthocenter of the medial triangle of $ABC$. Prove also that the centroid of $ABC$ is the centroid of the medial triangle.

**Problem 4.10.** Prove that the orthocenter $H$ and the circumcenter $O$ of a triangle $ABC$ are isogonal conjugates: that is, the line $AH$ is the reflection of the line $AO$ in the internal bisector of the angle $A$, and similarly at the vertices $B$ and $C$.

**Problem 4.11.** Prove that the orthocenter of an acute triangle $ABC$ is the incenter of its orthic triangle.

**Problem 4.12.** Let $G$ be the centroid of a triangle $ABC$. The line $XY$ through $G$ parallel to $BC$ meets $AB$ and $AC$ at $X$ and $Y$ respectively. Let $P$ be the intersection point of $BG$ and $CX$, and let $Q$ be the intersection point of $CG$ and $BY$. If $M$ is the midpoint of $BC$, show that the triangles $ABC$ and $MQP$ are similar.

The next two problems concern one more cevian through each vertex. The reflection of the median from $A$ in the internal bisector of the angle $A$ is called the *$A$-symmedian* of the triangle, and similarly at $B$ and $C$.

**Problem 4.13.** Let the tangents to the circumcircle of a triangle $ABC$ at $B$ and at $C$ meet at $T$, and let $M$ be the midpoint of $BC$. Prove that $\angle BAM = \angle CAT$; in other words, $AT$ is the $A$-symmedian of $ABC$.

**Problem 4.14.** Prove that the three symmedians of a triangle are concurrent.

## 4.3 The orthocenter and the circumcircle

The orthocenter of a triangle and the circumcircle are tied together by a short list of reflection facts. Once these are available, a surprising number of configurations become routine.

**Problem 4.15.** Let $H$ be the orthocenter of a triangle $ABC$ and let $D$ be the foot of the altitude from $A$. Show that the reflection of $H$ in the line $BC$ lies on the circumcircle $(ABC)$. Prove also that $DH \cdot DA = DB \cdot DC$.

**Problem 4.16.** Let $H$ be the orthocenter of a triangle $ABC$ and let $M$ be the midpoint of $BC$. Show that the reflection of $H$ in $M$ lies on the circumcircle $(ABC)$, and that this point is the antipode of $A$ on $(ABC)$.

**Problem 4.17.** Let $ABC$ be a triangle with circumcenter $O$ and orthocenter $H$, and let $M$ be the midpoint of $BC$. Show that $AH = 2 \cdot OM$.

**Problem 4.18.** A triangle $ABC$ has $AH = AO$, where $H$ and $O$ are its orthocenter and circumcenter. Find $\angle A$.

**Problem 4.19.** Let $H$ be the orthocenter of a triangle $ABC$. Prove that the circles $(ABC)$ and $(BHC)$ have the same radius.

**Problem 4.20.** Let $ABC$ be a triangle with $\angle A = 60°$, with orthocenter $H$, incenter $I$ and circumcenter $O$. Prove that $B$, $C$, $H$, $I$, $O$ are concyclic.

**Problem 4.21.** Let $\ell$ be a line through the orthocenter of a triangle $ABC$. Prove that the reflections of $\ell$ in the three sides of $ABC$ meet at a point of the circumcircle $(ABC)$.

**Problem 4.22.** Let $M$ be the midpoint of the side $BC$ of a triangle $ABC$, and let $P$ be a point on the segment $AM$ such that $\angle BAC + \angle BPC = 180°$. Show that $BC$ is a common tangent of the circles $(APB)$ and $(APC)$.

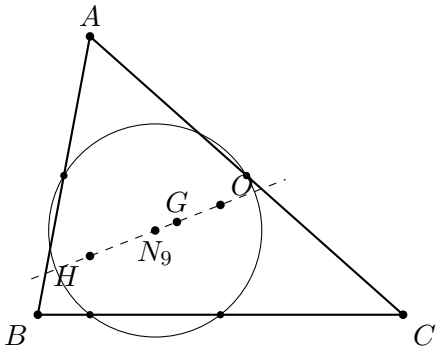
**Problem 4.23.** Let $ABC$ be a triangle with orthocenter $H$ and let $M$ be the midpoint of $BC$. Let $P$ be the second point of intersection of the circle $(BHC)$ with the line $AM$. Prove that $\angle APH = 90°$, and prove that $BC$ is a common tangent of the circles $(APB)$ and $(APC)$.[†]

**Problem 4.24.** In the configuration of the previous problem, let the lines $PH$ and $BC$ meet at $T$. Prove that the lines $TA$ and $HM$ meet on the circumcircle $(ABC)$.

**Problem 4.25.** We are given an acute triangle $ABC$. Let $D$ be the point on its circumcircle such that $AD$ is a diameter. Suppose that points $K$ and $L$ lie on the segments $AB$ and $AC$, respectively, and that $DK$ and $DL$ are tangent to the circle $(AKL)$. Show that the line $KL$ passes through the orthocenter of triangle $ABC$.

## 4.4 The Euler line and the nine-point circle

Three of the four classical centers always lie on one line, and nine naturally occurring points always lie on one circle. Both facts are proved with the reflection lemmas of the previous section.




**Problem 4.26.** Let $O$, $H$ and $G$ be the circumcenter, the orthocenter and the centroid of a triangle. Prove that $O$, $H$ and $G$ are collinear. The line through them is called the *Euler line* of the triangle.

**Problem 4.27.** Let $ABC$ be a triangle with orthocenter $H$. Prove that the three feet of the altitudes, the three midpoints of the sides, and the midpoints of the segments $AH$, $BH$, $CH$ all lie on one circle. This circle is called the *nine-point circle* of the triangle. Prove also that its center, denoted $N_9$, lies on the Euler line.

**Problem 4.28.** Let $D$ be the foot of the altitude from $A$ to the side $BC$ of a triangle $ABC$, let $X$ be the point of the circumcircle $(ABC)$ with $AX \parallel BC$, and let $G$ be the centroid of $ABC$. Prove that $D$, $G$ and $X$ are collinear.

## 4.5 The incenter, the excenters and the excircles

The incircle has three companions, each tangent to one side of the triangle and to the extensions of the other two. This section builds the whole family from scratch, starting from a single point of the circumcircle.

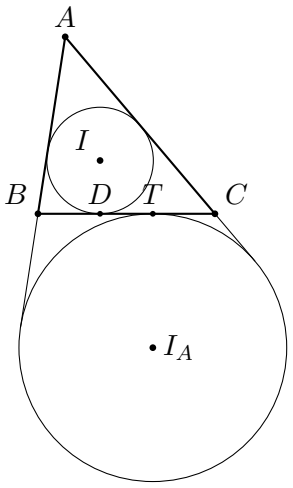
**Problem 4.29.** Prove that the internal bisector of the angle $A$ of a triangle $ABC$ and the perpendicular bisector of $BC$ meet at a point of the circumcircle $(ABC)$. Call this point $M$.

**Problem 4.30.** With $M$ as in the previous problem and $I$ the incenter of $ABC$, prove that $M$ is the center of the circle $(BIC)$.

**Problem 4.31.** Let $I_A$ be the reflection of the incenter $I$ of a triangle $ABC$ in the point $M$ of the two previous problems. Prove that $I_A$ lies on the external bisectors of the angles $B$ and $C$ and on the internal bisector of the angle $A$.

**Problem 4.32.** Prove that there is a circle tangent to the side $BC$ of a triangle $ABC$ and tangent to the lines $AB$ and $AC$ outside the triangle. This circle is called the *$A$-excircle*, and its center is called the *$A$-excenter*.

**Remark.** The $A$-, $B$- and $C$-excenters of a triangle $ABC$ are denoted $I_A$, $I_B$ and $I_C$. In the picture below, the $A$-excircle touches the side $BC$ at $T$ and the incircle touches it at $D$.




**Problem 4.33.** Prove that the incenter of a triangle $ABC$ is the orthocenter of the triangle $I_AI_BI_C$ formed by its three excenters.

**Problem 4.34.** Let $I$ and $I_A$ be the incenter and the $A$-excenter of a triangle $ABC$. Prove that the triangles $ABI_A$ and $AIC$ are similar.

**Problem 4.35.** Prove that the tangent segments from $A$ to the $A$-excircle of a triangle $ABC$ have length equal to half the perimeter of $ABC$.

**Problem 4.36.** Let $X$ and $Y$ be the points at which the incircle and the $A$-excircle of a triangle $ABC$ touch the side $BC$. Prove that $BX = CY$.

**Problem 4.37.** Prove that the incenter of a triangle $ABC$ is the orthocenter of the triangle formed by the midpoints of the arcs $BC$, $CA$, $AB$ of the circumcircle that do not contain $A$, $B$, $C$, respectively.

**Problem 4.38.** Let $ABC$ be an acute triangle. Show that $A$, $B$ and $C$ are the excenters of the orthic triangle of $ABC$.

**Problem 4.39.** The incircle of a triangle $ABC$, with incenter $I$, touches $BC$, $CA$, $AB$ at $D$, $E$, $F$ respectively. Let $M$ and $N$ be the midpoints of $BC$ and $AC$, and let $K$ be the intersection point of the lines $BI$ and $EF$. Prove that $BK \perp KC$, and prove in addition that $K$ lies on the line $MN$.

**Problem 4.40.** Let $ABC$ be a triangle with $CA > AB > BC$, and let $R$ and $E$ be variable points on the sides $AB$ and $AC$ respectively such that $BR + CE = BC$. Prove that the circle $(REA)$ passes through a fixed point.

**Problem 4.41.** Let $ABC$ be a triangle with incenter $I$. The circle through $B$ tangent to $AI$ at $I$ meets the side $AB$ again at $P$. The circle through $C$ tangent to $AI$ at $I$ meets the side $AC$ again at $Q$. Prove that $PQ$ is tangent to the incircle of $ABC$.

## 4.6 Harder configuration problems

The problems below combine several of the earlier facts at once. The starred ones are those the camp handout left unsolved in class.

**Problem 4.42.** The side $BC$ of a triangle $ABC$ is extended beyond $C$ to a point $D$ with $CD = BC$. The side $CA$ is extended beyond $A$ to a point $E$ with $AE = 2\,CA$. Prove that if $AD = BE$, then the triangle $ABC$ is right-angled.

**Problem 4.43.** Let $ABCD$ be a tangential quadrilateral, that is, a convex quadrilateral having an inscribed circle. Let $\ell$ be a line cutting $ABCD$ into two polygons of equal area and equal perimeter. Prove that $\ell$ passes through the center of the inscribed circle of $ABCD$.

**Problem 4.44.** Let $ABC$ be an acute triangle and let $D$ be the foot of the altitude from $C$. The bisector of the angle $\angle ABC$ meets $CD$ at $E$ and meets the circumcircle $\omega$ of triangle $ADE$ again at $F$. Given that $\angle ADF = 45^\circ$, show that $CF$ is tangent to $\omega$.

**Problem 4.45.** A trapezoid $ABCD$ with $AB \parallel CD$ and $AB > CD$ is circumscribed about a circle. The incircle of the triangle $ABC$ touches the lines $AB$ and $AC$ at the points $M$ and $N$ respectively. Prove that the incenter of the trapezoid $ABCD$ lies on the line $MN$. [JBMO]

**Problem 4.46.** Let $ABC$ be a triangle with incenter $I$. A point $P$ in the interior of the triangle satisfies

$$\angle PBA + \angle PCA = \angle PBC + \angle PCB.$$

Show that $AP \geq AI$, and that equality holds if and only if $P = I$. [IMO]

**Problem 4.47.** ★ In a triangle $ABC$ with incenter $I$ and $A$-excenter $I_A$, let $D$ and $T$ be the points at which the incircle and the $A$-excircle touch $BC$. Let $M$ be the midpoint of the altitude from $A$. Prove that $M$, $I$, $T$ are collinear and that $M$, $D$, $I_A$ are collinear.

**Problem 4.48.** ★ Let $ABC$ be a triangle with orthocenter $H$, circumcenter $O$ and centroid $G$. Let $E$ and $F$ be the midpoints of the sides $AB$ and $AC$ respectively, and let $M$ be the midpoint of $EF$. Prove that the line $AO$, the line $HM$ and the line through $G$ perpendicular to $EF$ are concurrent.

**Problem 4.49.** ★ Let $ABC$ be an acute triangle in which $BC < AB$ and $BC < CA$. Let a point $P$ lie on the segment $AB$ and a point $Q$ lie on the segment $AC$ such that $P \neq B$, $Q \neq C$ and $BQ = BC = CP$. Let $T$ be the circumcenter of triangle $APQ$, let $H$ be the orthocenter of triangle $ABC$, and let $S$ be the point of intersection of the lines $BQ$ and $CP$. Prove that $T$, $H$ and $S$ are collinear.

**Problem 4.50.** ★ A triangle $ABC$ is such that $AB < AC$. The perpendicular bisector of the side $BC$ meets the lines $AB$ and $AC$ at the points $P$ and $Q$ respectively. Let $H$ be the orthocenter of the triangle $ABC$, and let $M$ and $N$ be the midpoints of the segments $BC$ and $PQ$ respectively. Prove that the lines $HM$ and $AN$ meet on the circumcircle of $ABC$.

**Problem 4.51.** ★ Let $ABC$ be an acute triangle with $AB < AC$ and let $O$ be the center of its circumcircle $\omega$. Let $D$ be a point on the segment $BC$ such that $\angle BAD = \angle CAO$. Let $E$ be the second point of intersection of $\omega$ with the line $AD$. If $M$, $N$ and $P$ are the midpoints of the segments $BE$, $OD$ and $AC$ respectively, show that the points $M$, $N$ and $P$ are collinear.

**Problem 4.52.** ★ Let $ABC$ be an acute triangle with $AC > AB$ and let $D$ be the foot of the bisector from $A$ on $BC$. The reflections of the lines $AB$ and $AC$ in the line $BC$ meet $AC$ and $AB$ at the points $E$ and $F$ respectively. A line through $D$ meets $AC$ and $AB$ at $G$ and $H$ respectively, in such a way that $G$ lies strictly between $A$ and $C$ while $H$ lies strictly between $B$ and $F$. Prove that the circumcircles of the triangles $EDG$ and $FDH$ are tangent to each other.

**Problem 4.53.** ★ Consider a convex quadrilateral $ABCD$ and a point $P$ in its interior such that

$$\angle PAD : \angle PBA : \angle DPA = 1 : 2 : 3 = \angle CBP : \angle BAP : \angle BPC.$$

Prove that the following three lines meet at one point: the internal bisector of the angle $\angle ADP$, the internal bisector of the angle $\angle PCB$, and the perpendicular bisector of the segment $AB$.

**Problem 4.54.** ★ Let $ABC$ be an acute triangle with $AB \neq AC$. The circle with diameter $BC$ meets the sides $AB$ and $AC$ at $M$ and $N$ respectively. Denote by $O$ the midpoint of the side $BC$. The bisectors of the angles $\angle BAC$ and $\angle MON$ meet at $R$. Prove that the circumcircles of the triangles $BMR$ and $CNR$ have a common point lying on the side $BC$. [IMO]

# Part II

# Level 2

# Chapter 5

# Algebra

**Sources for this chapter:** Camp handouts by Marko Radovanović (March 2022, November 2025), Dušan Djukić (February 2023), Asaad Mohammedsaleh (Winter Camp 2021), Navid Safaei (November 2024), Łukasz Bożyk (February 2022, March–April 2023), Nikola Petrović (JBMO preparation, 2021).

Junior olympiad algebra rests on a small number of tools used again and again: a well chosen substitution, an identity that factors an expression, and above all the inequalities of the arithmetic–geometric mean and of Cauchy–Schwarz. The sections below follow the order in which the Saudi camps introduce them, and the last two sections open the two topics that a JBMO contestant meets for the first time: the floor function and functional equations.

## 5.1 Equations, systems, and identities

The first thing to try with a system is to look for structure: symmetry, a cyclic pattern, or one equation that can be rewritten so that a familiar inequality applies to it.

**Problem 5.1.** Solve the equations $(x+1)^4 = 8x^2$ and $(x+1)^4 = 8x$.

**Problem 5.2.** Solve the equation $x = 4 - (4 - x^2)^2$.

**Problem 5.3.** Solve the equation $\sqrt[5]{3+x} + \sqrt[5]{3-x} = 2$ in the real numbers.

**Problem 5.4.** Find all real solutions of the system of equations

$$ab + a + b = c, \qquad bc + b + c = a, \qquad ca + c + a = b.$$

**Problem 5.5.** Find all solutions of the system of equations

$$y + 2 = (3 - x)^2, \qquad (2z - y)(y + 2) = 9 + 4y, \qquad x^2 + z^2 = 4x.$$

**Problem 5.6.** Solve the system of equations $x + y + z = 0$, $x^2 + y^2 + z^2 = 6$, $x^3 + y^3 + z^3 = 6$.

**Problem 5.7.** Find all nonnegative real solutions of the system of equations

$$a\sqrt{b} - c = a, \qquad b\sqrt{c} - a = b, \qquad c\sqrt{a} - b = c.$$

**Problem 5.8.** Find all positive real solutions of the system of equations

$$\sqrt{x - \tfrac{1}{y}} = \sqrt{y - \tfrac{1}{z}} = \sqrt{z - \tfrac{1}{x}}.$$

†

**Problem 5.9.** Find all positive real solutions of the system of equations

$$x_1 + \frac{1}{x_2} = 4, \quad x_2 + \frac{1}{x_3} = 1, \quad x_3 + \frac{1}{x_4} = 4, \quad \ldots, \quad x_{99} + \frac{1}{x_{100}} = 4, \quad x_{100} + \frac{1}{x_1} = 1,$$

where the right-hand sides alternate between 4 and 1.

**Problem 5.10.** Find all triples $(x, y, z)$ of real numbers that satisfy the system of equations

$$\frac{x^3}{yz} = z - 2y, \qquad \frac{y^3}{zx} = x - 2z, \qquad \frac{z^3}{xy} = y - 2x.$$

**Problem 5.11.** Find all real solutions of the system of equations

$$x^4 + 6x^2y^2 + y^4 = x^3 + y^3 + 4x^2 + 4y^2 + x + y - 1, \qquad x^2 + y^2 = \tfrac{3}{4}(x + y) + 2.$$

**Problem 5.12.** Let $x_k = \dfrac{k}{101}$ for $1 \leqslant k \leqslant 101$. Calculate the sum

$$\frac{x_1^3}{1 - 3x_1 + 3x_1^2} + \frac{x_2^3}{1 - 3x_2 + 3x_2^2} + \cdots + \frac{x_{101}^3}{1 - 3x_{101} + 3x_{101}^2}.$$

**Problem 5.13.** Prove that the number

$$\sqrt{\underbrace{44\ldots4}_{2026} - 11 \cdot \underbrace{44\ldots4}_{1013} + 9}$$

is an integer.

**Problem 5.14.** Find the 2025th digit after the decimal point of the number $\sqrt{a}$, where $a = 0.\underbrace{44\ldots4}_{2025}$.

**Problem 5.15.** Let $a$ be a real number such that $a^5 - a^3 + a = 2$. Show that $3 < a^6 < 4$.

**Problem 5.16.** Let $a$ and $b$ be real numbers such that $a^3 + b^3 - 6ab = -11$. Prove that $-\frac{7}{3} < a + b < -2$.

**Problem 5.17.** Ana thinks of four real numbers and on the table writes all products of some two of them (six numbers in total). Then Banana erases one number, and the five numbers that remain on the table are consecutive positive integers. Determine the number that Banana erased.

**Problem 5.18.** Ana wants to find the smallest $2n + 1$ consecutive positive integers such that the sum of the squares of the $n + 1$ smallest ones is equal to the sum of the squares of the $n$ biggest ones. If such numbers exist, Ana writes them in the $n$-th row of her pyramid; if they do not exist, the $n$-th row remains empty. For example, the first three rows of Ana's pyramid are

$$3^2 + 4^2 = 5^2, \qquad 10^2 + 11^2 + 12^2 = 13^2 + 14^2, \qquad 21^2 + 22^2 + 23^2 + 24^2 = 25^2 + 26^2 + 27^2.$$

Thus the numbers $3, 4, 5$ are in the first row; the numbers $10, 11, 12, 13, 14$ are in the second row; and the numbers $21, \ldots, 27$ are in the third row. Does the number 2011 belong to some row of Ana's pyramid?

**Problem 5.19.** Let $a, b, c, d$ be real numbers such that $abcd = 1$ and

$$a + b + c + d = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}.$$

Prove that some two of the numbers $ab$, $ac$, $ad$, $bc$, $bd$, $cd$ are equal.

**Problem 5.20.** Let $a$ and $b$ be two positive numbers such that each of the equations

$$(a + b - x)^2 = a - b \qquad \text{and} \qquad (ab + 1 - x)^2 = ab - 1$$

has two distinct solutions. If the bigger solutions are equal, prove that the smaller solutions are also equal.

**Problem 5.21.** Let $a, b, c$ be positive real numbers. Is it possible that among the six numbers $a$, $b$, $c$, $\frac{a^2}{b}$, $\frac{b^2}{c}$, $\frac{c^2}{a}$ there are exactly three different values?

**Problem 5.22.** Let $x, y, z, t$ be nonzero real numbers. Prove that the difference between some two of the twelve fractions

$$\frac{x}{y}, \frac{x}{z}, \frac{x}{t}, \frac{y}{x}, \frac{y}{z}, \frac{y}{t}, \frac{z}{x}, \frac{z}{y}, \frac{z}{t}, \frac{t}{x}, \frac{t}{y}, \frac{t}{z}$$

is at most $\frac{11}{60}$.†

**Problem 5.23.** Prove that there do not exist an integer $n > 1$ and real numbers $a, b, c$ such that

$$\{\, a + b + c,\ ab + bc + ca,\ abc \,\} = \{\, n,\ n + 1,\ n + 2 \,\}.$$

**Problem 5.24.** Let $x_1, x_2, \ldots, x_{2025} \in \{-1, 0, 1, 2\}$ be such that

$$x_1 + x_2 + \cdots + x_{2025} = 111 \qquad \text{and} \qquad x_1^2 + x_2^2 + \cdots + x_{2025}^2 = 999.$$

Find the maximum value of $x_1^3 + x_2^3 + \cdots + x_{2025}^3$.

## 5.2 Vieta's formulas and symmetric expressions

Everything that a polynomial does to its coefficients, it also does to its roots. The dictionary between the two is provided by the elementary symmetric polynomials.

**Definition 5.1 (Elementary symmetric polynomials).** The *elementary symmetric polynomials* in $n$ variables $x_1, \ldots, x_n$ are the polynomials $\sigma_1, \sigma_2, \ldots, \sigma_n$ defined by

$$\sigma_k = \sigma_k(x_1, x_2, \ldots, x_n) = \sum x_{i_1} x_{i_2} \cdots x_{i_k},$$

where the summation runs over all $k$-element subsets $\{i_1, \ldots, i_k\}$ of $\{1, 2, \ldots, n\}$.

**Theorem 5.2 (Vieta's relations).** *If $x_1, x_2, \ldots, x_n$ are the $n$ zeros of $P(x) = x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_n$, then*

$$a_k = (-1)^k \sigma_k(x_1, \ldots, x_n) \qquad \textit{for } k = 1, 2, \ldots, n.$$

**Theorem 5.3 (Symmetric polynomials).** *Every symmetric polynomial $P(x_1, \ldots, x_n)$ can be expressed as a polynomial in $\sigma_1, \sigma_2, \ldots, \sigma_n$. This polynomial will have integer coefficients if $P$ does.*

**Theorem 5.4 (Rational root theorem).** *If a rational number $\frac{p}{q}$ with $q \neq 0$ and $(p, q) = 1$ is a zero of the polynomial $P(x) = a_n x^n + \cdots + a_0$ with integer coefficients, then $p \mid a_0$ and $q \mid a_n$.*

**Lemma 5.5 (Integer differences).** *If $P$ is a polynomial with integer coefficients and $a, b$ are different integers, then $P(a) - P(b)$ is divisible by $a - b$.*

**Theorem 5.6 (Cardano's formula).** *The cubic equation $x^3 + px + q = 0$ has the solution*

$$x = \sqrt[3]{-\frac{q}{2} + \sqrt{D}} + \sqrt[3]{-\frac{q}{2} - \sqrt{D}}, \qquad \text{where} \qquad D = \left(\frac{q}{2}\right)^2 + \left(\frac{p}{3}\right)^3$$

*is the discriminant.*

**Problem 5.25.** Cardano's formula above produces one solution of $x^3 + px + q = 0$. What are the other two solutions?

**Problem 5.26.** Prove that the polynomial $x^3 - 3x + 1$ has three distinct real zeros. What is the product of these zeros?

**Problem 5.27.** All four roots of the polynomial $f(x) = x^4 + px^3 + qx^2 + rx + s$ are negative integers. If $p + q + r + s = 99$, find $s$.

**Problem 5.28.** A monic polynomial $F(x)$ of degree 4 satisfies $F(1) = 10$, $F(3) = 30$ and $F(5) = 50$. Determine $F(2) + F(4)$.

**Problem 5.29.** A polynomial $P$ of degree $n$ is such that $P(x) = 2^x$ for $x = 0, 1, 2, \ldots, n$. How much is $P(n + 1)$?

**Problem 5.30.** If $P$ is a polynomial of degree $n$ such that $P(k) = \dfrac{1}{k}$ for $k = 1, 2, \ldots, n + 1$, what is $P(0)$?

**Problem 5.31.** Find constants $A, B, C, D, E$ such that

$$\frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3} + \frac{D}{x-5} + \frac{E}{x-7} = \frac{x}{(x-1)(x-2)(x-3)(x-5)(x-7)},$$

if they exist.

**Problem 5.32.** Find a cubic polynomial with integer coefficients having $\sqrt[3]{2 + \sqrt{3}} + \sqrt[3]{2 - \sqrt{3}}$ as a zero.

**Problem 5.33.** If a polynomial $P(x) = x^3 + ax^2 + bx + c$ with real coefficients has three real zeros in the interval $[0, 2]$, prove that $-2 \leqslant a + b + c \leqslant 0$.

**Problem 5.34.** Find all real parameters $a$ for which the equation $x^4 - ax^3 + x^2 + a = 0$ has two zeros (possibly complex) whose sum is 1.

**Problem 5.35.** Prove that the polynomial $x^4 + px + q$ never has four real zeros.

**Problem 5.36.** If $u, v, w$ are the zeros of the polynomial $x^3 - x - 1$, compute $u^{11} + v^{11} + w^{11}$.

**Problem 5.37.** Find all polynomials all of whose coefficients are 1 or $-1$ and all of whose zeros are real.

**Problem 5.38.** Suppose $a, b, c, d$ are the four roots of $P(x) = x^4 - 4x - 1$. Form a cubic polynomial whose roots are $ab + cd$, $ac + bd$ and $ad + bc$. Then find $a, b, c, d$.

**Problem 5.39.** A polynomial $P$ with integer coefficients takes values $\pm 1$ at three different integer points. Prove that this polynomial has no integer zeros.

**Problem 5.40.** Is there a polynomial $P(x)$ with integer coefficients such that $P(1) = P(5) = 3$ and $P(3) = 2021$?

**Problem 5.41.** Let $P$ be a polynomial with integer coefficients. If $P(3) = 7$ and $P(P(0)) = 100$, what is $P(0)$?

**Problem 5.42.** A polynomial $P$ is such that $P(x)$ is an integer whenever $x$ is an integer. Moreover, $a - b$ divides $P(a) - P(b)$ for all integers $a \neq b$. Can $P$ have non-integer coefficients?

**Problem 5.43.** Let $P(x)$ be a polynomial with integer coefficients and let $a$ be an integer. Suppose that $P(P(\cdots P(a) \cdots)) = a$, where $P$ is applied $n$ times. Prove that $P(P(a)) = a$.

**Problem 5.44.** Is there a polynomial $P(x)$, taking integer values at integer points, such that $2 \mid P(x)$ if and only if $4 \mid x$? Or such that $2 \mid P(x)$ if and only if $3 \mid x$?

**Problem 5.45.** Let $f(x)$ be a quadratic polynomial and $g(x)$ a cubic polynomial. Can $f(g(x))$ have six distinct roots which are all powers of two?

**Problem 5.46.** Let $f, g$ be two quadratic polynomials with positive leading coefficients that assume the same minimal value. Prove that there are real numbers $a$ and $b$ such that $f(x) = g(ax + b)$.

**Problem 5.47.** Let $P(x)$, $Q(x)$, $R(x)$ be three monic quadratic polynomials such that

$$P(Q(x)) = (x-1)(x-3)(x-5)(x-7), \qquad Q(R(x)) = (x-2)(x-4)(x-6)(x-8).$$

Determine $P(0) + Q(0) + R(0)$.

**Problem 5.48.** Let $x, y, z$ be positive real numbers such that $x^4 - 23x^2 + 1 = 0$, $y^4 - 223y^2 + 1 = 0$ and $z^4 - 2023z^2 + 1 = 0$. Find all positive integers $n$ such that

$$x^2y^2z^2 - nxyz + 1 = (xy + 1)(yz + 1)(zx + 1).$$

## 5.3 Classical inequalities: AM–GM and Cauchy–Schwarz

### 5.3.1 First estimates

**Technique 5.7 (Elementary estimates).** The following remarks solve a surprising number of junior problems.

- $x^2 \geqslant 0$ for each $x \in \mathbb{R}$;
- $x^2 \leqslant x$ if and only if $0 \leqslant x \leqslant 1$;
- if $a > 0$ and $x > y > 0$, then $\dfrac{a}{x} < \dfrac{a}{y}$ and $\dfrac{x}{a} > \dfrac{y}{a}$;
- several useful identities (every $\pm$ should be replaced by the same sign): $(1 \pm x)(1 \pm y) = 1 \pm x \pm y + xy$ and $(x \pm y)^2 = x^2 \pm 2xy + y^2$.

**Problem 5.49.** Prove that if real numbers $x, y$ satisfy $x^2 + x \leqslant y$, then $y^2 + y \geqslant x$.

**Problem 5.50.** Prove that if $a, b, c$ are positive real numbers, then

$$\frac{a}{\sqrt{a+b}} + \frac{b}{\sqrt{b+c}} + \frac{c}{\sqrt{c+a}} > \sqrt{a+b+c}.$$

**Problem 5.51.** Prove that if $a, b, c$ are positive real numbers, then

$$\frac{a}{a+1} + \frac{b}{(a+1)(b+1)} + \frac{c}{(a+1)(b+1)(c+1)} < 1.$$

**Problem 5.52.** Real numbers $a, b, c, d$ satisfy $a + b = cd$ and $c + d = ab$. Prove that $(a+1)(b+1)(c+1)(d+1) \geqslant 0$.

**Problem 5.53.** Given are real numbers $a, b \in (0, 1)$. Prove that $a\sqrt{b} + b\sqrt{a} + 1 > 3ab$.

**Problem 5.54.** Given are real numbers $x, y \in (0, 1)$. Prove that $x(1-y)^2 + y(1-x)^2 < (1-xy)^2$.

**Problem 5.55.** Positive numbers $a, b, c$ satisfy $a \leqslant 1$, $b \leqslant 2$, $c \leqslant 3$. Prove that $a + b + c \geqslant abc$.

**Problem 5.56.** Distinct positive numbers $a, b$ satisfy $a + b = 1$. Prove that

$$\left| \frac{a-b}{\sqrt{1-a^2} - \sqrt{1-b^2}} \right| < \sqrt{3}.$$

**Problem 5.57.** Prove that if $a, b, c \in [0, 1]$, then $\dfrac{a}{1+bc} + \dfrac{b}{1+ca} + \dfrac{c}{1+ab} \leqslant 2$.

**Problem 5.58.** Prove that

$$\frac{1}{2\sqrt{n}} \leqslant \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdot \ldots \cdot \frac{2n-1}{2n} < \frac{1}{\sqrt{2n+1}}$$

for every integer $n \geqslant 1$.

### 5.3.2 Squares, AM–GM and Cauchy–Schwarz

**Theorem 5.8 (Basic inequalities).** *For all $x, y, z$ as indicated:*

- *if $x, y > 0$, then $\dfrac{x+y}{2} \geqslant \sqrt{xy}$, with equality only for $x = y$;*
- *if $x, y, z \in \mathbb{R}$, then $x^2 + y^2 + z^2 \geqslant xy + yz + zx$, with equality only for $x = y = z$;*
- *if $x, y > 0$, then $\dfrac{x}{y} + \dfrac{y}{x} \geqslant 2$, $\quad \dfrac{1}{x} + \dfrac{1}{y} \geqslant \dfrac{4}{x+y}$, $\quad \sqrt{2(x+y)} \geqslant \sqrt{x} + \sqrt{y}$, $\quad \sqrt{\dfrac{x^2+y^2}{2}} \geqslant$*

$\dfrac{x+y}{2}$.

**Theorem 5.9 (Sedrakyan's inequality).** *If $a_1, \ldots, a_n \in \mathbb{R}$ and $b_1, \ldots, b_n > 0$, then*

$$\frac{a_1^2}{b_1} + \frac{a_2^2}{b_2} + \cdots + \frac{a_n^2}{b_n} \geqslant \frac{(a_1 + a_2 + \cdots + a_n)^2}{b_1 + b_2 + \cdots + b_n},$$

*with equality exactly when $(a_1, a_2, \ldots, a_n) = (\lambda b_1, \lambda b_2, \ldots, \lambda b_n)$ for some $\lambda \in \mathbb{R}$. This is a form of the Cauchy–Schwarz inequality, also called its Engel form.*

**Problem 5.59.** Prove that for all positive real numbers $a, b, c$ one has $(a+b)(b+c)(c+a) \geqslant 8abc$.

**Problem 5.60.** Prove that for all real numbers $a, b, c$ one has $a^4 + b^4 + c^4 \geqslant abc(a + b + c)$.

**Problem 5.61.** Prove that if $a, b \in \mathbb{R}$ and $x, y > 0$, then $\dfrac{a^2}{x} + \dfrac{b^2}{y} \geqslant \dfrac{(a+b)^2}{x+y}$.

**Problem 5.62.** Prove that if $a, b > 0$ satisfy $a + b = 1$, then $\left(a + \dfrac{1}{a}\right)^2 + \left(b + \dfrac{1}{b}\right)^2 \geqslant \dfrac{25}{2}$.

**Problem 5.63.** Prove that if $a \geqslant b > 0$, then

$$\frac{(a-b)^2}{8a} \leqslant \frac{a+b}{2} - \sqrt{ab} \leqslant \frac{(a-b)^2}{8b}.$$

**Problem 5.64.** Prove that for all real numbers $a, b, c, d$ one has $a^2+b^2+c^2+d^2+1 \geqslant a+b+c+d$.

**Problem 5.65.** Prove that for all real numbers $a, b, c$ one has $a^2+b^2+c^2 \geqslant ab+bc+ca+\frac{3}{4}(a-b)^2$.

**Problem 5.66.** Suppose that positive real numbers $a, b, c$ satisfy $abc(a + b + c) = 1$. What is the minimum value of $(a + b)(a + c)$?

**Problem 5.67.** Prove that for all positive real numbers $a, b, c$ one has

$$abc(a + b + c) \geqslant 2a^2b^2 + 2b^2c^2 + 2c^2a^2 - a^4 - b^4 - c^4.$$

**Problem 5.68.** Real numbers $a, b, c, d$ satisfy $0 \leqslant a \leqslant b \leqslant c \leqslant d$ and $a + b + c + d \geqslant 1$. Prove that $a^2 + 3b^2 + 5c^2 + 7d^2 \geqslant 1$.

### 5.3.3 Substitution and homogenization

**Technique 5.10 (Substitution).** It is often useful to introduce new variables — for differences, for denominators, for ratios — and to restate the desired inequality in terms of them.

**Definition 5.11 (Homogeneous expression).** An expression $\Phi(x_1, \ldots, x_n)$ is *homogeneous of degree $d$* if $\Phi(\lambda x_1, \ldots, \lambda x_n) = \lambda^d \Phi(x_1, \ldots, x_n)$ for every $\lambda > 0$; intuitively, scaling every variable by the same constant scales the whole expression uniformly.

**Technique 5.12 (Homogenization).** Rewriting an inequality in an equivalent homogeneous form $\Phi(x_1, \ldots, x_n) \geqslant 0$ is particularly useful when the variables are tied together by a non-homogeneous condition, which can then be used to eliminate the constants.

**Problem 5.69.** Prove Nesbitt's inequality: for all positive real numbers $a, b, c$,

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geqslant \frac{3}{2}.$$

**Problem 5.70.** Prove that if positive real numbers $a, b, c$ satisfy $abc = 1$, then

$$\frac{1}{a+ab} + \frac{1}{b+bc} + \frac{1}{c+ca} \geqslant \frac{3}{2}.$$

**Problem 5.71.** Prove that if positive real numbers $a, b, c$ satisfy $a + b + c = 1$, then $a^2 + b^2 + c^2 + \sqrt{12abc} \leqslant 1$.

**Problem 5.72.** Prove that for all positive real numbers $a, b, c$ one has $2^{a+b} + 2^{b+c} + 2^{c+a} < 2^{a+b+c+1} + 1$.

**Problem 5.73.** Prove that for all real numbers $a, b, c$ one has

$$\max(a, b, c) - \min(a, b, c) \leqslant \sqrt{\tfrac{4}{3}\,(p^2 - 3q)},$$

where $p = a + b + c$ and $q = ab + bc + ca$.

**Problem 5.74.** Given are $n \geqslant 2$ positive real numbers $a_1, a_2, \ldots, a_n$ such that $a_1 a_2 \cdots a_n = 1$. Prove that

$$1 \leqslant \frac{1}{1+a_1} + \frac{1}{1+a_2} + \cdots + \frac{1}{1+a_n} \leqslant n - 1.$$

**Problem 5.75.** Positive real numbers $a, b, c$ satisfy $a+b+c = 1$. Prove that $(1+a)(1+b)(1+c) \geqslant 8(1-a)(1-b)(1-c)$.

**Problem 5.76.** Positive real numbers $a, b, c$ satisfy $a^2+b^2+c^2 = 1$. Prove that $\frac{ab}{c}+\frac{bc}{a}+\frac{ca}{b} \geqslant \sqrt{3}$.

**Problem 5.77.** Positive real numbers $a, b, c$ satisfy $a+b+c = 1$. Prove that $\sqrt{a+bc}+\sqrt{b+ca}+\sqrt{c+ab} \leqslant 2$.

**Problem 5.78.** Real numbers $a, b, c$ satisfy $a^2 + b^2 + c^2 = 2$. Prove that $a + b + c \leqslant abc + 2$.

**Problem 5.79.** Let $a, b, c, d$ be four real numbers such that $a \geqslant b \geqslant c \geqslant d > 0$ and $a+b+c+d = 1$. Prove that

$$(a + 2b + 3c + 4d)\left(a^2 + b^2 + c^2 + d^2\right) < 1.$$

**Problem 5.80.** Let $a, b, c, d$ be four real numbers such that $a \geqslant b \geqslant c \geqslant d > 0$ and $a+b+c+d = 1$. Prove that

$$(a + 2b + 3c + 4d)\, a^a b^b c^c d^d < 1.$$

[IMO 2020]

### 5.3.4 Constrained inequalities and competition problems

**Problem 5.81.** Positive real numbers $a, b, c$ satisfy $a+b+c = 1$. Prove that $\sqrt{1-a^2}+\sqrt{1-b^2}+\sqrt{1-c^2} \leqslant 2\sqrt{2}$.

**Problem 5.82.** Prove that for every pair of real numbers $x, y$ one has $(x^2+1)(y^2+1) \geqslant 2(xy-1)(x+y)$.

**Problem 5.83.** Prove that for all positive real numbers $a, b, c$,

$$\frac{a^2+b^2}{a+b} + \frac{b^2+c^2}{b+c} + \frac{c^2+a^2}{c+a} \geqslant a+b+c.$$

**Problem 5.84.** Positive real numbers $a, b, c$ satisfy $abc = 1$. Prove that $\frac{1+ab}{1+a} + \frac{1+bc}{1+b} + \frac{1+ca}{1+c} \geqslant 3$.

**Problem 5.85.** Positive real numbers $a, b, c$ satisfy $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$. Prove that

$$\frac{1}{a-1} + \frac{1}{b-1} + \frac{1}{c-1} + \frac{a+b+c}{2} > \sqrt{abc}.$$

**Problem 5.86.** Prove that if positive real numbers $a, b, c$ satisfy $\frac{1}{a^2+1} + \frac{1}{b^2+1} + \frac{1}{c^2+1} = 2$, then

$$a^2+b^2+c^2 \geqslant \frac{3}{2} \geqslant ab+bc+ca.$$

**Problem 5.87.** Positive real numbers $a, b, c$ satisfy $abc = 1$. Prove that

$$\left(a+\frac{1}{b}\right)^2 + \left(b+\frac{1}{c}\right)^2 + \left(c+\frac{1}{a}\right)^2 \geqslant 3(a+b+c+1).$$

[JBMO 2014]

**Problem 5.88.** Let $a, b, c$ be positive real numbers. Prove that

$$\frac{8}{(a+b)^2+4abc} + \frac{8}{(b+c)^2+4abc} + \frac{8}{(a+c)^2+4abc} + a^2+b^2+c^2 \geqslant \frac{8}{a+3} + \frac{8}{b+3} + \frac{8}{c+3}.$$

[JBMO 2016]

**Problem 5.89.** Let $a, b, c$ be positive real numbers such that $abc = 1$. Prove that

$$\frac{bc}{b^5+c^5+bc} + \frac{ca}{c^5+a^5+ca} + \frac{ab}{a^5+b^5+ab} \leqslant 1.$$

**Problem 5.90.** Let $x, y, z$ be positive integers such that $x \neq y$, $y \neq z$ and $z \neq x$. Prove that

$$(x+y+z)(xy+yz+zx-2) \geqslant 9xyz.$$

When does equality hold? [JBMO 2017]

**Problem 5.91.** Show that

$$\left(a+2b+\frac{2}{a+1}\right)\left(b+2a+\frac{2}{b+1}\right) \geqslant 16$$

for all positive real numbers $a$ and $b$ such that $ab \geqslant 1$. [JBMO 2013]

**Problem 5.92.** Positive real numbers $a, b, c, d$ satisfy $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}=4$. Prove that

$$\sqrt[3]{\frac{a^3+b^3}{2}}+\sqrt[3]{\frac{b^3+c^3}{2}}+\sqrt[3]{\frac{c^3+d^3}{2}}+\sqrt[3]{\frac{d^3+a^3}{2}} \leqslant 2(a+b+c+d)-4.$$

[Poland 2007]

### 5.3.5 Choosing the weights

**Technique 5.13 (Weighted AM–GM).** For all $x_1, \ldots, x_n > 0$ and all $\alpha_1, \ldots, \alpha_n > 0$ with $\alpha_1 + \cdots + \alpha_n = 1$,

$$\alpha_1 x_1 + \cdots + \alpha_n x_n \geqslant x_1^{\alpha_1} \cdots x_n^{\alpha_n}.$$

In practice one first looks at the equality case of the inequality to be proved, and then chooses the weights $\alpha_i$ (or the number of copies of each term in a plain AM–GM) so that equality occurs exactly there.

**Problem 5.93.** Let $x$ and $y$ be real numbers such that $x > y > 0$. Prove the inequality

$$x+\frac{4}{(x-y)(y+1)^2} \geqslant 3.$$

**Problem 5.94.** Find all positive real numbers $a$ and $b$ such that $(1+a)(8+b)(a+b)=27ab$.

**Problem 5.95.** Let $x, y, z$ be positive real numbers such that $xy+yz+zx=1$. Prove that $10x^2+10y^2+z^2 \geqslant 4$.

**Problem 5.96.** Let $x, y, z$ be positive real numbers such that $xy+yz+zx=1$. Find the minimum value of $8x^2+8y^2+z^2$.†

**Problem 5.97.** Let $x, y, z, t$ be positive real numbers such that $xy+yz+zt+tx=1$. Find the minimum of the expression $5x^2+4y^2+5z^2+t^2$.

**Problem 5.98.** Let $a, b, c$ be positive real numbers. Prove the inequality

$$\frac{a}{b}+\sqrt{\frac{b}{c}}+\sqrt[3]{\frac{c}{a}} > 2.$$

**Problem 5.99.** Let $x, y, z$ be positive real numbers. Prove the inequality

$$x^3y^2+y^3z^2+z^3x^2 \geqslant x^2y^2z+y^2z^2x+z^2x^2y.$$

**Problem 5.100.** Let $a, b, c, d$ be positive real numbers such that $a^2+2b^3+3c^4+4d^5<1$. Prove that $2a+3b^2+4c^3+5d^4<3$.

**Problem 5.101.** Let $x_1, x_2, \ldots, x_n$ be real numbers with sum equal to 0. If the smallest among them is equal to $m$ and the largest is equal to $M$, prove the inequality

$$x_1^2 + x_2^2 + \cdots + x_n^2 \leqslant -nmM.$$

**Problem 5.102.** Let $x_1, x_2, \ldots, x_n$ be real numbers satisfying

$$x_1 + x_2 + \cdots + x_n = 0 \qquad \text{and} \qquad x_1^2 + x_2^2 + \cdots + x_n^2 = 1.$$

If the smallest among them is equal to $m$ and the largest is equal to $M$, prove that $-nmM \geqslant 1$.
[IMO 2019 Shortlist A2]

**Problem 5.103.** Let $x_0 > x_1 > \cdots > x_n$ be positive integers. Prove the inequality

$$x_0 + \frac{1}{x_0 - x_1} + \frac{1}{x_1 - x_2} + \cdots + \frac{1}{x_{n-1} - x_n} \geqslant x_n + 2n.$$

**Problem 5.104.** Let $a_1, a_2, \ldots, a_n$ be positive real numbers such that $a_1 + a_2 + \cdots + a_n < 1$. Prove the inequality

$$\frac{a_1 a_2 \cdots a_n \left[1 - (a_1 + a_2 + \cdots + a_n)\right]}{(a_1 + a_2 + \cdots + a_n)(1 - a_1)(1 - a_2) \cdots (1 - a_n)} \leqslant \frac{1}{n^{n+1}}.$$

**Problem 5.105.** Let $0 < a_1 < a_2 < \cdots$ be an unbounded sequence of real numbers. Prove that there is a positive integer $k$ such that

$$\frac{a_1}{a_2} + \frac{a_2}{a_3} + \cdots + \frac{a_k}{a_{k+1}} < k - 1985.$$

**Problem 5.106.** Positive real numbers $x, y, z$ satisfy the inequality $x + y + z \geqslant \dfrac{1}{x} + \dfrac{1}{y} + \dfrac{1}{z}$. Prove that then also $x + y + z \geqslant \dfrac{3}{xyz}$.

**Problem 5.107.** Let $a, b, c$ be positive real numbers such that $abc = 8$. Prove the inequality

$$\frac{a^2}{\sqrt{(1 + a^3)(1 + b^3)}} + \frac{b^2}{\sqrt{(1 + b^3)(1 + c^3)}} + \frac{c^2}{\sqrt{(1 + c^3)(1 + a^3)}} \geqslant \frac{4}{3}.$$

**Problem 5.108.** Let $a, b, c$ be positive real numbers such that $abc = 1$. Prove the inequality

$$\frac{a}{a^2 + 2} + \frac{b}{b^2 + 2} + \frac{c}{c^2 + 2} \leqslant 1.$$

**Problem 5.109.** Find the greatest value of $A$ such that the inequality

$$\frac{x}{\sqrt{y^2 + z^2}} + \frac{y}{\sqrt{z^2 + x^2}} + \frac{z}{\sqrt{x^2 + y^2}} \geqslant A$$

holds for all positive real numbers $x, y, z$.

**Problem 5.110.** If $a, b, c$ are positive real numbers such that $abc = 1$, prove the inequality

$$\frac{1}{\sqrt{b+\frac{1}{a}+\frac{1}{2}}}+\frac{1}{\sqrt{c+\frac{1}{b}+\frac{1}{2}}}+\frac{1}{\sqrt{a+\frac{1}{c}+\frac{1}{2}}} \geqslant \sqrt{2}.$$

**Problem 5.111.** ★ Let $a_1, a_2, \ldots, a_n$ be positive real numbers. Prove the inequality

$$a_1+\sqrt{a_1a_2}+\sqrt[3]{a_1a_2a_3}+\cdots+\sqrt[n]{a_1a_2\cdots a_n} < 3(a_1+a_2+\cdots+a_n).$$

**Problem 5.112.** ★ Let $a_1, a_2, \ldots, a_n$ be positive real numbers. Prove the inequality

$$\frac{1}{a_1}+\frac{2}{a_1+a_2}+\cdots+\frac{n}{a_1+a_2+\cdots+a_n} < 2\left(\frac{1}{a_1}+\frac{1}{a_2}+\cdots+\frac{1}{a_n}\right).$$

**Problem 5.113.** Let $a_2, a_3, \ldots, a_n$, where $n \geqslant 3$, be positive real numbers such that $a_2a_3\cdots a_n = 1$. Prove the inequality

$$(1+a_2)^2(1+a_3)^3\cdots(1+a_n)^n > n^n.$$

[IMO 2012]

### 5.3.6 Ordering and the discriminant

**Technique 5.14 (The discriminant).** In order to prove that $Ax^2+Bx+C \geqslant 0$ for all real $x$, one needs $A > 0$ and $B^2-4AC \leqslant 0$. Many inequalities in several variables can be attacked by regarding one chosen variable as the unknown of a quadratic and estimating the resulting discriminant.

**Technique 5.15 (Ordering the variables).** Some tips for inequalities that are not directly amenable to the standard means:

- use AM–GM in a fruitful way;
- try to establish some ordering between the variables (for instance, that $b$ lies between $a$ and $c$, so that $(b-a)(b-c) \leqslant 0$);
- use elementary algebra to factorize some expressions;
- adopt the functional point of view: use linear and second degree expressions and investigate their behaviour.

**Problem 5.114.** Let $a, b$ be real numbers. Prove that $3(1-a+a^2)(1-b+b^2) \geqslant 2(1-ab+a^2b^2)$.

**Problem 5.115.** Let $a, b, c$ and $x, y, z$ be real numbers. Prove that:
(i) if $ab+bc+ca > 0$, then $[(b+c)x+(c+a)y+(a+b)z]^2 \geqslant 4(ab+bc+ca)(xy+yz+zx)$;
(ii) if $a, b, c \geqslant 0$, then $[(b+c)x+(c+a)y+(a+b)z]^2 \geqslant 4(a+b+c)(ayz+bzx+cxy)$.

**Problem 5.116.** Let $a, b, c$ be real numbers. Prove that $3(1-a+a^2)(1-b+b^2)(1-c+c^2) \geqslant 1+abc+a^2b^2c^2$.

**Problem 5.117.** Let $x, y > 0$ satisfy $x+y^2 = y^3+1$. Prove that $y+x^2 \leqslant x^3+1$.

**Problem 5.118.** Elyas was tasked with building a computer that would calculate the square of the sum of three non-negative numbers not greater than 1. Instead, Elyas chose functions $f, g, h$,

and when the numbers $x, y, z$ are entered into the computer, it outputs the sum $f(x)+g(y)+h(z)$. The result of the calculation is, of course, not always exact, but Muhammad says that the calculation error (that is, the absolute value of the difference between the desired square and the result obtained) never exceeds a certain number $m$. Find the smallest possible $m$.

**Problem 5.119.** Let $a, b, c > 0$ with $a + b + c = 3$. Prove that $a^2 + b^2 + c^2 + abc \geqslant 4$.

**Problem 5.120.** Let $x, y, z > 0$ satisfy $xy + yz + zx + 2xyz = 1$. Prove that $4x + y + z \geqslant 2$.

**Problem 5.121.** Let $a, b, c \in [1, 2]$. Find the maximum value of $(a - b)^4 + (b - c)^4 + (c - a)^4$.

**Problem 5.122.** Let $a, b, c \in [2, 4]$. Prove that

$$\frac{c}{a^2b^2 + c^2} + \frac{b}{a^2c^2 + b^2} + \frac{a}{b^2c^2 + a^2} \leqslant \frac{1}{a^2 + b^2} + \frac{1}{b^2 + c^2} + \frac{1}{c^2 + a^2}.$$

**Problem 5.123.** Let $a, b, c \in [1, 2]$. Prove that

$$\frac{1}{1 + a + b^2} + \frac{1}{1 + b + c^2} + \frac{1}{1 + c + a^2} \leqslant \frac{3}{a + b + c}.$$

**Problem 5.124.** Let $a, b, c \geqslant 0$ with $a + b + c = 3$. Prove that $ab^2 + bc^2 + ca^2 \leqslant 4$.

**Problem 5.125.** If $a, b, c$ are positive real numbers such that $a + b + c = 3$, prove that $(ab + bc + ca)(ab^2 + bc^2 + ca^2) \leqslant 9$.

**Problem 5.126.** Let $a, b, c$ be non-negative real numbers such that $a + b + c = 1$. Find the maximum value of

$$a(b - c)^3 + b(c - a)^3 + c(a - b)^3.$$

**Problem 5.127.** Let $x, y, z > 0$ be such that $x + y^2 + z^3 = 1$. Prove that

$$\frac{x}{2 + xy} + \frac{y}{2 + yz} + \frac{z}{2 + xz} > \frac{1}{2}.$$

**Problem 5.128.** If $a, b, c$ are non-negative real numbers such that $a^2 + b^2 + c^2 = 3$, prove that $ab^2 + bc^2 + ca^2 \leqslant abc + 2$.

### 5.3.7 Symmetric and cyclic inequalities

**Theorem 5.16 (Muirhead's inequality).** *For a triple $(x_1, x_2, x_3)$ of real numbers write*

$$T_{x_1,x_2,x_3} = \sum_{(y_1,y_2,y_3)} a^{y_1} b^{y_2} c^{y_3},$$

*the sum running over all permutations $(y_1, y_2, y_3)$ of $(x_1, x_2, x_3)$; here $a, b, c > 0$. Then $T_{x_1,x_2,x_3} \geqslant T_{y_1,y_2,y_3}$ if and only if $(x_1, x_2, x_3)$ majorizes $(y_1, y_2, y_3)$, that is,*

$$x_1 \geqslant y_1, \qquad x_1 + x_2 \geqslant y_1 + y_2, \qquad x_1 + x_2 + x_3 = y_1 + y_2 + y_3$$

*(the triples being written in decreasing order).*

**Problem 5.129.** Let $a, b, c$ be positive real numbers such that $abc = 1$. Prove that

$$\frac{a^3}{(1+b)(1+c)} + \frac{b^3}{(1+a)(1+c)} + \frac{c^3}{(1+a)(1+b)} \geqslant \frac{3}{4}.$$

**Problem 5.130.** Let $a \leqslant b \leqslant c \leqslant d$ be positive real numbers. Prove that

$$ab^3 + bc^3 + cd^3 + da^3 \geqslant a^2b^2 + b^2c^2 + c^2d^2 + d^2a^2.$$

**Problem 5.131.** Let $x, y, z$ be positive real numbers such that $x + y + z = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$. Prove that

$$x + y + z \geqslant \sqrt{\frac{xy+1}{2}} + \sqrt{\frac{yz+1}{2}} + \sqrt{\frac{zx+1}{2}}.$$

**Problem 5.132.** Let positive real numbers $x_1, x_2, \ldots, x_n$ satisfy

$$\frac{1}{1+x_1} + \frac{1}{1+x_2} + \cdots + \frac{1}{1+x_n} = 1.$$

Show that $x_1x_2\cdots x_n \geqslant (n-1)^n$.

**Problem 5.133.** Let $0 < a, b, c < 1$ be real numbers. Prove that

$$a + b + c + 2abc > ab + bc + ca + 2\sqrt{abc}.$$

**Problem 5.134.** Let $a, b, c$ be positive real numbers. Prove that

$$abc\,(a^2+2b^2)(b^2+2c^2)(c^2+2a^2) \leqslant (a+b+c)^3(a^2-ab+b^2)(a^2-ac+c^2)(b^2-bc+c^2).$$

**Problem 5.135.** Let $a, b, c$ be positive real numbers. Prove that

$$\frac{a}{9bc+1} + \frac{b}{9ac+1} + \frac{c}{9ab+1} \geqslant \frac{a+b+c}{1+(a+b+c)^2}.$$

**Problem 5.136.** Let $a, b, c$ be positive real numbers such that $ab + bc + ca = 3$, and let $n$ be a positive integer. Prove that

$$\frac{a}{\sqrt{a^n+2n-1}} + \frac{b}{\sqrt{b^n+2n-1}} + \frac{c}{\sqrt{c^n+2n-1}} \leqslant \frac{3}{\sqrt{2n}}.$$

**Problem 5.137.** ★ Let $x, y, z$ be three positive reals such that $xyz \geqslant 1$. Prove that

$$\frac{x^5-x^2}{x^5+y^2+z^2} + \frac{y^5-y^2}{x^2+y^5+z^2} + \frac{z^5-z^2}{x^2+y^2+z^5} \geqslant 0.$$

**Problem 5.138.** ★ Let $a, b, c$ be positive real numbers. Prove that

$$(a^2+ac+c^2)\left(\frac{1}{a+b+c} + \frac{1}{a+c}\right) + b^2\left(\frac{1}{b+c} + \frac{1}{a+b}\right) > a+b+c.$$

## 5.4 The floor function

Recall that $\lfloor x \rfloor$ denotes the greatest integer not exceeding $x$, and that $\{x\} = x - \lfloor x \rfloor$ is the fractional part of $x$. Two numbers have the same fractional part exactly when their difference is an integer; this simple observation is behind several of the problems below.

**Problem 5.139.** For $n \in \{1, 2, \ldots, 2022\}$ define $a_n = \left\lfloor \dfrac{n^2}{2022} \right\rfloor$. Determine the number of distinct members of this sequence.

**Problem 5.140.** Calculate

$$\left\lfloor \underbrace{\sqrt{2022 + \sqrt{2022 + \cdots + \sqrt{2022 + \sqrt{2022}}}}}_{\text{2022 square roots}} \right\rfloor .$$

**Problem 5.141.** Determine all possible values of the expression $\lfloor x \rfloor \cdot \left\lfloor \dfrac{2000}{x} \right\rfloor$ for positive real $x$.

**Problem 5.142.** Find all real numbers $a$ and $b$ such that the equality

$$\lfloor ax + by \rfloor + \lfloor bx + ay \rfloor = (a + b) \lfloor x + y \rfloor$$

is true for all real numbers $x$ and $y$.

**Problem 5.143.** Let $a$ and $n$ be positive integers such that $\left\{ \sqrt{n + \sqrt{n}} \right\} = \{\sqrt{a}\}$. Prove that $4a + 1$ is a perfect square.

**Problem 5.144.** Let $a$ be a positive odd integer which is not a perfect square, and let $m, n$ be strictly positive integers. Prove that:
(a) $\{m(a + \sqrt{a})\} \neq \{n(a - \sqrt{a})\}$;
(b) $\lfloor m(a + \sqrt{a}) \rfloor \neq \lfloor n(a - \sqrt{a}) \rfloor$.

**Problem 5.145.** For $n \geqslant 2$ let $A_n$ denote the set of real solutions of the equation

$$x = \left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x}{3} \right\rfloor + \cdots + \left\lfloor \frac{x}{n} \right\rfloor .$$

(a) Determine the set $A_2 \cup A_3$.
(b) Prove that the set $A = \bigcup_{n \geqslant 2} A_n$ is finite, and find its maximum.

**Problem 5.146.** Consider the equation $\{x^2\} + \{x\} = c$ in positive rational numbers $x$.
(a) Prove that there are infinitely many solutions when $c = 0.99$.
(b) Prove that there are no solutions when $c = 1$.

**Problem 5.147.** Let $r_1, r_2, \ldots, r_m$ be rational numbers from the interval $(0, 1)$ such that $r_1 + r_2 + \cdots + r_m = 1$. For every $n \in \mathbb{N}$ define

$$f(n) = n - \sum_{i=1}^{m} \lfloor r_i n \rfloor .$$

Find the minimum and the maximum value of $f(n)$ for $n \in \mathbb{N}$.

**Problem 5.148.** Do there exist positive integers $m$ and $n$ such that $n \leqslant 2022$ and $\left\lfloor \dfrac{am}{n} \right\rfloor = \lfloor a\pi \rfloor$ for all $a \in \{1, 2, \ldots, 2022\}$?

**Problem 5.149.** Given real numbers $x_1, x_2, \ldots, x_n$, prove that there exists a real number $y$ such that

$$\{y - x_1\} + \{y - x_2\} + \cdots + \{y - x_n\} \leqslant \frac{n-1}{2}.$$

## 5.5 First functional equations

The first four problems of this section form a small guided course: Problem 5.152 is Cauchy's equation together with an extra hypothesis that forces the linear solution, and the three problems after it replace that hypothesis by other conditions leading to the same conclusion. The remaining problems are unrestricted; the last seven were left unsolved at the camp where the list was used.

**Problem 5.150.** Find all functions $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ satisfying the conditions

$$f(1) = 1, \qquad f\left(\frac{1}{x+y}\right) = f\left(\frac{1}{x}\right) + f\left(\frac{1}{y}\right), \qquad (x+y)f(x+y) = xy\, f(x)f(y)$$

for all $x, y$ with $xy(x+y) \neq 0$.

**Problem 5.151.** Find all functions $f : \mathbb{Q}^+ \to \mathbb{Q}^+$ satisfying

$$f(x+1) = f(x) + 1 \qquad \text{and} \qquad f(x^2) = f(x)^2$$

for all $x \in \mathbb{Q}^+$.

**Problem 5.152.** (Cauchy's equation with monotonicity.) Let $f : \mathbb{R} \to \mathbb{R}$. If $f(x+y) = f(x)+f(y)$ for all $x, y$ and $f$ is increasing, prove that $f(x) = f(1)x$ for all $x \in \mathbb{R}$.

**Problem 5.153.** Let $f : \mathbb{R} \to \mathbb{R}$ be additive and satisfy $f(x^2) = xf(x)$. Prove that $f(x) = f(1)x$ for all $x \in \mathbb{R}$.

**Problem 5.154.** Let $f : \mathbb{R} \to \mathbb{R}$ be additive and satisfy $f(x^2) = f(x)^2$. Prove that $f(x) = f(1)x$ for all $x \in \mathbb{R}$.

**Problem 5.155.** Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be additive and satisfy $f(x) = x^2 f\left(\frac{1}{x}\right)$. Prove that $f(x) = f(1)x$ for all $x \in \mathbb{R} \setminus \{0\}$.

**Problem 5.156.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying $f(x^2 - y) = xf(x) - f(y)$ for all $x, y \in \mathbb{R}$.

**Problem 5.157.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x^4 + f(y)) = y + f(x)^4$ for all $x, y \in \mathbb{R}$.

**Problem 5.158.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(2x^2 + y + f(y)) = 2y + 2f(x)^2$ for all $x, y \in \mathbb{R}$.

**Problem 5.159.** Find all functions $f : \mathbb{R} \setminus \{0\} \to \mathbb{R} \setminus \{0\}$ such that $f(x+y) = x^2 f\left(\frac{1}{x}\right) + y^2 f\left(\frac{1}{y}\right)$.

**Problem 5.160.** Let $f : \mathbb{R} \to \mathbb{R}$ be such that $f(x + y + xy) = f(x) + f(y) + f(xy)$ for all $x, y$. Prove that $f$ is additive.

**Problem 5.161.** Find all functions $f : \mathbb{N} \to \mathbb{N}$ satisfying $f(m+n) = f(m) + f(n) + mn$.

**Problem 5.162.** Find all functions $f : \mathbb{Q} \to \mathbb{Q}$ such that $f(x+y) + f(x-y) = 2f(x) + 2f(y)$ for all $x, y \in \mathbb{Q}$.

**Problem 5.163.** Find all $f : \mathbb{R} \to \mathbb{R}$ such that $f\big(f(x+y)f(x-y)\big) = x^2 - yf(y)$ for all $x, y \in \mathbb{R}$.

**Problem 5.164.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying $f(x^2 + y) = f(f(x) - y) + 4f(x)y$ for all $x, y \in \mathbb{R}$.

**Problem 5.165.** Find all $f : \mathbb{N} \to \mathbb{N}$ such that $f(m) + f(n) \mid m + n$ for all $m, n \in \mathbb{N}$.

**Problem 5.166.** Let $P$ be the set of all prime numbers. Find all functions $f : P \to P$ such that

$$f(p)^{f(q)} + q^p = f(q)^{f(p)} + p^q \qquad \text{for all } p, q \in P.$$

[BMO 2019]

**Problem 5.167.** Find all functions $f : \mathbb{Z} \to \mathbb{R}$ such that $f(m)+f(n) = f(mn)+f(m+n+mn)$ for all $m, n \in \mathbb{Z}$. [Japan 2013]

**Problem 5.168.** Find all real functions $f$ defined on $\mathbb{R}$ such that $f(f(x) + y) = f(f(x) - y) + 4f(x)y$ for all $x, y \in \mathbb{R}$. [BMO 2007]

**Problem 5.169.** Determine all functions $f : \mathbb{Z}^+ \to \mathbb{Z}^+$ such that $a^2 + f(a)f(b)$ is divisible by $f(a) + b$ for all positive integers $a, b$. [APMO 2019]

**Problem 5.170.** Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that $n + f(m) \mid f(n) + nf(m)$ for all $m, n \in \mathbb{N}$. [BMO 2017]

**Problem 5.171.** ★ Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $x\,[f(x+y) - f(x-y)] = 4yf(x)$ for all $x, y \in \mathbb{R}$. [Saudi TST 2016]

**Problem 5.172.** ★ Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(y+f(x)) = f(x)^2+2yf(x)+f(y)$ for all $x, y \in \mathbb{R}$.

**Problem 5.173.** ★ Find all $f : \mathbb{Z}^+ \to \mathbb{Z}^+$ such that $f(n+1) > f(f(n))$ for all $n \in \mathbb{Z}^+$.

**Problem 5.174.** ★ Find all $f : \mathbb{Q}^+ \to \mathbb{Q}^+$ such that $f(x + xy) = f(x) + f(xy) + 2y$.

**Problem 5.175.** ★ Find all $f : \mathbb{N}_0 \to \mathbb{N}_0$ such that $f(1) = 1$ and $f(m^2 + n^2) = f(m)^2 + f(n)^2$.

**Problem 5.176.** ★ Find all $f : \mathbb{N}_0 \to \mathbb{N}_0$ such that $mf(n) + nf(m) = (m + n)f(m^2 + n^2)$.

**Problem 5.177.** ★ Find all $f : \mathbb{Z}^+ \to \mathbb{Z}^+$ such that $f\big(f(m)^2 + 2f(n)^2\big) = m^2 + 2n^2$. [BMO 2009]

# Chapter 6

# Combinatorics

**Sources for this chapter:** Camp handouts by Nikola Petrović (Feb/March 2022), Antropov Alexander (March/April 2020), Bojan Bašić (March 2023), Łukasz Bożyk (KAUST and Mawhiba training material; May 2023), Emil Kolev (February 2024), Dušan Djukić (April–June 2021), Rafael Kazuhiro Miyazaki (March/April 2023), Majid Almarhoumi (December 2019), Asaad Mohammedsaleh (Winter Camp 2021), and the January 2025 camp leaflets.

Combinatorics at this level is less about formulas than about habits: count a set in two different ways, match it against another set, find a quantity that never changes, colour the board so that every tile looks the same, or look at the very first or very last moment of a process. The sections below train these habits one at a time.

## 6.1 Counting and bijections

### 6.1.1 Basic enumeration

**Problem 6.1.** How many rectangles can be seen in the picture of a $4 \times 4$ square divided into unit squares?

**Problem 6.2.** Let $n \geq 1$ be a positive integer. A square of side length $n$ is divided by lines parallel to its sides into $n^2$ unit squares. (a) Find the number of rectangles whose vertices are vertices of the unit squares. (b) How many of these rectangles are squares? (c) Find the number of parallelograms with vertices among these points, both of whose sides have length at most 2 and whose area equals 2. †

**Problem 6.3.** Two parallel lines are given, with $n$ green points marked on each of them. Determine the number of triangles all three of whose vertices are green.

**Problem 6.4.** On a line the points corresponding to the numbers $1, 2, \ldots, n$ are marked. In how many ways can we choose three of these points so that one of them lies exactly in the middle between the other two?

**Problem 6.5.** In the plane there are $n$ lines, no two of them parallel and no three of them concurrent. These lines divide the plane into regions, some of finite area and some of infinite area. Find the number of (a) all regions; (b) bounded regions.

**Problem 6.6.** Given a regular $n$-gon ($n \geq 3$), how many isosceles triangles are formed by three of its vertices?

**Problem 6.7.** On the boundary of a circle $n$ green points are given. All segments with both endpoints green are drawn, and no three of them meet at a common non-green point. Into how many pieces do the drawn segments divide the disc?

**Problem 6.8.** On the boundary of a circle $n$ green points are given. All segments with both endpoints green are drawn, and no three of them meet at a common non-green point. How many triangles can be seen in the resulting picture?

**Problem 6.9.** You are given a grid consisting of $n$ vertical lines and $n$ horizontal lines. How many ways are there of placing red and blue markers on the intersections of the lines so that every small square contains exactly two red markers and two blue markers?

**Problem 6.10.** Let $n > 3$ be a positive integer. An equilateral triangle $ABC$ is divided into $n^2$ smaller congruent equilateral triangles by lines parallel to its sides. How many parallelograms are formed by the lines of this picture?

**Problem 6.11.** In the picture of the previous problem, how many rhombi consisting of two small triangles are there? How many rhombi consisting of eight small triangles are there?

**Problem 6.12.** With the notation of the previous problem, let $m$ be the number of rhombi that contain two small equilateral triangles and $d$ the number of rhombi that contain eight small equilateral triangles. Find the difference $m - d$ in terms of $n$. [JBMO 2011, Problem 3]

**Problem 6.13.** How many ways are there to represent $21 = a_1 + a_2 + \cdots + a_k$ (where $k$ may vary) such that $a_1 = 1$, $a_{i+1} \in \{1, 2, 3, 4\}$ and $a_{i+1} \equiv a_i \pm 1 \pmod 4$ for $i = 1, 2, \ldots, k - 1$?

**Problem 6.14.** How many ways are there to colour the vertices of a regular pentagon with 4 colours, if two colourings are considered the same whenever one can be obtained from the other by a rotation?

**Problem 6.15.** Let $p$ be a prime and let $n$ be a positive integer. How many ways are there to colour the vertices of a regular $p$-gon with $n$ colours, if two colourings are considered the same whenever one can be obtained from the other by a rotation?

**Problem 6.16.** An equilateral triangle of side length $n$ is divided into unit triangles. Let $f(n)$ be the number of paths from the triangle in the top row to the middle triangle in the bottom row such that adjacent triangles in a path share a common edge, and such that the path never travels up (from a lower row to a higher row) and never revisits a triangle. Determine the value of $f(2005)$. [Canada 2005]

### 6.1.2 Bijective proofs

**Definition 6.1 (Partition of an integer).** A *partition* of a positive integer $n$ is a decomposition of $n$ into a sum of (not necessarily distinct) positive integers, called the *parts* or *terms*; for example $9 = 4 + 3 + 1 + 1$ is a partition of 9 into 4 parts. Two partitions that differ only in the order of the terms are considered equal.

**Problem 6.17.** Show that the number of subsets of $\{1, 2, \ldots, n\}$ with an even number of elements equals the number of subsets of $\{1, 2, \ldots, n\}$ with an odd number of elements.

**Problem 6.18.** Show that the number of subsets of $\{1, 2, \ldots, n\}$ with an even sum of elements equals the number of subsets of $\{1, 2, \ldots, n\}$ with an odd sum of elements.

**Problem 6.19.** Which subsets of $\{1, 2, \ldots, n\}$ are more numerous: those in which no two elements are consecutive numbers, or those which can be broken up into disjoint pairs of consecutive numbers?

**Problem 6.20.** Let $S_n$ be the set of vertices of a regular $n$-gon. One of these vertices is coloured black and all the others white. Let $V_n$ be the set of all $k$-gons, $k = 3, 4, \ldots, n$, whose vertices belong to $S_n$. Are there more elements of $V_n$ that do contain the black point as a vertex, or more that do not?

**Problem 6.21.** Let $p$ be a prime number and $k$ a positive integer smaller than $p$. Show that the number of subsets of $\{1, 2, \ldots, p\}$ whose sum is divisible by $p$ is larger by 2 than the number of subsets whose sum leaves remainder $k$ upon division by $p$. Use this to find the number of subsets of $\{1, 2, \ldots, p\}$ whose sum is divisible by $p$.

**Problem 6.22.** ★ Let $A_i$ be the number of subsets of $\{1, 2, \ldots, 3n\}$ whose sum leaves remainder $i$ upon division by 3. Show that $A_0 - 2^n = A_1 = A_2$ and use this to find $A_0$, $A_1$ and $A_2$.

**Problem 6.23.** Show that for $n \geq k$ the number of partitions of $n$ into $k$ parts equals the number of partitions of $n + \binom{k}{2}$ into $k$ distinct parts.

**Problem 6.24.** Let $n$ be a positive integer. Prove that the number of partitions of $n$ equals the number of partitions of $2n$ into exactly $n$ parts.

**Problem 6.25.** Let $n$ and $k$ be positive integers. Prove that the number of partitions of $n$ into at most $k$ parts equals the number of partitions of $n$ into parts none of which is larger than $k$.

**Problem 6.26.** ★ Let $n$ be a positive integer. Show that the number of partitions of $n$ into odd parts equals the number of partitions of $n$ into distinct parts.

## 6.2 Induction

**Problem 6.27.** Prove that $1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2$.

**Problem 6.28.** Find the error in the following proof by induction: "Let us prove for each $n$ that in every group of $n$ people all people are of the same height. This is obviously true for $n = 1$. Assume it is true for $n - 1$ and let us prove the statement for $n$ people $a_1, a_2, \ldots, a_n$. By the induction hypothesis $a_1, a_2, \ldots, a_{n-1}$ are of the same height, but it also follows that $a_2, a_3, \ldots, a_n$ are of the same height, and therefore $a_1, a_2, \ldots, a_n$ are all of the same height."

**Problem 6.29.** A tree of order $n$ is a connected graph with $n$ vertices which contains no cycles. Prove that it has exactly $n - 1$ edges.

**Problem 6.30.** There are $n$ lines in the plane which divide the plane into a number of regions. Define a boundary of two regions to be a common line segment. Prove that the regions can be coloured in two colours so that no two regions of the same colour have a common boundary. (Having only a common vertex does not count as having a common boundary.)

**Problem 6.31.** The famous Tower of Hanoi consists of three rods attached to the ground onto which disks with a hole in the centre can be placed. Initially, on one rod, disks labelled $1, 2, \ldots, n$ are placed from top to bottom. In each move one is allowed to remove the topmost disk from a rod and place it onto another rod, with the restriction that a disk may not be placed onto a disk

with a smaller number. Find the number of moves needed to move all the disks from one rod onto another rod.

**Problem 6.32.** Prove that the set of edges of a complete graph on $n$ vertices can be covered by disjoint paths of lengths $1, 2, \ldots, n-1$. (A path of length $k$ is a collection of $k$ edges arranged in a sequence so that each edge shares a common vertex with the following one; a path is allowed to visit a vertex several times.)

**Problem 6.33.** Show that for any $n \geq 6$ one can divide a unit square into $n$ squares.

**Problem 6.34.** A finite collection of natural numbers is written on a board. In each move you erase a single natural number $k$ from the board and in its place write an arbitrary finite collection of natural numbers smaller than $k$. (If you erase 1, you write nothing.) Show that eventually the board will be empty.

**Problem 6.35.** Peano's axioms, which establish the natural numbers, are as follows: 1 is a natural number; every natural number $n$ has a follower $s(n)$ (thus $s(1) = 2$, $s(s(1)) = 3$ and so on); 1 is not the follower of any number; if a set contains 1 and, for each natural number it contains, also its follower, then it contains all natural numbers; $s(m) = s(n)$ if and only if $m = n$; and the axioms of addition $n + 1 = s(n)$ and $n + s(m) = s(n + m)$. Use Peano's axioms to show that $m + n = n + m$ for any two natural numbers $m$ and $n$.

**Problem 6.36.** ★ (a) Prove that for each integer $n \geq 2$ a $2^n \times 2^n$ board with one field cut out can be tiled by L-trominoes. (b) Prove that if $n \geq 3$, then the number of such tilings is at least $3^{4^{n-3}}$. †

**Problem 6.37.** ★ Show that if an $n$-gon ($n \geq 3$) is partitioned into triangles by mutually non-intersecting diagonals (the $n$-gon need not be convex), then it is possible to colour its vertices in 3 colours so that the three vertices of each triangle of the partition receive different colours.

**Problem 6.38.** ★ Let $S_n$ be the collection of non-empty subsets of $\{1, 2, \ldots, n\}$ which contain no two consecutive numbers. For a set $A$ let $f(A)$ be the square of the product of all its elements, and let $b_n$ be the sum of $f(A)$ over all $A \in S_n$. Show that $b_n = (n + 1)! - 1$.

**Problem 6.39.** ★ At a science conference there are $2n$ scientists; each pair of scientists is either friendly or not, and a friendly pair counts as a single friendship. Show that if there are at least $n^2 + 1$ friendships, then there exist 3 scientists at the conference who are mutually friends.

**Problem 6.40.** ★ Prove that for each positive integer $k$ a $(3k + 1) \times (3k + 1)$ board with one field cut out can be tiled by trominoes, at most one of which is the I-tromino (the $3 \times 1$ rectangle).

**Problem 6.41.** ★ There are $n$ points in the interior of a unit square such that no two of them determine a line parallel to a side of the square. Let $P$ be a partition of the unit square into rectangles such that each of the $n$ points lies on the boundary of a rectangle. Prove that $P$ contains at least $n + 1$ rectangles.

## 6.3 Pigeonhole and the extreme principle

**Problem 6.42.** There are 40 balls in a basket. If we take any 17 balls from the basket, at least one of them will be red. If we take any 25 balls from the basket, at least one of them will be black. How many red balls can be in the basket? (Give all possible answers.)

**Problem 6.43.** Fifteen magazines lie casually on a table and completely cover it (magazines may lie on top of each other and hang over the edge of the table). Prove that one can remove 7 magazines so that the remaining ones still cover at least 8/15 of the table.

**Problem 6.44.** In the cells of a $10 \times 10$ table zeros and ones are placed, and it is known that out of any four rows of the table some two coincide. Prove that the table has two identical columns.

**Problem 6.45.** Forty-one different natural numbers less than 1000 are given. It is known that among any three of them there are two whose product is a perfect square. Prove that among these numbers there is a perfect square.

**Problem 6.46.** There are 25 points in the plane. It is known that out of any three of them there are two whose distance does not exceed 1. (a) Prove that two circles of radius one can cover all these points. (b) Prove that two $1 \times 2$ rectangles can cover all these points.

**Problem 6.47.** In the cells of a $7 \times 7$ square there are 100 crosses. There are three horizontal lines whose cells contain in total at least 70 crosses, and three such vertical lines. Prove that either there is a cell without a single cross in it, or there is a cell containing at least seven crosses. (Both situations occurring at once is also acceptable.)

**Problem 6.48.** A school has 298 children. All children have different weights and also different heights. It is known that no matter which 150 children you take, the tallest of them will at the same time be the heaviest of them. Prove that you can choose 150 children in such a way that the shortest of them is at the same time the lightest of them.

**Problem 6.49.** A set of $4n$ positive numbers is given. It is known that an arithmetic progression can be made up of any four pairwise different numbers of this set. Prove that there are at least $n$ identical numbers in the set.

**Problem 6.50.** In the small county of Lipshire there are only 5 estates, some of them connected by roads. It is known that any two roads have a common end. Prove that there are three estates no two of which are connected by a road.

**Problem 6.51.** Sixty people came to a party. It is known that among any three of them there is one who is acquainted with the other two. What is the smallest possible number of pairs of acquaintances among these 60 people?

**Problem 6.52.** Ten volleyball teams held a tournament in one round. It turned out that among any three of these teams there is a team that won against the other two. Prove that among these teams there is one that lost to everyone.

**Problem 6.53.** Seventy-eight people of different ages played tennis, and 310 games were played. (Each pair played at most one game with each other and there are no draws in tennis.) Prove that there are 4 people among whom the oldest, or the youngest, won against the remaining three.

**Problem 6.54.** In Martian basketball each team consists of exactly six players. The coach of the Mars national team can form a lineup from any six players among 100 candidates. Some lineups are considered cohesive by the coach, while others are not (at least one cohesive lineup exists). A group of five candidates is called *promising* if it can be completed with one more candidate to form a cohesive lineup. A candidate is called *universal* if they can complete any promising group of five candidates (that does not already include them) into a cohesive lineup. The coach assembled a team of six universal candidates. Must this team necessarily be cohesive?

**Problem 6.55.** Each cell of a $4 \times 4$ board contains either 0 or 1. We calculate ten sums: in the four rows, in the four columns, and on the two diagonals. Prove that at least three out of these ten sums are equal.

**Problem 6.56.** In every cell of a $9 \times 9$ board a positive integer is written. Then for every row and every column of this board the sum of all its entries is calculated. Is it possible that the eighteen calculated sums are eighteen consecutive integers in some order?

**Problem 6.57.** On each face of a cube an integer is written. Then for every edge the sum of the numbers from the two faces sharing this edge is calculated. Prove that among the twelve calculated sums there are at least four even numbers.

**Problem 6.58.** Can the vertices of a regular 30-gon be labelled with the numbers $1, 2, \ldots, 30$, each used exactly once, in such a way that the sum of the labels of every pair of neighbouring vertices is a perfect square?

**Problem 6.59.** Can the vertices of a regular 20-gon be labelled with the numbers $1, 2, \ldots, 20$, each used exactly once, in such a way that for every four consecutive vertices the sum of their labels is less than 43?

**Problem 6.60.** We are given $n$ identical candles. On the first day we light one of them for exactly one hour. On the second day we pick two candles and light them also for exactly one hour. In general, on the $k$-th day some $k$ candles are lit for exactly one hour. Suppose that on the $n$-th day all the candles burnt out simultaneously, right after burning for one hour. Determine all values of $n$ for which this is possible.

**Problem 6.61.** In a group of 6 people there are exactly 9 friendships (every friendship is mutual). Does it follow that (a) someone has at least four friends? (b) someone has exactly three friends? (c) everyone has at least one friend?

**Problem 6.62.** Suppose that in a group of $n \geq 4$ people everyone has a piece of information, and all these pieces are different. If two people make a phone call, they share all the information that they have at that moment. Prove that $2n - 4$ calls are enough to make everyone know everything.

**Problem 6.63.** At a party there are $4n$ people, each wearing a hat. Every two of them want to greet each other by exchanging the hats worn by the two greeters at that moment, and each pair of people should exchange hats exactly once. Prove that one can order the greetings in such a way that after all exchanges everyone has their own hat back.

**Problem 6.64.** In a group of $n \geq 5$ people no one is friends with everyone else, and every two people who are not friends have exactly one common friend. Prove that five of these people can sit at a round table in such a way that each of them has exactly two friends at the table and is sitting between them.

**Problem 6.65.** There are $n \geq 2$ guests at a party. Every person likes all other guests apart from exactly one (this is not necessarily mutual, i.e. it may happen that $A$ likes $B$ although $B$ does not like $A$). Prove that all guests can be seated at at most three tables in such a way that everyone likes all the people at their own table.

**Problem 6.66.** Let $n \geq 1$ be an odd number. Suppose $n$ arrows are arranged in a row so that each arrow points either to the left or to the right. Prove that there is an arrow which is

pointed at by precisely as many arrows as it points at. (For example, for $n = 5$ and the sequence $\leftarrow\leftarrow\rightarrow\rightarrow\leftarrow$, the consecutive arrows from left to right point at $0, 1, 2, 1, 4$ arrows respectively.)

**Problem 6.67.** A strange calculator has only two buttons, $-1$ and $\times 2$, whose effects are subtracting one from the number on the screen and doubling the number on the screen. Initially the number 1 is displayed. A positive integer is called *$n$-clickable* if the smallest number of button presses needed to obtain it on the screen is exactly $n$ (so 1 is 0-clickable and 8 is 3-clickable). (a) Find $n$ such that 2023 is $n$-clickable. (b) Express, in terms of $n$, the smallest $n$-clickable number. (c) Find all $n$ such that $n$ itself is $n$-clickable. (d) Let $F_n$ be the number of all $n$-clickable positive integers. Prove that $F_{n+2} = F_{n+1} + F_n$ for every $n \geq 1$.

**Problem 6.68.** The natural numbers from 1 to 50 are written on a blackboard. At least how many of them must be deleted so that the sum of any two of the remaining numbers is not a prime?

**Problem 6.69.** ★ The integers $1, 2, \ldots, 2n$ are arbitrarily assigned to boxes labelled $1, 2, \ldots, 2n$, one number per box. For each box we add the number assigned to it to the number on its label. Show that two of these sums give the same remainder modulo $2n$.

**Problem 6.70.** ★ Let $n \geq 5$ be a positive integer. Prove that the set $\{1, 2, \ldots, n\}$ can be partitioned into two non-empty subsets $S_n$ and $P_n$ such that the sum of the elements of $S_n$ equals the product of the elements of $P_n$.

**Problem 6.71.** ★ Let $S$ be a set of 100 positive integers with the following property: among every four numbers of $S$ there is a number which divides each of the other three, or there is a number which is equal to the sum of the other three. Prove that $S$ contains a number which divides all the other 99 numbers of $S$.

**Problem 6.72.** In the land of Heptanomisma four different coins and three different banknotes are used, and their denominations are seven different natural numbers. The denomination of the smallest banknote is greater than the sum of the denominations of the four coins. A tourist has exactly one coin of each denomination and exactly one banknote of each denomination, but he cannot afford the book on numismatics he wishes to buy. However, the mathematically inclined shopkeeper offers to sell the book to the tourist at a price of the tourist's choosing, provided that he can pay this price in more than one way. (A price can be paid in more than one way if there are two different subsets of his coins and notes whose denominations both add up to this price.) (a) Prove that the tourist can purchase the book if the denomination of each banknote is smaller than 49. (b) Show that the tourist may have to leave the shop empty-handed if the denomination of the largest banknote is 49. [Benelux 2018]

### 6.3.1 The extreme principle in processes

**Technique 6.2 (Choosing the right moment).** If a problem asks you to find some interesting moment of a process, it is often useful to look at an extreme moment: the last action performed, the first participant to finish, or the last one to start. Extreme moments carry information that an arbitrary moment does not. After the last action nothing happens any more; the first participant to finish has stopped acting although everybody else still has moves left; and so on. Looking at the very first move or the very last one is usually less useful, since these are just the ends of the process.

**Problem 6.73.** A circle contains 2021 nuts. A squirrel eats the nuts one by one in some order, eating each nut exactly once. Prove that at some moment the squirrel will eat a nut which is located between an already eaten nut and an uneaten nut.

**Problem 6.74.** Igor places checkers one by one on a $10 \times 10$ board. Prove that at some moment one of the checkers will be able to capture another.

**Problem 6.75.** During a class each student falls asleep exactly once. For every pair of students there was a moment when they were both asleep. Prove that at some (probably very boring) moment all the students were asleep simultaneously.

**Problem 6.76.** In an infinite sequence of natural numbers each number is obtained by adding to the previous number one of its nonzero digits. Prove that there will be an even number in this sequence.

**Problem 6.77.** On a $100 \times 100$ board there are 10 kings. Each minute one of them makes a move; a king cannot move into a square occupied by another king. Eventually every king visits every square exactly once and returns to its starting position. Prove that there was a moment when no king was on its starting square.

**Problem 6.78.** Each of 18 people knows a unique piece of news. Each person can send a letter to another person, sharing all the news they know. After some exchanges everyone knows all 18 pieces of news. What is the minimum number of letters that could have been sent?

**Problem 6.79.** At a festival several children bring cookies with them. Every minute one child gives each of their friends exactly one cookie, provided he has enough cookies to do so. At some point the sharing process stops, i.e. there is no child with enough cookies to continue sharing. Prove that there is at least one child who never gave any cookies to anyone.

**Problem 6.80.** There are 2023 dwarves arranged in a circle, each holding 10 emeralds. In one move each dwarf distributes all their emeralds to their neighbours. We say that a dwarf made an *exclusive transfer* if he gave all his emeralds (a nonzero amount) to only one of his neighbours. After several moves all the emeralds end up with a single dwarf. Prove that there were at least 4044 exclusive transfers.

**Problem 6.81.** In a $17 \times 17$ square table 80 cells are coloured black and the rest are white. It is allowed to paint an entire row or column black if the majority of the cells in that row or column are already black. Prove that it is impossible to make the entire table black using such operations.

### 6.3.2 Extremal problems

**Problem 6.82.** Each cell of a $5 \times 5$ board is either white or green. Suppose that in every $2 \times 2$ square there are at least 2 green cells. What is the minimum number of green cells in the entire board?

**Problem 6.83.** We are given a marker and two standard six-sided dice. We want to put more dots on their faces so that it is no longer possible to obtain 7 as a total when we roll both dice. What is the minimum possible number of extra dots?

**Problem 6.84.** Each cell of an $8 \times 8$ grid is either green or white. Suppose that in each $3 \times 3$ square there is an even number of white cells. What is the smallest possible total number of green cells?

**Problem 6.85.** On the perimeter of an $n \times n$ square ($n \geq 2$ an integer) several points are marked, different from the vertices of the square, dividing the sides into segments of integer lengths. What is the greatest number of marked points, assuming that no four of them are vertices of a parallelogram?

**Problem 6.86.** A subset $S$ of $\{1, 2, 3, \ldots, 2n\}$ has the following property: if $a, b \in S$ and $a \neq b$, then neither $a/b$ nor $b/a$ is an integer. What is the maximum possible size of $S$?

**Problem 6.87.** Each element of the set $\{1, 2, 3, \ldots, 2^n - 1\}$ is coloured with one of $c$ colours. It turns out that there are no two distinct numbers of the same colour with one being a divisor of the other. What is the minimum possible value of $c$?

**Problem 6.88.** Each integer from 1 to $n$ is coloured either red or blue. Suppose that no three numbers of the same colour form an arithmetic progression. What is the maximum possible value of $n$?

**Problem 6.89.** Find the smallest positive integer $k$ with the following property: it is possible to fill a $10 \times 10$ table with all the integers from 1 to 100 in such a way that every 10 consecutive numbers are contained in some $k \times k$ square of whole cells.

**Problem 6.90.** Each point of the plane is to be coloured with one of $n$ colours in such a way that each straight line contains points of at most two different colours and each colour is used at least once. Find the largest $n$ for which such a colouring is possible.

**Problem 6.91.** Given 2000 positive integers all of whose prime divisors are less than 25, prove that one can always find four distinct ones among them whose product is a fourth power.

**Problem 6.92.** At most how many numbers can we choose from the set $\{1, 2, \ldots, 15\}$ so that the product of no three of them is a perfect square?

**Problem 6.93.** At most how many numbers can we choose from the set $\{105, 106, \ldots, 210\}$ so that every two of them are coprime?

**Problem 6.94.** Let $A$ be an 8-element subset of the set $\{1, 2, \ldots, 17\}$. Prove that there exists $k > 0$ such that the equation $x - y = k$ has at least three solutions with $x, y \in A$.

**Problem 6.95.** At most how many numbers from the set $\{1, 2, \ldots, 2000\}$ can we choose so that no two of them differ by 4 or by 7?

**Problem 6.96.** Can all positive integers be divided into 100 nonempty classes so that there are no three different numbers $a$, $b$, $c$ from three different classes satisfying $a + 99b = c$?

**Problem 6.97.** Is it possible to write the numbers $1, 2, \ldots, 101$ around a circle in some order so that for every two adjacent numbers $x$ and $y$ we have $25 \leq |x - y| \leq 49$?

**Problem 6.98.** Let $N = 1000!$. Is it possible to order the numbers $1, 2, \ldots, N$ around a circle so that every number equals the previous number plus 17 or plus 28, modulo $N$?

**Problem 6.99.** Prove that there is a permutation $a_1, a_2, \ldots, a_{2021}$ of the numbers $1, 2, \ldots, 2021$ such that there are no indices $i < j < k$ for which $a_i - a_j = a_j - a_k$.

**Problem 6.100.** For each integer $n \geq 4$ we consider $m$ subsets $A_1, A_2, \ldots, A_m$ of $\{1, 2, 3, \ldots, n\}$ such that $A_1$ has exactly one element, $A_2$ has exactly two elements, $\ldots$, $A_m$ has exactly $m$

elements, and none of these subsets is contained in any other. Find the maximum value of $m$. [Cono Sur 2018]

**Problem 6.101.** Let $M = \{1, 2, \ldots, 49\}$. Determine the maximum integer $k$ such that $M$ has a subset of $k$ elements containing no 6 consecutive integers. For this value of $k$, find the number of subsets of $M$ with $k$ elements having the given property. [Ibero 2003]

**Problem 6.102.** In a table consisting of $2021 \times 2021$ unit squares, some unit squares are coloured black in such a way that if we place a mouse in the centre of any square of the table, it can walk in a straight line (up, down, left or right along a column or a row) and leave the table without walking on any black square, other than the initial one if it is black. What is the maximum number of squares that can be coloured black? [Centro American 2021]

**Problem 6.103.** In two bowls there are in total $N$ balls, numbered from 1 to $N$. One ball is moved from one of the bowls into the other. The average of the numbers in the bowls increases in both bowls by the same amount $x$. Determine the largest possible value of $x$. [Nordic 2002]

**Problem 6.104.** Let $n$ be a positive integer. There are $3n$ women's volleyball teams playing in a tournament in which no more than one match is played between any pair of teams. We know that $3n^2$ matches are played in this tournament. Prove that there exists a team that wins at least $n/4$ matches and loses at least $n/4$ matches. [China Girls MO 2022]

**Problem 6.105.** In a $12 \times 12$ grid each unit square is coloured either black or white in such a way that there is at least one black unit square in any $3 \times 4$ and any $4 \times 3$ rectangle bounded by grid lines. Determine, with proof, the minimum possible number of black unit squares. [China Girls MO 2015]

**Problem 6.106.** In a group of $m$ girls and $n$ boys, any two persons either know each other or do not know each other. For any two boys and any two girls, there are at least one boy and one girl among them who do not know each other. Prove that the number of unordered pairs {boy, girl} who know each other does not exceed $m + \frac{n(n-1)}{2}$. [China Girls MO 2013, Problem 3]

### 6.3.3 Ramsey numbers

**Definition 6.3 (Ramsey number).** The *Ramsey number* $R(m, n)$ is the smallest number of vertices $N$ such that in every colouring of the edges of the complete graph on $N$ vertices with two colours, red and blue, there always exists either a complete graph on $m$ vertices all of whose edges are red, or a complete graph on $n$ vertices all of whose edges are blue.

**Remark.** The first two problems below together determine the value $R(3, 3) = 6$; the fifth one asks for $R(3, 4)$. In general one shows by induction that $R(m, n)$ exists for all $m$ and $n$.

**Problem 6.107.** Prove that in any group of 6 people there either exist three, any two of whom are friends, or there exist three, no two of whom are friends.

**Problem 6.108.** Show that there exists a colouring of the edges of the complete graph on 5 vertices with two colours which contains no monochromatic triangle.

**Problem 6.109.** Prove that in a complete graph on 6 vertices whose edges are coloured with two colours there always exist at least two monochromatic triangles.

**Problem 6.110.** Show that in a complete graph on 7 vertices whose edges are coloured with two colours there exist at least four monochromatic triangles.

**Problem 6.111.** Prove that $R(3,4) = 9$.

## 6.4 Invariants and monovariants

**Technique 6.4 (How to construct an invariant).** A few standard patterns cover most problems. If one quantity decreases by $n$ while another increases by $n$, their sum is unchanged; if all quantities increase by $n$, all differences are unchanged. If some quantity always changes by $n$, its remainder modulo $n$ is invariant; if one operation decreases a quantity by $n$ and another increases it by $m$, work modulo $n+m$. If one object is always replaced by several objects, give the objects *weights*, choosing them so that the weight of an object equals the total weight of its replacements; the total weight is then invariant. A quantity that is not invariant but only ever moves in one direction (a *monovariant*) proves that a process must terminate.

**Problem 6.112.** Initially the numbers 2 and 5 are written on a board. A move consists of replacing one of the two numbers on the board by their sum. Is it possible to obtain, in a finite number of moves, a situation in which the two integers written on the board are consecutive?

**Problem 6.113.** There are three kinds of chameleons: green, red and blue. When two chameleons of two different colours meet, they both change their colour to the third colour. Can we start with 7 green, 8 red and 15 blue chameleons and end up with 10 chameleons of each colour after a finite number of steps?

**Problem 6.114.** There are amoebas of three colours. When two amoebas of different colours meet, they turn into six amoebas of the third colour. Initially there are 10 amoebas of the first colour, 20 of the second and 30 of the third. Can the numbers of all three colours ever become equal?

**Problem 6.115.** The numbers $\frac{1}{1}, \frac{1}{2}, \ldots, \frac{1}{2020}$ are written on a board. The operation we perform is to take any two numbers $a$ and $b$ and replace them by $ab + a + b$. After we perform this operation 2019 times, what is the final number left on the board?

**Problem 6.116.** The numbers from 1 to 1000 are written on a piece of paper. At each step we select any two numbers and replace them by their difference. Is it possible to have 243 in the end as the only remaining number?

**Problem 6.117.** The number $8^n$ is written on a board. The sum of its digits is calculated, then the sum of the digits of the result is calculated, and so on, until we get a single digit. What is this digit if $n = 2020$?

**Problem 6.118.** We have three sacks of stones containing 49, 51 and 5 stones. We are allowed two operations on the sacks: combine two sacks into one big sack; or divide a sack containing an even number of stones into two equal sacks. Is it possible to end up with 105 sacks each containing exactly one stone?

**Problem 6.119.** There is a heap of 1001 stones on a table. You are allowed to perform the following operation: you choose a heap containing more than one stone, throw away one stone

from that heap, and divide the rest into two smaller (not necessarily equal) heaps. Is it possible to reach a situation in which all the heaps on the table contain exactly 3 stones?

**Problem 6.120.** There is a pile of 600 shells on a table. One shell is removed and the pile is split into two non-empty parts, not necessarily equal. This process repeats: from any pile containing more than two shells, one shell is removed and the pile is split again. Is it possible that after several steps only piles of 2 and of 3 shells remain, with an equal number of each?

**Problem 6.121.** The vertices of a cube contain numbers: one vertex has 1 and all the others have 0. It is allowed to add the same number (not necessarily an integer) to both vertices connected by an edge. Can all the numbers be made equal?

**Problem 6.122.** Some sixth-graders initially have the same number of cones. Occasionally one of them gives an equal number of cones to each of the others. In the end Vasya has 30 cones and Petya has 13. How many sixth-graders were there?

**Problem 6.123.** In the children's game "Chips" trades occur as follows: one plastic chip can be exchanged for 3 Pokémon chips and 1 Chupa Chups chip, while one Pokémon chip can be exchanged for 2 Chupa Chups chips. Starting with 1000 Chupa Chups chips, is it possible to eventually have only plastic chips?

**Problem 6.124.** A single bacterium starts at $(0, 0)$. In each move a bacterium at $(i, j)$ disappears and new bacteria appear at $(i + 1, j)$ and $(i, j + 1)$, but only if both these positions were empty. Prove that at least one bacterium always remains in a cell where the sum of coordinates is at most 3.

**Problem 6.125.** A $2 \times 3$ grid is filled with the numbers $1, 2, 3$ in the first row and $4, 5, 6$ in the second row. In one move you can add the same number to any two adjacent cells. Can you transform the grid so that it contains five zeros and one one?

**Problem 6.126.** The numbers $10, 20, 30, 40, 50$ are written on a board. In one step you may pick any number and divide it by a prime number, and pick a (possibly different) number and multiply it by a (possibly different) prime number. You may choose different primes in different steps and even within the same step, and the numbers on the board need not remain integers during the process. Can you obtain the numbers $20, 30, 40, 50, 60$?

**Problem 6.127.** There are 43 trees in a circle. Every tree has one swift on it, except for one tree which has two. Every hour two swifts fly: one moves to the next tree clockwise and the other moves to the next tree counterclockwise. After some time all the swifts gather on the same tree. Can this be a different tree from the one where the two swifts originally were?

**Problem 6.128.** A vase contains 100 balls of four colours: red, blue, yellow and green. You may: add or remove one red and one blue ball together; add or remove one yellow and one green ball together; add two red and one yellow ball; add two blue and one green ball; swap two red balls for one blue; swap two yellow balls for one green. In the end only one ball remains. Could it be that initially the number of red balls was equal to the number of green balls, and the number of blue balls was equal to the number of yellow balls?

**Problem 6.129.** A computer stores the number 123456789. Two viruses affect it. The first chooses any digit except the first or the last and either adds 1 to it and subtracts 1 from each of its two neighbours, or subtracts 1 from it and adds 1 to each of its two neighbours. The second chooses four consecutive digits and either adds 1 to the first and the last of them, or subtracts 1

from the first and the last of them. Can the number 987654321 appear on the computer? (You cannot subtract from 0 or add to 9.)

**Problem 6.130.** A $2 \times 100$ board has 2101 candies in the bottom-left cell. Every minute Vasya takes two candies from any cell, moves one of them to the right or up and eats the other. Prove that at least one candy will eventually reach the top-right cell, no matter what Vasya does.

**Problem 6.131.** On Mars there are two currencies, rubles and tugriks. A machine works as follows: if you insert $n^2$ rubles, for any positive integer $n \leq 26$, you get $n$ tugriks; if you insert $m$ tugriks, for any integer $m \geq 27$, you get $m^2$ rubles. Sasha starts with 89 tugriks. Can he end up with 2007 rubles and zero tugriks after some exchanges?

**Problem 6.132.** Vasya and Petya inherit many boats: Petya has more boats than Vasya, but both have fewer than 1000. Each day Petya makes 5 new boats and Vasya makes 7 new boats. Sometimes they visit a club where each donates either half or two thirds of his boats (both must donate the same fraction). Prove that they will visit the club at most 12 times.

**Problem 6.133.** A circle is divided into six parts, like a pizza, and the numbers $1, 0, 1, 0, 0, 0$ are placed in order on the parts. The operation we perform is to take two adjacent parts and increase the number on each of them by 1. Can we end up with all six numbers equal?

**Problem 6.134.** There is a machine that can perform one of the three following operations on pairs of integers: take $(a, b)$ and return $(a + 1, b + 1)$; take $(a, b)$ with $a$ and $b$ even and return $(a/2, 2b)$; take the pairs $(a, b)$ and $(b, c)$ and return $(a, c)$. After performing an operation the machine also returns the original pair or pairs. Is it possible to obtain $(1, 2020)$ starting from $(9, 23)$?

**Problem 6.135.** We can change a natural number $n$ in three ways: (a) if the number $n$ has at least two digits, we erase the last digit and subtract that digit from the remaining number (for example, from 123 we get $12 - 3 = 9$); (b) if the last digit is different from 0, we may write the digits in the opposite order (for example, from 123 we get 321); (c) we may multiply the number $n$ by a number from the set $\{1, 2, 3, \ldots, 2010\}$. Can we get the number 21062011 from the number 1012011? [JBMO 2011, C3]

**Problem 6.136.** In a house there are $2n + 1$ rooms (corridors also count as rooms). Each room has an odd number of doors. Show that the house has an odd number of doors leading to the outside.

**Problem 6.137.** You are given the set $\{3, 4, 12\}$. In each move one may select two numbers $a$ and $b$ from the set and replace them by $\frac{3a-4b}{5}$ and $\frac{4a+3b}{5}$. Will it be possible after finitely many moves to end up with the set $\{4, 6, 12\}$?

**Problem 6.138.** Let $n$ be a positive integer and consider an arrangement of $2n$ blocks in a straight line, $n$ of them red and the rest blue. A *swap* means choosing two consecutive blocks and interchanging their positions. Let $A$ be the minimum number of swaps needed to make the first $n$ blocks all red, and let $B$ be the minimum number of swaps needed to make the first $n$ blocks all blue. Show that $A + B$ is independent of the starting arrangement and determine its value.

**Problem 6.139.** Show that if it is possible to tile a rectangle using $m$ squares $2 \times 2$ and $n$ rectangles $1 \times 4$, then it is impossible to tile that same rectangle using $m + 1$ squares $2 \times 2$ and $n - 1$ rectangles $1 \times 4$.

**Problem 6.140.** ★ On an infinite two-dimensional grid we place 4 coins at the origin $(0, 0)$. In each move we may remove a coin at $(a, b)$ and instead place two coins, one at $(a + 1, b)$ and the other at $(a, b + 1)$. Prove that, however we make the moves, there will always be a cell with at least 2 coins.

**Problem 6.141.** ★ On an infinite one-dimensional grid we place an arbitrary number of coins (more than one coin may be placed on a cell). In each move we take two coins from a cell and move them to the two adjacent cells, one coin on each. Show that we will eventually run out of moves.

**Problem 6.142.** ★ A sequence begins $1, 0, 1, 0, 1, 0$ and thereafter each term is equal to the final digit of the sum of the previous six terms (so the sequence continues $3, 5, 0, 9, \dots$). Is it possible for the block $0, 1, 0, 1, 0, 1$ to occur in this sequence?

**Problem 6.143.** ★ A group of children has been given $2^n$ coins so that each child has at least one coin. Whenever a child has more than half of all the coins, that child gives to each other child exactly as many coins as that child already has. What is the largest possible number of such moves?

**Problem 6.144.** ★ Three numbers 0, 1, $\sqrt{2}$ are given. In each move one is permitted to add to one of the numbers an integer multiple of the difference of the remaining two numbers. Will it be possible to obtain the numbers 0, 2, $\sqrt{2}$?

**Problem 6.145.** ★ The vertices of an $n$-gon are labelled with real numbers $x_1, \dots, x_n$. Let $a$, $b$, $c$, $d$ be labels of four consecutive vertices along the $n$-gon. If $(a - d)(b - c) < 0$ we are allowed to swap $b$ and $c$. Determine whether this operation can be performed infinitely many times.

**Problem 6.146.** In four boxes there are in total 2024 balls. A move consists of taking two balls from two different boxes and placing both of them in a third box. Prove that all the balls can be gathered in one box.

**Problem 6.147.** A frog jumps around on the grid points in the plane, from one grid point to another. The frog starts at the point $(0, 0)$. Then it makes successively a jump of one step horizontally, a jump of two steps vertically, a jump of three steps horizontally, a jump of four steps vertically, and so on. Determine all $n > 0$ such that the frog can be back at $(0, 0)$ after $n$ jumps. [Netherlands 2021]

**Problem 6.148.** Consider an arrangement of tokens in the plane, not necessarily at distinct points. We are allowed to apply a sequence of moves of the following kind: select a pair of tokens at points $A$ and $B$ and move both of them to the midpoint of $A$ and $B$. We say that an arrangement of $n$ tokens is *collapsible* if it is possible to end up with all $n$ tokens at the same point after a finite number of moves. Prove that every arrangement of $n$ tokens is collapsible if and only if $n$ is a power of 2. [Canada 2018]

**Problem 6.149.** Let $n \geq 3$ be an integer. Put $n^2$ cards labelled $1, 2, \dots, n^2$ in any order into $n$ empty boxes so that there are exactly $n$ cards in each box. One can perform the following operation: first select 2 boxes, take out any 2 cards from each of the selected boxes, and then return these cards to the other selected box. Prove that for any initial order of the $n^2$ cards in the boxes one can perform the operation finitely many times so that the labels in each box are consecutive integers. [China Girls MO 2016]

## 6.5 Colourings and tilings

**Definition 6.5 (Small tiles and their names).** A *domino* is a $1 \times 2$ rectangle. The *I-tromino* is the $1 \times 3$ rectangle and the *L-tromino* is a $2 \times 2$ square with one cell removed. Among the tetrominoes, the *T-tetromino* consists of a $1 \times 3$ row of cells together with one cell attached to the middle cell of that row; the *L-tetromino* consists of a $1 \times 3$ row together with one cell attached perpendicularly at one of its ends; the *S-tetromino* is a $2 \times 3$ rectangle with two opposite corner cells removed, and the *Z-tetromino* is its mirror image. The *X-pentomino* is a cell together with its four neighbours, and the *P-pentomino* is a $2 \times 3$ rectangle with one corner cell removed. In a triangular grid, a *lozenge* consists of two unit triangles sharing an edge. Unless stated otherwise, tiles may be rotated and reflected.

**Technique 6.6 (Balanced tiles).** Colour the cells of a region with two colours and write $\#\square$ and $\#\blacksquare$ for the numbers of cells of each colour. Call a tile *balanced* if every placement of it satisfies $\#\square = \#\blacksquare$. If a tile is balanced, then every region that can be tiled by copies of it is balanced as well, so an imbalanced region admits no tiling. More generally, if the quantity $\#\square - \#\blacksquare$ is divisible by $d$ for every placement of every available tile, then it is divisible by $d$ for any tileable region. Beware of the converse: a balanced region need not be tileable, so to prove that a tiling exists one must always exhibit a construction.

**Theorem 6.7 (Gomory's theorem).** *An $8 \times 8$ chessboard from which any two cells of different colours have been removed can always be tiled with dominoes.*

**Technique 6.8 (Coarser colourings).** When a tile is itself a large rectangular block, colour the board in blocks rather than in single cells: use a chessboard pattern whose "cells" are boxes matching the dimensions of the tile. For example, in order to study $1 \times 2 \times 4$ bricks in a cube, colour the cube in a chessboard pattern with cells of size $2 \times 2 \times 2$; every brick is then balanced, and it suffices to check whether the cube is. This is the mechanism behind de Bruijn's theorem on packing boxes with bricks.

**Technique 6.9 (Make a wish).** Instead of trying out all the colourings you happen to know, decide first what property you want the colouring to have, and only then look for a colouring with that property. Typical wishes are: every placement of the tile covers exactly one cell of each colour; every placement covers an even number of marked cells; every placement has sum 0, or 1, or a nonnegative sum, for a suitable numbering of the cells with integers or real numbers. The wish may involve two colours, several colours, or arbitrary weights, and it is usually easiest to find a colouring of the infinite board first and then place the given finite region on top of it.

**Technique 6.10 ($k$-colourings for $k$-cell tiles).** If a tile consists of $k$ cells, look for a periodic colouring of the plane with $k$ colours such that every placement of the tile covers exactly one cell of each colour. Then the number of disjoint tiles that fit into a region is at most the number of cells of the least frequent colour occurring in that region, which often

gives a sharp bound. Rotating the colouring gives further restrictions, and intersecting the conclusions of several colourings pins down the possible positions of the leftover cells.

**Problem 6.150.** Can an $8 \times 8$ board with two opposite corners removed be tiled with dominoes?

**Problem 6.151.** A polygonal region is coloured like a chessboard and has equally many white and black cells. Does it follow that this region can be covered with dominoes?

**Problem 6.152.** We want to divide an $8 \times 8$ board into twenty-one rectangles of size $1 \times 3$ and a single unit square. Where can the unit square be?

**Problem 6.153.** Find all $n$ for which a $2n \times 2n$ board can be tiled with equally many vertical and horizontal dominoes.

**Problem 6.154.** Several $1 \times 4$ tiles, $2 \times 2$ tiles and S-tetrominoes cover an $8 \times 8$ board. Prove that the number of used $2 \times 2$ tiles is even.

**Problem 6.155.** Does there exist an odd positive integer $n$ such that an $n \times n$ board can be covered with X-pentominoes and dominoes?

**Problem 6.156.** Can a $9 \times 9$ square be covered by $1 \times 5$ and $1 \times 6$ tiles?

**Problem 6.157.** Call a square of side $k$ a $k$-square. Suppose that a $(2n + 1)$-square $(n \geq 1)$ is tiled with 2-squares and 1-squares. Determine, in terms of $n$, the smallest possible number of 1-squares used in such a tiling.

**Problem 6.158.** Can a $10 \times 10$ board be tiled with T-tetrominoes?

**Problem 6.159.** Is it possible to tile an $8 \times 8$ chessboard with 7 L-tetrominoes and 9 T-tetrominoes?

**Problem 6.160.** Let $k$ be a positive integer. Which $m \times n$ boards can be tiled with $1 \times k$ tiles?

**Problem 6.161.** Is it possible to cover a $4 \times 11$ board using L-tetrominoes? What about a $98 \times 98$ board?

**Problem 6.162.** We have a board of black and white tiles. The operation we perform is to take one row or one column and switch the colour of every tile in that row or column. If we start with a $4 \times 4$ board whose four corners are black and all of whose other tiles are white, can we end up with a board all of whose tiles have the same colour after finitely many operations?

**Problem 6.163.** Is it possible for a knight to pass through all the squares of a $4 \times 8$ board, visiting each square exactly once? Can it also return to the initial square?

**Problem 6.164.** Can a $7 \times 7 \times 7$ cube with the central cell removed be built from $1 \times 1 \times 2$ bricks?

**Problem 6.165.** A $7 \times 7$ square is tiled with 24 dominoes and one $1 \times 1$ square. Determine all possible positions of the single $1 \times 1$ tile.

**Problem 6.166.** Here the cells are equilateral triangles of side length 1. Suppose that an equilateral triangle of side length $n$ is tiled with a collection of unit triangles and lozenges. Find the smallest possible number of unit triangles used in such a tiling.

**Problem 6.167.** Can a $6 \times 6 \times 6$ cube be built from $1 \times 2 \times 4$ bricks?

**Problem 6.168.** (a) Can a $10 \times 10$ board be tiled with $1 \times 4$ rectangles? (b) Can a $13 \times 13$ board with the central cell removed be tiled with $1 \times 4$ rectangles?

**Problem 6.169.** We have a large supply of S-tetrominoes and L-trominoes. Determine the smallest total number of such tiles needed to cover a $7 \times 7$ board.

**Problem 6.170.** A rectangle is tiled with L-tetrominoes and S-tetrominoes. Prove that the number of L-tetrominoes used is even.

**Problem 6.171.** (a) Find all positive integers $n$ with the property that an $n \times n$ square can be covered with $2 \times 2$ and $3 \times 3$ tiles. (b) For each such $n$ determine the smallest possible number of $3 \times 3$ squares used.

**Problem 6.172.** (a) A honeycomb composed of 37 hexagonal cells (a regular hexagonal array with four cells along each side) is to be dissected into 12 sticks of three neighbouring collinear cells and a single cell. Find all possible positions of the single cell. (b) How would the answer change if the honeycomb had one more external layer of cells, i.e. had 61 cells and was to be covered with 20 sticks and a single cell?

**Problem 6.173.** Consider the three faces of a $5 \times 5 \times 5$ cube having a common vertex, each divided into unit cells. Can they be covered with twenty-five $1 \times 3$ stickers (tiles that may be folded over an edge of the cube), each covering whole cells?

**Problem 6.174.** What is the maximum number of $2 \times 3 \times 3$ bricks that can be placed inside an $8 \times 8 \times 9$ box?

**Problem 6.175.** A $7 \times 7$ board is to be covered with one-sided P-pentominoes (they may be rotated but not flipped over) in such a way that exactly one cell is covered by two tiles and all the remaining cells are covered by exactly one tile. Determine all possible placements of the doubly covered cell.

**Problem 6.176.** Here we allow rotating tiles but we do not allow flipping them over, so that S-tetrominoes and Z-tetrominoes are considered different. Suppose that a figure $F$ can be tiled with S-tetrominoes. Prove that no matter how we tile $F$ using only S- and Z-tetrominoes, we always use an even number of Z-tetrominoes. [IMO Shortlist 2014 C4]

**Problem 6.177.** Call an equilateral triangle of side $k$ a $k$-triangle. Suppose that a $(2n+1)$-triangle ($n \geq 1$) is tiled with 2-triangles and 1-triangles. Determine, in terms of $n$, the smallest possible number of 1-triangles used in such a tiling.

**Problem 6.178.** Is it possible to cover a $5 \times 7$ rectangle with several layers of L-trominoes, not crossing its borders, so that each cell of the rectangle is covered by the same number of tiles? [Russia 1996]

**Problem 6.179.** Consider an infinite board with square cells, each containing a light bulb. Initially only four light bulbs are switched on, and they form a $2 \times 2$ square. In one move one can change the state (on/off) of all the bulbs within any $3 \times 3$ or any $4 \times 4$ square. Is it possible to switch the light off completely in a finite number of moves?

**Problem 6.180.** Is it possible to tile a $5 \times 5$ square with sixteen $3 \times 1$ rectangles in such a way that each unit square is covered at most twice?

**Problem 6.181.** A $6 \times 6$ board is covered with dominoes. Show that there is a straight line that cuts the board into two parts without cutting any domino.

**Problem 6.182.** An $m \times n$ checkerboard has a white square in its upper-left corner. Starting in that corner, a piece moves diagonally; each time the piece comes to an edge of the board it continues along the reflected diagonal path, and the process ends when the piece hits a corner. For which values of $m$ and $n$ will the piece visit all of the white squares?

**Problem 6.183.** A board $2n \times 2n$ is filled by the numbers from 1 to $4n^2$ in the natural order (left to right, top to bottom). Half of the cells are coloured white and the other half black, so that each row contains exactly $n$ white and $n$ black cells, and the same holds for each column. Show that, no matter how the colouring looks, the sum of the numbers in the white cells equals the sum of the numbers in the black cells.

**Problem 6.184.** A square grid consisting of $n^2$ points is given in a plane. By a broken line of length $l$ we mean the union of segments $A_0A_1, A_1A_2, \ldots, A_{l-1}A_l$ in that plane. Find the smallest positive integer $l$ for which there exists a broken line of length $l$ containing all the $n^2$ given points.

**Problem 6.185.** ★ A configuration of a tetromino is an arrangement of a tetromino in which the sides of the unit squares are aligned horizontally and vertically. Two configurations are considered equal if and only if one can be brought into the other by a translation; rotations and reflections are not allowed, so for example a $4 \times 1$ rectangle is not the same configuration as a $1 \times 4$ rectangle. Find the set of all possible configurations of tetrominoes and prove that one cannot tile any rectangle with this set, each configuration being allowed to be moved only by translation.

**Problem 6.186.** ★ Let $n$ be an odd positive integer. The unit squares of an $n \times n$ chessboard are coloured alternately black and white, with the four corners coloured black. For which values of $n$ is it possible to cover all the black squares with non-overlapping L-trominoes? When it is possible, what is the minimum number of trominoes needed?

**Problem 6.187.** ★ A rectangle $R$ with odd integer side lengths is divided into small rectangles with integer side lengths. Prove that there is at least one among the small rectangles whose distances from the four sides of $R$ are either all odd or all even.

**Problem 6.188.** ★ A rectangle $R$ is tiled with rectangles each of which has an integer side. Prove that $R$ also has an integer side.

**Problem 6.189.** ★ Let $n$ be a positive integer. On a $2n \times 2n$ board, $1 \times n$ and $n \times 1$ pieces are arranged without overlap. Call an arrangement *maximal* if it is impossible to put a new piece on the board without overlapping the previous ones. Find the least $k$ for which there is a maximal arrangement that uses $k$ pieces.

**Problem 6.190.** ★ Let $a, b > 1$ be odd positive integers. A board with $a$ rows and $b$ columns, from which the fields $(2, 1)$, $(a - 2, b)$ and $(a, b)$ have been removed, is tiled with $2 \times 2$ squares and $2 \times 1$ dominoes (which may be rotated). Prove that the number of dominoes used is at least $\frac{3(a+b)-6}{2}$.

**Problem 6.191.** Fix an integer $n \geq 2$. An $n \times n$ *stencil* is an $n \times n$ array with $n$ cells removed so that exactly one cell is removed from every row and every column. A *stick* is a $1 \times k$ or $k \times 1$ array for a positive integer $k$. For any stencil $A$ let $m(A)$ be the minimal number of sticks required to partition $A$. Find all possible values of $m(A)$ as $A$ varies over all possible $n \times n$ stencils. [RMM 2017]

**Problem 6.192.** Given positive integers $m$ and $n$ with $n \geq m$, determine the largest number of dominoes ($1 \times 2$ or $2 \times 1$ rectangles) that can be placed on a rectangular board with $m$ rows and $2n$ columns consisting of unit cells so that: each domino covers exactly two adjacent cells of the board; no two dominoes overlap; no two dominoes form a $2 \times 2$ square; and the bottom row of the board is completely covered by $n$ dominoes. [RMM 2016]

**Problem 6.193.** Let $n$ be a natural number. A cube of edge $n$ may be divided into 1996 cubes whose edge lengths are also natural numbers. Find the minimum possible value of $n$. [Ibero 1996]

## 6.6 Games and strategies

**Problem 6.194.** One pile with $a$ candies is given. Two players in turn make moves, and a move consists of eating 1, 2 or 3 candies from the pile. The player who takes the last candy wins. Determine, in terms of $a$, who has a winning strategy.

**Problem 6.195.** Two piles with $a$ and $b$ candies are given. Two players in turn make moves, and a move consists of eating as many candies as the player wishes but only from one of the piles. The player who takes the last candy wins. Determine, in terms of $a$ and $b$, who has a winning strategy.

**Problem 6.196.** Two piles with 135 and 240 candies are given. Two players in turn make moves, and a move consists of eating all the candies from one pile and dividing the other pile into two (not necessarily equal) piles. A player who cannot move loses. Determine who has a winning strategy.

**Problem 6.197.** There are $n$ coins in a pile and two players $A$ and $B$ take turns taking either 1, 2 or 6 coins from the pile. The winner is the person who takes the last coin, and player $A$ plays first. Determine for which $n$ player $A$ has a winning strategy and for which $n$ player $B$ does.

**Problem 6.198.** Consider a strip of 13 empty cells in a row. Two players ($A$ first) put dominoes on the strip, each domino covering two adjacent empty cells. The player who cannot make a move loses the game. Who has a winning strategy?

**Problem 6.199.** On a strip of 2023 cells two players play the following game. On each turn a player writes the letter S or the letter O in an empty cell. The player after whose move three consecutive cells contain the letters S, O, S in this order wins the game. Who has a winning strategy?

**Problem 6.200.** On an $n \times m$ board a coin is placed in the upper-right corner. Players take turns moving the coin up to 3 spaces either down or to the left (but not both in one move), and whoever reaches the lower-left corner of the board wins. Player $A$ has the first move. Determine for which $(n, m)$ player $A$ has a winning strategy and for which $(n, m)$ player $B$ does.

**Problem 6.201.** There are $n$ coins in one pile and two players $A$ and $B$ take turns with the following game. In each move the player whose turn it is may either take a coin from a pile and discard it, or divide a pile into at least two equal piles. Player $A$ starts. The winner is the player who takes the last coin off the table. For which $n$ does player $A$ have a winning strategy, and for which $n$ does player $B$?

**Problem 6.202.** Two players $A$ and $B$ take turns placing the symbols X and O respectively on an empty square of an infinite two-dimensional grid, with player $A$ beginning. The first player

to place $n$ consecutive symbols of his own kind horizontally or vertically wins (diagonals do not count). For which values of $n$ does $A$ have a winning strategy?

**Problem 6.203.** The cells of an $8 \times 8$ table are initially white. Alice and Bob play a game. First Alice paints $n$ of the fields red. Then Bob chooses 4 rows and 4 columns of the table and paints all fields in them black. Alice wins if there is at least one red field left. Find the least value of $n$ such that Alice can win the game no matter how Bob plays.

**Problem 6.204.** Given a positive integer $n$, all of its positive divisors are written on a board. Two players $A$ and $B$ play the following game. Each turn, a player colours one of these divisors either red or blue; a player may choose whichever colour they wish, but may only colour numbers that have not been coloured before. The game ends once every number has been coloured. Player $A$ wins if the product of all the red numbers is a perfect square or if no number has been coloured red; otherwise $B$ wins. If $A$ goes first, determine who has a winning strategy for each $n$. [Ibero 2017]

**Problem 6.205.** A game is played on an $n \times n$ chessboard. At the beginning there are 99 stones on each square. Two players $A$ and $B$ take turns, and in each turn the player chooses either a row or a column and removes one stone from each square in the chosen row or column. A player is only allowed to choose a row or a column if it has at least one stone on each square. The first player who cannot move loses the game, and player $A$ takes the first turn. Determine all $n$ for which player $A$ has a winning strategy. [Nordic 2014]

**Problem 6.206.** ★ On a $5 \times 5$ board, $n$ white markers are positioned, each marker on a distinct field. A child is given the task of colouring as many markers as possible black in the following manner: a white marker is taken from the board, coloured black, and put back on the board on an empty field such that none of the neighbouring fields contains a white marker (two fields are neighbouring if they share a common side). If it is possible for the child to succeed in colouring all the markers black, we say that the initial positioning of the markers was *good*. (a) Prove that if $n = 20$, then a good initial positioning exists. (b) Prove that if $n = 21$, then a good initial positioning does not exist.

**Problem 6.207.** ★ We have two piles with 2000 and 2017 coins respectively. Ann and Bob take alternate turns making the following move: the player whose turn it is picks a pile with at least two coins, removes from that pile $t$ coins for some $2 \leq t \leq 4$, and adds to the other pile 1 coin. The players may choose a different $t$ at each turn, and the player who cannot make a move loses. If Ann plays first, determine which player has a winning strategy.

**Problem 6.208.** ★ Alice and Bob play the following game. Initially there are three piles, each containing 2020 stones. The players take turns to make a move, with Alice going first. Each move consists of choosing one of the available piles, removing the unchosen pile or piles from the game, and then dividing the chosen pile into 2 or 3 non-empty piles. A player loses the game if they are unable to make a move. Prove that Bob can always win the game, no matter how Alice plays.

**Problem 6.209.** Let $k$ and $n$ be integers with $1 \leq k < n$. Alice and Bob play a game with $k$ pegs in a line of $n$ holes. At the beginning of the game the pegs occupy the $k$ leftmost holes. A legal move consists of moving a single peg to any vacant hole that is further to the right. The players alternate moves, with Alice playing first. The game ends when the pegs are in the $k$ rightmost holes, so whoever is next to play cannot move and therefore loses. For what values of $n$ and $k$ does Alice have a winning strategy?

**Problem 6.210.** ★ A rook dipped in red paint is initially placed at a corner square of an $m \times n$ rectangular board. Players $A$ and $B$ take turns moving the same rook to a different square in the same row or the same column, with player $A$ having the first turn. As it has been dipped in red paint, the rook colours every square it passes over red, including both the square it lands on and the squares in between. A player loses if he is forced to move the rook over a square which has already been coloured red. For each value of $m$ and $n$, which player has a winning strategy?

**Problem 6.211.** ★ A set of $n$ points in space is given, no three of which are collinear and no four of which are coplanar, and each pair of points is connected by a line segment. Initially all the line segments are colourless. A positive integer $b$ is given, and Alice and Bob play the following game. In each turn Alice colours one segment red and then Bob colours up to $b$ segments blue; this is repeated until there are no colourless segments left. If Alice ever colours a red triangle she wins; if there are no colourless segments left and Alice has not succeeded in colouring a red triangle, Bob wins. Neither player is allowed to colour over an already coloured segment. (a) Prove that if $b < \sqrt{2n-2} - \frac{3}{2}$, then Alice has a winning strategy. (b) Prove that if $b \geq 2\sqrt{n}$, then Bob has a winning strategy.

## 6.7 Logic, riddles and information

**Problem 6.212.** On a certain island there are two tribes: people are either truth-tellers, who always tell the truth, or liars, who always lie. A traveller to the island has the following encounters. (a) Two persons approach, $A$ and $B$. $A$ says: "At least one of us is a liar." What are $A$ and $B$? (b) Two other persons approach. $A$ says: "Either I am a liar or $B$ is a truth-teller." What are $A$ and $B$? (c) Now three persons $A$, $B$ and $C$ approach. $A$ says: "All of us are liars." $B$ says: "Exactly one of us is a truth-teller." What are $A$, $B$ and $C$? (d) The traveller runs into a fork in the road splitting the path in two, and only one of these paths leads to the main town. He notices a native of the island approaching. How can he find the true path with just one question? (e) The following day a boat arrives with people from a neighbouring island, whose inhabitants are *semi-liars*: they alternately tell the truth and lie. The traveller meets three people $A$, $B$ and $C$. $A$ says: "I am a semi-liar." $B$ says: "This is true." $C$ says: "I am not a semi-liar." A trusted guide tells the traveller that exactly one of them is a truth-teller, one a liar and one a semi-liar. What are $A$, $B$ and $C$? (f) The traveller meets four people $A$, $B$, $C$ and $D$. $A$ says: "Among us there are more truth-tellers than semi-liars." $C$ and $D$ disagree while $B$ agrees. Then $A$ says: "Among us there are as many liars as semi-liars." This time $B$ and $C$ disagree while $D$ agrees. What are $A$, $B$, $C$ and $D$?

**Problem 6.213.** You and your five friends have entered a magical cave that opens once every thousand years, where there are 60 bottles on display. One bottle contains a magical potion that grants perfect health and long life to those who drink it, but turns their skin blue for a few hours on the morning after they drink it. The other 59 bottles contain water. The cave closes the next day, so there is only one chance to identify the potion, and you can leave the cave with just one bottle. Assuming the magical potion cannot be distinguished from water by taste or smell, how will the six of you find the magic potion with certainty and leave with it before the cave closes?

**Problem 6.214.** An emperor has told 100 wise men that tomorrow he will assemble them in a row where each wise man will only be able to see the caps of those in front of him, and will place on each one a cap of one of $k$ given colours. Each minute one wise man is supposed to guess the colour of his own cap. The emperor will award all the wise men who guess correctly. What strategy can the wise men agree upon in advance in order to maximize the guaranteed number of awarded wise men, and what is this number?

**Problem 6.215.** A guard has assembled 100 prisoners and told them he will place a number on each prisoner's cap from 1 to 100; he may place the same number on more than one prisoner. Each prisoner can see only the numbers on the other prisoners' caps, not his own, and is not allowed to communicate with the others once the caps are placed. Each prisoner writes on a piece of paper, which the others cannot see, what he guesses the number on his cap to be, and hands it to the guard. If at least one prisoner is correct in his guess, the guard will release all the prisoners. The prisoners have time to discuss a strategy beforehand, but the guard will listen in and exploit any strategy that is not foolproof. With what strategy can the prisoners ensure their release?

**Problem 6.216.** ★ A group of 100 prisoners, all together in the prison dining area, are told that they will all be put in isolation cells and then interrogated one by one in a room containing a light with an on/off switch. The prisoners may communicate with one another only by toggling the light switch. There is no fixed order of interrogation and no fixed interval between interrogations, and the same prisoner may be interrogated again at any stage. When interrogated, a prisoner can either do nothing, or toggle the light switch, or announce that all prisoners have been interrogated. If that announcement is true, all the prisoners will be set free; if it is false, they will all be punished. While still in the dining room, can the prisoners agree on a protocol that will set them free, assuming that at any stage every prisoner will be interrogated again sometime? Solve the problem if (a) the prisoners know that the light is initially switched off; (b) the prisoners do not know the initial state of the light.

**Problem 6.217.** A prisoner is locked in a cell. Next to the lock there is a vertical cylinder divided into four vertical chambers, and there is one fish in each of the chambers. The lock opens when all four fish are oriented in the same direction. The prisoner cannot see the fish, but he can choose any two fish and take them out, reverse them if he wants to (he can also reverse only one) and put them back in. He can do this an arbitrary number of times, but after every such step the cylinder spins with a high speed and it is impossible for the prisoner to follow the positions of the fish he manipulated. Is there a strategy for the prisoner to ensure that he will set himself free in a finite number of moves if (a) he can see the fish when he takes them out; (b) he cannot see the fish when he takes them out?

**Problem 6.218.** ★ A gang of $n$ pirates has found a treasure chest with 100 coins and now has to split the coins. The pirates are all ordered by strength and everyone is aware of the ordering. Under the pirate rules, the strongest pirate suggests a way to split the coins; if at least half of the pirates vote for the split, the coins are split in that way, and if less than half vote for it there is a mutiny, the strongest pirate is thrown overboard and the next strongest pirate becomes the captain and suggests a split. The pirates behave rationally and know that the others also behave rationally. The first priority of each pirate is to remain alive, the second is to maximize the number of coins he gets, and the third is to eliminate rivals. How many pirates will survive and how will the coins be divided among the survivors if (a) $n = 100$; (b) $n = 2020$?

**Problem 6.219.** You are at a market, wanting to buy two chickens, one male and one female. The villager shows you a box with 12 chickens, but they all look the same. However, the villager is somehow able to distinguish them and offers you help: you show him any set of chickens and he will tell you how many of them are male, without specifying which. At least how many questions do you need in order to determine a pair of chickens of the kind you need?

**Problem 6.220.** We have $n$ boxes put next to each other in a line. In one of them there is a mouse. You try to kill the mouse by throwing bombs into boxes. If the mouse is in the box you throw a bomb in, it dies; otherwise it moves to the adjacent box on its right, and from the

rightmost box it cannot move. How many bombs do you need in order to be sure of killing the mouse?

**Problem 6.221.** A group of thieves hides their treasure in one of 101 caves, numbered 1 to 101. During the day the treasure lies in a cave, but at night the thieves move it to one of the two adjacent caves. Ali Baba knows this and enters one cave every day, trying to retrieve the treasure. Does he have a strategy that guarantees him to find the treasure?

**Problem 6.222.** In one of 25 rooms arranged in a row there is a piece of gold; any two consecutive rooms are connected by a corridor. Every morning we check one of the rooms, and every evening the gold is moved into one of the adjacent rooms. Prove that it is possible to find the gold.

**Problem 6.223.** Seven cards, numbered 1 through 7, have been arbitrarily divided between three people $A$, $B$ and $C$, where $A$ and $B$ have three cards each and $C$ has only one. $A$ and $B$ want to tell each other which cards they have, but $C$ is a spy, hears everything they say, and must not learn the location of any card. How can $A$ and $B$ do the job?

**Problem 6.224.** There is a $7 \times 7$ board filled with the numbers $1, \ldots, 49$ and we must guess the arrangement of the numbers. In each question we choose a set of cells and ask which numbers are inside it. At least how many questions do we need?

**Problem 6.225.** The numbers from 1 to 100 are arranged in a sequence. With one question we can find out the mutual arrangement of any 50 of the numbers. Show that 5 questions are enough to find out the order of all 100 numbers.

**Problem 6.226.** Alice and Bob play a game against Charlie. Charlie starts by placing cards numbered 1 through 100 in a line on the table. Then Alice may choose two cards and switch them if she wants. Charlie then turns the cards face down and Bob comes to the table. Charlie chooses a number and asks Bob to find it; Bob turns the cards over one by one, but is given only 50 guesses. Can Alice and Bob win?

**Problem 6.227.** There are ten indistinguishable coins in a line. It is known that two of them are false and occupy consecutive positions in the line. For each set of positions you may ask how many false coins it contains. Is it possible to identify the false coins by asking only two such questions, without knowing the answer to the first question before asking the second? [Ibero American 2010]

**Problem 6.228.** Two thousand people are standing in a line. Each one of them is either a liar, who always lies, or a truth-teller, who always tells the truth. Each one of them says: "There are more liars to my left than truth-tellers to my right." Determine, if possible, how many people of each kind are in the line. [Argentina 2013]

**Problem 6.229.** Each of 5 numbers is 0 or 1. For any two of them you may find out their product, and using this operation you must determine all the numbers. What is the largest possible number of zeros among the original numbers?

**Problem 6.230.** In each cell of a $5 \times 5$ table a real number is written. You may find out the sum of the numbers in any $1 \times 2$ rectangle. Prove that the numbers could initially have been arranged in such a way that the sum of all the numbers in the table cannot be deduced from the answers received.

**Problem 6.231.** A teacher wrote a two-digit number on the board and asked Dima whether it is divisible by 2, by 3, ..., by 9. Dima answered all eight questions correctly, and the answers were "yes" and "no" equally often (four "yes" and four "no"). (a) Can you answer at least one of the teacher's questions without knowing the number itself? (b) Can you answer at least two of the teacher's questions without knowing the number itself?

**Problem 6.232.** (a) All plant species of one country were numbered in a row by the integers from 2 to 2018, each number appearing exactly once. For each pair of plant species the greatest common divisor of their numbers was remembered, and then the numbers themselves were forgotten. Is it possible to restore the number of each species? (b) The same question with 2018 replaced by 1010000.

**Problem 6.233.** Find all ten-digit numbers $a_0a_1 \dots a_9$ such that for every $i$ the digit $a_i$ is equal to the number of times the digit $i$ appears in the number.

**Problem 6.234.** In a football tournament with 5 teams every two teams played one game against each other. The points of all the teams turned out to be 5 consecutive integers. Find the outcome of the game between the second and the last team in the final standings.

## 6.8 Combinatorial geometry first steps

**Problem 6.235.** We have $n$ planets in space. We say that planet $a$ *looks at* planet $b$ if $b$ is the planet closest to $a$. If all the distances between planets are different, prove that there are two planets $i$ and $j$ such that $i$ looks at $j$ and $j$ looks at $i$ at the same time.

**Problem 6.236.** We place 5 points inside an equilateral triangle with side 2. Prove that there are two of them whose distance is less than 1.

**Problem 6.237.** We place 6 points inside a $3 \times 4$ rectangle. Prove that there are two of them whose distance is at most $\sqrt{5}$.

**Problem 6.238.** We place 9 points in a $1 \times 1$ square. Prove that there are three of them that form a triangle of area at most $\frac{1}{8}$.

**Problem 6.239.** Determine the minimum number of lines that can be drawn in the plane so that they intersect in exactly 200 distinct points.

**Problem 6.240.** We put a total of $n$ blue and red points in a plane such that every segment between two points of the same colour contains a point of the opposite colour. Prove that all the points lie on a line.

**Problem 6.241.** There are $n$ points in the plane such that any three of them form a right triangle. What is the maximum value of $n$?

**Problem 6.242.** There are $n$ points in the plane such that any triangle formed by three of them has area at most one. Prove that there is a triangle of area at most 4 containing all $n$ points.

**Problem 6.243.** Prove that any convex polygon of area at most 1 can be contained in a rectangle of area at most 2.

**Problem 6.244.** Points $X$ and $Y$ are given inside a rectangle $ABCD$. Find the point $Z$ on the side $CD$ such that $\angle XZD = \angle YZC$.

**Problem 6.245.** Given 100 points inside a unit circle, prove that some two of them are at distance less than $\frac{2}{9}$.

**Problem 6.246.** (Sylvester's problem) Given a finite set $S$ of points in the plane, not all collinear, prove that there is a line containing exactly two of these points.

**Problem 6.247.** Given a convex polygon $P$ of area 1, prove that there is a triangle of area $\frac{1}{4}$ that is fully contained in $P$. Now prove that there is also such a triangle of area $\frac{3}{8}$.

**Problem 6.248.** (a) Prove that among any four points inside a unit circle there are two at distance less than $\sqrt{2}$. (b) Prove that among any five points inside a unit sphere there are two at distance less than $\sqrt{2}$.

**Problem 6.249.** A plane figure of area greater than 51 is placed within a $1 \times 101$ rectangle. Prove that this figure must contain two points at mutual distance 1.

**Problem 6.250.** We are given $n$ blue and $n$ red points in the plane such that no three of them are collinear. Prove that we can draw $n$ pairwise disjoint segments, each connecting a red point to a blue point.

**Problem 6.251.** A finite set of points in the plane is given, and the distance between any two of them is at most 1. Prove that it is always possible to remove one point and divide the rest into two parts such that the distance between any two points in the same part is strictly less than 1.

**Problem 6.252.** Given a positive integer $n$, let $f$ be a map from the plane to the real numbers with the property that $f(A_1) + f(A_2) + \cdots + f(A_n) = 0$ whenever $A_1A_2 \ldots A_n$ is a regular $n$-gon. Prove that $f$ is identically zero.

**Problem 6.253.** An even number of unit segments in the plane form a closed, non-degenerate polygonal line. Prove that some two of its vertices are at distance greater than 1.

**Problem 6.254.** A set $S$ consists of $n$ points in the plane, no three of which are collinear. Prove that there is a set $P$ of $2n - 5$ points such that every triangle with vertices in $S$ contains at least one point of $P$ in its interior.

**Problem 6.255.** A *corner* is a polygonal line in the coordinate plane consisting of one vertical and one horizontal segment. We are given $n$ red and $n$ blue points in the plane, all of whose projections onto the coordinate axes are distinct. Prove that we can draw $n$ pairwise disjoint corners, each connecting a red point to a blue point.

**Problem 6.256.** A set of $n > 2$ vectors is given in the plane. Call a vector *lengthy* if its length is not less than that of the sum of all the other vectors. If each vector in the set is lengthy, prove that the sum of all these vectors is zero.

**Problem 6.257.** Prove that among any 111 unit vectors in the plane with sum zero, one can find 55 vectors whose sum has length not exceeding 1.

**Problem 6.258.** We are given $n^2$ points inside or on the edges of a unit square. Prove that we can always connect them all by a polygonal line of length less than $2n + 1$.

**Problem 6.259.** We are given four congruent right-angled triangles made of paper. In each step we may cut one triangle along its altitude to the hypotenuse. Can we ever obtain a set of pairwise incongruent triangles, and how many cuts do we need at least?

**Problem 6.260.** Finitely many unit equilateral triangles are given in the plane, all of which are translates of each other, and every two of them have a common point. Prove that there are three points such that each triangle contains at least one of them.

**Problem 6.261.** By *clipping* a convex polygon we mean choosing two adjacent sides, say $AB$ and $BC$, taking their midpoints $M$ and $N$, and removing the triangle $BMN$. If we start with a regular hexagon of area 1 and clip it finitely many times, prove that the remaining area will always be greater than $\frac{2}{3}$.

**Problem 6.262.** A *strip* is a part of the plane lying between two parallel lines. Suppose that $S$ is a set of points in the plane such that any three of them can be covered by a strip of width 1. Prove that the entire set $S$ can be covered by a strip of width 2.

**Problem 6.263.** A regular $n$-gon with $n$ odd is dissected into triangles by non-intersecting diagonals. Prove that exactly one of these triangles is acute.

**Problem 6.264.** For which $n$ is it possible to dissect a regular $n$-gon into isosceles triangles by non-intersecting diagonals?

**Problem 6.265.** A regular $n$-gon is dissected into triangles by non-intersecting diagonals. At most how many of these triangles can be pairwise incongruent?

**Problem 6.266.** Let $n > 1$ and let $3n$ lines be drawn in the plane such that no two of them are parallel and no three of them are concurrent. Prove that if $2n$ of the lines are coloured red and the other $n$ lines blue, then there are at least two regions of the plane all of whose borders are red. (For each region, all of its borders are contained in the original set of lines, and no line passes through the region.) [Cono Sur 2015]

**Problem 6.267.** We colour the plane with 2020 colours. Prove that there are infinitely many triangles of the same area whose vertices all have the same colour.

**Problem 6.268.** Given are 50 points in the plane, no three of them belonging to the same line. Each of these points is coloured using one of four given colours. Prove that there is a colour and at least 130 scalene triangles with vertices of that colour. [JBMO 2007, C2]

**Problem 6.269.** Along a round table are arranged 11 cards with the distinct names of the 11 members of a committee. The cards are arranged in a regular polygon. Assume that in the first meeting of the committee none of its 11 members sits in front of the card with his own name. Is it possible to rotate the table by some angle so that at the end at least two members sit in front of the card with their names? [JBMO 2012, C1]

**Problem 6.270.** Suppose there are $n$ points in a plane, no three of which are collinear, with the property that if we label these points as $A_1, A_2, \ldots, A_n$ in any way whatsoever, the broken line $A_1A_2 \ldots A_n$ does not intersect itself. Find the maximum value of $n$. [JBMO 2012, Problem 2]

**Problem 6.271.** Decide whether it is possible to have 2011 points in a plane such that the distance between every two of these points is different from 1 and each unit circle centred at one of these points leaves exactly 1005 of the points outside the circle. [JBMO 2011, C7]

**Problem 6.272.** On a board there are $n$ nails, each two connected by a rope. Each rope is coloured in one of $n$ given distinct colours. For each three distinct colours there exist three nails connected with ropes of these three colours. (a) Can $n$ be 6? (b) Can $n$ be 5? [JBMO 2012, Problem 3]

**Problem 6.273.** A flea is initially at the point $(0, 0)$ in the Cartesian plane. It makes $n$ jumps, the direction of each jump being one of the four cardinal directions. The first jump has length 1, the second length 2, the third length 4, and in general the $k$-th jump has length $2^{k-1}$. Is it possible for the flea to return to its original position after a finite number of jumps?

**Problem 6.274.** With the notation of the previous problem, prove that if you know the final position of the flea then it is possible to determine uniquely its position after each of the $n$ jumps.

**Problem 6.275.** ★ Cover each of the following sets with mutually disjoint line segments. (Open sets are denoted with round brackets and closed sets with square brackets.) (a) a closed triangular area without one interior point, $[ABC] \setminus \{E\}$ where $E \in (ABC)$; (b) a closed triangle $[ABC]$; (c) a closed $n$-gon $[A_1A_2 \dots A_n]$; (d) a closed disc $[k]$; (e) an open triangular area $(ABC)$; (f) an open $n$-gon $(A_1A_2 \dots A_n)$; (g) an open disc $(k)$.

**Problem 6.276.** ★ Each point in space is coloured with one of two colours in such a way that whenever an isosceles triangle $ABC$ with $AB = AC$ has all three vertices of the same colour $c$, the midpoint of $BC$ is also coloured with $c$. Prove that there exists a rectangular box all of whose vertices have the same colour.

**Problem 6.277.** ★ The faces of a polyhedron are coloured black or white so that there are more black faces than white faces and no two black faces share an edge. Prove that it is impossible to inscribe a sphere inside the polyhedron.

**Problem 6.278.** ★ Each point in space is coloured with one of three colours. Prove that there exists a colour $c$ such that for every positive real number $r$ there is a triangle of area $r$ all of whose vertices are coloured with $c$.

## 6.9 Proof by contradiction

**Technique 6.11 (Negate precisely).** Do not insert the assumption of the contrary everywhere without thinking: assume the negation of the statement that is most useful to negate. As a rule of thumb, the left-hand assumption below is a more useful thing to have than the right-hand one, so it is the right-hand statements that one should aim to assume: "for every" rather than "for some"; "existence" rather than "nonexistence"; "rational" rather than "irrational"; "finite" rather than "infinite"; "countable" rather than "of the cardinality of the continuum". Classical examples of this style of argument are the irrationality of $\sqrt{2}$, Euclid's theorem that the set of primes is infinite, and Cantor's theorem that the set of real numbers is not countable.

**Problem 6.279.** Every point of three-dimensional space is coloured red, green or blue. Prove that one of the colours attains all distances, meaning that for any positive real number there exist two points of that colour whose distance apart equals that number.

**Problem 6.280.** Each point of the plane is coloured with one of (a) 2; (b) 3 colours. Prove that there exist two points at distance 1 that are coloured the same.

**Problem 6.281.** The natural numbers are coloured with three colours. Prove that there is at least one colour such that, for every natural number $k$, among the numbers of that colour there exists a number divisible by $k$.

**Problem 6.282.** Prove that the number of primes that divide at least one number of the form $t^2 + t + 1$, for some integer $t$, is infinite.

**Problem 6.283.** Consider a grid made up of $1 \times 1$ squares. Each vertex of the grid is coloured in one of four given colours in such a way that the vertices of every $1 \times 1$ square have four different colours. Prove that there exists a straight line formed by vertices of the grid such that the vertices lying on it are coloured with only two colours.

**Problem 6.284.** Every positive integer is given some colour. Suppose that for every pair of integers $a, b \geq 2$ the numbers $a + b$ and $a^b$ have the same colour. Prove that all integers greater than 4 have the same colour.

**Problem 6.285.** Let $S$ be a set of real numbers that is closed under multiplication, that is, if $a$ and $b$ are in $S$ then so is $ab$. Let $T$ and $U$ be disjoint subsets of $S$ whose union is $S$. Given that the product of any three (not necessarily distinct) elements of $T$ is in $T$, and that the product of any three (not necessarily distinct) elements of $U$ is in $U$, show that at least one of the two subsets $T$ and $U$ is closed under multiplication.

**Problem 6.286.** Let $F = \{E_1, E_2, \ldots, E_s\}$ be a family of subsets, each with $r$ elements, of some set $X$. Show that if the intersection of any $r + 1$ (not necessarily distinct) sets in $F$ is nonempty, then the intersection of all the sets in $F$ is nonempty.

**Problem 6.287.** In a school there are 2018 students, and the school cafeteria sells 25 types of pie. Every student likes at least one type of pie, and every pie is liked by at least one student. A group of students is called *omnivorous* if it contains at least one fan of every type of pie. A *control group* is a group of students that includes at least one representative of every omnivorous group. It turned out that from the control group $G$ one cannot remove any nonempty subset of students while still keeping it a control group. Prove that there is at least one type of pie that is liked by every student in $G$.

## 6.10 How to write a proof

**Technique 6.12 (How to write a proof).** Organize the proof into discrete steps, giving each step its own paragraph unless it is one sentence long. Optionally open with a topic sentence for the whole proof, such as "We will solve the problem by induction on $n$" or "Let us assume the contrary". Before the main body, define all the terms you will use; a good definition (calling a suitable class of objects *good*, say) shortens everything that follows. Start each paragraph with a topic sentence describing what that paragraph proves. Use connecting phrases to guide the reader: "Thus", "Hence", "It follows that" when a consequence is drawn; "However", "On the other hand" for a contrast; "First", "Second", ..., "Finally" when listing cases; "Without loss of generality" when an arbitrary choice is made and the remaining cases follow analogously; "It suffices to prove" when you replace the claim by a stronger or equivalent one; "Note that" for a fact that will matter only later; and "This completes the proof" when the end is not obvious. Pictures are allowed but must be described in words as well ("We colour the squares of the chessboard as in the picture"). Any named theorem may be used, but say explicitly to which objects you are applying it and how. When in doubt, include a step even if it looks obvious, putting its one-line justification in brackets if it would otherwise interrupt the flow. Finally, write legibly, and summarize the conclusion in a closing sentence.

## 6.11 Further problems from the 2021–2022 courses

The Level-2 combinatorics course of the November 2021 camp and the Level-2 board-colouring classes of the January 2022 KAUST camp worked through their own problem lists. What follows is what those two lists add to the themes above: a second run at the pigeonhole principle, a systematic treatment of counting formulas, a long sequence of two-player games, and a first look at graphs.

**Problem 6.288.** Show that among any $n+1$ different numbers from the set $\{1, 2, \ldots, 2n\}$ it is possible to find two numbers that differ by $n$. Show that the statement no longer holds if $n+1$ is replaced by $n$.

**Problem 6.289.** Each square of a $3 \times 7$ board is coloured either black or white. Prove that there are four squares of the same colour whose centres are the vertices of a rectangle with sides parallel to the sides of the board.

**Problem 6.290.** Show that for any 5 points on a sphere one can find a closed hemisphere (that is, a hemisphere together with the corresponding great circle) containing 4 of the 5 points.

**Problem 6.291.** The lattice points of three-dimensional Euclidean space are the points all of whose coordinates $(a, b, c)$ are integers. Show that among any 9 lattice points there are two such that the segment joining them contains a lattice point in its interior (not necessarily one of the 9 selected points).

**Problem 6.292.** Show that for any $n$ integers $a_1, a_2, \ldots, a_n$ there exist indices $i$ and $j$ with $1 \le i \le j \le n$ such that $a_i + a_{i+1} + \cdots + a_j$ is divisible by $n$.

**Problem 6.293.** Show that there exists a number of the form $\underbrace{11\cdots1}_{k}$ that is divisible by 2021.

**Problem 6.294.** On each of the 20 faces of an icosahedron a non-negative integer is written, and the total sum of the numbers written is 39. Prove that there are two faces sharing a vertex on which the same number is written.

**Problem 6.295.** A collection of subsets of $\{1, 2, \ldots, n\}$ is given such that every two subsets in the collection have at least one element in common. Prove that the collection contains at most $2^{n-1}$ subsets.

**Definition 6.13 (Variations, combinations and permutations).** A *variation* of $k$ elements out of a set of $n$ elements is an ordered selection of $k$ of them; the number of variations is

$$V_k^n = n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}.$$

In a *variation with repetition* the same element may be selected several times; the number of these is $\overline{V}_k^n = n^k$. A *combination* is an unordered selection of $k$ elements out of $n$; the number of combinations is

$$C_k^n = \frac{V_k^n}{k!} = \frac{n(n-1)\cdots(n-k+1)}{k!} = \frac{n!}{k!\,(n-k)!} = \binom{n}{k}.$$

In a *combination with repetition* the same element may be selected several times; the number of these is $\overline{C}_k^n = \binom{n+k-1}{k}$. A *permutation* of $n$ elements is an ordering of all of them; the number of permutations is $P_n = n!$. Finally, in a *permutation with repetition* certain groups of elements are deemed identical: if the $n$ elements contain classes of identical elements of

sizes $\alpha_1, \alpha_2, \ldots, \alpha_k$ with $\alpha_1 + \cdots + \alpha_k \leq n$, the number of distinguishable orderings is

$$P_{n;\,\alpha_1,\ldots,\alpha_k} = \frac{n!}{\alpha_1! \cdots \alpha_k!}.$$

**Problem 6.296.** Give a combinatorial interpretation explaining why $P_{n;\,k,\,n-k} = C_k^n$.

**Problem 6.297.** In how many ways can 7 different balls be placed in 3 boxes if at most 1 ball may be placed in the first box, at most 2 balls in the second box and at most 6 balls in the third box?

**Problem 6.298.** Prove the formula for the number of combinations with repetition.

**Problem 6.299.** Let $n$ be a positive integer. A positive integer $d$ is said to be *ascending* if its decimal representation $\overline{d_m d_{m-1} \cdots d_2 d_1}$ satisfies $0 < d_m \leq d_{m-1} \leq \cdots \leq d_2 \leq d_1$; for example 1337 and 2455566799 are ascending. Find the number of ascending integers smaller than $10^n$.

**Problem 6.300.** In a mathematical committee there are $2n$ mathematicians, $n \geq 3$, and each of them is in a quarrel with exactly one other mathematician. They are to be divided into three committees for the upcoming competition: one for composing the problems, one for grading them and one for organizing the event. In how many ways can this be done so that each committee has at least two members and no two mathematicians who are in a quarrel sit on the same committee?

**Problem 6.301.** Find the number of integer sequences whose first term is 1, whose last term is 25, and in which every term after the first is larger than the preceding one by 2 or by 3.

**Problem 6.302.** In how many ways can 6 volumes of an encyclopedia be arranged on a shelf so that volume 1 is neither in the first nor in the last position, volumes 2 and 3 stand next to each other, and volumes 4 and 5 do not stand next to each other?

**Problem 6.303.** Find the number of ways to colour the cells of a $3 \times 3$ board in 4 colours so that no two cells in the same row or in the same column have the same colour.

**Problem 6.304.** There are 14 books arranged consecutively on a bookshelf. Find the number of ways of selecting 5 of these 14 books so that no two selected books are adjacent.

**Problem 6.305.** Find the number of permutations of the set $\{1, 2, 3, \ldots, n\}$ such that for each $i = 1, 2, \ldots, n$ the first $i$ numbers of the permutation are consecutive integers.

**Problem 6.306.** Let $k$, $l$ and $m$ be positive integers, and let $S$ be the set of integer points $(x, y, z)$ with $1 \leq x \leq k$, $1 \leq y \leq l$ and $1 \leq z \leq m$. A flea starts at the point $(1, 1, 1)$ of $S$ and wants to reach the point $(k, l, m)$, jumping only between points of $S$. From a point $(x, y, z)$ it can jump to one of the points $(x + 1, y, z)$, $(x, y + 1, z)$, $(x, y, z + 1)$, $(x, y, z - 1)$, but it may never jump to a point it has already visited. In how many different ways can the flea reach $(k, l, m)$?

**Problem 6.307.** A pawn is located on the second row and the first column (the a2 square) of an $8 \times 8$ chessboard. In how many ways can the pawn reach the eighth row (ignoring the moves of the other pieces) if it does not move two squares on its first move? A pawn moves from a square $(x, y)$ to one of the squares $(x, y + 1)$, $(x - 1, y + 1)$ and $(x + 1, y + 1)$, the latter two only when it captures a piece.

**Technique 6.14 (Winning and losing states).** A game is typically a large set of possible states, some of which are declared winning (the player who reaches such a state wins) and some losing. A *winning strategy* for one of the two players is a rule for choosing moves that guarantees that player the victory. Finding one usually means splitting the set of all states into two sets $W$ and $L$: the set $W$ of states from which the player to move will win, and the set $L$ of states from which the player to move will lose. For such a splitting to prove what is wanted, four conditions must hold: the winning final state lies in $W$, or the losing final state lies in $L$; from each state in $W$ some move leads to a state of $L$; from each state in $L$ every move leads to a state of $W$; and the rules of the game guarantee that a final state is reached after finitely many moves — which is usually proved by exhibiting a semi-invariant.

**Problem 6.308.** Alice and Bob play the following game, taking turns with Alice starting first. On a table there are 2020 coins. In the first move Alice takes some of the coins from the table, but not all of them. In each subsequent move a player may take only a number of coins that divides the number of coins taken in the previous move (a positive integer is regarded as divisible by itself). The winner is the player who takes the last coin. Which player has a winning strategy, and how should that player play?

**Problem 6.309.** A king is placed on the lower-left corner of an $m \times n$ chessboard. Players $A$ and $B$ take turns moving the king, with the restriction that the king may not move to the left, downwards or diagonally down-left; the remaining five directions are allowed. The winner is the player who places the king on the upper-right corner. For which $m$ and $n$ does which player have a winning strategy?

**Problem 6.310.** Initially there is a pile of $m$ coins and a pile of $n$ coins. In each move a player splits one of the existing piles into several (at least two) piles, which need not be of equal size. Players $A$ and $B$ move alternately, with $A$ starting first, and the player who cannot move loses. Determine, in terms of $m$ and $n$, which player has a winning strategy.

**Problem 6.311.** Alice and Bob play the following game. They arrange 2017 cookies in a row on a table and label them consecutively from 1 to 2017. In the $k$-th turn the player to move eats $k$ consecutive cookies still left on the table, where Alice takes the odd-numbered turns and Bob the even-numbered ones. The game ends when one cookie is left on the table: if the number of that cookie is odd then Alice wins, otherwise Bob wins. Which player has a winning strategy, and how should that player play?

**Problem 6.312.** Will and Bob play the following game. Will places his 8 white coins on the first row of an $8 \times 8$ chessboard and Bob places his 8 black coins on the eighth row of the same chessboard. They take turns moving their coins, with Will starting first. On his turn a player moves one of his coins along the column that coin stands in, in either direction and by as many steps as he wishes (but at least one), with the restriction that he may not land on the square occupied by an opponent's coin, and may not jump over it. The player who is first unable to make a move loses. Which player has a winning strategy, and how should that player play?

**Problem 6.313.** Alice and Bob play the following game. There are two piles of coins on a table, the first with $m$ coins and the second with $n$ coins, where $m \geq n$. In each turn a player may take an arbitrary number of coins from one of the piles, or an equal number of coins from both piles, subject to the restriction that the number of coins in the first pile may never become smaller than the number of coins in the second. The winner is the player who takes the last coin off the table. For which values of $m$ and $n$ does Alice have a winning strategy, and for which does Bob?

**Problem 6.314.** Alice and Bob play the following game. A pile of $n$ coins is given. Alice divides the pile into either two or three piles, not necessarily equal. Thereafter each player divides one of the piles obtained in the previous turn into two or three piles, again not necessarily equal. The player who is unable to make a move loses. For which values of $n$ does Alice have a winning strategy, and for which does Bob?

**Problem 6.315.** Let $n$ be a positive integer. Two players $A$ and $B$ play the following game with a pile of $s$ stones: they move alternately with $A$ going first, and on each turn the player to move takes either one stone, or a prime number of stones, or a positive multiple of $n$ stones. The winner is the one who takes the last stone. Assuming that both players play perfectly, for how many values of $s$ can player $A$ not win?

**Problem 6.316.** Given an initial integer $n_0 > 1$, two players $A$ and $B$ choose integers $n_1, n_2, n_3, \ldots$ alternately according to the following rules:

- knowing $n_{2k}$, player $A$ chooses any integer $n_{2k+1}$ such that $n_{2k} \leq n_{2k+1} \leq n_{2k}^2$;
- knowing $n_{2k+1}$, player $B$ chooses any integer $n_{2k+2}$ such that $n_{2k+1}/n_{2k+2}$ is a prime raised to a positive integer power.

Player $A$ wins the game by choosing the number 2021, and player $B$ wins by choosing the number 1. For which $n_0$ does (a) player $A$ have a winning strategy; (b) player $B$ have a winning strategy; (c) neither player have a winning strategy?

**Problem 6.317.** Call two squares of a board *knight-adjacent* if a knight can move from one to the other in one move. Find all pairs $(m, n)$ with $m, n \geq 2$ for which a knight can make a sequence of moves passing between each pair of knight-adjacent squares of an $m \times n$ board exactly once.

**Problem 6.318.** Colour $n$ of the vertices of a regular $2n$-gon red and the remaining $n$ vertices blue. Sort by length all the segments (sides as well as diagonals) whose endpoints are both red, and do the same for all the segments whose endpoints are both blue. Show that the two sorted sequences are identical, that is, that each pair of corresponding segments has equal length.

**Problem 6.319.** Call an $a \times b$ tile with both $a$ and $b$ even an *even rectangle*. Suppose that an $n \times n$ board with $n$ odd is tiled with even rectangles and unit squares. Determine the smallest possible number of unit squares used in such a tiling.

**Problem 6.320.** Some cells of an $m \times n$ chessboard are occupied by rooks. One rook attacks another if the two stand in the same row or in the same column and there is no other rook between them. Suppose that each rook is attacked by at most two other rooks. Find, in terms of $m$ and $n$, the largest number of rooks for which such a situation is possible.

# Chapter 7

# Geometry

**Sources for this chapter:** Camp handouts by Regis Barbosa (Feb 2025), Dominik Burek (June 2022, May 2025), Dren Neziri (Nov 2024), Łukasz Bożyk (Nov 2021, May 2023), Nikola Petrović (2021).

Olympiad geometry at this level is built from a small number of tools used over and over: similar triangles, the angle bisector theorems, the inscribed angle theorem, the ratio theorems of Menelaus and Ceva, equal tangent segments, and the power of a point. The sections below introduce each tool with the statements used in the camps, and then supply the problems on which the Saudi teams practised them. Nothing here needs coordinates or trigonometry beyond the law of sines and the law of cosines.

## 7.1 Angles, similar triangles, and the bisector theorems

**Theorem 7.1 (Criteria for similarity).** *If triangles $ABC$ and $DEF$ are similar, then $\widehat{A} = \widehat{D}$, $\widehat{B} = \widehat{E}$, $\widehat{C} = \widehat{F}$ and*

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD}.$$

*Two triangles are similar as soon as one of the following holds:*

*(a) two pairs of corresponding angles are equal (AA);*

*(b) two pairs of corresponding sides are proportional and the angles between them are equal (SAS);*

*(c) all three pairs of corresponding sides are proportional (SSS).*

**Theorem 7.2 (Thales's theorem).** *Let a family of parallel lines cut two transversals, the first in segments of lengths $a, b, c, \ldots$ and the second in the corresponding segments of lengths $x, y, z, \ldots$. Then*

$$\frac{a}{x} = \frac{b}{y} = \frac{c}{z} = \cdots = \frac{a+b}{x+y} = \cdots.$$

**Theorem 7.3 (Internal angle bisector theorem).** *Let $ABC$ be a triangle whose internal bisector from $A$ meets $BC$ at $D$. Then*

$$\frac{BD}{DC} = \frac{BA}{AC}.$$

**Theorem 7.4 (External angle bisector theorem).** *Let $ABC$ be a triangle whose external bisector at $A$ meets the line $BC$ at $E$. Then*

$$\frac{BE}{EC} = \frac{BA}{AC}.$$

### 7.1.1 Computing lengths

**Problem 7.1.** Point $D$ on side $AB$ of an equilateral triangle $ABC$ is such that $AD = 3$ and $BD = 5$. Compute $CD$.

**Problem 7.2.** Point $P$ inside a rectangle $ABCD$ is such that $AP = 4$, $BP = 1$ and $CP = 7$. Compute $DP$.

**Problem 7.3.** Given is a triangle $ABC$ with $AB = n + 1$, $BC = n + 2$ and $CA = n$, where $n > 3$. The circle centred at $C$ with radius $n$ meets the segment $AB$ at the points $A$ and $D$. Compute $AD$.

**Problem 7.4.** Circle $o_1$ of radius 4 and circle $o_2$ of radius 9 are externally tangent, and the line $l$ is one of their common external tangents. A circle $\omega$ is tangent to $l$, to $o_1$ and to $o_2$. Compute all possible radii of $\omega$.

**Problem 7.5.** A segment $AB$ of length 2 is a diameter of a circle $\gamma$ and a radius of a circle $\Gamma$. Find the radius of a circle that is externally tangent to $\gamma$, internally tangent to $\Gamma$, and tangent to the line $AB$.

**Problem 7.6.** Points $A$ and $D$ are distinct and lie on the same side of the line $BC$, with $AB = BC = CD$ and $AD \perp BC$. The lines $AD$ and $BC$ meet at $E$. Prove that

$$|BE - CE| < AD \cdot \sqrt{3}.$$

**Problem 7.7.** (The angle bisector theorem; compare Theorem 7.3.) Let $D$ be a point on the side $AB$ of a triangle $ABC$. Prove that $\angle ACD = \angle DCB$ if and only if

$$\frac{AC}{CB} = \frac{AD}{DB}.$$

### 7.1.2 Midpoints and congruences

**Problem 7.8.** Given is an acute triangle $ABC$ with $\angle ACB = 60°$. Segments $AK$ and $BL$ are two of its altitudes. Prove that $AB = 2 \cdot KL$.[†]

**Problem 7.9.** Given is a convex quadrilateral $ABCD$ in which $M$ and $N$ are the midpoints of $BC$ and $DA$, respectively. Prove that $AB + CD \geq 2 \cdot MN$, and describe the equality case.

**Problem 7.10.** Given is an acute triangle $ABC$. Points $D$ and $E$ are the projections of $A$ and $B$ onto $BC$ and $CA$, respectively. Points $P$ and $Q$ are the projections of $A$ and $B$ onto the line $DE$. Prove that $PE = DQ$.

**Problem 7.11.** In a triangle $ABC$ the point $D$ is the midpoint of the side $BC$ and the point $E$ is the midpoint of the segment $AD$. Prove that if $\angle ABE = \angle CAD$, then $AB = AD$.

**Problem 7.12.** Given is a convex pentagon $ABCDE$ with $\angle B = \angle D = 90°$. Prove that the perimeter of the triangle $ACE$ is not greater than $2 \cdot BD$.

**Problem 7.13.** Prove that if the diagonals of a trapezoid are perpendicular, then the sum of the lengths of the bases of this trapezoid is not greater than the sum of the lengths of its legs.

**Problem 7.14.** Given is a convex quadrilateral $ABCD$ in which $M$ is the midpoint of $CD$. Prove that if $\angle AMB = 90°$, then $AD + BC \geq AB$.

**Problem 7.15.** Given is a triangle $ABC$ with $AB = 3 \cdot BC$. Points $P$ and $Q$ divide the side $AB$ into three equal parts, so that $AP = PQ = QB$. Let $M$ be the midpoint of $AC$. Prove that $\angle PMQ = 90°$.

**Problem 7.16.** Let $M$ be the midpoint of the side $AB$ of a triangle $ABC$. Right triangles $ACK$ and $BCL$ are built outside $ABC$ so that

$$\angle AKC = \angle BLC = 90° \quad \text{and} \quad \angle CAK = \angle CBL.$$

Prove that $MK = ML$.

**Problem 7.17.** A point $P$ lies inside a parallelogram $ABCD$ and satisfies $PC = BC$. Prove that the line joining the midpoints of the segments $AP$ and $CD$ is perpendicular to $BP$.

### 7.1.3 Rotations and equilateral triangles

**Problem 7.18.** Let $P$ be a point inside an equilateral triangle $ABC$, and let $D$, $E$, $F$ be the projections of $P$ onto the sides $BC$, $CA$, $AB$, respectively.

(a) Prove that $PD + PE + PF$ does not depend on the choice of $P$.
(b) Prove that $[AFP] + [BDP] + [CEP] = [AEP] + [BFP] + [CDP]$.

**Problem 7.19.** On the sides $BC$ and $CD$ of a parallelogram $ABCD$, equilateral triangles $BCP$ and $CDQ$ are constructed outwards. Prove that the triangle $APQ$ is equilateral.

**Problem 7.20.** Let $ABC$ be an equilateral triangle.

(a) Let $P$ be an arbitrary point inside $ABC$. Prove that there exists a triangle with side lengths $AP$, $BP$, $CP$.
(b) Let $P$ lie inside $ABC$ so that $\angle APB = 150°$. Prove that $AP^2 + BP^2 = CP^2$.
(c) Let $P$ lie outside $ABC$ so that $\angle APB = 120°$. Prove that $AP + BP = CP$.

**Problem 7.21.** Equilateral triangles $ABC$ and $CDE$ lie on the same side of the line $AE$. Prove that $C$ and the midpoints of $AD$ and $BE$ are the vertices of an equilateral triangle.

**Problem 7.22.** On the sides $AB$ and $AC$ of a triangle $ABC$, equilateral triangles $ABP$ and $ACQ$ are constructed outside. Prove that the midpoints of the segments $BC$, $AP$, $AQ$ are the vertices of an equilateral triangle.

**Problem 7.23.** A line $l$ passes through the centre of an equilateral triangle $ABC$. Prove that among the distances of the points $A$, $B$, $C$ from $l$, one is the sum of the other two.

**Problem 7.24.** Given is a rhombus $ABCD$ with $\angle BAD = 60°$. A point $P$ lies inside this rhombus and satisfies $BP = 1$, $DP = 2$, $CP = 3$. Find $AP$.

**Problem 7.25.** Suppose that inside an equilateral triangle $ABC$ there exists a point $P$ with $AP = 3$, $BP = 4$, $CP = 5$. Find $AB$.

**Problem 7.26.** Given is an equilateral triangle $ABC$. Points $D$, $E$, $F$ lie on the sides $BC$, $CA$, $AB$, respectively, and satisfy $AF = BD$ and $DF = EF \neq DE$. Prove that $\angle CDE = 90°$.

**Problem 7.27.** Let $ABC$ be an equilateral triangle and let $l$ be a line not passing through any of its vertices, meeting the lines $AB$, $BC$, $CA$ at $M$, $K$, $L$, respectively. Prove that there exists a point $P$ such that $AK = PK$, $BL = PL$ and $CM = PM$.

## 7.2 The trigonometric toolbox

**Theorem 7.5 (Law of sines).** *In a triangle $ABC$ with circumradius $R$ and side lengths $a = BC$, $b = CA$, $c = AB$,*

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R.$$

**Theorem 7.6 (Law of cosines).** *In a triangle $ABC$ with side lengths $a$, $b$, $c$ as above,*

$$c^2 = a^2 + b^2 - 2ab\cos C.$$

**Theorem 7.7 (Area and projection formulas).** *For a triangle $ABC$ with side lengths $a$, $b$, $c$,*

$$[ABC] = \tfrac{1}{2}ab\sin C,$$

*and the three sides satisfy the projection formulas*

$$a^2 = ac\cos B + ab\cos C, \qquad b^2 = ba\cos C + bc\cos A, \qquad c^2 = ca\cos B + cb\cos A;$$

*equivalently $a = c\cos B + b\cos C$ and cyclically.*

**Lemma 7.8 (Products to sums).** *For all angles $A$ and $B$,*

$$\sin A\cos B = \frac{\sin(A+B) + \sin(A-B)}{2}, \qquad \sin A\sin B = \frac{\cos(A-B) - \cos(A+B)}{2},$$

$$\cos A\cos B = \frac{\cos(A-B) + \cos(A+B)}{2}.$$

**Lemma 7.9 (Equal sine ratios).** *Let $\alpha, \beta, \gamma, \delta$ be angles with $\alpha + \beta = \gamma + \delta = S$ and $\sin S \neq 0$. If*

$$\frac{\sin\alpha}{\sin\beta} = \frac{\sin\gamma}{\sin\delta},$$

*then $\alpha = \gamma$ and $\beta = \delta$.*

**Theorem 7.10 (The ratio lemma).** *Let $CD$ be a cevian of a triangle $ABC$, with $D$ on the line $AB$. Then*

$$\frac{AD}{BD} = \frac{AC \cdot \sin \angle ACD}{BC \cdot \sin \angle BCD}.$$

**Theorem 7.11 (Trigonometric form of Ceva's theorem).** *For cevians $AD$, $BE$, $CF$ of a triangle $ABC$, the lines $AD$, $BE$, $CF$ are concurrent if and only if*

$$\frac{\sin \angle BAD}{\sin \angle DAC} \cdot \frac{\sin \angle ACF}{\sin \angle FCB} \cdot \frac{\sin \angle CBE}{\sin \angle EBA} = 1.$$

**Definition 7.12 (Isogonal conjugates).** Two cevians through a vertex are called *isogonal* if they are reflections of one another in the internal bisector at that vertex. If $P$ and $P'$ are points such that $AP$ is isogonal to $AP'$ and $BP$ is isogonal to $BP'$, then $CP$ and $CP'$ are isogonal as well; the points $P$ and $P'$ are then called *isogonal conjugates*.

**Problem 7.28.** Consider an acute triangle $ABC$ with orthocentre $H$ and circumradius $R$. Given that $AH = R$, what is the measure of the angle $\angle BAC$?

**Problem 7.29.** Let $ABC$ be a triangle. Assume that

$$\sin^2 \angle A + \sin^2 \angle B + \sin^2 \angle C = 2.$$

Prove that the triangle $ABC$ is right-angled.

**Problem 7.30.** In a triangle $ABC$ we have $AB = BC$ and $\angle B = 20°$. A point $M$ on $AC$ is such that $AM : MC = 1 : 2$, and $H$ is the projection of $C$ onto $BM$. Find the angle $\angle AHB$.

**Problem 7.31.** Let $ABC$ be an acute triangle with $AB < BC$, and let $D$ and $E$ be points on the sides $AB$ and $BC$, respectively, such that $AD = DE = EC$. The segments $AE$ and $DC$ intersect at the point $X$. If $AX = XC$, determine the measure of the angle $\angle ABC$. [Rioplatense 2015]

**Problem 7.32.** Let $a$, $b$, $c$, $d$ be lines passing through a point $P$, and let a line $l$ meet them at the points $A$, $B$, $C$, $D$, respectively. Prove that the quantity

$$\frac{AC \cdot BD}{BC \cdot AD}$$

does not depend on the choice of the line $l$.

**Problem 7.33.** Let $ABC$ be a triangle such that $AC^2$ is equal to the arithmetic mean of $BC^2$ and $AB^2$. Prove that

$$\cot \frac{\angle B}{2} \geq \cot \angle A \cdot \cot \angle C.$$

†

**Problem 7.34.** Let $I$ be the incentre of a triangle $ABC$ with side lengths $a$, $b$, $c$. Prove that

$$\frac{AI^2}{bc} + \frac{BI^2}{ca} + \frac{CI^2}{ab} = 1.$$

**Problem 7.35.** Let $ABC$ be a triangle with semiperimeter $s$ and $c = AB$. Suppose that $D$ lies on $AB$ and that the inradii of the triangles $ADC$ and $CDB$ are equal. Prove that

$$CD^2 = s(s-c) = [ABC] \cdot \cot\left(\frac{\angle C}{2}\right).$$

**Problem 7.36.** A line tangent to the incircle of an equilateral triangle $ABC$ cuts the sides $AB$ and $AC$ at $D$ and $E$, respectively. Prove that

$$\frac{AD}{DB} + \frac{AE}{EC} = 1.$$

**Problem 7.37.** Given a triangle $ABC$, let $D$ and $E$ be points on the side $BC$ such that $\angle BAD = \angle CAE$. If $M$ and $N$ are, respectively, the points of tangency of the incircles of the triangles $ABD$ and $ACE$ with the line $BC$, prove that

$$\frac{1}{MB} + \frac{1}{MD} = \frac{1}{NC} + \frac{1}{NE}.$$

**Problem 7.38.** Let $D$, $E$, $F$ be the points where the incircle touches the sides $BC$, $CA$, $AB$ of a triangle $ABC$, and let $M$ be the midpoint of $EF$. Prove that $\angle BMC > 90°$.

**Problem 7.39.** (Construction of the symmedian.) Let $ABC$ be a triangle and let $S$ be the meeting point of the tangents to the circumcircle at $B$ and at $C$. Show that $AS$ is the symmedian from $A$, that is, that $AS$ is the isogonal of the median from $A$.

**Problem 7.40.** Let $ABC$ be an acute triangle with $AB \neq AC$, let $V$ be the intersection of the internal bisector of the angle $\angle A$ with $BC$, and let $D$ be the foot of the altitude from $A$. If $E$ and $F$ are the intersections of the circumcircle of the triangle $ADV$ with the lines $CA$ and $AB$, respectively, show that the lines $AD$, $BE$ and $CF$ are concurrent.

**Problem 7.41.** A circle cuts the sides $AB$, $BC$ and $CA$ of a triangle $ABC$ at the points $D_1, D_2$, $E_1, E_2$, $F_1, F_2$, respectively (they occur in this order on the circle). The segments $D_1E_1$ and $D_2E_2$ meet at $L$, the segments $E_1F_1$ and $E_2F_2$ meet at $M$, and the segments $F_1D_1$ and $F_2D_2$ meet at $N$. Prove that $AN$, $BL$ and $CM$ are concurrent.

**Problem 7.42.** Let $ABCD$ be a convex quadrilateral such that $\angle DAB = 90°$. Let $M$ be the midpoint of $BC$. If $\angle ADC = \angle BAM$, prove that $\angle ADB = \angle CAM$. [Russia]

**Problem 7.43.** Let $ABCD$ be a convex quadrilateral with the properties $AD = CD$ and $\angle DAB = \angle ABC < 90°$. The line through $D$ and the midpoint of $BC$ meets $AB$ at the point $E$. Prove that $\angle BEC = \angle DAC$. [Bulgaria 1998]

**Problem 7.44.** Let $ABC$ be an acute triangle with altitudes $AD$, $BE$, $CF$, and let $O$ be the centre of its circumcircle. Show that the segments $OA$, $OB$, $OC$, $AD$, $BE$, $CF$ dissect the triangle $ABC$ into three pairs of triangles that have equal areas. [APMO 2013]

**Problem 7.45.** Let $A_1$ be the centre of the square inscribed in an acute triangle $ABC$ with two vertices of the square on the side $BC$; thus one of the two remaining vertices of the square is on $AB$ and the other is on $AC$. Points $B_1$ and $C_1$ are defined in a similar way, for inscribed squares with two vertices on the sides $CA$ and $AB$, respectively. Prove that the lines $AA_1$, $BB_1$, $CC_1$ are concurrent. [IMO Shortlist 2001]

**Problem 7.46.** Let $ABC$ be an acute triangle with circumcentre $O$. The line $AO$ meets $BC$ at $D$. Points $E$ and $F$ lie on $AB$ and $AC$, respectively, and are such that $A$, $E$, $D$, $F$ are concyclic. Prove that the length of the projection of the segment $EF$ on the side $BC$ does not depend on the positions of $E$ and $F$. [CGMO 2004]

**Problem 7.47.** Let $ABC$ be a triangle with $AB = AC$. The angle bisectors of $\angle CAB$ and $\angle ABC$ meet the sides $BC$ and $CA$ at $D$ and $E$, respectively. Let $K$ be the incentre of the triangle $ADC$. Suppose that $\angle BEK = 45°$. Find all possible values of $\angle CAB$. [IMO 2009]

## 7.3 Cyclic quadrilaterals

**Theorem 7.13 (Central and inscribed angles).** *Let $A$, $B$, $C$ be points on a circle with centre $O$. The central angle satisfies $m(\angle AOB) = m(\overset{\frown}{AB})$, and the inscribed angle satisfies*

$$m(\angle ACB) = \frac{m(\overset{\frown}{AB})}{2} = \frac{m(\angle AOB)}{2}.$$

*In particular, if $O$ lies on $AB$, then $m(\angle AOB) = 180°$ and $m(\angle ACB) = 90°$.*

**Theorem 7.14 (Tangent–chord angle).** *If $BC$ is tangent at $B$ to the circle through $A$ and $B$ with centre $O$, then the semi-inscribed angle satisfies*

$$m(\angle ABC) = \frac{m(\overset{\frown}{AB})}{2} = \frac{m(\angle AOB)}{2}.$$

**Theorem 7.15 (Two tangents from a point).** *If the two tangent lines from a point $P$ touch a circle at $A$ and at $B$, then $PA = PB$.*

**Definition 7.16 (Cyclic quadrilateral).** A quadrilateral $ABCD$ is called *cyclic* (or *inscribed*) when there exists a circle passing through its four vertices. The main tools it provides are that opposite angles add up to 180°, i.e. $\angle ABC + \angle ADC = 180°$, and that angles subtending the same side from the same side are equal, i.e. $\angle BAC = \angle BDC$.

**Theorem 7.17 (Criteria for concyclicity).** *Let $ABCD$ be a quadrilateral.*
*(a) If $\angle BAC = \angle BDC$ (with $A$ and $D$ on the same side of $BC$), then $ABCD$ is cyclic.*
*(b) If $\angle BAD + \angle BCD = 180°$, then $ABCD$ is cyclic.*

**Theorem 7.18 (Perpendicular bisector and circumcircle).** *The perpendicular bisector of a segment $XY$ is the line perpendicular to $XY$ through the midpoint of $XY$; it is exactly the locus of points $P$ of the plane with $PX = PY$. Consequently, for every triangle there is exactly one circle passing through its three vertices. This circle is the* circumcircle *of the triangle and its centre is the* circumcentre.

**Problem 7.48.** In a cyclic quadrilateral $PQRS$ with $\angle PSR = \frac{\pi}{2}$, let $H$ and $K$ be the feet of the perpendiculars from $Q$ to the sides $PR$ and $RS$. Prove that $HK$ bisects the segment $SQ$. [Hong Kong 1998]

**Problem 7.49.** Let $ABCD$ be a cyclic quadrilateral such that $BC = CD$. Show that

$$AC^2 = BC^2 + AB \cdot AD.$$

**Problem 7.50.** Let $ABCD$ be a convex quadrilateral such that the diagonals $AC$ and $BD$ are perpendicular, and let $P$ be their intersection. Prove that the reflections of $P$ with respect to $AB$, $BC$, $CD$, $DA$ lie on a circle. [USAMO 1993]

**Problem 7.51.** Given a triangle $ABC$, let $P$ and $Q$ be points on the segments $AB$ and $AC$, respectively, such that $AP = AQ$. Let $S$ and $R$ be distinct points on the segment $BC$ such that $S$ lies between $B$ and $R$, $\angle BPS = \angle PRS$, and $\angle CQR = \angle QSR$. Prove that $P$, $Q$, $R$, $S$ are concyclic.

**Problem 7.52.** Let $\Gamma$ be the circumcircle of a triangle $ABC$ and let $D$ be a point on the side $BC$. The tangent to $\Gamma$ at $A$ meets the line through $D$ parallel to $BA$ at the point $E$. The segment $CE$ meets $\Gamma$ again at $F$. Suppose that $B$, $D$, $F$, $E$ are concyclic. Prove that $AC$, $BF$, $DE$ are concurrent.

**Problem 7.53.** Let $ABCD$ be a parallelogram with $AC = BC$. A point $P$ is chosen on the extension of the ray $AB$ beyond $B$. The circumcircle of $ACD$ meets the segment $PD$ again at $Q$, and the circumcircle of the triangle $APQ$ meets the segment $PC$ at $R$. Prove that the lines $CD$, $AQ$, $BR$ are concurrent.

**Problem 7.54.** Consider a triangle $ABC$ where $\angle A$ is the smallest of the three interior angles, and let $\Gamma$ be the circle passing through the vertices of $ABC$. Let $D$ be a point on the arc $BC$ of $\Gamma$ that does not contain $A$. Let $E$ be a point on the arc $AC$ of $\Gamma$ that does not contain $B$ such that $CE = BD$, and let $F$ be a point on the arc $AB$ of $\Gamma$ that does not contain $C$ such that $BF = CD$. The segment $EF$ cuts the sides $AB$ and $AC$ at the points $P$ and $Q$, respectively. It is also known that $DF$ cuts the side $AB$ at $Y$ and $DE$ cuts the side $AC$ at $X$. Let $R$ be the intersection of the lines $PX$ and $QY$. Prove that the quadrilateral $APRQ$ is cyclic.

## 7.4 Menelaus and Ceva

**Theorem 7.19 (Menelaus's theorem).** *Let $X$, $Y$ and $Z$ be points on the lines $AB$, $BC$ and $CA$ of a triangle $ABC$, respectively. Then $X$, $Y$, $Z$ lie on one line if and only if*

$$\frac{AX}{XB} \cdot \frac{BY}{YC} \cdot \frac{CZ}{ZA} = 1.$$

**Theorem 7.20 (Ceva's theorem).** *Let $X$, $Y$ and $Z$ be points on the lines $BC$, $CA$ and $AB$ of a triangle $ABC$, respectively. The lines $AX$, $BY$ and $CZ$ are concurrent if and only if*

$$\frac{BX}{XC} \cdot \frac{CY}{YA} \cdot \frac{AZ}{ZB} = 1.$$

**Problem 7.55.** (Ceva's theorem; compare Theorem 7.20.) Points $D$, $E$, $F$ lie on the sides $BC$, $CA$, $AB$ of a triangle $ABC$, respectively. Prove that the segments $AD$, $BE$, $CF$ have a common point (are concurrent) if and only if

$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1.$$

**Problem 7.56.** In a triangle $ABC$, the median and the altitude from the vertex $A$ divide the angle $B\widehat{A}C$ into three angles of the same measure. Determine the measures of the three angles of the triangle $ABC$. [Brazil]

**Problem 7.57.** Let the external bisector of $\angle BAC$ of a triangle $ABC$ meet the line $BC$ at $A'$, and define $B'$ and $C'$ analogously. Prove that $A'$, $B'$, $C'$ are collinear.

**Problem 7.58.** Let $ABCD$ be a trapezoid with $AB \parallel CD$ and let $X$ be a point on the segment $AB$. Put $P = BC \cap AD$, $Y = CD \cap PX$, $R = AY \cap BD$ and $T = PR \cap AB$. Prove that

$$\frac{1}{AT} = \frac{1}{AX} + \frac{1}{AB}.$$

**Problem 7.59.** In a triangle $ABC$ let $D$ be the point on the segment $BC$, and $E$ the point on the segment $CA$, for which $BD = CE = AB$. Let $l$ be the line through $D$ parallel to $AB$. If $M = l \cap BE$ and $F = CM \cap AB$, prove that

$$AE \cdot BF \cdot CD = (AB)^3.$$

**Problem 7.60.** Let $ABCD$ be a convex quadrilateral. A line $k$ meets $DA$, $AB$, $BC$ and $CD$ at $X$, $Y$, $Z$ and $T$, respectively. Prove that

$$\frac{DX}{XA} \cdot \frac{AY}{YB} \cdot \frac{BZ}{ZC} \cdot \frac{CT}{TD} = 1.$$

**Problem 7.61.** Let $ABCD$ be a parallelogram. Points $K$ and $L$ lie on the sides $AB$ and $AD$, respectively. The segments $DK$ and $BL$ intersect at $P$. A point $Q$ is chosen so that $AKQL$ is a parallelogram. Prove that $P$, $Q$, $C$ are collinear.

**Problem 7.62.** Let $ABC$ be a triangle with $\angle A = 100^\circ$ and $\angle B = 60^\circ$, and let $M \in BC$ and $N \in AC$ be points for which $\angle BAM = 30^\circ$ and $\angle ABN = 20^\circ$. Prove that the lines $AM$, $BN$ and the bisector of $\angle ACB$ are concurrent.

**Problem 7.63.** Let $ABC$ be a right triangle with the right angle at $C$. On the sides $BC$ and $CA$ squares $BEFC$ and $CGHA$ are built outwards. Let $D$ be the foot of the altitude from $C$ to $AB$. Prove that $AE$, $BH$ and $CD$ are concurrent.

**Problem 7.64.** A circle meets the sides $BC$, $CA$ and $AB$ of a triangle $ABC$ at the points $A_1, A_2$, $B_1, B_2$ and $C_1, C_2$, respectively. Prove that the lines $AA_1$, $BB_1$, $CC_1$ are concurrent if and only if the lines $AA_2$, $BB_2$, $CC_2$ are concurrent.

**Problem 7.65.** Let $ABC$ be a triangle. Prove that the lines joining the midpoints of the sides with the midpoints of the corresponding altitudes pass through a single point.

**Problem 7.66.** Let $ABCDEF$ be a hexagon inscribed in a circle $\omega$. Show that the diagonals $AD$, $BE$, $CF$ are concurrent if and only if

$$AB \cdot CD \cdot EF = BC \cdot DE \cdot FA.$$

**Problem 7.67.** Prove that in a triangle $ABC$ the interior bisector of the angle $A$, the median from $B$ and the altitude from $C$ are concurrent if and only if

$$\tan A = \frac{\sin C}{\cos B}.$$

**Problem 7.68.** In an acute triangle $ABC$ a semicircle $\omega$ centred on the side $BC$ is tangent to the sides $AB$ and $AC$ at the points $F$ and $E$, respectively. If $X$ is the intersection of $BE$ and $CF$, show that $AX \perp BC$.

**Problem 7.69.** Prove that in a regular 30-gon $A_1A_2 \ldots A_{30}$ the diagonals $A_1A_{19}$, $A_3A_{24}$ and $A_8A_{28}$ are concurrent.

**Problem 7.70.** In a triangle $ABC$ the internal angle bisectors $t_a$, $t_b$, $t_c$ meet $BC$, $CA$, $AB$ at $U$, $V$, $W$, respectively, and the medians $m_a$, $m_b$, $m_c$ meet $BC$, $CA$, $AB$ at $L$, $M$, $N$, respectively. Let $m_a \cap t_b = P$, $m_b \cap t_c = Q$ and $m_c \cap t_a = R$. Prove that

$$\frac{AR}{RU} \cdot \frac{BP}{PV} \cdot \frac{CQ}{QW} \ \geq \ 8.$$

**Problem 7.71.** Let $P$ be a point inside an equilateral triangle $ABC$. Let $AP$, $BP$, $CP$ meet the sides $BC$, $CA$, $AB$ at $A_1$, $B_1$, $C_1$. Prove that

$$A_1B_1 \cdot B_1C_1 \cdot C_1A_1 \ \geq \ A_1B \cdot B_1C \cdot C_1A.$$

When does equality hold?

**Problem 7.72.** Let $ABC$ be a triangle. The lines $r$ and $s$ are the internal bisectors of $\angle ABC$ and $\angle BCA$, respectively. The points $E$ on $r$ and $D$ on $s$ are such that $AD \parallel BE$ and $AE \parallel CD$. The lines $BD$ and $CE$ intersect at $F$. Let $I$ be the incentre of the triangle $ABC$. Show that if the points $A$, $F$ and $I$ are collinear, then $AB = AC$. [Brazil]

## 7.5 Tangents and tangential polygons

**Problem 7.73.** In a triangle $ABC$ the incircle and the excircle opposite to $A$ touch the segment $BC$ at the points $D$ and $T$, respectively. Prove that $BD = CT$.

**Problem 7.74.** In a triangle $ABC$ the excircles opposite to $B$ and to $C$ touch the segments $CA$ and $AB$ at the points $E$ and $F$, respectively.

(a) Prove that $BF = CE$.

(b) Prove that $AE + AF = BC$.

**Problem 7.75.** Two excircles of a triangle $ABC$ are tangent to the sides $BC$ and $AC$ at the points $D$ and $E$, respectively. Prove that $AE = BD$.

**Problem 7.76.** In a quadrilateral $ABCD$ the internal angle at $C$ is greater than $180°$. Suppose that there exists a circle tangent to the sides $BC$, $CD$ and to the extensions of the sides $AB$ and $DA$. Prove that $AB + BC = CD + DA$.

**Problem 7.77.** In a triangle $ABC$ the point $I$ is the incentre and $M$ is the midpoint of the side $BC$. Prove that if $AI = IM$, then one of the sides of $ABC$ is twice as long as another side of this triangle.

**Problem 7.78.** Let $ABCD$ be a unit square. Points $E$ and $F$ lie on the sides $BC$ and $CD$, respectively, so that the triangle $CEF$ has perimeter 2. Determine the distance from $A$ to the line $EF$.

**Problem 7.79.** A point $X$ is chosen on the side $BC$ of a triangle $ABC$. The common tangent of the incircles of the triangles $ABX$ and $ACX$ which is different from $BC$ meets the segment $AX$ at $Y$. Prove that the length of $CY$ does not depend on the choice of $X$.

**Problem 7.80.** Given two circles $\omega_1$ and $\omega_2$, a common transversal tangent $l$ touches the circles $\omega_1$ and $\omega_2$ at the points $B$ and $C$, respectively. The line $l$ meets the common external tangents of $\omega_1$ and $\omega_2$ at the points $A$ and $D$. Prove that $AB = CD$.

**Problem 7.81.** A variable point $D$ lies on the side $AB$ of a fixed triangle $ABC$. The common external tangent of the incircles of the triangles $ADC$ and $BDC$ (different from the line $AB$) meets the line $CD$ at the point $E$. Find the locus of the points $E$ as $D$ varies on the segment $AB$.

**Problem 7.82.** Let $ABC$ be a triangle with inradius $r$ and let $\omega$ be a circle of radius $a < r$ inscribed in the angle $BAC$. The tangents from $B$ and from $C$ to $\omega$ (different from the sides of the triangle) meet at a point $X$. Show that the incircle of the triangle $BCX$ is tangent to the incircle of the triangle $ABC$.

**Problem 7.83.** Choose a point inside an equilateral triangle and drop the perpendiculars from it to the sides of this triangle. They divide the triangle into six triangles. Prove that the sum of the inradii of three alternate ones among these six triangles equals the sum of the inradii of the remaining three.

**Problem 7.84.** In a trapezoid $ABCD$ the bisectors of the angles $A$ and $D$ intersect at a point lying on $BC$. These bisectors split the trapezoid into three triangles, in each of which a circle is inscribed. One of these circles touches the base $AB$ at $K$, while the other two are tangent to the bisector $DE$ at the points $M$ and $N$. Prove that $BK = MN$.

**Problem 7.85.** A convex quadrilateral $ABCD$ is cut into nine convex quadrilaterals arranged in a $3 \times 3$ pattern. Prove that if the four corner pieces and the central piece all possess incircles, then $ABCD$ is circumscribed about a circle.[†]

**Problem 7.86.** Prove that a quadrilateral $ABCD$ has an incircle if and only if the incircles of the triangles $ABD$ and $BCD$ are tangent to each other.

**Problem 7.87.** Let $ABCD$ be a circumscribed quadrilateral and let the point $P$ lie on the side $CD$. Prove that there exists a common tangent to the incircles of the triangles $ABP$, $BCP$ and $ADP$.

**Problem 7.88.** Points $P$ and $Q$ lie on the sides $AB$ and $AD$ of a convex quadrilateral $ABCD$. The lines $DP$ and $BQ$ meet at $S$. Prove that if the quadrilaterals $APSQ$ and $BCDS$ have incircles, then the quadrilateral $ABCD$ has an incircle as well.

**Problem 7.89.** Let $ABCD$ be a tangential quadrilateral with incentre $I$, and let $K$, $L$, $M$, $N$ be the tangency points of the incircle with the segments $AB$, $BC$, $CD$, $DA$, respectively. Suppose that the lines $KL$ and $MN$ meet at $P$. Prove that $BD \perp PI$.

## 7.6 Power of a point and the radical axis

**Definition 7.21 (Power of a point).** The power of a point $P$ with respect to a circle $\Gamma$ with centre $O$ and radius $r$ is

$$\operatorname{Pot}_\Gamma P = PO^2 - r^2.$$

**Theorem 7.22 (Secants and tangents).** *If $P$ lies inside the circle $\Gamma$ and a line through $P$ cuts $\Gamma$ at $A$ and $B$, then*

$$PA \cdot PB = -\operatorname{Pot}_\Gamma P.$$

*If $P$ lies outside $\Gamma$, a line through $P$ cuts $\Gamma$ at $A$ and $B$, and another line through $P$ is tangent to $\Gamma$ at $T$, then*

$$PA \cdot PB = PT^2 = \operatorname{Pot}_\Gamma P.$$

**Theorem 7.23 (Radical axis).** *Given two circles $\Gamma_1$ and $\Gamma_2$ with different centres, the locus of the points $P$ of the plane for which*

$$\operatorname{Pot}_{\Gamma_1} P = \operatorname{Pot}_{\Gamma_2} P$$

*is a line perpendicular to the line through the centres of $\Gamma_1$ and $\Gamma_2$.*

### 7.6.1 Power of a point

**Problem 7.90.** Chords $AB$ and $CD$ of a circle intersect at a point $P$. It is known that $PA = 3$, $PB = 4$ and $PC = 2$. What is the length of $PD$?

**Problem 7.91.** Let $\omega$ and $\gamma$ be two circles intersecting at $P$ and $Q$. Let their common external tangent touch $\omega$ at $A$ and $\gamma$ at $B$. Prove that $PQ$ passes through the midpoint $M$ of $AB$.

**Problem 7.92.** Consider two non-overlapping circles with centres $O_1$ and $O_2$. Let $O_2E$ be tangent to the circle $(O_1)$ at $E$, and let $O_1G$ be tangent to the circle $(O_2)$ at $G$. Suppose that $EG$ meets the two circles again at $J$ and $L$, respectively. Prove that $EJ = GL$.

**Problem 7.93.** Let $\omega$ be the incircle of a square $ABCD$. The diagonal $AC$ meets the circle $\omega$ at a point $E$ with $AE < EC$. The segment $BE$ meets $\omega$ at the point $F$. Prove that $EF = 2 \cdot BF$.

**Problem 7.94.** Determine the ratio of the lengths of the sides of a triangle in which the incircle divides one of the medians into three equal parts.

**Problem 7.95.** Let $ABC$ be an equilateral triangle. A circle meets the segments $AB$, $BC$ and $CA$ at the points $K, L$; $M, N$; $P, Q$, where the points $K, L, M, N, P, Q$ lie on the circle in this order. Prove that

$$AK + BM + CP = BL + CN + AQ.$$

**Problem 7.96.** A line $k$ is tangent to a circle $o$ at a point $A$. A chord $CD$ of the circle $o$ is parallel to the line $k$. The tangent to the circle $o$ at the point $D$ meets the line $k$ at $B$. The segment $BC$ meets the circle $o$ at $E$. Prove that the line $DE$ bisects the segment $AB$.

**Problem 7.97.** Semicircles with centres $O_1$ and $O_2$ are drawn on chords $AB$ and $CD$ of a circle $\Gamma$ in such a way that they are tangent at a point $T$. The line through $O_1$ and $O_2$ meets $\Gamma$ at $E$ and $F$. Prove that

$$O_1A - O_2C = O_1E - O_2F.$$

**Problem 7.98.** Given are two circles $\omega_1$, $\omega_2$ with different centres. Their common external tangents $AB$ and $CD$ are such that $A, C \in \omega_1$ and $B, D \in \omega_2$. The segment $AD$ meets $\omega_1$ and $\omega_2$ at $E$ and $F$, respectively. Prove that $AE = DF$.

**Problem 7.99.** Let $BE$ and $CF$ be altitudes of a triangle $ABC$ and let their intersection be $H$. If $M$ and $N$ are the midpoints of $BC$ and $EF$, respectively, prove that $MN \perp EF$.

**Problem 7.100.** Let a triangle $ABC$ have orthocentre $H$. Points $D$ and $E$ lie on the sides $AB$ and $AC$, respectively. Prove that $H$ lies on the radical axis of the circle with diameter $CD$ and the circle with diameter $BE$. Prove furthermore that if $DE \parallel BC$, then the two circles meet on one of the altitudes of the triangle $ABC$.

**Problem 7.101.** Let $O$ and $I$ be the circumcentre and the incentre of a triangle $ABC$, and let $R$ and $r$ be the radii of its circumcircle and incircle. Prove that

$$OI = \sqrt{R(R - 2r)}.$$

**Problem 7.102.** Suppose that non-intersecting chords $AB$ and $CD$ are drawn in a circle and that a point $G$ belongs to the arc $AB$ opposite to the chord $CD$. Suppose further that $CG$ meets $AB$ at the point $P$ and that $DG$ meets $AB$ at the point $Q$. Prove that

$$\frac{AP \cdot QB}{PQ}$$

is constant.

**Problem 7.103.** Let $ABC$ be a scalene triangle with incentre $I$ and circumcircle $\omega$. The lines $AI$, $BI$, $CI$ meet $\omega$ for the second time at the points $D$, $E$, $F$, respectively. The parallels through $I$ to the sides $BC$, $AC$, $AB$ meet $EF$, $DF$, $DE$ at the points $K$, $L$, $M$, respectively. Prove that the points $K$, $L$, $M$ are collinear.

**Problem 7.104.** Let $D$ be a point on the side $BC$ of a triangle $ABC$ and let $E$ be a point on the segment $AD$. The line $BE$ meets the circumcircle of $ACD$ at $K$ and $L$, and the line $CE$ meets the circumcircle of $ABD$ at $M$ and $N$. Prove that $K$, $L$, $M$, $N$ are concyclic.

**Problem 7.105.** Let $D$ and $E$ be points on the sides $AB$ and $AC$ of a triangle $ABC$ such that $DE \parallel BC$. Let $P$ be an interior point of the triangle $ADE$. The lines $PB$ and $PC$ meet $DE$

at $F$ and $G$, respectively. Prove that $AP$ is the radical axis of the circumcircles of the triangles $PDG$ and $PFE$.

**Problem 7.106.** Let $O$ and $H$ be the circumcentre and the orthocentre of an acute scalene triangle $ABC$. Points $P$ and $Q$ belong to the segments $AB$ and $AC$, respectively, and are such that $APHQ$ is a parallelogram. Prove that $OP = OQ$.

**Problem 7.107.** Let $ABC$ be a given acute triangle and let $M$ be the midpoint of $BC$. Draw the perpendicular $HP$ from the orthocentre $H$ of $ABC$ to $AM$. Show that $AM \cdot PM = BM^2$. [Japan 2011]

**Problem 7.108.** In an acute triangle $ABC$ the points $P$ and $Q$ are the feet of the perpendiculars from $C$ to $AB$ and from $B$ to $AC$, respectively. The line $PQ$ meets the circumcircle of $\triangle ABC$ in two distinct points $X$ and $Y$. Suppose that $XP = 10$, $PQ = 25$ and $QY = 15$. The value of $AB \cdot AC$ can be written in the form $m\sqrt{n}$, where $m$ and $n$ are positive integers and $n$ is not divisible by the square of any prime. Find $m + n$.

**Problem 7.109.** Let $ABC$ be an acute triangle with its circumcircle $\omega$. Let $D$ be the foot of the altitude of the triangle $ABC$ from the point $A$, and let $E$ and $F$ be the midpoints of the sides $AB$ and $AC$, respectively. Let $P$ and $Q$ be the second intersections of the circle $\omega$ with the circumcircles of the triangles $BDE$ and $CDF$, respectively. Suppose that $A$, $P$, $B$, $Q$, $C$ lie on $\omega$ in this order. Show that the lines $EF$, $BQ$ and $CP$ are concurrent.

**Problem 7.110.** In a triangle $ABC$ the points $M$, $N$, $K$ are the midpoints of the sides $BC$, $AC$, $AB$, respectively. Two semicircles with diameters $AB$ and $AC$ are constructed outside the triangle $ABC$. The lines $MK$ and $MN$ meet these semicircles at $X$ and $Y$, respectively. The tangents to the semicircles at $X$ and $Y$ meet at a point $Z$. Prove that $AZ \perp BC$.

**Problem 7.111.** Let $\omega$ be a circle and let $A$ be a point lying outside $\omega$. The tangents from $A$ to $\omega$ touch $\omega$ at the points $B$ and $C$. Let $M$ be the midpoint of $BC$ and let $D$ be a point on the side $BC$ different from $M$. The circle with diameter $AD$ meets $\omega$ at the points $X$ and $Y$, and meets the circumcircle of $\triangle ABC$ again at $E$. Prove that $AD$, $EM$ and $XY$ are concurrent.

**Problem 7.112.** Two circles $G_1$ and $G_2$ intersect at two points $M$ and $N$. Let $AB$ be the line tangent to these circles at $A$ and $B$, respectively, so that $M$ lies closer to $AB$ than $N$. Let $CD$ be the line parallel to $AB$ passing through the point $M$, with $C$ on $G_1$ and $D$ on $G_2$. The lines $AC$ and $BD$ meet at $E$; the lines $AN$ and $CD$ meet at $P$; the lines $BN$ and $CD$ meet at $Q$. Show that $EP = EQ$.

**Problem 7.113.** Let $ABC$ be an acute triangle with $\angle B > \angle C$. Let $M$ be the midpoint of $BC$, and let $D$ and $E$ be the feet of the altitudes from $C$ and $B$, respectively. Let $K$ and $L$ be the midpoints of $ME$ and $MD$, respectively. If $KL$ meets the line through $A$ parallel to $BC$ at $T$, prove that $TA = TM$. [Iran TST 2011]

**Problem 7.114.** $ABC$ is a triangle. A circle through $B$ and $C$ meets the side $AB$ again at $C'$ and meets the side $AC$ again at $B'$. Let $H$ be the orthocentre of $ABC$ and $H'$ the orthocentre of $AB'C'$. Show that the lines $BB'$, $CC'$ and $HH'$ are concurrent. [IMO Shortlist 1995]

**Problem 7.115.** Let $ABC$ be an acute-angled triangle with orthocentre $H$, and let $W$ be a point on the side $BC$, lying strictly between $B$ and $C$. The points $M$ and $N$ are the feet of the altitudes from $B$ and $C$, respectively. Denote by $\omega_1$ the circumcircle of $BWN$, and let $X$ be the point on $\omega_1$ such that $WX$ is a diameter of $\omega_1$. Analogously, denote by $\omega_2$ the circumcircle of

$CWM$, and let $Y$ be the point on $\omega_2$ such that $WY$ is a diameter of $\omega_2$. Prove that $X$, $Y$ and $H$ are collinear. [IMO 2013]

**Problem 7.116.** Let $ABC$ be a triangle with $AB < AC$ and let $\omega$ be its circumcircle. The tangent line to $\omega$ at $A$ meets the line $BC$ at $D$, and let $E$ be a point on $\omega$ such that $BE$ is parallel to $AD$. The line $DE$ meets the segment $AB$ and the circle $\omega$ at $F$ and $G$, respectively. The circumcircle of $BGF$ meets $BE$ at $N$. The line $NF$ meets the lines $AD$ and $EA$ at $S$ and $T$, respectively. Prove that $DGST$ is cyclic.

**Problem 7.117.** Let $ABC$ be an acute-angled triangle, and let $M$ and $N$ be the midpoints of the sides $AC$ and $BC$, respectively. The altitudes of the triangle $ABC$ from the vertices $A$ and $B$ meet the circumcircle of the triangle $ABC$ at the points $X$ and $Y$, respectively. Show that the line passing through the centres of the circumcircles of the triangles $CNX$ and $CMY$ is parallel to the line $AB$.

**Problem 7.118.** Let the incircle of $\triangle ABC$ touch $BC$, $CA$ and $AB$ respectively at $D$, $E$ and $F$. Let $K$ be a point on the same side of $FE$ as $A$ such that $\angle KFE = \angle ACB$ and $\angle KEF = \angle ABC$. Show that $KD \perp BC$.

**Problem 7.119.** Given a triangle $ABC$, let the bisector of $\angle BAC$ meet $BC$ at $D$, and let $M$ be the midpoint of $BD$. A circle through $A$ and tangent to the line $BC$ at $D$ meets $AM$ a second time at $P$ and meets $AC$ at $Q$ (with $P, Q \neq A$). Prove that $B$, $P$ and $Q$ are collinear.

**Problem 7.120.** In an acute triangle $ABC$, let $O$ be the circumcentre and let $P$ be a point on the opposite side of $BO$ with respect to the point $A$ such that $AO = AP$ and $AP \perp BO$. Let $Q$ be the midpoint of $OP$, and let $R$ be the point on the opposite side of $AB$ with respect to $C$ such that $AR = RB$ and $\angle ARB = 90^\circ$. Prove that $PR = \sqrt{2} \cdot BQ$.

### 7.6.2 The radical axis

**Problem 7.121.** Two circles are given that are externally disjoint. For each of their four common tangents, consider the midpoint of the segment joining the two points of tangency. Prove that these four midpoints lie on a straight line.

**Problem 7.122.** Two circles $\omega_1$ and $\omega_2$ are externally disjoint. The segment $AC$ is one of their common external tangents and the segment $BD$ is one of their common internal tangents, where the points $A$ and $B$ lie on the circle $\omega_1$ and the points $C$ and $D$ lie on the circle $\omega_2$. Prove that the lines $AB$ and $CD$ meet on the line joining the centres of $\omega_1$ and $\omega_2$.

**Problem 7.123.** Given is an acute triangle $ABC$. A line passing through the point $B$ and perpendicular to the line $AC$ meets the circle with diameter $AC$ at the points $K$ and $L$. A line passing through the point $A$ and perpendicular to the line $BC$ meets the circle with diameter $BC$ at the points $M$ and $N$. Prove that the points $K$, $L$, $M$, $N$ lie on one circle.

**Problem 7.124.** In a quadrilateral $ABCD$ the angles $A$ and $C$ are right. Two circles with diameters $AB$ and $CD$ meet at the points $X$ and $Y$. Prove that the line $XY$ passes through the midpoint of $AC$.

**Problem 7.125.** Given is a trapezoid $ABCD$ with bases $AB$ and $CD$, whose diagonals intersect at $X$. Suppose that the circles with diameters $BC$ and $AD$ intersect at two distinct points $P$ and $Q$. Prove that $P$, $Q$, $X$ are collinear.

**Problem 7.126.** Let $A$, $B$, $C$, $D$ be four distinct points on a line, in that order. The circles with diameters $AC$ and $BD$ intersect at $X$ and $Y$. The line $XY$ meets $BC$ at $Z$. Let $P$ be a point on the line $XY$ other than $Z$. The line $CP$ meets the circle with diameter $AC$ at $C$ and $M$, and the line $BP$ meets the circle with diameter $BD$ at $B$ and $N$. Prove that the lines $AM$, $DN$, $XY$ are concurrent.

**Problem 7.127.** Let $DJGO$ be a parallelogram. Points $X$ and $Y$ move on the segments $JD$ and $JG$ so that $XY$ is parallel to $DG$. Prove that the radical axis of the circumcircle of $\triangle JXY$ and the circumcircle of $\triangle DGO$ always passes through a fixed point while $X$ and $Y$ move.

**Problem 7.128.** Let $H$ be the orthocentre of an acute-angled triangle $ABC$. The circle $\Gamma_A$ centred at the midpoint of $BC$ and passing through $H$ meets the sideline $BC$ at the points $A_1$ and $A_2$. Similarly, define the points $B_1$, $B_2$, $C_1$ and $C_2$. Prove that the six points $A_1$, $A_2$, $B_1$, $B_2$, $C_1$, $C_2$ are concyclic.

**Problem 7.129.** Let $\triangle ABC$ have circumcentre $O$, orthocentre $H$, and altitudes $AD$, $BE$, $CF$. Let $EF$ meet $BC$ at $X$, let $FD$ meet $CA$ at $Y$, and let $DE$ meet $AB$ at $Z$. Prove that $X$, $Y$ and $Z$ are collinear on a line perpendicular to $OH$.

**Problem 7.130.** Given circles $\omega_1$ and $\omega_2$ intersecting at the points $X$ and $Y$, let $l_1$ be a line through the centre of $\omega_1$ meeting $\omega_2$ at the points $P$ and $Q$, and let $l_2$ be a line through the centre of $\omega_2$ meeting $\omega_1$ at the points $R$ and $S$. Prove that if $P$, $Q$, $R$ and $S$ lie on a circle, then the centre of this circle lies on the line $XY$.

**Problem 7.131.** Given is a convex hexagon $ABCDEF$ with $AB = BC$, $CD = DE$ and $EF = FA$. Prove that the lines through $A$, $C$, $E$ perpendicular to $FB$, $BD$, $DF$, respectively, are concurrent.

**Problem 7.132.** Let $ABCDEF$ be a convex hexagon in which $AB = AF$, $BC = CD$, $DE = EF$ and $\angle ABC = \angle EFA = 90°$. Prove that $AD \perp CE$.

**Problem 7.133.** Let $ABCD$ be a cyclic quadrilateral with $AB$ and $CD$ not parallel. Let $M$ be the midpoint of $CD$ and let $P$ be a point inside $ABCD$ such that $PA = PB = CM$. Prove that $AB$, $CD$ and the perpendicular bisector of $MP$ are concurrent.

**Problem 7.134.** Let $ABCD$ be a cyclic quadrilateral with $AB \neq CD$. The quadrilaterals $AKDL$ and $CMBN$ are rhombi with equal side lengths. Prove that the points $K$, $L$, $M$, $N$ lie on a single circle.

**Problem 7.135.** The lines $b$ and $c$ passing through the vertices $B$ and $C$ of a triangle $ABC$ are perpendicular to the sideline $BC$. The perpendicular bisectors of $AC$ and $AB$ meet $b$ and $c$ at the points $P$ and $Q$, respectively. Prove that the line $PQ$ is perpendicular to the median $AM$ of the triangle $ABC$.

**Problem 7.136.** Let $ABCD$ be a circumscribed quadrilateral with $BC = 2AB$. Suppose that the perpendicular bisector of $BC$ and the bisector of the angle $DCB$ meet at $X$. Prove that $AX$ and $BD$ are perpendicular.

**Problem 7.137.** Let $ABC$ be a scalene triangle with $BC$ its longest side, and let $I$ and $O$ be the incentre and the circumcentre, respectively. Points $D$ and $E$ are chosen on the rays $BA$ and $CA$, respectively, so that $BD = BC = EC$. Prove that $DE \perp OI$.

**Problem 7.138.** Let $\omega$ be a circle with centre $O$ and let $AB$ be a chord of $\omega$ whose midpoint $M$ is different from $O$. The tangents to $\omega$ at $A$ and $B$ meet at $C$. A circle passing through $O$ and $M$ meets $\omega$ at two different points $D$ and $E$. Prove that $C$, $D$, $E$ are collinear.

**Problem 7.139.** Let $ABC$ be a triangle and let $D$, $E$, $F$ be the tangency points of its incircle with the sides $BC$, $CA$, $AB$, respectively. Consider the three lines joining the midpoints of the segments $AE$ and $AF$, of $BF$ and $BD$, and of $CD$ and $CE$. Prove that the circumcentre of the triangle formed by these three lines coincides with the circumcentre of $ABC$.

## 7.7 Homothety and centres of similitude

**Definition 7.24 (Homothety).** A *homothety* with centre $Z$ and non-zero ratio $k$ is the map sending each point $P$ of the plane to the point $P'$ on the line $ZP$ with $\overrightarrow{ZP'} = k \cdot \overrightarrow{ZP}$. It sends every line to a parallel line and every circle to a circle, and multiplies all lengths by $|k|$.

**Theorem 7.25 (Homothetic triangles).** *Two triangles are homothetic if and only if their pairs of corresponding sides are parallel and the triangles are not congruent with the same orientation.*

**Lemma 7.26 (Tangent circle and arc midpoint).** *Let $AB$ be a chord of a circle $\Omega$ and let $\omega$ be a circle tangent to the chord $AB$ at $K$ and internally tangent to $\Omega$ at $T$. Then the ray $TK$ passes through the midpoint $M$ of the arc $AB$ of $\Omega$ not containing $T$. Moreover $MA^2 = MB^2$ is the power of $M$ with respect to $\omega$.*

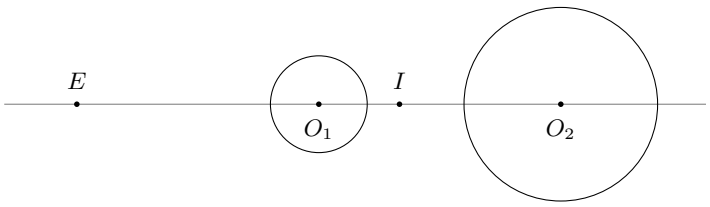
**Theorem 7.27 (Pascal's theorem).** *Given six points (which may coincide) on a circle, labelled $A$, $C$, $E$, $B$, $F$, $D$ in this order around the circle, the intersection points of $AB$ with $DE$, of $AF$ with $CD$, and of $BC$ with $EF$ are collinear.*

**Definition 7.28 (Exsimilicentre and insimilicentre).** Let $\omega_1$ and $\omega_2$ be circles with centres $O_1$, $O_2$ and radii $r_1$, $r_2$. The *exsimilicentre* of $\omega_1$ and $\omega_2$ is the point $E$ on the line $O_1O_2$ but not on the segment $O_1O_2$ with

$$\frac{EO_1}{EO_2} = \frac{r_1}{r_2},$$

and the *insimilicentre* is the point $I$ on the segment $O_1O_2$ with

$$\frac{IO_1}{IO_2} = \frac{r_1}{r_2}.$$



**Theorem 7.29 (Monge–D'Alembert).** *The pairwise exsimilicentres of three distinct circles are collinear. In particular, for three circles none of which lies inside another, the pairwise intersection points of their common external tangents are collinear.*

**Problem 7.140.** Two circles of radius 5 are externally tangent to each other and are internally tangent to a circle of radius 13 at the points $A$ and $B$. The distance $AB$ can be written in the form $\frac{m}{n}$, where $m$ and $n$ are relatively prime positive integers. What is $m+n$? [AMC 10A 2018]

**Problem 7.141.** A triangle $ABC$ has $AB = 13$, $BC = 14$ and $AC = 15$. The points $D$, $E$ and $F$ are the midpoints of $AB$, $BC$ and $AC$, respectively. Let $X \neq E$ be the intersection of the circumcircles of $\triangle BDE$ and $\triangle CEF$. What is $XA + XB + XC$? [AMC 12B 2011]

**Problem 7.142.** In $\triangle ABC$ with side lengths $AB = 13$, $AC = 12$ and $BC = 5$, let $O$ and $I$ denote the circumcentre and the incentre, respectively. A circle with centre $M$ is tangent to the legs $AC$ and $BC$ and to the circumcircle of $\triangle ABC$. What is the area of $\triangle MOI$? [AMC 12B 2018]

**Problem 7.143.** Four circles $\omega$, $\omega_A$, $\omega_B$, $\omega_C$ with the same radius are drawn in the interior of a triangle $ABC$ in such a way that $\omega_A$ is tangent to the sides $AB$ and $AC$, $\omega_B$ to $BC$ and $BA$, $\omega_C$ to $CA$ and $CB$, and $\omega$ is externally tangent to $\omega_A$, $\omega_B$ and $\omega_C$. If the sides of the triangle $ABC$ are 13, 14 and 15, the radius of $\omega$ can be written in the form $\frac{m}{n}$, where $m$ and $n$ are relatively prime positive integers. Find $m+n$. [AIME II 2007]

**Problem 7.144.** Three congruent circles have a common point $O$ and lie inside a given triangle. Each circle touches a pair of sides of the triangle. Prove that the incentre and the circumcentre of the triangle and the point $O$ are collinear. [IMO 1981]

**Problem 7.145.** A non-isosceles triangle $A_1A_2A_3$ is given with sides $a_1$, $a_2$, $a_3$, where $a_i$ is the side opposite $A_i$. For all $i = 1, 2, 3$, let $M_i$ be the midpoint of the side $a_i$, and let $T_i$ be the point where the incircle touches the side $a_i$. Denote by $S_i$ the reflection of $T_i$ in the interior bisector of the angle $A_i$. Prove that the lines $M_1S_1$, $M_2S_2$ and $M_3S_3$ are concurrent. [IMO 1982]

**Problem 7.146.** In $\triangle ABC$ we have $AB = AC$. A circle is tangent internally to the circumcircle of $ABC$ and also to the sides $AB$, $AC$ at $P$, $Q$, respectively. Prove that the midpoint of the segment $PQ$ is the incentre of $\triangle ABC$. [IMO 1978]

**Problem 7.147.** Let $A_1A_2 \dots A_{13}$ and $B_1B_2 \dots B_{13}$ be two regular 13-gons in the plane such that the points $B_1$ and $A_{13}$ coincide and lie on the segment $A_1B_{13}$, and both polygons lie in the same half-plane with respect to this segment. Prove that the lines $A_1A_9$, $B_{13}B_8$ and $A_8B_9$ are concurrent. [Sharygin 2017]

**Problem 7.148.** The diagonals of a trapezoid $ABCD$ intersect at the point $P$. A point $Q$ lies between the parallel lines $BC$ and $AD$ such that $\angle AQD = \angle CQB$, and the line $CD$ separates the points $P$ and $Q$. Prove that $\angle BQP = \angle DAQ$. [IMO Shortlist 2007]

**Problem 7.149.** In a triangle $ABC$ the excentre relative to $A$ is the point of intersection of the external bisectors of $\angle B$ and $\angle C$. Let $I_A$, $I_B$ and $I_C$ be the excentres relative to $A$, $B$ and $C$, respectively, of a scalene triangle $ABC$. Let $X$, $Y$ and $Z$ be the midpoints of $I_BI_C$, $I_CI_A$ and $I_AI_B$, respectively. The incircle of $ABC$ touches the sides $BC$, $CA$ and $AB$ at the points $D$, $E$ and $F$, respectively. Prove that the lines $DX$, $EY$ and $FZ$ pass through a common point on the line $OI$, where $O$ and $I$ are the circumcentre and the incentre of the triangle $ABC$. [Brazil 2012]

**Problem 7.150.** In the plane let $C$ be a circle, $L$ a line tangent to the circle $C$, and $M$ a point on $L$. Find the locus of all points $P$ with the following property: there exist two points $Q$, $R$ on $L$ such that $M$ is the midpoint of $QR$ and $C$ is the inscribed circle of $\triangle PQR$. [IMO 1992]

**Problem 7.151.** Let $I$, $G$ and $N$ be the incentre, the centroid and the Nagel point of a triangle $ABC$; here the excircles touch the sides $BC$, $CA$, $AB$ at the points $X$, $Y$, $Z$ and the Nagel point is the common point of $AX$, $BY$ and $CZ$. Prove that $I$, $G$ and $N$ are collinear and that

$$\frac{\overrightarrow{GN}}{\overrightarrow{GI}} = -2.$$

**Problem 7.152.** Let $A$ be one of the two distinct points of intersection of two unequal coplanar circles $C_1$ and $C_2$ with centres $O_1$ and $O_2$, respectively. One of the common tangents to the circles touches $C_1$ at $P_1$ and $C_2$ at $P_2$, while the other touches $C_1$ at $Q_1$ and $C_2$ at $Q_2$. Let $M_1$ be the midpoint of $P_1Q_1$ and let $M_2$ be the midpoint of $P_2Q_2$. Prove that $\angle O_1AO_2 = \angle M_1AM_2$. [IMO 1983]

**Problem 7.153.** Two circles $\Gamma_1$ and $\Gamma_2$ are inside the circle $\Gamma$ and are tangent to $\Gamma$ at the distinct points $M$ and $N$, respectively. The circle $\Gamma_1$ passes through the centre of $\Gamma_2$. The line passing through the two points of intersection of $\Gamma_1$ and $\Gamma_2$ meets $\Gamma$ at $A$ and $B$. The lines $MA$ and $MB$ meet $\Gamma_1$ at $C$ and $D$, respectively. Prove that $CD$ is tangent to $\Gamma_2$. [IMO 1999]

**Problem 7.154.** Let $ABC$ be a triangle. Let $M$ and $N$ be the points in which the median and the angle bisector at $A$, respectively, meet the side $BC$. Let $Q$ and $P$ be the points in which the perpendicular at $N$ to $NA$ meets $MA$ and $BA$, respectively, and let $O$ be the point in which the perpendicular at $P$ to $BA$ meets $AN$ produced. Prove that $QO$ is perpendicular to $BC$. [APMO 2000]

**Problem 7.155.** Let $\Omega$ be the circumcircle of a triangle $ABC$ and let $\omega_a$ be the circle tangent to the segment $CA$, the segment $AB$ and $\Omega$. Define $\omega_b$ and $\omega_c$ similarly. Let $\omega_a$, $\omega_b$ and $\omega_c$ touch $\Omega$ at the points $A'$, $B'$ and $C'$, respectively. Prove that the lines $AA'$, $BB'$ and $CC'$ meet on the line $OI$, where $O$ and $I$ are the circumcentre and the incentre of $ABC$, respectively.

**Problem 7.156.** Let $ABCD$ be a trapezoid with parallel sides $AB > CD$. Points $K$ and $L$ lie on the line segments $AB$ and $CD$, respectively, so that $AK/KB = DL/LC$. Suppose that there are points $P$ and $Q$ on the line segment $KL$ satisfying $\angle APB = \angle BCD$ and $\angle CQD = \angle ABC$. Prove that the points $P$, $Q$, $B$ and $C$ are concyclic. [IMO Shortlist 2006]

**Problem 7.157.** Let $\Gamma$ be the circumcircle of a triangle $ABC$. A circle $\Omega$ is tangent to the line segment $AB$ and is tangent to $\Gamma$ at a point lying on the same side of the line $AB$ as $C$. The angle bisector of $\angle BCA$ meets $\Omega$ at two different points $P$ and $Q$. Prove that $\angle ABP = \angle QBC$. [EGMO 2018]

**Problem 7.158.** Given a triangle $ABC$, let $\Gamma_A$ be a circle tangent to $AB$ and $AC$, let $\Gamma_B$ be a circle tangent to $BC$ and $BA$, and let $\Gamma_C$ be a circle tangent to $CA$ and $CB$. Suppose that $\Gamma_A$, $\Gamma_B$ and $\Gamma_C$ are tangent to one another. Let $D$ be the tangency point of $\Gamma_B$ and $\Gamma_C$, let $E$ be the tangency point of $\Gamma_A$ and $\Gamma_C$, and let $F$ be the tangency point of $\Gamma_A$ and $\Gamma_B$. Prove that the lines $AD$, $BE$ and $CF$ are concurrent. [Romania TST 2007]

**Problem 7.159.** Let $ABC$ be an acute triangle and let $A_1$, $B_1$, $C_1$ be the feet of its altitudes. The incircle of the triangle $A_1B_1C_1$ touches its sides at the points $A_2$, $B_2$, $C_2$. Prove that the Euler lines of the triangles $ABC$ and $A_2B_2C_2$ coincide.

**Problem 7.160.** Let $ABC$ be an acute-angled triangle. Let $A_1$ be the midpoint of the arc $BC$ which contains the point $A$, and let $A_2$ and $A_3$ be the feet of the perpendiculars from the point $A_1$ to the lines $AB$ and $AC$, respectively. Define $B_2$, $B_3$, $C_2$, $C_3$ analogously.

(a) Prove that the line $A_2A_3$ cuts $BC$ at its midpoint.

(b) Prove that the lines $A_2A_3$, $B_2B_3$ and $C_2C_3$ are concurrent.

**Problem 7.161.** In a triangle $ABC$, let $r_A$ be the line that passes through the midpoint of $BC$ and is perpendicular to the internal bisector of $\angle BAC$. Define $r_B$ and $r_C$ similarly. Let $H$ and $I$ be the orthocentre and the incentre of $ABC$, respectively. Suppose that the three lines $r_A$, $r_B$, $r_C$ define a triangle. Prove that the circumcentre of this triangle is the midpoint of $HI$.

**Problem 7.162.** Let $ABC$ be an acute-angled triangle with $AB < AC$. Denote its circumcircle by $\Omega$ and denote the midpoint of the arc $CAB$ by $S$. Let the perpendicular from $A$ to $BC$ meet $BS$ and $\Omega$ at $D$ and $E \neq A$, respectively. Let the line through $D$ parallel to $BC$ meet the line $BE$ at $L$ and denote the circumcircle of the triangle $BDL$ by $\omega$. Let $\omega$ meet $\Omega$ again at $P \neq B$. Prove that the line tangent to $\omega$ at $P$ and the line $BS$ meet on the internal bisector of $\angle BAC$. [IMO 2023]

## 7.8 Areas

Throughout this section $[\mathcal{P}]$ denotes the area of a polygon $\mathcal{P}$. Two ideas dominate: replacing a triangle by another one with the same base and the same height, and comparing two triangles that share an angle or a base by the ratio of the remaining data.

**Problem 7.163.** Let $ABCD$ be a rectangle with a point $E$ on the side $AB$ and a point $F$ on the segment $CE$. Prove that if $[ADE] = [CDF]$, then $[BCE] = [DEF]$.

**Problem 7.164.** Let $ABCD$ be a rectangle with points $E$ and $F$ on the sides $BC$ and $CD$, respectively. Suppose that $[ABE] = [ECF] = [FDA] = 1$. Calculate $[AEF]$.

**Problem 7.165.** Let $ABCDE$ be a convex pentagon in which $AB \parallel CE$ and $BC \parallel DA$. Prove that $[ABE] = [BCD]$.

**Problem 7.166.** Let $ABCD$ be a convex quadrilateral and let $K$ and $L$ be the midpoints of the sides $AB$ and $CD$, respectively. Prove that $[AKCL] = \frac{1}{2}[ABCD]$.

**Problem 7.167.** On the sides $AB$ and $AD$ of a convex quadrilateral $ABCD$ lie the points $K$ and $L$, respectively, in such a way that $AKCL$ is a parallelogram. The diagonal $BD$ meets the segments $CK$ and $CL$ at the points $P$ and $Q$, respectively. Prove that $[APQ] = [BCD]$.

**Problem 7.168.** The diagonals of a convex quadrilateral $ABCD$ intersect at $P$. Prove that

$$\sqrt{[ABCD]} \geq \sqrt{[ABP]} + \sqrt{[CDP]}$$

and describe the equality case.

**Problem 7.169.** Given is a convex hexagon $ABCDEF$ with $AB \parallel DE$, $BC \parallel EF$ and $CD \parallel FA$. Prove that $[ACE] = [BDF]$.

**Problem 7.170.** Let $ABCD$ be a convex quadrilateral and let $K$, $L$, $M$, $N$ be the midpoints of the sides $AB$, $BC$, $CD$, $DA$, respectively. The segment $AL$ meets $DK$ and $BM$ at $P$ and $Q$, respectively, and the segment $CN$ meets $DK$ and $BM$ at $S$ and $R$, respectively. Prove that

$$[APK] + [BQL] + [CRM] + [DSN] = [PQRS].$$

**Problem 7.171.** Points $K$ and $L$ lie on the sides $AC$ and $BC$ of a triangle $ABC$, respectively, so that $\dfrac{AK}{KC} = \dfrac{CL}{LB}$.

(a) The segments $AL$ and $BK$ intersect at $P$. Prove that $[ABP] = [CKPL]$.

(b) Let $M$ be the midpoint of the side $AB$. Prove that $[CKML] = \frac{1}{2}[ABC]$.

**Problem 7.172.** Points $K$ and $L$ lie on the sides $AC$ and $BC$ of a triangle $ABC$, respectively, and the segments $AL$ and $BK$ intersect at $P$. It is known that among the four regions $ABP$, $APK$, $BPL$ and $CKPL$ three have unit area. What are the possible areas of the fourth region?[†]

**Problem 7.173.** Given is a convex quadrilateral $ABCD$ which is not a parallelogram. Suppose that the line joining the midpoints of its diagonals meets the segment $BC$ at $P$. Prove that $[ABP] + [CDP] = [ADP]$.

**Problem 7.174.** The diagonals of a convex quadrilateral $ABCD$ intersect at $P$. The line joining the midpoint of the side $AB$ with the point $P$ meets the segment $CD$ at $Q$. Prove that

$$\frac{CQ}{DQ} = \frac{[BCP]}{[ADP]}.$$

**Problem 7.175.** A point $D$ lies on the side $BC$ of an acute triangle $ABC$, and the points $E$, $F$ are the projections of $D$ onto $AC$ and $AB$, respectively. The point $O$ is the circumcentre of $ABC$. Prove that the broken line $EOF$ bisects the area of $ABC$.

**Problem 7.176.** The point $M$ is the midpoint of the side $BC$ of an acute triangle $ABC$. The circle centred at $M$ and passing through $A$ meets the lines $AB$ and $AC$ again at the points $P$ and $Q$, respectively. A point $R$ lies on the segment $BC$ and satisfies $[BPR] = [CQR]$. Prove that $AR \perp PQ$.

**Problem 7.177.** (Gauss's theorem.) Let $ABCD$ be a convex quadrilateral such that $AB$ and $CD$ meet at $E$ and $AD$ and $BC$ meet at $F$. Prove that the midpoints of the segments $AC$, $BD$, $EF$ are collinear.

# Chapter 8

# Number Theory

**Sources for this chapter:** Camp handouts by Dominik Burek (April 2020), Dušan Djukić (2021–2025), Miroslav Marinov (Nov 2024, May 2025, Sep–Oct 2025), Mikhail (Nov 2019), Bojan Bašić (March 2022, February 2024), Majid Almarhoumi (Dec 2019), Nikola Petrović (2021).

Number theory is the part of the JBMO syllabus that rewards patient bookkeeping: a single well-chosen modulus, a factorisation nobody expected, or a comparison of the exponents of one prime on the two sides of an equation. The theory boxes below collect exactly the facts the camps state as known; everything else is a problem.

## 8.1 Divisibility, the GCD and the Euclidean algorithm

We begin with the arithmetic of divisibility itself: what one may deduce from a relation $a \mid b$, and how the greatest common divisor is computed and used.

**Definition 8.1 (Greatest common divisor).** The greatest common divisor $\gcd(a, b)$ of two positive integers $a$ and $b$ is their common divisor that is divisible by every common divisor of $a$ and $b$.

**Theorem 8.2 (Main property of the gcd; Euclidean algorithm).** *The sets of common divisors of the pair $(a, b)$ and of the pair $(a \pm b, b)$ coincide. Consequently, if $a = qb + r$ then $\gcd(a, b) = \gcd(b, r)$; iterating this replacement produces a strictly decreasing sequence of remainders $r_1 > r_2 > \cdots > r_k$ that must terminate, and then $\gcd(a, b) = r_k$.*

**Theorem 8.3 (Bezout's lemma).** *For any positive integers $a$ and $b$ there exist integers $x$ and $y$ with $ax + by = \gcd(a, b)$. Such $x, y$ are produced by running the Euclidean algorithm backwards.*

**Lemma 8.4 (Euclid's lemma).** *If $p$ is a prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.*

**Lemma 8.5 (Size bound for divisibility).** *Let $a, b$ be integers with $a \mid b$. Then $|a| \le |b|$ or $b = 0$; in particular, if $a, b > 0$ then $a \le b$. Moreover, if $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$.*

**Technique 8.6 (Bounding the quotient).** When a problem states $a \mid b$ for expressions $a, b$ in the unknowns, write $b = ka$ and prove an inequality of the form $|k| \le N$ for a small $N$. This leaves finitely many values of $k$, each of which turns the divisibility into an equation.

**Theorem 8.7 (Linear Diophantine equations).** *The equation $ax + by = c$ has integer solutions if and only if $\gcd(a, b) \mid c$. If $\gcd(a, b) = 1$ and $(x_1, y_1)$ is one solution, then all solutions are given by $x = x_1 + kb$, $y = y_1 - ka$ for $k \in \mathbb{Z}$.*

**Problem 8.1.** Let $a, b$ be integers such that $a + b \mid a^2$. Prove that $a + b \mid b^2$.

**Problem 8.2.** Let $m, n$ be integers such that $m + n \mid m + n^2$. Prove that $m + n \mid m + n^3$.

**Problem 8.3.** Let $m, n, d$ be integers such that $d \mid mn^2 + 1$ and $d \mid m^2n + 1$. Prove that $d \mid m^3 + 1$ and $d \mid n^3 + 1$.

**Problem 8.4.** Prove that $133 \mid 11^{n+2} + 12^{2n+1}$ for every nonnegative integer $n$.

**Problem 8.5.** An $n \times m$ rectangle is given. Every second one cuts off from a corner the square of the largest possible size, until a square remains. Find the side length of that square.

**Problem 8.6.** A diagonal is drawn in an $n \times m$ rectangle whose sides are divided into unit segments. How many lattice nodes does the diagonal pass through?

**Problem 8.7.** Prove that the fraction $\dfrac{2n + 13}{n + 7}$ is irreducible for all positive integers $n$.

**Problem 8.8.** Find all integers $n$ for which $\dfrac{n^4 + 1}{n^2 + n + 1}$ is an integer.

**Problem 8.9.** Prove the identity $\operatorname{lcm}(a, b) \cdot \gcd(a, b) = ab$.

**Problem 8.10.** Prove that for all positive integers $a, b, c$,

$$\frac{\operatorname{lcm}(a,b) \cdot \operatorname{lcm}(b,c) \cdot \operatorname{lcm}(c,a)}{\gcd(a,b) \cdot \gcd(b,c) \cdot \gcd(c,a)} = \frac{\operatorname{lcm}(a,b,c)^2}{\gcd(a,b,c)^2}.$$

**Problem 8.11.** Prove that $\gcd\big(\operatorname{lcm}(a,b), \operatorname{lcm}(b,c), \operatorname{lcm}(c,a)\big) = \operatorname{lcm}\big(\gcd(a,b), \gcd(b,c), \gcd(c,a)\big)$.

**Problem 8.12.** Prove that $\gcd(a^n - 1, a^m - 1) = a^{\gcd(m,n)} - 1$ for every integer $a > 1$ and all positive integers $m, n$.

**Problem 8.13.** Let $a > 1$ be an integer. Prove that: (a) $a^n - 1 \mid a^m - 1$ if and only if $n \mid m$; (b) $a^n + 1 \mid a^m + 1$ if and only if $n \mid m$ and $\frac{m}{n}$ is odd.

**Problem 8.14.** Prove that if $a, b, c$ have pairwise distinct remainders modulo 3, then the polynomial $x^a + x^b + x^c$ is divisible by $x^2 + x + 1$.

**Problem 8.15.** Find all positive integers $n$ and $m$ such that $n^5 + n^4 = 7^m - 1$.

**Problem 8.16.** Find all possible values of $\gcd(x^3 - 3x + 5,\ x^2 + 1)$, where $x$ is an integer.

**Problem 8.17.** If $a$ and $b$ are positive integers, prove that $\gcd(a+b+1,\ ab+1,\ a^2+b^2)=1$.

**Problem 8.18.** Solve the equation $x(x-y)=8y-7$ in positive integers. [JBMO Shortlist 2008]

**Problem 8.19.** Determine all positive integers $x$ and $y$ such that $xy \mid x^2+2y-1$. [European Mathematical Cup 2018, Junior Group]

**Problem 8.20.** Let $a, b$ be positive integers with $a+b \mid ab$. Prove that $\gcd(a,b) > \sqrt{a+b}$.

**Problem 8.21.** Let $a > b > 1$ be integers with $a+b \mid ab+1$ and $a-b \mid ab-1$. Prove that $a < \sqrt{3}\,b$.

**Problem 8.22.** Let $x, y, z$ be positive integers such that $\dfrac{x+1}{y}+\dfrac{y+1}{z}+\dfrac{z+1}{x}$ is an integer, and let $d=\gcd(x,y,z)$. Prove that $d \le \sqrt[3]{xy+yz+zx}$.

**Problem 8.23.** If $n$ is a positive integer, prove that at least one of the numbers $n, n+1, \ldots, n+5$ is coprime to each of the remaining five.

**Problem 8.24.** If $a$ and $b$ are positive integers and $ab \mid a^2+b^2$, prove that $a=b$.

**Problem 8.25.** Positive integers $a, b, c, d$ satisfy $ab=cd$. Prove that there exist positive integers $x, y, u, v$ such that $a=xu$, $b=yv$, $c=xv$, $d=yu$.

**Problem 8.26.** If $a, b, c, d$ are positive integers with $ab=cd$, prove that $a+b+c+d$ is composite.

**Problem 8.27.** Let $a, b, c, d$ be positive integers with $ab=cd$. Prove that $a^{2020}+b^{2020}+c^{2020}+d^{2020}$ is composite.

**Problem 8.28.** Let $n > 1$. If $n^2+1$ has a divisor $d > n$, prove that in fact $d > n+\sqrt{n}$.

**Problem 8.29.** Find all positive integers $a$ for which $a^2+1$ has a divisor $b$ satisfying $a \le b \le a+\sqrt{a}$.

**Problem 8.30.** Find all pairs $(n,d)$ of positive integers such that $d$ is a divisor of $n$ and $nd+1$ is a divisor of $n^2+d^2$.

**Problem 8.31.** Find all triples of positive integers $a, b, c$ such that $a \mid 2b+1$, $b \mid 2c+1$ and $c \mid 2a+1$.

**Problem 8.32.** Find all triples of positive integers $a, b, c$ such that $a \mid bc+1$, $b \mid ac+1$ and $c \mid ab+1$.

**Problem 8.33.** Find all pairs $(m,n)$ of positive integers such that $mn-1$ divides $n^3+1$.

**Problem 8.34.** Find all triples of positive integers $(x,y,z)$ such that each of the numbers $x^2-1$, $y^2-2$, $z^2-4$ is divisible by $x+y+z$.

**Problem 8.35.** Let $a_1, a_2, \ldots, a_n$ be an arithmetic progression of integers such that $i \mid a_i$ for $i=1,2,\ldots,n-1$ and $n \nmid a_n$. Prove that $n$ is a prime power.

**Problem 8.36.** Do there exist four different positive integers $a, b, c, d$ with $ad=bc$ and $n^2 \le a, b, c, d < (n+1)^2$ for some positive integer $n$?

**Problem 8.37.** Suppose that $a < b < c < d$ are odd positive integers such that $ad = bc$. If both $a + d$ and $b + c$ are powers of 2, prove that $a = 1$.

**Problem 8.38.** Integers $a > b > 1$ are such that $a^2 + b - 1$ is divisible by $b^2 + a - 1$. Prove that $b^2 + a - 1$ cannot be a power of a prime.

**Problem 8.39.** If each of the positive integers $a, b, c, d$ is divisible by $ab - cd$, what is $|ab - cd|$?

**Problem 8.40.** If $a, b, c$ are positive integers, prove that the numbers $a^3b + 1$, $b^3c + 1$, $c^3a + 1$ cannot all be divisible by $a^2 + b^2 + c^2$.

**Problem 8.41.** Suppose $a$ and $b$ are positive integers such that $\gcd(an + 2,\ bn + 3) > 1$ for every positive integer $n$. Prove that $b = \frac{3}{2}a$.

**Problem 8.42.** If $a, b, c$ are positive integers, prove that

$$\gcd(a, b - 1) \cdot \gcd(b, c - 1) \cdot \gcd(c, a - 1) \leq a(b - 1) + b(c - 1) + c(a - 1) + 1,$$

and show that equality occurs for infinitely many triples $(a, b, c)$.

**Problem 8.43.** Given $n$ positive integers, denote by $d_k$ the greatest common divisor of all products of $k$ of them. Prove that $d_k^2 \mid d_{k-1}d_{k+1}$ for $2 \leq k \leq n - 1$.

**Problem 8.44.** Find all positive integers $n$ that cannot be written in the form $n = \operatorname{lcm}(a, b) + \operatorname{lcm}(b, c) + \operatorname{lcm}(c, a)$ for positive integers $a, b, c$.

**Problem 8.45.** If $a$ and $b$ are positive integers such that $\operatorname{lcm}(a, b) + \operatorname{lcm}(a + 2, b + 2) = 2\operatorname{lcm}(a + 1, b + 1)$, prove that $a \mid b$ or $b \mid a$.

**Problem 8.46.** Find all pairs of positive integers $a$ and $b$ such that $\operatorname{lcm}(a + 1, b + 1) = a^2 - b^2$.

**Problem 8.47.** Let $a$ and $b$ be positive integers with $\dfrac{\operatorname{lcm}(a, b)}{\gcd(a, b)} = a - b$. Prove that $\operatorname{lcm}(a, b) = \gcd(a, b)^2$.

**Problem 8.48.** Find all triples of positive integers $a, b, c$ such that $ab + bc + ca = 4\operatorname{lcm}(a, b, c)$.

**Problem 8.49.** If $a < b < c < d < e$ are positive integers, prove that

$$\frac{1}{\operatorname{lcm}(a, b)} + \frac{1}{\operatorname{lcm}(b, c)} + \frac{1}{\operatorname{lcm}(c, d)} + \frac{2}{\operatorname{lcm}(d, e)} \leq 1,$$

and find all cases of equality.

**Problem 8.50.** Find all positive integers $n$ with the following property: whenever $n \mid xy + 1$ for some integers $x, y$, it also holds that $n \mid x + y$.

**Problem 8.51.** Find all positive integers $n$ such that, for every divisor $d > 1$ of $n$, the number $d - 1$ is a divisor of $n - 1$.

**Problem 8.52.** Find all positive integers $n$ with the property that whenever $a$ and $b$ are coprime divisors of $n$, the number $a + b - 1$ is also a divisor of $n$.

**Problem 8.53.** A proper divisor of a positive integer $n$ is a divisor different from 1 and $n$. Determine all positive integers $n$ having at least two proper divisors and such that the difference of any two proper divisors divides $n$. [Romania JBMO TST 2023; Azerbaijan Junior TST 2024]

**Problem 8.54.** Find all values of $n$ for which there are positive integers $a, b, c, d$ with $a+b+c+d = n$ and $n \mid abc + abd + acd + bcd$.

**Problem 8.55.** Let $a, b, c$ be positive integers such that $\dfrac{a\sqrt{2}+b}{b\sqrt{2}+c}$ is rational. Prove that $a+b+c \mid ab+bc+ca$.

**Problem 8.56.** Determine all pairs $(a, n)$ of positive integers such that $a^n + 2^n + 1$ divides $a^{n+1} + 2^{n+1} + 1$. [Romania 1998]

**Problem 8.57.** Determine all positive integers $k$ for which there exist positive integers $x, y, z$ with

$$\frac{x}{y} + \frac{y}{zx - y} = k.$$

[Bulgaria Regional Round 2020, Grade 11]

**Problem 8.58.** Determine all positive integers $n$ such that $a+b+c$ divides $a^{2n} + b^{2n} + c^{2n} - n(a^2b^2 + b^2c^2 + c^2a^2)$ for all pairwise distinct positive integers $a, b, c$. [Bulgaria Autumn Mathematical Tournament 2024, Grade 8]

**Problem 8.59.** Solve the equation $a = a^3 - 8a^2b + 21ab^2 - 18b^3$ in integers. [Bulgaria Winter Mathematical Competition 2022, Grade 8]

**Problem 8.60.** Find the number of sequences of 2022 positive integers such that: each number after the first is greater than or equal to the previous one; at least one of the numbers equals 2022; and the sum of every 2020 of the numbers is divisible by each of the other two. [Bulgaria Autumn Mathematical Tournament 2022, Grade 8]

**Problem 8.61.** Let $p$ be a prime such that $\frac{p+1}{2}$ is also a prime, and suppose there is a positive integer $n \neq p$ with $p + n^2 \mid n + p^2$. Prove that $p = 5$ or $p = (n-1)^2 + 1$. [JBMO Shortlist 2023]

**Problem 8.62.** ★ The positive integers $a$ and $b$ are distinct and such that $3^a + 2$ divides $3^b + 2$. Prove that $a > b^2$. [Balkan MO 2024/3]

**Problem 8.63.** ★ Determine all positive integers $a$ and $b$ such that $ab^2+b+7$ divides $a^2b+a+b$. [IMO 1998/4]

**Problem 8.64.** ★ Determine all positive integers $a$ and $b$ such that $7^a - 3^b$ divides $a^4 + b^2$. [IMO Shortlist 2007 N1]

### 8.1.1 Linear Diophantine equations and the Frobenius number

**Problem 8.65.** Positive integers $a$ and $b$ satisfy $19 \mid a + 17b$ and $17 \mid a + 19b$. Find the smallest possible value of $a + b$.

**Problem 8.66.** Solve the equation $105x + 211y = 106z$ in positive integers.

**Problem 8.67.** Let $a, b$ be coprime positive integers. (a) Prove that $ab - a - b$ cannot be expressed as $ax + by$ with integers $x, y \geq 0$. (b) Prove that every integer greater than $ab - a - b$ can be so expressed.

**Problem 8.68.** Find the largest positive integer $n$ that cannot be written in the form $99x + 100y + 101z$ for nonnegative integers $x, y, z$.

**Problem 8.69.** Find the largest positive integer that cannot be written as a sum of three pairwise coprime numbers greater than 1.

## 8.2 Primes, factorisation and the exponent $v_p$

**Theorem 8.8 (Fundamental theorem of arithmetic).** *Every integer greater than 1 can be written as a product of powers of distinct primes, and this representation is unique up to the order of the factors.*

**Theorem 8.9 (Euclid: infinitude of primes).** *There are infinitely many prime numbers: if $p_1 < p_2 < \cdots < p_k$ are the first $k$ primes, then $p_1 p_2 \cdots p_k - 1$ has a prime divisor different from all of them.*

**Definition 8.10 (The exponent $v_p$).** For a prime $p$ and a nonzero integer $n$, $v_p(n)$ denotes the exponent of $p$ in the prime factorisation of $n$. If $n = \prod p_i^{a_i}$ and $m = \prod p_i^{b_i}$ (zero exponents allowed), then

$$\gcd(m,n) = \prod p_i^{\min(a_i,b_i)}, \qquad \operatorname{lcm}(m,n) = \prod p_i^{\max(a_i,b_i)}.$$

Two positive integers $a, b$ are equal if and only if $v_p(a) = v_p(b)$ for every prime $p$; hence to prove $a \neq b$ it suffices to exhibit one prime $p$ with $v_p(a) \neq v_p(b)$.

**Lemma 8.11 (Valuation of a sum).** *For any prime $p$ and integers $a, b$ one has $v_p(a+b) \geq \min(v_p(a), v_p(b))$, with equality whenever $v_p(a) \neq v_p(b)$.*

**Problem 8.70.** Prove that there exist infinitely many primes of the form $6k-1$, $k \in \mathbb{Z}$.

**Problem 8.71.** Prove that the equation $x^2 + x + 1 = py$ has integer solutions $x, y$ for infinitely many primes $p$.

**Problem 8.72.** Prove that for each positive integer $n$ there exist $n$ consecutive composite integers. Is it true that for each positive integer $n$ there exist $n$ consecutive integers among which there is exactly one prime?

**Problem 8.73.** Prove that there are infinitely many primes of the form $4k+3$, and infinitely many of the form $4k+1$.

**Problem 8.74.** If $n > 1$, prove that $1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$ is never an integer.

**Problem 8.75.** Let $x, y, z$ be positive integers such that $\dfrac{xy^2}{z} + \dfrac{y^3z^4}{x} + \dfrac{z^5x^6}{y}$ is an integer. Prove that each summand is an integer.

**Problem 8.76.** Suppose that $a$, $b$, $c$ and $\dfrac{a}{b} + \dfrac{b}{c} + \dfrac{c}{a}$ are integers. Prove that $abc$ is a perfect cube.

**Problem 8.77.** Suppose $x, y, z$ are rational numbers with $xyz = 1$ such that both $x+y+z$ and $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ are integers. Prove that $|x| = |y| = |z| = 1$.

**Problem 8.78.** Suppose that positive integers $x, y$ are such that $\dfrac{x^2+y^2+x}{xy}$ is an integer. Prove that $x$ is a perfect square.

**Problem 8.79.** Prove that: (a) if $2^n-1$ is prime, then $n$ is prime; (b) if $2^n+1$ is prime, then $n$ is a power of 2.

**Problem 8.80.** Prove that $19 \cdot 8^n + 17$ is composite for every positive integer $n$.

**Problem 8.81.** If $p > 3$ is a prime, prove that $\dfrac{4^p+1}{5}$ is a composite number.

**Problem 8.82.** Prove that $2^{58}+1$ has at least three distinct prime divisors.

**Problem 8.83.** Prove that $302^{39} + 239^{30}$ is not prime.

**Problem 8.84.** If $a, b > 2$ are integers, prove that $2^b - 1$ never divides $2^a + 1$.

**Problem 8.85.** Find all primes $p$ for which $p^2-2$, $p^2+6$ and $p^2+10$ are also primes.

**Problem 8.86.** Denote by $[x]$ the integer part of $x$. Find all positive integers $n$ for which $\left[\frac{n^2}{5}\right]$ is a prime number.

**Problem 8.87.** Can all numbers greater than $10^{100}$ be written as the sum of a prime and a perfect square?

**Problem 8.88.** Find all triples of primes $p, q, r$ with the property that $p^2+q$ and $p^2-r$ are both perfect squares.

**Problem 8.89.** Find all primes $p$ and $q$ such that $p^2+q = 37q^2+p$.

**Problem 8.90.** (a) Find all primes $p, q$ such that $p^2 - pq - q^3 = 1$. (b) What if we do not require $p$ to be prime?

**Problem 8.91.** Find all primes $p, q$ such that $p^3 + 3q^3 - 32$ is also a prime.

**Problem 8.92.** Find all primes $p, q$ such that $p^{q+1} + q^{p+1}$ is a perfect square.

**Problem 8.93.** Find all primes $p, q, r$ such that $p^3 = p^2 + q^2 + r^2$.

**Problem 8.94.** If primes $p$ and $q$ satisfy $p \mid q-1$ and $q \mid p^3-1$, prove that $q = p^2+p+1$.

**Problem 8.95.** Find all primes $p, q$ such that $p^3 - q^5 = (p+q)^2$.

**Problem 8.96.** Find all primes $p$ for which $\dfrac{11^{p-1}-1}{p}$ is a perfect square.

**Problem 8.97.** Find all positive integers $x$ for which $x^5 + x^4 + 1$ has no prime divisors other than 3 and 7.

**Problem 8.98.** Let $n$ be a positive integer and let $x < n^2$ be a positive integer coprime to $n^2(n^2+1)(n^2+2)\cdots(n^2+n-2)$. Prove that $x$ is prime.

**Problem 8.99.** Define $p_1 = 2$ and, for $n \geq 2$, let $p_n$ be the largest prime factor of $p_1p_2\cdots p_{n-1}+1$. Prove that no term $p_n$ equals 5.

**Problem 8.100.** Let $p$ and $q$ be primes with $p < q < 2p$. Prove that there exist two consecutive positive integers whose largest prime divisors are $p$ and $q$ in some order.

**Problem 8.101.** Let $p(n)$ denote the largest prime divisor of $n^2 + n + 1$. Prove that there are infinitely many positive integers $n$ for which $p(n + 1) \leq p(n)$.

**Problem 8.102.** Is there an infinite sequence of primes $p_1, p_2, p_3, \ldots$ such that $p_{i+1} = 2p_i \pm 1$ for all $i$?

**Problem 8.103.** Prove that for every positive integer $n$ the product $\prod_{i=1}^{n}(3i^2 + 1)$ has at most $n$ distinct prime divisors. [Cyberspace Mathematical Competition 2020]

**Problem 8.104.** Denote by $\omega(x)$ the number of distinct prime divisors of a positive integer $x$. Let $a, b, c$ be arbitrary positive integers. Prove that there is a positive integer $n$ such that $\omega(an + c) > \omega(bn + c)$.

**Problem 8.105.** Determine all primes $p$ and positive integers $n$ such that $p^2 - p = 2(n^3 + 1)$. [Belarus 2013 National Round]

**Problem 8.106.** Determine all pairs $(p, q)$ of primes such that $p - q$ and $pq - q$ are perfect squares of nonnegative integers. [USAMO and USAJMO 2022]

**Problem 8.107.** Determine all pairs $(p, q)$ of primes such that $|p^2 - pq + q^2 - 7|$ is also a prime. [Kosovo National Round 2023, Grade 9]

**Problem 8.108.** Determine all primes $p, q$ and positive integers $n$ such that $p^{2n} + p^{2n-1} + \cdots + p^2 + p = q^2 + q$. [Bulgaria JBMO 2024 Preparation]

**Problem 8.109.** Determine all pairs $(p, n)$ of a prime $p$ and a positive integer $n$ such that $p^3 + 5^n - 1 = n(p^2 + 2p + 6^n)$. [Bulgaria Autumn Tournament 2019, Grade 8]

**Problem 8.110.** Determine all pairs $(a, b)$ of positive integers such that $a^2 + b$ and $b^2 + a$ are powers of the same prime. [Bulgarian Festival of Young Mathematicians 2013, Senior Group]

**Problem 8.111.** Find all pairs of positive integers $a$ and $b$ such that $b^2 + a = p^k$ and $a^2 + b = n\,p^k$ for some prime $p$ and positive integers $k$ and $n$.

**Problem 8.112.** Find all collections of (not necessarily distinct) primes whose product is ten times larger than their sum. [Hungary 1995]

**Problem 8.113.** Suppose $p < q$ are primes such that $pq(p + 1)(q + 1) + 1$ is a perfect square. Find all possible values of $q - p$. [Turkey JBMO TST 2025]

**Problem 8.114.** Maria writes in red 8 distinct positive integers and then writes in blue the 28 sums of each two red integers. At most how many of the blue integers can be prime? What happens if instead there are 9 red integers and correspondingly 36 blue sums? [Al-Khwarizmi International Junior Olympiad 2025]

**Problem 8.115.** For a prime $p$, a sequence $a_1, a_2, \ldots$ is defined by $a_1 = p + 1$ and $a_{n+1} = a_1 a_2 \cdots a_n + p$ for all $n \geq 1$. Find all values of $p$ for which this sequence contains a perfect square.

**Problem 8.116.** Find all pairs of primes $p$ and $q$ such that $pq \mid p^p + q^q + 1$.

**Problem 8.117.** Find all pairs of primes $p$ and $q$ such that $pq \mid p^q + q^p + 7$.

**Problem 8.118.** Find all primes $p$ that do not have a multiple of the form $4x^2 + 9y^2 + 1$.

**Problem 8.119.** Suppose that $f \in \mathbb{Z}[X]$. Prove that the set of primes dividing at least one term of the sequence $(f(n))_{n\geq 1}$ is infinite. [Schur's lemma]

**Problem 8.120.** Let $f$ be a nonconstant polynomial with integer coefficients and let $n$ and $k$ be positive integers. Prove that there is a positive integer $a$ such that $f(a), f(a+1), \ldots, f(a+n-1)$ has at least $k$ distinct prime divisors.

**Problem 8.121.** Let $P$ be a polynomial with integer coefficients such that there are two distinct integers at which $P$ takes coprime values. Show that there exists an infinite set of integers at which the values of $P$ are pairwise coprime.

**Problem 8.122.** I chose several (possibly just one) different integers, each between 2 and 50 inclusive. I raised each of them to the third power; for each result I calculated the remainder upon division by 79, and in doing so I obtained each of the original integers once again, in some order. What is the minimum number of integers I could have chosen? [JBMO Shortlist 2023]

**Problem 8.123.** ★ Let $n \geq 3$ be an integer. On a board we write in red all primes not exceeding $n$, and we write in blue all sums of two red numbers (some blue numbers may be equal). Determine all $n$ for which $n!$ divides the product of all blue numbers. [IMO Shortlist 2022 N2]

**Problem 8.124.** ★ For a composite integer $n \geq 6$ let $a_n$ be the smallest positive integer $k$ such that $k!$ is divisible by $n$. Determine the largest possible value of $\dfrac{a_n}{n}$. [Saudi Arabia EGMO TST 2025]

**Problem 8.125.** ★ Determine all primes $p$ and all positive integers $x, y$ satisfying $x^3 + y^3 = p(xy + p)$. [RMM 2023/1]

**Problem 8.126.** ★ Find all polynomials $f$ with integer coefficients and leading coefficient 1 satisfying the following condition: there exists a positive integer $N$ such that $p$ divides $2\,(f(p)!)+1$ for every prime $p > N$ for which $f(p)$ is a positive integer. [Balkan MO 2016/3]

## 8.3 Digits and decimal representation

**Theorem 8.12 (Divisibility rules).** *A positive integer written in decimal is divisible by* 2, 4, 8, *and more generally by* $2^n$, *exactly when its last one, two, three, respectively* $n$ *digits form a number divisible by* 2, 4, 8, *respectively* $2^n$*; by* 5, 25, *and more generally by* $5^n$, *exactly when its last one, two, respectively* $n$ *digits form a number divisible by* 5, 25, *respectively* $5^n$*; by* 3 *or by* 9 *exactly when its sum of digits is; by* 11 *exactly when the alternating sum of its digits is; by* 7 *or by* 13 *exactly when the alternating sum of its three-digit groups is.*

**Problem 8.127.** Let $A$ be the sum of digits of $4444^{4444}$ and let $B$ be the sum of digits of $A$. What is the sum of digits of $B$?

**Problem 8.128.** Let $A = 2023^{2024}$, let $B$ be the sum of digits of $A$, let $C$ be the sum of digits of $B$, and let $D$ be the sum of digits of $C$. Find $D$. [IMO 1975 P4, modified]

**Problem 8.129.** From the number $2018^{2018}$ the quadruple sum of its digits is subtracted; this operation is repeated until a non-positive number is obtained. Find the last positive result.

**Problem 8.130.** Find all positive integers $n$ such that the sum of digits of $n!$ equals 9.

**Problem 8.131.** Find the largest positive integer whose digits are all nonzero and pairwise distinct and which is divisible by each of its digits.

**Problem 8.132.** Find all positive integers $n$ whose product of digits equals $n - 16$.

**Problem 8.133.** Prove that there is a number divisible by $2^{100}$ that has sum of digits at most 100 and ends in a nonzero digit.

**Problem 8.134.** Prove that a square of an integer cannot end with four equal nonzero digits. Can it end with three equal nonzero digits?

**Problem 8.135.** The last $k$ digits of some perfect square are equal and nonzero. At most how large can $k$ be, and for which digit is this possible?

**Problem 8.136.** Is there a perfect square that ends with the digits (a) 987654321; (b) 987654329?

**Problem 8.137.** Is there a perfect square that starts with the digits 987654321?

**Problem 8.138.** Prove that there exists a multiple of $5^{100}$ all of whose digits are odd.

**Problem 8.139.** Using each of the digits $1, 2, \ldots, 9$ exactly once, we form nine nine-digit numbers, not necessarily distinct. Their sum ends in $n$ zeroes. Find the maximum possible value of $n$.

**Problem 8.140.** Find all positive integers $n$ with the property that $n, 2n, 3n, \ldots, n^2$ all have the same sum of digits.

**Problem 8.141.** Prove that every integer $n > 1$ has a multiple less than $n^4$ whose decimal expansion contains at most four distinct digits.

**Problem 8.142.** Denote by $S(n)$ the sum of decimal digits of $n$. Prove that there are infinitely many positive integers $k$ such that $S(2^k) > S(2^{k+1})$.

**Problem 8.143.** If $a, b, c, d, e, f, g, h, i$ are the numbers $1, 2, \ldots, 9$ in some order, what is the minimum possible value of $\dfrac{\overline{abc} - \overline{def}}{\overline{ghi}}$?

**Problem 8.144.** Is there a positive integer greater than $10^{100}$, having no zero digits, such that switching two of its digits yields a number with the same set of prime divisors?

**Problem 8.145.** We are given $n > 3$ consecutive odd three-digit numbers. Prove that they can be ordered in a sequence $b_1, b_2, \ldots, b_n$ so that the number $\overline{b_1 b_2 \cdots b_n}$, obtained by writing them one after another in the decimal system, is composite.

**Problem 8.146.** Denote by $b_n$ the number of binary digits equal to 1 of a positive integer $n$. Call $n$ *lively* if $b_n \mid n$. Prove that (a) there are no 5 consecutive positive integers that are all lively; (b) there are infinitely many instances of 4 consecutive positive integers that are all lively.

**Problem 8.147.** Prove that every multiple of $2^n - 1$ has at least $n$ digits equal to 1 in its binary representation.

**Problem 8.148.** Denote by $S(n)$ the sum of digits of a positive integer $n$. Positive integers $a, b, c$ are such that $S(a+b) \leq 4$, $S(b+c) \leq 4$ and $S(c+a) \leq 4$. What is the largest possible value of $S(a+b+c)$?

**Problem 8.149.** Find all positive integers $n$ such that the rightmost four digits of $2012^n$ are $2, 0, 1, 2$ in that order from left to right.

**Problem 8.150.** Given that $(10!)^3 = \overline{4778472583ab72000000}$, determine the digits $a$ and $b$.

**Problem 8.151.** Find the last digit of $2017^{2017^{2017}}$.

**Problem 8.152.** Find the units and tens digits of $19^{19^{19}}$.

**Problem 8.153.** A positive integer is called a *repunit* if all of its decimal digits equal 1; the repunit with $n$ digits is denoted $\underbrace{11\cdots1}_{n}$. Prove that (a) the repunit with $n$ digits is divisible by 37 if and only if $3 \mid n$; (b) there exists a positive integer $k$ such that the repunit with $n$ digits is divisible by 41 whenever $k \mid n$. [JBMO 2015 NT2]

**Problem 8.154.** Prove that for any prime $p \geq 3$ the number

$$\underbrace{1\cdots1}_{p}\,\underbrace{2\cdots2}_{p}\cdots\underbrace{9\cdots9}_{p} - 123456789$$

is divisible by $p$.

**Problem 8.155.** Let $a$ be an eight-digit number and let $b$ be the number obtained by moving the last digit of $a$ to the front. Prove that if $101 \mid a$, then $101 \mid b$.

**Problem 8.156.** A natural number $n$ is called *nice* if it has four decimal digits, its first and third digits are equal, its second and fourth digits are equal, and the product of its digits divides $n^2$. Determine all nice numbers.

**Problem 8.157.** Determine all pairs $(a, b)$ of positive integers such that the sum of the digits of $ab+1$ in its decimal representation equals $ab$. [Baltic Way 2023]

## 8.4 Arithmetic functions: $\tau$, $\sigma$, $\varphi$ and Legendre's formula

**Definition 8.13 (Number and sum of divisors).** Let $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$, where $p_1, \ldots, p_k$ are distinct primes and $\alpha_1, \ldots, \alpha_k > 0$. Then the number of positive divisors of $n$ is

$$\tau(n) = (\alpha_1 + 1)(\alpha_2 + 1)\cdots(\alpha_k + 1),$$

and their sum is

$$\sigma(n) = (1 + p_1 + \cdots + p_1^{\alpha_1})(1 + p_2 + \cdots + p_2^{\alpha_2})\cdots(1 + p_k + \cdots + p_k^{\alpha_k}).$$

(The function $\tau$ is also written $d$.)

**Theorem 8.14 (Properties of $\tau$ and $\sigma$).** *The functions $\tau$ and $\sigma$ are multiplicative: $\tau(mn)=\tau(m)\tau(n)$ and $\sigma(mn)=\sigma(m)\sigma(n)$ whenever $m$ and $n$ are coprime. Moreover $\tau(n)$ is odd if and only if $n$ is a perfect square, and $\sigma(n)$ is odd if and only if $n$ or $n/2$ is a perfect square. Finally $\tau(a)\tau(b)\geq\tau(ab)$, with equality if and only if $\gcd(a,b)=1$.*

**Lemma 8.15 (Divisor bound).** *For every positive integer $n$ one has $\tau(n)\leq 2\sqrt{n}$.*

**Definition 8.16 (Euler's totient function).** For $n=p_1^{\alpha_1}\cdots p_k^{\alpha_k}$, Euler's totient function is

$$\varphi(n)=\prod_{i=1}^{k} p_i^{\alpha_i-1}(p_i-1)=n\Big(1-\frac{1}{p_1}\Big)\cdots\Big(1-\frac{1}{p_k}\Big).$$

It counts the positive integers up to $n$ that are coprime to $n$; equivalently, a reduced residue system modulo $n$ has $\varphi(n)$ elements.

**Theorem 8.17 (Euclid–Euler theorem).** *A positive integer $n$ is called* perfect *if $\sigma(n)=2n$. An even positive integer $n$ is perfect if and only if $n=2^{p-1}(2^p-1)$, where $p$ is a prime such that $2^p-1$ is also prime.*

**Theorem 8.18 (Legendre's formula).** *Let $p$ be a prime and $n$ a positive integer. The largest $k$ with $p^k\mid n!$ is*

$$k=v_p(n!)=\left\lfloor\frac{n}{p}\right\rfloor+\left\lfloor\frac{n}{p^2}\right\rfloor+\left\lfloor\frac{n}{p^3}\right\rfloor+\cdots.$$

*For instance, the largest $k$ with $3^k\mid 100!$ is $33+11+3+1=48$.*

**Theorem 8.19 (Bertrand's postulate).** *For every real $x\geq 2$ there is a prime strictly between $x$ and $2x$.*

### 8.4.1 Counting and summing divisors

**Problem 8.158.** Prove that a positive integer has an odd number of divisors if and only if it is a perfect square.

**Problem 8.159.** A positive integer has exactly 2017 divisors. Show that it is not a multiple of 2018.

**Problem 8.160.** For any two positive integers $a$ and $b$ prove the inequality $\tau(ab)\geq\tau(a)+\tau(b)-1$.

**Problem 8.161.** Prove that $\sqrt{n}\,\tau(n)\leq 2\sigma(n)\leq(n+1)\tau(n)$ for every positive integer $n$.

**Problem 8.162.** Prove that $\tau(n)<2\sqrt{n}$ for every positive integer $n$.

**Problem 8.163.** Determine all positive integers $k$ such that there exist positive integers $a$ and $b$ with $k=\tau(a)=\tau(b)=\tau(2a+3b)$.

**Problem 8.164.** For a positive integer $n$ denote by $\omega(n)$ the number of distinct prime divisors of $n$. Prove that for each positive integer $n$ there exist positive integers $k$ and $m$ with $k - m = n$ and $\omega(k) - \omega(m) = 1$.

**Problem 8.165.** Find all positive integers $n$ such that $n^2$ has three times as many divisors as $n$.

**Problem 8.166.** Denote by $\tau(n, k)$ the number of divisors of $n$ that are greater than or equal to $k$. Find

$$\tau(1001, 1) + \tau(1002, 2) + \cdots + \tau(2000, 1000).$$

**Problem 8.167.** Denote by $\tau_k(n)$ the number of divisors of $n$ that are not less than $k$. Evaluate

$$\tau_1(2021) + \tau_2(2022) + \cdots + \tau_{2020}(4040).$$

**Problem 8.168.** How many pairs of positive integers $(x, y)$ are there for which $\operatorname{lcm}(x, y) = 20!$?

**Problem 8.169.** Find all primes $p$ for which $p^2 + 11$ has fewer than 12 divisors.

**Problem 8.170.** Given a positive integer $n$, does there always exist a positive integer divisible by $n$ that has exactly $n$ divisors?

**Problem 8.171.** Let $1 = d_1 < d_2 < d_3 < \cdots < d_k = 4n$ be all divisors of $4n$, where $n$ is a positive integer. Prove that there exists an index $i$ with $d_i - d_{i-1} = 2$.

**Problem 8.172.** Let $1 = d_1 < d_2 < \cdots < d_k = n$ be all divisors of a positive integer $n$. Find all $n$ for which $d_1^2 + d_2^2 + d_3^2 + d_4^2 = n$.

**Problem 8.173.** Given a positive integer $n$, define a sequence $(a_k)$ by $a_0 = n$ and $a_{k+1} = \tau(a_k)$. Find all $n$ for which no term $a_k$ is a perfect square.

**Problem 8.174.** Find all positive integers $n$ for which $\tau(n) = \sqrt{n+1}$.

**Problem 8.175.** Suppose that all divisors of $n$ have been divided into pairs so that the sum in each pair is a prime. Prove that all these sums are distinct.

**Problem 8.176.** For a positive integer $n$ define $f(n) = \lfloor \frac{n}{1} \rfloor + \lfloor \frac{n}{2} \rfloor + \cdots + \lfloor \frac{n}{n} \rfloor$. Prove that there are infinitely many $n$ for which $\dfrac{f(n+1)}{n+1} < \dfrac{f(n)}{n}$.

**Problem 8.177.** Prove that $\displaystyle\sum_{i=1}^{n} \left\lfloor \frac{n}{i} \right\rfloor^2 = \sum_{i=1}^{n} (2i-1) \left\lfloor \frac{n}{i} \right\rfloor$ for every positive integer $n$.

**Problem 8.178.** Prove that there are infinitely many positive integers $k$ for which the equation $\dfrac{x}{\tau(x)} = k$ has no solution $x \in \mathbb{N}$.

**Problem 8.179.** Prove that $\sigma(n) = \displaystyle\sum_{k=1}^{n} k \left( \left\lfloor \frac{n}{k} \right\rfloor - \left\lfloor \frac{n-1}{k} \right\rfloor \right)$ for every integer $n > 2$.

**Problem 8.180.** If $n$ is a composite positive integer, prove that $\sigma(n) > n + \sqrt{n} + 1$.

**Problem 8.181.** Prove the inequality $\sigma(n!) > n!\left(1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{n}\right)$ for all positive integers $n$, and find all cases of equality.

**Problem 8.182.** Prove the inequality $\sigma(n!) \leq \dfrac{(n+1)!}{2}$ for all positive integers $n$, and find all cases of equality.

**Problem 8.183.** Prove that $\sigma(n) > \tau(n)\sqrt{n}$ for all integers $n > 1$.

**Problem 8.184.** Prove that $\sigma(n) < n\sqrt{2\tau(n)}$ for all integers $n > 1$. [Belarusian Mathematical Olympiad 1999]

**Problem 8.185.** Prove that $\tau((n-1)!) > \dfrac{\tau(n!)}{2}$ for all integers $n > 1$.

**Problem 8.186.** If $a \mid b$ and $a < b$, prove that $\dfrac{\sigma(a)}{a} < \dfrac{\sigma(b)}{b}$.

**Problem 8.187.** Prove that there are infinitely many pairs of different positive integers $m$ and $n$ such that $\sigma(m^2) = \sigma(n^2)$.

**Problem 8.188.** Find all $n \in \mathbb{N}$ that satisfy $\sigma(n) + \sigma(2n) = \sigma(3n)$.

**Problem 8.189.** Prove that $\sigma(n)\varphi(n) < n^2$ for every positive integer $n$.

**Problem 8.190.** We call a positive integer $n$ *venerable* if all its positive divisors less than $n$ (including 1) add up to $n-1$. Find all venerable numbers some power of which (with exponent at least 2) is also venerable.

**Problem 8.191.** Let $m, n, k$ be positive integers with $n > 1$. Show that $\sigma(n)^k \neq n^m$. [St. Petersburg City Mathematical Olympiad 2001]

**Problem 8.192.** Prove that every even perfect number is of the form $n = 2^{k-1}(2^k - 1)$ for a positive integer $k$, and conversely that this number is perfect whenever $2^k - 1$ is prime.

**Problem 8.193.** Find all perfect numbers that are divisible by 7 but not by 49.

**Problem 8.194.** For $n \in \mathbb{N}$ denote by $f(n)$ the smallest positive integer having exactly $n$ divisors; thus $f(5) = 16$ and $f(6) = 12$. Prove that $f(2^k) \mid f(2^{k+1})$ for every $k \in \mathbb{N}$.

**Problem 8.195.** Is there a positive integer $n$ such that both $n - 2015$ and $\dfrac{n}{2015}$ are positive integers having exactly 2015 divisors?

**Problem 8.196.** Determine the smallest positive integer having three distinct divisors whose sum is 2022. [IMO Shortlist 2022 N1]

**Problem 8.197.** Determine all positive integers $n$ with exactly 8 positive divisors such that the sum of all divisors of $n$, except 1 and $n$, equals 14133. [Bulgaria JBMO TST 2017]

**Problem 8.198.** Determine all positive integers $n$ with exactly 2022 divisors $1 = d_1 < d_2 < \cdots < d_{2022} = n$ such that $3d_{2017} + 2d_{2019} = n$. [Bulgaria Winter Mathematical Competition 2022, Grade 11]

**Problem 8.199.** Determine all positive integers $n$ for which $|n^2 - 81|$ has exactly $n$ positive divisors. [North Macedonia, Memorial Aleksandar Blazhevski, 2021]

**Problem 8.200.** Prove that there are infinitely many positive integers $n$ that are divisible by $\left\lfloor \sqrt{3}\,\tau(n) \right\rfloor$.† [Baltic Way 2020]

**Problem 8.201.** Determine all integers $n \geq 2$ with the following property: the largest divisor of $n$ different from $n$ equals the number of positive divisors of $n$. [Balkan MO Shortlist 2016]

**Problem 8.202.** (a) Let $(a_n)_{n\geq 1}$ be a sequence of positive integers with $a_{n+1} = a_n + \tau(n)$ for all positive integers $n$. Is it possible that some two consecutive terms of the sequence are perfect squares? (b) The same question if $a_{n+1} = a_n + 2\tau(n)$. [Balkan MO Shortlist 2012; Bulgaria National Round 2012]

**Problem 8.203.** ★ Given an integer $n > 1$, define $f(n) = \tau(n!) - \tau((n-1)!)$. Prove that there are infinitely many composite numbers $n$ such that $f(m) < f(n)$ for all $m$ with $1 < m < n$. [China TST 2015]

**Problem 8.204.** ★ Determine all positive integers $m$ for which there exists a positive integer $n$ with $\dfrac{\tau(n^2)}{\tau(n)} = m$. [IMO 1998/3]

**Problem 8.205.** ★ Find all positive integers $n$ for which all positive divisors of $n$ can be put into the cells of a rectangular table such that each cell contains a distinct divisor and the sums of all rows are equal and the sums of all columns are equal. [IMO Shortlist 2016 C2]

### 8.4.2 Euler's totient function

**Problem 8.206.** Find all positive integers $n$ for which $\varphi(n)$ divides $n$.

**Problem 8.207.** Prove that for every positive integer $n$ there exists a positive integer $x$ such that $\varphi(x) = n!$.

**Problem 8.208.** Suppose that $n$ is odd and both $\varphi(n)$ and $\varphi(n+1)$ are powers of 2. Prove that either $n = 5$, or $n+1$ is itself a power of two.

**Problem 8.209.** Let $a_1 < a_2 < \cdots < a_{\varphi(n)}$ be the positive integers not exceeding $n$ and coprime to $n$. Find all $n$ for which none of the sums $a_i + a_{i+1}$ is divisible by 3.

**Problem 8.210.** Let $n$ be a positive integer. Find the smallest $k$ for which $\varphi(\varphi(\cdots\varphi(5^n)\cdots)) = 1$, where $\varphi$ is applied $k$ times.

**Problem 8.211.** (a) Do there exist infinitely many positive integers $n$ whose number of divisors is a perfect square and whose number of positive integers not exceeding $n$ and coprime to $n$ is also a perfect square? (b) Do there exist infinitely many positive integers $n$ whose number of divisors is a fifth power of an integer and for which the number of positive integers not exceeding $n$ and coprime to $n$ is a 357th power of an integer? [RMM Shortlist 2020; Bulgarian Festival of Young Mathematicians 2023]

### 8.4.3 Legendre's formula and factorials

**Problem 8.212.** Count the number of zeroes at the end of 2019!.

**Problem 8.213.** Prove that the largest power of 2 dividing $\dfrac{(2n)!}{n!}$ is $2^n$.

**Problem 8.214.** Find $\gcd(n!+1,\ (n+1)!)$.

**Problem 8.215.** Prove that there are infinitely many pairs of positive integers $n$ and $k$ greater than 1 such that $\gcd(n!-1,\ k!-1)>1$.

**Problem 8.216.** Prove that there exist infinitely many positive integers $n$ for which $n!$ is divisible by $n^2+1$.

**Problem 8.217.** Let $a,b\in\mathbb{N}$ be such that $a!\,b!$ is divisible by $a!+b!$. Prove that $3a>2b+2$.

**Problem 8.218.** Determine all positive integers $n$ such that $n!+5$ is a power of a prime.

**Problem 8.219.** Determine all pairs $(m,n)$ of positive integers such that $2\,n!=m!\,(m!+2)$.

**Problem 8.220.** Find all pairs $(k,n)$ of positive integers such that $(2^n-1)(2^n-2)(2^n-2^2)\cdots(2^n-2^{n-1})=k!$.

**Problem 8.221.** Find all triples $(x,y,z)$ of positive integers such that $x\cdot y!+2y\cdot x!=z!$. [Kolmogorov Cup 2014]

**Problem 8.222.** Determine all pairs $(a,b)$ of positive integers such that $a!+b!=a^b$. [Middle European Mathematical Olympiad 2007]

**Problem 8.223.** Solve in nonnegative integers the equation $2^a+3^b+5^c=n!$. [USA TSTST 2017]

**Problem 8.224.** ★ Let $n$ be a positive integer. We wish to erase some (possibly none) of the numbers $1!,2!,\ldots,n!$ so that the product of all the remaining ones is a perfect square. (a) Suppose $n=1000$. What is the minimum number of numbers that have to be erased, and what are all the possibilities achieving this minimum? (b) Prove that for any $n$ at most 3 erasures suffice. (c) Prove that for any $n\geq 2$ at least one number must be erased. (d) For each $s\in\{1,2,3\}$ prove that there are infinitely many $n$ for which $s$ is the minimum number of erasures required. [Bulgaria Regional Round 2017, Grade 9]

## 8.5 Congruences: Fermat, Euler and Wilson

**Theorem 8.20 (Fermat's little theorem).** *Let $p$ be a prime and $a$ an integer. Then $a^p\equiv a \pmod p$; in particular, if $p\nmid a$, then $a^{p-1}\equiv 1 \pmod p$.*

**Theorem 8.21 (Euler's theorem).** *If $n$ is a positive integer and $a$ an integer coprime to $n$, then $a^{\varphi(n)}\equiv 1 \pmod n$.*

**Theorem 8.22 (Wilson's theorem).** *If $p$ is a prime, then $(p-1)! \equiv -1 \pmod{p}$. The variant $(p-2)! \equiv 1 \pmod{p}$ is often useful.*

**Remark.** If $n \geq 6$ is composite, then $(n-1)! \equiv 0 \pmod{n}$; the number $n = 4$ is an exception, since $3! \equiv 2 \pmod{4}$.

**Definition 8.23 (Multiplicative order).** The multiplicative order of an integer $a$ modulo a positive integer $n$, where $\gcd(a, n) = 1$, is the smallest positive integer $d$ for which $a^d \equiv 1 \pmod{n}$; equivalently, it is the smallest period of the sequence $1, a, a^2, \ldots$ modulo $n$.

**Theorem 8.24 (The order divides $\varphi(n)$).** *Let $n$ be a positive integer and $a$ an integer coprime to $n$. Then $\operatorname{ord}_n(a)$ divides $\varphi(n)$.*

**Problem 8.225.** Prove Fermat's little theorem.

**Problem 8.226.** Prove that if $\gcd(a, p) = 1$ for a prime $p$, then $a^{p(p-1)} \equiv 1 \pmod{p^2}$.

**Problem 8.227.** Prove that if $\gcd(a, p) = 1$ for a prime $p$, then $a^{p^n(p-1)} \equiv 1 \pmod{p^{n+1}}$ for every positive integer $n$.

**Problem 8.228.** Let $p$ be a prime and let $\gcd(m, p-1) = 1$. Prove that $\{0^m, 1^m, 2^m, \ldots, (p-1)^m\}$ is a complete residue system modulo $p$.

**Problem 8.229.** Prove that $2222^{5555} + 5555^{2222}$ is a multiple of 7.

**Problem 8.230.** The sum of the numbers $a$, $b$ and $c$ is divisible by 30. Prove that $a^5 + b^5 + c^5$ is also divisible by 30.

**Problem 8.231.** Find all positive integers $x$ such that $2^{2x+1} + 2$ is divisible by 17.

**Problem 8.232.** Solve the congruences $x^{3333} \equiv 4 \pmod{11}$ and $x^{86} \equiv 6 \pmod{29}$.

**Problem 8.233.** Show that the equation $5x^2 - 7 = 11y$ has no solutions in integers $x$ and $y$.

**Problem 8.234.** Prove that a perfect cube is always congruent to $-1$, 0 or 1 modulo 9.

**Problem 8.235.** If $p$ is a prime and $x \equiv y \pmod{p}$, prove that $x^p \equiv y^p \pmod{p^2}$.

**Problem 8.236.** Prove that $x^2 + 5x + 2$ cannot be divisible by 289 for any integer $x$.

**Problem 8.237.** Prove that $5^{2^n} - 1$ is divisible by $2^{n+2}$, but not by $2^{n+3}$.

**Problem 8.238.** Find all $n$ such that $5^n - 4^n$ is divisible by 61.

**Problem 8.239.** Find the last two digits of the power towers $2^{2^{\cdot^{\cdot^{\cdot^2}}}}$ and $3^{3^{\cdot^{\cdot^{\cdot^3}}}}$, each with 100 levels.

**Problem 8.240.** Find the largest positive integer $n$ such that, for all primes $p$ and $q$, $n$ divides $p^8q^4 - p^4q^8$.

**Problem 8.241.** How many positive integers $n$ with $1 \leq n \leq 2020$ satisfy $210 \mid n^n - 1$?

**Problem 8.242.** Let $a, b$ be positive integers and let $p, q$ be distinct primes such that $aq \equiv 1 \pmod{p}$ and $bp \equiv 1 \pmod{q}$. Prove that $\frac{a}{p} + \frac{b}{q} > 1$.

**Problem 8.243.** Let $p$ be a prime and $n = 2^p - 1$. Prove that $2^{n-1} \equiv 1 \pmod{n}$.

**Problem 8.244.** Prove that there also exist composite numbers $n$ such that $a^{n-1} \equiv 1 \pmod{n}$ whenever $a$ is coprime to $n$.

**Problem 8.245.** Prove that $7^n - 1$ is never divisible by $6^n - 1$.

**Problem 8.246.** Can $2^n + 1$ ever be divisible by 247?

**Problem 8.247.** Find all primes $p$ for which $\dfrac{2^{p-1} - 1}{p}$ is a perfect square.

**Problem 8.248.** Find the greatest common divisor of the numbers $A_n = 2^{3n} + 3^{6n+2} + 5^{6n+2}$ for $n = 0, 1, \ldots, 1999$. [JBMO 2001]

**Problem 8.249.** Find all positive integers $n$ for which $2^n - 1$ is divisible by $n$.

**Problem 8.250.** (a) If $n$ is a positive integer with $n > 1$ and $n \mid 2^n + 1$, prove that $n$ is a multiple of 3. (b) Prove that there exist infinitely many positive integers $n$ for which $n \mid 2^n + 1$.

**Problem 8.251.** Find all positive integers $n$ with the property that $n \mid 2^n$, $n - 1 \mid 2^n - 1$ and $n - 2 \mid 2^n - 2$.

**Problem 8.252.** If $m$ and $n$ are positive integers such that $2^m - 1$ is divisible by $(2^n - 1)^2$, prove that $m$ is divisible by $2^n - 1$.

**Problem 8.253.** What is the order of 2 modulo 1001?

**Problem 8.254.** Find the remainder when the power tower $3^{3^{\cdot^{\cdot^{\cdot^3}}}}$ with 100 threes is divided by 101.

**Problem 8.255.** The sequence $(a_n)$ is defined by $a_1 = 1$, $a_2 = 3$ and $a_{n+2} = (n+3)a_{n+1} - (n+2)a_n$. Find all values of $n$ for which $a_n$ is divisible by 11.

**Problem 8.256.** If $a$ and $b$ are any coprime positive integers, prove that there are infinitely many exponents $n$ for which $a^n + b$ is composite.

**Problem 8.257.** Prove that there is an infinite set of pairwise coprime integers of the form $2^n - 3$.

**Problem 8.258.** Define $a_1 = 2$ and $a_{n+1} = 2^{a_n} + 2$ for all $n$. Prove that $a_m$ divides $a_n$ whenever $m < n$.

**Problem 8.259.** Let $p$ be a prime. For each of the numbers $1^p, 2^p, 3^p, \ldots, (p^2)^p$ we write down its remainder upon division by $p^2$. Prove that exactly $p$ distinct remainders are written.

**Problem 8.260.** Let $p$ be a prime. For a positive integer $n$ denote $s_n = 1^n + 2^n + \cdots + (p-1)^n$. Prove that $s_n \equiv 0 \pmod{p}$ if $p - 1 \nmid n$, while $s_n \equiv -1 \pmod{p}$ if $p - 1 \mid n$.

**Problem 8.261.** Let $p$ be a prime and $a$ a positive integer with $\gcd(a, p!) = 1$. Prove that $p! \mid a^{(p-1)!} - 1$. [Kazakhstan National Olympiad, Juniors]

**Problem 8.262.** Let $p > 3$ be a prime with $p \equiv 2 \pmod 3$ and let $a_k = k^2 + k + 1$ for $k = 1, 2, \ldots, p-1$. Prove that $a_1 a_2 \cdots a_{p-1} \equiv 3 \pmod p$. [Poland 2010]

**Problem 8.263.** Let $p > 3$ be a prime of the form $3k + 2$. Prove that $1^3, 2^3, \ldots, (p-1)^3$ give pairwise different residues modulo $p$.

**Problem 8.264.** Let $p$ be a prime. If the equation $x^3 + px^2 = y^3$ has a solution in integers, prove that $3 \mid p - 1$.

**Problem 8.265.** Let $n \geq 2$ and let $a$ be an integer coprime to $n$. Prove that $n \mid a^{n-1} + (n-1)!$ if and only if $n$ is prime.

**Problem 8.266.** Let $p$ be an odd prime. Prove that $(p-3)! \equiv \dfrac{p-1}{2} \pmod p$.

**Problem 8.267.** Let $p$ be an odd prime and let $a_1, a_2, \ldots, a_p$ be an arithmetic sequence whose common difference is not divisible by $p$. Prove that there exists an index $i$ with $p^2 \mid a_i + a_1 a_2 \cdots a_p$.

**Problem 8.268.** Let $p$ be an odd prime and let $\sigma$ be a permutation of the set $\{1, 2, \ldots, p-1\}$. Prove that there exist $i \neq j$ such that $p \mid i\sigma(i) - j\sigma(j)$.

**Problem 8.269.** Find all positive integers $n$ such that $(n-1)! + 1$ is a power of $n$.

**Problem 8.270.** Let $p = 4k + 1$ be a prime and let $x = (2k)!$. Prove that $p \mid x^2 + 1$.

**Problem 8.271.** Let $a$ and $b$ be integers with $a \neq 0$ such that $3 + a + b^2$ is divisible by $6a$. Prove that $a$ is negative.

**Problem 8.272.** Prove that for every positive integer $n$, every prime divisor $p$ of $n^4 - n^2 + 1$ is of the form $12k + 1$.

**Problem 8.273.** Suppose that for a prime $p$ and integers $a, b, c$ we have $6 \mid p + 1$, $p \mid a + b + c$ and $p \mid a^4 + b^4 + c^4$. Prove that $p \mid a$, $p \mid b$ and $p \mid c$.

**Problem 8.274.** Let $x_1 = 7$ and $x_{n+1} = 2x_n^2 - 1$ for $n \geq 1$. Prove that 2003 does not divide any term of the sequence.

**Problem 8.275.** Let $p$ be a prime. Prove that there exists $x \in \mathbb{Z}$ with $p \mid x^2 - x + 3$ if and only if there exists $y \in \mathbb{Z}$ with $p \mid y^2 - y + 25$.

**Problem 8.276.** Prove that the number $2^n + 1$ has no prime divisor of the form $8k - 1$.

**Problem 8.277.** Find all positive integers $n$ for which there is an integer $m$ with $2^n - 1 \mid m^2 + 9$.

**Problem 8.278.** Determine all primes $p$ such that $(x + y)^{19} - x^{19} - y^{19}$ is divisible by $p$ for all integers $x, y$. [JBMO Shortlist 2020]

**Problem 8.279.** Determine all positive integers $n$ such that $2^n - 1$ is not divisible by any prime greater than or equal to 11. [South Korea 2013 National Round]

**Problem 8.280.** Determine all positive integers that are relatively prime to every number of the form $2^n + 3^n + 6^n - 1$, $n \geq 1$. [IMO 2005 P4]

**Problem 8.281.** Prove that there are infinitely many primes that divide at least one integer of the form $2^{n^3+1} - 3^{n^2+1} + 5^{n+1}$, where $n$ is a positive integer. [ITAMO 2010]

**Problem 8.282.** Let $n$ be an odd positive integer. Prove that there exists a positive integer $m$ such that $n$ divides $2^m + m$. [IMO Shortlist 2006 N7]

**Problem 8.283.** For each positive integer $a_0$, the sequence $a_0, a_1, a_2, \ldots$ of positive integers is defined by

$$a_{n+1} = \begin{cases} \sqrt{a_n} & \text{if } \sqrt{a_n} \text{ is an integer,} \\ a_n + 3 & \text{otherwise.} \end{cases}$$

Determine all $a_0 > 1$ for which there is a number $A$ such that $a_n = A$ for infinitely many $n$. [IMO 2017 P1]

**Problem 8.284.** Find all positive integers $n$ for which there exist primes $p$ and $q = p + 2$ such that $2^n + p$ and $2^n + q$ are both prime. [Belarus 2005]

**Problem 8.285.** Lamine and Yamal play the following game. First Lamine names a positive integer $a$; then Yamal names a positive integer $b$; then Lamine names a positive integer $n$; finally Yamal names a prime $p$ and the game ends. Yamal wins if and only if $p \mid a^n + b$ and $p^2 \nmid a^n + b$. Who has a winning strategy? [Vojtech Jarnik IMC 2025, Category I]

**Problem 8.286.** Let $a$ and $b$ be distinct positive integers. Prove that there is a positive integer $n$ such that $a^n - b^n$ is not a perfect power.

**Problem 8.287.** ★ Let $p$ be a prime and write $1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{p-1} = \frac{m}{n}$ in lowest terms. (a) If $p \geq 3$, prove that $p \mid m$. (b) If $p \geq 5$, prove that $p^2 \mid m$.

**Problem 8.288.** ★ If $p > 5$ is a prime, prove that $\binom{2p}{p} \equiv 2 \pmod{p^2}$. [Wolstenholme's theorem]

**Problem 8.289.** ★ Let $p$ be a prime. Prove that there exists a prime $q$ such that $x^p \equiv p \pmod{q}$ has no solutions.

## 8.6 The Chinese Remainder Theorem and constructions

**Theorem 8.25 (Chinese Remainder Theorem).** *Let $n_1, n_2, \ldots, n_k$ be pairwise coprime positive integers and let $a_1, a_2, \ldots, a_k$ be any integers. Then the system of congruences*

$$x \equiv a_i \pmod{n_i}, \qquad i = 1, 2, \ldots, k,$$

*has a unique solution modulo $n_1 n_2 \cdots n_k$.*

**Technique 8.26 (Solving a system by auxiliary values).** Write the solution as a sum $x = x_1 + x_2 + \cdots + x_k$ of auxiliary values chosen so that $x_i \equiv a_i \pmod{n_i}$ while $x_i \equiv 0$ modulo the product of all the other moduli. Each $x_i$ is easy to find separately, and their sum then satisfies all the congruences at once.

**Problem 8.290.** Find all $x$ such that $x \equiv 1 \pmod{2019}$, $x \equiv 2 \pmod{2021}$ and $x \equiv 3 \pmod{2023}$.

**Problem 8.291.** Prove that there exists a positive integer $x$ such that the smallest prime divisor of $1000x + 1$ is 601.

**Problem 8.292.** Prove that there exist 201 consecutive positive integers, each of which has a prime divisor not exceeding 103.

**Problem 8.293.** Show that there are 1000 consecutive positive integers, none of which is a perfect power.

**Problem 8.294.** Prove that there exist 2020 consecutive integers each of which is divisible by the square of some integer greater than 1.

**Problem 8.295.** Do there exist 2024 consecutive positive integers none of which is squarefree? (A number is squarefree if it is not divisible by any perfect square larger than 1.)

**Problem 8.296.** Do there exist 2024 consecutive positive integers none of which is a prime or a prime power?

**Problem 8.297.** Prove that there are arbitrarily many consecutive integers none of which can be written as a sum of perfect squares.

**Problem 8.298.** Is there a positive integer $n$ such that $n$, $2n$ and $3n$ are all perfect powers? Find the smallest such $n$ if it exists.

**Problem 8.299.** Prove that for all $n \geq 1$ there is a positive integer $a$ such that $a, 2a, 3a, \ldots, na$ are all perfect powers.

**Problem 8.300.** Let $a_1, a_2, \ldots, a_k$ be given positive integers. Prove that there exists a positive integer $b$ such that $ba_1, ba_2, \ldots, ba_k$ are all perfect powers.

**Problem 8.301.** Does there exist an arithmetic progression of length 2024 all of whose terms are perfect powers?

**Problem 8.302.** Does there exist a set of 2024 positive integers such that the sum of the elements of each of its nonempty subsets is a perfect power?

**Problem 8.303.** Show that for any $n$ one can find a set $X$ of $n$ distinct integers greater than 1 such that the average of the elements of any subset of $X$ is a perfect power.

**Problem 8.304.** Does there exist a set of 2022 positive integers such that the product of any three of them is divisible by the sum of those three?

**Problem 8.305.** Does there exist a set of 2022 positive integers such that their sum is divisible by each of them?

**Problem 8.306.** Does there exist a set of 2022 positive integers such that the sum of any two of them is divisible by their difference?

**Problem 8.307.** Does there exist a set of 2022 positive integers such that the arithmetic mean of any 1012 of them is never an integer?

**Problem 8.308.** Prove that every arithmetic progression contains infinitely many terms that are composed of the same set of prime factors (possibly with different exponents).

**Problem 8.309.** Prove that for any positive integer $k$ there exists an arithmetic progression $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \ldots, \frac{a_k}{b_k}$ of rational numbers, where $a_i, b_i$ are coprime positive integers, such that the $2k$ integers $a_1, b_1, a_2, b_2, \ldots, a_k, b_k$ are all distinct.

**Problem 8.310.** Prove that for each positive integer $n$ there are pairwise coprime integers $k_0, k_1, \ldots, k_n$, all greater than 1, such that $k_0 k_1 \cdots k_n - 1$ is the product of two consecutive integers.

**Problem 8.311.** Let $k$ be a positive integer. Prove that there exist distinct integers $x, y$ such that $\gcd(x+i,\, y+j) > 1$ for all $1 \le i, j \le k$.

**Problem 8.312.** Prove that there is a positive integer $n$, not divisible by any of the numbers from 2 to 1000, such that the numbers $n^2 - 1, n^2 - 2, \ldots, n^2 - 1000$ are all composite.

**Problem 8.313.** Is there a positive integer $n$ such that $n! + 1$ is divisible by $n + 100$?

**Problem 8.314.** Does there exist an integer $x$ such that $x^2 + 2$ is divisible by $3^{2022}$?

**Problem 8.315.** Given $a_0 = a$, define $a_{n+1} = 2^{2a_n} + 1$. Is it possible to choose $a$ so that $a_{2021}$ is divisible by 2021?[†]

**Problem 8.316.** Denote $m = 2^{100}$ and $n = 3^{100}$. Prove that there exist positive integers $a, b, c, d$ such that $am - bn = cm - dn = ad - bc = 1$.

**Problem 8.317.** We call a positive integer $n$ *nice* if there exist positive integers $a, b, c$ such that

$$n = \gcd(b,c)\gcd(a,bc) + \gcd(c,a)\gcd(b,ca) + \gcd(a,b)\gcd(c,ab).$$

Prove that there exist 2020 consecutive nice positive integers.

## 8.7 Squares and higher powers: squeezing, descent, sums of two squares

**Theorem 8.27 (Remainders of powers).** *Modulo small numbers, the powers $x^k$ take only the following residues: $x^2$ modulo 3: $0, 1$; modulo 4: $0, 1$; modulo 5: $0, 1, 4$; modulo 7: $0, 1, 2, 4$; modulo 8: $0, 1, 4$; modulo 10: $0, 1, 4, 5, 6, 9$; modulo 11: $0, 1, 3, 4, 5, 9$; modulo 13: $0, 1, 3, 4, 9, 10, 12$; modulo 16: $0, 1, 4, 9$. $x^3$ modulo 7: $0, 1, -1$; modulo 9: $0, 1, -1$; modulo 13: $0, 1, -1, 5, -5$. $x^4$ modulo 5: $0, 1$; modulo 16: $0, 1$; modulo 17: $0, 1, -1, 4, -4$. $x^5$ modulo 11: $0, 1, -1$.*

**Technique 8.28 (Guessing the modulus).** To show an equation has no solutions, look for a modulus $m$ for which the powers occurring take few residues: the tables above suggest $m = 4, 8, 9, 16$ for squares and fourth powers, $m = 7, 9, 13$ for cubes, and $m = p$ or $p^2$ when a factor $x^{p-1}$ appears.

**Technique 8.29 (Squeezing between consecutive powers).** If an expression $E(n)$ is trapped strictly between two consecutive $k$-th powers, say $a^k < E(n) < (a+1)^k$, then $E(n)$

cannot be a $k$-th power. Multiplying the equation by a constant to complete a square or a cube is the usual way to create such a trap.

**Technique 8.30 (Fermat's method of infinite descent).** Suppose a homogeneous equation has a nontrivial solution; choose one minimising $|x| + |y| + \cdots$ or $\gcd(x, y, \dots) = 1$. Congruence considerations then force each unknown to be divisible by a fixed prime, producing a strictly smaller solution – a contradiction. Hence only the trivial solution exists.

**Theorem 8.31 (Fermat's two-square theorem).** *Every prime $p \equiv 1 \pmod 4$ can be represented in exactly one way as a sum of the squares of two integers, up to the order and the signs of the terms.*

**Lemma 8.32 (Primes $p \equiv 3 \pmod 4$).** *Let $p \equiv 3 \pmod 4$ be a prime and $a, b$ integers. If $p \mid a^2 + b^2$, then $p \mid a$ and $p \mid b$.*

**Remark.** A positive integer is a sum of three squares if and only if it is not of the form $4^s(8l + 7)$.

### 8.7.1 When is an expression a perfect power?

**Problem 8.318.** Find all positive integers $n$ for which $n(n - 10)$ is a perfect square.

**Problem 8.319.** Find all positive integers $n$ for which (a) $n^3 - n$; (b) $n^3 - 3n$ is a perfect square.

**Problem 8.320.** Find all positive integers $n$ for which $n^3 - 7n$ is a perfect square.

**Problem 8.321.** Find all positive integers $x$ for which $x(x + 11)$ is a perfect square.

**Problem 8.322.** Find all solutions of the equation $x^2 + x + 1 = y^2$ in integers.

**Problem 8.323.** Find all positive integers $n$ for which $n \cdot 2^n + 4$ is a perfect square.

**Problem 8.324.** Find all positive integers $n$ for which $2^n + 5^n$ is a perfect square.

**Problem 8.325.** Find all positive integers $n$ for which $1 + 2^n + 2^{2n+1}$ is a perfect square.

**Problem 8.326.** Find all triples of positive integers $(a, b, c)$ for which $3^a + 3^b + 3^c$ is a perfect square.

**Problem 8.327.** Find all positive integers $x$ for which $3x^4 + 10x^2 + 3$ is a perfect square.

**Problem 8.328.** Determine all primes $p$ such that both $\dfrac{p+1}{2}$ and $\dfrac{p^2+1}{2}$ are perfect squares.
[Bundeswettbewerb Mathematik 1997]

**Problem 8.329.** Determine all positive integers $n$ for which $1! + 2! + \cdots + n!$ is a perfect square.

**Problem 8.330.** Find all integers $n > 1$ for which $1! + 2! + \cdots + n!$ is a perfect power.

**Problem 8.331.** Solve the equation $x! + 76 = y^2$ in positive integers.

**Problem 8.332.** Find all $n$-tuples $(a_1, a_2, \ldots, a_n)$ of positive integers such that $(a_1! - 1)(a_2! - 1) \cdots (a_n! - 1) - 16$ is a perfect square.

**Problem 8.333.** Prove that $x^8 + 1 = n!$ has only finitely many solutions in nonnegative integers.

**Problem 8.334.** Find all positive integers $n$ such that there exist $n$ consecutive positive integers whose sum is a perfect square.

**Problem 8.335.** Determine the smallest integer $n \geq 41$ such that the sum of all integers from 41 to $n$ inclusive is a perfect square. [Bulgaria Regional Round 2025, Grade 8]

**Problem 8.336.** Find all pairs $(k, m)$ of positive integers for which $k^2 + 4m$ and $m^2 + 5k$ are perfect squares.

**Problem 8.337.** Find all pairs $(x, y)$ of positive integers such that $x^2 + 3y$ and $y^2 + 3x$ are simultaneously perfect squares.

**Problem 8.338.** Find all positive integers $x, y$ for which both $x^2 - 4y$ and $y^2 - 4x$ are perfect squares.

**Problem 8.339.** Find all pairs $(a, b)$ of positive integers such that $b \mid 2a^2$ and $a^2 + b$ is a perfect square.

**Problem 8.340.** Find all pairs of positive integers $(a, b)$ for which $a$ is odd, $b$ is a power of 2, and $a^2 - ab + b^2$ is a perfect square.

**Problem 8.341.** Let $a$ and $b$ be odd positive integers such that $a^b b^a$ is a perfect square. Prove that $ab$ is a perfect square.

**Problem 8.342.** Suppose that $a$ and $b$ are integers such that $2^n a + b$ is a perfect square for every $n \in \mathbb{N}$. Prove that $a = 0$.

**Problem 8.343.** Let $x, y > 1$ be integers such that $x^2 + xy - y$ is a perfect square. Prove that $x + y + 1$ is composite.

**Problem 8.344.** Prove that $4xy - x - y$ is never a perfect square for positive integers $x, y$.

**Problem 8.345.** Prove that there are no positive integers $m, n$ such that $4mn - m - n$ is a perfect square.

**Problem 8.346.** Prove that no positive integers $a, b$ exist such that $2a^2+1$, $2b^2+1$ and $2(ab)^2+1$ are all perfect squares.

**Problem 8.347.** Is there a positive integer $x$ such that each of $2x - 1$, $5x - 1$ and $13x - 1$ is a perfect square?

**Problem 8.348.** Let $a$ and $b$ be integers of different parity. Prove that there exists an integer $c$ such that the three numbers $c + a$, $c + b$ and $c + ab$ are all perfect squares.

**Problem 8.349.** Let $p > 2$ be a prime and let $x, y \in \left\{1, 2, \ldots, \frac{p-1}{2}\right\}$. Prove that if $x(p-x)y(p-y)$ is a perfect square, then $x = y$.

**Problem 8.350.** Let $a, b, c$ be positive integers with $\gcd(a, b, c) = 1$ and $a^2 + b^2 + c^2 = 2(ab + bc + ca)$. Prove that $a$, $b$ and $c$ are all perfect squares.

**Problem 8.351.** Suppose that $a$ and $b$ are positive integers satisfying $2a^2 + a = 3b^2 + b$. Prove that $a - b$ is a perfect square.

**Problem 8.352.** If the difference of two consecutive cubes equals $n^2$ for a positive integer $n$, prove that $2n - 1$ is a perfect square.

**Problem 8.353.** Positive integers $a, b, c$ satisfy $\frac{1}{a} + \frac{1}{b} = \frac{1}{c}$; let $d = \gcd(a, b, c)$. Prove that $abcd$ is a perfect square.

**Problem 8.354.** Can $n(n + 1)(n + 2)(n + 3)$ be a perfect cube for some $n \in \mathbb{N}$?

**Problem 8.355.** Prove that the product of five consecutive integers is never a perfect square. [IMO Shortlist 1986]

**Problem 8.356.** Find all primes $p$ such that $\dfrac{p^2 - p - 2}{2}$ is a perfect cube.

**Problem 8.357.** Find all triples $(a, b, c)$ of positive integers such that $b^2 + 1$ and $c^2 + 1$ are both prime and $a^2 + 1 = (b^2 + 1)(c^2 + 1)$.

**Problem 8.358.** Prove that if $p$ is prime, then $7^p + 3^p - 4$ is not a perfect square. [JBMO 2007 P4]

**Problem 8.359.** Find all positive integers $n$ for which there exist primes $p, q, r$ with $p^n + q^n = r^2$. [Romanian Mathematical Olympiad 2004]

**Problem 8.360.** Prove that there are no positive integers $a, b$ such that $ab$ and $a + b$ are perfect squares and $16a - 9b$ is a prime number. [JBMO Shortlist 2014]

**Problem 8.361.** Are there integers $a$ and $b$ such that $a^5b + 3$ and $ab^5 + 3$ are both perfect cubes? [USAJMO 2013]

**Problem 8.362.** Find all positive integers $(a, b)$ for which $a^3 + 6ab + 1$ and $b^3 + 6ab + 1$ are both perfect cubes.

**Problem 8.363.** Prove that $9n^4 + 170n^2 + 289$ is not a perfect cube for any integer $n$. [JBMO Shortlist 2020]

**Problem 8.364.** Determine all integers $1 \le n \le 6$ for which there exist integers $a, b, c$ such that $a^n + b^n - c^n = n$. [Baltic Way 2016]

**Problem 8.365.** A sequence $(a_n)$ satisfies $a_{n+1} = a_n^3 + 103$ for all $n \in \mathbb{N}$. Prove that this sequence contains at most one perfect square.

**Problem 8.366.** Let $x_1 = 4$ and $x_{n+1} = x_1x_2 \cdots x_n + 5$ for $n \ge 1$. Find all pairs $\{a, b\}$ of positive integers such that $x_a x_b$ is a perfect square. [ITAMO 2014]

**Problem 8.367.** Find all quadratic polynomials $P(x) = ax^2 + bx + c$ with integer coefficients such that $P(1), P(2), \ldots, P(6)$ are all powers of two.

**Problem 8.368.** Find all triples $(m, n, p)$ of positive integers, where $p$ is prime, such that $\dfrac{5^m + 2^n p}{5^m - 2^n p}$ is a perfect square. [Turkey JBMO TST 2012]

**Problem 8.369.** Determine all positive integers $c$ for which there exists a positive integer $m$ such that $m^3 - m = c!$ and $m^2 - 1$ has at most 11 positive divisors. [JBMO Shortlist 2024]

**Problem 8.370.** ★ Determine all positive integers $n$ such that $\lfloor \sqrt{n} \rfloor^2 + 2$ divides $n^2 + 1$. [APMO 2013]

### 8.7.2 Diophantine equations solved by factoring and squeezing

**Problem 8.371.** Solve the equation $y^4 = x^3 + 7$ in integers.

**Problem 8.372.** Solve the equation $x^3 + 7 = y^2$ in integers.

**Problem 8.373.** Solve the equation $x^5 = y^2 + 4$ in integers.

**Problem 8.374.** Solve the equation $x^2 + 4 = y^5$ in integers.

**Problem 8.375.** Solve the equation $2x^6 + y^7 = 11$ in integers.

**Problem 8.376.** Find all integer solutions of $x^4 + y = x^3 + y^2$.

**Problem 8.377.** Find all integers $x$ and $y$ that satisfy $x^2 + x = y^3 + y^2 + y$.

**Problem 8.378.** Solve the equation $x^2 + x = y^4 + y^3 + y^2 + y$ in positive integers.

**Problem 8.379.** Find all pairs of integers $x, y$ such that $x^2(1 - y) + y^2(1 - x) = 7$.

**Problem 8.380.** Find all rationals $a, b$ such that $a^2 + ab + b^2 = 2$.

**Problem 8.381.** Find all positive integers $k$ for which the equation $x(x + k) = y(y + 1)$ has a solution in positive integers.

**Problem 8.382.** Find all pairs $(a, b)$ of positive rational numbers with $a < b$ such that $a^b = b^a$.

**Problem 8.383.** Find all pairs of positive integers $(a, b)$ such that $a^{b^2} = b^a$.

**Problem 8.384.** Find all triples of positive integers $(x, y, z)$ such that $x^3+y^3+z^3 = 3(xyz+3^{100})$.

**Problem 8.385.** Solve in integers the system $x^2 + 6y^2 = z^2$, $6x^2 + y^2 = t^2$.

**Problem 8.386.** Are there nonzero integers $a, b, c, d$ such that $ac + bd = 16$ and $ad - bc = 1$?

**Problem 8.387.** Find all positive integers $a, b, c, d$ satisfying $ac + bd = 100$ and $ad - bc = 1$.

**Problem 8.388.** Find all triples of positive integers $a, b, c$ such that $a^2 + b^2 = c^2$ and $a^3 + b^3 = (c - 1)^3 - 1$.

**Problem 8.389.** Find all integers $m$ and $n$ such that $m^5 - n^5 = 16mn$. [JBMO 2018 NT1]

**Problem 8.390.** Determine all pairs $(x, y)$ of positive integers such that $x^3+y^3 = x^2+42xy+y^2$. [Balkan MO 2017/1]

**Problem 8.391.** Find all pairs $(x, y)$ of integers such that $x^2 + 5y^2 = 2021y$. [JBMO Shortlist 2021]

**Problem 8.392.** Does the equation $x^4 + y^3 = z! + 7$ have infinitely many solutions in positive integers?

**Problem 8.393.** Call an integer an *almost-square* if it is a product of two consecutive integers. Prove that every almost-square can be written as a quotient of two other almost-squares.

**Problem 8.394.** Prove that $\dfrac{x^2+1}{y^2-5}$ is not an integer for any integers $x, y > 2$.

**Problem 8.395.** Prove that there exist infinitely many positive integers that cannot be represented in the form $x^2 + y^3$ with $x, y$ positive integers.

**Problem 8.396.** ★ Prove that the equation $(m^2+n)(n^2+m) = 2(m-n)^3$ has no solutions in positive integers. [IMO Shortlist 2012 N4]

### 8.7.3 Descent

**Problem 8.397.** Solve in integers the equation $x^2 + y^2 + z^2 - 2xyz = 0$.

**Problem 8.398.** Prove that the following equations have no solutions in nonzero integers: (a) $x^2 + y^2 = 3z^2$; (b) $x^3 + 10y^3 = 100z^3$.

**Problem 8.399.** Find all integer solutions of $x^3 + 3y^3 + 9z^3 - 3xyz = 0$.

**Problem 8.400.** Find all integer solutions of $a^2 + b^2 + c^2 = 7d^2$.

**Problem 8.401.** Solve in integers the equation $x^4 + y^4 + z^4 = 9u^4$.

**Problem 8.402.** Find all integer solutions of the equation $6(6a^2 + 3b^2 + c^2) = 5d^2$.

**Problem 8.403.** Solve in integers the equation $x^2 + y^2 + z^2 = x^2y^2$.

**Problem 8.404.** Let $a, b, c, d$ be nonzero integers such that the only quadruple of integers $(x, y, z, t)$ satisfying $ax^2 + by^2 + cz^2 + dt^2 = 0$ is $x = y = z = t = 0$. Does it follow that $a, b, c, d$ all have the same sign?

**Problem 8.405.** Prove that there are infinitely many positive integers $n$ that cannot be written in the form $n = x_1^4 + x_2^4 + \cdots + x_{15}^4$ for integers $x_1, \ldots, x_{15}$.

**Problem 8.406.** Can every positive integer greater than $100^{100}$ be written as a sum of 15 fourth powers (some of which may be zero)?

**Problem 8.407.** Let $a, b, c$ be positive integers such that $\dfrac{a^2+b^2+c^2}{ab+bc+ca}$ is an integer. Prove that this integer is not divisible by 3.

### 8.7.4 Sums of two squares

**Problem 8.408.** Let $p$ be a prime dividing $x^2+y^2$ for some integers $x, y$ with $p \nmid xy$. Prove that $p \equiv 1 \pmod 4$ or $p = 2$.

**Problem 8.409.** Prove that any odd divisor of a number of the form $n^2+1$ is of the form $4k+1$.

**Problem 8.410.** Prove that if $p \equiv 3 \pmod 4$ is a prime, then no perfect square is congruent to $p-1$ modulo $p$.

**Problem 8.411.** If a prime $p$ divides $x^2+xy+y^2$ for some integers $x, y$ with $p \nmid xy$, prove that $p \equiv 1 \pmod 3$ or $p = 3$.

**Problem 8.412.** Prove that a positive integer can be written as the sum of two perfect squares if and only if it can be written as the sum of the squares of two rational numbers.

**Problem 8.413.** Prove that for any prime $p > 3$ there exist integers $x, y, k$ with $0 < 2k < p$ and $kp + 3 = x^2 + y^2$.

**Problem 8.414.** Prove that the equation $3^k - 1 = m^2 + n^2$ has infinitely many solutions in positive integers.

**Problem 8.415.** Prove that the equation $x^4 - 4 = y^2 + z^2$ has no integer solutions.

**Problem 8.416.** Prove that if $a$ is an integer, then $2a^2 - 1$ has no divisor of the form $b^2 + 2$.

**Problem 8.417.** Let $S$ be the set of all numbers of the form $x^2 + 2y^2$ with $x, y$ integers. If $3n \in S$ for some integer $n$, prove that $n \in S$.

**Problem 8.418.** Find the number of different remainders that the numbers $1^2, 2^2, 3^2, \ldots$ give when divided by $n$, where (a) $n$ is a prime; (b) $n$ is a product of two primes; (c) $n = 2^k$.

**Problem 8.419.** Prove that $\dfrac{1}{2n} < \{n\sqrt{7}\} < 1 - \dfrac{1}{6n}$ for all positive integers $n$, where $\{x\}$ denotes the fractional part of $x$.

**Problem 8.420.** ★ Let $n$ be a positive integer. Define

$$A = \left\lfloor \frac{1^2}{n} \right\rfloor + \left\lfloor \frac{2^2}{n} \right\rfloor + \cdots + \left\lfloor \frac{(n-1)^2}{n} \right\rfloor, \qquad B = \lfloor \sqrt{n} \rfloor + \left\lfloor \sqrt{2n} \right\rfloor + \cdots + \left\lfloor \sqrt{(n-1)n} \right\rfloor.$$

Prove that $A + B \geq (n-1)^2$, and determine when equality holds.

## 8.8 Exponential Diophantine equations

The standard attack is to bound the exponents by a well-chosen modulus: reduce modulo a power of one base to pin down an exponent in an arithmetic progression, then reduce modulo a prime for which the corresponding order is small.

**Problem 8.421.** Solve the equation $2^x = 3^y + 5$ in positive integers. [IMO Longlist 1964]

**Problem 8.422.** Solve the equation $2^x = 7^y + 25$ in positive integers.

**Problem 8.423.** Solve the equation $4^x = 3^y + 253$ in positive integers.

**Problem 8.424.** Solve the equation $2^a - 5^b = 7$ in positive integers $a, b$.

**Problem 8.425.** Solve the equation $2^a - 5^b = 3$ in positive integers $a, b$.

**Problem 8.426.** Find all positive integers $a, b, c$ such that $2^a + 3^b = 5^c$.

**Problem 8.427.** Find all integers $a, b, c$ such that $2^a + 3^b + 1 = 6^c$.

**Problem 8.428.** Show that there are no positive integers $a, b, c, d$ such that $2^a + 4^b + 5^c = 2014^d$.

**Problem 8.429.** Find all positive integer solutions of the equation $3^x + 4^y = 5^z$.

**Problem 8.430.** Find all positive integers $x, y, z$ such that $x^2 = 2^y + 2001^z$.

**Problem 8.431.** Find all positive integers $x, y, z$ such that $20^x + 13^y = 2013^z$.† [JBMO Shortlist 2013]

**Problem 8.432.** Solve the equation $(2^{2015} + 1)^x + 2^{2015} = 2^y + 1$ in nonnegative integers. [Serbia 2015]

**Problem 8.433.** Solve the equation $2^x + 17 = y^4$ in integers.

**Problem 8.434.** Find all ordered pairs of positive integers $(m, n)$ such that $125 \cdot 2^n - 3^m = 271$. [JBMO 2018 NT2]

**Problem 8.435.** Find all triples of primes $(p, q, r)$ satisfying $3p^4 - 5q^4 - 4r^2 = 26$. [JBMO 2014 NT2]

**Problem 8.436.** Find all triples of positive integers $a, b, n$ such that $3^a 5^b = n^3 + 125$.

**Problem 8.437.** Find all pairs of primes $(p, q)$ satisfying $2^p = 2^{q-2} + q!$.

**Problem 8.438.** Find all pairs of primes $p, q$ for which $5pq + 1$ is a fifth power of an integer.

**Problem 8.439.** Solve in positive integers the equation $x! + 123 = y^z$, where $z \geq 2$.

**Problem 8.440.** Solve in positive integers the equation $10 \cdot x! + 3 = 3(x + 1)^y$.

**Problem 8.441.** Are there primes $p$ and $q$ such that $p^2 + 2012pq + q^2$ is a perfect square?

**Problem 8.442.** Find all integers $n$ such that $n^3 + 9n^2 - 11n - 11$ is a power of 2. [Serbia IMO TST 2024]

**Problem 8.443.** Determine all triples $(a, b, c)$ of nonnegative integers with $a! + 5^b = 7^c$. [JBMO Shortlist 2019]

**Problem 8.444.** Determine all positive integers $k$ for which $3^k - 1$ is a power of a positive integer with exponent at least 2. [Folklore; a particular case of Mihailescu's theorem]

**Problem 8.445.** Solve in positive integers the equation $4^a \cdot 5^b - 3^c \cdot 11^d = 1$. [Romania JBMO TST 2015]

**Problem 8.446.** Determine the smallest prime $p$ of the form $8k+1$ that cannot be represented as $x^2 - 2^n$ for positive integers $x$ and $n$. [Russian Festival of Young Mathematicians 2015; Bulgaria Spring Mathematics Competition 2016]

**Problem 8.447.** Determine the smallest prime that cannot be represented as $|3^a - 2^b|$ with $a, b$ nonnegative integers. [China IMO TST 1995]

**Problem 8.448.** Determine all nonnegative integers $n$ such that $441 \cdot 410^{n+1} - 1$ has exactly two distinct prime divisors. [Bulgarian Festival of Young Mathematicians 2025, Grades 8–9]

**Problem 8.449.** Find all positive integers $a, b, c$ and primes $p$ such that $p+2$ is prime and $2^a p^b = (p+2)^c - 1$. [Canada National Round 2025]

**Problem 8.450.** Find all nonnegative integers $n$ for which there exist primes $p, q$ and a nonnegative integer $m$ such that $10^{4n} - 1 = pqm^2$. [JBMO 2025/2]

**Problem 8.451.** Determine all quadruples $(k, m, n, p)$ of nonnegative integers such that $p$ is prime and $|4^m - 7^n| = p^k$. [JBMO Shortlist 2024]

**Problem 8.452.** Determine all primes $p$ and positive integers $a, b$ such that $a! + b$ and $b! + a$ are powers of $p$. [JBMO 2023/1, generalisation]

**Problem 8.453.** Determine all positive integers $a$ such that $3^a + a^4$ is a perfect square. [IMO Shortlist 2023 N2]

**Problem 8.454.** ★ Determine all positive integers $a$ and $b$, odd positive integers $c$, and primes $p$ and $q$ such that $p^a q^b - 1 = (p+4)^c$. [Bulgaria JBMO TST 2023]

## 8.9 Size, bounding and pigeonhole arguments

**Technique 8.33 (Size arguments).** If $m \mid n$ for positive integers $m, n$, then $1 \le m \le n$. Turning a divisibility or an equation into an inequality between the two sides – and then checking the finitely many surviving cases – solves a surprising number of Diophantine problems.

**Problem 8.455.** Solve in integers the equation $x^3(y+1) + y^3(x+1) = 19$. [JBMO Shortlist 2019]

**Problem 8.456.** Solve in integers the equation $\dfrac{1}{x^2} + \dfrac{2}{xy} + \dfrac{3}{y^2} = \dfrac{1}{4}$. [Bulgaria, Ivan Salabashev Tournament 2011 and 2022, Grades 10–12]

**Problem 8.457.** Solve in positive integers the equation $\dfrac{x^2+y^2}{z!} = \dfrac{1}{x!} + \dfrac{1}{y!}$. [Bulgaria Winter Mathematical Competition 2013, Grade 9]

**Problem 8.458.** Determine all integers $n \ge 3$ for which the sum of all composite positive integers less than $2n$ equals $n^2$. [Bulgaria Regional Round 2021, Grades 11–12]

**Problem 8.459.** The positive integers $a$ and $b$ are such that $\sqrt{11} > \dfrac{a}{b}$. Prove that $\sqrt{11} - \dfrac{a}{b} > \dfrac{1}{2ab}$. [Baltic Way 2018]

**Problem 8.460.** Find all 6-tuples of positive integers $a, b, c, d, e, f$ satisfying $a + b + c = def$ and $d + e + f = abc$.

**Problem 8.461.** Integers $a > n > 0$ have the property that each of the numbers $n^2 + 1, n^2 + 2, \ldots, n^2 + n$ divides one of the numbers $a + 1, a + 2, \ldots, a + n$. Prove that $a > n^4 - n^3$.

**Problem 8.462.** Find the largest positive integer $n$ that cannot be written as $n = a + b$, where $a$ and $b$ are positive integers such that $ab$ has no prime divisor greater than $\sqrt{n}$.

**Problem 8.463.** Find all positive integers $n$ that cannot be written in the form $n = \dfrac{a}{b} + \dfrac{a+1}{b+1}$ for positive integers $a$ and $b$.

**Problem 8.464.** Let $a, b, c, d$ be positive integers such that $b < c$ and $a + b + c + d = ab - cd$. Prove that $a + c$ is a composite number.

**Problem 8.465.** Let $n$ be a positive integer and $A = \{n, n + 1, \ldots, 2n - 1\}$. Prove that there is a number $a \in A$ such that the sum of the elements of $A \setminus \{a\}$ is not divisible by any element of $A$.

**Problem 8.466.** Find the smallest integer $n > 1$ for which there exist positive integers $a_1, a_2, \ldots, a_n$ such that $a_1^2 + a_2^2 + \cdots + a_n^2$ divides $(a_1 + a_2 + \cdots + a_n)^2 - 1$. [Kazakhstan 2011]

**Problem 8.467.** Let $p > 5$ be a prime and $X = \{p - n^2 : 0 < n < \sqrt{p}\}$. Prove that $X$ contains two distinct elements $x$ and $y$ with $x \neq 1$ and $x \mid y$. [Balkan MO 1996]

**Problem 8.468.** Let $A$ be a finite set of positive integers with at least three elements. Prove that there are two elements of $A$ whose sum does not divide the sum of the other elements of $A$. [Romanian JBMO TST 2003]

**Problem 8.469.** Let $A$ be a set of positive integers such that (i) if $a \in A$, then all positive divisors of $a$ are also in $A$; and (ii) if $a, b \in A$ and $1 < a < b$, then $1 + ab \in A$. Prove that if $A$ has at least three elements, then $A$ is the set of all positive integers. [Romanian JBMO TST 2004]

**Problem 8.470.** Find all positive integers $n > 2$ having the following property: there is a permutation $(a_1, a_2, \ldots, a_n)$ of $\{1, 2, \ldots, n\}$ such that the partial sums $a_1 + a_2 + \cdots + a_k$, $k = 1, 2, \ldots, n$, form a complete residue system modulo $n$. [Romanian JBMO TST 2013]

**Problem 8.471.** For a permutation $a_1, a_2, \ldots, a_{100}$ of the numbers $1, 2, \ldots, 100$ denote $S_k = a_1 + a_2 + \cdots + a_k$. At most how many of the numbers $S_1, S_2, \ldots, S_{100}$ can be perfect squares?

**Problem 8.472.** Are there 20 positive integers $a_1, \ldots, a_{10}, b_1, \ldots, b_{10}$ with the following property: for every subset of indices $S \subseteq \{1, \ldots, 10\}$, the sum $\sum_{i \in S} a_i$ divides $12 + \sum_{i \in S} b_i$?

**Problem 8.473.** Suppose that positive integers $a_1, a_2, \ldots, a_n$ (indices taken cyclically, so $a_0 = a_n$ and $a_{n+1} = a_1$) are such that each of the quotients $k_i = \dfrac{a_{i-1} + a_{i+1}}{a_i}$, $i = 1, \ldots, n$, is an integer. Prove that $2n \leq k_1 + k_2 + \cdots + k_n \leq 3n$.

**Problem 8.474.** Given $n + 1$ positive integers not exceeding $2n$, prove that there exist two of them, say $a$ and $b$, such that $a \mid b$. (Give proofs with and without induction.)

**Problem 8.475.** Given any $n$ integers, prove that one can always choose several of them (at least one) whose sum is divisible by $n$.

**Problem 8.476.** Prove that among any 23 integers there exist 12 whose sum is divisible by 12.

**Problem 8.477.** Prove that from any set of 71 integers one can choose 36 numbers whose sum is a multiple of 36.

**Problem 8.478.** The number 1 is written on the screen of a calculator. Each second the following operation is performed: if the number on the screen is divisible by $2^k$, the calculator adds to it a random integer between 1 and $k+1$. Prove that every power of 2 will occur on the screen at some moment.

**Problem 8.479.** Determine the smallest integer $k$ with the following property: no matter which $k$ integers among $1, 2, \ldots, 2024$ we choose, there will be two distinct chosen integers $a, b$ such that $ab+1$ is a perfect square. [JBMO Shortlist 2024]

**Problem 8.480.** On the board is written a positive integer $n$. In one move one may choose a number on the board, delete it, and write all of its positive divisors except itself; equal numbers may appear on the board. For which $n$ is it possible that at some point there are exactly $n^2$ (not necessarily distinct) numbers on the board? [All-Russian Olympiad 2017, Grade 11]

**Problem 8.481.** On the board are written 999 consecutive integers. Maria has 999 labels of the form "This number is not divisible by $i$" for $i \in \{2, 3, \ldots, 1000\}$. She places each label next to a number on the board, so that each number gets exactly one label. For every true statement Maria gets a piece of candy. How many pieces of candy can Maria guarantee to win, regardless of the numbers written on the board, if she plays optimally? [Zhautykov Olympiad 2025/4]

**Problem 8.482.** ★ Determine the smallest positive integer $a$ for which there exist infinitely many positive integers $n$ such that $1000n^2+an+1$ has a divisor, different from 1 and itself, which gives remainder 1 when divided by $n$. [Bulgaria JBMO TST 2025]

**Problem 8.483.** ★ Solve in positive integers the equation $a^3+b^3+c^3 = a^2b^2c^2$. [IMO Shortlist 2019 N2]

**Problem 8.484.** ★ Determine all pairs $(a, b)$ of positive integers with $a > b$ such that $(a-b)^{ab} = a^b b^a$. [Taiwan IMO TST 2016]

# Chapter 9

# Mixed Practice

**Sources for this chapter:** Weekly e-training problem sheets, Level 2, from three seasons of the program: 2019–2020 (weeks 1–16), September–November 2021, and 2022–2023 (weeks 1–14).

Each week of the program's e-training course mixed algebra, number theory, combinatorics and geometry into a single short problem set. This chapter reproduces that training form: a sequence of practice sets, drawn directly from the weekly sheets and merged where a week ran thin, arranged in roughly increasing difficulty.

## Practice set 1

**Problem 9.1.** There are four points $A, B, C, D$ on the plane, such that any three points are not collinear. Prove that among the triangles $ABC$, $ABD$, $ACD$ and $BCD$ there is at least one triangle which has an interior angle not greater than $45^\circ$.

**Problem 9.2.** Find all triples $(a, b, c)$ such that $a = (b + c)^2$, $b = (c + a)^2$, $c = (a + b)^2$.

**Problem 9.3.** Solve in integers: $x! + 13 = y^2$.

**Problem 9.4.** Triangles $ABC$ and $ABD$ are isosceles with $AB = AC = BD$, and $AC$ intersects $BD$ at $E$. Also $AC$ is perpendicular to $BD$. Find $\angle C + \angle D$ in degrees.

**Problem 9.5.** Let $a$, $b$, $c$ be pairwise different numbers. Solve the system of equations

$$\begin{cases} z + ay + a^2x + a^3 = 0 \\ z + by + b^2x + b^3 = 0 \\ z + cy + c^2x + c^3 = 0. \end{cases}$$

**Problem 9.6.** Find the number of acute triangles whose perimeter is less than 100 and whose sides are three consecutive positive integers.

## Practice set 2

**Problem 9.7.** In triangle $ABC$, $\angle A = 70^\circ$. The point $D$ is chosen on segment $AC$ such that the bisector $AE$ of $\angle A$ meets $BD$ at $H$, with $AH : HE = 3 : 1$ and $BH : HD = 5 : 3$. Find $\angle C$.

**Problem 9.8.** Let $ABCD$ be a quadrilateral with $AD = BC$ and $\angle A + \angle B = 120^\circ$, and let $P$ be a point exterior to $ABCD$ such that $P$ and $A$ lie on opposite sides of line $DC$ and triangle $DPC$ is isosceles. Prove that triangle $APB$ is also isosceles.

**Problem 9.9.** In triangle $ABC$, $\angle A = 96^\circ$. Segment $BC$ is extended to a point $D$. The bisectors of $\angle ABC$ and $\angle ACD$ meet at $A_1$; the bisectors of $\angle A_1BC$ and $\angle A_1CD$ meet at $A_2$; and so on, until the bisectors of $\angle A_4BC$ and $\angle A_4CD$ meet at $A_5$. Find $\angle BA_5C$ in degrees.

**Problem 9.10.** Prove that for all $n \geq 4$, $n! > 2^n$ and $2^n \geq n^2$.

**Problem 9.11.** It is known that $a < 1$, $b < 1$ and $a + b \geq 0.5$. Prove that $(1 - a)(1 - b) \leq \dfrac{9}{16}$.

**Problem 9.12.** Read the proof of Bernoulli's inequality. Conclude that $8^{91} > 7^{92}$.

## Practice set 3

**Problem 9.13.** By using Bernoulli's inequality, prove that for $n \geq 1$ the following inequality holds

$$1 + \frac{5}{6n - 5} \leq 6^{1/n} \leq 1 + \frac{5}{n}.$$

**Problem 9.14.** Let

$$a_n = \frac{1}{n+1} + \frac{1}{n+2} + \ldots + \frac{1}{2n}.$$

Prove that for $n \geq 3$ one has $a_{n+1} \geq a_n$, and based on this conclude that $a_{2019} > \dfrac{3}{5}$.

**Problem 9.15.** Let $a$, $b$, $c$ be positive and less than 1. Prove that

$$1 - (1 - a)(1 - b)(1 - c) > \max(a, b, c).$$

**Problem 9.16.** In the square $ABCD$ let $K$ be a point on the side $BC$ and let the bisector of $\angle KAD$ meet the side $CD$ at $M$. Prove that $AK = DM + BK$.

**Problem 9.17.** Let $ABCD$ be a square, $P$ an inner point such that $PA : PB : PC = 1 : 2 : 3$. Find $\angle APB$ in degrees.

**Problem 9.18.** In square $ABCD$, $M$ is the midpoint of $AD$ and $N$ is the midpoint of $MD$. Prove that $\angle NBC = 2\angle ABM$.

**Problem 9.19.** Let $ABC$ be an isosceles triangle with $AB = AC = 2$. There are 100 points $P_1, P_2, \ldots, P_{100}$ on side $BC$. Let $m_i = AP_i^2 + BP_i \cdot CP_i$. Find $m_1 + m_2 + \cdots + m_{100}$.

## Practice set 4

**Problem 9.20.** Show that for all $n \geq 1$,

$$\frac{1}{1^2} + \frac{1}{2^2} + \cdots + \frac{1}{n^2} < 2.$$

**Problem 9.21.** Quadrilateral $ABCD$ is given such that $\angle DAC = \angle CAB = 60^\circ$ and $AB = BD - AC$. Lines $AB$ and $CD$ meet at $E$. Prove that $\angle ADB = 2\angle BEC$.

**Problem 9.22.** There are $n > 2$ lines in the plane in general position (any two meet, but no three are concurrent). All their intersection points are marked, and then the lines themselves are erased, leaving only the marked points. It is not known which marked point belongs to which two lines. Is it always possible to determine which line each point belongs to, and so restore all the lines?

**Problem 9.23.** Find all quadrilaterals $ABCD$ such that all four triangles $DAB$, $CDA$, $BCD$, $ABC$ are similar to one another.

**Problem 9.24.** Find all positive integers $n$ and $k$ for which $n^2 + 7 = 2 \cdot 3^k$.

**Problem 9.25.** There are two irreducible fractions with denominators 600 and 700. Find the minimal possible value of the denominator of their sum.

**Problem 9.26.** Solve in integers: $x^2 + y^2 + z^2 = 2xyz$.

## Practice set 5

**Problem 9.27.** Prove that $40^{1963} + 1963^{40}$ is a composite number.

**Problem 9.28.** Let $n$ and $q$ be positive integers such that all prime divisors of $q$ are greater than $n$. Show that

$$(q-1)(q^2-1)\cdots(q^{n-1}-1) \equiv 0 \pmod{n!}.$$

**Problem 9.29.** Find all pairs of integers $(m, n)$ such that $\binom{n}{m} = 1984$.

**Problem 9.30.** Prove that for every positive integer $n$,

$$\frac{2n-1}{2} - \frac{2n-2}{3} + \cdots - \frac{2}{2n-1} + \frac{1}{2n} = \frac{1}{n+1} + \frac{3}{n+2} + \cdots + \frac{2n-1}{2n}.$$

**Problem 9.31.** Let $a, b, c$ be positive real numbers with $abc = 8$. Prove that

$$\frac{ab+4}{a+2} + \frac{bc+4}{b+2} + \frac{ca+4}{c+2} \geq 6.$$

**Problem 9.32.** A sequence of positive integers is given with $a_1 = 1$ and $a_{n+1} \leq 1 + a_1 + a_2 + \cdots + a_n$ for all $n$. Prove that every positive integer $N$ can be written as a sum of distinct terms of the sequence $\{a_n\}$.

## Practice set 6

**Problem 9.33.** In triangle $ABC$, $\angle C = 90^\circ$ and $D$ is the midpoint of $AB$. Points $E$, $F$ lie on $AC$, $BC$ respectively so that $DE \perp DF$. Prove that $EF^2 = AE^2 + BF^2$.

**Problem 9.34.** Segments $AB$ and $CD$ lie on the two sides of an angle with vertex $O$, with $A$ between $O$ and $B$, and $C$ between $O$ and $D$. The line joining the midpoints of $AD$ and $BC$ meets $AB$ at $M$ and $CD$ at $N$. Prove that $\dfrac{OM}{ON} = \dfrac{AB}{CD}$.

**Problem 9.35.** Solve the equation

$$\left\lfloor x^3 \right\rfloor + \left\lfloor x^2 \right\rfloor + \lfloor x \rfloor = \{x\} - 1,$$

where $\lfloor x \rfloor$ is the floor function and $\{x\}$ is the fractional part.

**Problem 9.36.** Calculate the sum

$$\left\lfloor \frac{1}{3} \right\rfloor + \left\lfloor \frac{2}{3} \right\rfloor + \left\lfloor \frac{2^2}{3} \right\rfloor + \left\lfloor \frac{2^3}{3} \right\rfloor + \cdots + \left\lfloor \frac{2^{2019}}{3} \right\rfloor.$$

**Problem 9.37.** Let triangle $ABC$ be given, and let $D$ be the midpoint of $BC$. Points $E, F$ lie on sides $AB, AC$ respectively so that $\angle EDF = 90^\circ$. Prove that $BE + CF > EF$.

**Problem 9.38.** Let triangle $ABC$ satisfy $\angle A : \angle B : \angle C = 1 : 2 : 4$. Prove that

$$\frac{1}{AB} + \frac{1}{AC} = \frac{1}{BC}.$$

## Practice set 7

**Problem 9.39.** A set of integers $\{a, b, c\}$ is, each minute, replaced by $\{a + b - c,\ b + c - a,\ c + a - b\}$. Originally we have $\{2000, 2002, 2003\}$. Is it possible that after some time we obtain $\{2001, 2002, 2003\}$?

**Problem 9.40.** The 4 corner cells of an $8 \times 8$ board are colored black and the other 60 cells white. At each step one may choose a row or column and invert all its colors. Is it possible, after some steps, to make all cells white?

**Problem 9.41.** Countries Alistan and Babastan have currencies ali and baba. In Alistan, 1 ali can be exchanged for 10 baba; in Babastan, 1 baba can be exchanged for 10 ali. Omar may travel between the two countries and starts with 1 ali. Prove that his number of alis will never equal his number of babas (only integer amounts occur).

**Problem 9.42.** There are 13 gray, 15 red and 17 blue chameleons on an island. Whenever two chameleons of different colors meet, they both change to the third color. Can all the chameleons on the island ever end up the same color?

**Problem 9.43.** The numbers 1 and 2 are written on a board. Every morning, Professor Ali erases the two numbers and writes their arithmetic mean and their harmonic mean. (a) At some

point $\dfrac{941664}{665857}$ was written on the board; determine whether it is the arithmetic or the harmonic mean, and find the other number written at that time. (b) Determine whether $\dfrac{35}{24}$ can ever appear on the board.

**Problem 9.44.** A rectangle is tiled by $2 \times 2$ and $1 \times 4$ bricks. Haider removes one $2 \times 2$ brick and replaces it with a $1 \times 4$ brick. Prove that he cannot re-tile the original rectangle.

## Practice set 8

**Problem 9.45.** A convex 10-gon is given, no three of whose diagonals are concurrent. At each vertex and at each intersection point of two diagonals, the number $+1$ is written. At each step, one may choose any diagonal or side and flip the sign of every number on it. Is it possible that after some steps all the numbers equal $-1$?

**Problem 9.46.** Several numbers, not all zero, are written on a board. At each step, one may choose two numbers $a$ and $b$ and replace them with $a - \frac{b}{2}$ and $b + \frac{a}{2}$. Prove that one can never reach a situation where all the numbers are zero.

**Problem 9.47.** A non-regular polygon is inscribed in a circle. At each step, one may choose a vertex $A$ that divides the arc between its two neighboring vertices into two unequal parts, and move $A$ to the midpoint of that arc. Is it possible that after 100 moves the polygon becomes congruent to the original one?

**Problem 9.48.** Five numbers are written on a board. Mohammed computes the sum of every pair and obtains the ten values $110, 112, 113, 114, 115, 116, 117, 118, 120, 121$. Find the five numbers.

**Problem 9.49.** Let $a, b, c$ be positive integers with $a^2 + b^2 = c^2$. Prove that $\frac{1}{2}(c-a)(c-b)$ is a perfect square.

**Problem 9.50.** The numbers $1, 2, 3, \ldots, 19, 20$ are written on a board. At each step, one may erase two numbers $a$ and $b$ and write $a + b - 1$. What number remains on the board after 19 steps?

**Problem 9.51.** Let 100 points be drawn in the plane such that the distance between any two of them is less than 1, and such that no three of them form an acute triangle. Prove that there exists a circle of radius 0.5 containing all 100 points.

## Practice set 9

**Problem 9.52.** Find the minimum possible value of

$$|a-1| + |b-2| + |c-3| + |3a+2b+c|\,.$$

**Problem 9.53.** Let $s$ be the number of divisors of a positive integer $n$. Express the product of all the divisors of $n$ in terms of $s$ and $n$.

**Problem 9.54.** Find the smallest positive integer $n$ such that $2019!$ is not divisible by $n^n$.

**Problem 9.55.** Find the maximum possible value of

$$\big|\cdots\big|\,|x_1 - x_2| - x_3\big| - x_4\big| - \cdots - x_{1990}\big|,$$

where $x_1, x_2, \ldots, x_{1990}$ is a permutation of $1, 2, \ldots, 1990$.

**Problem 9.56.** The numbers $12, 1, 10, 6, 8, 3$ are written, in this order, on the vertices of a regular hexagon. At each step, one may choose a side of the hexagon and either increase both its endpoint values by 1, or decrease both by 1. Is it possible to reach the configuration (a) $14, 6, 13, 4, 5, 2$? (b) $6, 17, 14, 3, 15, 3$?

**Problem 9.57.** There are $n$ regions in the plane, each colored red or blue. At each step, one may choose a region $X$ most of whose neighboring regions have a color different from $X$'s, and repaint $X$ the opposite color. Prove that this process must terminate after finitely many steps.

## Practice set 10

**Problem 9.58.** Let $S$ be the set of numbers $n^2 + n + 1$ for positive integers $n$. Prove that there exist infinitely many pairwise different $a, b, c, d \in S$ such that $\dfrac{a}{b} = \dfrac{c}{d} \in S$.

**Problem 9.59.** Find the smallest $n$ such that there is a table with 2 rows and $n$ columns, each cell containing one of 1, 50, 100, such that the two row sums are equal but the two row sums-of-squares are different.

**Problem 9.60.** For coprime positive integers $x, y$, find all possible integer values of

$$\frac{x+1}{y} + \frac{y+1}{x}.$$

**Problem 9.61.** For a positive integer $n$, list its divisors in increasing order $1 = d_1 < d_2 < \cdots < d_k = n$. Find the smallest positive integer $n$ such that $n = d_{19} \cdot d_{20}$.

**Problem 9.62.** Let

$$A = \frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n}.$$

Prove that $2^k A$ is an integer for some positive integer $k$.

## Practice set 11

**Problem 9.63.** Let integers $x$ and $y$ satisfy $5x + 7y = 111$. Prove that $x + y$ is odd.[1]

**Problem 9.64.** Is it possible to put signs $+$ and $-$ instead of the $*$'s to get a correct expression

$$1 * 2 * 3 * 4 * 5 * 6 * 7 * 8 * 9 * 10 = 0?$$

**Problem 9.65.** Find the number of 3-digit positive integers such that all of their digits are even.

[1]The source states this problem with "even" in place of "odd"; its own worked solution proves $x + y$ is odd. Indeed, the general solution is $x = 4 + 7t$, $y = 13 - 5t$, so $x + y = 17 + 2t$, which is always odd. The statement above gives the corrected, true claim.

**Problem 9.66.** During a contest, 10 students together solved 35 problems. It is known that some student solved exactly 1 problem, another student solved exactly 2 problems, and another student solved exactly 3 problems. Prove that some student solved at least 5 problems.

**Problem 9.67.** Recover the missing digits:

$$1 * \cdot * 1 = 1 * *1.$$

**Problem 9.68.** Which number is bigger: $(n-1)! \cdot (n+1)$ or $n! \cdot n$?

**Problem 9.69.** Let $AA'$, $BB'$, $CC'$ be the altitudes of triangle $ABC$. Let $A_1$ and $A_2$ be the projections of $A'$ on $AB$ and $AC$, respectively; $B_1$ and $B_2$ the projections of $B'$ on $BC$ and $BA$; and $C_1$ and $C_2$ the projections of $C'$ on $CA$ and $CB$. Prove that $B_2C_1 \parallel BC$, and that the hexagon $A_1B_2C_1A_2B_1C_2$ is cyclic.

## Practice set 12

**Problem 9.70.** Let $ABCD$ be a cyclic quadrilateral. Let the extensions of $BA$ and $CD$ meet at $X$, and the extensions of $AD$ and $BC$ meet at $Y$. Let the bisector of $\angle X$ meet $AD$, $BC$ at $E$, $F$, and let the bisector of $\angle Y$ meet $AB$, $CD$ at $G$, $H$. Prove that $EGFH$ is a rhombus.

**Problem 9.71.** Find all positive integers $n$ such that the product of its digits equals $n^2-10n-22$.

**Problem 9.72.** Prove that there exist infinitely many positive integers divisible by 2021, each containing an equal number of each digit $0, 1, \ldots, 9$.

**Problem 9.73.** Find all values of $a$ for which the equation $x^3 + ax^2 + 56x - 4 = 0$ has three roots forming consecutive terms of a geometric progression.

**Problem 9.74.** Let $f(x) = \dfrac{9^x}{9^x + 3}$. Evaluate the sum

$$\sum_{k=0}^{2021} f\Big(\frac{k}{2021}\Big).$$

## Practice set 13

**Problem 9.75.** A grid of size $8 \times 8$ is cut by a straight line. Find the maximal possible number of unit cells that are cut by the line.

**Problem 9.76.** In the cells of a $10 \times 10$ grid are written positive integers, all less than 11. It is known that the sum of the two numbers in any two cells sharing a common vertex is prime. Prove that some number is written in at least 17 cells.

**Problem 9.77.** Let $a, b, c, d$ be real numbers such that $a^4 + b^4 + c^4 + d^4 = 16$. Prove that $a^5 + b^5 + c^5 + d^5 \le 32$.

**Problem 9.78.** Find all pairs of positive integers $(x, y)$ such that

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{\operatorname{lcm}(x,y)} + \frac{1}{\gcd(x,y)} = \frac{1}{2}.$$

**Problem 9.79.** Ali has chosen 8 cells of a $8 \times 8$ chessboard, no two sharing a row or column. On each move, Baba chooses 8 cells, no two sharing a row or column, and puts coins on them; Ali then reveals how many of Baba's coins are not on Ali's chosen cells. If that number is even, Baba wins; otherwise Baba removes all coins and moves again. Find the minimal number of moves that guarantees Baba a win.

**Problem 9.80.** Let $1 \leq r \leq n$. Consider all $r$-element subsets of $\{1, 2, \ldots, n\}$; each has a minimum element. Prove that the average of these minima is $\dfrac{n+1}{r+1}$.

## Practice set 14

**Problem 9.81.** Let $a$ and $b$ be divisors of $n$ with $a > b$. Prove that

$$a > b + \frac{b^2}{n}.$$

**Problem 9.82.** Do there exist real numbers $a, b, c$ such that

$$|a| < |b - c|, \quad |b| < |c - a|, \quad |c| < |a - b|$$

hold simultaneously?

**Problem 9.83.** Let $x, y, z \geq 0$ with $x + y + z = 3$. Prove that

$$\sqrt{x} + \sqrt{y} + \sqrt{z} \geq xy + yz + zx.$$

**Problem 9.84.** Let $a, b, c > 0$. Prove that

$$\frac{a+b}{a^2+b^2} + \frac{b+c}{b^2+c^2} + \frac{c+a}{c^2+a^2} \leq \frac{1}{a} + \frac{1}{b} + \frac{1}{c}.$$

**Problem 9.85.** Let $P$ be a point inside $\triangle ABC$ such that $\angle PBA = \angle PCA$. Let $PD \perp AB$ at $D$ and $PE \perp AC$ at $E$. Show that the perpendicular bisector of $DE$ passes through the midpoint of $BC$.

## Practice set 15

**Problem 9.86.** Let the sequence $(x_n)$ satisfy $0 < x_1 < 1$ and $x_{k+1} = x_k - x_k^2$ for all $k \geq 1$. Prove that for all $n$,

$$x_1^2 + x_2^2 + \cdots + x_n^2 < 1.$$

**Problem 9.87.** Prove that

$$\sqrt{a+1} + \sqrt{2a-3} + \sqrt{50-3a} \leq 12.$$

**Problem 9.88.** The parabola $y = x^2 + px + q$ meets the coordinate axes in three distinct points. Consider the circumcircle of the triangle formed by these three points. Prove that this circle always passes through a fixed point, regardless of $p$ and $q$, and find that point.

**Problem 9.89.** Let $I$ be the incenter of $\triangle ABC$. Let $AI$ extended meet the circumcircle of $\triangle ABC$ at $P$. Let $ID \perp BP$ at $D$ and $IE \perp CP$ at $E$. Show that $ID + IE = AP \sin \angle BAC$.

**Problem 9.90.** Find all pairs of positive integers $a, b$ such that both

$$\frac{a^2+b}{b^2-a} \quad \text{and} \quad \frac{b^2+a}{a^2-b}$$

are integers.

**Problem 9.91.** Let $ABCD$ be a quadrilateral with $AB = AD$ and $BC \neq CD$. Let $CA$ bisect $\angle BCD$. Prove that $A, B, C, D$ are concyclic.

## Practice set 16

**Problem 9.92.** A real number $a$ is such that $a^2 + a$ and $a^3 + a$ are both rational. Prove that $a$ is rational.

**Problem 9.93.** Let $P(x)$ and $Q(x)$ be monic polynomials of degree 10 with real coefficients such that the equation $P(x) = Q(x)$ has no real solution. Prove that the equation $P(x+1) = Q(x-1)$ has a real solution.

**Problem 9.94.** Find all positive integers $n$ for which $n^5 + n^4 + 1$ is prime.

**Problem 9.95.** Let $n$ be a positive integer. Prove that there exist positive integers $a$ and $b$ such that

$$a^2 + a + 1 = (n^2 + n + 1)(b^2 + b + 1).$$

**Problem 9.96.** At a conference, it happens that any two scientists with the same number of friends have no common friend. Prove that some scientist has exactly 1 friend.

**Problem 9.97.** Let $\triangle ABC$ be a non-isosceles acute triangle with circumcenter $O$. Let $P$ be a point on ray $AO$ extended such that $\angle BPA = \angle CPA$. Let $PQ \perp AB$ at $Q$, $PR \perp AC$ at $R$, and $AD \perp BC$ at $D$. Show that $PQDR$ is a parallelogram.

**Problem 9.98.** Prove that the sum of the distances from any point of the base of an isosceles triangle to its two legs equals the altitude drawn to either leg.

**Problem 9.99.** In triangle $ABC$ the bisector of the angle at $B$ meets $AC$ at $D$ and the bisector of the angle at $C$ meets $AB$ at $E$. These bisectors intersect at $O$, and $OD = OE$. Prove that either $\angle BAC = 60^\circ$ or triangle $ABC$ is isosceles.

## Practice set 17

**Problem 9.100.** Let $a$, $b$, $c$ be solutions of the equation $x^3 + x^2 - 3x - 1 = 0$. Construct an equation whose roots are $a + 1$, $b + 1$, $c + 1$.

**Problem 9.101.** Let $ABCD$ be a quadrangle with $AD = BC$ and $\angle A + \angle B = 120^\circ$, and let $P$ be a point exterior to the quadrangle such that $P$ and $A$ lie on opposite sides of line $DC$ and triangle $DPC$ is equilateral. Prove that triangle $APB$ is also equilateral.

**Problem 9.102.** Does there exist a numeral system (for example, we normally use base 10) in which (a) $3+4=10$ and $3\cdot 4=15$; (b) $2+3=5$ and $2\cdot 3=11$?

**Problem 9.103.** Let $x_1, x_2, \ldots, x_n$ be positive integers. Assume that in their decimal representations no $x_i$ "prolongs" $x_j$ (for instance, 123 prolongs 12, and 459 prolongs 4, but 124 does not prolong 123). Prove that

$$\frac{1}{x_1}+\frac{1}{x_2}+\cdots+\frac{1}{x_n}<3.$$

**Problem 9.104.** Prove that for any polynomial $f$ with integer coefficients and any integer $n$, we have $f(n) \mid f(n+f(n))$.

**Problem 9.105.** Find all triples of positive integers $a, b, c$ such that the sum of the squares of any two of them is divisible by the square of the third one.

## Practice set 18

**Problem 9.106.** Let $x_1$ and $x_2$ be the roots of $x^2+ax-5a+1=0$. Find $a$ for which the value of $x_1^2+x_2^2$ is the smallest possible.[2]

**Problem 9.107.** The sum of the first 10 terms of an arithmetic progression $a_i$ equals 140, and $a_2a_9=147$. Find $a_1$.

**Problem 9.108.** Let 52 points be chosen inside a square of side length 10. Prove that one can choose 2 of them at distance at most $\sqrt{5}$.

**Problem 9.109.** Let $m$ and $n$ be coprime positive integers. Prove that $\gcd(5^m+7^m,\ 5^n+7^n) = 2$ or 12.

**Problem 9.110.** Is it possible to cover a $6\times 6$ square by 18 dominoes and then draw one diagonal in each domino, so that no two of the diagonals share a point?

**Problem 9.111.** Let the odd prime $p$ be given. Prove that there exist infinitely many positive integers $n$ for which $n\cdot 2^n+1$ is divisible by $p$.

**Problem 9.112.** Solve the equation

$$\{(x+1)^3\}=x^3,$$

where $\{\cdot\}$ denotes the fractional part.

**Problem 9.113.** Let $n>4$ points be chosen in the plane, no three of them collinear. It is known that every four of the points form a convex quadrilateral. Prove that all $n$ points form a convex $n$-gon.[3]

[2]The source asks for the value of $a$ making $x_1^2+x_2^2$ *biggest*; since $x_1^2+x_2^2=(a-5)^2-27$ is unbounded above, that reading is ill-posed. Its own worked solution instead finds the minimum, which is well-posed (constrained by the discriminant); that corrected reading is printed here.

[3]The source's closing clause literally reads "form a convex quadrilateral," which cannot be correct once $n>4$. Its own worked solution restates the claim as "form a convex $n$-gon," the standard result (a finite point set is in convex position once every 4-element subset is), and that corrected reading is printed here.

# Part III

# Level 3

# Chapter 10

# Algebra

**Sources for this chapter:** Camp and online handouts by Marko Radovanović (March 2022, January 2023, March–April 2023, November 2025), David Zmiaikou (May 2025), Navid Safaei (November 2024), Lu (November 2022), Łukasz Bożyk (June 2022), Dušan Djukić (November 2021, February–March 2022), Nikola Petrović (2023), Arsenii "Senya" (2023), and the KAUST February 2022 algebra notes.

The national-olympiad track asks for more than a stock of inequalities. This chapter collects the four themes the camps return to at Level 3: algebraic manipulation strong enough to survive a long computation, polynomials as objects with roots and divisibility, sequences solved in closed form, and functions determined by a single equation. Each section is arranged from the routine to the demanding.

## 10.1 Identities, systems, and algebraic manipulation

Almost every problem below is solved by finding the right thing to name: a factorisation, an auxiliary variable, a symmetric function.

### 10.1.1 Identities and factorisation

**Problem 10.1.** Decompose each of the following expressions into a product of at least two non-constant terms of lower degree: (a) $x^n - y^n$; (b) $x^{2n+1} + y^{2n+1}$; (c) $x^4 + 4$; (d) $(x+y+z)^3 - x^3 - y^3 - z^3$; (e) $x^3 + y^3 + z^3 - 3xyz$; (f) $(x-y)^3 + (y-z)^3 + (z-x)^3$; (g) $a^{10} + a^5 + 1$; (h) $a^4(b-c) + b^4(c-a) + c^4(a-b)$; (i) $x^4 + x^3 + x^2 + x + 12$.

**Problem 10.2.** Decompose $x^8 + x^4 + 1$ into a product of four non-constant terms of lower degree.

**Problem 10.3.** Express each of the following as a product: (a) $x^4+x^2+1$; (b) $x^5+x^4+1$; (c) $x^3+y^3+z^3-3xyz$; (d) $(x+y+z)^3-x^3-y^3-z^3$; (e) $(x+1)^4+x^4+1$; (f) $(x+y+z)^5-x^5-y^5-z^5$.

**Problem 10.4.** Present the expression

$$(a_1^2 + \cdots + a_n^2)(b_1^2 + \cdots + b_n^2) - (a_1 b_1 + \cdots + a_n b_n)^2$$

as a sum of squares.

**Problem 10.5.** Prove that if two numbers $m$ and $n$ can each be presented as a sum of squares of two integers, then their product $mn$ can be as well.

**Problem 10.6.** Let $a$ be a real number such that $a + \frac{1}{a}$ is an integer. Prove that $a^n + \frac{1}{a^n}$ is an integer for all $n \in \mathbb{N}$.

**Problem 10.7.** Show that any odd integer is a difference of squares of two integers.

**Problem 10.8.** Show that a product of four consecutive integers plus 1 is a perfect square.

**Problem 10.9.** Prove that if $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{a+b+c}$, then $\frac{1}{a^n} + \frac{1}{b^n} + \frac{1}{c^n} = \frac{1}{a^n+b^n+c^n}$ for all odd $n \in \mathbb{N}$.

**Problem 10.10.** Prove that if $\dfrac{a}{b-c} + \dfrac{b}{c-a} + \dfrac{c}{a-b} = 0$, then $\dfrac{a}{(b-c)^2} + \dfrac{b}{(c-a)^2} + \dfrac{c}{(a-b)^2} = 0$.

**Problem 10.11.** Let $x, y, z$ be pairwise distinct integers. Prove that the number $(x-y)^5 + (y-z)^5 + (z-x)^5$ is divisible by $5(y-z)(z-x)(x-y)$.

**Problem 10.12.** Prove that for every $x > 0$,

$$\sqrt{x + \sqrt{x + \sqrt{x + \sqrt{x + \cdots}}}} = 1 + \cfrac{x}{1 + \cfrac{x}{1 + \cfrac{x}{1 + \cdots}}}.$$

**Problem 10.13.** Prove that every integer $n$ can be expressed in the form $\pm 1^2 \pm 2^2 \pm 3^2 \pm \cdots \pm k^2$ for some positive integer $k$ and some choice of signs.

**Problem 10.14.** Given is a finite sequence of real numbers whose sum is zero but not all of whose members are zero. Prove that there exists a permutation $(a_1, a_2, \ldots, a_n)$ of this sequence such that

$$a_1a_2 + a_2a_3 + a_3a_4 + \cdots + a_{n-1}a_n + a_na_1 < 0.$$

### 10.1.2 Equations and systems

**Problem 10.15.** Given numbers $s$ and $p$, do there exist real numbers with sum $s$ and product $p$? If so, how does one find them?

**Problem 10.16.** Solve the system

$$x^2 + y^2 = 1, \qquad 4xy(2y^2 - 1) = 1.$$

**Problem 10.17.** Solve the equation

$$(x+1)^{63} + (x+1)^{62}(x-1) + (x+1)^{61}(x-1)^2 + \cdots + (x-1)^{63} = 0.$$

**Problem 10.18.** Let $a+b+c = 14$, $a^2+b^2+c^2 = 84$ and $a^3+b^3+c^3 = 584$. Find $\max\{a, b, c\}$.

**Problem 10.19.** Solve the following system in the set of positive real numbers:

$$x = \frac{1}{y^2 + y - 1}, \qquad y = \frac{1}{z^2 + z - 1}, \qquad z = \frac{1}{x^2 + x - 1}.$$

**Problem 10.20.** Solve the system

$$4xy + 4(x^2 + y^2) + \frac{3}{(x+y)^2} = \frac{85}{3}, \qquad 2x + \frac{1}{x+y} = \frac{13}{3}.$$

†

**Problem 10.21.** Let $x, y \in \mathbb{R}$ be such that $x = y(3-y)^2$ and $y = x(3-x)^2$. Find all possible values of $x + y$.

**Problem 10.22.** Everywhere below, if something appears in a denominator we assume it is nonzero, and we do not assume that the numbers are positive.

(a) Suppose that $a^3 + b^3 + 1 = 3ab$. Find all possible values of $a + b$.
(b) Suppose that $\left(a + \sqrt{a^2+1}\right)\left(b + \sqrt{b^2+1}\right) = 1$. Find all possible values of $a + b$.
(c) Suppose that $a + \frac{1}{b} = b + \frac{1}{c} = c + \frac{1}{a}$, where $a \neq b \neq c \neq a$. Find all possible values of $abc$.
(d) Suppose that $x + y + z = 0$ and $\frac{x}{y} + \frac{y}{z} + \frac{z}{x} = \frac{x}{z} + \frac{y}{x} + \frac{z}{y} + 1$. Find all possible values of both sides of this equality.
(e) Suppose that $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = 1$. Find all possible values of $\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b}$.
(f) Suppose that $x^2 - yz = y^2 - zx = z^2 - xy = 2$. Find all possible values of $xy + yz + zx$.
(g) Suppose that $x^2 - y = y^2 - z = z^2 - x$ and $x \neq y \neq z \neq x$. Find all possible values of $(x+y)(y+z)(z+x)$.
(h) Suppose that $a^2 + a = b^2$, $b^2 + b = c^2$, $c^2 + c = a^2$. Find all possible values of $(a-b)(b-c)(c-a)$.

**Problem 10.23.** Find all triples $(a, b, c)$ of real numbers such that the following system holds:

$$a + b + c = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}, \qquad a^2 + b^2 + c^2 = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}.$$

[JBMO 2020]

**Problem 10.24.** Positive numbers $a, b, c, d$ satisfy $abcd = 4$ and $a^2 + b^2 + c^2 + d^2 = 10$. Find the greatest possible value of $ab + bc + cd + da$.

**Problem 10.25.** On the board there are four positive integers. It is known that they are equal to

$$a + b + c, \qquad a + bc, \qquad b + ca, \qquad c + ab$$

for some positive integers $2 \leq a \leq b \leq c$, but it is not given which value corresponds to which expression. Can one uniquely determine the values of $a$, $b$, $c$ given only the numbers on the board?

**Problem 10.26.** Positive real numbers $a, b, c, d$ satisfy

$$a^2 + d^2 - ad = b^2 + c^2 + bc \qquad \text{and} \qquad a^2 + b^2 = c^2 + d^2.$$

Find all possible values of the expression $\dfrac{ab + cd}{ad + bc}$.

**Problem 10.27.** Prove that the following system has no solution in rational numbers:

$$xy + z = 3, \qquad xz + y = 1, \qquad yz + x = 0.$$

**Problem 10.28.** Let $(a, b, c)$ be any solution of the system

$$a^3 = 3b^2 + 3c^2 - 25, \qquad b^3 = 3c^2 + 3a^2 - 25, \qquad c^3 = 3a^2 + 3b^2 - 25.$$

Find the value of $abc$.

**Problem 10.29.** Let $0 < x, y, z < 1$ and

$$\left(x^2 - x + \tfrac{1}{2}\right)\left(y^2 - y + \tfrac{1}{2}\right)\left(z^2 - z + \tfrac{1}{2}\right) = (x - yz)(y - zx)(z - xy).$$

Find all such triples.

**Problem 10.30.** Let $x, y, z$ be positive numbers such that

$$\frac{x^2}{y} + \frac{y^2}{z} + \frac{z^2}{x} = \frac{x^2}{z} + \frac{z^2}{y} + \frac{y^2}{x}.$$

Prove that at least two of the numbers $x, y, z$ are equal to each other.

**Problem 10.31.** There are real numbers $a, b, c$ such that $a + b + c = 2$ and $a^2 + b^2 + c^2 = 2$. Prove that there are two numbers among them that differ by at least 1.

**Problem 10.32.** A real number $\alpha$ is given. Find all $\alpha$ for which the system

$$\frac{a^3}{b + c + \alpha} = \frac{b^3}{a + c + \alpha} = \frac{c^3}{a + b + \alpha}$$

has a solution consisting of three distinct numbers of $[-1, 1]$.

**Problem 10.33.** A real number $\alpha$ is given. Retaj and Lana call a pair of real numbers $x, y$ *friendly* if

$$\frac{xy}{x + y + 1} + x^2 + y^2 = \alpha.$$

[†]Retaj considered six distinct real numbers and told Lana that exactly 5 of the 15 pairs of these numbers are friendly. Prove that Lana can determine at least one of the numbers chosen by Retaj.

**Problem 10.34.** Does the system of equations

$$x_1 + x_2 = y_1 + y_2 + y_3 + y_4, \quad x_1^2 + x_2^2 = y_1^2 + y_2^2 + y_3^2 + y_4^2, \quad x_1^3 + x_2^3 = y_1^3 + y_2^3 + y_3^3 + y_4^3$$

admit a solution in integers such that the absolute value of each of these integers is greater than 2024?

**Problem 10.35.** There are $k$ real numbers, each between 100 and 200. Prove that there exist two of them, say $a$ and $b$ with $a > b$, such that

$$\frac{ab}{a - b} > 200 \cdot (k - 1).$$

**Problem 10.36.** Find the smallest real number $C$ such that for any positive integers $x \neq y$,

$$\min\left(\left\{\sqrt{x^2 + 2y}\right\}, \left\{\sqrt{y^2 + 2x}\right\}\right) < C,$$

where $\{t\}$ denotes the fractional part of $t$.

**Problem 10.37.** Point $P$ lies inside an equilateral triangle $ABC$. Lines $PA$, $PB$, $PC$ intersect the sides $BC$, $CA$, $AB$ at points $D$, $E$, $F$ respectively. Is it possible that exactly three among the six triangles $AFP$, $BFP$, $BDP$, $CDP$, $CEP$, $AEP$ have equal areas?

### 10.1.3 Algebra meets number theory

**Problem 10.38.** Prove that for all $n \in \mathbb{N}$ the number $n^2 + 3n + 5$ is not divisible by 121.

**Problem 10.39.** Prove that the expression

$$x^5 + 3x^4y - 5x^3y^2 - 15x^2y^3 + 4xy^4 + 12y^5$$

is not equal to 33 for any integers $x$ and $y$.

**Problem 10.40.** Prove that for all integers $n \geq 2$, the number $2^{4n+2} + 1$ is not a product of two primes.

**Problem 10.41.** Prove that the numbers $2^{32001} - 1$ and $2^{32001} + 1$ are composite.

**Problem 10.42.** Let the number $a^n + 1$ be prime, where $a, n \in \mathbb{N}$. Prove that $n$ is a power of 2.

**Problem 10.43.** Prove that every rational number can be presented as a sum of three cubes of rational numbers.

**Problem 10.44.** Solve in integers the equations (a) $x^2 = y^2 + 2y + 13$; (b) $xy + x - 3y = 4$; (c) $x^3 - y^3 = 91$.

**Problem 10.45.** Prove that $1^{1987} + 2^{1987} + \cdots + n^{1987}$ is not divisible by $n + 2$.

**Problem 10.46.** Find all triples $(a, b, c)$ of natural numbers such that $a^2 + b^2 - 33c^2 = 8bc$ and $a$ is a prime number.

**Problem 10.47.** Let $a, b, c$ be natural numbers such that $a^3b^3 + b^3c^3 + c^3a^3 = abc(a^3 + b^3 + c^3)$. Prove that one of the numbers $ab$, $bc$, $ac$ is a square.

**Problem 10.48.** Find a pair of natural numbers $(a, b)$ such that the number $(a + b)^7 - a^7 - b^7$ is divisible by $7^7$ and the number $ab(a + b)$ is not divisible by 7.

**Problem 10.49.** Initially, the number 1 is written on a board. If $a$ is on the board, then you may write on the board any $b \geq 1$ such that $a + b + 1 \mid a^2 + b^2 + 1$. For example, in your first move you can write 4 on the board because $1 + 4 + 1 \mid 1^2 + 4^2 + 1$, and in your second move you can write 37 because $4 + 37 + 1 \mid 4^2 + 37^2 + 1$. Can every positive integer eventually appear on the board?

**Problem 10.50.** Fix $c \in \mathbb{Z}^+$ and consider the previous problem with the divisibility condition replaced by $a + b + c \mid a^2 + b^2 + c$. Can every positive integer eventually appear on the board?

**Problem 10.51.** (a) Do there exist positive real numbers $a, b, c, x$ such that $a^2 + b^2 = c^2$ and $(a+x)^2 + (b+x)^2 = (c+x)^2$? (b) Suppose that positive integers $a, b, c$ satisfy $a^2 + b^2 = c^2$. Prove that $\frac{1}{2}(c - a)(c - b)$ is the square of an integer.

**Problem 10.52.** Positive integers $a, b$ satisfy $\dfrac{a}{b} > \sqrt{2}$. Prove that $\dfrac{a}{b} - \dfrac{1}{2ab} > \sqrt{2}$.

**Problem 10.53.** Does there exist an integer $n > 1$ such that $9n + 16$ and $16n + 9$ are both perfect squares?

**Problem 10.54.** The Fibonacci numbers are defined by starting with two ones, every next term being the sum of the previous two: $1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, \ldots$ Calculate

$$0.1 + 0.01 + 0.002 + 0.0003 + 0.00005 + 0.000008 + 0.0000013 + 0.00000021 + \cdots,$$

where the $i$-th summand has $i$ decimal places and ends with the $i$-th Fibonacci number.

**Problem 10.55.** Prove that $2^{10} + 5^{12}$ is composite.

**Problem 10.56.** Positive integers $a, b, c$ satisfy $a^2 + ab - 1 = c^2$. Prove that $b \geq \sqrt{4c - 3}$.

**Problem 10.57.** Let $n$ be a positive integer. Prove that $\sqrt{100^n + 2}$ has at least $n$ consecutive zeros after the decimal point.

**Problem 10.58.** Each positive integer was given some colour. Suppose that for every pair of integers $a, b > 1$ the numbers $a + b$ and $ab$ have the same colour. Prove that all integers greater than 4 have the same colour.

**Problem 10.59.** Each positive integer was given some colour. Suppose that for every pair of integers $a, b > k$ the numbers $a + b$ and $ab$ have the same colour, where $k \geq 1$ is fixed. Prove that for some $K$ all integers greater than $K$ have the same colour.

**Problem 10.60.** Prove that there are infinitely many 5-tuples of positive integers $(a, b, c, d, e)$ such that $a < b < c < d < e$ and $a^2 + b^2 + c^2 = d^2 + e^2$.

**Problem 10.61.** Prove that there are infinitely many triples of positive integers $(a, b, c)$ such that $a^3 + b^3 + 2 = c^3$.

**Problem 10.62.** Find all triples $(a, b, c)$ of positive integers such that

$$\frac{a^2 + b}{b^2 + a} = c \qquad \text{and} \qquad \frac{a^2 + c}{c^2 + a} = b.$$

**Problem 10.63.** Find all triples $(a, b, c)$ of integers satisfying the system of equations $a + b = c$, $a^2 + b^3 = c^2$.

**Problem 10.64.** Do there exist prime numbers $p, q, r$ such that $(p^2 + p)(q^2 + q)(r^2 + r)$ is a perfect square?

**Problem 10.65.** Positive integers $a$ and $b$ are such that both $ab$ and $(a + 1)(b + 1)$ are perfect squares. Prove that there exists an integer $n \geq 2$ such that $(a + n)(b + n)$ is also a perfect square.

**Problem 10.66.** Suppose that $1 + 3^n + 5^n$ is a prime number. Prove that $n$ is divisible by 12.

**Problem 10.67.** Prove that if for an integer $n \geq 2$ the number $n \cdot 2^n + 1$ is prime, then the numbers $n + 1$ and $n + 2$ are both composite.

**Problem 10.68.** Do there exist integers $a, b, c$ such that each of the quadratics $ax^2 + bx + c$ and $(a + 1)x^2 + (b + 1)x + (c + 1)$ has two integer roots?

**Problem 10.69.** Determine all possible values of $\gcd(n^2 + 2, n^3 + 3)$, where $n$ is an integer.

**Problem 10.70.** Find all pairs $(n, p)$ of positive integers such that $p > n$, $p$ is a prime number and $n^2 + np + p^2$ is a perfect square.

## 10.2 Polynomials I: roots, coefficients, irreducibility

A polynomial carries two kinds of information at once: its roots and its coefficients. Vieta's formulas translate between them, and divisibility of polynomials behaves so much like divisibility of integers that Diophantine reflexes transfer directly.

**Theorem 10.1 (Eisenstein's irreducibility criterion).** *Let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ be a polynomial with integer coefficients. Suppose there is a prime $p$ with*

$$p \nmid a_n, \qquad p \mid a_{n-1}, a_{n-2}, \ldots, a_1, a_0, \qquad p^2 \nmid a_0.$$

*Then $f$ is irreducible: there do not exist non-constant polynomials $g, h$ with integer coefficients such that $f = gh$.*

**Remark.** Problem 10.93 below is the *generalized Eisenstein criterion*: the hypotheses are only required up to the index $m \le n$, and the conclusion weakens accordingly. Eisenstein's criterion itself is the case $m = n$.

**Problem 10.71.** We know that $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$. But what about the coefficients of polynomials of the form

$$P_n(x) := x(x+1)(x+2)\cdots(x+n-1)\,?$$

Denote by $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]$ the coefficient of $x^k$ in $P_n(x)$; for example $P_4(x) = x^4 + 6x^3 + 11x^2 + 6x$, so $\left[\begin{smallmatrix} 4 \\ 2 \end{smallmatrix}\right] = 11$. Prove that if $1 \le k \le n$, then

$$\begin{bmatrix} n+1 \\ k \end{bmatrix} = n \begin{bmatrix} n \\ k \end{bmatrix} + \begin{bmatrix} n \\ k-1 \end{bmatrix}.$$

**Problem 10.72.** Let $a_0, a_1, \ldots, a_{r-1}, b_0, b_1, \ldots, b_{s-1}$ be non-negative real numbers such that

$$(a_0 + a_1 x + \cdots + a_{r-1}x^{r-1} + x^r)(b_0 + b_1 x + \cdots + b_{s-1}x^{s-1} + x^s) = 1 + x + x^2 + \cdots + x^{r+s-1} + x^{r+s}.$$

Prove that $a_i, b_j \in \{0, 1\}$ for all $0 \le i \le r-1$ and $0 \le j \le s-1$.

**Problem 10.73.** The polynomial $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + 1$ with non-negative real coefficients has all roots real. Prove that $f(2) \ge 3^n$.

**Problem 10.74.** Find all $a \in \mathbb{R}$ such that for all $x \in \mathbb{R}$,

$$(x+1)\left(x^3 + ax^2 + x + (a-2)^2\right) \ge 0.$$

**Problem 10.75.** Let $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ be a polynomial of degree $n \ge 5$ with integer coefficients that has $n$ distinct integer roots $0, \alpha_2, \ldots, \alpha_n$. Find all integers $k$ such that $P(P(k)) = 0$.

**Problem 10.76.** Find all polynomials $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ with $a_n \neq 0$ and $n \ge 3$ such that (i) $(a_0, a_1, \ldots, a_n)$ is a permutation of the numbers $(0, 1, 2, \ldots, n)$, and (ii) all roots of $P(x)$ are real numbers.

**Problem 10.77.** Find all polynomials $f(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ with $a_0 \neq 0$ and integer coefficients whose roots are exactly $a_0, a_1, \ldots, a_{n-1}$.

**Problem 10.78.** Let $f(z) = z^n + a_1 z^{n-1} + \cdots + a_n$ be a polynomial with complex coefficients. Prove that all roots of this polynomial lie inside the circle $|z| = 1 + \max\{|a_1|, |a_2|, \ldots, |a_n|\}$.

**Problem 10.79.** Prove that the polynomial $P(x) = (x^4 + x^3 - 3x^2 + x + 1)^{2008}$ has at least two negative coefficients.

**Problem 10.80.** Let

$$P(x) = \sum_{k=1}^{n} a_k x^k \qquad \text{and} \qquad Q(x) = \sum_{k=1}^{n} \frac{a_k}{2^k - 1} x^k$$

be two polynomials with real coefficients. If 1 and $2^{n+1}$ are roots of $Q(x)$, prove that $P(x)$ has a positive real root less than $2^n$.

**Problem 10.81.** Let $f(x) = x^n - b_1 x^{n-1} - \cdots - b_n$ be a polynomial with real coefficients, where $b_i \geq 0$ and at least one of them is nonzero. Prove that $f(x)$ has exactly one positive root $R$, and that $A^A \leq R^B$, where $A = \sum_{i=1}^{n} b_i$ and $B = \sum_{i=1}^{n} i b_i$.

**Problem 10.82.** Let $P(x)$ be a polynomial with real coefficients such that $P(x) \geq 0$ for all $x \in \mathbb{R}$. Prove that there exist polynomials $A(x)$ and $B(x)$ with real coefficients such that $P(x) = A(x)^2 + B(x)^2$.

**Problem 10.83.** Let $P(x)$ be a polynomial with integer coefficients. Prove that there do not exist distinct integers $x_1, x_2, \ldots, x_n$ with $n \geq 3$ such that

$$P(x_1) = x_2, \quad P(x_2) = x_3, \quad \ldots, \quad P(x_n) = x_1.$$

**Problem 10.84.** For $m \in \mathbb{N}$ let $P_m(x) = x^{m-1} + x^{m-2} + \cdots + x + 1$. Prove that for every positive integer $a$ there exist a positive integer $n$ and polynomials $R(x)$ and $Q(x)$ with integer coefficients such that

$$P_n(x) = \big(1 + ax + x^2 R(x)\big) \cdot Q(x).$$

**Problem 10.85.** Let $P(x)$ and $Q(x)$ be polynomials with integer coefficients of degrees $p$ and $q$ respectively. If $P(x)$ divides $Q(x)$ and each coefficient of these polynomials is equal to 1 or 2020, prove that $p + 1$ divides $q + 1$.

**Problem 10.86.** Find all polynomials $P$ and $Q$ with real coefficients such that

$$\frac{P(x)}{Q(x)} - \frac{P(x+1)}{Q(x+1)} = \frac{1}{x(x+2)}$$

holds for infinitely many $x \in \mathbb{R}$.

**Problem 10.87.** Find all polynomials $P(x)$ with real coefficients such that $P(a) \in \mathbb{Q}$ for every $a \in \mathbb{Q}$, and $P(b) \in \mathbb{R} \setminus \mathbb{Q}$ for every $b \in \mathbb{R} \setminus \mathbb{Q}$.

**Problem 10.88.** Let $A$ be the set of all polynomials in the variables $x, y, z$ with integer coefficients, and let $B$ be the set of polynomials from $A$ that can be written in the form

$$(x + y + z)P(x, y, z) + (xy + yz + zx)Q(x, y, z) + xyzR(x, y, z)$$

for some polynomials $P, Q, R$ from $A$. Determine the smallest positive integer $n$ such that $x^i y^j z^k \in B$ for all non-negative integers $i, j, k$ with $i + j + k \geq n$. [IMO 2020 Shortlist A2]

**Problem 10.89.** Prove that for every polynomial $P(x)$ with real coefficients there exist polynomials $Q(x)$ and $R(x)$ with real coefficients such that

$$P(x) = Q(x^2) + R((x+1)^2).$$

**Problem 10.90.** Prove that there exists exactly one polynomial $P(x) \in \mathbb{R}[X]$ such that the polynomial $(x+y)^{1000} - P(x) - P(y)$ is divisible by $xy - x - y$.

**Problem 10.91.** Find all pairs of integers $(a, b)$ for which there exists a polynomial $P(x) \in \mathbb{Z}[X]$ such that the product $(x^2 + ax + b)P(x)$ is a polynomial each of whose coefficients is equal to 1 or $-1$.

**Problem 10.92.** Let $f(x) = c_{2m+1}x^{2m+1} + \cdots + c_1x + c_0$ be a polynomial with integer coefficients. Suppose there exists a prime number $p$ such that

$$p \nmid c_{2m+1}, \quad p \mid c_{2m}, c_{2m-1}, \ldots, c_{m+1}, \quad p^2 \mid c_m, c_{m-1}, \ldots, c_1, c_0, \quad p^3 \nmid c_0.$$

Prove that $f(x)$ is irreducible, i.e. that there do not exist non-constant polynomials $g(x)$ and $h(x)$ with $f(x) = g(x)h(x)$.

**Problem 10.93.** Let $f(x) = a_nx^n + \cdots + a_1x + a_0$ be a polynomial with integer coefficients such that there exist a number $m \leq n$ and a prime number $p$ satisfying

$$p \nmid a_m, \qquad p \mid a_{m-1}, a_{m-2}, \ldots, a_1, a_0, \qquad p^2 \nmid a_0.$$

If $g(x)$ and $h(x)$ are polynomials with integer coefficients such that $f(x) = g(x)h(x)$, prove that the degree of at least one of $g(x)$ and $h(x)$ is at least $m$.

**Problem 10.94.** Let $n > 1$ be a positive integer. Prove that the polynomial $P(x) = x^n + 5x^{n-1} + 3$ is irreducible.

**Problem 10.95.** Find all integers $k$ such that the polynomial

$$P(x) = x^{n+1} + kx^n - 870x^2 + 1945x + 1995$$

is reducible in $\mathbb{Z}[X]$ for infinitely many positive integers $n$.

**Problem 10.96.** Let $p > 5$ be a prime number. Find the number of polynomials of the form

$$x^p + px^k + px^l + 1, \qquad k > l,\ k, l \in \{1, 2, \ldots, p-1\},$$

which are irreducible in $\mathbb{Z}[X]$.

**Problem 10.97.** Find all polynomials $P(x) \in \mathbb{R}[X]$ such that $xP(x-1) = (x-3)P(x)$.

**Problem 10.98.** Find all polynomials $P(x) \in \mathbb{R}[X]$ such that $P(x)P(x+1) = P(x^2)$.

**Problem 10.99.** Find all polynomials $P(x) \in \mathbb{R}[X]$ such that $P(x)P(2x^2) = P(2x^3 + x)$.

**Problem 10.100.** Find all non-constant rational functions $f(x)$ (a rational function is a quotient of two polynomials) and all real numbers $a$ such that

$$f(x)^2 - a = f(x^2)$$

holds for infinitely many $x \in \mathbb{R}$.

**Problem 10.101.** Let $P(x)$, $Q(x)$, $R(x)$ be polynomials with integer coefficients such that $P(x) = Q(x)R(x)$. Denote by $a$ and $b$ the largest absolute values of the coefficients of $P$ and $Q$ respectively. Does $b \le 2023a$ always hold?

**Problem 10.102.** Let $a_1, a_2, \dots, a_{10}$ and $b_1, b_2, \dots, b_{10}$ be real numbers such that the ten polynomials

$$x^2 + a_1 x + b_1, \quad x^2 + a_2 x + b_2, \quad \dots, \quad x^2 + a_{10} x + b_{10}$$

have 20 distinct roots, given by the set $X = \{\pm 1, \pm 2, \dots, \pm 10\}$ (not necessarily in that order). (a) What is the maximum number of odd values among $a_1, b_1, a_2, b_2, \dots, a_{10}, b_{10}$? (b) Find the minimum and the maximum value of the sum $T = b_1 + b_2 + \cdots + b_{10}$.

## 10.3 Sequences and recurrences

This section discharges a promise made in Chapter 8: the closed-form theory of linear recurrences, and the problems that live on it.

**Definition 10.2 (Properties of a sequence).** A sequence $(u_n)_{n \in \mathbb{N}}$ is a family of real numbers indexed by $\mathbb{N}$ (or by another countable index set such as $\mathbb{N} \cup \{0\}$ or $\mathbb{Z}$). One says that $(u_n)$ is *increasing* (resp. *decreasing*) if $u_{n+1} > u_n$ (resp. $u_{n+1} < u_n$) for every $n$; *non-increasing* (resp. *non-decreasing*) if $u_{n+1} \le u_n$ (resp. $u_{n+1} \ge u_n$) for every $n$; *monotonic* if it is either increasing or decreasing; *periodic* if $u_{n+k} = u_n$ for all $n$ and some $k \in \mathbb{N}$ (such a $k$ is a *period*); *stationary* if there is $n_0$ with $u_n = u_{n_0}$ for all $n \ge n_0$; *bounded from above* (resp. *below*) if $u_n \le M$ (resp. $u_n \ge M$) for all $n$ and some $M \in \mathbb{R}$; and *bounded* if it is bounded from above and from below.

**Definition 10.3 (Arithmetic and geometric sequences).** A sequence $u_n = an + b$ (with $a, b \in \mathbb{R}$) is *arithmetic* with common difference $a$; a sequence $v_n = ba^n$ is *geometric* with common ratio $a$. For such sequences,

$$\sum_{n=1}^{N} u_n = a\,\frac{N(N+1)}{2} + bN, \qquad \sum_{n=1}^{N} v_n = ba\,\frac{1 - a^N}{1 - a} \quad (a \neq 1).$$

**Definition 10.4 (Linear recursive sequence, characteristic polynomial).** A *linear recursive sequence* $(u_n)$ of order $k$ is a sequence for which there exist real numbers $c_1, \dots, c_k$ with

$$u_{n+k} = c_1 u_{n+k-1} + c_2 u_{n+k-2} + \cdots + c_k u_n \qquad \text{for all } n.$$

Its *characteristic polynomial* is $P(X) = X^k - c_1 X^{k-1} - \cdots - c_{k-1} X - c_k$.

**Theorem 10.5 (Closed form of a linear recurrence).** *Let $(u_n)$ be a linear recursive sequence with characteristic polynomial $P$, and let $r_1, \dots, r_j$ be the distinct (possibly non-real) roots of $P$, with respective multiplicities $m_1, \dots, m_j$. Then there exist polynomials $Q_1, \dots, Q_j$ of degrees at most $m_1 - 1, \dots, m_j - 1$ such that*

$$u_n = \sum_{i=1}^{j} Q_i(n)\, r_i^n \qquad \textit{for all } n.$$

*Conversely every such combination satisfies the recurrence. The polynomials $Q_i$ are determined by the first $k$ terms $u_1, \ldots, u_k$.*

**Technique 10.6 (Working with a sequence).** When a sequence is given by a recurrence and the question is not a closed form, the standard moves are: find the explicit formula anyway; prove monotonicity or boundedness; look for periodicity (in $\mathbb{R}$, or modulo a fixed integer); apply plain algebra to the recurrence (write it for $n+1$ as well and subtract, so as to eliminate a constant or a radical); and define an auxiliary sequence — a ratio, a difference, or a product of consecutive terms — which satisfies a simpler, often linear, recurrence.

**Problem 10.103.** A rabbit attempts to climb a stairway of 20 steps. It can only jump one or two steps at once. In how many ways can it get upstairs (the rabbit always moves up)?

**Problem 10.104.** Find a closed form for the sequence $(a_n)_{n\geq 0}$ defined by: (a) $a_0 = 1$, $a_1 = 1$ and $a_{n+2} - 3a_{n+1} + a_n = 0$ for $n \geq 0$; (b) $a_0 = 1$, $a_1 = 1$ and $a_{n+2} - 4a_{n+1} + 4a_n = 0$ for $n \geq 0$; (c) $a_0 = 3$, $a_1 = 3$, $a_2 = 24$, $a_3 = 9$, $a_4 = 186$ and $a_{n+5} - 3a_{n+4} - 5a_{n+3} + 27a_{n+2} - 32a_{n+1} + 12a_n = 0$ for $n \geq 0$.

**Problem 10.105.** Find $a_n$ in closed form if: (a) $a_0 = 0$, $a_1 = 1$ and $a_n = 2a_{n-1} + a_{n-2}$ for $n \geq 2$; (b) $a_0 = a_1 = 0$, $a_2 = 1$ and $a_n = 3a_{n-2} - 2a_{n-3}$ for $n \geq 3$.

**Problem 10.106.** Let $(a_n)_{n\geq 0}$ be defined by $a_0 = a$, $a_1 = b$, $a_2 = 2b - a + 2$ and

$$a_{n+3} = 3a_{n+2} - 3a_{n+1} + a_n \qquad \text{for all } n \geq 0.$$

If $a_n$ is the square of an integer for every $n \geq 2025$, prove that $2 \mid b - a - 1$ and that $a$ is the square of an integer.

**Problem 10.107.** The Fibonacci sequence is defined by $F_0 = 0$, $F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$. Find $F_n$ in closed form.

**Problem 10.108.** Let $(F_n)$ be the Fibonacci sequence, with $F_0 = 0$ and $F_1 = 1$. Prove that $\sum_{k=0}^{n} F_k = F_{n+2} - 1$, that $F_{m+n} = F_n F_{m-1} + F_m F_{n+1}$, and that if $i$ divides $j$ then $F_i$ divides $F_j$, for all $m, n, i, j \geq 1$.

**Problem 10.109.** Prove that for the Fibonacci sequence $(f_n)$ given by $f_1 = f_2 = 1$ and $f_{n+2} = f_{n+1} + f_n$ there exists a unique triple $a, b, c \in \mathbb{N}$ with $b < a$, $c < a$, such that $f_n - nb\,c^n$ is divisible by $a$ for all $n \in \mathbb{N}$. †

**Problem 10.110.** Find $a_n$ if the sequence $(a_n)_{n\geq 1}$ is defined by $a_1 = 0$ and $a_{n+1} = \dfrac{a_n - 1}{a_n + 3}$ for $n \geq 1$.

**Problem 10.111.** Let $a_0 = 1$ and $a_{n+1} = \dfrac{1 + a_n}{3 + a_n}$. Find $a_n$ in closed form.

**Problem 10.112.** Find $x_{2020}$ if the sequence $(x_n)_{n\geq 0}$ is defined by $x_0 = a$ and $x_{n+1} = \dfrac{x_n + 4}{x_n + 1}$ for $n \geq 0$.

**Problem 10.113.** Define a sequence $(u_n)$ by $u_1 = 3$ and $u_{n+1} = \sqrt{12 + u_n}$ for all $n \geq 1$. Study the behaviour of $(u_n)$.

**Problem 10.114.** How many $n$-digit strings are there such that every digit is 0, 1 or 2 and the block "00" never appears?

**Problem 10.115.** How many $n$-digit strings are there such that every digit is 0, 1, 2 or 3 and the number of 0's is even?

**Problem 10.116.** The same question as in the previous problem, but with the additional requirement that the number of 1's is also even.

**Problem 10.117.** How many $n$-digit strings are there such that every digit is 0, 1, 2 or 3 and there is never a 0 next to a 1?

**Problem 10.118.** How many $n$-digit strings are there such that every digit is 0, 1 or 2 and the difference between two consecutive digits is at most 1?

**Problem 10.119.** Let $a > 0$, $a_0 = a$ and $a_{n+1} = a_n + \dfrac{1}{a_n}$. Is it possible that $(a_n)$ is bounded?

**Problem 10.120.** The sequence $(a_n)$ is defined by $a_1 = 0$ and

$$a_{n+1} = \frac{a_1 + a_2 + \cdots + a_n}{n} + 1.$$

Prove that $a_{2024} > \frac{1}{2} + a_{1000}$.

**Problem 10.121.** Let $a_1 = \frac{1}{2}$ and $a_{n+1} = \sqrt{a_n^2 + 2a_n}$ for $n = 1, 2, \ldots$. Prove that $a_n < \dfrac{2n^2}{2n+1}$ for all $n$.

**Problem 10.122.** Let $(a_n)$ be a sequence of real numbers such that $a_{n+1} = \sqrt{a_n^2 + a_n - 1}$ for $n = 1, 2, \ldots$. Prove that $a_1 \notin (-2, 1)$.

**Problem 10.123.** Let $a_0, a_1, \ldots, a_n$ be a sequence such that $a_0 = a_n = 0$ and, for all $1 \le k \le n-1$, one has $2a_k - a_{k+1} - a_{k-1} \le 0$. Prove that $a_k \le 0$ for all $0 \le k \le n$.

**Problem 10.124.** Does there exist an increasing sequence $(u_n)$ of positive integers such that $u_{nm} = u_n + u_m$ for all positive integers $n, m \ge 2$? What if one replaces "increasing" by "non-decreasing"?

**Problem 10.125.** The sequence $(w_n)$ is defined by $w_1 = 1$ and $w_{n+1} = \dfrac{w_n}{n} + \dfrac{n}{w_n}$ for all $n \ge 1$. Show that for $n \ge 4$ one has $\lfloor w_n^2 \rfloor = n$, where $\lfloor x \rfloor$ is the largest integer not exceeding $x$.

**Problem 10.126.** For each integer $b_0 > 1$, define the sequence $(b_n)$ by letting $b_{n+1} = \sqrt{b_n}$ if $b_n$ is a perfect square and $b_{n+1} = b_n + 3$ otherwise. Find all possible values of $b_0$ for which there exists an integer $B$ with $B = b_n$ for infinitely many $n$.

**Problem 10.127.** Prove that the first hundred decimals of the number $(5 + \sqrt[3]{3})^{200}$ are 9's. †

**Problem 10.128.** Let $a_0 \in \mathbb{R}^+$ and $a_{n+1} = \sqrt{1 + a_n}$ for all $n \ge 0$. Prove that at least one term of the sequence is irrational.

**Problem 10.129.** Let $(a_n)_{n \in \mathbb{N}}$ be such that for all $m, n \ge 1$, $|a_{n+m} - a_n - a_m| \le \dfrac{1}{m+n}$. Prove that $(a_n)$ is an arithmetic sequence.

**Problem 10.130.** Calculate the sum $a_1 + \cdots + a_{99}$, where $a_n = \dfrac{1}{(n+1)\sqrt{n} + n\sqrt{n+1}}$.

**Problem 10.131.** Show that there exist numbers $A$ and $B$ such that for all $n \in \mathbb{N}$ one has $a_1 + \cdots + a_n = A\tan(n) + Bn$, where $a_k = \tan(k)\tan(k-1)$.

**Problem 10.132.** Let $a_n = \dfrac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n}$. Find $\lim_{n\to\infty} a_n$.

**Problem 10.133.** In a sequence $(a_n)$, every term $a_n$ equals either $a_{n-1}/2$ or $\sqrt{a_{n-1}}$. Could such a sequence have a limit in the interval $(0, 1)$?

**Problem 10.134.** For a given positive integer $n \geq 3$, find the maximum possible number of increasing arithmetic sequences with three terms which can be chosen from a finite sequence having $n$ distinct terms.

**Problem 10.135.** In the sequence $1, 9, 8, 1, \ldots$ every next term $a_n$ is equal to the last digit of the sum of the four previous terms. Does this sequence contain the four consecutive terms $1, 2, 3, 4$?

**Problem 10.136.** A sequence of positive integers $a_1 < a_2 < a_3 < \cdots$ satisfies $a_1 = 1$ and $a_{n+1} \leq 2n$ for all $n \in \mathbb{N}$. Prove that for all $n \in \mathbb{N}$ there exist terms $a_p$ and $a_q$ such that $a_p - a_q = n$.

**Problem 10.137.** Let $(a_n)$ and $(b_n)$ be two sequences of positive integers. Show that there are two indices $p < q$ such that $a_p \leq a_q$ and $b_p \leq b_q$.

**Problem 10.138.** A sequence $(a_n)$ of positive real numbers satisfies $a_{2n} \leq a_n - a_{n+1}$ for all $n \in \mathbb{N}$. Prove that $a_n < \dfrac{1}{n}$ for all $n \in \mathbb{N}$. †

**Problem 10.139.** Consider a finite sequence $a_0, a_1, \ldots, a_n$ such that $a_0 = \frac{1}{2}$ and $a_k = a_{k-1} + \frac{1}{n}a_{k-1}^2$ for $k = 1, \ldots, n$. Prove that $1 - \frac{1}{n} < a_n < 1$.

**Problem 10.140.** A sequence $(a_n)$ of real numbers satisfies $|a_{k+m} - a_k - a_m| \leq 1$ for all $k, m \in \mathbb{N}$. Prove that for all $p, q \in \mathbb{N}$ one has $\left|\dfrac{a_p}{p} - \dfrac{a_q}{q}\right| < \dfrac{1}{p} + \dfrac{1}{q}$.

**Problem 10.141.** For a given $a_1 \in \mathbb{R}$, construct the sequence $(a_n)$ recursively by $a_{n+1} = \dfrac{a_n^2 - 1}{2a_n}$ if $a_n \neq 0$, and $a_{n+1} = 0$ if $a_n = 0$. Show that the sequence contains infinitely many non-positive terms.

**Problem 10.142.** Find all numbers $a_0 \in \mathbb{R}$ for which the sequence $a_0, a_1, \ldots$ defined recursively by $a_{n+1} = 2^n - 3a_n$ is increasing.

**Problem 10.143.** Consider a number $a \in \mathbb{R}$ and a sequence $(a_n)$ such that $a_1, a_2 \in \mathbb{Z}$, $\dfrac{a_1^2 + a_2^2 + a}{a_1 a_2} \in \mathbb{Z}$ and $a_{n+2} = \dfrac{a_{n+1}^2 + a}{a_n}$ for $n \in \mathbb{N}$. Prove that all terms of the sequence are integers.

**Problem 10.144.** Show that all terms of the sequence $a_n = ((2+\sqrt{3})^n - (2-\sqrt{3})^n)/(2\sqrt{3})$, where $n \in \mathbb{Z}$, are integers. Find all indices $n \in \mathbb{Z}$ for which $a_n$ is divisible by 3.

**Problem 10.145.** Show that each term of the sequence

$$a_n = \left(\frac{3+\sqrt{5}}{2}\right)^n + \left(\frac{3-\sqrt{5}}{2}\right)^n - 2, \qquad n \in \mathbb{N},$$

is an integer that can be presented as $5m^2$ or as $m^2$ (with $m \in \mathbb{N}$) according as $n$ is even or odd.

**Problem 10.146.** Let $a \in \mathbb{N}$ and let $a_0, a_1, \ldots$ be the sequence with $a_0 = 0$, $a_1 = 1$ and $a_{n+1} = 2a_n + (a-1)a_{n-1}$ for $n \in \mathbb{N}$. For a given prime number $p_0 > 2$, find the least possible value of $a$ such that both of the following hold: (a) for every prime $p \le p_0$, the term $a_p$ is divisible by $p$; (b) for every prime $p > p_0$, the term $a_p$ is not divisible by $p$.

**Problem 10.147.** Prove that there exists exactly one sequence of integers $a_1, a_2, \ldots$ such that $a_1 = 1$, $a_2 > 1$ and $a_{n+1}^3 + 1 = a_n a_{n+2}$ for $n \in \mathbb{N}$. †

**Problem 10.148.** For a given prime $p$, find the number of distinct sequences of positive integers $a_0, a_1, \ldots$ such that $\dfrac{a_0}{a_1} + \dfrac{a_0}{a_2} + \cdots + \dfrac{a_0}{a_n} + \dfrac{p}{a_{n+1}} = 1$ for all $n \in \mathbb{N}$. †

**Problem 10.149.** Let $(a_n)_{n\ge1}$ be a sequence satisfying the recurrence relation $a_{n+2} + pa_{n+1} + qa_n = 0$. Prove that each of the sequences (a) $\{a_{n+1}^2 + pa_n a_{n+1} + qa_n^2\}_{n\ge1}$ and (b) $\{a_{n+1}^2 - a_n a_{n+2}\}_{n\ge1}$ is a geometric progression with quotient $q$.

**Problem 10.150.** Let $(a_n)_{n\ge1}$ be the sequence defined by $a_1 = 20$, $a_2 = 30$ and $a_{n+2} = 3a_{n+1} - a_n$ for $n \ge 1$. Find all $n$ such that $1 + 5a_n a_{n+1}$ is a perfect square.

**Problem 10.151.** Let $(x_n)_{n\ge1}$ be defined by $x_1 = 1$, $x_2 = 4$ and $x_{n+2} = 4x_{n+1} - x_n$ for $n \ge 1$. Find all positive integers $m$ such that $3x_n^2 + m$ is the square of an integer for every $n \ge 1$.

**Problem 10.152.** Prove that for every $n \in \mathbb{N}$ the sum

$$S_n = \binom{2n+1}{0}2^{2n} + \binom{2n+1}{2}2^{2n-2}\cdot 3 + \cdots + \binom{2n+1}{2n}3^n$$

can be written as the sum of two consecutive perfect squares.

**Problem 10.153.** Prove that $10^n \mid \left\lfloor (5+\sqrt{35})^{2n-1} \right\rfloor$ for every $n \in \mathbb{N}$.

**Problem 10.154.** Let $(a_n)$ be a sequence of positive integers with $a_1 = 2$, $a_2 = 7$ and

$$-\frac{1}{2} < a_{n+1} - \frac{a_n^2}{a_{n-1}} \le \frac{1}{2} \qquad \text{for } n \ge 2.$$

Prove that $a_n$ is odd for all $n \ge 2$.

**Problem 10.155.** Let $(a_n)_{n\ge1}$ be defined by $a_1 = 3$, $a_2 = 11$ and $a_n = 4a_{n-1} - a_{n-2}$ for $n \ge 3$. Prove that for every $n \in \mathbb{N}$ the number $a_n$ can be written in the form $a^2 + 2b^2$ for some $a, b \in \mathbb{N}$.

**Problem 10.156.** Let $P, Q, R \in \mathbb{C}[X]$ be nonzero polynomials and let $a, b, c$ be distinct nonzero complex numbers. Put $z_n = P(n)a^n + Q(n)b^n + R(n)c^n$ for $n \in \mathbb{N}$. If the set $S = \{z_n \mid n \in \mathbb{N}\}$ is finite, prove that there exists $p \in \mathbb{N}$ such that $z_{n+p} = z_n$ for all $n \in \mathbb{N}$.

**Problem 10.157.** Let $(x_n)_{n\ge0}$ be a sequence such that $x_{n+2} = \dfrac{x_{n+1}^2 + c}{x_n}$ for $n \ge 0$. If $x_0$, $x_1$ and $\dfrac{x_0^2 + x_1^2 + c}{x_0 x_1}$ are positive integers, prove that $x_n$ is an integer for all $n \ge 0$.

**Problem 10.158.** The sequence $(x_n)_{n\geq 0}$ is defined by $x_0 = 1$ and $x_{n+1} = 2x_n + \sqrt{3x_n^2 + 1}$ for $n \geq 0$. Prove that $x_n \in \mathbb{N}$ for every $n \in \mathbb{N}$.

**Problem 10.159.** Define $a_0 = 0$ and $a_{n+1} = 3a_n + \sqrt{8a_n^2 + 1}$. Prove that each term $a_n$ is an integer.

**Problem 10.160.** Let $(a_n)_{n\geq 1}$ be defined by $a_1 = a_2 = a_3 = 1$ and $a_{n+3} = \dfrac{1 + a_{n+2}a_{n+1}}{a_n}$ for $n \geq 1$. Prove that $a_n \in \mathbb{N}$ for every $n \in \mathbb{N}$.

**Problem 10.161.** Let $c \in \mathbb{N}$ and let $(a_n)_{n\geq 1}$ be the sequence defined by $a_1 = c$ and $a_{n+1} = a_n^2 + a_n + c^3$ for $n \in \mathbb{N}$. Find all $c$ for which there exist integers $k \geq 1$ and $m \geq 2$ such that $a_k^2 + c^3$ is a perfect $m$-th power.

**Problem 10.162.** The sequence $(x_n)_{n\geq 1}$ is defined by $x_1 = 5$ and $x_{n+1} = x_n^2 - 3x_n + 3$ for $n \geq 1$. Prove that $x_n > 3^{2^{n-1}}$ for all $n \in \mathbb{N}$.

**Problem 10.163.** Let $(a_n)_{n\geq 1}$ be the sequence defined by $a_1 = 1$, $a_2 = 2$, $a_3 = 24$ and

$$a_n = \frac{6a_{n-1}^2 a_{n-3} - 8a_{n-1}a_{n-2}^2}{a_{n-2}a_{n-3}} \qquad \text{for } n \geq 4.$$

Prove that $n! \mid a_n$ for all $n \in \mathbb{N}$.

**Problem 10.164.** Let $(a_n)_{n\geq 0}$ be a sequence of positive integers such that

$$(a_{k-1} + a_k)(a_k + a_{k+1}) = a_{k-1} - a_{k+1} \qquad \text{for } k \geq 1.$$

Prove that $a_n < \dfrac{1}{n-1}$ for all $n \geq 2$.

**Problem 10.165.** The sequences $(a_n)$ and $(b_n)$ are defined by $a_1 = 1$, $b_1 = 2$ and

$$a_{n+1} = \frac{1 + a_n + a_n b_n}{b_n}, \qquad b_{n+1} = \frac{1 + b_n + a_n b_n}{a_n} \qquad \text{for } n \geq 1.$$

Prove that $a_{2022} < 5$.

**Problem 10.166.** The sequence $(a_n)$ is defined by $a_2 = 1$, $a_3 = 1$ and

$$(n+1)(n-2)a_{n+1} = n(n^2 - n - 1)a_n - (n-1)^3 a_{n-1} \qquad \text{for } n \geq 3.$$

Prove that $a_n$ is an integer if and only if $n$ is a prime number.

**Problem 10.167.** Let $(a_n)_{n\geq 1}$ be the sequence of positive real numbers defined by $a_1 = 2$, $a_2 = 9$, $a_3 = 2023$ and

$$a_{n+3} = a_n - \frac{1}{a_{n+1}a_{n+2}} \text{ if } a_{n+1}a_{n+2} \neq 0, \qquad a_{n+3} = 0 \text{ otherwise.}$$

Prove that there is some $n \in \mathbb{N}$ with $a_n = 0$, and find the smallest such $n$.

**Problem 10.168.** Find all sequences $(a_n)_{n\geq 1}$ of positive integers satisfying the equality $a_n = a_{a_{n-1}} + a_{a_{n+1}}$ (a) for all $n \geq 2$; (b) for all $n \geq 3$.

**Problem 10.169.** Let $a_1 = 43$, $a_2 = 142$ and $a_{n+1} = 3a_n + a_{n-1}$ for all $n \geq 2$. Prove that for every positive integer $m$ there are infinitely many positive integers $n$ such that $a_n - 1$ and $a_{n+1} - 1$ are both divisible by $m$.

**Problem 10.170.** A sequence $(a_n)$ is defined by $a_{n+2} = a_{n+1} + \sqrt{a_n + a_{n+1}}$. Is it possible to choose $a_1, a_2$ in such a way that the first ten terms of the sequence are positive integers?

**Problem 10.171.** In a sequence $a_1, a_2, \ldots$ of real numbers the product $a_1 a_2$ is negative, and to define $a_n$ for $n > 2$ one pair $(i, j)$ is chosen among all pairs $(i, j)$ with $1 \leq i < j < n$ not chosen before, so that $a_i + a_j$ has minimum absolute value; then $a_n$ is set equal to $a_i + a_j$. Prove that $|a_i| < 1$ for some $i$.

**Problem 10.172.** There is an infinite sequence $(x_n)$ whose first term $x_1$ is a rational number larger than 1, and $x_{n+1} = x_n + \dfrac{1}{\lfloor x_n \rfloor}$. Prove that there must be an integer in this sequence.

**Problem 10.173.** Define the function $f : (0,1) \to (0,1)$ by

$$f(x) = \begin{cases} x + \frac{1}{2}, & \text{if } x < \frac{1}{2}, \\ x^2, & \text{if } x \geq \frac{1}{2}. \end{cases}$$

Let $a, b \in \mathbb{R}$ with $0 < a < b < 1$. Define sequences $a_n$ and $b_n$ by $a_0 = a$, $b_0 = b$ and $a_n = f(a_{n-1})$, $b_n = f(b_{n-1})$ for $n \geq 1$. Show that there exists a positive integer $n$ such that $(a_n - a_{n-1})(b_n - b_{n-1}) < 0$.

## 10.4 Functional equations I: Cauchy, injectivity and surjectivity

*Revisits first functional equations (see Chapter 5); the problems below push it further.*

**Definition 10.7 (Vocabulary for functions).** For sets $D$ and $C$, the notation $f : D \to C$ means that with each $x \in D$ we associate exactly one number $f(x) \in C$; the sets $D$ and $C$ are the *domain* and *codomain* of $f$. A function $f : D \to C$ is *increasing* (resp. *non-decreasing*) if $x < y$ implies $f(x) < f(y)$ (resp. $f(x) \leq f(y)$); *decreasing* and *non-increasing* are defined similarly, and all four kinds are called *monotonic*. If $D$ is symmetric about zero, $f$ is *even* if $f(-x) = f(x)$ for all $x \in D$, and *odd* if $f(-x) = -f(x)$ for all $x \in D$. Finally, $f$ is *periodic* if there is some $t > 0$ with $x + t \in D$ and $f(x + t) = f(x)$ for every $x \in D$.

**Definition 10.8 (Injective, surjective, bijective).** A function $f : D \to C$ is *injective* if $f(a) = f(b)$ implies $a = b$ for all $a, b \in D$; it is *surjective* if for every $y \in C$ there exists $x \in D$ with $f(x) = y$. A function that is both injective and surjective is *bijective*.

**Theorem 10.9 (Cauchy's equation).** *Let $D$ be one of $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$. The only functions $f : D \to D$ with*

$$f(x + y) = f(x) + f(y) \qquad \text{for all } x, y \in D$$

*are $f(x) = cx$ with $c \in D$. For $D = \mathbb{R}$ an extra assumption — for instance monotonicity or continuity — is needed to ensure that these are the only solutions; without it there exist*

*other, so-called pathological, solutions.*

**Technique 10.10 (Proving injectivity, surjectivity, and exploiting periodicity).** Fix all but one of the variables and play with the most convenient one. Every odd-degree polynomial expression in a free variable is surjective onto $\mathbb{R}$, and an even-degree one is surjective onto some half-line $[b, \infty)$; so a term "outside all the $f$'s" of that shape is often what proves surjectivity. Once surjectivity is available, do not stop at solving $f(b) = 0$ or $f(b) = 1$: look for the more complicated values that $f$ is now known to attain. A function that is both periodic and monotonic is constant; and if the set of periods of $f$ is itself surjective — even only beyond some point — then $f$ is constant.

**Technique 10.11 (Iteration).** Over $\mathbb{R}^+$ the standard opening moves are: try to prove a basic inequality such as $f(x) \geq x$ or $f(x) \geq 1$; try surjectivity, then injectivity; *iterate*, that is, substitute an expression of the form $f(t)$ (or the whole left-hand side of the equation) back into the equation; look for a period; try more elaborate substitutions; and try to prove monotonicity.

### 10.4.1 First substitutions and Cauchy's equation

**Problem 10.174.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x + y) = f(x) + y$ for all $x, y \in \mathbb{R}$.

**Problem 10.175.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x)f(y) = f(xy) + x + y$ for all $x, y \in \mathbb{R}$.

**Problem 10.176.** Prove that every function $f : \mathbb{R} \to \mathbb{R}$ can be uniquely represented as a sum of an even function and an odd function.

**Problem 10.177.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x + y) - f(x - y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$.

**Problem 10.178.** Find all functions $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ such that for every $x \neq 0$,

$$f\left(\frac{1}{x}\right) + \frac{f(-x)}{x} = x.$$

**Problem 10.179.** Suppose that $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(x) = f(2x) = f(1 - x)$ for every $x \in \mathbb{R}$. Prove that $f$ is periodic.

**Problem 10.180.** Find all functions $f : \mathbb{Q} \to \mathbb{Q}$ such that $f(1) = 2$ and $f(xy) + f(x + y) = f(x)f(y) + 1$ for all $x, y \in \mathbb{Q}$.

**Problem 10.181.** Find all functions $f : \mathbb{Q}^+ \to \mathbb{Q}^+$ such that for every $x \in \mathbb{Q}^+$,

$$f(x + 1) = f(x) + 1 \qquad \text{and} \qquad f(x^3) = f(x)^3.$$

**Problem 10.182.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that for every $x \in \mathbb{R}$,

$$f(-x) = -f(x), \qquad f(x + 1) = f(x) + 1, \qquad f\left(\frac{1}{x}\right) = \frac{f(x)}{x^2} \quad (x \neq 0).$$

**Problem 10.183.** Given is a function $f : \mathbb{R} \to \mathbb{R}$ and a real number $\alpha \neq 0$. Suppose that $f\left(\frac{x+y}{\alpha}\right) = \frac{f(x)+f(y)}{\alpha}$ for all $x, y \in \mathbb{R}$. Prove that $f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2}$ for all $x, y \in \mathbb{R}$.

**Problem 10.184.** A function $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(x+1) - f(x) = 2x + 1$ for every $x \in \mathbb{R}$, and $|f(x)| \leq 1$ for $x \in [0, 1]$. Prove that $|f(x)| \leq x^2 + 2$.

**Problem 10.185.** Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that

$$f(a)^2 + f(b)^2 + f(c)^2 = 2f(a)f(b) + 2f(b)f(c) + 2f(c)f(a)$$

for all integers $a, b, c$ satisfying $a + b + c = 0$. [IMO 2012, Problem 4]

**Problem 10.186.** Functions $f, g : \mathbb{N} \to \mathbb{N}$ satisfy $g(f(n)) = g(n) - n$ for every $n \in \mathbb{N}$. Find all possible values of $f(0)$.

**Problem 10.187.** Find all pairs of functions $f, g : \mathbb{R} \to \mathbb{R}$ such that $f(x)f(y) = g(x)g(y) + g(x) + g(y)$ for all $x, y \in \mathbb{R}$.

**Problem 10.188.** Functions $f, g : (0, 2) \to (0, 2)$ satisfy $f(g(x)) = g(f(x)) = x$ and $f(x) + g(x) = 2x$ for every $x \in (0, 2)$. Prove that $f(1) = g(1)$.

**Problem 10.189.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f\big(x + f(x + y)\big) = f(x - y) + f(x)^2$ for all $x, y \in \mathbb{R}$.

**Problem 10.190.** Determine all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying $f(x^2 - y^2) = xf(x) - yf(y)$ for all $x, y \in \mathbb{R}$.

**Problem 10.191.** Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $x + f(x) = f(f(x))$ for all $x \in \mathbb{R}$. Find all $x \in \mathbb{R}$ such that $f(f(x)) = 0$.

**Problem 10.192.** Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ satisfying $f(a + f(b)) = b + f(a) + 1$ for all $a, b \in \mathbb{Z}$.

**Problem 10.193.** Determine all functions $f : \mathbb{Z} \to \mathbb{Z}$ satisfying $f(2a) + 2f(b) = f(f(a + b))$ for all $a, b \in \mathbb{Z}$. [IMO 2019, Problem 1]

**Problem 10.194.** Let $n \in \mathbb{N}$. Find all monotone functions $f : \mathbb{R} \to \mathbb{R}$ satisfying $f(x + f(y)) = f(x) + y^n$ for all $x, y \in \mathbb{R}$.

**Problem 10.195.** Determine all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying $f\big(xf(x) + f(y)\big) = y + f(x)^2$ for all $x, y \in \mathbb{R}$.

**Problem 10.196.** Determine all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying $f(f(x) + y) = 2x + f(f(y) - x)$ for all $x, y \in \mathbb{R}$.

**Problem 10.197.** Determine all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying $f\big(\lfloor x \rfloor y\big) = f(x) \lfloor f(y) \rfloor$ for all $x, y \in \mathbb{R}$. [IMO 2010, Problem 1]

**Problem 10.198.** Determine all functions $f, g : \mathbb{R} \to \mathbb{R}$ satisfying $g(f(x + y)) = f(x) + (2x + y)g(y)$ for all $x, y \in \mathbb{R}$. [IMO 2011 Shortlist A3]

**Problem 10.199.** Is there a function $f : \mathbb{Q}^+ \to \mathbb{Q}^+$ such that $f(xf(y)) = \frac{f(x)}{y}$ for all rational $x, y > 0$?

**Problem 10.200.** Determine all functions $f : \mathbb{Q}^+ \to \mathbb{Q}^+$ satisfying $f(x^2 f(y)^2) = f(x)^2 f(y)$ for all $x, y \in \mathbb{Q}^+$.

**Problem 10.201.** Let $f : \mathbb{R} \to \mathbb{R}$ be a non-constant function that satisfies

$$f(x)f(x-y) + f(y)f(x+y) = f(x)^2 + f(y)^2 \qquad \text{for all } x, y \in \mathbb{R}.$$

Prove that $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$.

**Problem 10.202.** Determine all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying

$$\frac{f(p)^2 + f(q)^2}{f(r^2) + f(s^2)} = \frac{p^2 + q^2}{r^2 + s^2}$$

for all $p, q, r, s \in \mathbb{R}^+$ such that $pq = rs$. [IMO 2008, Problem 4]

**Problem 10.203.** Determine all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying

$$f(x + f(y) + f(f(z))) = y\, f(1 + f(f(y))(x + z))$$

for all $x, y, z \in \mathbb{R}^+$.

**Problem 10.204.** Determine all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying

$$f(x)f(y) = f(y)\, f(xf(y)) + \frac{1}{xy}$$

for all $x, y \in \mathbb{R}^+$.

**Problem 10.205.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x+y) = f(x)f(y)f(xy)$ for all $x, y \in \mathbb{R}$.

**Problem 10.206.** Assume that for a function $f : \mathbb{R}^+ \to \mathbb{R}^+$ it holds that $f(x+y) = f(xy)\left(f\left(\frac{1}{x}\right) + f\left(\frac{1}{y}\right)\right)$ for all $x, y \in \mathbb{R}^+$. Show that $f(xy) = f(x)f(y)$ for all $x, y \in \mathbb{R}^+$. †

**Problem 10.207.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x + f(y) + f(f(z))) = f(f(x)) + f(y) + z$ for all $x, y, z \in \mathbb{R}$.

**Problem 10.208.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(xf(y) + x^2 + y) = f(x)f(y) + xf(x) + f(y)$ for all $x, y \in \mathbb{R}$.

**Problem 10.209.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(xf(y)-y)+f(xy-x)+f(x+y) = 2xy$ for all $x, y \in \mathbb{R}$.

**Problem 10.210.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x-f(y)) = f(f(y))+xf(y)+f(x)-1$ for all $x, y \in \mathbb{R}$.

**Problem 10.211.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f(x)f(yf(x)) = f(x+y)$ for all $x, y \in \mathbb{R}^+$.

**Problem 10.212.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying $f(x+yf(x))+f(xy) = f(x)+f(2019y)$ for all real numbers $x$ and $y$.

### 10.4.2 Injectivity, surjectivity, iteration

**Problem 10.213.** (a) Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(x - f(x)) = x - f(x)$ for all $x \in \mathbb{R}$. Prove that $f(0) = 0$ and that $x = 0$ is the unique solution of $f(x) = 0$. †
(b) Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(x-1)f(y) = y\big(f(xy) - f(y)\big)$ for all $x, y \in \mathbb{R}$ and $f(2022) \neq 0$. Calculate $f(2022)$. †

**Problem 10.214.** (a) Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(xy) = xf(y) + yf(x)$ and $f(x^2) = f(f(x))$ for all $x, y \in \mathbb{R}$. Prove that $f(x) = 0$ for all $x \in \mathbb{R}$.
(b) Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(x+y) = f(x) + f(y)$ and $f(x^2) = f(x)^2 - xf(x)$ for all $x, y \in \mathbb{R}$. Given that $f(2022) \neq 0$, prove that $f(x) = x$ for all $x \in \mathbb{R}$. †

**Problem 10.215.** (a) Find all functions $f : \mathbb{R}^+ \to \mathbb{R}$ such that $f(1) = \frac{1}{2}$ and $f(xy) = f(x)f(y) + f(4/y)f(4/x)$ for all $x, y > 0$. †
(b) Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x+y) - f(x-y) = 2y\big(3x^2 + y^2\big)$ for all $x, y \in \mathbb{R}$. †

**Problem 10.216.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that (a) $f(y^2 + f(x)) = f(y)^2 + x^2$ for all $x, y \in \mathbb{R}$; †
(b) $f(x^2 + xy + f(y)) = f(x)^2 + xf(y) + y$ for all $x, y \in \mathbb{R}$. †

**Problem 10.217.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that (a) $f(x^2 + f(y)) = xf(x) + y$ for all $x, y \in \mathbb{R}$; †
(b) $f(x^2 + xy) = f(x)f(y) + yf(x) + xf(x+y)$ for all $x, y \in \mathbb{R}$. †

**Problem 10.218.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that (a) $f(f(x)+y) = f(x+y)+f(x)f(y)-xy$ for all $x, y \in \mathbb{R}$;
(a′) $f(f(x) + y) = f(x) + f(y) + f(x)f(y) - xy$ for all $x, y \in \mathbb{R}$;
(b) $f((x+y)^2) = f(x)f(x+2y) + yf(x)$ for all $x, y \in \mathbb{R}$;
(b′) $f((x+y)^2) = f(x)f(x+2y) + yf(y)$ for all $x, y \in \mathbb{R}$. †

**Problem 10.219.** ★ Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(0) \neq 0$ and

$$2f\big(xy + f(x)\big) = xf(y) + yf(x) \qquad \text{for all } x, y \in \mathbb{R}.$$

Prove that $f(f(2)) = f(2)$. †

**Problem 10.220.** (a) Prove that if $f(xf(x)) = x^2$ for all $x$ and $f$ is injective, then $f$ is bijective.
(b) Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that for all $x \in \mathbb{R}$, $f(x^3) = f(2x)+2$ and $f(x^3+2) = 3f(x)+2$. Prove that $f$ is not injective on $\mathbb{R}$. †

**Problem 10.221.** (a) Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f\big(xf(y) - f(x)\big) = 2f(x) + xy$ for all $x, y \in \mathbb{R}$. Prove that $f$ is injective and $f(0) = 1$, $f(1) = 0$. †
(b) Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f\big(|x| + y + f(x)\big) = 2y + |f(x)|$ for all $x, y \in \mathbb{R}$. Prove that $f$ is surjective and $f(0) = 0$. †

**Problem 10.222.** (a) Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f\big(2022x^3 + y + f(y)\big) = 2y + 2022x^2 f(x)$ for all $x, y \in \mathbb{R}$. Prove that $f$ is bijective. †
(b) Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f\big(xf(x) + 2f(y)\big) = x^2 + y + f(y)$ for all $x, y \in \mathbb{R}$. Prove that $f$ is bijective. †

**Problem 10.223.** Find all $c \in \mathbb{R}$ for which there exists a function $f : \mathbb{R} \to \mathbb{R}$ such that $f(1) = 2022$ and $f(x + y + f(y)) = f(x) + cy$ for all $x, y \in \mathbb{R}$.

**Problem 10.224.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(0) \neq 0$ and $f(2xy + f(x + y)) = xf(y) + yf(x)$ for all $x, y \in \mathbb{R}$. †

**Problem 10.225.** Find all injective functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(yf(x) + y^2) = x\,f(y) + f((x-1)f(y))$ for all $x, y \in \mathbb{R}$. †

**Problem 10.226.** ★ Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(xf(x) + 2f(y)) = x^2 + y + f(y)$ for all $x, y \in \mathbb{R}$.

**Problem 10.227.** Consider a function $f : \mathbb{R} \to \mathbb{R}$ such that

$$2(y+1)f(x) + f(y) = (y+1)f(xy) + 2f(x) \qquad \text{for all } x, y \in \mathbb{R}.$$

Find all possible values of $f(2022)$. †

**Problem 10.228.** Find all functions $f : [1, +\infty) \to [1, +\infty)$ such that for all $x, y \geq 1$ with $x^2 - y^2 \geq 1$,

$$f(x^2 - y^2) = f(xy).$$

**Problem 10.229.** Find all functions $f : (0, +\infty) \to \mathbb{R}$ such that for all real numbers $x, y$ with $x + y > 0$ one has $f(x + y) = f(x^2 + y^2)$.

**Problem 10.230.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ and $g : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f(x^2 + y^2) = g(xy)$ holds for all positive real numbers $x, y$.

**Problem 10.231.** Find all injective functions $f : \mathbb{Z} \to \mathbb{Z}$ such that $f(x+yf(x)) = f(x)(1+f(y))$ for all $x, y \in \mathbb{Z}$.

**Problem 10.232.** Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that for all integers $x$,

$$2023\,f(f(x)) + 2022\,x^2 = 2022\,f(x) + 2023\,f(x)^2 + 1.$$

**Problem 10.233.** Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that for all positive integers $a, b, c$,

$$f(a^2 + f(b) + c) = af(a) + f(b + c).$$

**Problem 10.234.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that for all $x, y \in \mathbb{R}$, $f(yf(x)+f(x)f(y)) = xf(y) + f(xy)$.

**Problem 10.235.** Let $a > -2$. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that for all $x, y > 0$, $f(x) + f(y) = f\left(\sqrt{x^2 + axy + y^2}\right)$.

**Problem 10.236.** Find all functions $f : \mathbb{Q}^+ \to \mathbb{R}^+$ such that for all $0 < y < x$, $f(x+y) - f(x-y) = 4\sqrt{f(x)f(y)}$.

**Problem 10.237.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x + y + f(xy)) = f(f(x + y)) + xy$ for all $x, y \in \mathbb{R}$.

**Problem 10.238.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(xf(x) + f(x)f(y) + y - 1) = f(xf(x) + xy) + y - 1$ for all $x, y \in \mathbb{R}$.

**Problem 10.239.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x - f(y)) = f(f(y)) - 2xf(y) + f(x)$ for all $x, y \in \mathbb{R}$.

**Problem 10.240.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(2y + f(x)) = 6x + f(f(y) - x)$ for all $x, y \in \mathbb{R}$.

**Problem 10.241.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(xf(y) + f(x)) = 2f(x) + xy$ for all $x, y \in \mathbb{R}$.

**Problem 10.242.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x + f(y)) - f(x) = (x + f(y))^4 - x^4$ for all $x, y \in \mathbb{R}$.

**Problem 10.243.** Prove that there are real numbers $a, b$ and a surjective function $f : \mathbb{R} \to \mathbb{R}$ such that $f(f(x)) = bxf(x) + a$ for all $x \in \mathbb{R}$.

**Problem 10.244.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x + f(y)) = f(x - f(y)) + 4xf(y)$ for all $x, y \in \mathbb{R}$.

**Problem 10.245.** Let $n \geq 2$ be a positive integer. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that for all positive real numbers $x, y$, $f(x + y^n + f(y)) = f(x)$.

**Problem 10.246.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f(y + f(x)) = f(x) + 3x + yf(y)$ for all $x, y \in \mathbb{R}^+$.

**Problem 10.247.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that for all $x, y > 0$, $f(3f(x) + 3y) = f(3x + y) + 2y$.

**Problem 10.248.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that for all $x, y > 0$, $f(x^2 + y + f(y)) = f(x)^2$.

**Problem 10.249.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that for all $x, y > 0$, $f(x + y^2 f(y)) = f(1 + yf(x))f(x)$.

## 10.5 Inequalities II: the classical arsenal

*Revisits AM–GM and Cauchy–Schwarz (see Chapter 5); the problems below push it further.*

Level 3 expects the named inequalities to be available by name. They are collected here as statements; the problems that follow are the camps' own drill on them, followed by two techniques — the interval trick and the *pqr* substitution — which turn many symmetric problems into routine ones.

**Theorem 10.12 (Bernoulli's inequality).** *$(1 + x)^n \geq 1 + nx$ for all $x \geq 0$ and $n \in \mathbb{N}$. The inequality also holds for all $x \geq -1$ and any real $n \geq 1$.*

**Theorem 10.13 (Rearrangement inequality).** *Let $a_1 \leq \cdots \leq a_n$ and $b_1 \leq \cdots \leq b_n$ be two non-decreasing sequences of real numbers. Then for any permutation $(c_i)$ of the numbers $(b_i)$,*

$$a_1 b_n + \cdots + a_n b_1 \;\leq\; a_1 c_1 + \cdots + a_n c_n \;\leq\; a_1 b_1 + \cdots + a_n b_n.$$

**Theorem 10.14 (AM–GM).** *For non-negative reals $x_1, \ldots, x_n$,*

$$\frac{x_1 + x_2 + \cdots + x_n}{n} \;\geq\; \sqrt[n]{x_1 x_2 \cdots x_n},$$

*with equality if and only if $x_1 = \cdots = x_n$.*

**Theorem 10.15 (Weighted power mean).** *Let $a_1, \ldots, a_n$ and $w_1, \ldots, w_n$ be positive reals with $w_1 + w_2 + \cdots + w_n = 1$. For a real number $r$ define*

$$P(r) = \begin{cases} \left(w_1 a_1^r + w_2 a_2^r + \cdots + w_n a_n^r\right)^{1/r}, & r \neq 0, \\ a_1^{w_1} a_2^{w_2} \cdots a_n^{w_n}, & r = 0. \end{cases}$$

*If $r > s$ then $P(r) \geq P(s)$, with equality if and only if $a_1 = a_2 = \cdots = a_n$. In particular, for $w_1 = \cdots = w_n = \frac{1}{n}$ the values $r = 2, 1, 0, -1$ give*

$$\sqrt{\frac{a_1^2 + \cdots + a_n^2}{n}} \;\geq\; \frac{a_1 + \cdots + a_n}{n} \;\geq\; \sqrt[n]{a_1 a_2 \cdots a_n} \;\geq\; \frac{n}{\frac{1}{a_1} + \cdots + \frac{1}{a_n}},$$

*that is, quadratic mean $\geq$ arithmetic mean $\geq$ geometric mean $\geq$ harmonic mean.*

**Theorem 10.16 (Cauchy–Schwarz inequality).** *For any real numbers $a_i$ and $b_i$,*

$$(a_1^2 + \cdots + a_n^2)(b_1^2 + \cdots + b_n^2) \;\geq\; (a_1 b_1 + \cdots + a_n b_n)^2.$$

**Corollary 10.17 (Cauchy–Schwarz in Engel form).** *For any reals $x_i$ and positive $y_i$,*

$$\frac{x_1^2}{y_1} + \cdots + \frac{x_n^2}{y_n} \;\geq\; \frac{(x_1 + \cdots + x_n)^2}{y_1 + \cdots + y_n}.$$

**Theorem 10.18 (Hölder's inequality).** *Let $\alpha, \beta, \ldots, \zeta$ be positive reals with $\alpha + \beta + \cdots + \zeta = 1$, and let $a_1, \ldots, a_n$, $b_1, \ldots, b_n$, $\ldots$, $z_1, \ldots, z_n$ be positive reals. Then*

$$(a_1 + \cdots + a_n)^\alpha (b_1 + \cdots + b_n)^\beta \cdots (z_1 + \cdots + z_n)^\zeta \;\geq\; \sum_{i=1}^{n} a_i^\alpha b_i^\beta \cdots z_i^\zeta,$$

*with equality when $a_1 : a_2 : \cdots : a_n = b_1 : b_2 : \cdots : b_n = \cdots = z_1 : z_2 : \cdots : z_n$.*

**Definition 10.19 (Majorization).** Suppose we have two sequences $x_1 \geq x_2 \geq \cdots \geq x_n$ and $y_1 \geq y_2 \geq \cdots \geq y_n$ such that

$$x_1 + x_2 + \cdots + x_n = y_1 + y_2 + \cdots + y_n$$

and $x_1 + x_2 + \cdots + x_k \geq y_1 + y_2 + \cdots + y_k$ for $k = 1, 2, \ldots, n-1$. Then we say that $(x_n)$ *majorizes* $(y_n)$, written $(x_n) \succ (y_n)$.

**Theorem 10.20 (Muirhead's inequality).** *If $a_1, a_2, \ldots, a_n$ are positive reals and $(x_n)$ majorizes $(y_n)$, then*

$$\sum_{\text{sym}} a_1^{x_1} a_2^{x_2} \cdots a_n^{x_n} \;\geq\; \sum_{\text{sym}} a_1^{y_1} a_2^{y_2} \cdots a_n^{y_n}.$$

**Theorem 10.21 (Chebyshev's inequality).** *Let $a_1 \leq \cdots \leq a_n$ and $b_1 \leq \cdots \leq b_n$ be two non-decreasing sequences of real numbers. Then*

$$\frac{a_1 b_1 + \cdots + a_n b_n}{n} \;\geq\; \frac{a_1 + \cdots + a_n}{n} \cdot \frac{b_1 + \cdots + b_n}{n} \;\geq\; \frac{a_1 b_n + \cdots + a_n b_1}{n}.$$

**Theorem 10.22 (Minkowski's inequality).** *For positive real numbers $x_i$, $y_i$ and $p \geq 1$,*

$$\left(\sum_{i=1}^{n} (x_i + y_i)^p\right)^{1/p} \;\leq\; \left(\sum_{i=1}^{n} x_i^p\right)^{1/p} + \left(\sum_{i=1}^{n} y_i^p\right)^{1/p}.$$

**Theorem 10.23 (Jensen's inequality).** *If $f : I \to \mathbb{R}$ is a convex function, then for any $x_1, \ldots, x_n \in I$,*

$$\frac{f(x_1) + \cdots + f(x_n)}{n} \;\geq\; f\left(\frac{x_1 + \cdots + x_n}{n}\right).$$

*The opposite inequality holds when $f$ is concave.*

**Theorem 10.24 (Karamata's inequality).** *If $f$ is convex and $(x_n)$ majorizes $(y_n)$, then*

$$f(x_1) + \cdots + f(x_n) \;\geq\; f(y_1) + \cdots + f(y_n).$$

*The opposite inequality holds when $f$ is concave.*

**Theorem 10.25 (Nesbitt's inequality).** *For positive real numbers $a, b, c$, $\dfrac{a}{b+c} + \dfrac{b}{c+a} + \dfrac{c}{a+b} \;\geq\; \dfrac{3}{2}$.*

**Theorem 10.26 (Schur's inequality).** *Let $a, b, c$ be positive real numbers and let $r$ be positive. Then*

$$a^r(a-b)(a-c) + b^r(b-c)(b-a) + c^r(c-a)(c-b) \;\geq\; 0,$$

*with equality if and only if $a = b = c$, or $a = b$ and $c = 0$ (and permutations).*

**Technique 10.27 (Substitutions).** Several substitutions turn a hard inequality into an easy one. *Normalization*: a homogeneous inequality may be rescaled, so one may assume

$abc = 1$, or $a+b+c = 1$, or the like. *Homogenization* is the reverse move; when the condition $abc = 1$ is given it is often useful to substitute $a = x/y$, $b = y/z$, $c = z/x$. *Ravi substitution*: if $a, b, c$ are the side lengths of a triangle, there are positive reals $x, y, z$ with $a = x + y$, $b = y + z$, $c = z + x$. *Trigonometric substitution*: for $x \in [-1, 1]$ set $x = \sin t$ or $x = \cos t$; for $x \in [0, 1]$ set $x = \sin^2 t$ or $x = \cos^2 t$.

**Theorem 10.28 (Sturm's principle).** *Let $f$ be a function defined on a set $M$ and let $x_0 \in M$. If $f$ has a maximum (minimum) on $M$, and no other point $x$ of $M$ is a maximum (minimum) of $f$, then $x_0$ is the maximum (minimum) of $f$. To decide whether $f$ has an extremum one uses: a continuous function on a compact set always attains its extrema; and a closed and bounded subset of $\mathbb{R}^n$ is compact.*

### 10.5.1 Drill on the classical inequalities

**Problem 10.250.** Prove that $3^n > n \cdot 2^n$ for all $n \in \mathbb{N}$.

**Problem 10.251.** Prove that $2^n \geq n^2$ for all natural $n > 3$.

**Problem 10.252.** Prove that for all $n \in \mathbb{N}$, $\dfrac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n} \leq \dfrac{1}{\sqrt{2n+1}}$.

**Problem 10.253.** Prove that for all $n \in \mathbb{N}$, $\dfrac{1}{2^2} + \dfrac{1}{3^2} + \cdots + \dfrac{1}{n^2} < 1$.

**Problem 10.254.** Prove that for all natural $n \geq 3$, $\dfrac{1}{n+1} + \dfrac{1}{n+2} + \cdots + \dfrac{1}{2n} > \dfrac{3}{5}$.

**Problem 10.255.** Prove that for all $n \in \mathbb{N}$, $1 + \dfrac{1}{\sqrt{2}} + \dfrac{1}{\sqrt{3}} + \cdots + \dfrac{1}{\sqrt{n}} < 2\sqrt{n}$.

**Problem 10.256.** The product of the positive real numbers $a_1, a_2, \ldots, a_n$ is equal to 1. Prove that $(1 + a_1)(1 + a_2) \cdots (1 + a_n) \geq 2^n$.

**Problem 10.257.** Prove the inequality $\frac{1}{4}a^2 + b^2 + c^2 \geq ab - ac + 2bc$ for all $a, b, c \in \mathbb{R}$.

**Problem 10.258.** Prove the inequality $x^4 + y^4 + z^2 + 1 \geq 2x(xy^2 - x + z + 1)$ for all $x, y, z \in \mathbb{R}$.

**Problem 10.259.** Show that $x^2 + y^2 + 1 \geq xy + x + y$ for all $x, y \in \mathbb{R}$.

**Problem 10.260.** Show that $ab + bc + ca \geq a\sqrt{bc} + b\sqrt{ac} + c\sqrt{ab}$ for all $a, b, c \geq 0$.

**Problem 10.261.** For $a, b, c > 0$ which are the lengths of sides of a triangle, prove that

$$\frac{3}{2} \leq \frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} < 2.$$

**Problem 10.262.** For $a, b, c, d > 0$ prove that $ab^2c^3d^4 \leq \left(\dfrac{a + 2b + 3c + 4d}{10}\right)^{10}$.

**Problem 10.263.** For $a, b, c > 0$ prove that $\dfrac{9}{2(a+b+c)} \leq \dfrac{1}{b+c} + \dfrac{1}{c+a} + \dfrac{1}{a+b}$.

**Problem 10.264.** Let $x, y, z > 1$ with $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = 2$. Prove that $\sqrt{x+y+z} \geq \sqrt{x-1} + \sqrt{y-1} + \sqrt{z-1}$.

**Problem 10.265.** For $a, b, c > 0$ prove that

$$\left(1+\frac{a}{b}\right)\left(1+\frac{b}{c}\right)\left(1+\frac{c}{a}\right) \geq 2\left(1+\frac{a+b+c}{\sqrt[3]{abc}}\right).$$

**Problem 10.266.** Show that for all real $a > 0$ and natural $n$, $a^n + \frac{1}{a^n} - 2 \geq n^2\left(a + \frac{1}{a} - 2\right)$.

**Problem 10.267.** For $a, b, c > 0$ prove that $\frac{ab}{c} + \frac{bc}{a} + \frac{ca}{b} \geq a + b + c$.

**Problem 10.268.** Let $a_1, \ldots, a_n$ be real numbers not less than 1. Show that $(1+a_1)\cdots(1+a_n) \geq \frac{2^n}{n+1}(1 + a_1 + \cdots + a_n)$.

**Problem 10.269.** Let $a, b, c > 0$. Prove that $a^a b^b c^c \geq \left(\frac{a+b+c}{3}\right)^{a+b+c}$.

**Problem 10.270.** Show that for any positive real numbers $a, b, c$,

$$\frac{a^3}{a^2+4b^2} + \frac{b^3}{b^2+4c^2} + \frac{c^3}{c^2+4a^2} > \frac{a+b+c}{5}.$$

**Problem 10.271.** For $a, b, c > 0$ prove that $\frac{1}{b+c} + \frac{1}{c+a} + \frac{1}{a+b} \leq \frac{1}{2}\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$.

**Problem 10.272.** Let $x_1, \ldots, x_n \geq 1$ be real numbers. Prove that $\frac{1}{1+x_1} + \cdots + \frac{1}{1+x_n} \geq \frac{n}{1+\sqrt[n]{x_1 \cdots x_n}}$.

**Problem 10.273.** Real numbers $a, b, c, d, e$ are such that $a + b + c + d + e = 8$ and $a^2 + b^2 + c^2 + d^2 + e^2 = 16$. What is the largest possible value of $e$?

**Problem 10.274.** Let $a, b, c$ be positive real numbers with $abc = 1$. Prove that

$$\frac{1}{a^3(b+c)} + \frac{1}{b^3(c+a)} + \frac{1}{c^3(a+b)} \geq \frac{3}{2}.$$

**Problem 10.275.** For $x \geq y \geq 1$ prove that

$$\frac{\sqrt{x}}{x+y} + \frac{\sqrt{y}}{y+1} + \frac{1}{x+1} \geq \frac{\sqrt{y}}{x+y} + \frac{\sqrt{x}}{x+1} + \frac{1}{y+1}.$$

**Problem 10.276.** Prove that for any real numbers $a, b, c, d \geq \frac{1}{3}$ the following inequality holds:

$$\frac{a^6}{b^4+c^3} + \frac{b^6}{c^4+d^3} + \frac{c^6}{d^4+a^3} + \frac{d^6}{a^4+b^3} \geq \frac{a+b+c+d}{4}.$$

**Problem 10.277.** Let $a, b, c$ be non-negative real numbers such that $a+b+c=1$. Prove that $4(ab+bc+ca)-9abc \leq 1$.

**Problem 10.278.** Let $x_1, x_2, \ldots, x_n$ $(n \geq 2)$ be positive numbers such that $x_1+x_2+\cdots+x_n=1$. Prove that

$$\left(1+\frac{1}{x_1}\right)\left(1+\frac{1}{x_2}\right)\cdots\left(1+\frac{1}{x_n}\right) \geq (n+1)^n.$$

**Problem 10.279.** Prove that a necessary and sufficient condition for a triangle inscribed in an ellipse to have maximum area is that its centroid coincide with the centre of the ellipse.

**Problem 10.280.** Let $n > 2$ be an integer. A convex $n$-gon of area 1 is inscribed in a circle. What is the minimum that the radius of the circle can be?

**Problem 10.281.** Let $a, b, c > 0$ with $a+b+c=1$. Prove that $0 \leq ab+bc+ca-2abc \leq \frac{7}{27}$.

**Problem 10.282.** Let $a, b, c, d$ be non-negative numbers such that $a \leq 1$, $a+b \leq 5$, $a+b+c \leq 14$ and $a+b+c+d \leq 30$. Prove that $\sqrt{a}+\sqrt{b}+\sqrt{c}+\sqrt{d} \leq 10$.

**Problem 10.283.** What is the maximal value of the expression $\sum_{i<j} x_i x_j$ if $x_1, x_2, \ldots, x_n$ are non-negative integers whose sum is equal to $m$?

**Problem 10.284.** Given the $n \times n$ array $(a_{ij})$ with $a_{ij} = i+j-1$, what is the smallest product of $n$ elements of the array, provided that no two of them lie in the same row or the same column?

**Problem 10.285.** Given a positive integer $n$, find the minimum value of $\dfrac{x_1^3+x_2^3+\cdots+x_n^3}{x_1+x_2+\cdots+x_n}$ subject to the condition that $x_1, x_2, \ldots, x_n$ be distinct positive integers.

**Problem 10.286.** Prove that for positive $a, b, c$ one has $3(a+b+c) \geq 8\sqrt[3]{abc}+\sqrt[3]{\dfrac{a^3+b^3+c^3}{3}}$.

[Taiwan TST Quiz]

**Problem 10.287.** Prove that for positive $a, b, c$ one has

$$\frac{1}{a(b+c)}+\frac{1}{b(c+a)}+\frac{1}{c(a+b)} \geq \frac{27}{2(a+b+c)^2}.$$

[Balkan]

**Problem 10.288.** Given $a+b+c=\dfrac{1}{a}+\dfrac{1}{b}+\dfrac{1}{c}$, prove that

$$\frac{1}{(2a+b+c)^2}+\frac{1}{(a+2b+c)^2}+\frac{1}{(a+b+2c)^2} \leq \frac{3}{16}.$$

[IMO 2009 Shortlist]

**Problem 10.289.** Let $n \geq 2$ and let $x_1, \ldots, x_n$ be positive real numbers satisfying

$$\frac{1}{x_1+1998}+\cdots+\frac{1}{x_n+1998}=\frac{1}{1998}.$$

Prove that $\dfrac{\sqrt[n]{x_1\cdots x_n}}{n-1} \geq 1998$. [Vietnam 1998]

**Problem 10.290.** Let $a, b, c$ be the side lengths of a triangle. Prove that $a^2(b+c-a)+b^2(c+a-b)+c^2(a+b-c) \le 3abc$. [IMO 1964]

**Problem 10.291.** Let $a, b, c$ be positive real numbers. Prove that $1 \le \sum_{\text{cyc}} \frac{a}{\sqrt{a^2+8bc}} < 2$. [IMO 2001]

**Problem 10.292.** Real numbers $x_1, \dots, x_n \ge -1$ are such that $x_1^3 + \cdots + x_n^3 = 0$. Prove that $x_1 + \cdots + x_n \le \frac{n}{3}$.

**Problem 10.293.** Prove the Cauchy–Schwarz inequality by considering the quadratic polynomial $f(x) = (a_1x + b_1)^2 + \cdots + (a_nx + b_n)^2$.

**Problem 10.294.** Give a geometric interpretation of Minkowski's inequality for $p = 2$.

**Problem 10.295.** Find a geometric proof of Jensen's inequality.

**Problem 10.296.** Let $a_1, a_2, \dots, a_n$ be positive real numbers, $n \ge 2$, such that $a_1+a_2+\cdots+a_n = 1$. Prove that

$$\frac{a_1}{2-a_1} + \frac{a_2}{2-a_2} + \cdots + \frac{a_n}{2-a_n} \ \ge \ \frac{n}{2n-1}.$$

**Problem 10.297.** For arbitrary positive reals $a \ge b \ge c$ prove the inequality

$$\frac{a^2+b^2}{a+b} + \frac{a^2+c^2}{a+c} + \frac{c^2+b^2}{c+b} \ \ge \ (a+b+c) + \frac{(a-c)^2}{a+b+c}.$$

### 10.5.2 Inequalities on an interval: the tangent-line trick

**Technique 10.29 (Inequality on an interval).** If $x \in [a, b]$ then $(x-a)(x-b) \le 0$. If $a$ and $a+1$ are two consecutive integers, then $(x-a)(x-a-1) \ge 0$ for every integer $x$. Applied to a power $x^n$ on an interval $[a, b]$, this produces the linear bound through the endpoints: the chord of $y = x^n$ over $[a, b]$ lies above the curve, so $x^n \le \lambda x + \mu$ there for suitable $\lambda, \mu$, and each variable can be handled separately.

**Problem 10.298.** Let $a, b, c$ be real numbers in $[1, 3]$ with $a + b + c = 7$. Find the maximum value of the following expressions: (a) $T = a^2+b^2+c^2$; (b) $T = a^3+b^3+c^3$; (c) $T = a^5+b^5+c^5$; (d) $T = a^n + b^n + c^n$ for a positive integer $n$.

**Problem 10.299.** (a) Let $x, y, z \in [1, 2]$ be such that $x^2+y^2+z^2 = 6$. Find the minimum value of $N = x + y + z$.
(b) Let $x, y, z \in [0, 1]$. Find the maximum value of $M = x + y^2 + z^3 - xy - yz - zx$. †

**Problem 10.300.** Let $a, b, c$ be real numbers in $[1, 3]$ with $a + b + c = 6$. Find the maximum and the minimum value of $P = a^2 + b^2 + c^2$.

**Problem 10.301.** On the plane there are 66 points and 16 lines. Denote by $m$ the number of pairs $(a, \ell)$ such that the point $a$ belongs to the line $\ell$. Prove that $m \le 159$.

**Problem 10.302.** Let $a, b, c \in [0, 2]$ with $a + b + c = 3$. Find the maximum value of $M = a^3 + b^3 + c^3 - 3abc$.

**Problem 10.303.** Let $x, y \in [1, 2]$. Find the minimum value of

$$T = \frac{x+2y}{x^2+3y+5} + \frac{2x+y}{y^2+3x+5} + \frac{1}{4(x+y-1)}.$$

†

### 10.5.3 Symmetric three-variable inequalities: *pqr* and Schur

**Technique 10.30 (The *pqr* substitution).** For three variables write $p = a+b+c$, $q = ab+bc+ca$, $r = abc$; every symmetric expression is a polynomial in $p, q, r$. A homogeneous inequality may first be standardized (normalized), for instance by setting $p = 3$ or $r = 1$, after which the problem often becomes a one-variable question about the remaining parameter. Two constraints are used constantly: $p^2 \geq 3q$ and $q^2 \geq 3pr$.

**Corollary 10.31 (Schur's inequality in *pqr* form).** *For non-negative $a, b, c$ with $p = a+b+c$, $q = ab+bc+ca$, $r = abc$, Schur's inequality with $r = 1$ reads*

$$p^3 + 9r \;\geq\; 4pq.$$

**Problem 10.304.** Let $x, y, z > 0$ with $xy+yz+zx = 3$. Prove that $(x^2+1)(y^2+1)(z^2+1) \geq 8$.

**Problem 10.305.** Let $x, y, z > 0$ with $x+y+z = 3$. Prove that $\dfrac{12}{xyz} + \dfrac{1}{x} + \dfrac{1}{y} + \dfrac{1}{z} \;\geq\; 15$. †

**Problem 10.306.** Let $a, b, c > 0$ with $abc = 1$. Prove that $\dfrac{1}{a^2} + \dfrac{1}{b^2} + \dfrac{1}{c^2} + 3 \;\geq\; 2(a+b+c)$.

**Problem 10.307.** Let $a, b, c > 0$. Prove that

(a) $(a^3+b^3+c^3)\left(\dfrac{1}{a}+\dfrac{1}{b}+\dfrac{1}{c}\right) + 6(ab+bc+ca) \;\geq\; 9(a^2+b^2+c^2)$;

(b) $2(a^2+b^2+c^2) + 3\sqrt[3]{a^2b^2c^2} \;\geq\; (a+b+c)^2$.

**Problem 10.308.**

(a) Find the maximum value of $k$ such that the following inequality is true for all non-negative $a, b, c$:

$$2(a^3+b^3+c^3) \;\geq\; k(a^3+b^3+c^3-3abc) + ab(a+b) + bc(b+c) + ca(c+a).$$

(b) Find the maximum value of $k$ such that the following inequality is true for all non-negative $a, b, c$ satisfying $a+b+c = ab+bc+ca \neq 0$:

$$a+b+c+k(abc-1) \;\geq\; 3.$$

(c) Find the minimum value of $k$ such that the following inequality is true for all non-negative $a, b, c$ satisfying $ab+bc+ca = 3$:

$$(a+k)(b+k)(c+k) \;\geq\; (k+1)^3.$$

**Problem 10.309.** Let $a, b, c$ be non-negative numbers such that $2(a^2+b^2+c^2)+3(ab+bc+ca) = 5(a+b+c)$. Prove that $4(a^2+b^2+c^2)+2(ab+bc+ca)+7abc \leq 25$. †

**Problem 10.310.** ★ Let $a, b, c > 0$ be such that $a^2+b^2+c^2 = 3$. Prove that $a+b+c \;\geq\; 3\sqrt[5]{abc}$. †

## 10.6 Functional inequalities and mixed problems

When the functional relation is an inequality rather than an equation, substitutions no longer pin the value down at once; one has to squeeze the function between two bounds, or push a single substitution to a contradiction.

**Problem 10.311.** Let the function $f : [0, +\infty) \to [0, +\infty)$ satisfy the conditions: (a) for any $x, y \geq 0$ one has $f(x)f(y) \leq y^2 f(x/2) + x^2 f(y/2)$; (b) if $0 \leq x \leq 1$, then $f(x) \leq 2025$. Prove that $f(x) \leq x^2$ for any $x \geq 0$.

**Problem 10.312.** Is there a bounded function $f : \mathbb{R} \to \mathbb{R}$ such that $f(1) > 0$ and for all $x, y \in \mathbb{R}$ it satisfies $f(x + y)^2 \geq f(x)^2 + 2f(xy) + f(y)^2$?

**Problem 10.313.** Is there a function $f : \mathbb{R} \to \mathbb{R}$ such that for arbitrary real numbers $x, y$ we have $f(x - f(y)) \leq x - yf(x)$? †

**Problem 10.314.** Show that for any function $f : (0, +\infty) \to (0, +\infty)$ there exist real numbers $x > 0$ and $y > 0$ such that $f(x + y) < yf(f(x))$.

**Problem 10.315.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that for arbitrary $a, b \in \mathbb{R}$ we have $f(a^2) - f(b^2) \leq \big(f(a) + b\big)\big(a - f(b)\big)$.

**Problem 10.316.** Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that three conditions are satisfied: (a) for all $x, y \in \mathbb{R}$, $f(x) + f(y) + 1 \geq f(x + y) \geq f(x) + f(y)$; (b) for all $x \in [0, 1)$, $f(0) \geq f(x)$; (c) $f(1) = -f(-1) = 1$. Find all such functions.

**Problem 10.317.** Is there any function $f : \mathbb{R} \to \mathbb{R}$ satisfying $f(0) > 0$ and $f(x + y) \geq f(x) + yf(f(x))$ for all $x, y \in \mathbb{R}$?

**Problem 10.318.** Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that for any $x, y \in \mathbb{R}$ we have $f(xy) \leq yf(x) + f(y)$.

**Problem 10.319.** Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(1) \geq 0$ and, for all real numbers $x, y$, $f(x) - f(y) \geq (x - y)f(x - y)$. †

**Problem 10.320.** A function $f : \mathbb{R} \to \mathbb{R}$ satisfies the inequality $f(x + y) \leq yf(x) + f(f(x))$ for all real numbers $x$ and $y$. Prove that $f(x) = 0$ for all $x \leq 0$.

**Problem 10.321.** Let $f$ be an arbitrary function that maps the set of real numbers to the set of real numbers. Prove that there exist real numbers $x$ and $y$ such that $f\big(x - f(y)\big) > yf(x) + x$.

**Problem 10.322.** Determine all functions $f : (0, +\infty) \to (0, +\infty)$ such that for arbitrary positive real numbers $x, y$ we have $f(x + y) \geq f(x) + y$ and $f(f(x)) \leq x$. †

**Problem 10.323.** Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(x^2 + y^2) \geq f(x + y)$ for all $x, y \in \mathbb{R}$. Prove that $f$ is constant on $(0, 2)$.

**Problem 10.324.** Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that $f(x^2 + 2y) \geq f(x^2 + 3y)$ for all $x, y \in \mathbb{R}$. Prove that $f$ is constant.

**Problem 10.325.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that for all real numbers $y$ and all $x \neq 0$, $f(x^2 + y) \geq \left(\frac{1}{x} + 1\right) f(y)$.

**Problem 10.326.** Let $f : \mathbb{R}^+ \to \mathbb{R}^+$ be an increasing function such that $f(2x) \le 4f(x+1)$ for all $x > 0$. Prove that $f(x) \le f(4)\, x^2$ for all $x > 1$.

**Problem 10.327.** Find all $a \in \mathbb{R}$ such that there exists a function $f : \mathbb{R} \to \mathbb{R}$ with $af(x) - x \le af(f(y)) - y$ for all $x, y \in \mathbb{R}$, and such that $f(s) = s$ for some real number $s$.

## 10.7 Classical inequalities: further practice

The June 2022 camp closed its Level-3 number-theory list with a short block of inequalities, all of them solved by the classical means: AM–GM applied to a carefully chosen set of terms, a squaring followed by Cauchy–Schwarz, or a single well-placed substitution.

**Problem 10.328.** Prove that for any non-negative real numbers $x_1, x_2, \ldots, x_n$,

$$x_1 + 2x_2 + \cdots + nx_n \;\le\; \frac{n(n-1)}{2} + x_1 + x_2^2 + \cdots + x_n^n.$$

**Problem 10.329.** For a given integer $n \ge 1$ find the smallest value of

$$x_1 + \frac{x_2^2}{2} + \frac{x_3^3}{3} + \cdots + \frac{x_n^n}{n},$$

where $x_1, x_2, \ldots, x_n$ are positive real numbers such that $\dfrac{1}{x_1} + \dfrac{1}{x_2} + \cdots + \dfrac{1}{x_n} = n$.

**Problem 10.330.** Let $a, b, c \in (0, 1)$. Prove that

$$\sqrt{abc} + \sqrt{(1-a)(1-b)(1-c)} < 1.$$

**Problem 10.331.** Prove that for any positive real numbers $a, b, c$,

$$a\sqrt{b^2 - bc + c^2} + c\sqrt{a^2 - ab + b^2} \;\ge\; b\sqrt{a^2 + ac + c^2}.$$

**Problem 10.332.** Let $a, b, c, d$ be positive real numbers. Prove that

$$\sqrt[3]{(a+b+c)(a+b+d)} \;\ge\; \sqrt[3]{ac} + \sqrt[3]{bd}.$$

**Problem 10.333.** Prove that for any positive real numbers $a_1, a_2, \ldots, a_n$,

$$\frac{a_1^3}{a_1^2 + a_1a_2 + a_2^2} + \frac{a_2^3}{a_2^2 + a_2a_3 + a_3^2} + \cdots + \frac{a_n^3}{a_n^2 + a_na_1 + a_1^2} \;\ge\; \frac{a_1 + a_2 + \cdots + a_n}{3}.$$

**Problem 10.334.** Let $a, b, c$ be real numbers. Prove that

$$\sqrt{2(a^2+b^2)} + \sqrt{2(b^2+c^2)} + \sqrt{2(c^2+a^2)} \;\ge\; \sqrt{3(a+b)^2 + 3(b+c)^2 + 3(c+a)^2}.$$

# Chapter 11

# Combinatorics

**Sources for this chapter:** Camp threads by Łukasz Bożyk (KAUST–Mawhiba website threads; e-training January 2023), Nikola Petrović (February–March 2022; online classes 2023), Emil Kolev (KAUST, February 2024), Mikhail (June 2020), David Zmiaikou (May 2025), Bojan Bašić (acceleration programme, autumn 2022), Arsenii "Senya" (remote lessons 2023–2024), and the January 2025 camp leaflets.

At Level 3 the combinatorial toolbox stops being a list of tricks and starts being a small number of general methods: bijections and generating functions for counting, double counting for inequalities, extremal and greedy arguments for existence, invariants for impossibility, and the language of graphs for everything else. Each section below opens with the statements the trainers used; the problems then push those statements as far as the olympiad requires.

## 11.1 Bijections, binomial coefficients and partitions

**Theorem 11.1 (Binomial theorem).** *For every positive integer $n$,*

$$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \cdots + \binom{n}{k}a^{n-k}b^k + \cdots + \binom{n}{n-1}ab^{n-1} + b^n.$$

**Definition 11.2 (Bijection, injection, surjection).** Let $X$ and $Y$ be sets. A *bijection* between $X$ and $Y$ is a pairing of the elements of $X$ with the elements of $Y$ such that every element of either set lies in exactly one pair; it is also called a one-to-one correspondence. A *surjection* from $X$ to $Y$ assigns to each element of $X$ an element of $Y$ in such a way that every element of $Y$ is assigned to at least one element of $X$. An *injection* from $X$ to $Y$ assigns to each element of $X$ an element of $Y$ in such a way that no two distinct elements of $X$ receive the same element of $Y$.

**Problem 11.1.** Prove that for any positive integer $n$ the following identity is true: $\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$.

**Problem 11.2.** Prove that for any positive integer $n$ the following identity is true: $\binom{n}{0} + \binom{n}{2} + \cdots + \binom{n}{2k} + \cdots = 2^{n-1}$.

**Problem 11.3.** Prove that for any positive integer $n$ the following identity is true: $\binom{n}{1} + 2\binom{n}{2} + \cdots + k\binom{n}{k} + \cdots + n\binom{n}{n} = n\,2^{n-1}$.

**Problem 11.4.** Prove that for any positive integer $n$ the following identity is true: $\binom{n}{0}^2 + \binom{n}{1}^2 + \cdots + \binom{n}{k}^2 + \cdots + \binom{n}{n}^2 = \binom{2n}{n}$.

**Problem 11.5.** Prove the identity $\sum_{k=1}^{n} k\binom{n}{k}^2 = n\binom{2n-1}{n-1}$.

**Problem 11.6.** Show that among the numbers less than 10000 there are as many whose digits sum to 16 as there are whose digits sum to 20.

**Problem 11.7.** In how many ways can one pick more than half of the people in a group of 100? (The answer should be in closed form.)

**Problem 11.8.** Are there more non-degenerate triangles with integer side lengths and perimeter 2016, or with perimeter 2019?

**Problem 11.9.** On a fixed number of vertices, are there more graphs which are connected or more graphs which are disconnected?

**Problem 11.10.** In a convex $n$-gon all diagonals are drawn and no three of them are concurrent. How many intersection points are there?

**Problem 11.11.** Given an $a \times b$ board: (a) how many rectangles are there on the board? (b) Call a set of cells a *plus* if it is the union of a $1 \times k$ rectangle and an $m \times 1$ rectangle (for some $k$ and $m$) that intersect in exactly one cell which is not at either end of either rectangle (the plus sign need not be symmetrical at all). How many different plus signs are there on the board?

**Problem 11.12.** Among all tilings of an $8 \times 8$ chessboard with dominoes, are there more tilings in which a single domino covers both the squares a1 and a2, or more tilings in which a single domino covers both the squares b2 and b3?

**Problem 11.13.** Find the number of subsets $S$ of the set $\{1, 2, 3, \ldots, 2n\}$ for which the equation $x + y = 2n + 1$ admits no solution with $x, y \in S$.

**Problem 11.14.** Student $A$ tosses a coin 2023 times and student $B$ tosses a coin 2024 times. Find the probability that student $B$ obtains more tails than student $A$.

**Problem 11.15.** A set $S$ is called *neighbouring* if it has the following two properties: (a) $S$ has exactly four elements; (b) for every $x \in S$, at least one of the numbers $x - 1$ or $x + 1$ belongs to $S$. Find the number of all neighbouring subsets of the set $\{1, 2, \ldots, n\}$. [JBMO SL 2018 C1]

**Problem 11.16.** Let $n$ be a positive integer and let $S = \{1, 2, \ldots, n\}$. A subset of $S$ is called *magical* if it has at least two elements and the arithmetic mean of all its elements is an integer. Prove that the number of magical subsets of $S$ is even.

**Problem 11.17.** Let $n$ be a positive integer. Show that the number of partitions of $n$ into parts not divisible by 3 is equal to the number of partitions of $n$ in which no three parts are equal.

**Problem 11.18.** Let $n$ be a positive integer. Each point $(x, y)$ of the plane whose coordinates are non-negative integers with $x+y < n$ is coloured red or blue, subject to the following condition: if a point $(x, y)$ is red, then so are all points $(x', y')$ with $x' \leq x$ and $y' \leq y$. Let $A$ be the number of ways to choose $n$ blue points with distinct $x$-coordinates, and let $B$ be the number of ways to choose $n$ blue points with distinct $y$-coordinates. Prove that $A = B$.

**Problem 11.19.** ★ A partition of $n$ is called *good* if, having sorted its parts into a non-increasing sequence, for each $k$ the $k$-th part is equal to the number of parts that are at least as large as

$k$. Show that the number of good partitions of $n$ is equal to the number of partitions of $n$ into distinct odd numbers.

**Problem 11.20.** ★ Let $\{a_n\}_{n\geq 0}$ be a non-decreasing, unbounded sequence of non-negative integers with $a_0 = 0$, and let $b_n$ be the number of members of the sequence not exceeding $n$. Prove that

$$(a_0 + a_1 + \cdots + a_m) + (b_0 + b_1 + \cdots + b_n) \geq (m+1)(n+1).$$

**Problem 11.21.** ★ An $n$-term sequence $(x_1, x_2, \ldots, x_n)$ in which each term is either 0 or 1 is called a binary sequence of length $n$. Let $a_n$ be the number of binary sequences of length $n$ containing no three consecutive terms equal to $0, 1, 0$ in that order, and let $b_n$ be the number of binary sequences of length $n$ containing no four consecutive terms equal to $0, 0, 1, 1$ or $1, 1, 0, 0$ in that order. Prove that $b_{n+1} = 2a_n$ for all positive integers $n$.

**Problem 11.22.** ★ Let $n$ be a positive integer. A *proper lattice path* is a path connecting two lattice points which uses only unit up-steps and unit right-steps. A proper path is called *good* if it does not pass above the line connecting its two endpoints (though it may contain points of that line). (a) Show that the number of proper paths from $(0, 0)$ to $(n, n)$ which are not good is equal to the number of proper paths from $(-1, 1)$ to $(n, n)$. (b) Use the result of the previous part to find the number of good proper paths from $(0, 0)$ to $(n, n)$.

**Problem 11.23.** ★ Show that the number determined in the previous problem is equal to each of the following.

(a) The number of binary trees with $n+1$ leaves. (A binary tree, defined inductively, consists of either a lone root point, called a leaf, or a root point connected to the roots of two smaller binary trees.)

(b) The number of ordered trees with $n + 1$ vertices. (An ordered tree, defined inductively, consists of either a lone root point, or a root point connected to the roots of one or more ordered trees which are arranged in a specific order; for example, if there are three branches from the root and one further branch, the cases where the extra branch is on the left, middle or right branch of the root are regarded as three different ordered trees.)

(c) The number of triangulations of an $(n + 2)$-gon.

(d) The number of ways to tile a stair-step shape of size $n$ (with $n, n-1, n-2, \ldots, 1$ fields respectively in each row, all rows starting from the same vertical line) with $n$ rectangles.

(e) The number of ways to pair $2n$ points on a circle into $n$ chords so that no two chords intersect.

(f) The number of ways to decompose the set of vertices of a regular $n$-gon into subsets whose convex hulls do not intersect each other.

(g) The number of ways of labelling a $2 \times n$ board with the numbers from 1 to $2n$ so that each row and each column is increasing.

## 11.2 Generating functions

**Definition 11.3 (Generating function).** The *generating function* of a sequence $(a_n)_{n\geq 0}$ is the formal power series $A(x) = \sum_{n\geq 0} a_n x^n$. Formal power series are added and multiplied term by term, and identities between them are equivalent to identities between the corresponding sequences.

**Theorem 11.4 (Generating functions for partitions).** *Let $p(n)$, $d(n)$ and $o(n)$ denote respectively the number of partitions of $n$, the number of partitions of $n$ into distinct parts, and the number of partitions of $n$ into odd parts. Then*

$$\sum_{n\geq 0} p(n)x^n = \prod_{k\geq 1} \frac{1}{1-x^k}, \qquad \sum_{n\geq 0} d(n)x^n = \prod_{k\geq 1} \left(1+x^k\right), \qquad \sum_{n\geq 0} o(n)x^n = \prod_{k\geq 1} \frac{1}{1-x^{2k-1}}.$$

*In each factor of the first product, the term $x^{jk}$ records that the part $k$ is used $j$ times; in the second product the factor $1+x^k$ records whether the part $k$ is used or not.*

**Definition 11.5 (Catalan numbers).** The *Catalan numbers* are defined by $c_0 = 1$ and $c_n = \sum_{i=0}^{n-1} c_i c_{n-1-i}$ for $n > 0$.

**Theorem 11.6 (The Catalan generating function).** *The generating function $C(x) = \sum_{n\geq 0} c_n x^n$ of the Catalan numbers satisfies the equation $C(x) = 1 + xC(x)^2$.*

**Technique 11.7 (The snake oil method).** To evaluate a sum whose value depends on a parameter, do not attack the individual sum. Instead form the generating function of the whole family of sums in that parameter, so that a generating function of two variables appears, reverse the order of summation, evaluate the inner sum in closed form, and finally read off the coefficient you wanted.

**Problem 11.24.** Find the generating functions of the following sequences in closed form: (a) $1, 1, 1, \ldots$; (b) $1, 2, 3, \ldots$; (c) $1^2, 2^2, 3^2, \ldots$; (d) $\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \ldots$, where $n$ is a given positive integer; (e) the Fibonacci sequence $0, 1, 1, 2, 3, 5, \ldots$.

**Problem 11.25.** By considering the function $f(x) = (1+x)^n$ for $x = -1, 1, 2$, find relations for the binomial coefficients.

**Problem 11.26.** By considering the equality $(1+x)^n(1+x)^k = (1+x)^{n+k}$, find a relation for the binomial coefficients.

**Problem 11.27.** Find closed forms of the expressions $\displaystyle\sum_{k=0}^{n} \binom{n}{k}^2$ and $\displaystyle\sum_{k=0}^{n} (-1)^k \binom{n}{k}^2$.

**Problem 11.28.** Find the generating function for the following sequences: (a) $a_n = \left\lfloor \frac{n+2}{2} \right\rfloor$; (b) $a_n = 3^n + n$.

**Problem 11.29.** Find the generating function of the Fibonacci sequence $F_0 = 0$, $F_1 = 1$, $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$, and use it to find a closed formula for $F_n$ (not depending on other terms) using the roots of $x^2 + x - 1 = 0$.

**Problem 11.30.** A sequence is defined by $a_1 = 1$ and $a_{n+1} = 2a_n + n$. Find the generating function of this sequence.

**Problem 11.31.** ★ Let $k$ be a fixed non-negative integer. Find the generating function of the sequence $a_n = \binom{n}{k}$, as a function of $k$.

**Problem 11.32.** By considering a generating function, show that any positive integer has a unique representation in the form (a) $a_0 + a_1 2 + \cdots + a_n 2^n$ with $a_0, a_1, \ldots, a_n \in \{0, 1\}$; (b) $a_0 + a_1 3 + \cdots + a_n 3^n$ with $a_0, a_1, \ldots, a_n \in \{0, 1, 2\}$; (c) $a_0 + a_1 3 + \cdots + a_n 3^n$ with $a_0, a_1, \ldots, a_n \in \{-1, 0, 1\}$.

**Problem 11.33.** Let $a(n)$ be the number of representations of a positive integer $n$ as a sum of 1's and 2's, taking the order into account. For example, since $4 = 1 + 1 + 2 = 1 + 2 + 1 = 2 + 1 + 1 = 2 + 2 = 1 + 1 + 1 + 1$, we have $a(4) = 5$. Let $b(n)$ be the number of representations of $n$ as a sum of integers greater than 1, again taking the order into account; for example, since $6 = 4 + 2 = 2 + 4 = 3 + 3 = 2 + 2 + 2$, we have $b(6) = 5$. Show that $a(n) = b(n + 2)$.

**Problem 11.34.** Six regular dice are rolled. What is the probability that the sum of the six numbers shown is equal to 21?

**Problem 11.35.** Objects $A$ and $B$ move simultaneously in the coordinate plane via a sequence of steps, each of length one. Object $A$ starts at $(0, 0)$, and each of its steps is either right or up, both equally likely; object $B$ starts at $(5, 7)$, and each of its steps is either left or down, both equally likely. What is the probability that the objects meet?

**Problem 11.36.** Claudia has two regular tetrahedra and she decides to make them into two fair dice. On each face of the dice some positive integer is written, in such a way that the numbers $1, 2, 3, 4$ do not all appear on one die. Claudia is interested in the distribution of the sums of the numbers on the bottom faces when the dice are thrown; she wants this distribution to be the same as that of two tetrahedral dice with the numbers $1, 2, 3, 4$ written on the faces of each die. Can Claudia achieve her goal?

**Problem 11.37.** Let $n$ be a positive integer with $n \geq 3$. Each of $n$ students has a different height. In how many ways can they line up in a row so that the heights of any three of them (not necessarily next to each other), read from left to right, are not in the order medium, tall, short?

**Problem 11.38.** Adrian has the coins $C_1, C_2, \ldots, C_n$. For each $k$, the coin $C_k$ is biased so that, when tossed, it has probability $\frac{1}{2k+1}$ of showing heads. If the $n$ coins are tossed, what is the probability that the number of heads is odd? Express the answer as a rational function of $n$.

**Problem 11.39.** The bottom right unit square is removed from a $2 \times 2$ unit grid to form a tromino. Find the number of ways in which $k$ trominoes can be placed, without being rotated and without overlapping, on a $3 \times n$ rectangular unit grid. (Each tromino covers exactly three unit squares of the grid.)

**Problem 11.40.** For a set $A$ let $s(A)$ denote the sum of the elements of $A$; if $A = \emptyset$ then $|A| = s(A) = 0$. Let $S = \{1, 2, \ldots, 1999\}$ and, for $r = 0, 1, 2, \ldots, 6$, define $T_r = \{T : T \subseteq S \text{ and } s(T) \equiv r \pmod 7\}$. For each $r$, find the number of elements of $T_r$.

**Problem 11.41.** Let $p$ be an odd prime number. How many $p$-element subsets $A$ of $\{1, 2, \ldots, 2p\}$ are there such that the sum of the elements of $A$ is divisible by $p$?

**Problem 11.42.** Let $(a_1, a_2, \ldots, a_n)$ and $(b_1, b_2, \ldots, b_n)$ be two different unordered $n$-tuples of integers such that the sequences of all pairwise sums

$$a_1 + a_2,\ a_1 + a_3,\ \ldots,\ a_{n-1} + a_n \qquad \text{and} \qquad b_1 + b_2,\ b_1 + b_3,\ \ldots,\ b_{n-1} + b_n$$

coincide up to permutation. Prove that $n$ is a power of two.

**Problem 11.43.** Let $a_0, a_1, a_2, \ldots$ be an increasing sequence of non-negative integers such that every non-negative integer can be expressed uniquely in the form $a_i + 2a_j + 4a_k$, where $i$, $j$ and $k$ are not necessarily distinct. Determine $a_{1998}$.

**Problem 11.44.** Determine whether there is a subset $X$ of the integers with the following property: for any integer $n$ there is exactly one solution of $a + 2b = n$ with $a, b \in X$.

**Problem 11.45.** Use generating functions to prove that the number of ways of representing $n$ as a sum of distinct natural numbers is equal to the number of ways of representing $n$ as a sum of odd numbers.

**Problem 11.46.** ★ Let $p$ be an odd prime number. Use generating functions to find the number of subsets of $\{1, 2, \ldots, p\}$ whose sum of elements is divisible by $p$.

**Problem 11.47.** ★ Find the generating function for the Catalan numbers and use it to find a formula for $c_n$ as a function of $n$.

**Problem 11.48.** ★ Show that each of the numbers described in parts (a)–(f) of Problem 11.23 satisfies the recurrence relation of the Catalan numbers.

**Problem 11.49.** ★ Let $n$ be a positive integer and fix $2n$ distinct points on a circle. Determine the number of ways to connect the points with $n$ arrows (oriented segments) such that all of the following conditions hold: each of the $2n$ points is a startpoint or an endpoint of an arrow; no two arrows intersect; and there are no two arrows $\overrightarrow{AB}$ and $\overrightarrow{CD}$ such that $A$, $B$, $C$ and $D$ appear in clockwise order around the circle (not necessarily consecutively).

## 11.3 Double counting

**Technique 11.8 (Double counting).** Choose a finite set of objects — most often a set of incidences, that is, of pairs (object, feature) — and count it in two different ways. The two answers must agree, and the resulting identity (or, when one of the two counts is only an estimate, the resulting inequality) is the content of the proof. Typical incidence sets are (segment, triangle), (vertex, edge), (person, pair of acquaintances) and (candidate, pair of judges).

**Definition 11.9 (Triangulation of a polygon).** A dissection of a (not necessarily convex) $n$-gon into triangles by non-crossing diagonals, where the diagonals are allowed to share endpoints, is called a *triangulation* of the $n$-gon.

**Theorem 11.10 (Size of a triangulation).** *Every triangulation of an $n$-gon consists of exactly $n-2$ triangles and uses exactly $n-3$ diagonals.*

**Problem 11.50.** In a tournament $2n+1$ people took part; each two participants played exactly one game and no ties were allowed. A 3-element subset of participants is called a *triangle* if these players can be denoted $A$, $B$, $C$ so that $A$ defeated $B$, $B$ defeated $C$ and $C$ defeated $A$. Suppose that everyone won and lost equally many times, so exactly $n$ games each. Prove that the total number of triangles is $\frac{n(n+1)(2n+1)}{6}$.

**Problem 11.51.** A rectangle is divided by segments parallel to its sides into small rectangles. Every point which is a vertex of exactly two small rectangles is called a *$T$-point*. Prove that the total number of $T$-points is even.

**Problem 11.52.** We are given a triangulated $n$-gon with $n \geq 4$. Colour green all triangles sharing exactly two sides with the border of the $n$-gon, and colour red all triangles all of whose sides are diagonals of the $n$-gon. Prove that there are exactly two more green triangles than red ones.

**Problem 11.53.** Given is a group of $2n$ people. Call three people a *cherry* if among them there is exactly one who knows the remaining two. Determine the largest possible number of cherries in the group.

**Problem 11.54.** There are $n$ points in the plane such that no three of them are collinear. Prove that the number of triangles of area 1 with vertices among these points is not greater than $\frac{2}{3}(n^2-n)$.

**Problem 11.55.** Let $n$ and $k$ be positive integers and let $S$ be a set of $n$ points in the plane such that: no three points of $S$ are collinear, and for every $P \in S$ there are at least $k$ points of $S$ equidistant from $P$. Prove that $k < \frac{1}{2} + \sqrt{2n}$.

**Problem 11.56.** Given are positive integers $n$ and $k$. At a party $n$ guests met, some of whom are friends. It is known that nobody has more than $2k$ friends, but every pair of guests who are not friends have at least $k$ common friends. Prove that $n \leq 6k$.

**Problem 11.57.** Show that a graph with $n$ vertices and $m$ edges has at least $\frac{m}{3n}(4m-n^2)$ triangles.

**Problem 11.58.** Exactly $2n+1$ points of a circle are painted green, where $n \geq 1$ is an integer. Determine the greatest possible number of acute triangles with three green vertices.

**Problem 11.59.** There are $n$ people at a party. Prove that it is possible to choose two of them in such a way that among the remaining $n-2$ people there are at least $\lfloor \frac{n}{2} \rfloor - 1$, each of whom is either a common friend or a common stranger of the two chosen people.

**Problem 11.60.** At a contest there are $m$ candidates and $n \geq 3$ judges, where $n$ is an odd integer. Each candidate is evaluated by each judge as either passing or failing. Suppose that each pair of judges agrees on at most $k$ candidates. Prove that $\frac{k}{m} \geq \frac{n-1}{2n}$.

**Problem 11.61.** Given are $n$ pairwise intersecting convex $k$-gons in the plane such that for every pair of these $k$-gons at least one of them contains a vertex of the other. Prove that there is a point of the plane belonging to at least $1 + \frac{n-1}{2k}$ of these $k$-gons.

**Problem 11.62.** A table with $m$ rows and $n$ columns is given, where $n$ is even and where every entry is either 0 or 1. Every two rows differ in at least $1 + \frac{n}{2}$ cells. Prove that $m \leq 1 + \frac{n}{2}$ and that equality cannot hold when $n$ is divisible by 4.

**Problem 11.63.** At a competition of $n$ students, for each group of 3 students there is a student whom they conspire to get rid of. Any student who conspires to get rid of student $A$ in some group of three is called an *enemy* of $A$. Prove that there is a student with at least $\sqrt[3]{(n-1)(n-2)}$ enemies.

**Problem 11.64.** All vertices of a regular 2023-gon are coloured in two colours, blue and red, and it is known that there are $n$ red vertices. Find the number of isosceles triangles having all three vertices of one and the same colour. (In particular, this number depends only on $n$ and not on the way the colouring is made.)

**Problem 11.65.** ★ At a social gathering of students, every boy knows at least one girl. Prove that there exists a set $S$ containing at least half the students at the gathering such that every boy in $S$ knows an odd number of girls in $S$.

## 11.4 The extremal principle and greedy algorithms

*Revisits the pigeonhole principle and the extremal principle (see Chapter 6); the problems below push it further.*

**Technique 11.11 (Take the most suitable object).** When you are asked to find something very small or very large — for instance "nothing" or "everything" — it is often useful to consider the situation in which that quantity is already minimised or maximised, and then to show that this extremal configuration must have the required property: if it did not, one could improve it, contradicting extremality.

**Technique 11.12 (Greedy algorithms).** Build the required object step by step, taking at each step whatever is best available at that moment, and never revising an earlier choice. Such a construction is analysed either by showing that the process cannot get stuck before the goal is reached, or by comparing the greedy object with an arbitrary competitor and proving that the greedy one is at least half as good.

**Problem 11.66.** In a certain country, every pair of cities is connected by exactly one transportation route: either by train or by bus. Prove that there exists a mode of transportation (either train or bus) that allows one to travel from any city in the country to any other city using only that mode.

**Problem 11.67.** A country has 1000 cities, some of which are connected by roads, and it is possible to travel from any city to any other city by roads. What is the smallest $k$ for which the following statement holds: in any such country it is possible to designate $k$ cities as capitals so that every non-capital city is connected by a road to at least one capital?

**Problem 11.68.** There are 100 floppy disks and 100 labels of different colours. Prove that it is possible to attach the labels to the floppy disks in such a way that all "duplicates" — that is, situations where a floppy disk and its label have the same colour — are of the same colour.

**Problem 11.69.** In a country, some pairs of cities are connected by one-way direct flights (there is at most one flight between any two cities). A city $A$ is said to be *accessible* from a city $B$ if it is possible to travel from $B$ to $A$, possibly with layovers. It is known that for any two cities $P$ and $Q$ there exists a city $R$ such that both $P$ and $Q$ are accessible from $R$. Prove that there exists a city from which all cities in the country are accessible. (A city is always accessible from itself.)

**Problem 11.70.** In a graph, the degree of each vertex is at least 10. Prove that the graph contains a simple path of length at least 10, the length of a path being measured in edges. Conclude that there is a simple cycle of length at least 11.

**Problem 11.71.** In a group of 1000 people speaking different languages, any three people can communicate, possibly with one of them acting as a translator for the other two. Prove that it is possible to divide the group into pairs such that in each pair the two people share a common language.

**Problem 11.72.** In a country, some cities are connected by two-way direct flights. No city has more than 10 flights departing from it, and the country has more than 195 flights in total. Prove that there are 11 flights, no two of which share a common endpoint.

**Problem 11.73.** There are 46 points in a plane, no four of which lie on the same straight line. Prove that among them one can select 10 points such that no three of them lie on the same straight line.

**Problem 11.74.** In a graph with $2n$ vertices there are no loops or multiple edges, and the degree of each vertex is at most 19. Prove that the vertices of the graph can be coloured with 5 colours in such a way that no more than $3n$ edges have both endpoints of the same colour.

**Problem 11.75.** You are given 2000 numbers with an even sum, each of which is equal to one of the numbers from 1 to 100, and every one of these hundred values appears at least once. Prove that these numbers can be split into two groups with equal sums.

**Problem 11.76.** Some islands in an archipelago are connected by water-taxi routes. It is possible to travel from any island to any other, possibly with transfers, and among any 100 islands there are at least two which are directly connected by a water taxi. Prove that the islands can be divided among 98 travel agencies in such a way that each travel agency can organise a tour using water taxis which visits only its own assigned islands, without visiting any island more than once.

**Problem 11.77.** Given $n$ subsets $A_1, \ldots, A_n$ of a 2020-element set, each containing at least four elements, such that the intersection of any two of them contains at most two elements. Prove that it is possible to select a 24-element subset that does not contain any $A_i$ as a subset.

**Problem 11.78.** Twenty-six teams participated in the League of Discoveries. Each team played against every other team exactly once and no match ended in a draw. Prove that it is possible to select 4 teams such that every remaining team lost to at least one of the selected teams.

**Problem 11.79.** At a convention 5000 film enthusiasts gathered, each of whom had watched at least one film. They are to be divided into groups of two types: either groups in which all members have seen the same film, or groups in which each member discusses a film they watched that no one else in the group has seen. Prove that it is possible to organise everyone into exactly 100 groups. (Groups with a single person are allowed; such a person writes a review of a film they watched.)

**Problem 11.80.** Masha has 7 double-sided cards, on each of which all the numbers from 1 to 100 are written exactly once, some on one side and the rest on the other. Prove that Masha can always arrange the cards so that all numbers from 1 to 100 appear on the visible sides.

**Problem 11.81.** A total of 200 participants solved 6 problems in an olympiad, and each problem was solved by at least 120 participants. Prove that there exist two participants who solved all the problems.

**Problem 11.82.** A set of positive numbers $a_1, a_2, \ldots, a_n$ is given. You need to select some numbers from this set so that their sum is less than 1000. It is known that there exists a way to choose numbers whose sum equals $k < 1000$. Prove that a greedy algorithm will find a sum that is at least $k/2$.

**Problem 11.83.** There are 50 stones of different weights lying in a row on a table. Bob and Alice take turns picking one stone from either end of the row, with Alice going first. (a) Give an example to show that if Alice always picks the heaviest available stone (using a greedy strategy), she may fail to collect a total weight greater than or equal to Bob's. (b) Prove that Alice can guarantee collecting a total weight that is at least equal to Bob's.

**Problem 11.84.** Thirty-six students agreed to chat after lights out, and each student sent a message to exactly three other students. Prove that it is possible to select 6 students such that none of them sent a message to the other five.

**Problem 11.85.** Sixteen students participated in five olympiads, and each of them received a diploma of first, second or third degree in each olympiad. Prove that there exist three students such that in every olympiad at least two of them received the same diploma.

**Problem 11.86.** Some $n$ people formed $n$ secret societies, each consisting of exactly three people. Prove that a government can arrest $\lfloor n/3 \rfloor$ people in such a way that every secret society has at least one member who is not arrested.

**Problem 11.87.** A company has several employees, and the salary of each employee is an integer (different employees may have different salaries). Collectors brought to the company 10 coins of 1 unit, 10 coins of 2 units, ..., 10 coins of 200 units, and the total amount of money brought matches the total sum of all the employees' salaries. What is the maximum number of employees for which it is guaranteed that each of them can receive their exact salary?

**Problem 11.88.** There are 100 numbers on a board: the first one is between 0 and 99, the second one is between $-1$ and 98, ..., the hundredth one is between $-99$ and 0. Prove that the sum of some of these numbers is equal to 0.

**Problem 11.89.** An alpaca chooses a set $A$ consisting of $m$ non-negative integers and informs the badger about this set. Then the alpaca and the badger play the following game. They have an expression

$$E = v_0 \cdot 2^0 + v_1 \cdot 2^1 + \cdots + v_{99} \cdot 2^{99},$$

where each $v_i$ is initially undefined. On each turn a player replaces one undefined $v_i$ with either 0 or 1. The players take turns, with the badger going first. After 100 turns the value of the expression $E$ is calculated. What is the smallest $m$ such that the alpaca can ensure that the value of $E$ belongs to $A$?

**Problem 11.90.** Consider an empty table with $2^{100}$ rows and 100 columns. Alice and Eve take turns filling empty cells in the first row of the table, with Alice going first: on each turn a player

selects an empty cell and fills it with either a 0 or a 1. After the first row is filled they move to the second row, and so on, and in each new row Alice moves first. The game ends when all rows are filled. Alice wants to maximise the number of distinct rows in the table, while Eve wants to minimise this number. How many distinct rows will there be if both play optimally?

## 11.5 Processes: step by step

**Technique 11.13 (Start the process yourself).** A statement about a finished configuration is often much easier to prove in a dynamic form: instead of treating the whole configuration at once, build it up one object at a time and check that the desired property survives every single step. Compare "100 rooks can be 5-coloured" with "the rooks are placed one by one and each one is coloured as it arrives": the second version only ever asks about one rook. If the problem does not come with a process of its own, invent one — and choose the order of the steps to your advantage.

**Technique 11.14 (Step-by-step reduction).** It is sometimes useful to split an algorithm into similar steps, each of which reduces some quantity. After a while this quantity becomes small and the remaining situation can be handled by hand. Writing the solution up as an induction may help — or sometimes make things harder.

**Problem 11.91.** (a) On a grid plane there are 100 rooks. Prove that they can be coloured with 5 colours so that no two rooks of the same colour attack each other. (b) Petya places rooks one by one on a chessboard, and Vasya colours each rook as it is placed with one of 5 colours. Prove that Vasya can always colour them in such a way that no two rooks of the same colour attack each other.

**Problem 11.92.** Rooks are placed on an infinite grid board. Prove that they can be coloured with 3 colours so that no two rooks of the same colour attack each other.

**Problem 11.93.** A single-round tennis tournament took place with 100 participants. Each match victory earned 1 point, while a loss gave 0 points, and draws do not exist in tennis. Sergey can see the total number of points each participant scored, and he is trying to determine the results of all the matches. To do this he selects a pair of participants he has not chosen before and makes a guess about the result of their match; he is then told whether he was correct. He continues this process until all possible pairs have been considered. How should Sergey proceed in order to predict correctly the outcome of more than half of all the matches?

**Problem 11.94.** The numbers $1, 2, \ldots, 100$ are written on a board and Petya erases them one by one. Prove that he can do this in such an order that the sum of the remaining numbers is always a composite number.

**Problem 11.95.** A school has 17 clubs, and for any two clubs there are at least two students who attend both. Prove that it is possible to colour some students red and some blue so that each club has students of both colours.

**Problem 11.96.** There are 100 lines in general position in a plane. The points where these lines intersect are called *nodes*. Prove that the nodes can be coloured with 3 colours so that no node has a neighbour on the same line with the same colour.

**Problem 11.97.** A square is divided into 100 smaller squares (not necessarily of equal size). Prove that these squares can be coloured with (a) 9 colours; (b) 8 colours, so that any two squares sharing at least one point have different colours.

**Problem 11.98.** Is it possible to colour all natural numbers with two colours so that the sum of any two distinct numbers of the same colour is never a power of two?

**Problem 11.99.** Let $A$ be a non-empty subset of $\{1, 2, \ldots, 2023\}$. Prove that there exists a subset $B$ of $\{1, 2, \ldots, 2023\}$ such that $A$ consists of exactly those elements of $\{1, 2, \ldots, 2023\}$ which divide an odd number of elements of $B$.

**Problem 11.100.** Let $m < n$ be natural numbers. A *pattern* is a triple of natural numbers $x < y < z$ such that $\{z - y, y - x\} = \{m, n\}$. Prove that the natural numbers can be partitioned into such patterns.

**Problem 11.101.** (a) A collection of squares has a total area of at most $1/2$. Prove that they can be packed without overlap inside a square of side length 1. (b) A collection of squares has a total area of $S$. What is the smallest $S$ that guarantees that they can cover a square of side length 1?

**Problem 11.102.** Amina has a deck of cards numbered from 1 to 101. Initially she holds 50 cards in her hand, while the remaining cards are laid face up on the table. In one move she can swap all 50 cards in her hand for any 50 cards from the table. Amina wants to obtain all the cards numbered from 1 to 50 in her hand in at most $N$ moves. What is the minimum $N$ that guarantees that she can always achieve this, regardless of her starting hand?

**Problem 11.103.** There are two bags: one contains $m$ stones, the other contains $n$ stones, where $m$ and $n$ are natural numbers. Two operations are allowed: remove the same number of stones from both bags; or double the number of stones in one of the bags. Is it possible to empty both bags?

**Problem 11.104.** Two natural numbers $p$ and $q$ are written on a board. The following operations are allowed: increase both numbers by 1; or, if one of the numbers is a perfect square, replace it by its square root. For which initial values of $p$ and $q$ can we eventually make them equal?

**Problem 11.105.** Each of five bags contains at least 2019 toy cats. In one move it is allowed to pick two non-empty bags, take one cat from each, and place both cats into a third bag. Prove that all the cats can be gathered into a single bag.

**Problem 11.106.** The students locked their teacher in a circular room with 2017 doors, of which 2016 are locked. The teacher can check whether three consecutive doors are open or locked. If at least one of the three is open, the teacher escapes; if all three are locked, the students lock the previously open door and unlock one of its neighbours. Prove that the teacher will eventually escape.

**Problem 11.107.** The prankster Vasya lives in an $N$-floor building with an elevator. The elevator has buttons for every floor from 1 to $N$, but it does not always respond: if Vasya is on floor $a$ and presses the button for floor $b$, the elevator moves there only if $a + b$ is divisible by 2017, or $a - b$ is divisible by 2018. Vasya claims that it is still possible to travel between any two floors using this elevator. What is the smallest $N$ for which this is true?

**Problem 11.108.** An adventurer arrives at the Forbidden Island, which is an $11 \times 11$ square of cells. Every night, two unflooded cells not containing the adventurer sink under water. Every

day, the adventurer moves to a neighbouring unflooded cell (adjacent by a side or a diagonal) and drains all flooded cells adjacent to his new cell by a side. If the adventurer cannot move, he perishes. Initially all cells are dry. Can the adventurer survive on the island for an entire year, regardless of how the flooding occurs?

**Problem 11.109.** One hundred wizards are arranged in a circle. Some of them are knights, while the others are liars, and every wizard knows who is who. Any wizard can declare: "I am changing the nature of my neighbours! Abracadabra!" If the wizard is a knight, their statement is true and each of their two neighbours is transformed into the opposite type (a knight becomes a liar and a liar becomes a knight); if the wizard is a liar, nothing actually happens. Prove that, regardless of the initial setup, the wizards can act in such a way that at most two knights remain in the end.

## 11.6 Invariants and monovariants revisited

*Revisits invariants and monovariants (see Chapter 6); the problems below push it further.*

**Problem 11.110.** A group of people is present in a building, and each person has at most three enemies. Prove that they can be divided into two rooms in such a way that everybody has at most one enemy in his room.

**Problem 11.111.** There are $2n$ people in a building, and each person has at most $n-1$ enemies. Prove that they can be seated around a circular table so that nobody sits next to his enemy.

**Problem 11.112.** Five integer numbers are assigned to the vertices of a pentagon so that their sum is positive. If $x$, $y$, $z$ are numbers assigned to three consecutive vertices and $y < 0$, then it is allowed to replace the triple $(x, y, z)$ by $(x + y, -y, y + z)$. We repeat this procedure as long as there are negative numbers. Is it possible that this process lasts indefinitely? [IMO 1986 P3]

**Problem 11.113.** There are $n$ real numbers written on a blackboard. In one step it is allowed to choose any two numbers, say $a$ and $b$, erase them, and write $\frac{a+b}{4}$ instead. After $n-1$ steps only one number remains. Assume that in the beginning all the numbers are equal to 1. Prove that the number at the end is no less than $\frac{1}{n}$.

**Problem 11.114.** There are $n$ coins placed around a circle, equally spaced. In one step it is allowed to choose any two coins and move them by one place in opposite directions. The goal is to move all the coins to the same position. For which $n$ can this be achieved?

**Problem 11.115.** A game is played on the squares $(x, y)$ with $x, y \geq 0$. If there is a chip on $(x, y)$ while $(x + 1, y)$ and $(x, y + 1)$ are empty, then the chip on $(x, y)$ can be removed and each of $(x + 1, y)$ and $(x, y + 1)$ receives a chip. The goal is to reach a position with no chips on any of the squares $(0, 0)$, $(0, 1)$, $(0, 2)$, $(1, 0)$, $(1, 1)$, $(2, 0)$. Is it possible to achieve this if the initial position is: (a) one chip on each of the six mentioned squares; (b) only one chip, on $(0, 0)$?

**Problem 11.116.** A game is played on the squares $(x, y)$ with $x, y \in \mathbb{Z}$. In the initial position there are $n^2$ chips forming an $n \times n$ square. A move consists of choosing three consecutive squares in a horizontal or vertical direction, such that the first two of them have a chip each and the third one is empty, and then removing the two chips and placing a chip on the third square. Find all values of $n$ for which it is possible to finish the game with exactly one chip remaining. [IMO 1993 P3]

**Problem 11.117.** A board $2005 \times 2005$ is given, and a chip is placed on each square. In one move it is allowed to remove a chip from the board, provided that the number of its neighbours is even and positive; by neighbours we mean squares with at least one common vertex. (a) Find the minimal possible number of chips that can remain on the board. (b) Prove that if that minimal number of chips remained, then no two of them are neighbours.

**Problem 11.118.** A chip is placed on the square $(1, 1)$. In one step it is allowed either to double one of its coordinates, or to decrease the larger coordinate by subtracting the smaller coordinate from it. Find the set of all squares that the chip can reach.

**Problem 11.119.** A red and a blue chip are given on a line, in that order. In one move it is allowed either to add or to remove two neighbouring chips of the same colour. Is it possible to reach a position with a blue and a red chip in that order, with no other chips present?

**Problem 11.120.** One digit (0 or 1) is written on each vertex of a regular 12-gon. A move consists of choosing a vertex carrying a 1 and changing each of the two neighbouring digits, from 0 to 1 or from 1 to 0. Find all initial configurations from which it is possible, after some moves, to have only one digit 1 left.

**Problem 11.121.** In a group of $n$ people, each one had a different ball. They performed a sequence of swaps; in each swap, two people swapped the ball they had at that moment. Each pair of people performed at least one swap, and in the end each person had the ball they had at the start. Find the least possible number of swaps if (a) $n = 5$; (b) $n = 6$. [JBMO SL 2011]

**Problem 11.122.** There is an infinite cell board, the side of each cell having length 1. Each of the unit segments (the sides of the cells) is coloured blue or yellow, and there are only finitely many blue ones. Two unit segments are called *neighbouring* if they share an end. Each second, all of the unit segments that have an even number of blue neighbours become yellow, whereas all the unit segments that have an odd number of blue neighbours become blue. Prove that there will always be a blue unit segment if and only if there is one after the first second.

## 11.7 Graphs: first steps

**Definition 11.15 (Graph).** A *graph* is a pair $(V, E)$, where $V$ is the set of *vertices* and $E$ is a set of unordered pairs of vertices, called the *edges* of the graph. The *degree* of a vertex is the number of edges meeting at it.

**Corollary 11.16 (Handshake lemma).** *In any graph the number of vertices of odd degree is even.*

**Theorem 11.17 (Existence of cycles).** *In any graph with $n$ vertices and at least $n$ edges there exists a cycle. In particular, if every vertex of a graph has degree at least two, then the graph has a cycle.*

**Definition 11.18 (Tree).** A graph $G$ with $n$ vertices is called a *tree* if $G$ is connected and has no cycles.

**Theorem 11.19 (The three properties of a tree).** *Consider the following three properties of a graph $G$ with $n$ vertices: (1) $G$ has $n-1$ edges; (2) $G$ is connected; (3) $G$ has no cycles. Then any two of these properties imply the third one: (1) and (2) imply (3); (1) and (3) imply (2); (2) and (3) imply (1).*

**Definition 11.20 (Perfect matching).** In a graph with $2n$ vertices, a set of $n$ edges no two of which have a common vertex is called a *perfect matching*.

**Problem 11.123.** Prove that any graph with at least two vertices has two vertices of the same degree, that is, two vertices which are connected to the same number of other vertices.

**Problem 11.124.** Let a graph be given such that there are no closed paths in it and one can reach any vertex from any other going along the edges of the graph. Prove that one can colour the vertices of the graph in 2 colours so that no two vertices of the same colour are connected. (Such a graph is called a *bipartite* graph.)

**Problem 11.125.** Prove that a graph is bipartite if and only if all its cycles are of even length.

**Problem 11.126.** There are $n$ towns in a country and some of them are connected by roads in such a way that any town can be reached from any other town. Prove that it is possible to build hospitals in at most $\frac{n}{2}$ towns in such a way that from any town without a hospital there is a road to a town with a hospital.

**Problem 11.127.** In a connected graph all of whose vertices have even degree, one edge is deleted. Prove that the graph remains connected.

**Problem 11.128.** Prove that in any tournament of $n$ players there exists a chain of players $A_1, A_2, \ldots, A_n$ such that $A_i$ wins the game against $A_{i+1}$ for $i = 1, 2, \ldots, n-1$.

**Problem 11.129.** In a tournament everyone played exactly one game with everyone else and there were no ties. Prove that there exists a player $K$ with the following property: every player $A \neq K$ was defeated by $K$ directly or indirectly, meaning that either $K$ won the game against $A$, or $K$ won against some $B$ who won against $A$.

**Problem 11.130.** Prove that the maximum number of edges in a graph with $n$ vertices and no triangles is achieved by the complete bipartite graph with almost equal numbers of vertices in each part.

**Problem 11.131.** Every vertex of a graph $G$ with $2n$ vertices has degree at least $n$. Prove that $G$ has a perfect matching.

**Problem 11.132.** Ivan has 28 classmates, all of whom have different numbers of friends in the class. Determine the number of friends of Ivan.

**Problem 11.133.** For $n \geq 2$ let $0 \leq d_1 \leq d_2 \leq \cdots \leq d_n$ be integers. Show that $(d_i)_{i=1}^n$ is the degree sequence of a tree if and only if $\sum_{i=1}^n d_i = 2n - 2$.

**Problem 11.134.** In a group of $n$ people it is true that in any subgroup of 4 people there is one who knows the other three people. Prove that there is a person in this group who knows everyone else.

**Problem 11.135.** There is a bipartite graph with maximum degree $\Delta$. Prove that it is possible to colour its edges in $\Delta$ colours in such a way that no two edges of the same colour share a vertex.

**Problem 11.136.** There is a group of people in which every two people have precisely 5 common friends (friendships are mutual). Prove that the total number of friendships is divisible by 3.

**Problem 11.137.** There is a country where exactly 3 railways come out of each city. Prove that it is possible to distribute the railways between two companies in such a way that each of the cities has railways of both companies coming out of it.

**Problem 11.138.** Suppose there is a graph on $2n$ vertices with strictly more than $n^2$ edges. Show that it must have at least $n/3$ triangles.

**Problem 11.139.** Ali went to a country in which there are more than 100 cities and the price of flying between different pairs of cities is different (the price of flying from $A$ to $B$ is the same as the price of flying from $B$ to $A$). However, according to Ali, the price of any cyclic route around all the cities, visiting each city exactly once, is the same. Is it even theoretically possible that Ali was not lying?

**Problem 11.140.** There is a graph where each vertex carries a lamp that can be either off or on. In one move it is possible to change the state of some lamp together with the states of all lamps that are connected to it by an edge. Initially all the lamps are switched off. Madi managed to switch them all on in $m$ steps, while Hadi managed to switch them all on in $h$ steps. Prove that $m - h$ is even.

**Problem 11.141.** There is a connected graph with 100 vertices and 199 edges. Prove that it is possible to erase all the edges of some cycle in such a way that the remaining graph will still be connected.

**Problem 11.142.** There is a graph in which each vertex has degree 100. Prove that it is possible to erase some of the edges so that the remaining graph has only vertices of degree 2.

**Problem 11.143.** Let $A$ and $B$ be two countries such that each city is connected to exactly $k \geq 2$ cities in the other country via an interstate road. Interstate roads connect a city of country $A$ with a city of country $B$ with no stops in between, and no pair of cities is connected by more than one interstate road. Assuming that each city is reachable from every other city using the interstate roads, is it possible to destroy a single road so that this no longer holds?

**Problem 11.144.** In each of three schools there are $n$ students, and each student knows at least $n + 1$ students from the other two schools. Prove that there are three students, one from each school, who all know each other.

**Problem 11.145.** Let a *$k$-group* be a collection of $k$ people who all know each other (knowing someone is a symmetric relation). Assume that at a party there are no 5-groups and that no two 3-groups are disjoint, in other words every two 3-groups have at least one common person. Prove that there are two people whose departure would result in there being no 3-group left at the party.

**Problem 11.146.** A graph is called *good* if among any three vertices there are at least two which are connected with each other. A complete graph of order 3 will be called a triangle, and also a *windmill of order* 1; a windmill of order $n > 1$ is a collection of $n$ triangles that intersect only in a unique vertex common to all of them, in other words a collection of $2n + 1$ distinct

vertices $a, b_1, b_2, \ldots, b_n, c_1, c_2, \ldots, c_n$ such that $ab_i$, $ac_i$ and $b_ic_i$ are all edges of the graph for every $i = 1, 2, \ldots, n$. Find $f(n)$, the smallest number such that every good graph with $f(n)$ vertices contains a windmill of order $n$.

**Problem 11.147.** Let a complete graph on $n$ vertices be given, that is, a graph in which every pair of vertices is connected. In each move we select a cycle of length 4, that is four vertices $A_1, A_2, A_3, A_4$ such that $A_1A_2$, $A_2A_3$, $A_3A_4$ and $A_4A_1$ all belong to the graph, and remove one of these edges, say $A_1A_2$. We continue making moves until there are no more cycles of length 4 in the graph. Find the lowest number of edges we can obtain.

**Problem 11.148.** On the continent of Graphia there are $n \geq 3$ countries. In year 1 of the New Mathematical Era we find that some pairs of countries are at war. Each country is at war with at least one other country, and it is impossible to divide the countries into two groups such that no two countries in opposing groups are at war. Starting from year 2, each year two countries that are at war sign a peace treaty; unfortunately, each country that is at war with exactly one of the two countries that signed the peace treaty will then also declare war on the other one. Additionally, if at any point in time we divide the countries into two arbitrary non-empty groups, after a finite amount of time two countries in the opposite groups will sign a peace treaty. Prove that after a finite amount of time there will be a country that is at war with all the other countries.

**Problem 11.149.** At a party there are $n$ guests, $n \geq 5$. It is known that among every three guests there are two people who are friends. Prove that at least $n/2$ guests can sit at a round table in such a way that every two people sitting next to each other are friends.

**Problem 11.150.** There are $2n$ pools ($n \geq 2$) arranged in a circle and a hoop between every pair of neighbouring pools. Initially there is one dolphin in each pool. During an acrobatic show the dolphins perform a series of stunts; in a single stunt two dolphins from neighbouring pools swap places by jumping simultaneously through the hoop between their pools. Suppose that each pair of dolphins performed exactly one stunt together. Prove that some hoop has never been used.

**Problem 11.151.** In a tournament every two participants played exactly one game. There were no ties and each person won at least once. Suppose there exist four participants $A$, $B$, $C$, $D$ such that $A \to B \to C \to A$ and $A \to D$, $B \to D$, $C \to D$, where $X \to Y$ means that $X$ won against $Y$. Prove that there exist participants $A'$, $B'$, $C'$, $D'$ (possibly some of them among $A$, $B$, $C$, $D$) such that $A' \to B' \to C' \to A'$ and $D' \to A'$, $D' \to B'$, $D' \to C'$.

**Problem 11.152.** Determine all positive integers $n$ with the following property: there exists a permutation $(x_1, x_2, \ldots, x_n)$ of the set $\{1, 2, \ldots, n\}$ satisfying, for every $i \in \{1, 2, \ldots, n\}$, the condition that $x_{i+1}$ is one of the numbers $2x_i$, $2x_i - 1$, $2x_i - n$, $2x_i - n - 1$, where $x_{n+1} = x_1$.

**Problem 11.153.** Every Martian has three hands and some number of antennae. Each Martian took some other Martian's hand in each of their three hands. It turned out that between any two Martians holding hands, one of them has 6 times as many antennae as the other. Can the total number of antennae be 2006?

**Problem 11.154.** Each child can be either noisy or quiet. Consider a group $M$ of children in which each noisy child has at least one friend who is quiet. It is known that during a lecture a noisy child sits quietly if and only if there is an odd number of his friends present who are quiet. Prove that it is possible to choose a subgroup of $M$ for the lecture such that all noisy children in it will sit quietly.

**Problem 11.155.** ★ A crazy physicist discovered a new kind of particle which he called an *imon*, after some of them mysteriously appeared in his lab. Some pairs of imons in the lab can be entangled, and each imon can participate in many entanglement relations. The physicist has found a way to perform the following two kinds of operations with these particles, one operation at a time.

- If some imon is entangled with an odd number of other imons in the lab, then the physicist can destroy it.
- At any moment, he may double the whole family of imons in the lab by creating a copy $I'$ of each imon $I$. During this procedure, the two copies $I'$ and $J'$ become entangled if and only if the original imons $I$ and $J$ are entangled, and each copy $I'$ becomes entangled with its original imon $I$; no other entanglements occur or disappear at this moment.

Prove that the physicist may apply a sequence of such operations resulting in a family of imons, no two of which are entangled.

## 11.8 Colourings, markings and boards

**Problem 11.156.** Each cell of a $5 \times 5$ board is either white or green. Suppose that in each $2 \times 2$ subsquare of this board there are at least two green cells. What is the smallest possible number of green cells in the entire board?

**Problem 11.157.** In each cell of a $5 \times 5$ board there is one of the three numbers $-1$, 0, 1. Suppose that in each $2 \times 2$ subsquare of this board three of the four numbers have sum 0. What is the greatest possible sum of all the numbers in the entire board?

**Problem 11.158.** A king is a chess piece which can move to any of the (at most) 8 neighbouring squares of the chessboard, by either a side or a corner, in a single move. Suppose that an $8 \times 8$ chessboard is given with a king initially standing on the square c3, the intersection of the third column and the third row. What is the smallest number of squares of the chessboard, different from c3 and f6, that one needs to remove in order to make it impossible for the king to reach the square f6?

**Problem 11.159.** A $5 \times 100$ table is divided into 500 cells, of which $n$ are coloured black and the rest are coloured white. Two cells are called adjacent if they share a common side. Suppose that each cell has at most two adjacent black cells. Find the largest possible value of $n$. [JBMO 2019]

**Problem 11.160.** In how many ways can the cells of an $8 \times 8$ table be coloured in two colours such that in any $2 \times 2$ square there exist two adjacent cells of the same colour?

**Problem 11.161.** Exactly 2024 points on a circle are coloured in 10 different colours. It is known that among any 100 consecutive points there are points of all 10 colours. Prove that there exist 90 consecutive points such that among them there are points of all 10 colours.

**Problem 11.162.** All points in the plane with integer coordinates are coloured in two colours, red and blue. Prove that there exists an infinite set of points of the same colour which is symmetric with respect to a certain point.

**Problem 11.163.** Find the least number of colours $k$ such that for any colouring of the two-digit numbers in $k$ colours there exist three numbers $a$, $b$ and $c$ such that: $a$, $b$ and $c$ are of different colours; and $a$ and $b$ have the same units digit, while $b$ and $c$ have the same tens digit.

**Problem 11.164.** Let there be $n$ light bulbs in one row, and let the light bulb in the $k$-th position be on and all the others off. In one move we are allowed to select three consecutive light bulbs and change their states, turning them on if they are off and off if they are on. For which values of $n$ and $k$ can we reach a state where all the light bulbs are off?

**Problem 11.165.** On a $29 \times 29$ board, 99 squares of size $2 \times 2$ are placed without overlap. Prove that at least one more $2 \times 2$ square can be placed without overlap.

**Problem 11.166.** Let $n$ be a positive integer. A spider travels on a $(2n+1) \times (2n+1)$ board. In every move it can travel two squares vertically, up or down, two squares diagonally, or one square horizontally, left or right. Find the minimum number of moves the spider needs in order to pass through every single square. The spider is considered to have visited the starting square, the ending square after every move, and the square between the starting and the ending square of a move when the spider moves two squares. The starting point can be chosen arbitrarily.

**Problem 11.167.** Can a $10 \times 10$ board be covered by L-tetrominoes?

**Problem 11.168.** Let $n$ be a positive integer. On a $2n \times 2n$ board certain squares are marked in such a way that each square is adjacent to at least one marked square. Two squares are adjacent if they share exactly one side; two diagonally connected squares are not adjacent, and a square is not adjacent to itself. Find the smallest possible number of marked squares. [IMO 1999 P3]

**Problem 11.169.** Let $n$ and $k$ be positive integers such that $1 < k \leq n^2$. Labelling the centres of the cells of the plane by integer coordinates $(x, y)$, let an *$n$-diamond* be the collection of cells for which $|x| + |y| \leq n$; for example, the X-pentomino is a 1-diamond. Show that the number of ways of placing $k$ dominoes ($2 \times 1$ or $1 \times 2$) on an $n$-diamond is larger than the number of ways of placing $k$ squares of size $2 \times 2$ on a $(2n+1) \times (2n+1)$ square, in both cases without any overlap.

**Problem 11.170.** A queen sits at the centre of a $(2n+1) \times (2n+1)$ board. Find the number of ways for the queen to reach the edge of the board if in each move the queen must move closer to the edge. The distance to the edge is the distance from the centre of the queen's square to the centre of the closest square adjacent to the edge.

**Problem 11.171.** An $n \times n$ grid is the collection of edges of all unit squares on an $n \times n$ board. Let an $n \times n$ grid be tiled by unit angles, that is, by pairs of unit edges meeting at a right angle, each oriented up-left, up-right, down-left or down-right; thus a unit square is tiled by two unit angles. Prove that the number of up-left unit angles is equal to the number of down-right unit angles.

**Problem 11.172.** A king starts from one square of an $8 \times 8$ board and visits each square exactly once before returning to the original square, without ever moving diagonally. Prove that the number of horizontal moves is not equal to the number of vertical moves.

**Problem 11.173.** An ant is a chess piece that can move one square to the right, one square up, or one square diagonally down-left. Is it possible for the ant to start at a square of an $8 \times 8$ chessboard, visit each other square exactly once and return to its starting square?

**Problem 11.174.** Find all unit squares which can be taken out of a $7 \times 7$ square such that the remaining squares can be tiled by $1 \times 3$ rectangles.

**Problem 11.175.** On a $1 \times (m+1)$ board the squares are labelled from 0 to $m$. Initially there are $n$ stones at position zero. In each move we can move a stone to the right by at most as many

squares as the number of stones on the square that this stone currently occupies; for instance, a lone stone can only be moved one square. Show that the number of moves needed for all the stones to reach square $m$ is at least $\lceil \frac{m}{1} \rceil + \lceil \frac{m}{2} \rceil + \cdots + \lceil \frac{m}{n} \rceil$.

**Problem 11.176.** An $m \times n$ board without one corner unit square, where both $m$ and $n$ are odd, is tiled with $2 \times 1$ dominoes and $2 \times 2$ squares. Show that the number of dominoes is at least $\frac{m+n}{2} - 1$. †

**Problem 11.177.** On each cell of a $4 \times 4$ board a coin is placed, with one side black and the other white, so that initially the white side is on top for all coins. You are allowed to select an arbitrary $3 \times 3$ square or $2 \times 2$ square and flip all the coins in the selected square. Is it possible to achieve any combination of colours shown by the coins after a finite sequence of moves?

**Problem 11.178.** Consider an $n \times n \times n$ cube divided into $n^3$ unit cells. A $1 \times 1 \times n$ cuboid consisting of $n$ cells is called a *line*. A set $S$ of cells is *line-breaking* if every line contains at least one cell of $S$. What is the minimum size of a line-breaking set, in terms of $n$?

**Problem 11.179.** There is a board $20 \times 20$ whose cells are coloured black and white in such a way that the 76 cells next to the border of the board are all black. It turns out that there is no $2 \times 2$ square all of whose cells have the same colour. Prove that there exists a $2 \times 2$ square which is coloured in the chessboard pattern.

**Problem 11.180.** Taras plays the following computer game: there is a board $7 \times 7$ on which there may or may not be one battleship of size $2 \times 2$, and the battleship is invisible. He is allowed to place a few detectors in some cells of the board. Once they are placed and he presses the button "Go", each detector tells him, all at once, whether the battleship contains the cell with this detector. What is the smallest number of detectors required in order to be sure to be able to tell whether the battleship is present or not, and, if it is, to tell its exact position?

**Problem 11.181.** A square is divided into a number of rectangles, probably of different sizes. It turns out that every line which intersects the interior of the square and is parallel to a side of the square also intersects the interior of one of the rectangles. Prove that among those rectangles there is one that does not touch the boundary of the square.

**Problem 11.182.** There are several bulbs, all of them on initially, and there are several buttons. Each button is connected to a subset of the bulbs via wires, and pressing this button changes the states of exactly those bulbs. It turns out that for any subset of the bulbs there is a button that is connected to an odd number of the bulbs from this subset. Prove that it is possible to turn all of the bulbs off by pressing some of the buttons.

**Problem 11.183.** Let $n$ be an odd integer. Consider a square lattice of size $n \times n$, consisting of $n^2$ unit squares and $2n(n+1)$ edges. All edges are painted red or blue so that the number of red edges does not exceed $n^2$. Prove that there is a cell that has at least three blue edges.

**Problem 11.184.** ★ A *snake* is a set of cells of a board such that they can be arranged in a sequence in which each cell of the snake is to the right of or above the previous cell. Find the number of ways to tile an $m \times n$ board with snakes.

**Problem 11.185.** ★ Some squares of an $n \times n$ board ($n > 2$) are black and the rest are white. In every white square we write the number of all the black squares having at least one common vertex with it. Find the maximum possible sum of all these numbers.

**Problem 11.186.** Every integer was painted with some colour in such a way that the sum of every two integers of different colours is always green. What is the maximum possible number of colours used?

**Problem 11.187.** Every positive integer is to be coloured green or red in such a way that: for every red $n$, the sum of $n$ (not necessarily distinct) red numbers is red; and for every green $n$, the sum of $n$ (not necessarily distinct) green numbers is green. Find all such colourings.

**Problem 11.188.** Let $n \geq 5$ be an integer. Prove that the elements of the set $\{1, 2, \ldots, n\}$ can be coloured red and blue so that the sum of all blue numbers is equal to the product of all red numbers.

**Problem 11.189.** Determine all possible partitions of the set of all positive integers into two non-empty subsets $A$ and $B$ with the following properties: for every $a \in A$ and every $b \in B$ we have $a + b \in A$ and $a \cdot b \in B$.

**Problem 11.190.** The positive integers are coloured with two colours. Prove that, given a positive integer $n$, there exist distinct integers $a, b > n$ such that the set $\{a, b, a+b\}$ is monochromatic, that is, all its elements have the same colour.

**Problem 11.191.** Two disjoint subsets of the set $\{1, 2, \ldots, n\}$ have equal sums of elements. Prove that each of these subsets has fewer than $\frac{n}{\sqrt{2}}$ elements. †

**Problem 11.192.** The elements of the set $\{1, 2, \ldots, 4n\}$ were coloured with some number of colours so that no two different numbers whose sum is divisible by 4 have the same colour. What is the minimum possible number of colours used?

**Problem 11.193.** A set $S$ of natural numbers is called *good* if for each element $x \in S$, $x$ does not divide the sum of the remaining numbers in $S$. Find the maximal possible number of elements of a good set which is a subset of the set $\{1, 2, 3, \ldots, 63\}$. [JBMO SL 2011 C5]

**Problem 11.194.** Suppose that $P_1, P_2, \ldots, P_n$ are $n$ different two-element subsets of the set $\{1, 2, \ldots, n\}$ with the following property: for every $i \neq j$ the sets $P_i$ and $P_j$ have an element in common if and only if $\{i, j\}$ is one of the given sets. Prove that every number from 1 to $n$ appears in precisely two of the given sets.

## 11.9 Combinatorial geometry

**Definition 11.21 (Convex hull).** The *convex hull* of a set $M$ of points is the smallest convex set that contains $M$. If $M$ is finite, then its convex hull is a convex polygon whose vertices belong to $M$.

**Theorem 11.22 (Helly's theorem in the plane).** *Let $S$ be a finite family of convex figures in the plane such that any three figures of $S$ have a common point. Then all the figures of $S$ have a common point.*

**Theorem 11.23 (Young's theorem).** *If $n$ points in the plane are such that the distance between any two of them does not exceed 1, then all these points can be covered by a circle of radius $\frac{1}{\sqrt{3}}$.*

**Problem 11.195.** Prove that every polygon has at least one diagonal, that is, a segment joining two of its vertices and contained entirely inside it.

**Problem 11.196.** There were 31 children in a room at a party, and all the distances between them were pairwise distinct. At some moment each child threw a piece of cake at the person nearest to him. Prove that there is a person at whom no cake was thrown.

**Problem 11.197.** Prove that it is impossible to draw a convex 2020-gon all of whose angles have an integer number of degrees.

**Problem 11.198.** Inside a triangle $n$ points are marked. All these points are connected by non-intersecting segments with each other and with the vertices of the triangle, so that the triangle is partitioned into $m$ smaller triangles (each marked point is a vertex of some of these triangles and cannot lie on the side of any of them). Find all possible values of $m$.

**Problem 11.199.** From some point inside a convex polygon the perpendiculars to all the sides are drawn. Prove that at least one foot of these perpendiculars lies on the side itself and not on its extension.

**Problem 11.200.** From each vertex of a polygon the perpendiculars to all sides for which this vertex is not an endpoint are drawn. Prove that at least one foot of these perpendiculars lies on the side itself and not on its extension.

**Problem 11.201.** Prove that in each convex $2n$-gon there exists a diagonal which is not parallel to any side of this $2n$-gon.

**Problem 11.202.** On the sides of a convex quadrilateral, as diameters, four circles are constructed. Prove that these circles cover the whole quadrilateral.

**Problem 11.203.** Prove that there is no convex polyhedron all of whose faces have different numbers of sides.

**Problem 11.204.** A polygonal line of length 200 is drawn inside a square with side length 1. Prove that there exists a line parallel to one of the sides of the square which intersects the polygonal line in at least 101 points.

**Problem 11.205.** Find all $n > 3$ such that it is possible to draw $n$ points in the plane and connect some of them by segments so that each point is the endpoint of exactly 3 segments and no two segments intersect.

**Problem 11.206.** Each of nine given lines partitions a square into two quadrilaterals with the ratio of their areas equal to $2 : 3$. Prove that at least three of these lines pass through the same point.

**Problem 11.207.** Find the minimal possible number of points which it is enough to mark inside a convex $n$-gon so that each triangle whose vertices are vertices of the $n$-gon contains inside it at least one marked point.

**Problem 11.208.** Given $n$ points in the plane, not all of which lie on the same line, prove that there exist at least $n$ different lines connecting these points.

**Problem 11.209.** A finite number of pairwise non-parallel lines is drawn in the plane so that for any two lines the point of their intersection lies on at least one other line. Prove that all the lines pass through one point.

**Problem 11.210.** A convex polygon $P$ lies inside another convex polygon $Q$. Prove that the perimeter of $Q$ is greater than the perimeter of $P$.

**Problem 11.211.** Given $n > 3$ points in the plane so that not all of them lie on the same line, prove that there exists a circle passing through some three of these points and not containing inside it any other given point.

**Problem 11.212.** Prove that in every convex $n$-gon it is impossible to choose more than $n$ diagonals such that any two of them have a common point.

**Problem 11.213.** Given a convex polygon $A_1A_2 \dots A_n$, prove that the circumcircle of some triangle $A_iA_{i+1}A_{i+2}$ contains the whole polygon.

**Problem 11.214.** On a squared plane $n$ cells are marked. Prove that it is possible to choose at least $n/4$ of them such that they do not have common points.

**Problem 11.215.** Given $n$ segments on a line such that any two of them have a common point, prove that all the segments have a common point.

**Problem 11.216.** There is a finite number of parallel segments in the plane. It is given that for any three of them there is a line that intersects all three. Prove that there is a line that intersects all of them.

**Problem 11.217.** Given $n$ points in the plane such that any 4 of them are the vertices of a convex polygon, prove that all the given points are the vertices of a convex $n$-gon.

**Problem 11.218.** Is it possible to draw 2020 segments in the plane so that the endpoint of each segment lies inside some other segment?

**Problem 11.219.** Give an example of four non-convex sets in the plane for which the statement of Helly's theorem is false.

**Problem 11.220.** In the plane a finite set of (not necessarily convex) polygons is given, and each two of them have a common point. Prove that there exists a line having a common point with all these polygons.

**Problem 11.221.** Given $n$ points in the plane such that any three of them can be covered by a circle of radius $R$, prove that all these points can be covered by a circle of radius $R$.

**Problem 11.222.** There are 2017 points marked on a line. Consider a set $M$ containing 2017 intervals whose ends are some of the marked points. Prove that there are two intervals in $M$ such that one of them fully contains the other one.

**Problem 11.223.** Prove that in each convex polygon which is not a parallelogram it is possible to choose 3 sides such that the triangle formed by the intersections of the lines passing through these sides contains the whole polygon.

**Problem 11.224.** Given a convex polygon such that for any 3 of its sides there exists a point $O$ such that the feet of the perpendiculars drawn from $O$ to these sides lie inside the sides, prove that such a point exists for all the sides simultaneously.

**Problem 11.225.** The plane is lit by $n > 3$ lamps, each of which lights a half-plane. Prove that we can keep 3 of them so that they still light the whole plane.

**Problem 11.226.** There is a town called "Fencer" with $N$ houses in it. What is the largest number of fences that could be built around the houses such that: no two fences intersect; each fence surrounds at least one house; and no two fences surround the same set of houses?

**Problem 11.227.** The vertices of a convex $n$-gon are coloured in three colours such that no two neighbouring vertices are of the same colour and there is at least one vertex of each colour. Prove that it is possible to triangulate the $n$-gon in such a way that all the vertices of any triangle are of different colours.

**Problem 11.228.** On a plane 999 points are coloured blue. Prove that for any circle of radius 1 there is a point on it such that the sum of its distances to all of the 999 points is at least 999.

**Problem 11.229.** Prove that a polygon $P$ cannot be covered by two polygons that are homothetic to $P$ with coefficient 0.99.

**Problem 11.230.** There are 650 points chosen inside a circle of radius 16. Prove that there is a ring with inner radius 2 and outer radius 3 which contains at least 10 of the chosen 650 points.

**Problem 11.231.** On a plane there are initially $n$ green points. For every segment with two green endpoints we colour its midpoint blue; a green point can get recoloured as well. What is the smallest possible number of colourful, that is green or blue, points after this operation?

**Problem 11.232.** On a circle there are given $2n$ points, of which $n$ are blue and $n$ are red. The blue points are labelled $1, 2, \ldots, n$ clockwise and the red points are labelled $1, 2, \ldots, n$ anticlockwise. Prove that there exists an arc of this circle containing exactly $n$ points whose labels do not repeat.

**Problem 11.233.** Does there exist a finite set $S$ of points of a plane, not all collinear, with the following property: every triangle with all three vertices in $S$ has its circumcentre in $S$ as well?

**Problem 11.234.** Prove that every convex polygon of area 1 contains a convex hexagon of area $\frac{3}{4}$.

**Problem 11.235.** Given $n$ points in general position in the plane, the lines passing through each pair of points are drawn. Find the minimal possible number of pairwise non-parallel lines among all of them.

**Problem 11.236.** Given $2n + 3$ points in general position in the plane such that no 4 of them lie on a circle, prove that it is always possible to choose 3 of them so that $n$ of the remaining points lie inside their circumcircle and the other $n$ lie outside it.

**Problem 11.237.** Prove that inside each convex heptagon there exists a point which does not belong to any quadrilateral formed by 4 consecutive vertices.

**Problem 11.238.** During a day 100 people visited a library, each one of them exactly once. Among any 3 visitors there exist two which met at the library. Prove that it was possible to make 2 announcements during the day so that each visitor heard one of them.

**Problem 11.239.** Given $n$ points in general position in the plane, prove that it is always possible to connect them by a non-self-intersecting polygonal line.

**Problem 11.240.** Given $n$ points in general position in the plane, is it always possible to connect them by a closed non-self-intersecting polygonal line?

**Problem 11.241.** Given $2n$ points in general position in the plane, $n$ non-intersecting segments connect these points in pairs. Is it always possible to draw $n$ more segments so as to connect all the points into a closed non-self-intersecting polygonal line?

**Problem 11.242.** Given a convex $n$-gon $P$ with no four concyclic vertices, a triple of vertices $\{A, B, C\}$ of $P$ is called *fat* if $P$ is contained inside the circumcircle of the triangle $ABC$. Determine all possible numbers of fat triples, in terms of $n$.

**Problem 11.243.** Given a convex $n$-gon $P$ with no four concyclic vertices, prove that there exist two vertices of $P$ such that every circle passing through these two points contains at least $\frac{1}{3}(n-2)$ other vertices of $P$.

**Problem 11.244.** Given an integer $n \geq 1$ and a circle $\omega$, finitely many red points are marked on $\omega$ and some pairs of red points are joined by green chords. Suppose that each red point is an endpoint of at least $2n$ different green chords. Prove that there exists a green chord which intersects at least $n^2$ other green chords; two chords intersect if they share a point, possibly an endpoint.

**Problem 11.245.** Exactly $4n$ points of a plane are green and no three green points are collinear. Prove that there exists a point $P$ of this plane which belongs to the interior of at least $2n^3$ triangles with three green vertices.

**Problem 11.246.** Let $P$ be a plane. Does there exist a function $f : P \to P$ with the following property: for every convex quadrilateral $ABCD$, the points $f(A)$, $f(B)$, $f(C)$, $f(D)$ are vertices of a concave, that is non-convex, quadrilateral?

**Problem 11.247.** In a plane there are given $3n^2$ circles with equal radii. Prove that one can select $n$ of these circles which either mutually intersect or are pairwise disjoint.

**Problem 11.248.** Let $n$ be an integer greater than 1. Suppose $2n$ points are given in the plane, no three of which are collinear; $n$ of the given points are coloured blue and the other $n$ are coloured red. A line in the plane is called a *balancing line* if it passes through one blue and one red point and, for each side of the line, the number of blue points on that side is equal to the number of red points on the same side. Prove that there exist at least two balancing lines.

**Problem 11.249.** Let $n$ be a positive integer relatively prime to 6. We paint the vertices of a regular $n$-gon with three colours so that there is an odd number of vertices of each colour. Show that there exists an isosceles triangle whose three vertices are of different colours.

**Problem 11.250.** One hundred circles of radius one are given in the plane, and the triangle formed by the centres of any three of these circles has area at most 2017. Prove that there is a line intersecting at least three of the circles.

**Problem 11.251.** Let $n$ be a positive integer. The kingdom of Zoomtopia is a convex polygon with integer sides, perimeter $6n$, and $60^\circ$ rotational symmetry, that is, there is a point $O$ such that a $60^\circ$ rotation about $O$ maps the polygon to itself. The government of Zoomtopia would like to relocate its $3n^2 + 3n + 1$ citizens at $3n^2 + 3n + 1$ points in the kingdom so that every two

citizens have a distance of at least 1 from each other. Prove that this is possible. (The kingdom is assumed to contain its boundary.)

**Problem 11.252.** Does there exist a set $S$ of 100 points in a plane such that the centre of mass of any 10 points of $S$ is also a point of $S$?

**Problem 11.253.** ★ A square is partitioned into several convex polygons with pairwise distinct numbers of sides. Prove that one of these polygons is a triangle. [Sharygin Geometry Olympiad 2012]

**Problem 11.254.** ★ Let $n$ be a fixed odd positive integer. Consider $m+2$ distinct points $P_0, P_1, \ldots, P_{m+1}$ in the Cartesian plane, where $m$ is a non-negative integer, such that: (a) $P_0 = (0,1)$, $P_{m+1} = (n+1, n)$ and for each $i$ with $1 \le i \le m$ both coordinates of the point $P_i$ are integers from $[1, n]$; (b) for each $i$ with $0 \le i \le m$ the line $P_iP_{i+1}$ is parallel to the $x$-axis if $i$ is even, and parallel to the $y$-axis if $i$ is odd; (c) for each pair $(i, j)$ with $0 \le i < j \le m$ the segments $P_iP_{i+1}$ and $P_jP_{j+1}$ have at most one common point. Find the maximal possible value of $m$. [APMO 2011]

**Problem 11.255.** ★ Ann and Bob play a game making alternate moves. Initially Ann chooses a set $V$ of 2020 vectors with zero sum and draws one of them on the board. Then, starting from Bob, each player chooses a vector from $V$ which was not drawn before and draws it on the board; each player must put the beginning of his vector at the end of the previously drawn vector. After all 2019 moves are performed a closed polygonal line is obtained. Ann wins if it is non-self-intersecting, otherwise Bob wins. Determine who has a winning strategy. [Iran 2018]

**Problem 11.256.** ★ Is it true that in any convex $n$-gon with $n > 3$ there exists a vertex and a diagonal passing through this vertex such that the angles of this diagonal with both sides adjacent to this vertex are acute? [IGO 2019]

**Problem 11.257.** ★ There are 2021 lines in a plane such that no 3 of them go through the same point. Turbo the snail can slide along the lines in the following fashion: she initially moves on one of the lines and continues moving on a given line until she reaches an intersection of two lines; at the intersection she continues her journey on the other line, turning left or right, alternating the direction she chooses at each intersection point she passes. Can it happen that she slides through a line segment for a second time in her journey, but in the direction opposite to the one she did the first time? [EGMO 2017]

**Problem 11.258.** ★ Given $n > 2$ lines in the plane, these lines divide the plane into parts and some parts are coloured such that no two coloured parts share a side. Prove that the number of coloured parts does not exceed $\frac{1}{3}(n^2 + n)$. [USSR 1985]

**Problem 11.259.** ★ The set $L$ consists of 2020 lines in general position in the plane. Let us say that the line $\ell_1 \in L$ *bounds* another line $\ell_2 \in L$ if all the points of intersection of $\ell_2$ with the other lines from $L$ lie in one and the same half-plane with respect to $\ell_1$. Prove that there exist lines $\ell$ and $\ell'$ in $L$ such that $\ell$ bounds $\ell'$ and $\ell'$ does not bound $\ell$. [SRMC 2006]

**Problem 11.260.** ★ For $n > 5$ let $A_1A_2 \ldots A_n$ be a convex $n$-gon for which all internal angles are obtuse. For each $i$ with $1 \le i \le n$ let $O_i$ be the circumcentre of the triangle $A_{i-1}A_iA_{i+1}$, indices being taken modulo $n$. Prove that the polygonal line $O_1O_2 \ldots O_n$ is not a convex $n$-gon. [Turkey 2016]

**Problem 11.261.** ★ Given a convex $2n$-gon $H$ with pairwise parallel opposite sides. (a) Prove that there exists a pair of opposite sides of $H$ such that one can draw a line which is perpendicular

to both of these sides and intersects both of them. (b) Prove that for each $n > 2$ there exists a $2n$-gon $H$ for which there is exactly one pair of sides as in part (a). (c) Construct a non-convex hexagon for which the statement of part (a) is false. [Belarus 2017]

**Problem 11.262.** ★ Let $n$ be a positive integer, and consider a square of dimensions $2^n \times 2^n$. We cover this square by a number of at least two rectangles, without overlaps, in such a way that every rectangle has integer dimensions and a power of two as area. Show that two of the rectangles used must have the same dimensions. Two rectangles are said to have the same dimensions if they have the same height and the same width, without rotating them. [Bulgaria 2012]

**Problem 11.263.** ★ There are $n > 4$ points in the plane, no three of which are collinear, and each pair of points is connected by a segment. Find the maximal possible number of segments having no intersection with the other segments except at their endpoints. [Turkey 2016]

**Problem 11.264.** ★ Given $n > 3$ lines in general position in the plane, mark all their points of intersection. A marked point $A$ is called *internal* if for each line $\ell$ containing it there exist marked points $B$ and $C$ on $\ell$ such that $A$ lies between $B$ and $C$. Prove that the number of all internal points is at least $\frac{(n-2)(n-3)}{2}$.

**Problem 11.265.** ★ Let $N > 2$ be an integer. Some $N(N+1)$ soccer players, no two of the same height, stand in a row in some order. Coach Ralph wants to remove $N(N-1)$ people from this row so that in the remaining row of $2N$ players, no one stands between the two tallest ones, no one stands between the third and the fourth tallest ones, ..., and finally no one stands between the two shortest ones. Show that this is always possible. [IMO 2017]

**Problem 11.266.** ★ Let $n \geq 2$ be an integer. In the plane there are $n$ segments given in such a way that any two segments have an intersection point in the interior, and no three segments intersect at a single point. Jeff places a snail at one of the endpoints of each of the segments and claps his hands $n-1$ times. Each time when he claps his hands, all the snails move along their own segments and stay at the next intersection points until the next clap. Since there are $n-1$ intersection points on each segment, all snails will reach the furthest intersection points from their starting points after $n-1$ claps. (a) Prove that if $n$ is odd then Jeff can always place the snails so that no two of them ever occupy the same intersection point. (b) Prove that if $n$ is even then there must be a moment when some two snails occupy the same intersection point, no matter how Jeff places the snails. [IMO 2016]

**Problem 11.267.** ★ A diagonal of a regular 2006-gon is called *odd* if its endpoints divide the boundary into two parts, each composed of an odd number of sides; sides are also regarded as odd diagonals. Suppose the 2006-gon has been dissected into triangles by 2003 non-intersecting diagonals. Find the maximum possible number of isosceles triangles with two odd sides. [IMO 2006]

**Problem 11.268.** ★ Consider a rectangle $R$ partitioned into 2016 smaller rectangles such that the sides of each smaller rectangle are parallel to one of the sides of the original rectangle. Call the corners of each rectangle a *vertex*. For any segment joining two vertices, call it *basic* if no other vertex lies on it; the segments must be part of the partitioning. Find the maximum and the minimum possible number of basic segments over all possible partitions of $R$.

**Problem 11.269.** ★ Finitely many unit circles are given in a plane such that the area of their union is $S$. Prove that it is possible to select a subset of these circles such that no two circles in the subset intersect each other and the total area of these circles is at least $2S/9$.

**Problem 11.270.** ★ There are 2019 points given in the plane. A child wants to draw $k$ closed discs in such a manner that for any two distinct points there exists a disc that contains exactly one of these two points. What is the minimal $k$ such that for any initial configuration of points it is possible to draw $k$ discs with the above property?

## 11.10 Games II

**Problem 11.271.** Two players $A$ and $B$ play the following game. The players are given a chocolate bar of dimension $m \times n$, where the bottom left square of the chocolate is poisonous. The players take turns breaking off the chocolate horizontally or vertically into two pieces, without breaking apart the individual squares, and eating the piece that does not contain the poisonous square, with player $A$ making the first move. The person who is left with just the poisoned square loses. Determine, as a function of $m$ and $n$, which player has a winning strategy and what this strategy is.

**Problem 11.272.** In a box there are cards with the numbers from 1 to 25 written on them. Players take turns removing a card from the box, with player $A$ starting first. The player after whose turn the sum of the labels on the removed cards is divisible by 3 loses. Which player has a winning strategy and what is it?

**Problem 11.273.** Two piles of coins are given on a table, containing $m$ and $n$ coins respectively. Players $A$ and $B$ take turns removing coins from one of the piles, with $A$ making the first move. In each move a player removes from one pile a number of coins which divides the number of coins in the other pile. For example, if the piles have 5 and 4 coins, the player may remove 1, 2 or 4 coins from the first pile, but not 3 or 5. If all the coins have been taken from a pile, its number of coins is considered to be 0. The player who takes the last coin off the table wins. Determine, as a function of $m$ and $n$, which player has a winning strategy and what this strategy is.

**Problem 11.274.** On two piles there are initially $m$ and $n$ coins respectively. Two players $A$ and $B$ take turns removing at least one coin from one of the piles, with player $A$ starting first. For which values of $m$ and $n$ does player $A$ have a winning strategy, and for which values does player $B$, if the player who takes the last coin loses?

**Problem 11.275.** On a rectangular board of size $100 \times 2023$ two players $A$ and $B$ take turns placing unit circles so that they do not overlap with each other and the circles do not go outside of the board; player $A$ starts first. The person who cannot make a placement loses. Which player has a winning strategy and what is it?

**Problem 11.276.** Consider three piles of candies with $a$, $b$ and $c$ candies respectively, each having at least one candy. Two players $A$ and $B$ make moves one after another, $A$ being first. Each move consists of the following two steps: the player chooses a pile and eats all the candies from this pile; then the player splits all the candies from the other two piles into three new piles, each having at least one candy. The player after whose move the three piles have one candy each loses the game, the other player winning it. Find all triples $(a, b, c)$ for which player $B$ has a winning strategy.

**Problem 11.277.** On an infinite board two players take turns writing in X and O respectively, the winner being the one who connects nine consecutive symbols belonging to them, horizontally, vertically or diagonally. Prove that the second player can prevent the first player from ever winning.

**Problem 11.278.** Alice and Bob play a game on a $15 \times 15$ board. One by one each of them puts either a cross or a zero in one of the empty cells; there is no restriction forcing one person to use only crosses or only zeros throughout the game. Alice goes first. The first to obtain three consecutive identical symbols in a row, column or diagonal wins; if this does not happen the game is a draw. Does there exist a winning strategy for Alice?

**Problem 11.279.** On a $2022 \times 2022$ board players $A$ and $B$ play the following game, alternating turns with player $A$ having the first turn. In each turn a player selects a column and writes a digit, an arbitrary integer from 0 to 9, in the topmost cell of the column in which a digit has not yet been written. After the board has been filled with digits we observe the 4044 numbers with 2022 digits formed by reading the digits of a column from top to bottom, or of a row from left to right. If at least 2 of these 4044 numbers are prime, then $A$ wins, otherwise $B$ wins. Which player has a winning strategy and what is it?

**Problem 11.280.** A graph of $n$ points and no edges is given. The players $A$ and $B$ take turns drawing a single edge on the graph, with $A$ starting first; multiple edges between two points are not permitted, nor are edges from a point to itself. The player who completes a cycle of odd length loses. Determine, as a function of $n$, which player has a winning strategy and what this strategy is.

**Problem 11.281.** There are 10 wise men trying to overcome a challenge. Everyone gets a cap of some colour, and so everyone can see the colours of the caps of all the others, but not of his own. At some point, when the king says so, all of them together say a number; different people may say different numbers. If at least one of them names the total number of caps of his own colour, then the king will set everyone free; otherwise they are not so lucky. Find a way for the wise men to agree on what to do once they get the caps so that they are certain to be let free.

**Problem 11.282.** ★ You are cheating at a trivia contest. For each question, you can peek at each of the $n > 1$ other contestants' guesses before writing down your own. For each question, after all guesses are submitted, the host announces the correct answer. A correct guess is worth 0 points; an incorrect guess is worth $-2$ points for the other contestants, but only $-1$ point for you, since you hacked the scoring system. After announcing the correct answer the host proceeds to read the next question. Show that if you are leading by $2^{n-1}$ points at any time, then you can surely win first place.

## 11.11 Further problems from the 2021–2022 courses

Two courses of the 2021–2022 season worked through Level-3 problem lists of their own: the combinatorics course of the November 2021 camp, built around tilings, graphs and games, and the graph-theory thread of the January 2022 KAUST camp, which introduced the subject from its definitions and reached non-trivial olympiad material in three sessions. Their problems are collected here.

**Problem 11.283.** In a soccer tournament there are $2n$ teams, $n \geq 2$. On the first day $n$ matches were played, each team playing exactly one game. On the second day every team again played exactly one game, with a different opponent. Prove that after the two days one can still find $n$ teams no two of which have played each other.

**Problem 11.284.** At a party there are $n$ people. Initially everyone had exactly 3 friends among the others, but during the party some people got to know each other, so that when the party ended everyone had exactly 4 friends. Find all numbers $n$ for which this is possible.

**Problem 11.285.** At a party there are 99 people.

(a) Suppose that among every three of them there is one who knows the other two. Prove that at least one person knows everyone else at the party.

(b) Suppose that among every four of them there is one who knows the other three. Prove that at least 96 people know everyone else at the party.

**Problem 11.286.** The plane is divided by horizontal and vertical lines into unit squares, and finitely many of these squares are painted black. Suppose that each black square is adjacent (along a side) to exactly 2 other black squares. Determine all possible numbers of black squares.

**Problem 11.287.** In a group everyone has exactly $d$ friends, and every two people who are not friends have exactly one common friend. Prove that the group has at most $d^2 + 1$ members.

**Problem 11.288.** Find all $n$ for which there exists a regular graph of degree $n$ on $2n + 1$ vertices.

**Problem 11.289.** A bipartite graph has parts $A$ and $B$ of sizes $2n$ and $2n + 1$ respectively. Prove that if all the vertices of $A$ have equal degrees, then some two vertices of $B$ have equal degrees.

**Problem 11.290.** The *complement* of a graph $G$ is the graph on the same vertices in which two vertices are joined exactly when they are not joined in $G$.

(a) Prove that if a graph on $n$ vertices is isomorphic to its complement, then $n \equiv 0$ or $1 \pmod 4$.

(b) For every $k \geq 1$ construct a graph on $4k$ vertices isomorphic to its complement.

(c) Prove that a graph on $4k + 1$ vertices isomorphic to its complement has a vertex of degree $2k$.

**Problem 11.291.** In a triangle-free graph on $n$ vertices every pair of non-neighbours has exactly two common neighbours. Prove that the graph is regular.

**Problem 11.292.** A convex polyhedron has only triangular faces, and its vertices are coloured in three colours. Prove that the number of rainbow faces — faces having one vertex of each colour — is even.

**Problem 11.293.** A non-empty graph has an even number of vertices. Prove that some two of its vertices have an even number of common neighbours.

**Problem 11.294.** A graph $G$ with at least 5 vertices has the following two properties: every vertex has at least one non-neighbour, and every two non-neighbours have exactly one common neighbour. Prove that $G$ contains an induced cycle on 5 vertices, that is, a cycle of length 5 none of whose diagonals is an edge.

**Problem 11.295.** Let $G$ be a simple graph on $n \geq 3$ vertices in which the sum of the degrees of every pair of non-neighbours is at least $n$. Prove that $G$ contains a cycle of length $n$.

**Problem 11.296.** Let $G$ be a graph on $n \geq 5$ vertices whose complement is triangle-free. Prove that $G$ contains a cycle of length at least $n/2$.

**Problem 11.297.** ★ In the country of Graphia there are 100 towns, numbered from 1 to 100. Some pairs of towns are connected by a direct road, and such pairs of towns are called adjacent;

no two roads connect the same pair of towns. Peter, a foreign tourist, plans to visit Graphia 100 times. For each $i = 1, 2, \ldots, 100$, Peter starts his $i$-th trip by arriving in the town numbered $i$, and on each following day he travels from the town he is currently in to the adjacent town with the lowest number among those he has not already visited on the $i$-th trip, assuming such a town exists; otherwise he deems his $i$-th trip complete and returns home. It turns out that after all 100 trips Peter has visited each town of Graphia the same number of times. Find the largest possible number of roads in Graphia.

**Problem 11.298.** ★ Let $n$ and $k$ be positive integers. One morning a candy shop had $2n$ pieces of candy of each flavour, and during that day a total of $m$ children bought candy at the shop, each of them buying exactly 2 pieces, of two different flavours. It is known that among any $k+1$ of these children there are two who bought at least one flavour in common. Find the largest possible value of $m$. (The shop receives no new candy during the day.)

**Problem 11.299.** A regular $4n$-gon is partitioned into parallelograms. Prove that at least $n$ of these parallelograms are rectangles.

**Problem 11.300.** Find the minimum number of colours needed to colour the squares of an $8 \times 8$ chessboard so that no two squares covered by one and the same L-tetromino have the same colour.

**Problem 11.301.** ★ A rectangular board is tiled with $1 \times 2$ dominoes, placed horizontally or vertically. Show that there is a colouring of the cells of the board in two colours such that every domino of this tiling covers two cells of different colours, while in every other tiling of the board by dominoes some domino covers two cells of the same colour.

**Problem 11.302.** ★ Is it possible to partition a disc into mutually congruent shapes in such a way that some of the shapes do not contain the centre of the disc, neither as an interior point nor as a boundary point?

**Problem 11.303.** ★ A rectangle of dimensions $a \times b$, where $a \geq b$, is partitioned into right-angled triangles in such a way that any two adjacent triangles share a whole common side, and this side is the hypotenuse of one of the two triangles and a leg of the other. Prove that $a \geq 2b$.

**Problem 11.304.** ★ An isosceles trapezoid is called *proper* if exactly one pair of its sides is parallel (so parallelograms and rectangles are not proper). A partition of a rectangle into $n$ proper isosceles trapezoids is called *strict* if the union of any $i$ of the trapezoids of the partition $(2 \leq i \leq n)$ is not itself a proper isosceles trapezoid. Prove that for every $n \geq 9$ there is a strict partition of a $2020 \times 2021$ rectangle into $n$ proper isosceles trapezoids.

**Problem 11.305.** ★ An $n \times n$ board is filled with integers in such a way that the integers written on any two adjacent cells differ by at most 1. Prove that some integer appears at least $n$ times on the board. (Two cells are adjacent if they share a common side.)

**Problem 11.306.** ★ An L-tromino is a $2 \times 2$ square with one of its unit cells removed. For which values of $M$ and $N$ can an $M \times N$ rectangle be tiled by L-trominoes in such a way that no smaller rectangle is tiled by a subset of these L-trominoes?

**Problem 11.307.** An *$n$-staircase* is the figure consisting of $1 + 2 + \cdots + n$ unit squares, namely of all lattice unit squares whose top right corners $(x, y)$ satisfy $x, y \geq 1$ and $2 \leq x + y \leq n + 1$. Calculate (a) the number of rectangles contained in an $n$-staircase; (b) the sum of the areas of these rectangles.

**Problem 11.308.** At a party there are $n \geq 2$ people, each wearing a hat. Every two of them greeted each other, each greeting consisting of exchanging the hats worn by the two greeters at that moment. Find all numbers $n$ for which it could happen that after all the greetings everyone has their own hat back.

**Problem 11.309.** Players $A$ and $B$ play the following game. Initially they have one large piece of plastic. In his move $A$ cuts one of the existing pieces into three pieces of arbitrary sizes, and in his move $B$ glues two existing pieces into one. They repeat this procedure, and $A$ wins if at some moment there are among the pieces 100 pieces of equal weight. Can player $B$ prevent $A$ from winning?

**Problem 11.310.** ★ Let $n$ be a positive integer. There are $2n$ people in a spacecraft, and any two of them are either friends or enemies (the relation being reciprocal). Two aliens play the following game: they alternately choose one person each, subject to the rule that the person chosen in a round must be a friend of the person chosen by the other player in the previous round. In the first round the player may choose any person, each person may be chosen at most once, and the player who cannot move loses. Prove that the second player has a winning strategy if and only if the $2n$ people can be split into $n$ pairs so that the two people in each pair are friends.

# Chapter 12

# Geometry

**Sources for this chapter:** Camp handouts by Dominik Burek (Jan 2020, May 2025), Lê Phúc Lũ (June 2022), Regis Barbosa (Nov 2022, Feb 2025), Łukasz Bożyk (Nov 2021, May 2023), Mikhail (June 2020).

At this level the geometry of the circle is organised around one quantity, the power of a point, and around the transformations that move circles to circles: homothety and spiral similarity. The chapter first develops the radical axis and the radical centre into a working tool, then turns to the incircle and the excircles, and finally to the two named points — the Humpty and the Dumpty point — in which the median and the symmedian meet the circles of the configuration. A closing section collects the problems the camps used to train the habit of adding one well-chosen element to a picture.

## 12.1 Power of a point and the radical axis

*Revisits power of a point (see Chapter 7); the problems below push it further.*

**Definition 12.1 (Power of a point).** Let $\Gamma$ be a circle with centre $O$ and radius $r$, and let $P$ be a point of the plane. The *power* of $P$ with respect to $\Gamma$ is the number

$$\operatorname{pow}(P,\Gamma) = PO^2 - r^2.$$

If an arbitrary line through $P$ meets $\Gamma$ at $A$ and $B$, then $\operatorname{pow}(P,\Gamma) = \overline{PA}\cdot\overline{PB}$, where the segments are measured as signed lengths along that line: if $\overline{PA}$ and $\overline{PB}$ point in opposite directions their product is negative. Consequently

(a) $P$ lies outside $\Gamma$ if and only if $\operatorname{pow}(P,\Gamma) > 0$;
(b) $P$ lies inside $\Gamma$ if and only if $\operatorname{pow}(P,\Gamma) < 0$;
(c) $P$ lies on $\Gamma$ if and only if $\operatorname{pow}(P,\Gamma) = 0$.

**Theorem 12.2 (Chord theorem).** *Let $\Gamma$ be a circle and $P$ a point. Let a line through $P$ meet $\Gamma$ at $A$ and $B$, and let another line through $P$ meet $\Gamma$ at $C$ and $D$. Then*

$$PA\cdot PB = PC\cdot PD.$$

*If moreover $P$ lies outside $\Gamma$ and $PT$ is tangent to $\Gamma$ at $T$, then*

$$PA\cdot PB = PC\cdot PD = PT^2.$$

**Theorem 12.3 (Converse to the power of a point).** *Let $A$, $B$, $C$, $D$ be four distinct points and let the lines $AB$ and $CD$ meet at $P$. Assume that either $P$ lies on both of the segments $AB$ and $CD$, or $P$ lies on neither of them. Then $A$, $B$, $C$, $D$ are concyclic if and only if*

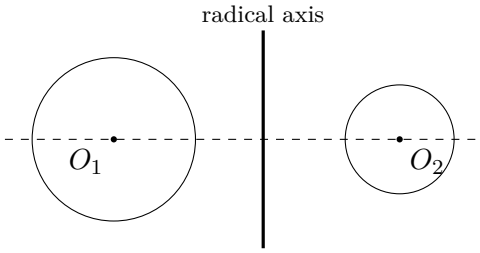
$$PA \cdot PB = PC \cdot PD.$$

**Theorem 12.4 (Radical axis).** *Given two circles $\Gamma_1$ and $\Gamma_2$ with different centres, the locus of the points $P$ with $\mathrm{pow}(P, \Gamma_1) = \mathrm{pow}(P, \Gamma_2)$ is a line perpendicular to the line joining the two centres. It is called the* radical axis *of $\Gamma_1$ and $\Gamma_2$. There are three configurations:*

*(1) if the circles meet at two points, the radical axis is the line through those two common points;*

*(2) if the circles are tangent, the radical axis is their common tangent line at the point of tangency (this holds both for internal and for external tangency);*

*(3) if the circles are disjoint, the radical axis is a line separating the plane into two half-planes, one containing each circle.*




**Remark.** Not every pair of circles has a radical axis. Two concentric circles of different radii have none at all: no point of the plane has the same power with respect to both.

**Definition 12.5 (Degenerate circle).** A circle of radius 0 is called a *degenerate circle*; it consists of its centre alone. Every property of circles stated above may be applied to it. In particular, if the centre $A$ of a degenerate circle lies on a circle $(O)$, then the radical axis of the two is the line through $A$ perpendicular to $AO$; if $A$ does not lie on $(O)$, the radical axis is still a line perpendicular to $AO$, and it can be constructed from any circle through $A$ meeting $(O)$.

**Theorem 12.6 (Euler's relation).** *In a triangle with circumcentre $O$, incentre $I$, circumradius $R$ and inradius $r$,*

$$OI^2 = R^2 - 2Rr.$$

**Problem 12.1.** In a unit square $ABCD$ the inscribed circle $\omega$ meets $CD$ at $M$, and the segment $AM$ meets $\omega$ at a point $P$ different from $M$. What is $AP$? [AMC 12B 2020]

**Problem 12.2.** In a triangle $ABC$ we have $AB = 86$ and $AC = 97$. The circle with centre $A$ and radius $AB$ meets $BC$ at the points $B$ and $X$, and both $BX$ and $CX$ have integer lengths. What is $BC$? [AMC 10A 2013]

**Problem 12.3.** In a convex quadrilateral $KLMN$ the side $MN$ is perpendicular to the diagonal $KM$, the side $KL$ is perpendicular to the diagonal $LN$, $MN = 65$ and $KL = 28$. The line through

$L$ perpendicular to the side $KN$ meets the diagonal $KM$ at $O$, and $KO = 8$. Find $MO$. [AIME I 2019]

**Problem 12.4.** Let $\Gamma$ be a circle and let $A$ be a point outside $\Gamma$. The tangent lines to $\Gamma$ through $A$ touch $\Gamma$ at $B$ and $C$. Let $M$ be the midpoint of $AB$. The segment $MC$ meets $\Gamma$ again at $D$, and the line $AD$ meets $\Gamma$ again at $E$. Given that $AB = a$ and $BC = b$, compute $CE$ in terms of $a$ and $b$. [Brazil 2013]

**Problem 12.5.** Let $AB$ be a chord of a circle $\omega$ and let $P$ be a point on the chord $AB$. The circle $\omega_1$ passes through $A$ and $P$ and is internally tangent to $\omega$; the circle $\omega_2$ passes through $B$ and $P$ and is internally tangent to $\omega$. The circles $\omega_1$ and $\omega_2$ meet at $P$ and $Q$, and the line $PQ$ meets $\omega$ at $X$ and $Y$. Suppose that $AP = 5$, $PB = 3$, $XY = 11$ and $PQ^2 = \dfrac{m}{n}$, where $m$ and $n$ are relatively prime positive integers. Find $m + n$. [AIME I 2019]

**Problem 12.6.** Let $ABCD$ be a quadrilateral, not necessarily convex, inscribed in a circle $\omega$, and let $P = AB \cap CD$ and $Q = AD \cap BC$. Prove that

$$\operatorname{pow}(P, \omega) + \operatorname{pow}(Q, \omega) = PQ^2.$$

**Problem 12.7.** Let $C_1$ and $C_2$ be concentric circles, with $C_2$ in the interior of $C_1$. Let $A$ be a point on $C_1$ and $B$ a point on $C_2$ such that $AB$ is tangent to $C_2$. Let $C$ be the second point of intersection of the line $AB$ with $C_1$, and let $D$ be the midpoint of $AB$. A line through $A$ meets $C_2$ at $E$ and $F$ in such a way that the perpendicular bisectors of $DE$ and $CF$ meet at a point $M$ on $AB$. Find, with proof, the ratio $AM/MC$. [USAMO 1998]

**Problem 12.8.** Consider a parallelogram $ABCD$ with an obtuse angle at $A$. Let $H$ be the foot of the perpendicular from $A$ to the side $BC$. The median from $C$ in the triangle $ABC$ meets the circumcircle of the triangle $ABC$ at the point $K$. Prove that the points $K$, $H$, $C$ and $D$ lie on one circle. [Russia 2012]

**Problem 12.9.** Let the incircle $\omega$ of a triangle $ABC$ touch $BC$, $CA$ and $AB$ at $D$, $E$ and $F$, respectively. Let $Y_1$, $Y_2$, $Z_1$, $Z_2$ and $M$ be the midpoints of $BF$, $BD$, $CE$, $CD$ and $BC$, respectively, and let $X = Y_1Y_2 \cap Z_1Z_2$. Prove that $MX \perp BC$.

**Problem 12.10.** Let $ABC$ be an acute triangle with $AB < AC$, and let $E$ and $F$ be the feet of its altitudes from $B$ and from $C$, respectively. The tangent at $A$ to the circumcircle of $ABC$ meets $BC$ at $P$, and the line through $A$ parallel to $BC$ meets $EF$ at $Q$. Prove that $PQ$ is perpendicular to the median of the triangle $ABC$ from $A$.

**Problem 12.11.** Let $AD$, $BE$, $CF$ be the altitudes of a triangle $ABC$, meeting at $H$. The line through $A$ parallel to $BE$ meets $CF$ at $P$, and the line through $A$ parallel to $CF$ meets $BE$ at $Q$. Prove that $PQ$ is perpendicular to the median of the triangle $ABC$ from $A$.

**Problem 12.12.** A circle $\Omega$, its chord $AB$ and the midpoint $W$ of the minor arc $AB$ are given. Take an arbitrary point $C$ on the major arc $AB$. The tangent to $\Omega$ at $C$ meets the tangents at $A$ and at $B$ at the points $X$ and $Y$, respectively. The lines $WX$ and $WY$ meet $AB$ at $N$ and $M$, respectively. Prove that the length of the segment $NM$ does not depend on the choice of $C$.

**Problem 12.13.** A triangle $ABC$ is given. The points $P$ and $Q$ are chosen so that

$$\angle PBC = \angle QCB = 90^\circ, \qquad AP = PB, \qquad AQ = CQ.$$

The tangent to the circumcircle of $ABC$ at the point $A$ meets the line $BC$ at $R$. Prove that $P$, $Q$ and $R$ are collinear.

**Problem 12.14.** Let $CX$ and $CY$ be the tangents from the vertex $C$ of a triangle $ABC$ to the circle passing through the midpoints of its sides. Prove that the line $XY$, the line $AB$ and the tangent at $C$ to the circumcircle of $ABC$ pass through one point.

**Problem 12.15.** Given are two circles and three lines. Each of the lines meets both circles, and cuts off chords of equal length in the two circles. Prove that the circumcircle of the triangle formed by the three lines passes through the midpoint of the segment joining the centres of the two circles.

**Problem 12.16.** The diagonals of a trapezoid $ABCD$ with bases $AB$ and $CD$ meet at the point $E$. A point $P$ inside the trapezoid satisfies

$$\angle APD = \angle BPC = 90^\circ.$$

Prove that $P$ and $E$ lie on a line perpendicular to the bases.

**Problem 12.17.** Let $M$ be a point in the interior of a triangle $ABC$. The tangent to the circle $(MBC)$ at $M$ meets $BC$ at $X$; the points $Y$ and $Z$ are defined analogously using the circles $(MCA)$ and $(MAB)$. Prove that $X$, $Y$ and $Z$ are collinear.

**Problem 12.18.** Let $ABC$ be an acute triangle whose altitudes $AD$, $BE$, $CF$ meet at $H$. The line through $H$ parallel to $EF$ meets $BC$ at $X$; the points $Y$ and $Z$ are defined analogously.

(a) Prove that $X$, $Y$ and $Z$ are collinear.

(b) Let $I$ be the incentre of $ABC$. The circle $(IAD)$ meets $BC$ again at $M \neq D$, and the points $N$ and $P$ are defined analogously. Prove that $M$, $N$ and $P$ are collinear.

**Problem 12.19.** A point $P$ is taken in the interior of a triangle $ABC$ so that

$$\angle PAB = \angle PCB = \tfrac{1}{4}\left(\angle A + \angle C\right).$$

Let $L$ be the foot of the bisector of $\angle B$. The line $PL$ meets the circumcircle of the triangle $APC$ at the point $Q$. Prove that $QB$ is the bisector of the angle $\angle AQC$. [Tuymaada 2012]

**Problem 12.20.** A trapezoid $ABCD$ with bases $AB$ and $CD$ is inscribed in a circle $\Omega$. A circle $\omega$ passes through $C$ and $D$ and meets the segments $CA$ and $CB$ at $A_1 \neq C$ and $B_1 \neq C$, respectively. The points $A_2$ and $B_2$ are symmetric to $A_1$ and $B_1$ with respect to the midpoints of $CA$ and $CB$, respectively. Prove that the points $A$, $B$, $A_2$, $B_2$ are concyclic. [Russia 2014]

**Problem 12.21.** Let $ABC$ be a triangle with incentre $I$ and circumcircle $(O)$. The line $AI$ meets $(O)$ again at $D$, and the line through $I$ perpendicular to $AD$ meets $BC$ at $K$. Let $H$ be the projection of $K$ onto the line $OI$. Prove that $K$, $H$, $A$ and $D$ lie on one circle.

**Problem 12.22.** Let $ABC$ be a triangle with $\angle B < \angle A$. On the segment $BC$ take $D$ with $\angle CAD = \angle ABC$. A circle with centre $O$ passes through $B$ and $D$ and meets $AB$ and $AD$ at $E$ and $F$, respectively. The lines $BF$ and $DE$ meet at $G$, and $M$ is the midpoint of $AG$. Prove that $CM$ is perpendicular to $AO$.

**Problem 12.23.** Let $ABCD$ be a parallelogram whose diagonals meet at $T$. A circle $\omega$ with centre $O$ passes through $D$ and is tangent to $BD$. Suppose that $\omega$ meets the segment $CD$ at $E$ and the ray $AD$ at $F$ in such a way that $B$, $E$, $F$ are collinear. Prove that $\angle ATD = \angle DOB$.

**Problem 12.24.** Let the circles $\omega_1$ and $\omega_2$, with centres $O_1$ and $O_2$, respectively, meet at two distinct points $P$ and $Q$. Their common tangent closer to $P$ touches the circles at $A$ and $B$, respectively. The perpendicular from $A$ to the line $BP$ meets $O_1O_2$ at $C$. Prove that $\angle APC = 90°$. [China TST 2011]

**Problem 12.25.** Let $ABC$ be a triangle with $\angle BCA = 90°$ and let $D$ be the foot of the altitude from $C$. Let $X$ be a point in the interior of the segment $CD$. Let $K$ be the point on the segment $AX$ with $BK = BC$, and let $L$ be the point on the segment $BX$ with $AL = AC$. Let $M$ be the intersection of $AL$ and $BK$. Show that $MK = ML$.

**Problem 12.26.** Two circles $K_1$ and $K_2$ of different radii meet at two points $A$ and $B$. Let $C$ and $D$ be points on $K_1$ and $K_2$, respectively, such that $A$ is the midpoint of the segment $CD$. The extension of $DB$ meets $K_1$ at another point $E$, and the extension of $CB$ meets $K_2$ at another point $F$. Let $l_1$ and $l_2$ be the perpendicular bisectors of $CD$ and $EF$, respectively.

(a) Show that $l_1$ and $l_2$ have a unique common point, which we denote by $P$.

(b) Prove that the lengths $CA$, $AP$ and $PE$ are the side lengths of a right triangle.

[China 2013]

**Problem 12.27.** Let $A$ be a point lying outside a circle $(O)$, and let $AB$, $AC$ be the tangent lines from $A$ to $(O)$. Take $D$, $E$, $M$ on $(O)$ with $MD = ME$. Suppose that $MB$ and $MC$ meet $DE$ at $R$ and $S$, respectively, and take $X \in OB$ and $Y \in OC$ with $RX \perp DE$ and $SY \perp DE$. Prove that $XY \perp AM$. [China TST 2008]

**Problem 12.28.** Let $A_1A_2A_3A_4$ be a non-cyclic quadrilateral. Let $O_1$ and $r_1$ be the circumcentre and the circumradius of the triangle $A_2A_3A_4$, and define $O_2, O_3, O_4$ and $r_2, r_3, r_4$ in a similar way. Prove that

$$\frac{1}{O_1A_1^2 - r_1^2} + \frac{1}{O_2A_2^2 - r_2^2} + \frac{1}{O_3A_3^2 - r_3^2} + \frac{1}{O_4A_4^2 - r_4^2} = 0.$$

[IMO Shortlist 2011]

## 12.2 The radical centre and tangent circles

**Theorem 12.7 (Radical centre).** *Given three circles, no two of them concentric, the three pairwise radical axes are either concurrent or all parallel. In the second case the three centres are collinear. When the three radical axes meet at a point, that point is called the* radical centre *of the three circles.*

**Remark.** Stated as a question about points: given three distinct circles, the number of points having the same power with respect to all three is infinitely many if the circles share one or two common points; zero if the three centres are collinear and the circles have no common point; and exactly one otherwise, which is the generic case.

**Problem 12.29.** Let $ABC$ be a triangle. Take points $D$, $E$, $F$ on the perpendicular bisectors of $BC$, $CA$, $AB$, respectively. Show that the lines through $A$, $B$, $C$ perpendicular to $EF$, $FD$, $DE$, respectively, are concurrent or parallel. [USAMO 1997]

**Problem 12.30.** An acute triangle $ABC$ is inscribed in a circle with centre $O$, and its altitudes are $AD$, $BE$ and $CF$. The line $EF$ meets the circle at $P$ and $Q$.

(a) Show that $OA$ is perpendicular to $PQ$.

(b) If $M$ is the midpoint of $BC$, show that $AP^2 = 2 \cdot AD \cdot OM$.

[IberoAmerican 1999]

**Problem 12.31.** Two circles $(O_1, R_1)$ and $(O_2, R_2)$ are tangent at $M$, with $R_1 < R_2$. A point $A$ moves on the circle $(O_2)$ and does not lie on the line $AO_1$. Let $AB$ and $AC$ be the two tangent lines from $A$ to the circle $(O_1)$. The lines $CM$ and $BM$ meet $(O_2)$ again at $F$ and $E$. Let $D$ be the intersection of $EF$ with the tangent line to $(O_2)$ at $A$. Prove that $D$ lies on the common tangent of the two circles.

**Problem 12.32.** Circles $k_1$, $k_2$, $k_3$, $k_4$ are such that $k_2$ and $k_4$ each touch $k_1$ and $k_3$. Show that the four points of tangency are collinear or concyclic.

**Problem 12.33.** Let $\omega$ be the semicircle with diameter $PQ$. A circle $k$ is tangent internally to $\omega$ and tangent to the segment $PQ$ at $C$. Let $AB$ be the tangent to $k$ perpendicular to $PQ$, with $A$ on $\omega$ and $B$ on the segment $CQ$. Show that $AC$ bisects the angle $\angle PAB$.

**Problem 12.34.** Let $p$ be the semiperimeter of a triangle $ABC$. Points $E$ and $F$ lie on the line $AB$ and satisfy $|CE| = |CF| = p$. Prove that the circumcircle of the triangle $CEF$ is tangent to the excircle of the triangle $ABC$ opposite to the side $AB$.

**Problem 12.35.** A circle $\omega$ lies inside a circle $\Omega$ and they are internally tangent at $S$. The segments $SA$ and $SB$ are diameters of $\omega$ and $\Omega$, respectively. A circle $o$ tangent to the segment $AB$, to $\omega$ and to $\Omega$ touches $AB$ at $C$. Prove that $SC$ is the harmonic mean of $SA$ and $SB$.

**Problem 12.36.** A triangle $ABC$ is right-angled at $A$. A circle $\omega$ touches the segment $AB$, touches the segment $BC$ at $D$, and is externally tangent to the circle with diameter $AC$. Prove that $AC = CD$.

**Problem 12.37.** Given is a trapezoid $ABCD$ with $AB \parallel CD$ in which $AC \cap BD = E$ and $BC \cap AD = F$. The circumcircles of the triangles $BCE$ and $ADE$ meet at the distinct points $E$ and $G$. Prove that $\angle EFA = \angle GFB$.

**Problem 12.38.** Let $ABC$ be an acute non-isosceles triangle. The perpendicular bisector of $AB$ meets $CA$ at $C_1$ and $BC$ at $C_2$; the perpendicular bisector of $BC$ meets $AB$ at $A_1$ and $CA$ at $A_2$; the perpendicular bisector of $CA$ meets $BC$ at $B_1$ and $AB$ at $B_2$. Prove that the circumcentres of the triangles $ABC$, $A_1B_1C_1$ and $A_2B_2C_2$ are collinear.

**Problem 12.39.** A circle $\omega$ lies inside a circle $\Omega$ and the two are disjoint. Consider all pairs of circles $o_1$, $o_2$ that are externally tangent to $\omega$ and internally tangent to $\Omega$. Prove that the radical axes of all such pairs of circles have a common point.

**Problem 12.40.** Circles $o_1$ and $o_2$ are internally tangent to a circle $\Omega$ at the points $B$ and $C$, respectively, and the tangents to $\Omega$ at these points meet at $A$. Two further circles $\alpha$ and $\beta$ are each externally tangent to $o_1$ and $o_2$ and internally tangent to $\Omega$, at $S$ and $T$, respectively. Prove that $A$, $S$ and $T$ are collinear.

**Problem 12.41.** Circles $o_1$ and $o_2$ meet at two different points $A$ and $B$. Circles $o_3$ and $o_4$ are both externally tangent to both $o_1$ and $o_2$, with $o_3$ touching them at $C$ and $E$, respectively, and $o_4$ touching them at $D$ and $F$, respectively. Prove that the circumcircles of the triangles $ADF$ and $ACE$ are tangent at $A$.

**Problem 12.42.** Points $A$, $B$, $C$ are given on a line in this order. Semicircles $\omega$, $\omega_1$, $\omega_2$ are drawn on $AC$, $AB$, $BC$ as diameters, all on the same side of the line. A sequence of circles $(k_n)$ is constructed as follows: $k_0$ is the circle determined by $\omega_2$, and for $n \geq 1$ the circle $k_n$ is tangent

to $\omega$, to $\omega_1$ and to $k_{n-1}$. Prove that the distance from the centre of $k_n$ to $AB$ is $2n$ times the radius of $k_n$.

**Problem 12.43.** Let $\omega$ be the circumcircle of a triangle $ABC$. A circle $\omega_1$ is inscribed in the angle $BAC$ and touches $\omega$ internally at $T$. Let $D$ be the point at which the $A$-excircle touches $BC$. Prove that $\angle BAT = \angle DAC$.

**Problem 12.44.** Let $\omega$ be a circle tangent internally to a circle $\Omega$. Let $A$ be a point on $\Omega$, and let $AX$ and $AY$ be the tangents from $A$ to $\omega$. Consider the circles $\omega_1$ and $\omega_2$ tangent internally to $\Omega$ and tangent to $\omega$ at $X$ and $Y$, respectively. Prove that $\omega$, $\omega_1$ and $\omega_2$ have a common exterior tangent.

**Problem 12.45.** In an acute triangle $ABC$ the segment $AD$ is an altitude, and $M$, $N$ are the projections of $D$ onto $AB$, $AC$, respectively. The line $MN$ meets the circumcircle of $ABC$ at $P$ and $Q$, and the line $AD$ meets it at $A$ and $R$. Prove that $D$ is the incentre of the triangle $PQR$.

**Problem 12.46.** Circles $o_1, o_2, o_3, o_4, o_5, o_6$ are internally tangent to a circle $\Omega$ at the points $A_1, A_2, A_3, A_4, A_5, A_6$, respectively. Moreover, for $i = 1, 2, 3, 4, 5, 6$ the circles $o_i$ and $o_{i+1}$ are externally tangent, where $o_7 = o_1$. Prove that the lines $A_1A_4$, $A_2A_5$, $A_3A_6$ are concurrent.

**Problem 12.47.** A circle $\omega$ lies inside a circle $\Omega$ and passes through its centre. Circles $o_1, o_2, o_3, o_4, o_5, o_6$ are internally tangent to $\Omega$ and externally tangent to $\omega$, and for $i = 1, 2, 3, 4, 5, 6$ the circles $o_i$ and $o_{i+1}$ are externally tangent, where $o_7 = o_1$. Find all possible values of the ratio of the diameters of $\Omega$ and $\omega$.

**Problem 12.48.** Circles $\omega_1$ and $\omega_2$ meet at the points $X$ and $Y$. A line $l$ is tangent to $\omega_1$ and $\omega_2$ at $A$ and $B$, respectively, with the line $AB$ closer to $X$ than to $Y$. A circle $\omega$ passes through $A$ and $B$, meeting $\omega_1$ again at $D \neq A$ and $\omega_2$ again at $C \neq B$. The three points $C$, $Y$, $D$ are collinear, $XC = 67$, $XY = 47$ and $XD = 37$. Find $AB^2$. [AIME I 2016]

**Problem 12.49.** Let $AB$ be a chord of a circle $(O)$ and let $M$ be the midpoint of one of the two arcs $AB$. A circle $(I)$ lying on the other side of $AB$ from $M$ is tangent to $AB$ and internally tangent to $(O)$. The lines through $M$ perpendicular to $AI$ and to $BI$ meet $AB$ at $C$ and $D$, respectively. Prove that $AB = 2 \cdot CD$.

**Problem 12.50.** Let $ABC$ be an acute triangle whose incircle touches the segments $AB$ and $AC$ at $Z$ and $Y$, respectively. Let $G$ be the intersection of $BY$ and $CZ$, and construct the points $R$ and $S$ so that the two quadrilaterals $BCYR$ and $BCSZ$ are parallelograms. Prove that $GR = GS$. [IMO Shortlist 2009 G3]

**Problem 12.51.** Let $ABC$ be an acute triangle with circumcircle $\Omega$. Let $B_0$ be the midpoint of $AC$ and let $C_0$ be the midpoint of $AB$. Let $D$ be the foot of the altitude from $A$, and let $G$ be the centroid of the triangle $ABC$. Let $\omega$ be a circle through $B_0$ and $C_0$ that is tangent to $\Omega$ at a point $X \neq A$. Prove that the points $D$, $G$ and $X$ are collinear. [IMO Shortlist 2011]

## 12.3 Incentre, excentres, and the Nagel point

**Definition 12.8 (Nagel point).** In any triangle the segments joining each vertex to the point at which the opposite excircle touches the corresponding side meet at one point. This point is called the *Nagel point* of the triangle.

**Problem 12.52.** The orthocentre of an isosceles triangle lies on its incircle. Find the angles of the triangle.

**Problem 12.53.** In a triangle $ABC$ the angle $A$ equals $60°$. Prove that $IO = IH$, where $I$, $O$ and $H$ are the incentre, the circumcentre and the orthocentre.

**Problem 12.54.** Let $ABCD$ be a cyclic quadrilateral such that the inradii of the triangles $ABC$ and $BCD$ are equal. Prove that the inradii of the triangles $CDA$ and $DAB$ are also equal.

**Problem 12.55.** A quadrilateral $ABCD$ satisfies $AD = BD = CD$. On the ray $DB$, beyond the point $B$, a point $E$ is chosen so that $2\angle ECB = \angle EDA$. Prove that $2\angle EAB = \angle EDC$.

**Problem 12.56.** Let $ABC$ be a non-isosceles triangle with $AB \neq BC$, and let $B_1$ be the midpoint of the arc $AC$ of its circumcircle that contains $B$. Points $A_0$ and $C_0$ are marked on the segments $BC$ and $BA$, respectively. Prove that the points $B$, $B_1$, $C_0$ and $A_0$ are concyclic if and only if $AC_0 = CA_0$. Consider also all possible positions of $C_0$ and $A_0$ on the lines $BC$ and $BA$.

**Problem 12.57.** Let $ABC$ be a non-isosceles triangle with $AB \neq BC$ and let $I$ be its incentre. Points $A_0$ and $C_0$ are marked on the segments $BC$ and $BA$, respectively. Prove that the points $B$, $I$, $C_0$ and $A_0$ are concyclic if and only if $AC_0 + CA_0 = AC$. Consider also all possible positions of $C_0$ and $A_0$ on the lines $BC$ and $BA$.

**Problem 12.58.** Let $ABC$ be a non-isosceles triangle with $AB \neq BC$, let $M$ be the midpoint of $AC$, and denote by $I_A$ and $I_C$ the incentres of the triangles $MAB$ and $MCB$. Let $B_1$ be the midpoint of the arc $AC$ of the circumcircle of $ABC$ that contains $B$. Prove that the points $B$, $B_1$, $I_A$ and $I_C$ are concyclic.

**Problem 12.59.** A point $M$ lies on the side $AB$ of a circumscribed quadrilateral $ABCD$. Let $I_1$, $I_2$ and $I_3$ be the incentres of the triangles $MBC$, $MCD$ and $MDA$, respectively. Prove that the points $M$, $I_1$, $I_2$ and $I_3$ lie on one circle.

**Problem 12.60.** Points $A$ and $B$ lie on a circle $\omega$. Points $C$ and $D$ move on the arc $AB$ in such a way that $CD$ has constant length. Let $I_1$ and $I_2$ be the incentres of the triangles $ABC$ and $ABD$, respectively. Prove that the segment $I_1I_2$ also has constant length.

**Problem 12.61.** Let $ABC$ be a triangle with incentre $I$, and let $P$ and $Q$ be the midpoints of the arcs $BC$ and $CA$ of its circumcircle not containing $A$ and $B$, respectively. Let $F$ be the point for which $CPFQ$ is a parallelogram. Prove that if $I \neq F$, then $\angle CIF = 90°$.

**Problem 12.62.** Let $ABCD$ be a convex quadrilateral inscribed in a circle such that $AB = BC$ and $\angle ABC = 60°$. Let $K$ be the midpoint of the shorter arc $AC$ of the circumcircle. Let $I$ and $J$ be the incentres of the triangles $ABD$ and $CBD$, respectively. Prove that the triangle $IJK$ is equilateral.

**Problem 12.63.** Let $ABCDEFG$ be a regular heptagon, and let $I$ and $J$ be the incentres of the triangles $ABC$ and $ADE$, respectively. Prove that $IG = JG$.

**Problem 12.64.** In a triangle $ABC$ the bisector of the angle $\angle BAC$ meets the circumcircle at the point $D$. Let $\omega_1$ and $\omega_2$ be the incircles of the triangles $ABD$ and $ACD$, respectively. Prove that one of the common external tangents of $\omega_1$ and $\omega_2$ is parallel to $BC$.

**Problem 12.65.** Let $ABCD$ be a cyclic quadrilateral and let $M$ and $N$ be the midpoints of the arcs $AB$ and $CD$, respectively. Prove that $MN$ bisects the segment joining the incentres of the triangles $ABC$ and $ADC$.

**Problem 12.66.** Let $I$ be the incentre of a triangle $ABC$, let $M$ be the midpoint of $AC$, and let $W$ be the midpoint of the arc $AB$ of its circumcircle not containing $C$. It is known that $\angle AIM = 90^\circ$. Find the ratio $CI : IW$.

**Problem 12.67.** The incircle of a triangle $ABC$ touches the sides $BC$, $CA$ and $AB$ at the points $D$, $E$ and $F$, respectively. Let $M$, $N$ and $J$ be the incentres of the triangles $AEF$, $BDF$ and $DEF$, respectively. Prove that the points $F$ and $J$ are symmetric with respect to the line $MN$.

**Problem 12.68.** Let $I$ be the incentre of a triangle $ABC$. The line $AI$ meets the line $BC$ at $D$ and the circumcircle of $ABC$ at $S \neq A$. Let $K$ be the incentre of the triangle $DSB$ and let $L$ be the incentre of the triangle $DSC$. Let $P$ be the reflection of $I$ in the line $KL$. Prove that the angle $\angle BPC$ is right.

**Problem 12.69.** Let $K$ and $L$ be points on the arcs $AB$ and $BC$, respectively, of the circumcircle of a triangle $ABC$, such that $KL \parallel AC$. Prove that the incentres of the triangles $ABK$ and $CBL$ are equidistant from the midpoint of the arc $AC$ of the circumcircle of $ABC$ that contains $B$.

**Problem 12.70.** In a triangle $ABC$ let $I$ and $J$ be the incentre and the $A$-excentre, respectively. Let $O_B$ be the centre of the circle passing through $B$ and tangent to the line $IC$ at $I$, and let $O_C$ be the centre of the circle passing through $C$ and tangent to the line $IB$ at $I$. The lines $O_BO_C$ and $IJ$ meet at $K$. Determine the ratio $IK : KJ$.

**Problem 12.71.** Let $I$ be the incentre of an acute triangle $ABC$. A point $X$ lies on the segment $BC$, on the same side of the line $AI$ as the point $B$, and a point $Y$ lies on the shorter arc $AB$ of the circumcircle of $ABC$. The equalities

$$\angle AIX = \angle XYA = 120^\circ$$

hold. Prove that the line $YI$ is the bisector of the angle $\angle XYA$.

**Problem 12.72.** Let $M$, $N$, $P$ be the points where the incircle of a scalene triangle $ABC$ touches the sides $BC$, $CA$, $AB$, respectively. Prove that the orthocentre of the triangle $MNP$, the incentre $I$ of the triangle $ABC$ and the circumcentre $O$ of the triangle $ABC$ are collinear.

## 12.4 Homothety II

*Revisits homothety and the centres of similitude of two circles (see Chapter 7); the problems below push it further.*

**Problem 12.73.** Circles $o_1$ and $o_2$ are externally tangent at a point $A$. The common external tangent of $o_1$ and $o_2$ meets the line joining their centres at $B$. A line through $B$ meets $o_1$ at $D$ and $o_2$ at $E$. Prove that $\angle DAE = 90^\circ$.

**Problem 12.74.** Circles $o_1$ and $o_2$ are inscribed in an angle with vertex $P$. A line through $P$ meets the circles at the points $A$, $B$, $C$, $D$, lying on the line in that order. The tangents to the circles at $A$ and at $D$ meet at $X$. Prove that $AX = DX$.

**Problem 12.75.** Through a point inside a triangle, lines are drawn parallel to its sides. These lines divide the triangle into three smaller triangles and three quadrilaterals. Prove that the sum of the inradii of the three small triangles equals the inradius of the original triangle.

**Problem 12.76.** Let $o$ be the incircle of a triangle $ABC$. The tangents to $o$ parallel to the lines $BC$, $CA$, $AB$ cut off from $ABC$ three triangles $AKL$, $BMN$ and $CPQ$. Prove that the sum of the inradii of the triangles $AKL$, $BMN$, $CPQ$ equals the inradius of $ABC$, and express the area of $ABC$ in terms of the areas of $AKL$, $BMN$ and $CPQ$.

**Problem 12.77.** Let $ABC$ be a triangle. Three circles of radius $\rho$ pass through a point $P$, and each of them is tangent to two sides of the triangle $ABC$. Prove that $P$, the circumcentre $O$ and the incentre $I$ of the triangle $ABC$ lie on one line, and determine $\rho$ in terms of the inradius $r$ and the circumradius $R$.

**Problem 12.78.** Let $ABC$ be a triangle and let $D$, $E$, $F$ be the midpoints of the sides $BC$, $CA$, $AB$, respectively. The line $AD$ meets the circumcircle of the triangle $DEF$ again at $P$ and the circumcircle of the triangle $ABC$ again at $Q$. Prove that $PD = AP + DQ$.

**Problem 12.79.** Two circles are internally tangent at $N$. The chords $BA$ and $BC$ of the larger circle are tangent to the smaller circle at $K$ and $M$, respectively. Let $Q$ and $P$ be the midpoints of the arcs $AB$ and $BC$, respectively. The circumcircles of the triangles $BQK$ and $BPM$ meet at $L$. Show that $BPLQ$ is a parallelogram. [Russia 2000]

**Problem 12.80.** In a triangle $ABC$ let $H$ be the orthocentre, $I$ the incentre, $O$ the circumcentre, and $K$ the point at which the incircle touches $BC$. Given that $IO$ is parallel to $BC$, prove that $AO$ is parallel to $HK$. [Tournament of Towns 2003]

**Problem 12.81.** Let $ABCD$ be an isosceles trapezoid with $AB \parallel CD$. The inscribed circle $\omega$ of the triangle $BCD$ meets $CD$ at $E$. Let $F$ be a point on the internal bisector of $\angle DAC$ such that $EF \perp CD$. The circumscribed circle of the triangle $ACF$ meets the line $CD$ at $C$ and $G$. Prove that the triangle $AFG$ is isosceles. [USAMO 1999]

**Problem 12.82.** A circle $\omega$ with centre $I$ is inscribed in a segment of a disk formed by an arc and a chord $AB$. Let $M$ be the midpoint of this arc $AB$ and let $N$ be the midpoint of the complementary arc. The tangents from $N$ touch $\omega$ at the points $C$ and $D$. The opposite sidelines $AC$ and $BD$ of the quadrilateral $ABCD$ meet at $X$, and the diagonals of $ABCD$ meet at $Y$. Prove that $X$, $Y$, $I$ and $M$ are collinear. [Sharygin 2012]

**Problem 12.83.** Let $ABCD$ be a trapezoid with bases $AD$ and $BC$. Let $E$ and $F$ be points on the segments $AB$ and $CD$, respectively. The circumcircle of the triangle $AEF$ meets the segment $AD$ at $A_1$, and the circumcircle of the triangle $CEF$ meets the segment $BC$ at $C_1$. Prove that the lines $A_1C_1$, $BD$ and $EF$ are concurrent.

**Problem 12.84.** Points $A$, $B$, $C$, $D$, $E$, $F$ lie in this order on a circle $\omega$ of radius $r$, and the chords $AD$, $BE$, $CF$ meet at a common point $P$. A circle of radius $r_1$ is tangent to the segments $AP$, $BP$ and to the circle $\omega$; a circle of radius $r_2$ is tangent to the segments $CP$, $DP$ and to $\omega$; a circle of radius $r_3$ is tangent to the segments $EP$, $FP$ and to $\omega$. Prove that $r_1 + r_2 + r_3 \leq r$.

**Problem 12.85.** Let $I$ be the incentre of a triangle $ABC$, and let $M$ and $N$ be the midpoints of the arcs $ABC$ and $BAC$ of its circumcircle, that is, of the arcs containing the third vertex. Prove that the points $M$, $I$ and $N$ are collinear if and only if

$$AC + BC = 3AB.$$

**Problem 12.86.** Let $ABCD$ be a circumscribed quadrilateral with incircle $\omega$, and let the lines $AB$ and $CD$ meet at $E$. Let $\omega_1$ be the circle tangent to the side $BC$ at the point $K$ and to the extensions of the sides $AB$ and $CD$, and let $\omega_2$ be the circle tangent to the side $AD$ at the point $L$ and to the extensions of the sides $AB$ and $CD$. Prove that if the points $E$, $K$ and $L$ lie on one line, then the midpoints of the segments $BC$ and $AD$ and the centre of $\omega$ are collinear.

**Problem 12.87.** Let $ABCDEF$ be a convex hexagon such that

$$\angle FAB + \angle BCD + \angle DEF = 360^\circ \qquad \text{and} \qquad \angle AEB = \angle ADB,$$

and assume that the segments $AB$ and $DE$ are not parallel. Prove that the centres of the circumcircles of the triangles $AFE$ and $BCD$ and the intersection point of the lines $AB$ and $DE$ lie on one line.

**Problem 12.88.** In a convex pentagon $ABCDE$ the lines $AE$ and $BC$ are parallel and $\angle ADE = \angle BDC$. The diagonals $AC$ and $BE$ meet at $P$. Prove that

$$\angle EAD = \angle BDP \qquad \text{and} \qquad \angle CBD = \angle ADP.$$

**Problem 12.89.** Let $ABCD$ be a convex quadrilateral, not a trapezoid, inscribed in a circle and circumscribed about a circle $\omega$. Let the points of tangency of $\omega$ with the sides $AB$, $BC$, $CD$, $DA$ be $K$, $L$, $M$, $N$, respectively. Let the circle with centre $I_K$, distinct from $\omega$, be tangent to the side $AB$ and to the lines $AD$ and $BC$; let the circle with centre $I_L$, distinct from $\omega$, be tangent to the side $BC$ and to the lines $AB$ and $CD$; let the circle with centre $I_M$, distinct from $\omega$, be tangent to the side $CD$ and to the lines $AD$ and $BC$; and let the circle with centre $I_N$, distinct from $\omega$, be tangent to the side $DA$ and to the lines $AB$ and $CD$. Prove that the lines $I_K K$, $I_L L$, $I_M M$ and $I_N N$ are concurrent.

**Problem 12.90.** A circle $\omega$ lies inside a triangle $ABC$ and is tangent to the sides $AC$ and $BC$ at the points $K$ and $L$, respectively. A point $P$ lies on the circle $\omega$ and satisfies

$$\frac{AP}{AK} = \frac{BP}{BL}.$$

Prove that the circumcircle of the triangle $ABP$ is tangent to the circle $\omega$.

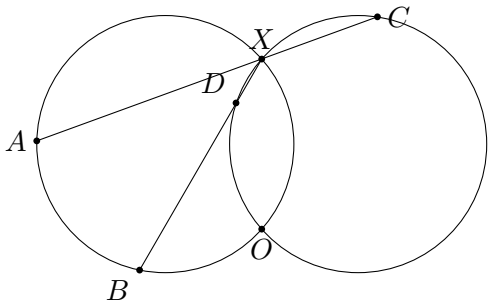
## 12.5 Spiral similarity and the Miquel point

**Definition 12.9 (Spiral similarity).** A *spiral similarity* (also called a *roto-homothety*) is the composition of a rotation about a point $O$ with a homothety centred at the same point $O$. The point $O$ is its centre.

**Lemma 12.10 (Spiral similarity lemma).** *If a spiral similarity with centre $O$ takes $A$ to $A'$ and $B$ to $B'$, then*

$$\triangle OAA' \sim \triangle OBB' \qquad \text{and} \qquad \triangle OAB \sim \triangle OA'B'.$$

**Theorem 12.11 (Centre of a spiral similarity; the Miquel point).** *Let $AC$ and $BD$ be two non-parallel segments and let the lines $AC$ and $BD$ meet at $X$. The circumcircles of the triangles $ABX$ and $CDX$ meet again at a point $O \neq X$. Then $O$ is the centre of the unique spiral similarity taking $AC$ to $BD$. This point $O$ is the Miquel point of $ABCD$.*




**Problem 12.91.** Let $ABCD$ be a square labelled clockwise and let $P$ be any point inside the segment $BC$. Construct the square $APRS$, also labelled clockwise. Show that the line $CR$ is tangent to the circumcircle of the triangle $ABC$. [Cono Sur]

**Problem 12.92.** Let $ABCD$ be a convex quadrilateral with $AD = DC$, $AC = AB$ and $\angle ADC = \angle CAB$. If $M$ and $N$ are the midpoints of the sides $AD$ and $AB$, prove that the triangle $MNC$ is isosceles. [Brazil Junior 2011]

**Problem 12.93.** Consider a triangle $ABC$. Let $D$ be a point on the circumcircle of the triangle and let $E$ and $F$ be the feet of the perpendiculars from $A$ to $DB$ and $DC$, respectively. Let $N$ be the midpoint of $EF$ and let $M$ be the midpoint of the side $BC$. Assuming that $N$ is distinct from $M$, prove that the lines $NA$ and $NM$ are perpendicular. [Brazil Junior 2013]

**Problem 12.94.** Two circles $\Gamma_1$ and $\Gamma_2$ meet at $A$ and $B$. A straight line through $B$ meets $\Gamma_1$ at $C \neq B$ and $\Gamma_2$ at $E \neq B$. Another straight line through $B$ meets $\Gamma_1$ at $D \neq B$ and $\Gamma_2$ at $F \neq B$. Suppose further that $B$ lies between $C$ and $E$ and between $D$ and $F$. Let $M$ and $N$ be the midpoints of $CE$ and $DF$, respectively. Prove that the triangles $ACD$, $AEF$ and $AMN$ are similar. [Germany]

**Problem 12.95.** The sidelines of an acute-angled triangle $T$ are coloured red, green and blue. These lines are rotated about the circumcentre of $T$ clockwise by $120°$, each line keeping its colour. Prove that the three points determined by the pairs of lines of the same colour are the vertices of a triangle congruent to $T$. [Russia 2023]

**Problem 12.96.** In a triangle $ABC$ we have $\angle BAC = 45°$. Points $P$ and $Q$ inside $ABC$ satisfy

$$\angle CBP = \angle PBQ = \angle QBA = \tfrac{1}{3}\angle ABC, \qquad \angle BCP = \angle PCQ = \angle QCA = \tfrac{1}{3}\angle ACB.$$

Let $D$ and $E$ be the orthogonal projections of $P$ onto $AB$ and $AC$, respectively. Prove that $Q$ is the orthocentre of the triangle $ADE$. [Rioplatense]

**Problem 12.97.** The points $A_1$, $B_1$ and $C_1$ lie on the sides $BC$, $CA$ and $AB$ of a triangle $ABC$, respectively, and satisfy

$$AB_1 - AC_1 = CA_1 - CB_1 = BC_1 - BA_1.$$

Let $O_A$, $O_B$ and $O_C$ be the circumcentres of the triangles $AB_1C_1$, $A_1BC_1$ and $A_1B_1C$, respectively. Prove that the incentre of the triangle $O_AO_BO_C$ is the incentre of the triangle $ABC$. [Russia 2012]

**Problem 12.98.** Let $ABCD$ be a fixed convex quadrilateral with $BC = DA$ and $BC$ not parallel to $DA$. Let two variable points $E$ and $F$ lie on the sides $BC$ and $DA$, respectively, and satisfy $BE = DF$. The lines $AC$ and $BD$ meet at $P$, the lines $BD$ and $EF$ meet at $Q$, and the lines $EF$ and $AC$ meet at $R$. Prove that the circumcircles of the triangles $PQR$, as $E$ and $F$ vary, have a common point other than $P$. [IMO 2005]

**Problem 12.99.** Let $ABC$ be an acute scalene triangle with circumcentre $O$. A point $P$ lies inside the triangle $ABC$ and satisfies $\angle PAB = \angle PBC$ and $\angle PAC = \angle PCB$. A point $Q$ lies on the line $BC$ with $QA = QP$. Prove that $\angle AQP = 2\angle OQB$. [USA TST 2005]

**Problem 12.100.** Let $ABC$ be a triangle. Triangles $PAB$ and $QAC$ are constructed outside the triangle $ABC$ so that $AP = AB$, $AQ = AC$ and $\angle BAP = \angle CAQ$. The segments $BQ$ and $CP$ meet at $R$. Let $O$ be the circumcentre of the triangle $BCR$. Prove that $AO \perp PQ$. [USA TST 2006]

**Problem 12.101.** Let the points $A_1$, $A_2$, $A_3$ lie on a circle $\Gamma$ in counter-clockwise order, and let $P$ be a point in the same plane. For $i \in \{1, 2, 3\}$ let $\tau_i$ denote the counter-clockwise rotation of the plane centred at $A_i$ whose angle equals the angle at the vertex $A_i$ in the triangle $A_1A_2A_3$. Define $P_i$ to be the point $\tau_{i+2}(\tau_i(\tau_{i+1}(P)))$, where the indices are taken modulo 3, that is, $\tau_4 = \tau_1$ and $\tau_5 = \tau_2$. Prove that the circumradius of the triangle $P_1P_2P_3$ is at most the radius of $\Gamma$. [India 2024]

## 12.6 Humpty and Dumpty points

**Definition 12.12 (Humpty point).** Let $ABC$ be a triangle. Let $\Gamma_B$ be the circle through $A$ and $B$ tangent to $BC$, and let $\Gamma_C$ be the circle through $A$ and $C$ tangent to $BC$. Let $X_a$ be the second intersection of $\Gamma_B$ and $\Gamma_C$, so $X_a \neq A$. The point $X_a$ is called the *Humpty point* of $ABC$ related to $A$.

**Definition 12.13 (Dumpty point).** Let $ABC$ be a triangle. Let $\Omega_B$ be the circle through $A$ and $B$ tangent to $AC$, and let $\Omega_C$ be the circle through $A$ and $C$ tangent to $AB$. Let $Y_a$ be the second intersection of $\Omega_B$ and $\Omega_C$, so $Y_a \neq A$. The point $Y_a$ is called the *Dumpty point* of $ABC$ related to $A$.

**Theorem 12.14 (Properties of the Humpty and Dumpty points).** *Let $ABC$ be a triangle with circumcentre $O$ and orthocentre $H$; let $M_a$, $M_b$, $M_c$ be the midpoints of the sides and let $D$, $E$, $F$ be the feet of the altitudes. Let $X_a$ and $Y_a$ be the $A$-Humpty and the $A$-Dumpty point, and let $AS_a$ denote the symmedian from $A$. Then*

*(a) $X_a$ lies on the median $AM_a$;*
*(b) $BCHX_a$ is cyclic;*
*(c) $A$, $F$, $H$, $X_a$, $E$ are concyclic;*
*(d) $HX_a \perp AM_a$;*
*(e) $Y_a$ lies on the symmedian $AS_a$;*
*(f) $BCOY_a$ is cyclic;*

*(g)* $A$, $M_b$, $M_c$, $O$, $Y_a$ *are concyclic;*
*(h)* $OY_a \perp AS_a$*;*
*(i)* $Y_a$ *is the centre of the spiral similarity taking* $BA$ *to* $AC$*;*
*(j)* $X_a$ *and* $Y_a$ *are isogonal conjugates.*

**Problem 12.102.** Let $ABC$ be a triangle and let $M$, $N$ be the midpoints of $AB$, $AC$, respectively. The perpendicular bisectors of $AB$ and $AC$ meet the median from $A$ at $D$ and $E$, respectively, and the lines $BD$ and $CE$ meet at $F$. Prove that $A$, $M$, $F$, $N$ are concyclic. [USAMO 2008]

**Problem 12.103.** Let $ABC$ be a scalene triangle. Let $K_a$, $L_a$, $M_a$ be the intersections of $BC$ with the interior angle bisector, the exterior angle bisector and the median from $A$, respectively. The circle $AK_aL_a$ meets $AM_a$ again at $X_a$; define $X_b$ and $X_c$ similarly. Prove that the circumcentre of the triangle $X_aX_bX_c$ lies on the Euler line of $ABC$. [USA TSTST 2015]

## 12.7 Enriching the picture: auxiliary constructions

**Technique 12.15 (Enriching the picture).** Many problems whose solutions use no advanced machinery become transparent once one extra, consciously chosen element is added to the figure. The elements worth trying first are:

- two equilateral triangles sharing a vertex, which produce many congruences;
- parallelograms, which produce many equalities, of both angles and segments;
- equal segments sharing a point, or even overlapping;
- in the presence of a right triangle, the midpoint of its hypotenuse.

**Problem 12.104.** Given is a right triangle $ABC$ with $\angle B = 90^\circ$. The point $M$ is the midpoint of $AC$ and the point $P$ lies on $BC$ in such a way that $\angle APB = \angle CPM$. Prove that $PC = 2 \cdot PB$.

**Problem 12.105.** Let $M$ be the midpoint of the side $BC$ of a triangle $ABC$. Suppose that $\angle ABC + \angle MAC = 90^\circ$. Prove that $ABC$ is right-angled or isosceles.

**Problem 12.106.** Squares $ABDE$ and $ACFG$ are built outside a triangle $ABC$. Express the distance between the midpoints of the segments $DG$ and $EF$ in terms of the side lengths of $ABC$.

**Problem 12.107.** Given is a convex hexagon $ABCDEF$ in which $ABC$ and $DEF$ are equilateral triangles. Prove that $AD + CF \geq BE$.

**Problem 12.108.** In a triangle $ABC$ with $AB > AC$ the point $P$ is the projection of $C$ onto the bisector of the angle $A$, and $M$ is the midpoint of $BC$. Find $PM$ in terms of the side lengths of $ABC$.

**Problem 12.109.** Given is a right triangle $ABC$ with $\angle ACB = 90^\circ$. On the sides $BC$, $CA$, $AB$ lie the points $D$, $E$, $F$, respectively, so that

$$\angle ADC = \angle BDF, \qquad \angle BFD = \angle AFE, \qquad \angle AEF = \angle BEC.$$

Prove that $AD + DF = BE + EF$.

**Problem 12.110.** The point $M$ is the midpoint of the hypotenuse $BC$ of a right triangle $ABC$. The perpendicular bisector of $AM$ meets $AB$ and $AC$ at $K$ and $L$, respectively. Prove that there exists a right triangle with side lengths $BK$, $CL$, $KL$.

**Problem 12.111.** Given is a triangle $ABC$ with $AC = BC$. The points $D$ and $E$ belong to the segments $AC$ and $BC$, respectively, and satisfy

$$\angle EAB = \angle ABD = \angle ACB \qquad \text{and} \qquad AE + ED + DB = AC.$$

Find $\angle ACB$.

**Problem 12.112.** Given is a triangle $ABC$ with $AB = AC$. On the sides $AB$ and $AC$ the points $D$ and $E$ are chosen, respectively, so that $BC = CD = DE$ and $AE = BD$. Find $\angle A$.

**Problem 12.113.** Given is a triangle $ABC$ with $\angle A = 120°$, and let $M$ be the midpoint of $BC$. On the sides $AB$ and $AC$ lie the points $P$ and $Q$, respectively, so that $BP = PQ = QC$. Prove that $\angle PMQ = 90°$.

**Problem 12.114.** Given is a convex pentagon $ABCDE$ all of whose sides are equal and in which $\angle ABC + \angle CDE = 180°$. The segments $AD$ and $BE$ meet at $P$. Prove that the length of $CP$ equals the side length of the pentagon.

**Problem 12.115.** In a convex hexagon $ABCDEF$ we have $\angle BCD = \angle EFA = 90°$. Prove that the perimeter of the quadrilateral $ABDE$ is at least $2 \cdot CF$.

**Problem 12.116.** Given is a circle $\omega$ of diameter 1 and a broken line $z$ of length less than 1 whose endpoints lie on $\omega$. Prove that some diameter of $\omega$ is disjoint from $z$.

**Problem 12.117.** Given is an acute triangle $ABC$ in which $AB < AC$. Let $M$ be the midpoint of the arc $BAC$ of the circumcircle of $ABC$, and let $O$ be the centre of this circle. The circumcircle of $AMO$ meets the segments $AB$ and $AC$ at $K$ and $L$, both different from $A$. Let $P$ and $Q$ be the projections of $K$ and $L$ onto $BC$, respectively. Prove that $BC = 2 \cdot PQ$.

**Problem 12.118.** Let $ABCDEFG$ be a regular 7-gon and let the lines $AB$ and $CE$ meet at $P$. Find $\angle PDG$.

**Problem 12.119.** Let $ABCDEFG$ be a regular 7-gon of side length 1 and let the lines $CF$ and $DG$ meet at $P$. Find $AP$.

**Problem 12.120.** Let $ABC$ be an acute triangle and let $H$ be the foot on $AB$ of the altitude through $C$. Suppose that $AH = 3BH$. Let $M$ and $N$ be the midpoints of the segments $AB$ and $AC$, respectively, and let $P$ be a point such that $NP = NC$, $CP = CB$, and $B$ and $P$ lie on opposite sides of the line $AC$. Prove that $\angle APM = \angle PBA$.

**Problem 12.121.** ★ Triangles $ABC$, $DEF$, $GHI$ and $CFI$ are equilateral and consistently oriented. Prove that the midpoints of $BD$, $EG$, $AH$ are the vertices of an equilateral triangle.

**Problem 12.122.** ★ Given is an acute triangle $ABC$ with $BC < CA < AB$. Points $K$ and $L$ lie on the segments $AC$ and $AB$ and satisfy $AK = AL = BC$. The perpendicular bisectors of the segments $CK$ and $BL$ meet the line $BC$ at the points $P$ and $Q$, respectively, and the segments $KP$ and $LQ$ meet at $M$. Prove that $CK + KM = BL + LM$.

**Problem 12.123.** ★ Given is a cyclic quadrilateral $ABCD$. The points $K$, $L$, $M$, $N$ belong to the segments $AB$, $BC$, $CD$, $DA$, respectively, and satisfy

$$\angle ADK = \angle BCK, \quad \angle BAL = \angle CDL, \quad \angle CBM = \angle DAM, \quad \angle DCN = \angle ABN.$$

Prove that $KM \perp LN$.

### 12.7.1 The magic of the equilateral triangle

**Problem 12.124.** Let $P$ be a point inside an equilateral triangle $ABC$, and let $D$, $E$, $F$ be the projections of $P$ onto the sides $BC$, $CA$, $AB$, respectively. Prove that

$$AF + BD + CE = AE + BF + CD.$$

**Problem 12.125.** Points $E$ and $F$ lie on the sides $BC$ and $CD$ of a rectangle $ABCD$, respectively, in such a way that the triangle $AEF$ is equilateral. Prove that $[ABE] + [ADF] = [CEF]$.

**Problem 12.126.** An equilateral triangle $ABC$ is inscribed in a circle $\Gamma$. Consider an arbitrary point $P$ of $\Gamma$. Prove that the sum of the squares of the distances from $P$ to the lines $AB$, $BC$, $CA$ does not depend on the choice of $P$.

**Problem 12.127.** Let $ABC$ be an equilateral triangle and let $l$ be any line through the vertex $C$ not crossing the triangle. The circle $o_A$ touches $l$, the segment $AC$ and the line $AB$, and the circle $o_B$ touches $l$, the segment $BC$ and the line $AB$. Show that as the line $l$ varies, the sum of the radii of $o_A$ and $o_B$ remains constant.

**Problem 12.128.** Let $ABC$ be an equilateral triangle and let $l$ be any line through its centre meeting the segments $AB$ and $AC$ at the points $K$ and $L$, respectively. A point $P$ satisfies $KP = KC$ and $LP = LB$. Show that as the line $l$ varies, the distance from $P$ to $l$ remains constant.

## 12.8 Tangential quadrilaterals

**Problem 12.129.** Let $AD$, $BE$, $CF$ be cevians in a triangle $ABC$ meeting at $P$. Prove that if two of the quadrilaterals $AEPF$, $BFPD$, $CDPE$ are tangential, then so is the third one.

**Problem 12.130.** The internal bisector of the angle $C$ in a triangle $ABC$ meets $AB$ at $D$, and the external bisector of the angle $C$ meets $AB$ at $E$. A line through $E$ meets the segments $AC$, $BC$, $CD$ at $K$, $L$, $M$, respectively. Prove that if $ADMK$ is tangential, then so is $DBLM$.

**Problem 12.131.** A point $D$ lies on the side $AB$ of a triangle $ABC$. The incircle of $BCD$ has radius $r_1$. Externally tangent circles of radii $r_1$ and $r_2$ are placed inside the triangle $ACD$ in such a way that the line joining their centres is perpendicular to $AB$, the circle of radius $r_1$ is tangent to the segment $AD$, and the circle of radius $r_2$ is tangent to the segments $AC$ and $CD$. Prove that the inradius of $ABC$ equals $r_1 + r_2$.

**Problem 12.132.** Given is a triangle $ABC$ and points $D$, $E$ on the segment $BC$. Prove that the common external tangent of the incircles of $ABD$ and $ACE$ different from $BC$, and the common external tangent of the incircles of $ABE$ and $ACD$ different from $BC$, are parallel or meet on the line $BC$.

**Problem 12.133.** Given is a triangle $ABC$ with $AB = AC$, and a point $D$ on the segment $BC$. Prove that the angle between $AD$ and the common internal tangent, different from $BC$, to the incircles of the triangles $ABD$ and $ACD$ equals the angle between $AD$ and the common internal tangent, different from $BC$, to the $A$-excircles of these two triangles.

**Problem 12.134.** Given is a tangential quadrilateral $ABCD$. The circle $o_1$ is tangent to the segment $AB$ and to the extensions of the sides $AD$ and $BC$; the circles $o_2$, $o_3$, $o_4$ are defined

analogously for the sides $BC$, $CD$, $DA$. Prove that the lengths of the internal tangents of the pairs of circles $(o_1, o_3)$ and $(o_2, o_4)$ are equal.

**Problem 12.135.** Given is a tangential quadrilateral $ABCD$. Prove that there exists a circle simultaneously tangent to the four circles with diameters $AB$, $BC$, $CD$, $DA$.

# Chapter 13

# Number Theory

**Sources for this chapter:** Camp handouts by Dominik Burek (March 2020, November 2020), Dušan Djukić (April 2021, November 2021, February 2022, January 2025, November 2025), Miroslav Marinov (November 2024, May 2025), Mikhail (November 2019, June 2020), Bojan Bašić (February 2024), Arsenii (2024).

At this level number theory stops being a collection of tricks and becomes a small theory. Three tools carry most of the chapter: the multiplicative order of an element, which turns a divisibility of exponentials into a divisibility of exponents; the lifting-the-exponent lemma, which computes a $p$-adic valuation exactly; and Vieta jumping, which converts a divisibility into a quadratic and then descends. Around them sit the constructive techniques — the Chinese Remainder Theorem, Dirichlet's theorem, Pell's equation — and the size and growth-rate estimates that close most of the remaining problems.

## 13.1 Divisibility, the gcd and equivalence

*Revisits the arithmetic of divisibility and the greatest common divisor (see Chapter 8); the problems below push it further.*

**Definition 13.1 (Equivalence relation).** A relation $\sim$ between the elements of a set $M$ is called an *equivalence* if for all $a, b, c \in M$ it satisfies: (1) $a \sim a$ (reflexivity); (2) if $a \sim b$ then $b \sim a$ (symmetry); (3) if $a \sim b$ and $b \sim c$ then $a \sim c$ (transitivity). For $a \in M$ let $[a]$ be the set of all elements equivalent to $a$. Then $b \sim c$ for any $b, c \in [a]$, and $b \not\sim d$ for any $b \in [a]$ and $d \notin [a]$; hence $M$ is partitioned into such sets, called *equivalence classes*. Equality of numbers, similarity of triangles, congruence modulo a fixed positive integer and connectedness by a path in a graph are equivalences; their classes are single numbers, triangles with equal angles, all numbers with a given remainder, and connected components.

**Theorem 13.2 (Four numbers lemma).** *If $a, b, c, d$ are positive integers satisfying $ab = cd$, then there exist positive integers $x, y, z, w$ such that $a = xy$, $b = zw$, $c = xz$ and $d = yw$.*

**Technique 13.3 (Multiplicative inverses by the extended Euclidean algorithm).** Given a positive integer $n$ and an integer $a$ coprime to $n$, an *inverse* of $a$ modulo $n$ is an integer $b$ with $ab \equiv 1 \pmod{n}$; it is unique modulo $n$ and written $a^{-1}$. To compute it, first run the Euclidean algorithm on $n$ and $a$, recording each division; then express the final remainder 1 in terms of the boxed remainders, eliminating the smallest one at every step. For $n = 999$

and $a = 128$ the algorithm gives $999 - 7 \cdot 128 = 103$, $128 - 1 \cdot 103 = 25$, $103 - 4 \cdot 25 = 3$, $25 - 8 \cdot 3 = 1$; substituting backwards yields $1 = 320 \cdot 128 - 41 \cdot 999$, so $128^{-1} \equiv 320 \pmod{999}$.

**Problem 13.1.** Let $a, b, c$ be three positive integers. Prove that the least common multiple of $a$ and $b$ cannot be equal to the least common multiple of $a + c$ and $b + c$.

**Problem 13.2.** Prove that the number $\sqrt{8.\underbrace{00\cdots0}_{n}1}$ is never rational.

**Problem 13.3.** Let $a_1, a_2, \ldots, a_n$ be positive integers and let $N$ be their product. It is given that $N$ is a multiple of the square of each of $a_1, a_2, \ldots, a_n$. For each pair $(i, j)$ with $i < j$ the number $\gcd(a_i, a_j)$ is computed, and $M$ denotes the product of all these greatest common divisors. Prove that $N$ divides $M^2$.

**Problem 13.4.** A quadruple $(a, b, c, d)$ of integers is said to be *good* if $ad - bc = 2018$. Two good quadruples are said to be *dissimilar* if one cannot be obtained from the other by a finite number of applications of the following three operations:

$$(a, b, c, d) \mapsto (-c, -d, a, b), \quad (a, b, c, d) \mapsto (a, b, c + a, d + b), \quad (a, b, c, d) \mapsto (a, b, c - a, d - b).$$

Show that any set of pairwise dissimilar good quadruples has at most 3030 elements.

**Problem 13.5.** Let $a, b, c, d$ be positive integers such that $cn + d \mid an + b$ for every positive integer $n$. Prove that $a = kc$ and $b = kd$ for some integer $k$.

**Problem 13.6.** Let $m, n$ be positive integers such that the set $\{1, 2, \ldots, n\}$ contains exactly $m$ different prime numbers. Prove that if we choose any $m + 1$ different numbers from $\{1, 2, \ldots, n\}$, then one of the chosen numbers divides the product of the other $m$.

**Problem 13.7.** Let $k > 1$ be an integer. The sum of a divisor of $k$ and a divisor of $k - 1$ equals $l$, where $l > k + 1$. Prove that at least one of the numbers $l - 1$ and $l + 1$ is composite.

**Problem 13.8.** Let $k$ and $n$ be positive integers with $k > n!$. Prove that there exist distinct prime numbers $p_1, p_2, \ldots, p_n$ such that $p_i \mid k + i$ for all $i = 1, 2, \ldots, n$.

**Problem 13.9.** Consider the sequence $a_n = |n(n + 1) - 19|$ for integers $n \geq 0$. Prove that for any $n \neq 4$ the following holds: if $a_k$ and $a_n$ are coprime for all integers $k < n$, then $a_n$ is a prime.

**Problem 13.10.** Let $a < b \leq c < d$ be positive integers such that $\sqrt{d} - \sqrt{a} < 1$. Prove that $ad \neq bc$.

**Problem 13.11.** Let $p$ be a prime and let $r$ be an integer such that $p \mid r^7 - 1$. Prove that if there exist integers $a, b$ such that $p \mid r + 1 - a^2$ and $p \mid r^2 + 1 - b^2$, then there exists an integer $c$ such that $p \mid r^3 + 1 - c^2$.

## 13.2 Orders of elements and primitive roots

*Revisits the multiplicative order modulo $n$ (see Chapter 8); the problems below push it further.*

**Definition 13.4 (Multiplicative order).** Given coprime integers $a, n$ with $|a| > 1$ and $n \neq 0$, the *multiplicative order* of $a$ modulo $n$, written $\operatorname{ord}_n(a)$, is the smallest positive integer $r$ such that $a^r \equiv 1 \pmod{n}$. Equivalently, it is the smallest period of the sequence $1, a, a^2, \ldots$ modulo $n$.

**Theorem 13.5 (The order divides).** *If* $a^d \equiv 1 \pmod{n}$ *then* $\operatorname{ord}_n(a) \mid d$. *In particular* $\operatorname{ord}_n(a) \mid \varphi(n)$.

**Definition 13.6 (Multiplicative function).** A function $f\colon \mathbb{N} \to \mathbb{Z}$ is *multiplicative* if $\gcd(m, n) = 1$ implies $f(mn) = f(m)f(n)$. The study of such a function reduces to the study of its values at prime powers. Examples: $\tau(n)$, $\sigma(n)$, more generally $\sigma_x(n) = \sum_{d|n} d^x$, and $\varphi(n)$.

**Problem 13.12.** Find the order of 3 modulo 2021.

**Problem 13.13.** Find (a) the order of 2 modulo $3^n$; (b) the order of 3 modulo $2^n$.

**Problem 13.14.** Let $p > 2$ be a prime. Prove that every divisor of $2^p - 1$ is congruent to 1 modulo $2p$.

**Problem 13.15.** If $p$ is an odd divisor of $a^{2^n} + 1$, prove that $p \equiv 1 \pmod{2^{n+1}}$.

**Problem 13.16.** Prove that there are infinitely many primes of the form $2px + 1$, where $p$ is a fixed prime. Do not use Dirichlet's theorem.

**Problem 13.17.** If $a > 1$, prove that $\varphi(a^n - 1)$ is divisible by $n$.

**Problem 13.18.** Prove that $\sum_{d|n} \varphi(d) = n$ for every positive integer $n$.

**Problem 13.19.** Let $h(x)$ be a polynomial with integer coefficients and let $p$ be a prime. Prove that the congruence $h(x) \equiv 0 \pmod{p}$ has at most $\deg h$ solutions modulo $p$.

**Problem 13.20.** Let $\gcd(a, b) = 1$ and let $p$ be a prime. Prove that every prime divisor of $\dfrac{a^p - b^p}{a - b}$, with the possible exception of $p$ itself, is congruent to 1 modulo $p$.

**Problem 13.21.** Find all positive integers $n$ for which $n$ divides $2^{n-1} + 1$.

**Problem 13.22.** Prove that the number $3^n + 1$ has no prime divisor of the form $12k + 11$.

**Problem 13.23.** If $n > 1$ is an integer, prove that $4^n + 2^n + 1$ cannot be divisible by $3^n - 2^n$.

**Problem 13.24.** For a positive integer $n > 2$, let $S$ be the set of all integers $1 \leq x \leq n$ such that $n \mid x^2 - 1$. Find the remainder when the product of all elements of $S$ is divided by $n$.

**Problem 13.25.** Find all pairs of primes $(p, q)$ such that $2p - 1$, $2q - 1$ and $2pq - 1$ are all perfect squares.

**Problem 13.26.** Let $a > 3$ be an integer. Prove that there exist infinitely many positive integers $n$ such that $n^2 \mid a^n - 1$.

**Problem 13.27.** Find all positive integers $n$ for which the fraction $\dfrac{n^{3n-2}-3n+1}{3n-2}$ is an integer.†

**Problem 13.28.** Let $(a_n)$ be a sequence of positive integers and let $(p_n)$ be a sequence of primes such that for every $n \geq 1$ we have $p_n \mid a_n$ and $a_{n+1} = \dfrac{a_n}{p_n}(p_n^{1009}-1)$. Show that this sequence contains a term $a_n$ that is a multiple of 2018.†

**Problem 13.29.** Prove that there exist infinitely many positive integers $n$ such that at least one of the numbers $2^{2^n}+1$ and $2018^{2^n}+1$ is composite.

**Problem 13.30.** Find all pairs $(p,n)$ of a prime $p$ and a positive integer $n$ with $n \neq p$ such that $\dfrac{n^p+1}{p^n+1}$ is an integer. [APMO 2012]

**Problem 13.31.** Let $n > 2$ be a positive integer. Prove that $n$ divides $(1^{n-1}+2^{n-1}+\cdots+(n-1)^{n-1})+1$ if and only if for every prime divisor $p$ of $n$ both $p \mid \frac{n}{p}-1$ and $p-1 \mid \frac{n}{p}-1$ hold.

**Problem 13.32.** Show that for each squarefree integer $n > 1$ there exist a prime divisor $p$ of $n$ and an integer $m$ such that $n \mid p^2 + p\,m^p$.

**Problem 13.33.** Let $p \geq 5$ be a prime and let $a$ be an integer with $1 < a < p-1$. Let $a_1, a_2, \ldots, a_n$ be all the distinct remainders that the powers $1, a, a^2, a^3, \ldots$ give when divided by $p$. If $a_1, a_2, \ldots, a_n$ form an arithmetic sequence modulo $p$ in some order, prove that $n = p-1$. (For example, 6, 1 and 3 form an arithmetic sequence modulo 7.)

**Problem 13.34.** Define $a_0 = 0$, $a_1 = 1$ and $a_{n+1} = 2a_n + a_{n-1}$. Prove that $2^k \mid a_n$ if and only if $2^k \mid n$.

**Problem 13.35.** Find all positive integers $n$ for which $(1+\sqrt{2})^n + (1-\sqrt{2})^n$ is divisible by 5.

### 13.2.1 Primitive roots

**Definition 13.7 (Primitive root).** A number $a$ with $\operatorname{ord}_n(a) = \varphi(n)$ is called a *primitive root* modulo $n$. The definition implies that every number coprime to $n$ is congruent to some power of $a$. Primitive roots clearly exist modulo 2 and modulo 4; for a prime modulus their existence follows from the identity $\sum_{d \mid n} \varphi(d) = n$ together with the bound on the number of roots of a polynomial congruence.

**Problem 13.36.** Prove that primitive roots exist modulo $p^k$, where $p$ is an odd prime.

**Problem 13.37.** Prove that primitive roots exist modulo $2p^k$, where $p$ is an odd prime.

**Problem 13.38.** Prove that $a^{2^{k-2}} \equiv 1 \pmod{2^k}$ for every odd integer $a$ and every integer $k > 3$, and deduce that there is no primitive root modulo $2^k$ for $k > 2$.

**Problem 13.39.** Suppose that $n$ is not of the form $2^k$, $p^k$ or $2p^k$, where $p$ is an odd prime. Prove that $\operatorname{ord}_n(a) \mid \frac{\varphi(n)}{2}$ for every $a$ with $\gcd(a,n) = 1$, and conclude that there is no primitive root modulo $n$.

### 13.2.2 Carmichael's function

**Definition 13.8 (Carmichael's function $\lambda$).** For an odd prime $p$ and $k \geq 1$ set $\lambda(p^k) = p^{k-1}(p-1)$. Set $\lambda(2) = 1$, $\lambda(4) = 2$ and $\lambda(2^k) = 2^{k-2}$ for $k > 2$. Finally, for $n = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$ set

$$\lambda(n) = \operatorname{lcm}\left[\lambda(p_1^{\alpha_1}), \ldots, \lambda(p_k^{\alpha_k})\right].$$

**Theorem 13.9 (Carmichael's theorem).** *If $a$ is coprime to $n$, then $a^{\lambda(n)} \equiv 1 \pmod{n}$.*

**Problem 13.40.** Find all positive integers $n$ such that $n^2 \mid a^n - 1$ for every positive integer $a$ coprime to $n$.[†]

**Problem 13.41.** Suppose that a positive integer $a$ is such that for every positive integer $n$ the number $n^2a - 1$ has a divisor of the form $nx + 1$ with $x \in \mathbb{N}$. Prove that $a$ is a perfect square.

## 13.3 Lifting the exponent

**Theorem 13.10 (Lifting the exponent lemma).** *Let $p$ be a prime and let $a, b$ be integers not divisible by $p$, with $|a| \neq |b|$ and $p \mid a - b$. Then for every positive integer $n$:*
- *if $p > 2$, then $v_p(a^n - b^n) = v_p(a - b) + v_p(n)$;*
- *if $p = 2$ and $4 \mid a - b$, then $v_2(a^n - b^n) = v_2(a - b) + v_2(n)$;*
- *if $p = 2$ and $n$ is even, then $v_2(a^n - b^n) = v_2(a - b) + v_2(a + b) + v_2(n) - 1$.*

*For odd $n$ the lemma also computes $v_p(a^n + b^n)$, by replacing $b$ with $-b$.*

**Problem 13.42.** Let $p$ be an odd prime divisor of $a - b$ which divides neither $a$ nor $b$. (a) If $p \nmid n$, prove that $p \nmid \dfrac{a^n - b^n}{a - b}$. (b) Prove that $\dfrac{a^p - b^p}{a - b}$ is divisible by $p$ but not by $p^2$.

**Problem 13.43.** What is the largest power of 5 dividing $22^{99} + 21^{99} - 1$?

**Problem 13.44.** Prove that for every positive integer $n$ the number $2^{3^n} + 1$ is divisible by $3^{n+1}$, but not by $3^{n+2}$.

**Problem 13.45.** Find the largest (a) power of 2; (b) power of 11 that divides $3^{100!} - 1$.

**Problem 13.46.** Find all positive integers $n$ for which $n^2$ divides $2^n + 1$. [IMO 1990]

**Problem 13.47.** Find all positive integers $x$, $y$ and $n$ for which $x^3 + y^3 = 3^n$.

**Problem 13.48.** Find all positive integers $n$ for which there exist positive integers $x, y$ and $k > 1$ such that $\gcd(x, y) = 1$ and $x^k + y^k = 3^n$.

**Problem 13.49.** Find all quadruples $(x, y, p, n)$ of positive integers $x, y, n$ and a prime $p$ that satisfy $x^p + y^p = p^n$.

**Problem 13.50.** Find at least one pair $(a, b)$ of positive integers such that none of $a$, $b$ and $a + b$ is divisible by 7, but $(a + b)^7 - a^7 - b^7$ is divisible by $7^7$.

**Problem 13.51.** Let $n > 2$ be a positive integer. Prove that the number $2^{2^n - 1} - 2^n - 1$ is composite.

**Problem 13.52.** Find all primes $p$ and positive integers $m, n$ such that $(p-1)^m + 1 = p^n$.

**Problem 13.53.** Find the smallest integer $a > 1$ for which the numbers $1$, $1+a$, $1+a+a^2$, ... give all possible remainders modulo 10!.

**Problem 13.54.** Find all triples $(a, b, k)$ of positive integers with $k > 2$ such that $(a^k+b)(b^k+a)$ is a power of 2.

**Problem 13.55.** Find all pairs $(x, y)$ of positive integers that satisfy the equation $3^x - 8^y = 2xy + 1$.

**Problem 13.56.** If $n$ is a positive integer and $p$ a prime, prove that $n^p + \left(\frac{n-1}{2}\right)^2$ is not a power of $p$.

**Problem 13.57.** Find all primes $p$ and $q$ such that $3p^{q-1} + 1$ divides $11^p + 17^p$. [Balkan MO 2018]

**Problem 13.58.** Prove that $n^7 + 7$ is not a perfect square for any integer $n$.

**Problem 13.59.** Let $S$ be the set of all positive integers $n$ such that $n^4$ has a divisor in the range $n^2+1, n^2+2, \ldots, n^2+2n$. Prove that $S$ contains infinitely many elements of each of the forms $7m$, $7m+1$, $7m+2$, $7m+5$, $7m+6$, and no element of the form $7m+3$ or $7m+4$, where $m$ is an integer.

**Problem 13.60.** Prove that for each positive integer $m$ one can find $m$ consecutive positive integers $n$ such that $(1^3 + 2018^3)(2^3 + 2018^3) \cdots (n^3 + 2018^3)$ is not a perfect power.†

## 13.4 Vieta jumping

*Revisits Fermat's method of infinite descent (see Chapter 8); the problems below push it further.*

**Technique 13.11 (Vieta jumping (root flipping)).** Suppose a divisibility such as $ab+1 \mid a^2+b^2$ is given, and let $k$ denote the fixed value of the quotient. The equation $a^2+b^2 = k(ab+1)$ is a quadratic in $a$, so if $(a, b)$ is a solution then so is $(a', b)$, where $a'$ is the second root of $X^2 - kbX + b^2 - k = 0$; by Vieta's formulas $a' = kb - a = \frac{b^2-k}{a}$, which shows that $a'$ is an integer. Now choose among all solutions one with $a + b$ minimal, assume $a \geq b$, and derive a contradiction by showing that $a'$ is a nonnegative integer with $a' < a$; the excluded cases give the answer.

**Problem 13.61.** Let $a$ and $b$ be nonnegative integers such that $ab + 1 \mid a^2 + b^2$. Prove that $\frac{a^2+b^2}{ab+1}$ is a perfect square. [IMO 1988, Problem 6]

**Problem 13.62.** Let $a, b$ be positive integers such that $ab \mid a^2 + b^2 + 1$. Find all possible values of $\frac{a^2+b^2+1}{ab}$.

**Problem 13.63.** Let $a, b$ be positive integers with $ab \neq 1$ such that $ab - 1 \mid a^2 + b^2$. Find all possible values of $\frac{a^2+b^2}{ab-1}$. [USA TST 2002]

**Problem 13.64.** Suppose that $a, b$ are positive integers such that $4ab-1$ divides $(a-b)^2$. Prove that $a = b$. [IMO 2007]

**Problem 13.65.** Let $a$ and $b$ be positive integers such that $4ab - 1$ divides $(4a^2 - 1)^2$. Prove that $a = b$. [IMO 2007, Problem 5]

**Problem 13.66.** Suppose that $a$ and $b$ are odd positive integers such that $2ab + 1 \mid a^2 + b^2 + 1$. Prove that $a = b$. [Iran 2013]

**Problem 13.67.** Let $a, b$ be positive integers such that $ab$ divides $a^2 + b^2 + a + b + 1$. Prove that $\dfrac{a^2 + b^2 + a + b + 1}{ab} = 5$.

**Problem 13.68.** Find all positive integers $a, b$ such that $ab + a + b$ divides $a^2 + b^2 + 1$.

**Problem 13.69.** Find all positive integers $m, n$ such that $mn - 1$ divides $(n^2 - n + 1)^2$.

**Problem 13.70.** Find all positive integers $a, b$ such that $a \mid b^2 + 1$ and $b \mid a^2 + 1$.

**Problem 13.71.** Let $a$ and $b$ be positive integers such that $ab - 1$ divides $a^2 + b^2 + ab$. Prove that $\dfrac{a^2 + b^2 + ab}{ab - 1} \in \{4, 7\}$.

**Problem 13.72.** Let $a, b$ be positive integers such that $a^2 + b^2 + ab$ is divisible by $ab - 2$. Find all possible values of $\dfrac{a^2 + b^2 + ab}{ab - 2}$.

**Problem 13.73.** Let $a, b, c$ and $m$ be positive integers such that $abcm = 1 + a^2 + b^2 + c^2$. Prove that $m = 4$. [AMM]

**Problem 13.74.** Let $a, b$ be positive integers such that $a + b + 1$ is a prime divisor of $4ab - 1$. Prove that $a = b$.

**Problem 13.75.** Find all pairs $(a, b)$ of positive integers such that $2ab^2 - b^3 + 1$ divides $a^2$.

**Problem 13.76.** Show that the equation $x^2 + y^2 + z^2 + u^2 = xyzu$ has infinitely many integer solutions.

**Problem 13.77.** Show that for each real number $N$ the equation $x^2 + y^2 + z^2 + u^2 = xyz + xyu + xzu + yzu$ has an integer solution in which all the variables are greater than $N$.

**Problem 13.78.** Prove that there are infinitely many triples $(x, y, z)$ of positive integers such that $x^2 + y^2 + z^2 + xy + yz + zx = 6xyz$.

**Problem 13.79.** Let $n$ be a given positive integer. Prove that if the equation $x^3 - 3xy^2 + y^3 = n$ has an integer solution in $x$ and $y$, then it has at least three integer solutions. [IMO 1982]

**Problem 13.80.** Let $a_1, a_2, \ldots, a_{2n+1}$ be integers such that, if any one of them is removed, the remaining ones can be divided into two sets of $n$ integers with equal sums. Prove that $a_1 = a_2 = \cdots = a_{2n+1}$.

## 13.5 Constructions: the Chinese Remainder Theorem, Dirichlet and Pell

*Revisits the Chinese Remainder Theorem (see Chapter 8); the problems below push it further.*

**Technique 13.12 (Building the number you need).** Most number-theoretic existence problems are solved by making choices rather than by deduction. The standard moves are: pick a really large number with a single prime factor, or with a great many prime factors; impose modular conditions recklessly and let the Chinese Remainder Theorem combine them; rely on a heavy theorem such as Dirichlet's or Pell's; and, in a divisibility problem, force the divisor to be a prime or to be divisible by a suitable prime. If a construction uses none of these, it is either simple (a power of a prime) or iterative: if $n$ works, then so does something larger expressed in terms of $n$.

**Theorem 13.13 (Chinese Remainder Theorem).** *Let $a_1, \dots, a_k$ be integers and let $n_1, \dots, n_k$ be pairwise coprime positive integers. Then the system of congruences $x \equiv a_i \pmod{n_i}$, $i = 1, 2, \dots, k$, has a unique solution $x$ modulo $n_1 n_2 \cdots n_k$.*

**Remark.** The Chinese Remainder Theorem extends to the general case in which $\gcd(n_i, n_j)$ divides $a_i - a_j$ for all $i, j$; then a solution $x$ exists and is unique modulo $\operatorname{lcm}(n_1, \dots, n_k)$.

**Theorem 13.14 (Dirichlet's theorem).** *Let $a$ and $d$ be relatively prime positive integers. Then there are infinitely many primes $p \equiv a \pmod{d}$.*

**Theorem 13.15 (Pell's equation).** *Let $D$ be a positive integer which is not a perfect square. Then the equation $x^2 - Dy^2 = 1$ has infinitely many solutions in positive integers.*

**Problem 13.81.** Let $a, b, c$ be positive integers. Prove that there exists a positive integer $k$ such that $\gcd(a^k + bc,\ b^k + ca,\ c^k + ab) > 1$. [Ukraine National Round 2019]

**Problem 13.82.** Let $n$ and $a$ be given positive integers. Suppose that for every $m \in \mathbb{N}$ there is an integer $x$ such that $x^n \equiv a \pmod{m}$. Prove that $a = y^n$ for some integer $y$. [IMO Shortlist 2007 N2]

**Problem 13.83.** The positive integers $x$ and $y$ are such that $(ny)^2 + 1$ divides $x^{\varphi(n)} - 1$ for all positive integers $n$. Prove that $x = 1$. [Balkan MO Shortlist 2018]

**Problem 13.84.** Prove that there exist infinitely many pairs $(m, n)$ of positive integers such that $m + n$ divides $(m!)^n + (n!)^m + 1$. [Zhautykov Olympiad 2018]

**Problem 13.85.** Prove that there exist infinitely many positive integers $n$ such that the number of divisors of $n$ does not divide the number of divisors of $a^2 + b^2$ for any positive integers $a, b$ satisfying $a + b = n$. [EGMO 2014]

**Problem 13.86.** An integer $n$ is called *squarefull* if for every prime $p$ dividing $n$ it is the case that $p^2$ divides $n$. Prove that there are infinitely many pairs of consecutive squarefull positive integers.

**Problem 13.87.** Prove that there are infinitely many positive integers $n$ such that $2^{2^n+1}+1$ is divisible by $n$ but $2^n+1$ is not divisible by $n$. [RMM 2012]

**Problem 13.88.** Let $n \geq 2$ be an integer. Prove that there are infinitely many odd numbers of the form $\left\lfloor \frac{n^k}{k} \right\rfloor$. [IMO Shortlist 2014 N4]

**Problem 13.89.** A positive integer is called an *almost square* if it can be represented as a product $ab$ of two positive integers $a$ and $b$ such that $a \leq b \leq \frac{101}{100}a$. Prove that there are infinitely many quadruples of consecutive numbers all of which are almost squares.

**Problem 13.90.** Prove that there are infinitely many pairs of consecutive positive integers, neither of which has any prime factor of the form $4k+3$.

**Problem 13.91.** Prove that every integer can be uniquely written in the form $a_0 + a_1(-3) + a_2(-3)^2 + \cdots + a_k(-3)^k$ for some integer $k \geq 0$ and some $a_0, a_1, \ldots, a_k \in \{0, 1, 2\}$.

**Problem 13.92.** Denote $\varphi = \dfrac{-1-\sqrt{5}}{2}$. Can every integer be written as a sum of powers of $\varphi$, where each power occurs at most 1000 times?

**Problem 13.93.** Given a prime number $p > 2$, prove that one can find $p$ positive integers less than $2p^2$ such that all their pairwise sums are distinct. (For example, the pairwise sums of the numbers $1, 3, 6$ are $2, 4, 6, 7, 9, 12$.)

**Problem 13.94.** Prove that the equation $x^3 + y^3 + z^3 = 2$ has infinitely many solutions in integers.

**Problem 13.95.** A *perfect power* is a positive integer of the form $a^b$ with $a$ and $b \geq 2$ positive integers. Prove that there exist 1000000 consecutive positive integers exactly two of which are perfect powers.

**Problem 13.96.** Find all positive integers $n$ for which there is a permutation $(a_1, a_2, \ldots, a_n)$ of $1, 2, \ldots, n$ with the property that both $(a_1+1, a_2+2, \ldots, a_n+n)$ and $(a_1-1, a_2-2, \ldots, a_n-n)$ are also permutations of $1, 2, \ldots, n$ modulo $n$.

**Problem 13.97.** Do there exist positive integers $a, b, c, d$ such that $a^2 < b^3 < c^4 < d^5$ is an arithmetic progression?

**Problem 13.98.** The Director wants to arrange the students of his school into identical squares on holidays. He does not know how many of them are ill and will not come, but he knows that the number of ill students is between 1 and 37. Prove that there is a fixed number of students in the school for which the Director will manage in any case.

**Problem 13.99.** Given a polynomial $P(x) = a_m x^m + a_{m-1}x^{m-1} + \cdots + a_2 x^2 + a_0$ with $m > 2$ and positive integer coefficients, consider the sequence $(b_n)$ defined by $b_1 = a_0$ and $b_{n+1} = P(b_n)$ for $n \geq 1$. Prove that for each $n \geq 2$ the number $b_n$ has a prime divisor which does not divide the product $b_1 b_2 \cdots b_{n-1}$. [Bulgaria 2018]

**Problem 13.100.** ★ Find all positive integers $n > 1$ such that there exist positive integers $b_1, b_2, \ldots, b_n$ (some of them may be equal, but not all) with the property that for every positive integer $k$ the product $(b_1+k)(b_2+k)\cdots(b_n+k)$ is a perfect power. (The exponent may depend on $k$ but must exceed 1.) [Russia 2008]

**Problem 13.101.** Consider the polynomial $P(n) = n^2 + n + 1$. For positive integers $a$ and $b$, the set $\{P(a), P(a+1), \ldots, P(a+b)\}$ is said to be *fragrant* if none of its elements is relatively prime to the product of the other elements. (a) Determine the smallest size of a fragrant set. (b) Prove that there exists a fragrant set of every size greater than the minimal one. [IMO 2016]

**Problem 13.102.** Determine all positive integers $d$ for which there exists an integer $k \geq 3$ such that the numbers $d, 2d, \ldots, kd$ can be arranged in a sequence so that the sum of every two consecutive of them is a perfect square. [Bulgaria National Round 2019]

**Problem 13.103.** Call a set of positive integers *central* if its number of elements equals the arithmetic mean of its elements. Call an integer $n$ *nice* if one can partition the set $\{1, 2, \ldots, n\}$ into central subsets. Prove that there are infinitely many nice positive integers, as well as infinitely many positive integers which are not nice. [Caucasus Mathematical Olympiad 2021]

**Problem 13.104.** ★ Determine all integers $n \geq 2$ for which there exists a permutation $a_1, \ldots, a_{n-1}$ of the integers $1, 2, \ldots, n-1$ such that the remainders upon division by $n$ of $1 \cdot a_1, 2 \cdot a_2, \ldots, (n-1) \cdot a_{n-1}$ also form a permutation of $1, 2, \ldots, n-1$.

## 13.6 Arithmetic functions revisited

*Revisits the functions $\tau$, $\sigma$ and $\varphi$ (see Chapter 8); the problems below push it further.*

**Theorem 13.16 (Inclusion–exclusion principle).** *The cardinality of a union of finite sets $A_1, A_2, \ldots, A_n$ is*

$$\Big|\bigcup_{i=1}^{n} A_i\Big| = \sum_i |A_i| - \sum_{i<j} |A_i \cap A_j| + \sum_{i<j<k} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n-1} |A_1 \cap A_2 \cap \cdots \cap A_n|.$$

*Applied to the sets of multiples of the distinct primes $p_1, \ldots, p_k$ of $n$, it yields the product formula $\varphi(n) = n\left(1 - \frac{1}{p_1}\right)\left(1 - \frac{1}{p_2}\right)\cdots\left(1 - \frac{1}{p_k}\right)$ for the number of positive integers up to $n$ that are coprime to $n$.*

**Theorem 13.17 (The functions $\tau$ and $\sigma$).** *For a positive integer $n$ let $\tau(n)$ denote the number of positive divisors of $n$ and $\sigma(n)$ their sum. (a) If $m$ and $n$ are coprime positive integers, then $\tau(mn) = \tau(m)\tau(n)$ and $\sigma(mn) = \sigma(m)\sigma(n)$. (b) If $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$, then*

$$\tau(n) = (\alpha_1 + 1)(\alpha_2 + 1)\cdots(\alpha_k + 1), \qquad \sigma(n) = \frac{p_1^{\alpha_1+1} - 1}{p_1 - 1} \cdot \frac{p_2^{\alpha_2+1} - 1}{p_2 - 1} \cdots \frac{p_k^{\alpha_k+1} - 1}{p_k - 1}.$$

**Problem 13.105.** A function $f\colon \mathbb{N} \to \mathbb{N}$ is such that $f(f(n)) = \tau(n)$. Prove that if $p$ is prime, then $f(p)$ is prime.

**Problem 13.106.** Prove that there exist infinitely many positive integers $a$ for which there is no positive integer $n$ with $\sigma(n) = a$.

**Problem 13.107.** For every positive integer $n$ let $d(n)$ denote the number of divisors of $n$, and let $s(n)$ denote the number of divisors $d$ of $n$ such that $d + 1$ divides $n + 1$. Consider the set $M = \{2s(1) - d(1),\ 2s(2) - d(2),\ 2s(3) - d(3), \ldots\}$. Prove that $M$ has a maximal element and find its value.

**Problem 13.108.** Let $\tau(k)$ denote the number of positive divisors of a positive integer $k$. Prove that there exist infinitely many positive integers $M$ that cannot be written as $M = \dfrac{2\sqrt{n}}{\tau(n)}$ for any positive integer $n$.†

**Problem 13.109.** Find all positive integers $n$ with the following property: if $1 = d_1 < d_2 < \cdots < d_k = n$ are all the divisors of $n$, where $k \geq 7$, then $d_6^2 + d_7^2 - 1 = n$.

**Problem 13.110.** Suppose that $1 \leq a_1, a_2, \ldots, a_n \leq 2n$ are integers such that $\operatorname{lcm}(a_i, a_j) > 2n$ whenever $i < j$. Prove that $a_1 a_2 \cdots a_n$ divides $(n+1)(n+2)\cdots(2n)$.

**Problem 13.111.** Given a positive integer $n$, we write down the fractions $\frac{1}{n}, \frac{2}{n}, \ldots, \frac{n-1}{n}$, each in lowest terms. Define $f(n)$ to be the sum of the numerators of these fractions. Find all $n$ for which $f(n)$ and $f(999n)$ have opposite parities.

**Problem 13.112.** For any integer $N \geq 2$ let $f(N)$ denote the sum of $N$ and the greatest divisor of $N$ other than $N$. Prove that for any integer $A \geq 2$, by iterating $f$ on $A$ one can obtain a number divisible by $3^{2021}$.

**Problem 13.113.** Define $a_1 = 2$ and $a_{n+1} = a_n + l(a_n)$, where $l(x)$ denotes the largest prime divisor of $x$. Find all positive integers $m$ for which $m^2$ occurs in this sequence.

**Problem 13.114.** Define $p(x)$ to be the largest prime divisor of a positive integer $x$. A sequence is given by $a_0 \in \mathbb{N}$ and $a_{n+1} = a_n + p(a_n)$. Prove that some term $a_n$ is a perfect square.

**Problem 13.115.** Call a positive integer $n$ a *good number* if there exists a prime $p$ such that $p \mid n$ and $p^2 \nmid n$. Prove that 99% of the numbers among $1, 2, 3, \ldots, 10^{12}$ are good.

**Problem 13.116.** A triple of positive integers $(a, b, c)$ is *lame* if $c^2 + 1$ divides $(a^2+1)(b^2+1)$ but divides neither $a^2 + 1$ nor $b^2 + 1$. Given $c$, prove that if there is a lame triple $(a, b, c)$, then there is a lame triple in which $ab < c^3$.

**Problem 13.117.** The sequence $(a_n)$ is defined by $a_1 = 1$, $a_2 = 2$ and $a_{n+2} = a_n(a_{n+1} + 1)$ for all $n \geq 1$. Prove that $a_n$ divides $a_{a_n}$ for every $n > 100$.

**Problem 13.118.** Define $a_1 = 2021^{2021}$, and for $k \geq 2$ let $a_k$ be the remainder when $a_{k-1} - a_{k-2} + a_{k-3} - \cdots$ is divided by $k$. Find the $2021^{2022}$-th term of the sequence $(a_n)$.

**Problem 13.119.** Prove that there are infinitely many integers $n > 0$ for which $\lfloor\sqrt{n}\rfloor + \left\lfloor\sqrt{2n}\right\rfloor + \left\lfloor\sqrt{3n}\right\rfloor$ divides $n$.

### 13.6.1 Fibonacci numbers

**Problem 13.120.** The Fibonacci sequence is defined by $F_0 = 0$, $F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$. Prove that for each $n$ one of the numbers $5F_n^2 \pm 4$ is a perfect square.

**Problem 13.121.** Prove that for every $a \in \mathbb{N}$ there is a Fibonacci number that is divisible by $a$.

**Problem 13.122.** Prove that $\gcd(F_m, F_n) = F_{\gcd(m,n)}$.

**Problem 13.123.** Find all Fibonacci numbers that are powers of 2 or powers of 3.

**Problem 13.124.** If $n$ is a positive integer, prove that all prime divisors of the Fibonacci number $F_{2n+1}$ are congruent to 1 modulo 4.

## 13.7 Diophantine equations: squeezing, factoring and induction

**Problem 13.125.** Find all $n \in \mathbb{N}$ and primes $p, q$ such that $n^3 = p^3 + 2p^2q + 2pq^2 + q^3$. [Ukraine 2018]

**Problem 13.126.** Find all pairs of integers $(x, y)$ such that $y^2(x^2 + y^2 - 2xy - x - y) = (x + y)^2(x - y)$.

**Problem 13.127.** Determine all pairs $(x, y)$ of integers such that $1 + 2^x + 2^{2x+1} = y^2$. [IMO 2006]

**Problem 13.128.** Distinct prime numbers $p, q, r$ satisfy $rp^3 + p^2 + p = 2rq^2 + q^2 + q$. Find $pqr$.

**Problem 13.129.** Find all pairs of integers $(m, n)$ such that $n^{n-1} = 4m^2 + 2m + 3$.

**Problem 13.130.** Show that $2017^{2018}$ cannot be represented as the sum of two cubes of positive integers.

**Problem 13.131.** Determine whether there exists an infinite sequence $a_1, a_2, a_3, \ldots$ of positive integers which satisfies $a_{n+2} = a_{n+1} + \sqrt{a_{n+1} + a_n}$ for every positive integer $n$. [EGMO 2015]

**Problem 13.132.** (a) Do there exist positive integers $a$ and $b$ such that $a \cdot 2^n + b \cdot 5^n$ is a perfect square for each positive integer $n$? (b) Do there exist positive integers $a, b$ and $c$ such that $a \cdot 2^n + b \cdot 5^n + c$ is a perfect square for each positive integer $n$? [Belarus 2006]

**Problem 13.133.** Solve the equation $x^2 - y^2 = 2xyz$ in positive integers.

**Problem 13.134.** Find all pairs $(m, n)$ of positive integers such that $m^2 - 1 \mid 3^m + (n! - 1)^m$.

**Problem 13.135.** Solve the equation $x^2 = y^7 + 7$ in integers.

**Problem 13.136.** Solve the equation $y^3 - 9 = x^2$ in integers.

**Problem 13.137.** Find all integers $x, y$ such that $x^2 + 5 = y^3$.

**Problem 13.138.** Find all pairs $(x, y)$ of integers such that $y^2 = x^3 + 2x^2 + 2x + 1$. [Korea Junior Olympiad 2023]

**Problem 13.139.** Determine all triples $(a, b, c)$ of positive integers such that $a^2 + bc$, $b^2 + ca$ and $c^2 + ab$ are powers of 2. [Bulgarian Festival of Young Mathematicians 2023]

**Problem 13.140.** Prove that there are no positive integers $x$ and $y$ such that $x^2 + 6xy + y^2 = 36^{2023}$. [JBMO Shortlist 2023]

**Problem 13.141.** Determine all pairs $(a, b)$ of positive integers such that $a^2b - 1$ divides $ab^3 - 1$. [USA 2023]

**Problem 13.142.** Determine all pairs $(p, q)$ of primes such that $p^3 - q^3 = pq^3 - 1$. [Belarus IMO TST 2016]

**Problem 13.143.** Determine the smallest integer $n \geq 41$ such that the sum of all integers from 41 to $n$ inclusive is a perfect cube. [Bulgaria Regional Round 2025]

**Problem 13.144.** Let $a$ be a positive integer. Prove that there are only finitely many positive integers $n$ such that $n! - a$ is a perfect square. [Norway 2025]

**Problem 13.145.** Determine all pairs $(a, b)$ of positive integers such that $a^2 + b + 3$ is not divisible by the cube of any prime, and divides $ab + 3b + 8$. [IMO Shortlist 2021 N1]

**Problem 13.146.** Find all triples $(a, b, c)$ of positive integers such that $2^a - 4^b = 3^c + 1$. [Bulgaria Regional Round 2023]

**Problem 13.147.** Determine all pairs $(p, q)$ of primes such that $q^p$ divides $p + p^q + p^{q^p}$. [JBMO Shortlist 2024]

**Problem 13.148.** Determine all primes $p$ for which there exist primes $q, s$ and an integer $y \geq 2$ such that $2^s q = p^y - 1$. [Russia IMO TST 2014]

**Problem 13.149.** Determine all triples $(a, b, c)$ of positive integers such that $ab + c$, $bc + a$ and $ac + b$ are all powers of 2. [Romania JBMO TST 2025]

**Problem 13.150.** Determine all pairs $(m, n)$ of non-negative integers which satisfy $m^2 + 2 \cdot 3^n = m(2^{n+1} - 1)$. [IMO Shortlist 2010 N2]

**Problem 13.151.** Find all pairs $(p, q)$ of primes such that $p^5 + p^3 + 2 = q^2 - q$.

**Problem 13.152.** Find all pairs $(p, q)$ of primes such that $p^8 - p^4 = q^5 - q$.

**Problem 13.153.** Solve the equation $3^a - 2^b = 1$ in positive integers $a, b$.

**Problem 13.154.** Find all positive integers $a, b$ such that $7^a = 3 \cdot 2^b + 1$.

**Problem 13.155.** Find all triples $(x, y, z)$ of positive integers such that $2^x + 1 = 7^y + 2^z$.

**Problem 13.156.** Find all primes $p$ and positive integers $(x, y)$ such that $x^{p-1} + y$ and $y^{p-1} + x$ are powers of $p$.

**Problem 13.157.** Prove that there is no positive integer $x$ such that each of the numbers $2x^2 + 1$, $3x^2 + 1$ and $6x^2 + 1$ is a perfect square.

**Problem 13.158.** Find all positive integers $n$ for which $x^3 + x$ gives all possible remainders when divided by $n$.

**Problem 13.159.** Given a positive integer $n$, find the greatest common divisor of all numbers of the form $a^n + (a + 1)^n + (a + 2)^n$, where $a$ ranges over all positive integers.

**Problem 13.160.** Given a positive integer $n$, prove that there exists a positive integer $m > n^n$ such that $m + n$ divides $n^m - m^n$.

**Problem 13.161.** Prove that there exists an odd positive integer $a$ such that $2^n + a$ is composite for all nonnegative integers $n$.

**Problem 13.162.** Let $n > 2$ be an integer such that $n$ does not divide any element of the set $\{2^n - 1, 3^n - 1, \ldots, (n - 1)^n - 1\}$. (a) Prove that $n$ is squarefree. (b) Does it necessarily follow that $n$ is a prime number?

**Problem 13.163.** If a positive integer $n > 20$ is not squarefree, prove that there exist positive integers $a, b, c$ such that $ab + bc + ca = n$.

**Problem 13.164.** If $a, b$ are positive integers such that $a + b^3$ is divisible by $a^2 + 3ab + 3b^2 - 1$, prove that $a^2 + 3ab + 3b^2 - 1$ is divisible by a cube greater than 1.

**Problem 13.165.** If $a$ and $b$ are positive integers, can $(a + b)^{15}$ be divisible by $4ab - 1$?

**Problem 13.166.** Solve the following equation in positive integers: $2!! \cdot 4!! \cdot 6!! \cdots (2k)!! = (k(k + 1))!!$.

**Problem 13.167.** Solve the equation $p^3 + 41 = 7(7q! - r^3)$ in nonnegative integers.

**Problem 13.168.** Given that

$$23!! \cdot 40!! = 8{*}6\ 739\ 438\ {*}56\ 1{*}8\ 662\ 3{*}8\ 747\ 6{*}8\ {*}6{*}\ {***}\ {***},$$

find the missing digits.

**Problem 13.169.** Let $p$ be a prime. Prove that the number of pairs $m < n$ of integers from 1 to $p$ such that $p$ divides $(m^2 - n + 1)(n^2 - m + 1)$ is $p - 3$, $p - 2$, $p - 1$ or $p$. [IMO Shortlist 2020 N2]

### 13.7.1 Induction and iterated constructions

**Problem 13.170.** Prove that for all positive integers $n$ the equation $x^2 + y^2 + z^2 = 59^n$ is solvable in integers.

**Problem 13.171.** Prove that for all positive integers $n$ the equation $x^2 + xy + y^2 = 7^n$ is solvable in integers.

**Problem 13.172.** Prove that for all $n \geq 6$ the equation $\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n} = 1$ is solvable in distinct positive integers.

**Problem 13.173.** Prove that for all $n \geq 6$ the equation $\frac{1}{x_1^2} + \frac{1}{x_2^2} + \cdots + \frac{1}{x_n^2} = 1$ is solvable in positive integers.

**Problem 13.174.** Find all positive integers $n$ for which there exist positive integers $x_1, x_2, \ldots, x_n$ such that

$$\frac{1}{x_1^2} + \frac{2}{x_2^2} + \frac{2^2}{x_3^2} + \cdots + \frac{2^{n-1}}{x_n^2} = 1.$$

**Problem 13.175.** Find all positive integers $n$ for which there exist non-negative integers $a_1, a_2, \ldots, a_n$ such that

$$\frac{1}{2^{a_1}} + \frac{1}{2^{a_2}} + \cdots + \frac{1}{2^{a_n}} = \frac{1}{3^{a_1}} + \frac{2}{3^{a_2}} + \cdots + \frac{n}{3^{a_n}} = 1.$$

**Problem 13.176.** The sequence $(y_n)$ is defined by $y_1 = y_2 = y_3 = 1$ and $y_{n+1} = \frac{y_n y_{n-1} + 1}{y_{n-2}}$ for $n \geq 3$. Prove that all terms of this sequence are integers.

**Problem 13.177.** Find all constants $c$ for which there exists a strictly increasing sequence of positive integers satisfying $a_{2n-1} + a_{2n} = c \cdot a_n$.

**Problem 13.178.** Positive rational numbers $a$ and $b$ satisfy $a^3 + 4a^2b = 4a^2 + b^4$. Prove that the number $\sqrt{a} - 1$ is the square of a rational number.

## 13.8 Functional equations in number theory

**Problem 13.179.** In terms of the positive integer $k$, determine all functions $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that $m^2 + f(n^k) \mid mf(m) + n^k$ for all positive integers $m, n$. [Bulgaria Balkan MO TST 2014; generalises IMO Shortlist 2013 N1]

**Problem 13.180.** Find all functions $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that $n + f(m) \mid f(n) + nf(m)$ for all positive integers $m$ and $n$. [Balkan MO 2017]

**Problem 13.181.** Find all functions $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that $n! + f(m)! \mid f(n)! + f(m!)$ for all positive integers $m$ and $n$. [Balkan MO Shortlist 2018]

**Problem 13.182.** Find all functions $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that $a + f(b) \mid a^2 + bf(a)$ for all positive integers $a, b$ with $a + b > 2019$. [IMO Shortlist 2019 N4]

**Problem 13.183.** Determine all integers $k$ for which there exists a function $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}$ such that $f(2023) = 2024$ and $f(ab) = f(a) + f(b) + kf(\gcd(a, b))$ for all positive integers $a$ and $b$. [Bulgaria EGMO TST 2023; Taiwan National Round 1997]

**Problem 13.184.** Find all functions $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that for all positive integers $a$ and $b$ we have $f(ab) = f(a)f(b)$, and at least two of the numbers $f(a)$, $f(b)$, $f(a+b)$ are equal. [EGMO 2022]

**Problem 13.185.** Find all functions $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that $x^2 - y^2 + 2y(f(x) + f(y))$ is a perfect square for all positive integers $x$ and $y$. [European Mathematical Cup 2021]

**Problem 13.186.** Determine all functions $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that $(f(m) + n)(f(n) + m)$ is a perfect square for all positive integers $m, n$. [IMO 2010, Problem 3]

**Problem 13.187.** Let $k$ be a positive integer. Determine all functions $f\colon \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ such that: there are infinitely many primes $p$ for which there exists a positive integer $c$ (depending on $p$) with $f(c) = p^k$; and for all positive integers $m, n$ we have $f(m) + f(n) \mid f(m + n)$. [Iran National Round 2017]

**Problem 13.188.** A function $f\colon \mathbb{N} \to \mathbb{N}$ is such that $xy + 1$ divides $f(x)f(y) + 1$ for all $x, y$. Find all possible values of $f(101)$.

**Problem 13.189.** Let $n$ be an odd positive integer. Find all functions $f\colon \mathbb{Z} \to \mathbb{Z}$ such that $f(x) - f(y)$ divides $x^n - y^n$ whenever $x \neq y$.

## 13.9 Games and processes in number theory

Winning and losing positions do not classify these games, because the two players usually have different winning conditions; what carries over from combinatorics is the pairing strategy and the monotone quantity.

**Problem 13.190.** Alice and Bob play a game with the set $M = \{1, 2, \ldots, n-1\}$, where $n \geq 5$ is an odd integer. In each step Alice removes a number from $M$ and puts it in her set $A$, and Bob removes a number from $M$ and puts it in his set $B$ (both $A$ and $B$ are empty to begin with). When $M$ is empty, Alice picks two distinct numbers $x_1, x_2$ from $A$ and shows them to Bob; Bob then picks two distinct numbers $y_1, y_2$ from $B$. Bob wins if

$$(x_1x_2(x_1 - y_1)(x_2 - y_2))^{\frac{n-1}{2}} \equiv 1 \pmod{n},$$

otherwise Alice wins. In terms of $n$, who has a winning strategy? [Nordic Mathematical Competition 2022]

**Problem 13.191.** Alice and Bob play a game. At the beginning, Alice writes down a positive integer on the board. Then the players move in turn, Bob moving first. On any move of his, Bob replaces the number $n$ on the blackboard with a number of the form $n - a^2$, where $a$ is a positive integer. On any move of hers, Alice replaces the number $n$ on the blackboard with a number of the form $n^k$, where $k$ is a positive integer. Bob wins if the number on the board becomes zero. Can Alice prevent Bob's win? [RMM 2019, Problem 1]

**Problem 13.192.** Alice and Bob take turns (Alice starts) replacing the coefficients of the polynomial $a_{99}x^{99} + \cdots + a_1x + a_0$ with non-zero integers. Bob wants the polynomial to have as many different integer roots as possible. What is the largest number of roots he can always achieve? [239 Olympiad 2022, Junior League]

**Problem 13.193.** Alice and Bob play the following game. They alternate selecting distinct nonzero digits (from 1 to 9) until they have chosen seven such digits, and then consider the resulting seven-digit number obtained by concatenating the digits in the order selected. Alice wins if and only if the resulting number consists of the last seven decimal digits of some perfect seventh power. Determine which player has a winning strategy. [Taiwan IMO TST 2014]

**Problem 13.194.** Let $p$ be a prime number. Alice and Bob play the following game, making moves alternately: in each move the current player chooses an index $i$ in the set $\{0, 1, \ldots, p-1\}$ that was not chosen before by either of the players, and then chooses an element $a_i$ from the set $\{0, 1, 2, \ldots, 9\}$. Alice has the first move. The game ends after all the indices have been chosen, and then the number $M = \sum_{i=0}^{p-1} a_i 10^i$ is computed. The goal of Alice is to make $M$ divisible by $p$, and the goal of Bob is to prevent this. Who has a winning strategy? [IMO Shortlist 2017 N2]

**Problem 13.195.** Let $n > 1$ be an integer. The numbers $1, 2, \ldots, n$ are written on a board. Alice and Bob take turns circling an uncircled number on the board, with Alice going first. When the product of the circled numbers becomes a multiple of $n$, the game ends and the last player to have circled a number loses. Depending on $n$, which player has a winning strategy? [ELMO 2022, Problem 1]

**Problem 13.196.** Two players $A$ and $B$ play a game. For a given prime $p$, the numbers $1, 2, 3, \ldots, p-1$ are written on a blackboard. The players take turns making moves, with $A$ starting. A move consists of one of the players crossing out a number on the board that has not yet been crossed out. If the product of all currently crossed out numbers is 1 (mod $p$) after the move, the player whose move it was receives one point; otherwise zero points are awarded. The game ends after all numbers have been crossed out, and the player with the most points wins; equal scores mean a draw. For each $p$, determine which player (if any) has a winning strategy. [Baltic Way 2022]

**Problem 13.197.** Let $n$ be a positive integer. Alice and Bob play the following game, taking turns in an alternating fashion with Alice starting first. The player on turn fills in one of the stars in the equation

$$*x^n + *x^{n-1} + \cdots + *x^2 + *x + * = 0$$

with a non-zero rational number. Who has a winning strategy if the goals of the players are: Alice — that the resulting equation has a rational root; Bob — that it does not? [Iran National Round 2022]

**Problem 13.198.** Four integers are marked on a circle. On each step we simultaneously replace each number by the difference between this number and the next number on the circle, moving clockwise; that is, the numbers $a, b, c, d$ are replaced by $a-b$, $b-c$, $c-d$, $d-a$. Is it possible that after 2019 such moves we have numbers $a, b, c, d$ for which $|bc-ad|$, $|ac-bd|$ and $|ab-cd|$ are all primes?

**Problem 13.199.** Initially, some positive integer greater than 1 is written on the board. In every step, some prime divisor $p$ of the number $n$ on the board is chosen, and $n$ is replaced with $n+\frac{n}{p}$. If infinitely many steps are performed, prove that $p$ will be equal to 3 infinitely often.

**Problem 13.200.** Let $x>1$ be an integer. We are given the list of numbers $1, x+1, 2x+1, 3x+1, \ldots, x^{99}+1$. In each step we erase the rightmost number remaining on the board, along with all of its divisors. Which number will be deleted last?

## 13.10 Size and asymptotic reasoning

**Technique 13.18 (Compare growth rates).** Many existence and counting problems are settled by nothing more than the order in which the functions

$$2023, \quad \log_2 x, \quad x^2-1, \quad \text{a polynomial of degree 50}, \quad 1.1^x, \quad \lfloor x \rfloor !$$

grow. Estimate the number of objects a construction can produce and the number of objects that must be covered; if the first grows strictly more slowly than the second, the covering is impossible for all large parameters, and finitely many small cases remain.

**Problem 13.201.** Let $d$ and $d'>d$ be divisors of a positive integer $n$. Prove that $d' > d + \dfrac{d^2}{n}$.

**Problem 13.202.** Let $n$ be a positive integer. The positive integer $m$ is such that its divisors can be partitioned into $n$ groups, each with the same sum. Prove that $m > 2^n$. [Bulgaria Autumn Mathematical Tournament 2024]

**Problem 13.203.** Let $a, b>1$ be integers. For any positive integer $n$, let $r_n$ be the non-negative remainder that $b^n$ leaves upon division by $a^n$. Assume that there exists a positive integer $N$ such that $r_n < \dfrac{2^n}{n}$ for all integers $n \geq N$. Prove that $a$ divides $b$. [RMM 2024, Problem 4]

**Problem 13.204.** Find all positive integers $n$ with the following property: the $k$ positive divisors of $n$ have a permutation $(d_1, d_2, \ldots, d_k)$ such that for $i=1,2,\ldots,k$ the number $d_1+d_2+\cdots+d_i$ is a perfect square. [IMO Shortlist 2021 N3]

**Problem 13.205.** Let $n \geq 3$ be an integer and let $a_1, a_2, \ldots, a_n$ be distinct positive integers. Prove that there exist distinct indices $i$ and $j$ such that $a_i+a_j$ does not divide any of the numbers $3a_1, 3a_2, \ldots, 3a_n$. [IMO Shortlist 2008 N2]

**Problem 13.206.** (a) Let $n^2+1 = ab$ for positive integers $n, a, b$ with $a>b$. Prove that $a-b \geq \sqrt{4n-3}$. (b) Determine all positive integers $n$ for which the inequality in (a) becomes an equality.†

**Problem 13.207.** Let $a_1 < a_2 < \cdots < a_n$ be integers satisfying $1 < a_1$ and $a_n < 2a_1$. If $m$ is the number of distinct prime factors of $a_1 a_2 \cdots a_n$, prove that $(a_1 a_2 \cdots a_n)^{m-1} > (n!)^m$.

**Problem 13.208.** There are positive integers written in the cells of a $4 \times 4$ table (one number in each cell) in such a way that the sum of any two numbers in neighbouring cells is a factorial of some number. Prove that there are at least 4 equal numbers in the table.

**Problem 13.209.** There is a finite number of pairwise different palindromes. Prove that the sum of their reciprocals is less than 20.

**Problem 13.210.** Given an integer $N$, the numbers $N^3, N^3+1, \ldots, N^3+N$ are written on a board; $a$ of them are blue and $b$ of them are yellow. Prove that if the sum of the blue numbers is divisible by the sum of the yellow numbers, then $a$ is divisible by $b$.

**Problem 13.211.** Which quantity is bigger — the number of positive integers less than $10^{1000}$ that can be expressed as the sum of a perfect cube and a sixth power of a positive integer, or the number of positive integers less than $10^{1000}$ that are perfect squares?

**Problem 13.212.** We say that a positive integer $n$ is *friendly* if the equation $(x^2+y)(y^2+x) = n(x-y)^3$ has a solution in positive integers. Prove that (a) at least 500 positive integers $n \le 2020$ are friendly, and (b) $n = 2$ is not friendly.

**Problem 13.213.** What is the largest $n$ for which there exist $2n$ positive integers $a_1, \ldots, a_n$, $b_1, \ldots, b_n$ that satisfy $a_i b_j - a_j b_i = 1$ whenever $i < j$?

**Problem 13.214.** If $a, b, c$ are three distinct nonnegative integers, prove that $\gcd(ab+1,\, bc+1,\, ca+1) \le \dfrac{a+b+c}{3}$.

**Problem 13.215.** For which positive integers $n$ can we find integers $0 < a_1 < a_2 < \cdots < a_n$ such that $\sqrt[n]{a_n} - \sqrt[n]{a_1} < 1$ and $a_1 a_2 \cdots a_n$ is the $n$-th power of an integer?

**Problem 13.216.** ★ Is there an integer $n > 10^{10000}$ such that $n! + 1$ divides $(1000n)!$? [IMC 2012]

## 13.11 Congruences and Diophantine equations: the June 2022 course

The number-theory course of the June 2022 camp was a single long problem list, running from congruences and divisibility through factorisations to Diophantine equations and Pythagorean triples. It is reproduced here in its own order.

**Problem 13.217.** Let $a, b$ be positive integers and let $p, q$ be distinct primes such that $aq \equiv 1 \pmod{p}$ and $bp \equiv 1 \pmod{q}$. Prove that

$$\frac{a}{p} + \frac{b}{q} > 1.$$

**Problem 13.218.** Determine all positive integers $x, y, z$ such that

$$\frac{2}{x^2} + \frac{3}{y^2} + \frac{4}{z^2} = 1.$$

**Problem 13.219.** Prove that an integer $n \ge 2$ is composite if and only if there exist integers $a, b, x, y \ge 1$ such that $a + b = n$ and $\dfrac{x}{a} + \dfrac{y}{b} = 1$.

**Problem 13.220.** Let $n \geq 3$ and consider pairwise coprime positive integers $p_1, p_2, \ldots, p_n$. Suppose that for every $k \in \{1, 2, \ldots, n\}$ the remainder of $\prod_{i \neq k} p_i$ modulo $p_k$ is equal to one and the same number $r$. Prove that $r \leq n - 2$.

**Problem 13.221.** Let $x, y, z, t$ be positive integers such that $x^2 + y^2 + z^2 + t^2 = 2018!$. Prove that $x, y, z, t > 10^{250}$.

**Problem 13.222.** Find all primes $p$ for which the sequence $1, 2, \ldots, p$ (in that order) can be divided into some number, greater than one, of blocks of consecutive numbers such that the sum of the numbers in each block is the same.

**Problem 13.223.** Let $a, b, c, n$ be positive integers such that

(i) the numbers $a$, $b$, $c$, $a + b + c$ are pairwise coprime;

(ii) the number $(a + b)(b + c)(c + a)(a + b + c)(ab + bc + ca)$ is a perfect $n$-th power.

Prove that the product $abc$ can be expressed as a difference of two perfect $n$-th powers.

**Problem 13.224.** Let $a$, $b$ and $c$ be odd positive integers such that $a$ is not a perfect square and

$$a^2 + a + 1 = 3(b^2 + b + 1)(c^2 + c + 1).$$

Prove that at least one of the numbers $b^2 + b + 1$ and $c^2 + c + 1$ is composite.

**Problem 13.225.** Let $x, y$ be integers greater than 1 such that $x + y - 1$ divides $x^2 + y^2 - 1$. Prove that $x + y - 1$ is a composite number.

**Problem 13.226.** Let $m, n$ be two distinct positive integers such that $\operatorname{lcm}(m, n) = m^2 - n^2 + mn$. Prove that $mn$ is a perfect cube.

**Problem 13.227.** Find all primes $p$ and $q$ such that $p + q$, $p + q^2$, $p + q^3$ and $p + q^4$ are all prime.

**Problem 13.228.** Find all integer triples $(a, b, c)$ satisfying the equation $5a^2 + 9b^2 = 13c^2$.

**Problem 13.229.** Show that no non-zero integers $a, b, x, y$ satisfy

$$ax - by = 16, \qquad ay + bx = 1.$$

**Problem 13.230.** Let $a, b, c > 1$ be distinct integers with $\gcd(a, b, c) = 1$. Find all possible values of

$$\gcd\Big(a^2b + b^2c + c^2a,\ ab^2 + bc^2 + ca^2,\ a + b + c\Big).$$

**Problem 13.231.** Three pairwise distinct positive integers $a, b, c$ with $\gcd(a, b, c) = 1$ satisfy

$$a \mid (b - c)^2, \qquad b \mid (a - c)^2, \qquad c \mid (a - b)^2.$$

Prove that there is no non-degenerate triangle with side lengths $a$, $b$, $c$.

**Problem 13.232.** Find all triples of positive integers $(a, b, c)$ for which the number

$$\frac{(a + b)^4}{c} + \frac{(b + c)^4}{a} + \frac{(c + a)^4}{b}$$

is an integer and $a+b+c$ is a prime.

**Problem 13.233.** A *Pythagorean triple* $(a,b,c)$ is a triple of integers satisfying $a^2+b^2=c^2$; such a triple is called *primitive* if $\gcd(a,b,c)=1$. Let $d$ be an odd integer with exactly $n$ prime divisors. Show that there exist exactly $2^{n-1}$ primitive Pythagorean triples whose first element is $d$. For example, for $d=15$ the primitive triples are $(15,8,17)$ and $(15,112,113)$.

**Problem 13.234.** Let $p$ and $q$ be distinct odd primes. Prove that $(pq+1)^4-1$ has at least 4 distinct prime divisors.

**Problem 13.235.** Let $n$ be a positive integer. Prove that the equation $\sqrt{x}+\sqrt{y}=\sqrt{n}$ has a solution $(x,y)$ in positive integers if and only if $n$ is divisible by $m^2$ for some integer $m>1$.

**Problem 13.236.** All the prime numbers are written in increasing order, $p_1=2$, $p_2=3$, $p_3=5$ and so on. Find all pairs of positive integers $a$ and $b$ with $a-b\geq 2$ such that $p_a-p_b$ divides $2(a-b)$.

**Problem 13.237.** Find all pairs $(m,n)$ of integers that satisfy the equation

$$(m-n)^2=\frac{4mn}{m+n-1}.$$

# Chapter 14

# Mixed Practice

**Sources for this chapter:** Weekly e-training problem sheets, Level 3, from three seasons of the program: 2019–2020 (weeks 1–12), September–November 2021, and 2022–2023 (weeks 1–9).

Each week of the program's e-training course mixed algebra, number theory, combinatorics and geometry into a single short problem set. This chapter reproduces that training form: a sequence of practice sets, drawn directly from the weekly Level 3 sheets and merged where a week ran thin, arranged in roughly increasing difficulty.

## Practice set 1

**Problem 14.1.** Let $a \neq 0$, and let $x_1, x_2$ be the roots of $x^2 + ax - \dfrac{1}{2a^2} = 0$. Prove that $x_1^4 + x_2^4 \geq 2 + \sqrt{2}$.

**Problem 14.2.** Prove that $\operatorname{lcm}(1, 2, \ldots, 2n) = \operatorname{lcm}(n+1, n+2, \ldots, 2n)$.

**Problem 14.3.** In a school, more than 90% of the students speak both English and Armenian, and more than 90% speak both English and Arabic. Prove that among the students who speak both Armenian and Arabic, more than 90% speak English.

**Problem 14.4.** The endpoints of $N$ arcs split a circle into $2N$ equal arcs of length 1. Each of the $N$ arcs splits the circle into two parts of even length. Prove that $N$ is even.

**Problem 14.5.** Two lines through vertices $B$ and $C$ of triangle $ABC$ divide the triangle into four regions – three triangles and one quadrilateral. Given that three of these regions have equal area, prove that the quadrilateral is one of these three regions.

**Problem 14.6.** Let $ABCD$ be a square, with $K$ on side $AB$, $L$ on side $CD$, and $M$ a point on segment $KL$. Prove that the second intersection point of the circumcircles of triangles $AKM$ and $MLC$ lies on the diagonal $AC$.

## Practice set 2

**Problem 14.7.** For a positive integer $n$, let $f(n)$ denote the greatest odd divisor of $n$. Prove that for every positive integer $n$,

$$f(n+1) + f(n+2) + \cdots + f(2n) = n^2.$$

**Problem 14.8.** Find the smallest positive integer $m$ for which there exists a positive integer $n$ such that

$$\left|\frac{n}{m}-\frac{2}{5}\right|\le\frac{1}{100}.$$

**Problem 14.9.** Find $\gcd(5^{300}-1,\,5^{200}+6)$.

**Problem 14.10.** Ali has chosen 8 cells of an $8\times 8$ chessboard, no two in the same row or column (a *general configuration*). On each move, Baba chooses 8 cells forming a general configuration and places coins on them; Ali then reveals how many of Baba's coins lie outside Ali's own 8 cells. If that number is even, Baba wins; otherwise Ali removes the coins and Baba moves again. Find the minimum number of moves that guarantees Baba a win.

**Problem 14.11.** Let $S=\{1,2,3,4,5,6,7\}$. Find the number of functions $f:S\to S$ such that $f(f(x))=x$ for every $x\in S$.

**Problem 14.12.** In triangle $ABC$, the median $AM$ is drawn. Can the inradius of triangle $ABM$ be exactly twice the inradius of triangle $ACM$?

**Problem 14.13.** A point $M$ is chosen inside square $ABCD$ so that $\angle MAC=\angle MCD$. Find $\angle ABM$.

## Practice set 3

**Problem 14.14.** Solve the equation $x(7-x)(7+x^2)=12(x+1)^2$.

**Problem 14.15.** Find all $m$ for which the equations $x^3+mx+1=0$ and $x^4+mx^2+1=0$ have a common root.

**Problem 14.16.** Find the digits $x$ and $y$ if the number $\overline{12x3y4}$ is divisible by 599.

**Problem 14.17.** Prove that if $a+b$ and $ab$ are both divisible by $c$, then $a^6+b^6$ is divisible by $c^3$.

**Problem 14.18.** Points $A,B,C,D$ are given, no three of them collinear. Prove that the triangles $ABC$, $BCD$, $CDA$, $DAB$ cannot all be acute at the same time.

**Problem 14.19.** In triangle $ABC$, $\angle ABC=\angle ACB=40^\circ$. Let $BD$ bisect $\angle ABC$, with $D$ on $AC$. Prove that $BD+DA=BC$.[†]

## Practice set 4

**Problem 14.20.** Find all integer solutions of $1+x+x^2+x^3+x^4=y^4$.

**Problem 14.21.** Three primes $p,q,r$ and a positive integer $n$ are such that

$$\frac{p+n}{qr},\qquad\frac{q+n}{rp},\qquad\frac{r+n}{pq}$$

are all integers. Prove that $p=q=r$.

**Problem 14.22.** Let $a_1, a_2, \ldots, a_{100}$ be a permutation of $1, 2, \ldots, 100$, and let $S_k = a_1 + a_2 + \cdots + a_k$. Find the maximum possible number of perfect squares among $S_1, S_2, \ldots, S_{100}$.

**Problem 14.23.** Is it possible to fill the cells of a $7 \times 7$ table with positive integers so that the sum of the numbers in every $2 \times 2$ square and in every $3 \times 3$ square is odd?

**Problem 14.24.** The integers from 1 to 50 are written on a blackboard. What is the smallest number of them that must be erased so that the sum of any two remaining numbers is not a prime?

**Problem 14.25.** Let $ABCD$ be a parallelogram, and let $M$ be a point on line $AB$ such that $\angle MAD = \angle AMO$, where $O$ is the intersection of the diagonals. Prove that $MD = MC$.

## Practice set 5

**Problem 14.26.** Find the largest possible value of $\lfloor 7x + 11y \rfloor - 7\lfloor x \rfloor - 11\lfloor y \rfloor$, where $x, y$ are real numbers.

**Problem 14.27.** For $k \geq 1$ let $a_k = \underbrace{\sqrt{2 + \sqrt{2 + \cdots + \sqrt{2}}}}_{k \text{ radicals}}$. Prove that for every $n \geq 2$,

$$\frac{2 - a_n}{2 - a_{n-1}} > \frac{1}{4}.$$

**Problem 14.28.** Determine whether there exist rational numbers $r, q$ such that $r^2 + q^2 = 15$.

**Problem 14.29.** A rectangle is divided by segments parallel to its sides into smaller rectangles. Call a point a *$T$-point* if it is a vertex of exactly two of the small rectangles. Prove that the number of $T$-points is even.

**Problem 14.30.** Let $S$ be a set of 10 distinct positive real numbers. Show that there exist $x, y \in S$ such that

$$0 < x - y < \frac{(1 + x)(1 + y)}{9}.$$

**Problem 14.31.** Let $ABC$ be a right triangle with hypotenuse $AB$, and let $ABDE$ be a square with $C, D$ on opposite sides of $AB$. The bisector of $\angle C$ meets $DE$ at $F$. If $AC = 1$ and $BC = 3$, find $\dfrac{DF}{EF}$.

## Practice set 6

**Problem 14.32.** Characterize all integers $n > 1$ for which

$$(n-1)!\left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n-1}\right)$$

is not divisible by $n$.

**Problem 14.33.** Let $x(n)$ denote the largest prime divisor of $n$. Prove that there exist infinitely many $n$ such that $x(n) < x(n+1) < x(n+2)$.

**Problem 14.34.** Let $a_1, a_2, \ldots, a_{20}$ be a permutation of $1, 2, \ldots, 20$, arranged around a circle. Find the maximum possible value of

$$\min\{|a_2 - a_1|, |a_3 - a_2|, \ldots, |a_{20} - a_{19}|, |a_1 - a_{20}|\}.$$

**Problem 14.35.** Triangles $ABC$ and $A_1B_1C_1$ are symmetric to one another about the center of their common incircle, which has radius $r$. Prove that the product of the areas of $ABC$, $A_1B_1C_1$, and the six other triangles formed by the intersecting sides of $ABC$ and $A_1B_1C_1$ equals $r^{16}$.

**Problem 14.36.** Find all pairs of positive integers $a, b$ such that both

$$\frac{a^2+b}{b^2-a} \qquad \text{and} \qquad \frac{b^2+a}{a^2-b}$$

are integers.

**Problem 14.37.** The cells of an infinite grid are filled with positive integers so that every number equals the average of its four neighbors. Prove that all the numbers are equal.

**Problem 14.38.** Ten pairwise distinct numbers are written on a board. Ali lists the squared difference $(a-b)^2$ for every pair, and Bob lists the absolute difference of squares $|a^2 - b^2|$ for every pair. Can Ali's and Bob's lists coincide as collections?

## Practice set 7

**Problem 14.39.** Let $a, b, c$ be real numbers with

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = 1.$$

Prove that

$$\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} = 0.$$

**Problem 14.40.** The set $\{1, 2, \ldots, 10\}$ is partitioned into three subsets $A, B, C$. For each subset, its sum, its product, and the sum of the digits of its elements are computed. Is it possible for $A$ alone to have the largest sum, $B$ alone the largest product, and $C$ alone the largest digit sum?

**Problem 14.41.** Find all positive integers $n$ for which

$$3x^n + n(x+2) - 3 \geq nx^2$$

holds for every real number $x$.

**Problem 14.42.** Let $P(n)$ denote the greatest prime divisor of $n$. Find all integers $n \geq 2$ for which

$$P(n) + \lfloor \sqrt{n} \rfloor = P(n+1) + \left\lfloor \sqrt{n+1} \right\rfloor.$$

**Problem 14.43.** Two piles contain 10000 and 20000 tokens. A move consists of removing any positive number of tokens from one pile, or removing $x > 0$ tokens from one pile and $y > 0$ from the other with $2015 \mid x + y$. The player who cannot move loses. Which player has a winning strategy?

**Problem 14.44.** Three circles $\omega_1, \omega_2, \omega_3$ pass through a common point $P$. The tangent to $\omega_1$ at $P$ meets $\omega_2$ and $\omega_3$ again at $P_{1,2}$ and $P_{1,3}$; the points $P_{2,1}, P_{2,3}, P_{3,1}, P_{3,2}$ are defined analogously (tangent to $\omega_2$, resp. $\omega_3$, at $P$). Prove that the perpendicular bisectors of $P_{1,2}P_{1,3}$, $P_{2,1}P_{2,3}$, $P_{3,1}P_{3,2}$ are concurrent.

**Problem 14.45.** Let $ABC$ be a triangle with $\angle A = 60°$, and let $E, F$ be the feet of the internal bisectors from $B$ and $C$. Choose $P, Q$ so that $BFPE$ and $CEQF$ are parallelograms. Prove that $\angle PAQ > 150°$ (taking the angle not containing side $AB$).

## Practice set 8

**Problem 14.46.** Find 8 positive integers $n_1, \ldots, n_8$ such that every integer $n$ with $|n| < 2019$ can be written as $a_1 n_1 + \cdots + a_8 n_8$ with each $a_i \in \{0, 1, -1\}$.

**Problem 14.47.** Find the minimum value of

$$|x - 1| + |2x - 1| + |3x - 1| + \cdots + |119x - 1|.$$

**Problem 14.48.** Find all primes $p$ such that $p^2 + 11$ has exactly six divisors (including 1 and itself).

**Problem 14.49.** Find the number of odd coefficients of the polynomial $(x^2 + x + 1)^{33}$.

**Problem 14.50.** Integers $a, b$ are such that $2^n a + b$ is a perfect square for every nonnegative integer $n$. Show that $a = 0$.

**Problem 14.51.** Let $a, b$ be two sides of a triangle. How should the third side $c$ be chosen so that the points where the incircle and the excircle opposite $c$ touch $c$ divide it into three equal parts? (The excircle opposite $c$ is tangent to side $c$ and to the extensions of $a$ and $b$.)

## Practice set 9

**Problem 14.52.** Prove that for every positive integer $n$, at least one coefficient of

$$\left(x^4 + x^3 - 3x^2 + x + 2\right)^n$$

is negative.

**Problem 14.53.** Let

$$P(x) = \underbrace{\left(\left(\cdots\left(\left(}_{k} x - 2\right)^2 - 2\right)^2 - \cdots\right)^2 - 2\right)^2.$$

Find the coefficient of $x^2$.

**Problem 14.54.** Let $f(x) = x^2 - 6x + 5$. Sketch the set of points $(x, y)$ satisfying

$$\begin{cases} f(x) + f(y) \le 0 \\ f(x) - f(y) \ge 0. \end{cases}$$

**Problem 14.55.** Prove that for all positive integers $m > n$,

$$\operatorname{lcm}(m, n) + \operatorname{lcm}(m + 1, n + 1) > \frac{2mn}{\sqrt{m - n}}.$$

**Problem 14.56.** A convex $s$-gon is divided into $q$ quadrilaterals, of which $b$ are non-convex. Prove that

$$q \ge b + \frac{s - 2}{2}.$$

**Problem 14.57.** Positive numbers, all less than 1, are written around a circle. Prove that the circle can be split into three arcs so that, for any two of them, the sums of the numbers on them differ by at most 1.

**Problem 14.58.** In triangle $ABC$, points $D, E, F$ lie on $BC, CA, AB$ respectively with

$$\frac{BD}{CD} = \frac{CE}{AE} = \frac{AF}{BF}.$$

Show that if the circumcircle of $DEF$ coincides with the circumcircle of $ABC$, then $ABC$ is equilateral.

## Practice set 10

**Problem 14.59.** Ten students stand around a circle. In one move, a single student may relocate to any other position around the circle. Find the minimum number of moves that always suffices to arrange the students in clockwise increasing order of height, regardless of the starting arrangement.

**Problem 14.60.** A cake in the shape of a regular pentagon has a nut at its center. At each step, one may cut off a triangular piece whose cutting line does not pass through a vertex of the current polygon. Can the nut always be reached eventually?

**Problem 14.61.** Several distinct positive numbers are written on a board, and the sum of any two of them is a power of 2. Find the largest possible number of numbers on the board.

**Problem 14.62.** One hundred apples are placed in a row so that the weights of any two neighboring apples differ by less than 1 gram. Prove that they can be grouped into 50 pairs and rearranged in a row of pairs so that the total weights of any two neighboring pairs also differ by less than 1 gram.

**Problem 14.63.** A triangle is divided into three similar triangles. Prove that it can also be divided into two similar triangles.

## Practice set 11

**Problem 14.64.** A parabola $f(x)$ has two distinct roots and satisfies $f(a^2+b^2) \geq f(2ab)$ for all real $a, b$. Prove that at least one root of $f$ is negative.

**Problem 14.65.** Let $a > b$ be positive integers with $\gcd(a-b, ab+1) = 1$ and $\gcd(a+b, ab-1) = 1$. Prove that $(a-b)^2 + (ab+1)^2$ is not a perfect square.

**Problem 14.66.** Let $n$ be a positive integer and $p$ a prime such that $np+1$ is a perfect square. Prove that $n+1$ can be written as a sum of $p$ perfect squares.

**Problem 14.67.** Omar writes the sequence $x_1 = 20$, $x_2 = 18$, $x_{k+1} = x_{k-1} - \dfrac{1}{x_k}$. Prove that this process cannot continue forever: at some point a term equal to 0 is reached.

**Problem 14.68.** Determine whether there exists an infinite sequence of positive integers such that no term, and no sum of finitely many terms, is a perfect power.

**Problem 14.69.** Determine the maximal possible length of a run of consecutive integers each expressible in the form $x^3 + 2y^2$ with $x, y$ integers.

**Problem 14.70.** There are $n$ people, with at most one telephone call ever taking place between any two of them. The total number of calls among any $n-2$ of them equals $3^k$ for some non-negative integer $k$. Find all possible values of $n$.

## Practice set 12

**Problem 14.71.** Let $ABC$ be a non-isosceles acute triangle and $O$ the midpoint of $BC$. The circle with diameter $BC$ meets $AB, AC$ at $D, E$. The bisectors of $\angle A$ and $\angle DOE$ meet at $P$. If the circumcircles of $BPD$ and $CPE$ meet again at $Q$, show that $Q$ lies on $BC$.

**Problem 14.72.** Let $a, b, c, d$ be real numbers with $a^4 + b^4 + c^4 + d^4 = 16$. Prove that

$$a^5 + b^5 + c^5 + d^5 \leq 32.$$

**Problem 14.73.** Let $1 \leq r \leq n$. Consider all $r$-element subsets of $\{1, 2, \ldots, n\}$; each has a minimum element. Prove that the average of these minima equals $\dfrac{n+1}{r+1}$.

**Problem 14.74.** Let $ABC$ be an acute triangle with circumcenter $O$ and altitudes $AD, BE, CF$. Show that the segments $OA, OF, OB, OD, OC, OE$ dissect $ABC$ into three pairs of triangles of equal area.

**Problem 14.75.** Find an increasing sequence of natural numbers $1 \leq a_1 < a_2 < \cdots$ with the property that every positive integer $m$ can be written uniquely as $m = a_i - a_j$ with $i > j \geq 1$.

**Problem 14.76.** Prove the identity

$$\frac{n!}{x(x+1)(x+2)\cdots(x+n)} = \frac{\binom{n}{0}}{x} - \frac{\binom{n}{1}}{x+1} + \frac{\binom{n}{2}}{x+2} - \cdots + (-1)^n \frac{\binom{n}{n}}{x+n}.$$

**Problem 14.77.** Let $n > 3$ and $x_1, x_2, \ldots, x_n > 0$ with $x_1 x_2 \cdots x_n = 1$. Prove that

$$\frac{1}{1 + x_1 + x_1 x_2} + \frac{1}{1 + x_2 + x_2 x_3} + \cdots + \frac{1}{1 + x_n + x_n x_1} > 1.$$

**Problem 14.78.** Let $ABCD$ be a convex quadrilateral such that line $CD$ is tangent to the circle with diameter $AB$. Prove that line $AB$ is tangent to the circle with diameter $CD$ if and only if $BC \parallel AD$.

## Practice set 13

**Problem 14.79.** Given four positive integers, the square of the sum of any two of them is divisible by the product of the other two. Prove that at least three of the four numbers are equal.

**Problem 14.80.** Prove that for every prime $p > 2$ there is a unique pair of positive integers $(x, y)$ with $x > y$ such that

$$\frac{2}{p} = \frac{1}{x} + \frac{1}{y}.$$

**Problem 14.81.** Find all integers $n$ for which $x^2 - x + n$ divides $x^{13} + x + 90$.

**Problem 14.82.** Prove that the sequence $\left\lfloor 2^n \sqrt{2} \right\rfloor$ contains infinitely many composite numbers.

**Problem 14.83.** Find all positive integers $k$ such that the set $\{1990, 1991, \ldots, 1990 + k\}$ can be split into two subsets with equal sums.

**Problem 14.84.** It is known that a circle can be inscribed in trapezoid $ABCD$. Prove that the two circles constructed on its oblique (non-parallel) sides as diameters are tangent to each other.

## Practice set 14

**Problem 14.85.** Find the number of real solutions of $x^3 + 2x^2 - 6 = 0$.

**Problem 14.86.** Prove that for all real numbers $a, b, c$,

$$ab + bc + ca + |a - b| \leq 1 + \frac{1}{3}(a + b + c)^2.$$

**Problem 14.87.** Sixteen football teams play a round-robin tournament (a win is worth 3 points, a draw 1 point to each side, a loss 0). A team is called *lucky* if it earns more than half of the maximum number of points a team could possibly earn. Find the largest possible number of lucky teams.

**Problem 14.88.** Each cell of an $n \times n$ board is colored black or white. Suppose that any row and column meeting at a white cell together contain at least $n$ black cells. Prove that the total number of black cells is at least $\frac{n^2}{2}$.

**Problem 14.89.** Find all pairs of non-negative integers $(x, y)$ such that $x^2 + 3y$ and $y^2 + 3x$ are simultaneously perfect squares.

**Problem 14.90.** Prove that for any sequence of positive integers $a_1, a_2, \ldots, a_n$, there is a positive integer $k$ for which

$$s(ka_1) = s(ka_2) = \cdots = s(ka_n),$$

where $s(m)$ denotes the digit sum of $m$.

**Problem 14.91.** Find all real solutions $(a, b, c, d)$ of

$$a + b + c = d, \qquad \frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{d}.$$

**Problem 14.92.** At a birthday party, several married couples attended, each with between 1 and 10 children. The number of triples {father, mother, child}, no two members from the same family, is exactly 3630. Find the number of children at the party.

**Problem 14.93.** Twenty black and twenty white balls are placed in a row. In one move, two neighboring balls may be swapped. Find the minimum number of moves that always suffices to guarantee a position with 20 consecutive black balls.

## Practice set 15

**Problem 14.94.** Find all integer polynomials $P$ for which $(x^2+6x+10)P(x)^2-1$ is the square of an integer polynomial.

**Problem 14.95.** (a) Find the minimum number of elements that must be deleted from $\{1, 2, \ldots, 2018\}$ so that the remaining set contains no two elements together with their product. (b) Does the sequence $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots, \frac{1}{n}, \ldots$ contain an arithmetic progression of every length $k$?

**Problem 14.96.** Prove that not all roots of a polynomial of the form

$$x^n + 2nx^{n-1} + 2n^2x^{n-2} + \cdots$$

can be real.

**Problem 14.97.** Let $P(x) = a_{2n}x^{2n} + a_{2n-1}x^{2n-1} + \cdots + a_1x + a_0$ have all coefficients $a_i \in [100, 101]$. Find the smallest possible value of $n$ for which $P$ can have a root.

**Problem 14.98.** Let $ABC$ be a right triangle with $\angle A = 90°$, and let $AD$ be an altitude. Let $J, K$ be the incenters of triangles $ABD, ACD$. If line $JK$ meets $AB, AC$ at $E, F$, prove that $AE = AF$.

## Practice set 16

**Problem 14.99.** Find all integers $a, b, c$ with $1 < a < b < c$ such that $(a-1)(b-1)(c-1)$ divides $abc - 1$.

**Problem 14.100.** Prove that every integer can be written as a sum of 5 perfect cubes (of possibly negative integers).

**Problem 14.101.** Let $\varphi_n(m) = \varphi(\varphi_{n-1}(m))$, where $\varphi_1(m) = \varphi(m)$ is Euler's totient function, and let $\omega(m)$ be the least $n$ such that $\varphi_n(m) = 1$. Prove that if $m < 2^\alpha$ then $\omega(m) \leq \alpha$.

**Problem 14.102.** Consider cutting a rectangle along lattice lines only. Prove that for every integer $m > 12$ one can cut a rectangle of area greater than $m$ from which no sub-rectangle of area exactly $m$ can be cut.

**Problem 14.103.** Let $ABC$ be a triangle with midpoints $K, M, N$ of $BC, CA, AB$, and altitudes $AD, BE, CF$. Let $U, V, W$ be the midpoints of $FD, DE, EF$. Prove that $KW$, $MV$, $NU$ are concurrent.

## Practice set 17

**Problem 14.104.** Find all pairs of integers $(m, n)$ such that $\binom{n}{m} = 1984$.

**Problem 14.105.** Prove that for every integer $n > 1$, $2^n - 1$ does not divide $3^n - 1$.

**Problem 14.106.** Let $u_n$ be the least common multiple of the first $n$ terms of a strictly increasing sequence of positive integers $a_1, a_2, \ldots, a_{1000}$. Prove that

$$\sum_{k=1}^{1000} \frac{1}{u_k} \leq 2.$$

**Problem 14.107.** Let $\sigma(n)$ denote the sum of the divisors of $n$. Prove that there exist infinitely many integers $n$ with $\sigma(n) > 3n$, and also that $\sigma(n) < n(1 + \log_2 n)$ for every $n$.

**Problem 14.108.** Let $\sigma(n)$ denote the sum of the divisors of $n$. Show that $\sigma(n) = 2^k$ if and only if $n$ is a product of distinct Mersenne primes (primes of the form $2^k - 1$).

**Problem 14.109.** Let $a_1 = 1$, $a_{n+1} = a_n + \lfloor \sqrt{a_n} \rfloor$. Find all $n$ for which $a_n$ is a perfect square.

**Problem 14.110.** In an acute triangle $ABC$, let $D$ be the foot of the altitude from $A$, and let $O, H$ be the circumcenter and orthocenter. The perpendicular bisector of $AO$ meets line $BC$ extended at $E$. Show that the midpoint of $OH$ lies on the circumcircle of $ADE$.

## Practice set 18

**Problem 14.111.** Let $d$ be a positive integer with $d \neq 2, 5, 13$. Show that there exist distinct $a, b \in \{2, 5, 13, d\}$ such that $ab - 1$ is not a perfect square.

**Problem 14.112.** Let $x, y, z$ be non-negative reals with $x + y + z = 1$. Show that

$$0 \leq xy + yz + zx - 2xyz \leq \frac{7}{27}.$$

**Problem 14.113.** Find all ordered pairs $(a, b)$ of positive integers for which

$$\frac{1}{a} + \frac{1}{b} = \frac{3}{2018}.$$

**Problem 14.114.** Let $M$ be a set of 2019 distinct positive integers, none of which has a prime divisor greater than 23. Prove that $M$ contains a 4-element subset whose product is a perfect fourth power.

**Problem 14.115.** For $n \geq 1$, let $p_n(k)$ be the number of permutations of $\{1, \ldots, n\}$ with exactly $k$ fixed points. Prove that

$$\sum_{k=0}^{n} k \cdot p_n(k) = n!.$$

## Practice set 19

**Problem 14.116.** Prove that there is no function $f$ from the non-negative integers to itself with $f(f(n)) = n + 2019$ for every $n$.

**Problem 14.117.** Let all vertices of a polygon be lattice points. Prove that its area equals

$$I + \frac{1}{2}B - 1,$$

where $I$ is the number of interior lattice points and $B$ the number of boundary lattice points.

**Problem 14.118.** Let $ABC$ be equilateral, and let $\mathcal{E}$ be the union of segments $AB, BC, CA$ (including the vertices). Determine whether, for every partition of $\mathcal{E}$ into two disjoint subsets, at least one subset contains the vertices of a right triangle.

**Problem 14.119.** A circle has its center on side $AB$ of cyclic quadrilateral $ABCD$, and the other three sides are tangent to it. Prove that $AD + BC = AB$.

**Problem 14.120.** The vertices of a convex quadrilateral $ABCD$ lie on a circle with center $O$, and $F$ is the second intersection point of the circumcircles of triangles $ABO$ and $CDO$. Prove that the circle through $A, F, D$ also passes through the intersection point of the diagonals $AC$ and $BD$.

# Part IV

# Level 4

# Chapter 15

# Algebra

**Sources for this chapter:** Camp and online handouts by Marko Radovanović (March 2020, April–June 2021, March 2022, October 2022, March–April 2023, September–October 2023, November 2025), Dmytro Nomirovskii (November 2019), Dominik Burek (November 2020), Nikola Petrović (June 2020), Lê Phúc Lũ (June 2022, November 2022), Łukasz Bożyk (KAUST, February 2022), Dušan Djukić (February 2023), Navid Safaei (November 2024), Dima (March 2021), and the January 2020 camp algebra sheet.

This is the IMO-track chapter. Everything the earlier levels built — Vieta's formulas, the classical inequalities, Cauchy's equation — is assumed known and is used here as a tool rather than as a topic. The new material is of three kinds: structural theory of polynomials (interpolation, reduction modulo a prime, roots of unity, Chebyshev polynomials), techniques that turn a functional equation or a recurrence into something rigid enough to solve, and the "non-classical" inequalities, in which the difficulty lies in finding the right combinatorial or extremal reformulation rather than in applying a named inequality.

## 15.1 Polynomials II: structure, irreducibility, interpolation

*Revisits Vieta's formulas and the rational root theorem (see Chapter 5); the problems below push it further.*

*Revisits polynomial divisibility and Eisenstein's criterion (see Chapter 10); the problems below push it further.*

The facts collected below are the working toolkit of an olympiad polynomial problem. Two of them — the quotient–remainder theorem and Bézout's identity — come with a warning that is easy to forget: they are theorems about polynomials over a *field*, and both fail over $\mathbb{Z}$.

**Theorem 15.1 (Fundamental theorem of algebra).** *A polynomial $p \in \mathbb{C}[X]$ of degree $n$ has exactly $n$ complex roots counted with multiplicity, and hence factors into linear factors*

$$p(x) = c(x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_n), \qquad c, \alpha_1, \ldots, \alpha_n \in \mathbb{C}.$$

*In particular such a $p$ has at least one and at most $n$ distinct complex roots.*

**Theorem 15.2 (Factorisation of real polynomials).** *Let $p \in \mathbb{R}[X]$. If $\alpha \in \mathbb{C}$ is a root of $p$, then so is $\overline{\alpha}$. Consequently the roots of $p$ are $\beta_1, \ldots, \beta_k \in \mathbb{R}$ together with*

$\alpha_1, \overline{\alpha_1}, \ldots, \alpha_l, \overline{\alpha_l} \in \mathbb{C} \setminus \mathbb{R}$, *where* $k + 2l = \deg p$, *and*

$$p(x) = c \prod_{i=1}^{k} (x - \beta_i) \prod_{i=1}^{l} (x^2 - 2(\operatorname{Re} \alpha_i)x + |\alpha_i|^2).$$

*That is, every real polynomial is a product of real linear and real quadratic factors.*

**Theorem 15.3 (Quotient and remainder).** *Let* $K$ *be one of* $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$. *For* $f, g \in K[X]$ *with* $g \neq 0$ *there exist* $q, r \in K[X]$ *with*

$$f = gq + r, \qquad \deg r < \deg g.$$

*This fails over* $\mathbb{Z}$: *for* $f(x) = x^2$ *and* $g(x) = 2x$ *no such* $q, r \in \mathbb{Z}[X]$ *exist. It does hold over* $\mathbb{Z}$ *when* $g$ *is monic.*

**Theorem 15.4 (Bézout's identity for polynomials).** *Let* $K$ *be one of* $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$. *For* $f, g \in K[X]$ *there exist* $u, v \in K[X]$ *with*

$$uf + vg = \gcd(f, g).$$

*Again this fails over* $\mathbb{Z}$, *as the pair* $f(x) = x^2$, $g(x) = 2x$ *shows.*

**Lemma 15.5 (Even and odd polynomials).** *A polynomial* $f(x) = a_n x^n + \cdots + a_1 x + a_0$ *satisfies* $f(-x) = f(x)$ *for all* $x$ *if and only if* $a_k = 0$ *for every odd* $k$, *and satisfies* $f(-x) = -f(x)$ *for all* $x$ *if and only if* $a_k = 0$ *for every even* $k$.

**Theorem 15.6 (Polynomials with integer coefficients).** *Let* $p \in \mathbb{Z}[X]$. *Then:*

*(a)* $a - b \mid p(a) - p(b)$ *for all* $a, b \in \mathbb{Z}$;

*(b) if* $\frac{a}{b} \in \mathbb{Q}$ *with* $a, b \in \mathbb{Z}$, $b \neq 0$, $\gcd(a, b) = 1$, *is a zero of* $p$, *then* $a$ *divides the constant term of* $p$ *and* $b$ *divides its leading coefficient;*

*(c) if in addition* $m \in \mathbb{Z}$ *satisfies* $a \neq bm$, *then* $a - bm \mid p(m)$.

**Theorem 15.7 (Reduction modulo a prime).** *Let* $p$ *be a prime and* $f, g, h \in \mathbb{Z}[X]$ *with* $f = gh$. *Writing* $f_p, g_p, h_p$ *for the polynomials obtained by reducing every coefficient modulo* $p$, *one has* $f_p = g_p h_p$.

**Theorem 15.8 (Gauss's lemma).** *If* $f, g \in \mathbb{Z}[X]$ *and the greatest common divisor of the coefficients of* $g$ *is* $1$, *and if* $f = gh$ *for some polynomial* $h$ *with rational coefficients, then* $h$ *has integer coefficients.*

**Theorem 15.9 (The binomial basis).** *Let* $p \in \mathbb{C}[X]$ *have degree* $n$. *Then there are unique*

$c_0, c_1, \dots, c_n \in \mathbb{C}$ *with*

$$p(x) = c_n\binom{x}{n} + c_{n-1}\binom{x}{n-1} + \cdots + c_0, \qquad \binom{x}{k} := \frac{x(x-1)\cdots(x-k+1)}{k!}.$$

**Theorem 15.10 (Sums over roots of unity).** *Let $\varepsilon = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n}$ and $\varepsilon_k = \varepsilon^k$ for $0 \le k \le n-1$; these are exactly the $n$-th roots of unity. Then*

$$1 + \varepsilon_1^k + \varepsilon_2^k + \cdots + \varepsilon_{n-1}^k = \begin{cases} n, & n \mid k, \\ 0, & \text{otherwise}, \end{cases}$$

*and for $f(x) = \sum_{i=0}^{m} c_i x^i$ one has $f(x) + f(\varepsilon x) + \cdots + f(\varepsilon^{n-1}x) = n\sum_{n|i} c_i x^i$.*

**Theorem 15.11 (Power sums modulo a prime).** *Let $p$ be a prime. Then*

$$1^k + 2^k + \cdots + (p-1)^k \equiv \begin{cases} p-1 \pmod p, & (p-1) \mid k, \\ 0 \pmod p, & \text{otherwise}, \end{cases}$$

*and for $f(x) = \sum_{i=0}^{n} c_i x^i \in \mathbb{Z}[X]$,*

$$f(0) + f(1) + \cdots + f(p-1) \equiv -\sum_{(p-1)|i} c_i \pmod p.$$

**Theorem 15.12 (Lagrange's interpolation formula).** *Given $x_0, \dots, x_n$ pairwise distinct and $y_0, \dots, y_n$ arbitrary, there is a unique polynomial $p$ of degree at most $n$ with $p(x_i) = y_i$ for all $i$, namely*

$$p(x) = \sum_{i=0}^{n} y_i \prod_{\substack{j=0 \\ j\neq i}}^{n} \frac{x - x_j}{x_i - x_j}.$$

**Lemma 15.13 (A bound for the roots).** *If $c$ is a root of $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ with $a_n \neq 0$, then*

$$|c| \;\le\; 1 + \max_{0\le i\le n-1}\left|\frac{a_i}{a_n}\right|.$$

**Lemma 15.14 (Bézout's lemma for two variables).** *Let $P(x, y)$ be a polynomial in two variables such that $P(a, y) = 0$ for a fixed value $a$ and for infinitely many values of $y$. Then $x - a$ divides $P(x, y)$.*

### 15.1.1 Division, roots, and factorisation

**Problem 15.1.** Let $P(x)$ and $Q(x)$ be polynomials with $\deg P = n$ and $\deg Q = m$. Determine $\deg(P(x)Q(x))$, $\deg(P(x) + Q(x))$ and $\deg(P(x)^k)$ for a positive integer $k$.

**Problem 15.2.** Let $P(x)$ be a polynomial and $a$ a real number. Prove that there is a polynomial $Q(x)$ with $P(x) = Q(x)(x-a) + P(a)$ for all $x$.

**Problem 15.3.** Prove that for a polynomial $P(x)$ and a real number $a$ there exist a unique polynomial $Q(x)$ and a unique real number $b$ such that $P(x) = (x-a)Q(x) + b$ for all $x \in \mathbb{R}$.

**Problem 15.4.** Find a polynomial $Q(x)$ and a number $b$ with $x^5 - x + 1 = (x-1)Q(x) + b$.

**Problem 15.5.** Prove the following theorem. If $A(x)$ is a polynomial with integer coefficients and $B(x)$ is a monic polynomial with integer coefficients, then there exist unique polynomials $Q(x)$ and $R(x)$ with integer coefficients such that

$$A(x) = B(x)Q(x) + R(x), \qquad \deg R(x) < \deg B(x).$$

**Problem 15.6.** Let $x_1, x_2, \ldots, x_{2019}$ be the roots of a polynomial $P(x)$ of degree 2019 with leading coefficient 2019. Factorise $P(x)$ completely.

**Problem 15.7.** Let $P(x)$ be a polynomial of degree $n$. Prove that the number of roots of $P$ is at most $n$.

**Problem 15.8.** Let $P(x)$ and $Q(x)$ be polynomials of degree $n$. Prove that if $x_1, \ldots, x_{n+1}$ are distinct numbers with $P(x_i) = Q(x_i)$ for every $i$, then $P(x) = Q(x)$ for all $x \in \mathbb{R}$.

**Problem 15.9.** Find the greatest common divisor of $P(x) = (x-1)(2x+1)^2(x^2-3x+2)^3$ and $Q(x) = 3x^4 - 13x^3 + 19x^2 - 11x + 2$.

**Problem 15.10.** Let $P(x)$ and $Q(x)$ be coprime polynomials, that is $\gcd(P(x), Q(x)) = 1$. Prove that there exist polynomials $A(x)$ and $B(x)$ with $P(x)A(x) + Q(x)B(x) = 1$.

**Problem 15.11.** Solve the equation $x^3 - 3x^2 + 7x - 10 = 0$.

**Problem 15.12.** Solve the equation $x^4 - 5x^3 + 13x^2 - 16x + 10 = 0$.

**Problem 15.13.** Prove that every cubic polynomial with real coefficients has at least one real root.

**Problem 15.14.** Let $P(x)$ be a polynomial with real coefficients whose degree is odd. Prove that the equation $P(x) = 0$ has a real solution.

**Problem 15.15.** Let $P(x)$ be a polynomial with real coefficients. Prove that $P(x)$ can be factorised so that every factor is either a linear or a quadratic function.

**Problem 15.16.** Find real numbers $a$ and $b$ for which there exists a polynomial $Q(x)$ with $x^{120} + 2x^{65} + 2 = (x^2 - x + 1)Q(x) + ax + b$ for all $x \in \mathbb{R}$.

**Problem 15.17.** Find real numbers $a$ and $b$ for which there exists a polynomial $Q(x)$ with $x^{2020} - x^{101} + 5 = (x^2 + 1)Q(x) + ax + b$ for all $x \in \mathbb{R}$.

**Problem 15.18.** Find real numbers $a$ and $b$ such that $x^{2019} - 3x^{100} + 5 = (x^2 - 1)Q(x) + ax + b$ for all $x \in \mathbb{R}$, where $Q(x)$ is a polynomial.

**Problem 15.19.** If one expands the brackets in $(x^3 - 5x^2 + 4x - 1)^{2010}$ and collects like terms, one obtains a polynomial $P(x)$. Find the sum of all coefficients of $P(x)$.

**Problem 15.20.** Let $P(x)$ be a non-zero polynomial. Prove that there exist non-zero polynomials $Q(x)$ and $R(x)$ such that $P(x)Q(x) = R(x^{2019})$.

**Problem 15.21.** Let $P(x)$ be a polynomial for which the equation $P(x) = x$ has no real solutions. Is it possible that the equation $P(P(x)) = x$ has a solution?

**Problem 15.22.** Let $P(x)$ be a polynomial of degree 23 with real coefficients such that the equation $P(x) = 0$ has at least 8 distinct real roots. Prove that the equation $P(P(x)) = 0$ also has at least 8 distinct real roots.

**Problem 15.23.** Let $P$ be a polynomial with $P(0) = P(1)$. Show that there are two points $A$ and $B$ on the graph of $P$ such that the segment $AB$ is horizontal and $AB = \frac{1}{4}$.

**Problem 15.24.** Suppose the polynomial $P(x) = x^{2n} + a_{2n-1}x^{2n-1} + \cdots + a_1x + a_0$ has $2n$ distinct positive roots. Prove that there exists a polynomial $Q(x)$ with $P(x^2) = Q(x)Q(-x)$.

**Problem 15.25.** A polynomial $P(x)$ of degree $n$ has $n$ distinct real roots. Find the maximum possible number of its coefficients that are zero.

**Problem 15.26.** Find all polynomials $P(x)$ such that $(x+5)P(x-1) = (x-2)P(x)$ for all $x \in \mathbb{R}$.

**Problem 15.27.** Find all polynomials $P(x)$ such that $P(x) = \dfrac{P(x+2) + P(x-2)}{2}$ for all $x \in \mathbb{R}$.

**Problem 15.28.** Find all positive integers $n$ such that the polynomial $P(x) = x^{4n} + x^{4(n-1)} + \cdots + x^4 + 1$ is divisible by $Q(x) = x^{2n} + x^{2(n-1)} + \cdots + x^2 + 1$.

**Problem 15.29.** Find all $n \in \mathbb{N}$ for which the polynomial $P(x) = x^n + x + 90$ is divisible by $x^2 - x + 2$.

**Problem 15.30.** Find all $m, n, p \in \mathbb{N}$ for which $x^2 - x + 1$ divides the polynomial $x^{3m} - x^{3n+1} + x^{3p+2}$.

**Problem 15.31.** Find all $n \in \mathbb{N}$ and $r \in \mathbb{R}$ such that $2x^2 + 2x + 1$ divides $(x+1)^n - r$.

**Problem 15.32.** Find all triples $(a, b, c)$ of non-negative integers such that $x^a + x^b + x^c$ is divisible by $x^2 + x + 1$.

**Problem 15.33.** Find all polynomials $P \in \mathbb{R}[x]$ of odd degree such that $P(x^2 - 1) = (P(x))^2 - 1$ for every $x \in \mathbb{R}$.

**Problem 15.34.** Find all real polynomials $P$ satisfying $P(x^2) = P(x+1)P(x-1)$ for all $x$.

**Problem 15.35.** For every $n \geq 1$ determine the minimum value attained by the polynomial

$$P_n(x) = x^{2n} + 2x^{2n-1} + 3x^{2n-2} + \cdots + (2n-1)x^2 + 2nx, \qquad x \in \mathbb{R}.$$

**Problem 15.36.** Prove that for $n \in \mathbb{N}$ all roots of the polynomial $\sum_{k=0}^{n} 2^{k(n-k)}x^k$ are real.

**Problem 15.37.** Prove that there are no polynomials $f_1, f_2, f_3, f_4 \in \mathbb{Q}[x]$ such that $x^2 + 7 = (f_1(x))^2 + (f_2(x))^2 + (f_3(x))^2 + (f_4(x))^2$ for every $x \in \mathbb{R}$. ★

**Problem 15.38.** Let $P(x)$ be a polynomial with real coefficients such that $P(x) \geq 0$ for all $x \geq 0$. Prove that there exist polynomials $A(x)$ and $B(x)$ with real coefficients such that $P(x) = A(x)^2 + xB(x)^2$.

**Problem 15.39.** Let $f(x)$ be a polynomial with real coefficients such that $f(x) > 0$ for all $x \geq 0$. Prove that there exists a positive integer $n$ such that $(x+1)^n f(x)$ has non-negative coefficients.

**Problem 15.40.** Determine whether there exists a polynomial $P(x)$ with a negative coefficient such that all the coefficients of $(P(x))^n$ are positive for every integer $n > 1$.

**Problem 15.41.** Determine whether the polynomial $F(x) = 1 + x + \dfrac{x^2}{2!} + \cdots + \dfrac{x^{2017}}{2017!}$ can be factorised as $F(x) = P(x)Q(x)$ with $P$ and $Q$ non-constant polynomials with rational coefficients.

**Problem 15.42.** Determine whether there exists a function whose graph has a point of intersection with every parabola $y = ax^2 + bx + c$.

**Problem 15.43.** Let $P(x)$ and $Q(x)$ be polynomials with $P(Q(x)) = Q(P(x))$ for every $x \in \mathbb{R}$. Find as many such pairs $(P, Q)$ as you can.

**Problem 15.44.** Let $P(x)$ be a polynomial and suppose $P(x-1) = P(x+1)$ for all $|x| \leq 1$. Find $P(x)$ if $\deg P(x) \leq 3$. Can you solve the problem without the assumption $\deg P(x) \leq 3$?

**Problem 15.45.** Find a polynomial $P(x)$ such that the equation $P(x) = Q(x)$ has a real root for every polynomial $Q(x)$ of degree at most 3. Try to find several such polynomials $P(x)$.

**Problem 15.46.** Let $P(x)$ be a polynomial of degree 6 with $P(1) = P(-1)$, $P(2) = P(-2)$ and $P(3) = P(-3)$. Prove that $P(2022) = P(-2022)$.

**Problem 15.47.** Let $P(x)$ be a monic polynomial of degree 4 with $P(1) = 2$, $P(2) = 5$ and $P(3) = 10$. Prove that $P(0) + P(4)$ is a constant, and find it.

**Problem 15.48.** Find all monic polynomials of degree 3 such that $P(1)P(2)P(3) = P(5)P(6)P(7)$.

**Problem 15.49.** Find all quadratic polynomials $P(x)$ such that $P(13) = 2017$ and $x^2 + 2x \leq P(x^2 + 15x + 30) \leq x^2 + 16x$ for all $x$. †

**Problem 15.50.** Let $P(x)$ be a monic polynomial of degree 3 with three distinct roots, and let $Q(x) = x^2 + 2x + 2022$ be such that $P(Q(x))$ has no real root. Prove that $P(2022) > 1$.

**Problem 15.51.** Let $P(x) = (x^2 - 1)(x^2 - x + a) - 1$ with $a > 5$. (a) Prove that $P(x)$ has five distinct real roots $x_1, \ldots, x_5$. (b) Compute $f(x_1)f(x_2)f(x_3)f(x_4)f(x_5)$ in the two cases $f(x) = x^2 - 2022$ and $f(x) = x^2 + 1$. †

**Problem 15.52.** Let $a, b, c, d$ be real numbers such that

$$a + b + c + d > 0, \quad ab + ac + ad + bc + bd + cd > 0, \quad abc + abd + acd + bcd > 0, \quad abcd > 0.$$

Prove that $a, b, c, d$ are all positive.

**Problem 15.53.** (a) Let $P(x)$ be a monic polynomial of degree 2021 whose roots are $\frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{2022}$. Find the sum of the coefficients of all odd-degree terms of $P(x)$. (b) Find all positive integers $n$ such that, for the monic polynomial $f_n(x)$ of degree $n$ with roots $1, 2, \ldots, n$, the sum of the coefficients of all odd-degree terms of $f_n(x)$ is odd. †

**Problem 15.54.** Let $P(x) = x^4 + ax^3 + bx^2 + cx + 1$ with $a, b, c \in \mathbb{R}$ have four roots that are the side lengths of some parallelogram. Find the minimum value of $T = a + b + c$. †

**Problem 15.55.** Find all polynomials $P(x)$ of degree 3 such that $(x^3 - 2x + 1 - P(x))(2x^3 - 5x^2 + 4 - P(x)) \le 0$ for all $x \in \mathbb{R}$. ★ †

**Problem 15.56.** Let $P(x)$, $Q(x)$, $R(x)$ be non-constant real polynomials with $P(x)^2 + xQ(x)^2 = (x^2 + x - 4)R(x)^2$ for all $x$. (a) Prove that the equation $Q(x) = R(x - 3)$ has at least two distinct real roots. (b) Suppose the sum of the degrees of $P$, $Q$, $R$ is 5 and $R$ is monic. Find the minimum value of $M = P(0)^2 + 8Q(3)^2$. †

### 15.1.2 Symmetric functions, systems, and interpolation

**Problem 15.57.** Let $a, b, c$ be the roots of the cubic polynomial $P(x) = a_3x^3 + a_2x^2 + a_1x + a_0$. Prove that

$$a + b + c = -\frac{a_2}{a_3}, \qquad ab + bc + ca = \frac{a_1}{a_3}, \qquad abc = -\frac{a_0}{a_3},$$

and state the corresponding formulas for a polynomial of degree $n$.

**Problem 15.58.** Solve the system $a + b + c = 3$, $ab + bc + ca = -6$, $abc = -8$.

**Problem 15.59.** Real numbers $a, b, c$ satisfy $a + b + c = p$, $ab + bc + ca = q$, $abc = r$. Express in terms of $p, q, r$: $a^2 + b^2 + c^2$; $a^3 + b^3 + c^3$; $ab(a^2 + b^2) + bc(b^2 + c^2) + ca(a^2 + c^2)$.

**Problem 15.60.** Solve the system $x + y + z = 2$, $x^2 + y^2 + z^2 = 6$, $x^3 + y^3 + z^3 = 8$.

**Problem 15.61.** Real numbers $a_1 < a_2 < a_3$ and $b_1 < b_2 < b_3$ satisfy

$$a_1 + a_2 + a_3 = b_1 + b_2 + b_3, \quad a_1a_2 + a_1a_3 + a_2a_3 = b_1b_2 + b_1b_3 + b_2b_3, \quad a_1 < b_1.$$

Prove that $a_2 > b_2$ and $a_3 < b_3$.

**Problem 15.62.** Let $a_1, \ldots, a_{2019}$ and $b_1, \ldots, b_{2019}$ be two sets of distinct real numbers. Determine whether there exists a polynomial $P(x)$ with $P(a_i) = b_i$ for all $i \in \{1, \ldots, 2019\}$.

**Problem 15.63.** Let $f(x)$ be a cubic polynomial. Prove that for any distinct numbers $a, b, c, d$,

$$\frac{f(a)(x-b)(x-c)(x-d)}{(a-b)(a-c)(a-d)} + \frac{f(b)(x-a)(x-c)(x-d)}{(b-a)(b-c)(b-d)} + \frac{f(c)(x-a)(x-b)(x-d)}{(c-a)(c-b)(c-d)} + \frac{f(d)(x-a)(x-b)(x-c)}{(d-a)(d-b)(d-c)} =$$

**Problem 15.64.** Find a polynomial $P(x)$ with $P(1) = 2$, $P(2) = 3$, $P(3) = 4$ and $P(4) = 1$.

**Problem 15.65.** Let $P(x)$ be a polynomial such that $P(x) \in \mathbb{Z}$ for every $x \in \mathbb{Z}$. Determine whether every coefficient of $P(x)$ must be an integer.

**Problem 15.66.** Let $P(x)$ be a polynomial such that $P(x) \in \mathbb{Q}$ for every $x \in \mathbb{Q}$. Determine whether $P(x)$ can have irrational coefficients.

**Problem 15.67.** Prove that if $p(k)$ is an integer for every $k \in \mathbb{Z}$, then the coefficients $c_0, c_1, \ldots, c_n$ of $p$ in the binomial basis are all integers.

**Problem 15.68.** Find the monic polynomial $f \in \mathbb{Z}[X]$ of the smallest degree such that $f(n)$ is divisible by 100 for every integer $n$.

**Problem 15.69.** Let $m \in \mathbb{N}$ and let $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ be a polynomial with $m \mid P(n)$ for all $n \in \mathbb{Z}$. Prove that $m \mid a_n \cdot n!$.

**Problem 15.70.** Let $q_0, q_1, q_2, \ldots$ be a sequence of integers such that $i - j$ divides $q_i - q_j$ for any two distinct non-negative integers $i, j$, and suppose there is a polynomial $P(x)$ with $|q_n| < P(n)$ for all $n \geq 0$. Prove that there exists a polynomial $Q(x)$ with $Q(n) = q_n$ for all $n \geq 0$.

**Problem 15.71.** Find the smallest positive integer $n > 11$ for which there exists a polynomial $f(x)$ of degree $n$ satisfying (i) $f(k) = k^{11}$ for all $k \in \{1, 2, \ldots, n\}$; (ii) $f(0)$ is an integer; (iii) $f(-1) = 2003$.

**Problem 15.72.** Let $a_1, a_2, \ldots, a_{2002}$ be real numbers satisfying

$$\frac{a_1}{k+1} + \frac{a_2}{k+2} + \cdots + \frac{a_{2002}}{k+2002} = \frac{4}{2k+1} \qquad \text{for } k = 1, 2, \ldots, 2002.$$

Find the value of $\dfrac{a_1}{3} + \dfrac{a_2}{5} + \cdots + \dfrac{a_{2002}}{4005}$.

**Problem 15.73.** Let $a_1, \ldots, a_5$ be real numbers satisfying

$$\frac{a_1}{k^2+1} + \frac{a_2}{k^2+2} + \frac{a_3}{k^2+3} + \frac{a_4}{k^2+4} + \frac{a_5}{k^2+5} = \frac{1}{k^2} \qquad \text{for all } k \in \{1, 2, 3, 4, 5\}.$$

Find the value of $\dfrac{a_1}{37} + \dfrac{a_2}{38} + \dfrac{a_3}{39} + \dfrac{a_4}{40} + \dfrac{a_5}{41}$.

**Problem 15.74.** Let $a_1, \ldots, a_n, b_1, \ldots, b_n$ be $2n$ distinct real numbers, where $n$ is even. An $n \times n$ table is filled so that the entry in position $(i, j)$ is $a_i + b_j$. If the product of the numbers in each row equals 1, prove that the product of the numbers in each column equals $-1$.

### 15.1.3 Roots and coefficients: location and bounds

**Problem 15.75.** Let $P(x) = x^6 - 2x^5 + x^4 - 2x^3 + x^2 - 2x + 1$. Prove that exactly 4 roots of $P(x)$ have modulus equal to 1.

**Problem 15.76.** Let $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ with $a_n \neq 0$ be a polynomial with complex coefficients. Prove that the modulus of each zero of $f$ is at most

$$2 \max\left\{ \sqrt[n]{\left|\frac{a_0}{a_n}\right|},\ \sqrt[n-1]{\left|\frac{a_1}{a_n}\right|},\ \ldots,\ \sqrt{\left|\frac{a_{n-2}}{a_n}\right|},\ \left|\frac{a_{n-1}}{a_n}\right| \right\}.$$

**Problem 15.77.** Let $f(x) = a_n x^n + a_{n-r}x^{n-r} + \cdots + a_0$ with $a_n \neq 0$, $a_{n-r} \neq 0$, be a polynomial with complex coefficients. Prove that the modulus of each zero of $f$ is at most

$$\sqrt[r]{\left|\frac{a_{n-r}}{a_n}\right|} + \max\left\{ \sqrt[n-r]{\left|\frac{a_0}{a_{n-r}}\right|}, \ldots, \sqrt{\left|\frac{a_{n-r-2}}{a_{n-r}}\right|}, \left|\frac{a_{n-r-1}}{a_{n-r}}\right|, 1 \right\}.$$

**Problem 15.78.** Let $0 < a_0 \leq a_1 \leq \cdots \leq a_{n-1} \leq 1$ be real numbers. If $\alpha$ is a complex root of $P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ with $|\alpha| \geq 1$, prove that $\alpha^{n+1} = 1$.

**Problem 15.79.** For an integer $n \geq 2$ write $(1 + ix)^n = f(x) + ig(x)$, where $i$ is the imaginary unit and $f, g$ are polynomials with real coefficients. Prove that for every real $k$ the equation $f(x) + kg(x) = 0$ has only real solutions.

**Problem 15.80.** Let $f(z) = a_n z^n + a_{n-1}z^{n-1} + \cdots + a_0$ be a polynomial with real coefficients. If all roots of $f$ lie in the half-plane $\{z \in \mathbb{C} : \operatorname{Re} z < 0\}$, prove that $a_k a_{k+3} < a_{k+1}a_{k+2}$ for every $0 \le k \le n-3$.

**Problem 15.81.** Let $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ be a polynomial of even degree $n$ with real coefficients such that (i) $a_0 > 0$ and $a_n > 0$; (ii) $a_1^2 + a_2^2 + \cdots + a_{n-1}^2 \le 4\min\{a_0^2, a_n^2\}$. Prove that $f(x) \ge 0$ for every real $x$. [Romania TST 1987]

**Problem 15.82.** Find all polynomials with coefficients in $\{1, -1\}$ all of whose roots are real.

**Problem 15.83.** The polynomial $f(x) = x^n + a_1 x^{n-1} + \cdots + a_{n-1}x + 1$ has non-negative real coefficients and $n$ real roots. Prove that (a) $f(2) \ge 3^n$; (b) $f(x) \ge (x+1)^n$ for all $x \ge 0$; (c) $a_k \ge \binom{n}{k}$ for all $k = 1, 2, \ldots, n-1$.

**Problem 15.84.** Let $f_1, f_2, f_3, f_4$ be real polynomials such that the sum of any two of them has no real root. Prove that if $f_1 + f_2 + f_3 + f_4$ has a real root, then at least one of $f_1, f_2, f_3, f_4$ has no real root.

**Problem 15.85.** Suppose that $F$, $G$, $H$ are polynomials of degree at most $2n+1$ with real coefficients such that (i) $F(x) \le G(x) \le H(x)$ for all real $x$; (ii) there exist distinct real numbers $x_1, \ldots, x_n$ with $F(x_i) = H(x_i)$ for $i = 1, \ldots, n$; (iii) there exists a real number $x_0 \notin \{x_1, \ldots, x_n\}$ with $F(x_0) + H(x_0) = 2G(x_0)$. Prove that $F(x) + H(x) = 2G(x)$ for all real $x$.

**Problem 15.86.** Determine whether there exist non-constant polynomials $P(x)$ and $Q(x)$ with real coefficients satisfying $P(x)^{10} + P(x)^9 = Q(x)^{21} + Q(x)^{20}$.

**Problem 15.87.** Let $P$ and $Q$ be polynomials with real coefficients such that $P(P(x)) = \big(Q(x)\big)^2$ for every real $x$. Prove that there exists a polynomial $R(x)$ with real coefficients such that $P(x) = \big(R(x)\big)^2$. [Mszana 2019]

**Problem 15.88.** Given $n \ge 2$ distinct points $x_1, x_2, \ldots, x_n$ in $[-1, 1]$, let $\tau_k$ denote the product of the distances from $x_k$ to the other points. Prove that $\displaystyle\sum_{k=1}^{n} \frac{1}{\tau_k} \ge 2^{n-2}$.

**Problem 15.89.** Let $f$ be a real polynomial of degree $n$ with leading coefficient 1, and let $x_0 < x_1 < \cdots < x_n$ be integers. Prove that there exists $k \in \{0, 1, \ldots, n\}$ with $|f(x_k)| \ge \dfrac{n!}{2^n}$.

**Problem 15.90.** Let $a_1, a_2, \ldots, a_n$ be pairwise distinct positive integers. Prove that for every positive integer $k$ the number $\displaystyle\sum_{i=1}^{n} \frac{a_i^k}{\prod_{j \ne i}(a_i - a_j)}$ is an integer.

### 15.1.4 Polynomials with integer coefficients

**Problem 15.91.** Let $P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ be a polynomial with integer coefficients. Suppose $a$ is an integer root of $P$. Prove that $a$ divides $a_0$.

**Problem 15.92.** Let $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ be a polynomial with integer coefficients and suppose $x_0 = \frac{m}{k}$ with $\gcd(m, k) = 1$ is a rational root of $P$. Prove that $m$ divides $a_0$ and $k$ divides $a_n$.

**Problem 15.93.** Let $P(x)$ and $Q(x)$ be polynomials of degree two with integer coefficients. Prove that there exists a polynomial $R(x)$ with integer coefficients and degree at most two such that $R(8)R(12)R(2019) = P(8)P(12)P(2019)Q(8)Q(12)Q(2019)$.

**Problem 15.94.** Let $P \in \mathbb{Z}[X]$ be such that $|P(n)| = |P(m)| = 1$ for some $n, m \in \mathbb{Z}$ with $n \neq m$, and $P(a) = 0$ for some $a \in \mathbb{Q}$. Prove that $|n - m| \in \{1, 2\}$ and $a = \frac{n+m}{2}$.

**Problem 15.95.** Prove that there does not exist a non-constant polynomial $P(x)$ with integer coefficients such that $P(n)$ is prime for every $n \in \mathbb{Z}$.

**Problem 15.96.** Find all triples $(a, b, c)$ of distinct integers for which there exists $P(x) \in \mathbb{Z}[x]$ with $P(a) = b$, $P(b) = c$, $P(c) = a$.

**Problem 15.97.** Let $P(x)$ be a polynomial of degree $n > 1$ with integer coefficients and let $k \in \mathbb{N}$. Prove that the polynomial $Q(x) = P^k(x)$, the $k$-fold iterate of $P$, has at most $n$ integer fixed points.

**Problem 15.98.** Let $p$ be a prime and $f(x)$ a polynomial with integer coefficients of degree $d$ with $f(0) = 0$, $f(1) = 1$, and $f(n) \equiv 0$ or $1 \pmod{p}$ for every integer $n$. Prove that $d \geq p - 1$.

**Problem 15.99.** Let $P \in \mathbb{Z}[X]$ be a monic polynomial of degree $k$, where (a) $k = 2002$, or (b) $k = 7$. Prove that there exist positive integers $a$ and $b$ with $0 \leq a < b \leq 2002$ and $2003 \mid P(b) - P(a)$.

**Problem 15.100.** Let $f(x)$ be a monic polynomial of degree 1991 with integer coefficients. Prove that the polynomial $g(x) = f(x)^2 - 9$ has at most 1991 integer roots. [IMO Shortlist 1991]

**Problem 15.101.** Find all positive integers $k$ for which the following statement is true: if $F(x)$ is a polynomial with integer coefficients satisfying $0 \leq F(c) \leq k$ for each $c \in \{0, 1, \ldots, k + 1\}$, then $F(0) = F(1) = \cdots = F(k + 1)$. [IMO Shortlist 1997]

**Problem 15.102.** Let $P(x)$ be a polynomial with integer coefficients such that $P(n) > n$ for every positive integer $n$. Define $x_1 = 1$ and $x_n = P(x_{n-1})$ for $n \geq 2$. If for every positive integer $N$ some term of this sequence is divisible by $N$, prove that $P(x) = x + 1$. [Iran TST 2004]

**Problem 15.103.** For polynomials $A(x) = a_n x^n + \cdots + a_0$ and $B(x) = b_m x^m + \cdots + b_0$ we say that they are *similar* if $n = m$ and there is a permutation $\pi$ of $\{0, 1, \ldots, n\}$ with $b_i = a_{\pi(i)}$ for every $i$. Let $P(x)$ and $Q(x)$ be similar polynomials with integer coefficients. If $P(16) = 3^{2012}$, find the smallest possible value of $|Q(3^{2012})|$.

**Problem 15.104.** Prove that for every positive integer $n$ there exists a unique polynomial $Q(x)$ with coefficients in $\{0, 1, 2, \ldots, 9\}$ such that $Q(-2) = Q(-5) = n$. ★ [USAMO 1997]

**Problem 15.105.** Let $P(x) = a_d x^d + \cdots + a_1 x + a_0$ be a non-constant polynomial with non-negative integer coefficients having $d$ rational roots. Prove that

$$\operatorname{lcm}(P(m), P(m + 1), \ldots, P(n)) \;>\; m\binom{n}{m}$$

for all positive integers $n > m$.

**Problem 15.106.** Let $f(x)$ be a polynomial with real coefficients. Prove that there exists a constant $c$ such that for every polynomial $p(x)$ with integer coefficients the number of integer roots of $f(p(x))$ is at most $\deg p + c$.

**Problem 15.107.** Ana thinks of a polynomial $P(x)$ with coefficients in $\mathbb{N}_0$ and Banana wants to determine it. In each move Banana chooses an integer $k$ and Ana tells her the value of $P(k)$. What is the least number of moves Banana needs?

**Problem 15.108.** In each wagon of a train there is a number of coins. In each wagon all the coins are of one type, real or fake; all coins look the same, all real coins have the same weight, all fake coins have the same weight, and the weight of a real coin is known. How many weighings suffice to determine the weight of a fake coin and to identify all wagons containing fake coins?

**Problem 15.109.** A polynomial $P(x)$ with integer coefficients has the following property: whenever $F(x)$, $G(x)$, $Q(x)$ are polynomials with integer coefficients satisfying $P(Q(x)) = F(x) \cdot G(x)$, one of $F$ and $G$ is constant. Prove that $P(x)$ is a constant polynomial.

**Problem 15.110.** The sequence of integers $f_0, f_1, f_2, \ldots$ is defined by $f_0 = 0$, $f_1 = 1$ and $f_n = f_{n-1} + f_{n-2}$ for $n \geq 2$. Find all polynomials $P$ with integer coefficients such that for every natural number $n$ there is an integer $k$ with $P(k) = f_n$.

**Problem 15.111.** Let $P, Q, R$ be polynomials of degree at least 1 with integer coefficients such that $P(Q(x)) = Q(R(x)) = R(P(x))$ for every real $x$. Show that $P = Q = R$. [60th Polish MO]

**Problem 15.112.** Let $n$ be a positive integer and let $k_0, k_1, \ldots, k_{2n}$ be non-zero integers with $k_0 + k_1 + \cdots + k_{2n} \neq 0$. Is it always possible to find a permutation $(a_0, a_1, \ldots, a_{2n})$ of $(k_0, k_1, \ldots, k_{2n})$ such that the equation $a_{2n}x^{2n} + a_{2n-1}x^{2n-1} + \cdots + a_0 = 0$ has no integer root?

**Problem 15.113.** Given 2023 non-zero integers with non-zero sum, prove that there is a permutation $c_0, c_1, \ldots, c_{2022}$ of them such that $P(x) = c_{2022}x^{2022} + \cdots + c_1 x + c_0$ has no integer root.

**Problem 15.114.** Let $P(x)$ be a polynomial with integer coefficients such that $P(-1) = 29$, $P(-3) = 11$ and $P(-2) = 2022$. Prove that there is no $a \in \mathbb{Z}$ with $P(P(a)) = a^2$.

**Problem 15.115.** Let $b_1, b_2, \ldots, b_n$ be integers with sum 2022 and let $a$ be a divisor of $\dfrac{2022!}{3^{1006}}$. Prove that the polynomial $P(x) = ax^{2n} + b_1^2 x^{2n-2} + b_2^2 x^{2n-4} + \cdots + b_n^2 + a$ has no rational root. †

**Problem 15.116.** Find all monic integer polynomials $P(x)$ of degree two for which there exists an integer polynomial $Q(x)$ such that all coefficients of $P(x)Q(x)$ lie in $\{-1, 0, 1\}$.

**Problem 15.117.** Find all pairs of integers $(a, b)$ for which there exists a polynomial $P(x) \in \mathbb{Z}[X]$ such that the product $(x^2 + ax + b)P(x)$ is a polynomial each of whose coefficients equals 1 or $-1$.

**Problem 15.118.** Let $P(x)$ be an integer polynomial of degree 2022 with 2022 integer roots whose product equals 0. Find the number of integer solutions of $P(P(x)) = 0$.

**Problem 15.119.** For a positive integer $k$ suppose that there exist infinitely many monic integer polynomials of pairwise distinct degrees, all having only integer roots (not necessarily distinct), which share the same non-zero coefficients of $x^0, x^1, \ldots, x^{k-1}$. Find the largest possible value of $k$.

**Problem 15.120.** Find all non-constant integer polynomials $P(x)$ such that $P(P(n) + n)$ is prime for infinitely many positive integers $n$.

**Problem 15.121.** Find all integer polynomials $P(x)$ for which there exist infinitely many integers $a$ such that the sequence $u_0 = a$, $u_{n+1} = P(u_n)$ is periodic from the first term on.

**Problem 15.122.** Let $P(x)$ and $Q(x)$ be non-constant polynomials with non-negative integer coefficients, in which every coefficient of $P(x)$ is at most 2021 while $Q(x)$ has at least one coefficient exceeding 2021. Suppose $P(2022) = Q(2022)$ and that $P$ and $Q$ share a common rational root $\frac{p}{q}$ with $\gcd(p, q) = 1$. Prove that $|p| + n|q|$ divides $Q(n) - P(n)$ for all $n = 1, 2, \ldots, 2021$. ★ †

**Problem 15.123.** Given an integer $n \geq 2$ and a polynomial $P$ of degree $n$ with integer coefficients satisfying $P(P(k)) = P(k) + 1$ for $k = 1, 2, \ldots, n-1$, find all possible values of $P(n+1)$.

**Problem 15.124.** Let $T$ be a set of 2023 real numbers such that for every $a \in T$ the number $\Pi(T) - a$ is an odd integer, where $\Pi(T)$ is the product of all elements of $T$. Prove that $T \subseteq \mathbb{Z}$. †

**Problem 15.125.** Let $N$ be a positive integer. Consider all polynomials $x^2 + ax + b$ where $a, b$ are positive integers not exceeding $N$. Prove that the number of pairs of such polynomials having a common real root does not exceed $N^2$.

**Problem 15.126.** Given $n \in \mathbb{N}$, consider polynomials of degree $n$ of the form $P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ with non-negative integer coefficients not exceeding 100. Call such a polynomial *expandable* if it can be written as a product of two non-constant polynomials with non-negative integer coefficients, and *non-expandable* otherwise. Prove that there are at least twice as many non-expandable polynomials as expandable ones.

**Problem 15.127.** Find all pairs of positive integers $m, n \geq 3$ for which there exist infinitely many positive integers $a$ such that $\dfrac{a^m + a - 1}{a^n + a^2 - 1}$ is an integer.

**Problem 15.128.** Find all positive integers $n$ for which there are infinitely many positive integers $a$ such that $a^n + a^{n-1} + \cdots + a + 1$ divides $a^{n!} + a^{(n-1)!} + \cdots + a^{1!} + 1$.

**Problem 15.129.** Find all polynomials $P(x)$ and $Q(x)$ with integer coefficients such that the sequence $\{x_n\}_{n \geq 0}$ defined by $x_0 = 2014$, $x_{2n+1} = P(x_{2n})$, $x_{2n+2} = Q(x_{2n+1})$ for all $n \geq 0$ has the property that for every positive integer $m$ there is some $t \geq 0$ with $x_t \neq 0$ and $m \mid x_t$. [Vietnam TST 2014]

### 15.1.5 Roots of unity and counting arguments

**Problem 15.130.** Let $\varepsilon = \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$. Find all polynomials $p(x)$ with integer coefficients such that (i) $p(x) = p(\varepsilon x)$ for all $x \in \mathbb{C}$; (ii) $p(1) = 2012$ and $p(2) = 2020$.

**Problem 15.131.** How many non-zero coefficients can a polynomial $P(z)$ with integer coefficients have if $|P(z)| \leq 2$ for every complex number $z$ of unit modulus?

**Problem 15.132.** Suppose $n$ points are given on the unit circle so that the product of the distances from any point of the circle to these $n$ points is at most 2. Prove that the points are the vertices of a regular $n$-gon.

**Problem 15.133.** Multiply all the expressions of the form $\pm\sqrt{1} \pm \sqrt{2} \pm \cdots \pm \sqrt{100}$ over all $2^{100}$ choices of signs. Prove that the result is the square of an integer.

**Problem 15.134.** The sequence of polynomials $P_n(x)$ is defined by $P_0(x) = x^3 - 4x$ and $P_{n+1}(x) = P_n(1+x)P_n(1-x) - 1$ for $n \geq 0$. Prove that the polynomial $P_{2020}(x)$ is divisible by $x^{2020}$.

**Problem 15.135.** For a polynomial $p(x)$ and $n \in \mathbb{N}$ write $p^n(x) = p(p(\dots p(x)\dots))$ with $n$ applications of $p$. Prove that the polynomial $p^{2020}(x) - 2p^{2019}(x) + p^{2018}(x)$ is divisible by $p(x) - x$.

### 15.1.6 Two-variable polynomials

**Problem 15.136.** Does there exist a polynomial $P(x, y)$ such that $P(x, y) > 0$ for all $x, y \in \mathbb{R}$ and $P$ is surjective onto $\mathbb{R}^+$?

**Problem 15.137.** (a) Find all polynomials $P(x, y)$ such that $xP(z, y) + yP(x, z) = 2zP(x, y)$ for all $x, y, z$. (b) Find all polynomials $P(x, y)$ such that $P(x+y, x-y) = 2P(x, y)$ for all $x, y \in \mathbb{R}$. †

**Problem 15.138.** Let $P(x, y)$ be a polynomial with $P(0, 0) = 0$ and $P(x+2y, x+y) = -P(x, y)$ for all $x, y$. Prove that $(x^2 - 2y^2)$ divides $P(x, y)$ and that there exists a polynomial $Q$ with $P(x, y) = Q((x^2 - 2y^2)^2)$. †

**Problem 15.139.** (a) Find all polynomials $P(x, y)$ such that $P(x^2 + y^2, 2xy)$ divides $P(x, y)^2$. (b) Does there exist a polynomial $P(x, y)$ such that $P(x, y)^2 + 2021$ is divisible by $x^2 + y^2 + 2022$? †

**Problem 15.140.** Let $P(x, y)$ be a polynomial such that $P(xy, z^2+1) + P(yz, x^2+1) + P(zx, y^2+1) = 0$ for all $x, y, z$. Prove that $P \equiv 0$. ★

**Problem 15.141.** Let $P(x, y)$ be a non-constant polynomial such that $P(x, y)P(z, t) = P(xz + yt,\ xt + yz)$ for all $x, y, z, t$. Prove that $P(x, y)$ is divisible by $x - y$ or by $x + y$. †

**Problem 15.142.** Suppose that for some $m \in \mathbb{N}$ there exist real polynomials $P(x)$, $Q(x)$ and $R(x, y)$ such that for all $a, b$ with $a^m = b^2$ one has $P(R(a, b)) = a$ and $Q(R(a, b)) = b$. Prove that $m = 1$. †

**Problem 15.143.** Let $S = \{(a, b) : a, b \in \mathbb{Z},\ 1 \leq a \leq 4,\ 1 \leq b \leq 4,\ a + b \leq 5\}$ and let $f_0(x, y)$ be a non-constant integer polynomial of minimum degree with $f_0(a, b) = 0$ for all $(a, b) \in S$. Prove that $f_0\left(3, \frac{5+\sqrt{6}}{2}\right) = 0$. †

**Problem 15.144.** Prove that for every positive integer $n$ the polynomial $x^n + xy + y^n$ cannot be written as a product of two two-variable polynomials each of degree less than $n$. ★

## 15.2 Chebyshev polynomials

Among all monic polynomials of a given degree, one family is flattest on $[-1, 1]$; everything in this section flows from that single extremal property.

**Definition 15.15 (Chebyshev polynomials).** The *Chebyshev polynomials* $T_n(x)$, $n \in \mathbb{N} \cup \{0\}$, are the sequence of polynomials determined by

$$T_n(x) = \cos(n \arccos x), \qquad -1 \leq x \leq 1.$$

**Theorem 15.16 (Recurrence and basic properties).** *The Chebyshev polynomials satisfy $T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$. Consequently $T_n$ is a polynomial of the form $T_n(x) = 2^{n-1}x^n + b_1x^{n-1} + \cdots$ with integer coefficients, and $|T_n(x)| \le 1$ for all $|x| \le 1$.*

**Theorem 15.17 (The extremal property).** *Let $P(x) = x^n + a_1x^{n-1} + \cdots + a_n$ satisfy $|P(x)| \le \dfrac{1}{2^{n-1}}$ for all $|x| \le 1$. Then $P(x) = \dfrac{1}{2^{n-1}}\,T_n(x)$.*

**Theorem 15.18 (Explicit formulas).** *For every $n$,*

$$T_n(x) = \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} x^{\,n-2j}(x^2-1)^j,$$

*and outside the interval $[-1,1]$ the values of $T_n$ are given by*

$$T_n\!\left(\frac{1}{2}\Big(x + \frac{1}{x}\Big)\right) = \frac{1}{2}\Big(x^n + \frac{1}{x^n}\Big).$$

**Theorem 15.19 (Chebyshev polynomials modulo a prime).** *For every odd prime $p$ and every integer $a$ one has $T_p(a) \equiv a \pmod p$.*

**Problem 15.145.** Determine whether there exists a polynomial $P(x)$ of the form $P(x) = x^5 + a_1x^4 + \cdots + a_5$ such that $|P(x)| < \frac{1}{15}$ for all $|x| \le 1$. †

**Problem 15.146.** Find $T_0(x)$, $T_1(x)$, $T_2(x)$ and $T_3(x)$.

**Problem 15.147.** Draw the graphs of $T_0(x)$, $T_1(x)$, $T_2(x)$ and $T_3(x)$, and sketch the graph of $T_n(x)$.

**Problem 15.148.** Let $\alpha$ and $\cos(\alpha\pi)$ both be rational. Prove that $2\cos(\alpha\pi)$ is an integer, that is, $\cos(\alpha\pi)$ is one of $0, \pm\frac{1}{2}, \pm 1$.

**Problem 15.149.** If the polynomial $P(x) = a_nx^n + a_{n-1}x^{n-1} + \cdots + a_0$ satisfies $|P(x)| \le 1$ for all $|x| \le 1$, prove that $|P(x)| \le |T_n(x)|$ for all $|x| \ge 1$.

**Problem 15.150.** Let $P(x)$ be a monic polynomial. Prove that there exists $c \in [-2,2]$ with $|P(c)| \ge 2$.

**Problem 15.151.** Prove that $\cos 31^\circ$ is an irrational number.

## 15.3 Sequences II: recurrences, growth, and integrality

*Revisits linear recurrences and their closed forms (see Chapter 10); the problems below push it further.*

**Technique 15.20 (Telescoping a non-homogeneous recurrence).** To solve $x_{k+2} + px_{k+1} + qx_k = b_k$ one lowers the order of the recurrence one root at a time. Let $\alpha$ be a root of the characteristic polynomial $t^2 + pt + q$. Multiply the relation written for the index $k - i$ by $\alpha^i$ and add over $i = 0, 1, \dots, k$; the terms telescope and what is left is a first-order recurrence with a modified right-hand side. Repeating the step with the second root gives a closed form.

**Technique 15.21 (The conjugate radical).** For a number $a + c\sqrt{b}$ with $b$ not a perfect square, the number $a - c\sqrt{b}$ is called its *conjugate*. The two almost always occur together: if a sequence or an expression involves one of them, introduce the other. Their sum and product are rational, so the pair satisfies a linear recurrence with integer coefficients, and the conjugate is usually so small that it can be discarded inside a floor function.

**Lemma 15.22 (Brahmagupta–Fibonacci identity).** *For all $x, y, z, t$,*

$$(x^2 + y^2)(z^2 + t^2) = (xz + yt)^2 + (xt - yz)^2.$$

*In particular a product of two sums of two squares is again a sum of two squares.*

### 15.3.1 Closed forms, telescoping, and the conjugate trick

**Problem 15.152.** Let $a, k, m \in \mathbb{N}$ and let $b_0, b_1, \dots, b_k$ be such that $f(n) = a^n + b_0 + b_1 n + \cdots + b_k n^k$ is divisible by $m$ for all $n \in \{1, 2, \dots, k+2\}$. Prove that $f(n)$ is divisible by $m$ for all $n \in \mathbb{N}$.

**Problem 15.153.** Prove that the number $\left\lfloor (3+\sqrt{7})^{2022} \right\rfloor + 1$ is divisible by $2^{1012}$.

**Problem 15.154.** Let $m$ and $n$ be positive integers with $m$ odd. Prove that the number

$$a_m = \frac{1}{3^m n} \sum_{k=0}^{m} \binom{3m}{3k} (3n-1)^k$$

is an integer for all $m \in \mathbb{N}$. †

**Problem 15.155.** Prove that the sum

$$S_n = \binom{n}{1} + \binom{n}{3} \cdot 2021 + \binom{n}{5} \cdot 2021^2 + \cdots = \sum_{k=0}^{\lfloor (n-1)/2 \rfloor} \binom{n}{2k+1} 2021^k$$

is divisible by $2^{n-1}$ for every positive integer $n$.

**Problem 15.156.** Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that for every $n \in \mathbb{N}$

$$2n + 2001 \le f(f(n)) + f(n) \le 2n + 2002.$$

**Problem 15.157.** Let $a$ and $b$ be positive real numbers. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f(f(x)) + af(x) = b(a+b)x$ for all $x > 0$.

**Problem 15.158.** For all real numbers $x, y$ define $x \star y = \dfrac{x+y}{1+xy}$. Evaluate $(\cdots(((2 \star 3) \star 4) \star 5) \star \cdots) \star 2020$.

**Problem 15.159.** Let $a_1 = 4$ and $a_n = \dfrac{4n^2 a_{n-1} - 1}{a_{n-1} + 4n^2 - 2}$ for $n \geq 2$. Calculate $a_{2020}$ in closed form.

**Problem 15.160.** Let $(a_n)_{n\geq 0}$ be a sequence of real numbers defined by $a_0 = -1$ and $\displaystyle\sum_{k=0}^{n} \frac{a_{n-k}}{k+1} = 0$ for $n \geq 1$. Show that $a_n > 0$ for all $n \geq 1$.

**Problem 15.161.** We start with the sequence $(1, 1)$. In each step, between every pair of consecutive terms we insert their sum, so after the first step we have $(1, 2, 1)$, after the second $(1, 3, 2, 3, 1)$, and so on. For every $n \geq 1$ compute the sum of the cubes of the terms after the $n$-th step.

### 15.3.2 Integrality and divisibility in recurrences

**Problem 15.162.** Let $(a_n)_{n\geq 0}$ be the sequence with $a_0 = a_1 = 1$ and $a_{n+1} = 14a_n - a_{n-1}$ for $n \geq 1$. Prove that $2a_n - 1$ is a perfect square for every $n \geq 0$.

**Problem 15.163.** Let $(a_n)$ be defined by $a_1 = 1$, $a_2 = 4$ and $a_n = \sqrt{a_{n-1}a_{n+1} + 1}$ for all $n > 1$. (a) Prove that all terms of the sequence are positive integers. (b) Prove that $2a_n a_{n+1} + 1$ is a perfect square for every $n \in \mathbb{N}$.

**Problem 15.164.** Let $u_1 = u_2 = 1$ and $u_n = \dfrac{u_{n-1}^2 + 2}{u_{n-2}}$ for $n \geq 3$. Prove that $(u_n)$ is a sequence of integers. ★

**Problem 15.165.** Let $a, b, c \in \mathbb{Z}$ with $b$ odd, and let $(x_n)_{n\geq 0}$ be defined by $x_0 = 4$, $x_1 = 0$, $x_2 = 2c$, $x_3 = 3b$ and $x_n = ax_{n-4} + bx_{n-3} + cx_{n-2}$ for $n \geq 4$. Prove that for every prime $p$ and every positive integer $m$ the number $x_{p^m}$ is divisible by $p$.

**Problem 15.166.** Let $a \in \mathbb{N}$ and let $(a_n)_{n\geq 0}$ be defined by $a_0 = 0$ and

$$a_{n+1} = a(a_n + 1) + (a + 1)a_n + 2\sqrt{a(a + 1)a_n(a_n + 1)} \qquad \text{for } n \geq 0.$$

Prove that $a_n \in \mathbb{N}$ for all $n \geq 0$.

**Problem 15.167.** Prove that all members of the sequence $(a_n)$ defined by $a_1 = a_2 = a_3 = a_4 = a_5 = 1$ and $a_{n+6}a_{n+1} = a_{n+5}a_{n+2} + a_{n+4}a_{n+3}$ for $n \geq 1$ are positive integers.

**Problem 15.168.** Let $(a_n)$ be defined by $a_0 = 0$, $a_1 = 1$ and $a_n = 2a_{n-1} + a_{n-2}$ for $n \geq 2$. Prove that, written as an irreducible fraction, $\dfrac{a_n}{n}$ has both numerator and denominator odd. †

**Problem 15.169.** Let $(a_n)_{n\geq 1}$ be defined by $a_1 = -5$, $a_2 = -6$ and

$$a_{n+1} = a_n + (a_1 + 1)(2a_2 + 1)(3a_3 + 1) \cdots \big((n - 1)a_{n-1} + 1\big)\big((n^2 + n)a_n + 2n + 1\big)$$

for all integers $n \geq 2$. Prove that if a prime $p$ divides $na_n + 1$ for some $n \in \mathbb{N}$, then there exists an integer $m$ with $m^2 \equiv 5 \pmod{p}$.

**Problem 15.170.** Let $(a_n)_{n\geq 1}$ be defined by $a_1 = 0$ and $(n+1)^3 a_{n+1} = 2n^2(2n+1)a_n + 2(3n+1)$ for $n \geq 1$. Prove that infinitely many terms of this sequence are positive integers.

**Problem 15.171.** Prove that there does not exist a sequence $(a_n)_{n\geq 1}$ of positive integers with $a_{a_n} = a_{n+1}a_{n-1} - a_n^2$ for all $n \geq 2$. †

**Problem 15.172.** Let $(a_n)$ be a strictly increasing sequence of positive integers such that there is a constant $c$ with $u_{n+1}^2 - u_n^2 = c\,u_n$ for all $n$; find the least possible value of $c$. †

**Problem 15.173.** For a positive integer $a$ consider the sequence $(u_n)$ defined by $u_1 = a$ and $u_{n+1} = u_n + f(u_n)$ for $n \geq 1$, where $f(x)$ is the product of all digits of $x$. Prove that there exists $N$ such that $u_n = u_N$ for all $n \geq N$.

**Problem 15.174.** Let $(u_n)$ be a sequence with $u_1 = 2^{2023} - 2$ and $u_{n+1} = \varphi(u_n) - 1$ for every $n \geq 1$, where $\varphi$ is Euler's function. Prove that $u_{2^{2022}}$ is a prime greater than 3. †

**Problem 15.175.** Consider two arithmetic progressions of integers $(a_n)$ and $(b_n)$ in each of which the first term and the common difference are coprime. Suppose there are infinitely many positive integers $n$ for which one of the two products $(a_n^2 + a_{n+1}^2)(b_n^2 + b_{n+1}^2)$ and $(a_n^2 + b_{n+1}^2)(a_{n+1}^2 + b_n^2)$ is a perfect square. Prove that $a_n = b_n$ for all $n$. †

**Problem 15.176.** Let $X$ be a non-empty set of positive integers such that $x \in X$ implies $4x \in X$, and $x \in X$ implies $\lfloor \sqrt{x} \rfloor \in X$. Prove that $X = \mathbb{N}$.

**Problem 15.177.** Let $(a_n)_{n\geq 0}$ be the sequence defined by $a_0 = 1$ and $a_{n+1} = a_n + b_n - 1$ for $n \geq 0$, where $b_n$ is the smallest positive integer that does not divide $a_n$. Does there exist a positive integer $n$ with $a_n = 20!$?

### 15.3.3 Growth, bounds, and monotonicity

**Problem 15.178.** Prove that any interval $(a, b)$ contains a number of the form $\sqrt[3]{n} - \sqrt{m}$, where $m$ and $n$ are positive integers.

**Problem 15.179.** Determine whether there exist an irrational number $x > 1$ and an interval $(a, b)$ of length $\frac{1}{2019}$ such that $\{x^n\} \in (a, b)$ for all sufficiently large positive integers $n$.

**Problem 15.180.** Let $a$ be a positive integer. Prove that there exists a positive integer $n$ such that $2^n$, written in base 10, starts with exactly the same digits as $a$. For example, if $a = 102$ then $n = 10$ works, since the first three digits of $2^{10} = 1024$ are $1, 0, 2$.

**Problem 15.181.** Let $(a_i, b_i)$ be a sequence of intervals centred at the points $c_i = \frac{a_i + b_i}{2}$ such that $c_{i+1} - c_i < 2019$ and $c_i > i$ for all $i \in \mathbb{N}$. Let $M$ be the union of these intervals. Prove that there exists a real number $a$ such that $na \in M$ for infinitely many $n \in \mathbb{N}$.

**Problem 15.182.** Let $(a_n)$ be a sequence of distinct positive integers with $a_n \geq 2$ for all $n$. Prove that $a_n > n$ for infinitely many $n \in \mathbb{N}$.

**Problem 15.183.** An increasing sequence of integers $x_1 < x_2 < x_3 < \cdots$ is placed on the number line. When a wheel rolls along the number line from smaller to bigger numbers, every term $x_k$ makes a mark on the wheel as the wheel goes over it. Prove that one can choose a radius of the wheel so that every arc of one degree of the wheel will be marked by one of the numbers $x_k$.

**Problem 15.184.** Find all triples $(a, b, c)$ of real numbers such that $an^2 + bn + c$ is a perfect fourth power for every positive integer $n$.

**Problem 15.185.** Let $(a_n)$ be a sequence of natural numbers in which every positive integer appears, and such that

$$\frac{1}{1000} < \frac{|a_n - a_m|}{|n-m|} < 1000 \qquad \text{for all } n \neq m.$$

Prove that $|a_n - n| < 500\,000$ for all $n \in \mathbb{N}$.

**Problem 15.186.** The sequences $(a_n)$ and $(b_n)$ are defined by $a_0 = \frac{\sqrt{2}}{2}$, $b_0 = 1$ and

$$a_{n+1} = \frac{\sqrt{2}}{2}\sqrt{1 - \sqrt{1 - a_n^2}}, \qquad b_{n+1} = \frac{\sqrt{1 + b_n^2} - 1}{b_n}.$$

Prove that $2^{n+2}a_n < \pi < 2^{n+2}b_n$ for all $n$. †

**Problem 15.187.** Let $a > 2$ be a real number and let $(a_n)$ be defined by $a_0 = 1$, $a_1 = a$ and $a_{n+1} = \left(\frac{a_n^2}{a_{n-1}^2} - 2\right) a_n$ for $n \geq 1$. Prove that for every $k \in \mathbb{N}$,

$$\frac{1}{a_0} + \frac{1}{a_1} + \cdots + \frac{1}{a_k} < \frac{1}{2}\left(2 + a - \sqrt{a^2 - 4}\right).$$

**Problem 15.188.** The sequence $(a_n)$ is defined by $a_1 = 56$ and $a_{n+1} = a_n - \frac{1}{a_n}$ for $n \geq 1$. Prove that there exists $k$ with $1 \leq k \leq 2023$ such that $a_k < 0$.

**Problem 15.189.** Let $(a_n)$ be a bounded sequence of real numbers such that for every $n \in \mathbb{N}$,

$$a_n < \frac{1}{2n + 2007} + \sum_{k=n}^{2n+2006} \frac{a_k}{k+1}.$$

Prove that $a_n < \frac{1}{n}$ for every $n \in \mathbb{N}$.

**Problem 15.190.** A sequence of real numbers $a_1, a_2, \ldots$ satisfies $a_n = -\max_{i+j=n}(a_i + a_j)$ for all $n > 2017$. Prove that this sequence is bounded. [IMO Shortlist 2018 A4]

**Problem 15.191.** Suppose that a sequence $a_1, a_2, \ldots$ of positive real numbers satisfies $a_{k+1} \geq \frac{ka_k}{a_k^2 + k - 1}$ for every positive integer $k$. Prove that $a_1 + a_2 + \cdots + a_n \geq n$ for every $n \geq 2$.

**Problem 15.192.** Let $a_0, a_1, \ldots, a_{100}$ and $b_1, b_2, \ldots, b_{100}$ be real numbers such that for every $n \in \{0, 1, \ldots, 99\}$ either $a_{n+1} = \frac{a_n}{2}$ and $b_{n+1} = 1 - a_n$, or $a_{n+1} = 2a_n^2$ and $b_{n+1} = a_n$. If $a_{100} \leq a_0$, find the maximum value of $b_1 + b_2 + \cdots + b_{100}$.

**Problem 15.193.** Let $(a_n)_{n\geq 1}$ be a sequence of positive real numbers for which there exists a real number $c$ with $0 \leq a_i \leq c$ for all $i \geq 1$ and $|a_i - a_j| \geq \frac{1}{i+j}$ for all $i \neq j$. Prove that $c \geq 1$.

**Problem 15.194.** Let $(a_n)_{n\geq 1}$ be a sequence of positive real numbers. Show that the inequality $1 + a_n > a_{n-1}\sqrt[n]{2}$ holds for infinitely many positive integers $n$. [IMO Shortlist 2001 A2]

**Problem 15.195.** Let $(a_n)_{n\geq 0}$ be a sequence of real numbers with $a_n = |a_{n+1} - a_{n+2}|$ for all $n \geq 0$, where $a_0$ and $a_1$ are two distinct positive real numbers. Prove that the sequence is not bounded. [IMO Shortlist 2004 A2]

**Problem 15.196.** Let $(a_n)_{n\geq 1}$ be a sequence of positive real numbers with $a_{n+1}^2 + a_n a_{n+2} \leq a_n + a_{n+2}$ for all $n \geq 1$. Prove that $a_{2022} \leq 1$. [IMO Shortlist 2022 A1]

**Problem 15.197.** Let $\{a_n\}_{n\geq 0}$ be a sequence of real numbers defined by $a_{i+1} = \lfloor a_i \rfloor \cdot \{a_i\}$ for $i \geq 0$. Prove that $a_{i+2} = a_i$ for all sufficiently large $i$. [IMO Shortlist 2006 A2]

**Problem 15.198.** Let $\{a_n\}_{n\geq 0}$ and $\{b_n\}_{n\geq 0}$ be defined by $a_n = \sqrt{n+1} + \sqrt{n}$ and $b_n = \sqrt{4n+2}$ for $n \geq 0$. Prove that (a) $\lfloor a_n \rfloor = \lfloor b_n \rfloor$; (b) $0 < b_n - a_n < \dfrac{1}{16n\sqrt{n}}$.

**Problem 15.199.** Prove that there do not exist positive integers $x$ and $y$ with $\sqrt{n} + \sqrt{n+1} < \sqrt{x} + \sqrt{y} < \sqrt{4n+2}$.

**Problem 15.200.** Given any set $S$ of positive integers, show that at least one of the following holds: (1) there exist distinct finite subsets $F$ and $G$ of $S$ with $\sum_{x\in F} 1/x = \sum_{x\in G} 1/x$; (2) there exists a positive rational number $r < 1$ such that $\sum_{x\in F} 1/x \neq r$ for all finite subsets $F$ of $S$. ★

**Problem 15.201.** Let $a_0 < a_1 < a_2 < \cdots$ be an infinite sequence of positive integers. Prove that there exists a unique integer $n \geq 1$ such that $a_n < \dfrac{a_0 + a_1 + \cdots + a_n}{n} \leq a_{n+1}$.

**Problem 15.202.** ★ Let $N \geq 2$ be an integer and let $a = (a_1, \ldots, a_N)$ and $b = (b_1, \ldots, b_N)$ be sequences of non-negative integers, extended periodically by $a_i = a_k$ and $b_i = b_k$ whenever $i \equiv k \pmod N$. Say that $a$ is *$b$-harmonic* if

$$a_i = \frac{1}{2b_i + 1} \sum_{s=-b_i}^{b_i} a_{i+s} \qquad \text{for every } i.$$

Suppose that neither $a$ nor $b$ is constant, that $a$ is $b$-harmonic and that $b$ is $a$-harmonic. Prove that at least $N + 1$ of the numbers $a_1, \ldots, a_N, b_1, \ldots, b_N$ are zero.

**Problem 15.203.** Let $(u_n)$ be a strictly increasing sequence of positive integers such that among any 2022 consecutive positive integers there is at least one term of $(u_n)$. Prove that

$$\frac{1}{u_2^2 - u_1^2} + \frac{1}{u_3^2 - u_2^2} + \cdots + \frac{1}{u_{n+1}^2 - u_n^2}$$

is unbounded as $n \to \infty$.

**Problem 15.204.** ★ Let $(u_n)$ be a strictly increasing sequence of positive integers such that (i) $u_1 = 1$ and the terms are pairwise coprime; (ii) the sum $\dfrac{1}{u_1 u_2} + \dfrac{1}{u_3 u_4} + \cdots + \dfrac{1}{u_{2n-1} u_{2n}}$ is unbounded as $n \to \infty$. Prove that this sequence contains infinitely many prime numbers.

**Problem 15.205.** ★ Let $c \in \mathbb{R}^+$ and let $(x_n)$ be an infinite sequence with $0 \leq x_i \leq c$ for all $i$ and $|x_i - x_j| \geq \dfrac{1}{j}$ for all $1 \leq i < j$. Prove that $c \geq \ln 4$.

**Problem 15.206.** ★ Let $c \in \mathbb{R}^+$ and let $(x_n)$ be an infinite sequence with $0 \leq x_i \leq c$ for all $i$, such that for every $n$ the numbers $x_1, x_2, \ldots, x_n$ divide $(0, c)$ into subintervals each of length less than $\frac{1}{n}$. Prove that $c \leq \ln 2$. †

**Problem 15.207.** Prove that there does not exist a sequence $(x_n)$ of real numbers with $x_1 = 2$ and $x_{n+1} = \dfrac{x_n^2 + 2x_n + 2}{3x_n^2 + 3x_n + 2022}$ for all $n = 1, 2, 3, \ldots$ †

**Problem 15.208.** In the coordinate plane consider the sequence of points $A_n(x_n, y_n)$, where $(x_n)$ and $(y_n)$ are sequences of positive real numbers with $x_{n+1}^2 = x_n^2 + 2$ and $\sqrt{y_{n+1}} = \sqrt{y_n} + \frac{1}{2}$ for all $n \geq 1$. Suppose $O$, $A_1$, $A_{2022}$ lie on a line $d$ with $A_1 \neq A_{2022}$. Prove that all of $A_2, A_3, \ldots, A_{2021}$ lie on the same side of $d$. †

**Problem 15.209.** ★ Let $(a_n)$ be the sequence defined by $a_1 = a_2 = 1$ and $a_n = a_{n-a_{n-1}} + a_{n-a_{n-2}}$ for all $n \geq 3$. Prove that $a_{2n} \leq 2a_n$ for every $n$. †

**Problem 15.210.** Consider the sequence $(a_n)$ with $a_1, a_2 > 0$ and $a_{n+2} = |a_{n+1} - a_n|$ for $n \geq 1$. Find the conditions on $a_1, a_2$ under which this sequence contains the term 0.

**Problem 15.211.** ★ Suppose that the equation $x^3 + 2x = 1$ has the unique real root $r$. Find all strictly increasing sequences of positive integers $(a_n)$ such that $r^{a_1} + r^{a_2} + r^{a_3} + \cdots = \dfrac{1}{2}$. †

## 15.4 Functional equations II: the master class

*Revisits Cauchy's equation, injectivity and surjectivity (see Chapter 10); the problems below push it further.*

### 15.4.1 Cauchy's equation and additivity

**Theorem 15.23 (Additive functions and the conditions forcing linearity).** *Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(x+y) = f(x) + f(y)$ for all $x, y$. Then $f(qx) = qf(x)$ for every rational $q$, so $f(x) = f(1)x$ for every rational $x$, and $f(nx) = nf(x)$ for every integer $n$ and every real $x$. To conclude that $f(x) = ax$ for all real $x$ it suffices to know any* one *of the following:*

*(1) $f$ is continuous;*
*(2) $f$ is monotonic (increasing if $a > 0$, decreasing if $a < 0$);*
*(3) $f(x) \geq 0$ for all $x \geq 0$, or $f(x) \leq 0$ for all $x \geq 0$;*
*(4) $f(x^2) = f(x)^2$;*
*(5) $f$ is multiplicative, that is $f(xy) = f(x)f(y)$;*
*(6) $f(x^2) = xf(x)$;*
*(7) $f(1/x) = 1/f(x)$ for all $x \neq 0$;*
*(8) $f(x^3) = x^2 f(x)$, or $f(x^3) = xf(x^2)$;*
*(9) $f(x^n) = f(x)^n$ for some odd $n \geq 3$;*
*(10) $f(x) \geq c$ for all $x \geq 0$ and some constant $c$.*

*Being bijective is* not *enough. Two substitutions are almost always worth trying first: $(x, y) \mapsto \left(\frac{x+y}{2}, \frac{x-y}{2}\right)$ or $(x + y, 0)$, and $(x, y) \mapsto (\sqrt{xy}, \sqrt{xy})$ or $(xy, 1)$.*

**Problem 15.212.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(2022x^3 + y + f(y)) = 2y + 2022x^2 f(x)$ for all $x, y \in \mathbb{R}$. †

**Problem 15.213.** Prove that for each positive integer $n$ there exist at most three functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$2022\Big(f\big((x+y)^2\big) - f\big(x^2 + y^2\big)\Big) = x\, f^{\,n}(y) + y\, f^{\,n}(x) \qquad \text{for all } x, y \in \mathbb{R},$$

where $f^n(x) = f(f(\dots f(x)\dots))$ denotes the $n$-fold iterate. †

**Problem 15.214.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(xf(x) + 2f(y)) = x^2 + y + f(y)$ for all $x, y \in \mathbb{R}$.

**Problem 15.215.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f(3f(xy)^2 + (xy)^2) = (xf(y) + yf(x))^2$ for all $x, y > 0$.

**Problem 15.216.** Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ satisfying (i) $f(x) + f(y) + f(z) + 3xyz = 0$ whenever $x + y + z = 0$; (ii) $\max\{f(x) - f(y)\} = 2$ over all $x \geq y \geq 0$.

**Problem 15.217.** ★ Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $2f(x)f(x + y) - f(x^2) = \frac{x}{2}(f(2x) + 4f(f(y)))$ for all $x, y \in \mathbb{R}$. †

**Problem 15.218.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x^2 + y^3 + z) = f(x)^2 + f(y)^3 + f(z)$ for all $x, y, z \in \mathbb{R}$. †

**Problem 15.219.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x + y + f(y)) = 2y + f(x)$ for all $x, y \in \mathbb{R}$.

**Problem 15.220.** Find all functions $g : \mathbb{R} \to \mathbb{R}$ satisfying $g(x+y)+g(x)g(y) = g(xy)+g(x)+g(y)$ for all $x, y \in \mathbb{R}$.

**Problem 15.221.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x^3 + y^3) = (x + y)(f(x)^2 - f(x)f(y) + f(y)^2)$ for all $x, y \in \mathbb{R}$, and prove that $f(2025x) = 2025f(x)$ for all $x \in \mathbb{R}$.

**Problem 15.222.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}$ satisfying $\left(x + \frac{1}{x}\right) f(y) = f(xy) + f\left(\frac{y}{x}\right)$ for all $x, y > 0$.

**Problem 15.223.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying both
- $f(xyz) + f(x) + f(y) + f(z) = f(\sqrt{xy})f(\sqrt{yz})f(\sqrt{zx})$ for all $x, y, z > 0$;
- $f(x) < f(y)$ whenever $1 \leq x < y$.

**Problem 15.224.** Let $f : \mathbb{R} \to \mathbb{R}$ be a function with $f(x)^2 = f(f(f(x)))$ and $f(xy) = xf(y) + yf(x)$ for all $x, y$. Prove that $f(x) = 0$ for all $x$.

**Problem 15.225.** Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(x + y) = f(x) + f(y)$ for all $x, y$. Find $f$ in each of the following cases: (a) $f(x^2) = xf(x)$ and $f(1) = 2022$; (b) $f(x^2) = f(x)^2 + xf(x)$ for all $x \in \mathbb{R}$. †

**Problem 15.226.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(f(x) + y) = f(x)f(y) + f(x) + f(y) + xy$ for all $x, y \in \mathbb{R}$. †

### 15.4.2 Injectivity, surjectivity, and monotonicity

**Technique 15.24 (Injectivity at a single point).** Many equations do not force $f$ to be injective, but do force the much weaker statement that $f$ takes the value 0 (or 1, or whatever the natural distinguished value is) at most once. Establish this first: substitute $x = y = 0$ to find $f(0)$, look for an $a$ with $f(a) = 0$, and substitute $x = a$ or $y = a$. Once $f$ is known to be injective at that one point, every relation of the form $f(A) = f(B)$ that reduces to $f(\text{something}) = 0$ yields an equation, and full injectivity usually follows.

**Problem 15.227.** Prove that the following non-constant functions $f : \mathbb{R} \to \mathbb{R}$ are injective at 0: (a) $2f(x)f(x+y) - f(x^2) = \frac{x}{2}(f(2x) + 4f(f(y)))$ for all $x, y \in \mathbb{R}$; (b) $f(x + 2022f(xy)) = f(x) + 2022xf(y)$ for all $x, y \in \mathbb{R}$. †

**Problem 15.228.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x+y) \le x + f(y) + f(xy) + yf(x)$ for all $x, y \in \mathbb{R}$. †

**Problem 15.229.** Find all functions $f : [0, \infty) \to [0, \infty)$ such that (i) $f(f(x)) = x^4$ for all $x \ge 0$; (ii) there is a constant $c$ with $f(x) \le cx^2$ for all $x \ge 0$.

**Problem 15.230.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying (i) $f(2x) = 2f(x)$ for all $x > 0$; (ii) $f$ is strictly increasing on $(0, \infty)$; (iii) $f\big(f(x)f(y) + x\big) = f\big(xf(y)\big) + f(x)$ for all $x, y > 0$.

**Problem 15.231.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f\big(2xf(x) + 2f(y)\big) = 2x^2 + y + f(y)$ for all $x, y \in \mathbb{R}$.

**Problem 15.232.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ that satisfy $f(x)f\big(yf(x) - 1\big) = x^2 f(y) - f(x)$ for all $x, y \in \mathbb{R}$.

**Problem 15.233.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $xf(x^2)f(f(y)) + f(yf(x)) = f(xy)f(f(x^2)) + f(f(y^2))$ for all $x, y > 0$. [IMO Shortlist 2016 A4]

**Problem 15.234.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f(x + f(y)) = f(x+y) + f(y)$ for all $x, y > 0$. [IMO Shortlist 2007 A4]

**Problem 15.235.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f\big(f(x)f(y)\big) + f(x+y) = f(xy)$ for all $x, y \in \mathbb{R}$. [IMO 2017 P2]

**Problem 15.236.** Find all functions $f : \mathbb{Q} \to \mathbb{Q}$ such that $f\big(xf(x) + y\big) = f(y) + x^2$ for all $x, y \in \mathbb{Q}$. [EGMO 2021 P2]

**Problem 15.237.** Find all functions $f, g : \mathbb{R} \to \mathbb{R}$ such that $f(x + g(y)) = xf(y) - yf(x) + g(x)$ for all $x, y \in \mathbb{R}$.

**Problem 15.238.** Find all functions $f, g : \mathbb{R} \to \mathbb{R}$ such that (1) $f\big(xg(y+1)\big) + y = xf(y) + f\big(x + g(y)\big)$ for all $x, y \in \mathbb{R}$; (2) $f(0) + g(0) = 0$.

**Problem 15.239.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f(x) + f(y) = \big(f(f(x)) + f(f(y))\big)f(xy)$ for all $x, y > 0$, and such that only finitely many elements of the image of $f$ have more than one preimage.

**Problem 15.240.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f\big(x + f(x+y)\big) + f(xy) = x + f(x + y) + yf(x)$ for all $x, y \in \mathbb{R}$. [IMO 2015 P5]

**Problem 15.241.** Find all functions $f : [1, \infty) \to [1, \infty)$ satisfying (1) $f(x) \le 2(1+x)$ for all $x \in [1, \infty)$; (2) $xf(x+1) = f(x)^2 - 1$ for all $x \in [1, \infty)$.

**Problem 15.242.** Prove that there does not exist a function $f : \mathbb{R}^+ \to \mathbb{R}^+$ satisfying $f(x)^2 \ge f(x+y)\big(f(x) + y\big)$ for all $x, y > 0$.

**Problem 15.243.** Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that for all $x, y \in \mathbb{N}$ there is a non-degenerate triangle with side lengths $x$, $f(y)$ and $f(y + f(x) - 1)$. [IMO 2009 P2]

**Problem 15.244.** Consider functions $f : \mathbb{N} \to \mathbb{N}$ satisfying $f(m+n) \geq f(m) + f(f(n)) - 1$ for all $m, n \in \mathbb{N}$. Find all possible values of $f(2025)$.

**Problem 15.245.** For a function $f : \mathbb{N} \to \mathbb{N}$ define $f_1 = f$ and $f_{k+1} = f \circ f_k$ for $k \geq 1$; call $f$ *$k$-nice* if $f_k(n) = f(n)^k$ for all $n \in \mathbb{N}$. (a) For which $k$ is there an injective $k$-nice function? (b) For which $k$ is there a surjective $k$-nice function?

**Problem 15.246.** Find all functions $f : \mathbb{Q}^+ \to \mathbb{R}^+$ such that $f(xy) = f(x+y)\big(f(x) + f(y)\big)$ for all $x, y \in \mathbb{Q}^+$. ★

**Problem 15.247.** Find all functions $f : \mathbb{N}_0 \to \mathbb{N}_0$ such that $f\big(m + f(n)\big) = f(f(m)) + f(n)$ for all $m, n \in \mathbb{N}_0$. ★

**Problem 15.248.** ★ A function $f : \mathbb{N} \to \mathbb{N}$ satisfies $\dfrac{f(f(\ldots f(n) \ldots))}{f(f(n))} = n^2$, where $f$ is applied $f(n)$ times in the numerator. Determine all possible values of $f(1000)$.

**Problem 15.249.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that for any $a, b \in \mathbb{R}$, $f\big(a^2 + ab + f(b^2)\big) = af(b) + b^2 + f(a^2)$.

**Problem 15.250.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that for any $x \neq y$, $f\left(\dfrac{x+y}{x-y}\right) = \dfrac{f(x) + f(y)}{f(x) - f(y)}$. [Poreba 2019]

**Problem 15.251.** Construct an infinite family of functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f(x)f(f(x)) = 1$ for all $x \in \mathbb{R}^+$ and $f(x) \neq \frac{1}{x}$ for infinitely many positive reals $x$.

**Problem 15.252.** Construct a function $f : \mathbb{N} \to \mathbb{N}$ that is not strictly increasing, whose image contains every integer greater than 1000, and which satisfies $f(f(f(n))) = n + 2019$ for all $n \in \mathbb{N}$.

**Problem 15.253.** Does there exist a function $f : \mathbb{R} \to \mathbb{R}$ such that $f\big(f(x) + f(y) + x\big) = f(x) + f(y) + y$ for all $x, y \in \mathbb{R}$?

**Problem 15.254.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x)f(y) + f(x) + f(y) = f(xy) + f(x+y)$ for all $x, y \in \mathbb{R}$.

**Problem 15.255.** Find all functions $f : \mathbb{R} \to \mathbb{R}_{\geq 0}$ such that $f(2x) = f(x+y)f(y-x) + f(x-y)f(-x-y)$ for all $x, y \in \mathbb{R}$.

**Problem 15.256.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f\big(xy + f(x)\big) = f\big(f(x)f(y)\big) + x$ for all $x, y > 0$.

**Problem 15.257.** Let $f(x) = x^2 - 2$. Prove that for every positive integer $n$ the equation $f(f(\ldots f(x) \ldots)) = x$, with $n$ applications of $f$, has $2^n$ different real solutions.

**Problem 15.258.** Find all functions $f : [0, \infty) \to [0, \infty)$ satisfying $f(x) = \sqrt{1 + xf(x+1)}$ for every $x \geq 0$.

**Problem 15.259.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ with $f(-1) \neq 0$ such that $f(1+xy) - f(x+y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$. [IMO Shortlist 2012 A5]

**Problem 15.260.** Let $c \geq 1$ be an integer. Consider the graph on the vertex set $\mathbb{Z}^+$ in which an edge $\{a, b\}$ is present exactly when $a + b + c$ divides $a^2 + b^2 + c$. Prove that this graph is connected.

**Problem 15.261.** Let $f : [0, 1] \to \mathbb{R}$ satisfy $f(\frac{1}{n}) = (-1)^n$ for every integer $n \geq 1$. Prove that $f$ cannot be written as $g(x) - h(x)$ with $g, h : [0, 1] \to \mathbb{R}$ strictly increasing.

### 15.4.3 The Cauchy-difference technique

The three statements below turn a relation of the shape "shifting the argument by $g(x)$ shifts the value by $h(x)$" into rigid information about $g$ and $h$. They are the engine behind every problem in this subsection.

**Theorem 15.25 (The CDE lemma).** *Let $f, g, h : \mathbb{R}^+ \to \mathbb{R}^+$ be such that for all $x > 0$ and all $y > T(x)$,*

$$f(y + g(x)) = f(y) + h(x).$$

*Then the ratio $g(x)/h(x)$ is constant.*

**Theorem 15.26 (The CDE lemma, second form).** *Let $g : \mathbb{R}^+ \to \mathbb{R}^+$ and suppose that $g(x + a_1) + b_1 = g(x + c_1) + d_1$ for all $x$. Then the ratio $\dfrac{d_1 - b_1}{a_1 - c_1}$ is a non-negative constant: if also $g(x + a_2) + b_2 = g(x + c_2) + d_2$ for all $x$, then*

$$\frac{d_1 - b_1}{a_1 - c_1} = \frac{d_2 - b_2}{a_2 - c_2} \geq 0.$$

**Corollary 15.27 (Multiplicative form).** *Let $g : \mathbb{R}^+ \to \mathbb{R}^+$ be increasing and suppose $g(ax + b) = cg(x) + d$. Then $\dfrac{\log a}{\log c}$ is constant.*

**Problem 15.262.** Let $f, g, h : \mathbb{R}^+ \to \mathbb{R}^+$ satisfy $f(g(x) + y) = h(x) + f(y)$ for all $x, y > 0$. Prove that $\dfrac{g(x)}{h(x)}$ is constant.

**Problem 15.263.** Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that $f(x) + f(y) + f\big(f(x - f(y) + 1)\big) = 1$ for all integers $x, y$.

**Problem 15.264.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f(x^2 + f(y)) = xf(x) + y$ for all $x, y > 0$.

**Problem 15.265.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f\big(xf(x) + f(y)\big) = f(x)^2 + y$ for all $x, y > 0$.

**Problem 15.266.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f\big(2024x + f(x) + y\big) = f(2025x) + f(y)$ for all $x, y > 0$.

**Problem 15.267.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f(x^2 + y + f(y)) = 2y + f(x)^2$ for all $x, y > 0$.

**Remark.** The previous problem is the twin of a Level 3 problem in Chapter 10, in which the right-hand side is $f(x)^2$ rather than $2y + f(x)^2$; the extra term changes the answer, and the Cauchy-difference technique replaces the elementary argument that works there.

**Problem 15.268.** Find all functions $f : \mathbb{R}^+ \cup \{0\} \to \mathbb{R}$ such that the equation $f(x) = 0$ has only finitely many roots and $f(x + f(x) + 2y) = f(2x) + 2f(y)$ for all $x, y \geq 0$.

**Problem 15.269.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f(x + f(x) + y) = f(2x) + f(y)$ for all $x, y > 0$.

**Problem 15.270.** Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f(2021x + f(x) + y) = f(2022x) + f(y)$ for all $x, y > 0$.

**Problem 15.271.** Let $f : \mathbb{R}^+ \to \mathbb{R}^+$ be such that there exist positive reals $a, b, c$ with $f(x) \leq cx$ for all $x > 0$ and $f(x + a) = f(x) + b$ for all $x > 0$. Prove that $a = b$.

**Problem 15.272.** ★ Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that $f(f(x)f(f(x)) + y) = xf(x) + f(y)$ for all $x, y > 0$. †

**Problem 15.273.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that $2f(f(kxy)) = f(x+y)+xf(y)+yf(x)$ for all $x, y \in \mathbb{R}$, where $k$ is a real parameter. (a) Suppose $f(0) = 0$; find $k$. (b) Suppose $k = \frac{1}{2}$; prove that $f(f(2)) = f(2)$. †

### 15.4.4 Polynomial functional equations

**Problem 15.274.** Find all polynomials $P(x) \in \mathbb{R}[X]$ such that $P(x)P(2x^2-1) = P(x^2)P(2x-1)$ for all real $x$.

**Problem 15.275.** Find all polynomials $P(x) \in \mathbb{R}[X]$ such that $P(x^2+x+1)$ divides $P(x^3-1)$.

**Problem 15.276.** Find all polynomials $P(x)$ with real coefficients such that $P(x)P(x+1) = P(x^2+x+1)$ for every $x \in \mathbb{R}$.

**Problem 15.277.** Find all polynomials $P(x)$ with real coefficients such that $P(x)P(2x^2+1) = P(x^2)(P(2x+1) - 4x)$ for every $x \in \mathbb{R}$. [Romania TST 2001]

**Problem 15.278.** Find all polynomials $P(x)$ with real coefficients such that for all real $a, b, c$ with $ab + bc + ca = 0$, $P(a-b) + P(b-c) + P(c-a) = 2P(a+b+c)$. [IMO 2004 P2]

**Problem 15.279.** Find all nonconstant rational functions $f(x)$ and all real numbers $a$ such that $f(x)^2 - a = f(x^2)$ for all $x \in \mathbb{R}$. [Japan MO 1995]

**Problem 15.280.** If each of the polynomials $P, Q \in \mathbb{R}[X]$ has a real root and $P(1+x+Q(x)^2) = Q(1+x+P(x)^2)$ for all $x \in \mathbb{R}$, prove that $P(x) = Q(x)$ for all $x \in \mathbb{R}$.

**Problem 15.281.** Let $P \in \mathbb{R}[X]$ be a polynomial of degree 2012 such that for all $a, b, c \in \mathbb{R}$ with $a + b + c = 0$ one has $P(a)^3 + P(b)^3 + P(c)^3 > 3P(a)P(b)P(c)$. Is it possible that $P(x)$ has 2012 distinct real roots?

**Problem 15.282.** Find all polynomials $P(x)$ such that $P(2x+3) = P(2x) + x$ for all $x$. †

**Problem 15.283.** Find all polynomials $P(x)$ such that $P(P(x)) = (x^2+x+1)P(x)$ for all $x$. †

**Problem 15.284.** Find all polynomials $P(x)$ such that $(x-10)P(2x) = (8x-32)P(x-6)$ for all $x$ and $P(1) = 210$.

**Problem 15.285.** Find all polynomials $P(x)$ such that $(x^2-2x)P(x+1) = (x^2-4x+3)P(x) + 2x^2 - 2x$ for all $x$. †

**Problem 15.286.** Let $a, b \in \mathbb{R}\setminus\{0\}$ with $\frac{b}{a} \notin \mathbb{N}$. Prove that there is no non-constant polynomial $P(x)$ with $xP(x-a) = (x-b)P(x)$ for all $x$.

**Problem 15.287.** ★ For $c \in \mathbb{R}$, suppose there exists a polynomial $P(x)$ with $P(x^2) = P(x)^2 + cx^{2022}$ for all $x$. Prove that $c \le \frac{1}{4}$. †

**Problem 15.288.** Find all polynomials $P(x)$ satisfying $P(x+y) + P(y+z) + P(z+x) = 3\big(P(x) + P(y) + P(z)\big)$ for all $x, y, z$ with $x + y + z = 0$.

**Problem 15.289.** Find all polynomials $P(x)$ satisfying $(x-y)P(z)+(y-z)P(x)+(z-x)P(y) = 0$ for all $x, y, z$ with $x + y + z = 0$.

**Problem 15.290.** Given an integer $n > 1$, find all polynomials $P(x)$ satisfying $P(x)P(x^2)\cdots P(x^n) = P\big(x^{n(n+1)/2}\big)$ for all $x$.

### 15.4.5 Functional equations on point configurations

A different flavour of functional equation: the "variable" is a point of the plane, and the constraint is imposed by a geometric family of configurations rather than by an algebraic substitution.

**Problem 15.291.** Let $a_1, \ldots, a_n$ be real numbers such that $A_k = a_1^k + a_2^k + \cdots + a_n^k = 0$ for all odd positive $k \le n$. Prove that $A_k = 0$ for all odd positive $k$.

**Problem 15.292.** Let $p > 2$ be a prime number. Prove that the sum $1^{p+2}+2^{p+2}+\cdots+(p-1)^{p+2}$ is divisible by $p^2$.

**Problem 15.293.** (The main problem.) Let $B$ be a fixed finite set of points in the plane. Suppose that every point $A$ of the plane is assigned a real number $f(A)$ in such a way that for every set $C$ similar to $B$ the sum of the numbers $f(X)$ over $X \in C$ equals zero. Prove that $f(A) = 0$ for every point $A$ of the plane.

**Problem 15.294.** Solve the main problem when $B$ is a two-element set.

**Problem 15.295.** Solve the main problem when $B$ consists of three points on a straight line.

**Problem 15.296.** Solve the main problem when $B$ is a four-point set on the number line with coordinates 1, 2, 4 and 9.

**Problem 15.297.** Solve the main problem when $B$ is a four-point set on the number line with coordinates 1, 2, 4 and 5.

**Problem 15.298.** Solve the main problem when $B$ is a three-element set.

**Problem 15.299.** Solve the main problem when $B$ consists of the four vertices of a square.

**Problem 15.300.** Suppose that every point $A$ of the plane is assigned a real number $f(A)$ such that $f(M) = f(X)+f(Y)+f(Z)$ for every triangle $XYZ$ with centroid $M$. Prove that $f(A) = 0$ for all points $A$.

**Problem 15.301.** ★ Solve the main problem when $B$ is an arbitrary finite set.

**Problem 15.302.** Using the formulation of the centroid problem above and the solution of the main problem, propose a generalisation of the centroid problem and solve it.

**Problem 15.303.** Points $M_1(x_1, y_1)$, $M_2(x_2, y_2)$, …, $M_n(x_n, y_n)$ are given in the plane with $y_1, \dots, y_k > 0$ and $y_{k+1}, \dots, y_n < 0$. Points $A_1, \dots, A_{n+1}$ are given on the $x$-axis so that

$$\sum_{i=1}^{k} \angle M_i A_j B = \sum_{i=k+1}^{n} \angle M_i A_j B$$

holds for every $1 \le j \le n+1$, where $B$ is a point of the $x$-axis whose $x$-coordinate exceeds that of every $A_i$. Prove that the set $\{M_1, \dots, M_n\}$ is symmetric with respect to the $x$-axis.

## 15.5 Inequalities III: nonclassical and extremal

*Revisits the classical inequalities, the tangent-line trick and the pqr substitution (see Chapter 10); the problems below push it further.*

Karamata's inequality and Sturm's principle, stated at Level 3, are used freely below. What is new is the kind of problem: most of these are not solved by naming an inequality but by finding the right ordering, the right partition of the indices, or the right extremal configuration.

### 15.5.1 Sums, permutations, and partial sums

**Problem 15.304.** Let $x_1, x_2, \dots, x_n$ be real numbers such that for every $i \in \{1, 2, \dots, n\}$ there is a positive integer $k$ with $x_i + x_{i+1} + \cdots + x_{i+k-1} > 0$, indices taken modulo $n$. Prove that $x_1 + x_2 + \cdots + x_n > 0$.

**Problem 15.305.** Call a sequence of non-negative real numbers $a_1, a_2, \dots, a_k$ *embeddable* in the interval $[b, c]$ if there exist $x_0, x_1, \dots, x_k \in [b, c]$ with $|x_i - x_{i-1}| = a_i$ for $i = 1, \dots, k$; call the sequence *normalised* if none of its terms exceeds 1. For a given positive integer $n$ prove that (a) every normalised sequence of length $2n+1$ is embeddable in $\left[0,\, 2 - \frac{1}{2^n}\right]$; (b) there exists a normalised sequence of length $4n+3$ which is not embeddable in $\left[0,\, 2 - \frac{1}{2^n}\right]$. ★

**Problem 15.306.** Given a sequence $a_1, a_2, \dots, a_n$ of real numbers, for each $1 \le i \le n$ put $d_i = \max\{a_j : 1 \le j \le i\} - \min\{a_j : i \le j \le n\}$, and let $d = \max\{d_i : 1 \le i \le n\}$. (a) Prove that for arbitrary real numbers $x_1 \le x_2 \le \cdots \le x_n$ one has $\max\{|x_i - a_i| : 1 \le i \le n\} \ge \frac{d}{2}$. (b) Show that there exists a sequence $x_1 \le x_2 \le \cdots \le x_n$ of reals for which equality holds. [IMO 2007 P1]

**Problem 15.307.** Real numbers $x_1, \dots, x_n$ have sum zero and are not all zero. Suppose $c$ is a constant such that $x_i \le x_{i+1} + c\max\{0, x_{i+2}\}$ for every $i$, indices modulo $n$. Prove that $c \ge 2$, and that $c = 2$ is possible if and only if $n$ is even.

**Problem 15.308.** Let $n \ge 3$ be a positive integer and let $(a_1, a_2, \dots, a_n)$ be a strictly increasing sequence of $n$ positive real numbers with sum 2. Let $X$ be a subset of $\{1, 2, \dots, n\}$ for which the

value of $|1 - \sum_{i \in X} a_i|$ is minimal. Prove that there is a strictly increasing sequence $(b_1, \ldots, b_n)$ of positive reals with sum 2 such that $\sum_{i \in X} b_i = 1$. [IMO Shortlist 2019 A3]

**Problem 15.309.** Find all integers $n \geq 3$ with the following property: for all real numbers $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$ satisfying $|a_k| + |b_k| = 1$ for $1 \leq k \leq n$, there exist $x_1, \ldots, x_n$, each equal to $-1$ or 1, such that $|\sum_{k=1}^{n} x_k a_k| + |\sum_{k=1}^{n} x_k b_k| \leq 1$. [IMO Shortlist 2016 A3]

**Problem 15.310.** For a sequence $x_1, \ldots, x_n$ of real numbers define its *price* as $\max_{1 \leq i \leq n} |x_1 + \cdots + x_i|$. Given $n$ real numbers, Dave checks all possible arrangements and finds the minimum possible price $D$. George instead chooses $x_1$ with $|x_1|$ as small as possible, then among the remaining numbers chooses $x_2$ so that $|x_1 + x_2|$ is as small as possible, and so on; his sequence has price $G$. Find the least constant $c$ such that $G \leq cD$ always holds. [IMO Shortlist 2014 A3]

**Problem 15.311.** Let $x_1, \ldots, x_n$ be real numbers. Order them as $y_1, \ldots, y_n$ inductively: given $y_1, \ldots, y_{i-1}$, choose $y_i$ among the remaining numbers so that $|y_1 + \cdots + y_i|$ is minimal. Find the smallest $c$ such that one always has $\max_k |x_1 + \cdots + x_k| \geq c \cdot \max_k |y_1 + \cdots + y_k|$.

**Problem 15.312.** Let $x_1, x_2, \ldots, x_n$ be real numbers. Prove that $\displaystyle\sum_{i=1}^{n} \sum_{j=1}^{n} |x_i + x_j| \geq n \sum_{i=1}^{n} |x_i|$.

**Problem 15.313.** Let $a_1, a_2, \ldots, a_n$ be real numbers with $|a_i| \leq 1$ for $1 \leq i \leq n$ and $a_1 + a_2 + \cdots + a_n = 0$. Prove that there exists $k \in \{1, 2, \ldots, n\}$ with $|a_1 + 2a_2 + \cdots + ka_k| \leq \dfrac{2k+1}{4}$.

**Problem 15.314.** Let $x_1, x_2, \ldots, x_n$ be real numbers with $|x_i| \leq \dfrac{n+1}{2}$ for $1 \leq i \leq n$ and $|x_1 + x_2 + \cdots + x_n| = 1$. Prove that there exists a permutation $(y_1, \ldots, y_n)$ of $x_1, \ldots, x_n$ with $|y_1 + 2y_2 + \cdots + ny_n| \leq \dfrac{n+1}{2}$. ★

**Problem 15.315.** ★ Suppose that $-1 \leq x_1, x_2, \ldots, x_n \leq 1$ are real numbers with $x_1 + x_2 + \cdots + x_n = 0$. (a) Prove that there exists a permutation $\sigma$ of $\{1, 2, \ldots, n\}$ such that for every $1 \leq p \leq q \leq n$, $|x_{\sigma(p)} + \cdots + x_{\sigma(q)}| < 2 - \frac{1}{n}$. (b) Prove that the right-hand side cannot be replaced by $2 - \frac{4}{n}$.

**Problem 15.316.** Let $p, q, n$ be positive integers with $p + q < n$, and let $(x_0, x_1, \ldots, x_n)$ be an $(n+1)$-tuple of integers such that $x_0 = x_n = 0$ and, for each $1 \leq i \leq n$, either $x_i - x_{i-1} = p$ or $x_i - x_{i-1} = -q$. Prove that there exists a pair $(i, j)$ of distinct indices with $(i, j) \neq (0, n)$ such that $x_i = x_j$.

**Problem 15.317.** Let $a_0, a_1, a_2, \ldots$ be a sequence of real numbers with $a_0 = 0$, $a_1 = 1$, and such that for every $n \geq 2$ there is some $1 \leq k \leq n$ with $a_n = \dfrac{a_{n-1} + a_{n-2} + \cdots + a_{n-k}}{k}$. Find the maximum possible value of $a_{2018} - a_{2017}$. [IMO Shortlist 2018 A4]

**Problem 15.318.** Let $n$ be a fixed positive integer. Find the maximum possible value of $\displaystyle\sum_{1 \leq r < s \leq 2n} (s - r - n) x_r x_s$, where $-1 \leq x_i \leq 1$ for all $i = 1, \ldots, 2n$. [IMO Shortlist 2015 A3]

**Problem 15.319.** Let $n \geq 2$ and let $a_1, a_2, \ldots, a_n$ be real numbers with $a_1 + a_2 + \cdots + a_n = 0$. Define $A = \{(i, j) : 1 \leq i < j \leq n, |a_i - a_j| \geq 1\}$. Prove that if $A$ is non-empty then $\displaystyle\sum_{(i,j) \in A} a_i a_j < 0$. [IMO Shortlist 2019 A4]

**Problem 15.320.** Real numbers $x_1, \ldots, x_n$ are such that among the values $\dfrac{x_1 + \cdots + x_k}{k}$, $k = 1, \ldots, n$, the smallest is 0 and the largest is 1. Find the largest and the smallest possible value of $\max_{i,j}(x_i - x_j)$. †

**Problem 15.321.** Prove that any 101 real numbers $x_1, \ldots, x_{101}$ can be divided into two groups whose average values differ by at least $\dfrac{101}{200}\max_{i,j}(x_i - x_j)$.

**Problem 15.322.** Let $x_1, \ldots, x_n$ be $n$ real numbers with zero sum. At least how many pairs of indices $i < j$ satisfy $x_i + x_j \geq 0$?

**Problem 15.323.** Let $0 \leq \alpha \leq 1$ be a real number. For a 0–1 sequence $(\varepsilon_n)$ define

$$s_n = \frac{\varepsilon_1}{n(n+1)} + \frac{\varepsilon_2}{(n+1)(n+2)} + \cdots + \frac{\varepsilon_n}{(2n-1)2n}.$$

Prove that $(\varepsilon_n)$ can be chosen so that $\alpha - \dfrac{2}{n+1} \leq 2ns_n \leq \alpha$ for all $n \geq 2$.

**Problem 15.324.** For which $n$ is it possible to write $n$ distinct numbers around a circle so that every number is either greater than each of the next 100 numbers or smaller than each of the next 100 numbers, but so that this property is lost as soon as any one number is removed? †

**Problem 15.325.** If $x_1 + x_2 + \cdots + x_{2023} = 1$, find the largest possible value of $\sum x_i x_j$, where the sum runs over all pairs $i, j$ with $i > 2j$.

**Problem 15.326.** Let $x_1, x_2, \ldots, x_n$ be real numbers from the interval $[0, 2]$. Prove that $\displaystyle\sum_{i=1}^{n}\sum_{j=1}^{n} |x_i - x_j| \leq n^2$.

**Problem 15.327.** Let $a_1, a_2, \ldots, a_n$ be real numbers with $a_1 = 0$, $|a_2| = |a_1 + 1|$, $|a_3| = |a_2 + 1|$, $\ldots$, $|a_n| = |a_{n-1} + 1|$. Prove that $\dfrac{a_1 + a_2 + \cdots + a_n}{n} \geq -\dfrac{1}{2}$.

**Problem 15.328.** Let an $n \times n$ table be filled with non-zero real numbers, the entry in row $i$ and column $j$ being $a_{i,j}$; let $R_i$ be the sum of the entries in row $i$ and $C_j$ the sum of the entries in column $j$. Prove that there is a pair $(k, l)$ with either $a_{k,l} > 0$, $R_k \geq 0$, $C_l \geq 0$, or $a_{k,l} < 0$, $R_k \leq 0$, $C_l \leq 0$.

**Problem 15.329.** Let an $n \times n$ table be filled with real numbers $a_{i,j}$ such that $a_{i,i} = 0$ for all $1 \leq i \leq n$. Prove that there exists a subset $J$ of $\{1, 2, \ldots, n\}$ with

$$\sum_{i \in J,\ j \notin J} a_{i,j} + \sum_{i \notin J,\ j \in J} a_{i,j} \ \geq\ \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} a_{i,j}.$$

**Problem 15.330.** Let $x_1, \ldots, x_n$ be $n$ distinct natural numbers such that every subset of $\{x_1, \ldots, x_n\}$ has a different sum of elements, so that there are $2^n$ different sums. Prove that $x_1^2 + x_2^2 + \cdots + x_n^2 \geq \dfrac{4^n - 1}{3}$.

**Problem 15.331.** Let $x_1, x_2, \ldots, x_n$ with $n \geq 3$ be real numbers that add up to 1. Prove that there is a permutation $y_1, y_2, \ldots, y_n$ of these numbers such that $y_1y_2 + y_2y_3 + \cdots + y_ny_1 \leq \dfrac{1}{n}$.
[Mathematical Miniatures]

**Problem 15.332.** ★ Let $x_1, \ldots, x_n$ be real numbers with $x_1^2 + x_2^2 + \cdots + x_n^2 = 1$. Prove that for every integer $k \geq 2$ there are integers $a_1, \ldots, a_n$, not all zero, with $|a_i| \leq k-1$ for all $i$ and

$$|a_1 x_1 + a_2 x_2 + \cdots + a_n x_n| \leq \frac{(k-1)\sqrt{n}}{k^n - 1}.$$

**Problem 15.333.** Let $n \geq 2$ be an integer and let $0 < a_1 < a_2 < \cdots < a_{2n+1}$ be real numbers. Prove that

$$\sqrt[n]{a_1} - \sqrt[n]{a_2} + \sqrt[n]{a_3} - \cdots + \sqrt[n]{a_{2n+1}} < \sqrt[n]{a_1 - a_2 + a_3 - \cdots + a_{2n+1}}.$$

### 15.5.2 Extremal constants and optimisation

**Problem 15.334.** Find the least value of $k$ such that $k(a^2+1)(b^2+1)(c^2+1) \geq (a+b+c)^2$ for all real numbers $a, b, c$.

**Problem 15.335.** (a) Find the least $k \geq 0$ such that $(x+k)(y+k)(z+k) \geq (1+k)^3$ for all $x, y, z \geq 0$ with $xy + yz + zx = 3$. (b) Find the least $k \geq 0$ such that

$$\frac{1}{(a+k)(b+k)} + \frac{1}{(b+k)(c+k)} + \frac{1}{(c+k)(a+k)} \geq \frac{3}{(1+k)^2}$$

for all $a, b, c \geq 0$ with $a + b + c = ab + bc + ca$.

**Problem 15.336.** (a) Find the least value of $k$ such that $\sigma(n)\varphi(n) \leq kn^2$ for every positive integer $n > 1$. (b) Find the least value of $c$ such that $\{n\sqrt{17}\} \geq \dfrac{1}{cn}$ for every $n \in \mathbb{N}$.

**Problem 15.337.** Find the maximum integer $k$ such that

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{a} + k \cdot \frac{(a-b)(b-c)(c-a)}{(a+b)(b+c)(c+a)} \geq 3$$

for all positive real numbers $a, b, c$.

**Problem 15.338.** For $a, b, c > 0$ put $m = \dfrac{a}{b} + \dfrac{b}{c} + \dfrac{c}{a}$ and $n = \dfrac{a}{c} + \dfrac{c}{b} + \dfrac{b}{a}$. Find all values of $k$ for which $(k^2 + 2k - 8)m + (6 - k - k^2)n + 6 - 3k \leq 0$ holds for all $m, n$ arising in this way. †

**Problem 15.339.** ★ Find the least $k \geq 0$ such that $|x + ky| + |y + kz| + |z + kx| \geq 7$ for all real numbers $x, y, z$ with $x^2 + y^2 + z^2 = 21$. †

**Problem 15.340.** Find the maximum value of $c$ such that for every positive integer $n$ and every sequence $0 = x_0 < x_1 < x_2 < \cdots < x_n = 1$ one has $\displaystyle\sum_{i=1}^{n} x_i^3 \,(x_i - x_{i-1}) \geq c$. †

**Problem 15.341.** For an integer $n \geq 3$ let $a_1 \leq a_2 \leq \cdots \leq a_n$ be real numbers with sum 0. Prove that $a_1^2 + a_2^2 + \cdots + a_n^2 + n a_1 a_n \leq 0$.

**Problem 15.342.** Consider the positive integers $n$ for which there exist real numbers $x_1, x_2, \ldots, x_n$ with $x_i \geq i^2$ for $i = 1, 2, \ldots, n$ and

$$\frac{x_1^2 + x_2^2 + \cdots + x_n^2}{(x_1 + 2x_2 + 3x_3 + \cdots + nx_n)^2} = \frac{27}{4n(n+1)(2n+1)}.$$

(a) Prove that $n \equiv 0, 4, 8 \pmod 9$. (b) Find all such $n$. †

**Problem 15.343.** Let $x_1, x_2, \ldots, x_{2022}$ be real numbers with sum 0 and $|x_1|+|x_2|+\cdots+|x_{2022}| = 2022$. Find the maximum and the minimum value of $T = x_1x_2\cdots x_{2022}$. †

**Problem 15.344.** ★ For a positive integer $n$, consider $n$ red balls and $n$ green balls arranged on a line. For the ball $X$ in position $i$ let $x_i$ be the number of pairs of balls lying on the two sides of $X$ whose colour differs from the colour of $X$. Find the maximum value of $T = x_1 + x_2 + \cdots + x_{2n}$.

**Problem 15.345.** For an integer $n \in [2, 21]$ consider non-negative real numbers $x_1, \ldots, x_n$ with sum $\frac{7}{4}$. Find the maximum value of $S = 5(x_1^2 + x_2^2 + \cdots + x_n^2) - 2(x_1^3 + x_2^3 + \cdots + x_n^3)$.

**Problem 15.346.** For an integer $n \geq 3$ consider $4n$ non-negative real numbers $x_1, \ldots, x_{2n}$ and $y_1, \ldots, y_{2n}$ such that (i) $S = x_1 + \cdots + x_{2n} = y_1 + \cdots + y_{2n} > 0$; (ii) $x_ix_{i+1} \geq y_i + y_{i+1}$ for all $1 \leq i \leq 2n$, indices taken cyclically. (a) Prove that $S \geq 12$ when $n = 3$. (b) For $n = 4$, find the minimum value of $S$. †

**Problem 15.347.** For an integer $n \geq 3$ and numbers $a_1, a_2, \ldots, a_n$ not all equal to 0, prove that

$$a_1^2 + a_2^2 + \cdots + a_n^2 \;\geq\; \frac{n^3 - n}{12} \min_{1 \leq i < j \leq n} (a_i - a_j)^2.$$

†

**Problem 15.348.** ★ Prove that there exist 100 positive real numbers $a_1, a_2, \ldots, a_{100}$ such that (i) $a_ia_{i+1} = 2i + 1$ for all $i = 1, 2, \ldots, 99$; (ii) $a_ia_j \leq i + j$ for all $i \neq j$.

**Problem 15.349.** Find the smallest real constant $C$ with the following property: for any positive real numbers $a_1, a_2, a_3, a_4, a_5$, not necessarily distinct, one can find distinct subscripts $i, j, k, l$ such that $\left|\frac{a_i}{a_j} - \frac{a_k}{a_l}\right| \leq C$. [IMO Shortlist 2016 A2]

**Problem 15.350.** Find the smallest constant $c$ with the following property: for every $n \in \mathbb{N}$, all positive numbers $x_1, \ldots, x_n$ and $y_k = \frac{x_1 + x_2 + \cdots + x_k}{k}$, one has $y_1^2 + y_2^2 + \cdots + y_n^2 \leq c(x_1^2 + x_2^2 + \cdots + x_n^2)$.

**Problem 15.351.** Find the smallest positive integer $n$ for which there exist integers $a_1, a_2, \ldots, a_n$ such that the quadratic trinomial $x^2 - 2(a_1 + a_2 + \cdots + a_n)^2x + (a_1^4 + a_2^4 + \cdots + a_n^4 + 1)$ has at least one integer root.

**Problem 15.352.** Find the least positive real $a$ such that whenever $x, y, z$ are the side lengths of a triangle in which the angle opposite $z$ is greater than 120°, one has $x^3y^5 < az^8$. †

### 15.5.3 Classical shapes with a twist

**Problem 15.353.** Let $a, b, c, d$ be positive real numbers with $a, c > 1$ and $b, d < 1$. Prove that

$$\frac{a}{ab + c + 1} + \frac{b}{bc + d + 1} + \frac{c}{cd + a + 1} + \frac{d}{da + b + 1} > 1.$$

**Problem 15.354.** Let $a_1, a_2, \ldots, a_n$ be positive real numbers such that $a_1^2 + 2a_2^{3/2} + \cdots + na_n^{(n+1)/n} < 1$. Prove that $2a_1 + 3a_2 + \cdots + (n + 1)a_n < 3$.

**Problem 15.355.** Let $(a_0, a_1, a_2, \dots)$ and $(b_0, b_1, b_2, \dots)$ be sequences of non-negative real numbers such that $a_i^2 \le a_{i-1}a_{i+1}$ and $b_i^2 \le b_{i-1}b_{i+1}$ for every $i \ge 1$. Define $c_0 = a_0b_0$ and $c_n = \sum_{i=0}^{n} \binom{n}{i} a_i b_{n-i}$ for $n \ge 1$. Prove that $c_k^2 \le c_{k-1}c_{k+1}$ for every $k \ge 1$. ★

**Problem 15.356.** Let $x_1, x_2, \dots, x_n$ with $n \ge 2$ be real numbers greater than 1 with $|x_i - x_{i+1}| < 1$ for $i = 1, 2, \dots, n-1$. Prove that

$$\frac{x_1}{x_2} + \frac{x_2}{x_3} + \cdots + \frac{x_{n-1}}{x_n} + \frac{x_n}{x_1} < 2n - 1.$$

**Problem 15.357.** Suppose that $a, b, c$ are real numbers in $[-1, 1]$ with $1 + 2abc \ge a^2 + b^2 + c^2$. Prove that $1 + 2(abc)^n \ge a^{2n} + b^{2n} + c^{2n}$ for all positive integers $n$. [IMC 2010]

**Problem 15.358.** The positive real numbers $a_1, \dots, a_n$ and $k$ satisfy $a_1 + \cdots + a_n = 3k$, $a_1^2 + \cdots + a_n^2 = 3k^2$, $a_1^3 + \cdots + a_n^3 > 3k^3 + k$. Prove that the difference between some two of $a_1, \dots, a_n$ is greater than 1.

**Problem 15.359.** Let $x_1, x_2, \dots, x_n$ be positive real numbers whose product is 1. Prove that

$$\frac{1}{n - 1 + x_1} + \frac{1}{n - 1 + x_2} + \cdots + \frac{1}{n - 1 + x_n} \le 1.$$

**Problem 15.360.** Let $a_1, a_2, \dots, a_n$ be real numbers and let $S$ be a subset of $\{1, 2, \dots, n\}$. Prove that

$$\left(\sum_{i \in S} a_i\right)^2 \le \sum_{1 \le i \le j \le n} (a_i + a_{i+1} + \cdots + a_j)^2.$$

**Problem 15.361.** Let $a_1, a_2, \dots, a_n$, $k$ and $M$ be positive integers with $\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n} = k$ and $a_1a_2 \cdots a_n = M$. If $M > 1$, prove that the polynomial $P(x) = M(x+1)^k - (x + a_1)(x + a_2) \cdots (x + a_n)$ has no positive root. [IMO Shortlist 2017 A1]

**Problem 15.362.** Let $\lambda_1 \le \lambda_2 \le \lambda_3 \le \cdots$ be positive real numbers and let $K$ be a constant with $\sum_{k=1}^{n-1} \lambda_k^2 < K\lambda_n^2$ for all $n > 1$. Prove that there exists a constant $K'$ such that $\sum_{k=1}^{n-1} \lambda_k < K'\lambda_n$ for all $n > 1$.

**Problem 15.363.** For a real number $x$ let $\|x\|$ denote the distance from $x$ to the nearest integer. Let $0 \le x_n < 1$ for $n \ge 1$ and let $\varepsilon > 0$. Show that there exist infinitely many pairs $(n, m)$ of indices with $n \ne m$ and $\|x_n - x_m\| < \min\left\{\varepsilon, \frac{1}{2|n-m|}\right\}$. ★

**Problem 15.364.** Let $a$ and $b$ be positive integers such that $q = \sqrt{a} + \sqrt{b}$ is not an integer. Prove that $2q^3\{q\} > 3$.

**Problem 15.365.** Let $\lambda$ be a real number such that the inequality $0 < \left|\sqrt{2002} - \frac{a}{b}\right| < \frac{\lambda}{ab}$ holds for infinitely many pairs $(a, b)$ of positive integers. Prove that $\lambda \ge 5$.

**Problem 15.366.** Prove that for every positive integer $n$ one has $\{n\sqrt{2}\} > \frac{1}{2n\sqrt{2}}$.

**Problem 15.367.** Prove that for every positive integer $n$ one has $\{n\sqrt{7}\} > \dfrac{3}{2n\sqrt{7}}$.

**Problem 15.368.** Prove that among any 20 consecutive positive integers there is an integer $d$ such that for every positive integer $n$, $\{n\sqrt{d}\} > \dfrac{5}{2n\sqrt{d}}$. [Balkan MO 2015 P4]

**Problem 15.369.** Let $a, b, c$ be non-negative real numbers with $a + b + c = 1$. Prove that $a\sqrt{2b+1} + b\sqrt{2c+1} + c\sqrt{2a+1} \le \sqrt{2 - (a^2 + b^2 + c^2)}$, and determine when equality holds.

**Problem 15.370.** Let $a, b, c$ be positive real numbers. Prove that

$$abc(a^2 + 2b^2)(b^2 + 2c^2)(c^2 + 2a^2) \le (a + b + c)^3(a^2 - ab + b^2)(a^2 - ac + c^2)(b^2 - bc + c^2).$$

★

**Problem 15.371.** Let $x, y, z$ be positive real numbers with $xyz = 1$. Prove that

$$\frac{x^3}{(1+y)(1+z)} + \frac{y^3}{(1+z)(1+x)} + \frac{z^3}{(1+x)(1+y)} \ge \frac{3}{4}.$$

★

**Problem 15.372.** Let $x_1, x_2, \ldots, x_n$ be positive real numbers with $\dfrac{1}{1+x_1} + \dfrac{1}{1+x_2} + \cdots + \dfrac{1}{1+x_n} = 1$. Show that $x_1 x_2 \cdots x_n \ge (n-1)^n$. ★

**Problem 15.373.** Let $a, b, c, d$ be positive real numbers. Prove that

$$\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}\right)^2 \ge \frac{1}{a^2} + \frac{4}{a^2+b^2} + \frac{9}{a^2+b^2+c^2} + \frac{16}{a^2+b^2+c^2+d^2}.$$

†

**Problem 15.374.** Let $a, b, c, d$ be positive real numbers with $a + b + c + d = 4$. Prove that

(a) $\dfrac{1+a}{1+b^2} + \dfrac{1+b}{1+c^2} + \dfrac{1+c}{1+d^2} + \dfrac{1+d}{1+a^2} \ge 4$;

(b) $\dfrac{1}{1+a^2} + \dfrac{1}{1+b^2} + \dfrac{1}{1+c^2} + \dfrac{1}{1+d^2} \ge 2$;

(c) $\dfrac{a}{1+b^2c} + \dfrac{b}{1+c^2d} + \dfrac{c}{1+d^2a} + \dfrac{d}{1+a^2b} \ge 2$. ★

**Problem 15.375.** Let $x_1, \ldots, x_n$ be non-negative real numbers with $x_1 + \cdots + x_n = 1$. Find the smallest possible value of $-\dfrac{x_1^2}{2} + \displaystyle\sum_{i=1}^{n} x_i^2 - \sum_{i=1}^{n-1} x_i x_{i+1}$, and determine when it is attained.

**Problem 15.376.** Let $a, b, c$ be positive real numbers. Prove that

$$\frac{a^2 - bc}{2a^2 + b^2 + c^2} + \frac{b^2 - ac}{2b^2 + a^2 + c^2} + \frac{c^2 - ab}{2c^2 + a^2 + b^2} \ge 0.$$

**Problem 15.377.** Let $x, y, z \ge 1$ be real numbers with $\dfrac{1}{x} + \dfrac{1}{y} + \dfrac{1}{z} = 2$. Prove that $\sqrt{x+y+z} \ge \sqrt{x-1} + \sqrt{y-1} + \sqrt{z-1}$. ★

**Problem 15.378.** Let $a, b, c, d, e, f$ be positive real numbers. Prove that

$$\frac{a}{b+c}+\frac{b}{c+d}+\frac{c}{d+e}+\frac{d}{e+f}+\frac{e}{f+a}+\frac{f}{a+b} \geq 3.$$

★

**Problem 15.379.** Let $a, b, c$ be positive real numbers with $a+b+c=1$. Prove that

$$\frac{1+a}{1-a}+\frac{1+b}{1-b}+\frac{1+c}{1-c} \leq \frac{2a}{b}+\frac{2b}{c}+\frac{2c}{a}.$$

★

**Problem 15.380.** Let $a, b, c, d$ be non-negative real numbers with $a+b+c+d=4$. Prove that $a^2bc+b^2cd+c^2da+d^2ab \leq 4$, and find the cases of equality.

**Problem 15.381.** Let $a, b, c, d > 0$ with $a+b+c+d=4$. Show that $\frac{1}{ab}+\frac{1}{bc}+\frac{1}{cd}+\frac{1}{da} \geq a^2+b^2+c^2+d^2$.

**Problem 15.382.** Let $0 \leq x_1 \leq x_2 \leq \cdots \leq x_n \leq 1$ with $n \geq 2$. Prove that

$$\sqrt[n]{x_1x_2\cdots x_n}+\sqrt[n]{(1-x_1)(1-x_2)\cdots(1-x_n)} \leq \sqrt[n]{1-(x_1-x_n)^2}.$$

**Problem 15.383.** The positive reals $a_1, a_2, \ldots, a_n$ satisfy $a_1^2+a_2^2+\cdots+a_n^2 = a_1+a_2+\cdots+a_n$. Prove that $\sqrt[4]{a_1^4+1}+\sqrt[4]{a_2^4+1}+\cdots+\sqrt[4]{a_n^4+1} \leq n\sqrt[4]{2}$. †

**Problem 15.384.** Let $x_1, x_2, \ldots, x_n$ be non-negative reals with sum equal to 1. Find the minimal possible value of

$$\frac{1}{\sqrt{1+x_1}}+\frac{1}{\sqrt{1+2x_2}}+\cdots+\frac{1}{\sqrt{1+nx_n}}.$$

†

**Problem 15.385.** For real numbers $a, b, c$ not less than 1, prove that

$$\frac{a+b+c}{4} \geq \frac{\sqrt{ab-1}}{b+c}+\frac{\sqrt{bc-1}}{c+a}+\frac{\sqrt{ca-1}}{a+b}.$$

†

**Problem 15.386.** Let $ABC$ be a non-degenerate triangle. Show that $3\sqrt{3}\tan\frac{A}{2}\tan\frac{B}{2}\tan\frac{C}{2} \leq 1$.

**Problem 15.387.** Let $ABC$ be a triangle. Show that $\sin\frac{A}{2}+\sin\frac{B}{2}+\sin\frac{C}{2} \geq \cos A+\cos B+\cos C$.

## 15.6 Mixed master problems

The last section collects the problems that refuse to sit inside a single theme: irrationality arguments, identities that hide a polynomial, counting arguments that hide a vector space, and a handful of genuinely hard problems from the camps.

### 15.6.1 Rational and irrational numbers

**Problem 15.388.** Determine whether the number $\sqrt[3]{\sqrt{5}+2}-\sqrt[3]{\sqrt{5}-2}$ is rational or irrational.

**Problem 15.389.** Let $a_n = \dfrac{1}{10^n}$. Cross out all the terms $a_n$ for which $n$ is a perfect square. Prove that the sum of the remaining terms is an irrational number.

**Problem 15.390.** What are the possible values of $a \in \mathbb{R}$ for which

$$a\left(\left(\frac{1+\sqrt{17}}{2}\right)^n - \left(\frac{1-\sqrt{17}}{2}\right)^n\right)$$

is an integer for all $n \in \mathbb{N}$?

**Problem 15.391.** Let $x, y, z$ be real numbers such that $x+yz$, $y+zx$, $z+xy$ are all rational and $x^2+y^2=1$. Prove that $xyz^2$ is also rational.

**Problem 15.392.** Find the first 1000 digits after the decimal point of the number $(2+\sqrt{3})^{2019}$.

**Problem 15.393.** Determine whether there exist rational numbers $a, b, c, d$ such that $(a+b\sqrt{3})^4 + (c+d\sqrt{3})^4 = 1+\sqrt{3}$.

**Problem 15.394.** Let $x_1 = 0$ and $x_{n+1} = 5x_n + \sqrt{24x_n^2+1}$ for all $n \in \mathbb{N}$. Determine whether a term $x_n$ is rational or irrational.

**Problem 15.395.** Can every positive rational number be expressed as $\dfrac{a^2+b^3}{c^5+d^7}$, where $a, b, c, d$ are positive integers?

**Problem 15.396.** ★ Let $a_1 < a_2 < a_3 < \cdots$ be the sequence consisting of all positive integers of the form $x^y$ with $x, y \geq 2$ integers. Prove that $\displaystyle\sum_{i=1}^{\infty} \frac{1}{a_i - 1} = 1$.

### 15.6.2 Identities, manipulation, and quadratics

**Problem 15.397.** Prove that $\dfrac{a^2}{a+b} + \dfrac{b^2}{b+c} + \dfrac{c^2}{c+a} = \dfrac{b^2}{a+b} + \dfrac{c^2}{b+c} + \dfrac{a^2}{c+a}$.

**Problem 15.398.** Factorise $a^3+b^3+c^3-3abc$.

**Problem 15.399.** Factorise $(a^2-b^2)^3 + (b^2-c^2)^3 + (c^2-a^2)^3$.

**Problem 15.400.** Factorise $(a^2b+b^2c+c^2a) - (b^2a+c^2b+a^2c)$.

**Problem 15.401.** (a) Suppose that $a, b, c$ are real numbers with $\dfrac{1}{a}+\dfrac{1}{b}+\dfrac{1}{c} = \dfrac{1}{a+b+c}$. Prove that for every odd positive integer $n$, $\dfrac{1}{a^n}+\dfrac{1}{b^n}+\dfrac{1}{c^n} = \dfrac{1}{a^n+b^n+c^n}$. (b) Suppose that real numbers $a, b, c$ satisfy $\sqrt[3]{a-b}+\sqrt[3]{b-c}+\sqrt[3]{c-a} = 0$. Prove that at least two of $a, b, c$ are equal. (c) Real numbers $a, b$ satisfy $a^3+b^3 = 3ab-1$. Find all possible values of $a+b$. (d) Factor $(x+y+z)^3 - x^3 - y^3 - z^3$. (e) Factor $(x+y+z)^5 - x^5 - y^5 - z^5$.

**Problem 15.402.** Solve the equation $2x^4 + 2y^4 = 4xy - 1$.

**Problem 15.403.** Compute $\dfrac{1}{\sqrt{1}+\sqrt{2}} + \dfrac{1}{\sqrt{2}+\sqrt{3}} + \dfrac{1}{\sqrt{3}+\sqrt{4}} + \cdots + \dfrac{1}{\sqrt{99}+\sqrt{100}}$.

**Problem 15.404.** Compute $\dfrac{1}{1\cdot 3} + \dfrac{1}{3\cdot 5} + \dfrac{1}{5\cdot 7} + \cdots + \dfrac{1}{99\cdot 101}$.

**Problem 15.405.** Compute $\dfrac{1}{1\cdot 2\cdot 3} + \dfrac{1}{2\cdot 3\cdot 4} + \cdots + \dfrac{1}{99\cdot 100\cdot 101}$.

**Problem 15.406.** Simplify the product $(1+2^{2^0})(1+2^{2^1})(1+2^{2^2})\cdots(1+2^{2^{100}})$.

**Problem 15.407.** Compute $1\cdot 1! + 2\cdot 2! + 3\cdot 3! + \cdots + n\cdot n!$.

**Problem 15.408.** Simplify $\sqrt{11+6\sqrt{2}} + \sqrt{11-6\sqrt{2}}$.

**Problem 15.409.** Simplify $\sqrt{7+4\sqrt{3}} + \sqrt{7-4\sqrt{3}}$.

**Problem 15.410.** Simplify $\sqrt[3]{20+14\sqrt{2}} + \sqrt[3]{20-14\sqrt{2}}$.

**Problem 15.411.** Let $(u_n)$ be the sequence with $u_1 = u_2 = 1$ and $u_{n+2} = u_{n+1} + u_n$ for every $n \in \mathbb{N}$. Find the sum $u_1 + u_2 + u_3 + \cdots + u_n$.

**Problem 15.412.** Solve the following equations: $x^2 + 2020x + 2019 = 0$; $x(x-1)(x+3)(x-4) + 36 = 0$; $x^4 + 2x^3 - 3x^2 + 2x + 1 = 0$; $\dfrac{5x}{x^2-3x+1} - \dfrac{2x}{x^2+x+1} = 1$; $x^2 + \left(\dfrac{x}{x-1}\right)^2 = 8$; $x^4 - x + \dfrac{1}{2} = 0$. †

**Problem 15.413.** What is the minimum value of (a) $(x-1)^2 + \left(x-\frac{1}{2}\right)^2 + \left(x-\frac{1}{3}\right)^2$; (b) $(x-5)(x-1)(x-6)(x-2)+9$; (c) $\dfrac{2x^2+9x+11}{3x^2+11x+12}$? †

**Problem 15.414.** Prove that $b^2 > 4ac$ if $(a+b+c)c < 0$.

**Problem 15.415.** Find 100 quadratic functions $f_i(x) = a_i x^2 + b_i x + c_i$, $1 \le i \le 100$, such that every equation $f_i(x) = 0$ has two roots, but no equation $f_i(x) + f_j(x) = 0$ with $1 \le i < j \le 100$ has a root.

**Problem 15.416.** Solve the system

$$3\left(x+\frac{1}{x}\right) = 4\left(y+\frac{1}{y}\right) = 5\left(z+\frac{1}{z}\right), \qquad xy+yz+zx = 1.$$

[IMC 2010]

**Problem 15.417.** Solve the system $\dfrac{3x-y}{x-3y} = x^2$, $\dfrac{3y-z}{y-3z} = y^2$, $\dfrac{3z-x}{z-3x} = z^2$.

**Problem 15.418.** Find all ordered triples of pairwise distinct real numbers $(x, y, z)$ such that

$$\{x, y, z\} = \left\{\frac{x-y}{y-z}, \frac{y-z}{z-x}, \frac{z-x}{x-y}\right\}.$$

**Problem 15.419.** Nonzero real numbers $x, y, z$ satisfy $x + \dfrac{y}{z} = y + \dfrac{z}{x} = z + \dfrac{x}{y} = 2$. Find all possible values of $x + y + z$. ★

**Problem 15.420.** Prove that if $a_1, a_2, \ldots, a_n$ are integers such that $a_1 a_2 \cdots a_n \left(\dfrac{1}{a_1^2} + \dfrac{1}{a_2^2} + \cdots + \dfrac{1}{a_n^2}\right)$ is an integer, then $a_k^2$ divides the product $a_1 a_2 \cdots a_n$ for every $k = 1, 2, \ldots, n$. †

### 15.6.3 The linear-algebra method

The problems in this block all come from one idea: encode the objects as vectors, and use the fact that more than $\dim V$ vectors of a space $V$ must be linearly dependent.

**Problem 15.421.** Find the sum $100^2 \cdot 3^{100} + 99^2 \cdot 3^{99} + 98^2 \cdot 3^{98} + \cdots + 1^2 \cdot 3$.

**Problem 15.422.** Let $r$ be a positive real number. For every $(i, j) \in \mathbb{Z} \times \mathbb{Z}$ with $(i, j) \neq (0, 0)$ draw a circle of radius $r$ centred at the point $O(i, j)$ of the coordinate plane. Prove that any line through the origin meets at least one of the circles.

**Problem 15.423.** There are 17 boxes with coins inside; different boxes may contain different numbers of coins. You may choose 11 boxes and put one coin into each of them, and you may repeat this operation as often as you like. Is it always possible to make the numbers of coins in all boxes equal?

**Problem 15.424.** An $8 \times 8$ chessboard has a real number in each of its squares. You may pick a number $a \in \mathbb{R}$ and add $a$ to all the numbers in any $3 \times 3$ or $4 \times 4$ square, and you may repeat this with a different $a$ and a different square. Is it always possible to make all the numbers on the board equal to 0?

**Problem 15.425.** A $100 \times 100$ board has a $+$ or a $-$ in each cell. An admissible move is to select a row and reverse the signs of its cells, then select a column and reverse the signs of its cells. Is it always possible to reach the board with $+$ in every cell after several moves?

**Problem 15.426.** Let $p$ be a prime number and let integers $a_1, a_2, \ldots, a_p$ be arranged around a circle in this order. One replaces every $a_i$ by $b_i = a_{i+1} - a_i$ simultaneously for all $i = 1, \ldots, p$, where $a_{p+1} = a_1$. Prove that after several replacements all the numbers will be divisible by $p$.

**Problem 15.427.** Let $A_1, A_2, \ldots, A_{n+1}$ be non-empty subsets of $\{1, 2, \ldots, n\}$. Prove that one can always choose two disjoint sets of indices $\{i_1, \ldots, i_k\}$ and $\{j_1, \ldots, j_m\}$ such that $A_{i_1} \cup \cdots \cup A_{i_k} = A_{j_1} \cup \cdots \cup A_{j_m}$.

**Problem 15.428.** Let $A_1, A_2, \ldots, A_m$ be distinct subsets of $\{1, 2, \ldots, n\}$ such that $|A_i \cap A_j| = k$ for a fixed $k$ and all $i \neq j$. Prove that $m \leq n$.

### 15.6.4 Assorted hard problems

**Problem 15.429.** For a polynomial $p \in \mathbb{C}[X]$ let $R_p$ denote the set of all complex zeros of $p$. Find all polynomials $f \in \mathbb{C}[X]$ such that $R_f = R_{f \circ f}$.

**Problem 15.430.** Let $k \in \mathbb{N}$. Construct an infinite family $\mathcal{F}$ of subsets of $\mathbb{N}$ such that for any $k$-element subfamily $\{A_1, \ldots, A_k\} \subset \mathcal{F}$ one has $|A_1 \cap A_2 \cap \cdots \cap A_k| = 1$, while for every $(k+1)$-element subfamily $\{A_1, \ldots, A_{k+1}\} \subset \mathcal{F}$ one has $A_1 \cap A_2 \cap \cdots \cap A_{k+1} = \emptyset$.

**Problem 15.431.** Show that for every $x \in \left[\frac{1}{110}, 110\right]$ there are $a_0, a_1, \ldots, a_{100} \in \{-1, 1\}$ such that the sequence $(x_n)_{0 \le n \le 101}$ given by $x_0 = 1$, $x_{k+1} = (1 + x_k)^{a_k}$ for $0 \le k \le 100$, satisfies $|x_{101} - x| < \dfrac{1}{402}$. †

**Problem 15.432.** Let $m, n \in \mathbb{N}$ with $n \ge m \ge 4$ and let $A = \{a_1, a_2, \ldots, a_m\} \subseteq \{1, 2, \ldots, n\}$ be such that $a+b \in A$ whenever $a \ne b$ are in $A$ and $a+b \le n$. Show that $\dfrac{a_1 + a_2 + \cdots + a_m}{m} \ge \dfrac{n+1}{2}$.

**Problem 15.433.** Distinct positive integers have sum 2012. What is the maximum possible value of their product? †

**Problem 15.434.** Find all quadruples $X = (x, y, z, t) \in \mathbb{R}^4$ such that the sequence $X_n = (x_n, y_n, z_n, t_n)_{n \ge 0}$ of quadruples, defined by taking $X_0$ to be a permutation of $X$ and

$$x_{n+1} = |a_n - b_n|, \quad y_{n+1} = |b_n - c_n|, \quad z_{n+1} = |c_n - d_n|, \quad t_{n+1} = |d_n - a_n|,$$

where $(a_n, b_n, c_n, d_n)$ is a permutation of $X_n$, has the property that $X_n = X$ for some $n \in \mathbb{N}$.

**Problem 15.435.** Do there exist two infinite sets $S, T$ of positive integers such that every positive integer $n$ can be uniquely written in the form $n = s_1 t_1 + s_2 t_2 + \cdots + s_k t_k$, where $k$ depends on $n$, the numbers $s_1 < s_2 < \cdots < s_k$ are elements of $S$, and $t_1, \ldots, t_k$ are elements of $T$?

**Problem 15.436.** The equation $(x-1)(x-2)\cdots(x-2016) = (x-1)(x-2)\cdots(x-2016)$ is written on the board, with 2016 linear factors on each side. What is the least possible value of $k$ for which it is possible to erase exactly $k$ of these 4032 linear factors so that at least one factor remains on each side and the resulting equation has no real solution?

**Problem 15.437.** Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of real numbers such that for every $n \ge 2020$ the number $a_{n+1}$ is the smallest root of the polynomial $P_n(x) = x^{2n} + a_1 x^{2n-2} + \cdots + a_{n-1}x^2 + a_n$. Show that for some $N \in \mathbb{N}$ the sequence $a_N, a_{N+1}, a_{N+2}, \ldots$ is strictly decreasing.

**Problem 15.438.** The numbers $1, 2, \ldots, 49, 50$ are written on a blackboard. Ann repeatedly chooses three of the numbers $a, b, c$ on the board, replaces them by their sum $a + b + c$, and writes the number $(a+b)(b+c)(c+a)$ in her notebook; she stops when only two numbers remain on the board, so that she has performed 24 operations. She then adds up the 24 numbers in her notebook. Denote by $A$ and $B$ the minimum and the maximum values Ann can achieve. Determine $A/B$.

**Problem 15.439.** Let $x, y, z$ be rational numbers. In each step we replace them by $x + \dfrac{1}{yz}$, $y + \dfrac{1}{xz}$, $z + \dfrac{1}{xy}$. Prove that after a certain point none of the three numbers is an integer.

**Problem 15.440.** Prove that there are infinitely many positive integers $n$ that cannot be represented in the form $a^b + c^d$ for positive integers $a, c$ and integers $b, d$ with $b \ge 2$ and $d \ge 2$.

**Remark.** The next problem uses the following elementary estimate: for all sufficiently large $n$ one has $n > d(n)^4$, where $d(n)$ denotes the number of positive divisors of $n$.

**Problem 15.441.** Call a positive integer $k$ *unusual* if it cannot be represented in the form $n - d(n)^2$, where $d(n)$ is the number of divisors of $n$. Prove that there are infinitely many unusual numbers.

**Problem 15.442.** Does there exist a sequence of pairwise distinct integers $a_1, a_2, \dots$ satisfying both of the following conditions? (I) For all positive integers $k$ one has $a_{k^2} > 0$ and $a_{k^2+k} < 0$. (II) For all positive integers $n$ one has $|a_{n+1} - a_n| \leq 2023\sqrt{n}$.

**Problem 15.443.** An infinite set $S$ of positive numbers is called *dense* if for every interval of the form $\left[\frac{1}{n+1}, \frac{1}{n}\right]$ with $n$ a positive integer there is a number in that interval which can be written as the difference of two elements of $S$. Is there a dense set from which it is impossible to select several elements with sum greater than $10^{2024}$?

**Problem 15.444.** Let $n > 1$ be an integer. In space, consider the set $S = \{(x, y, z) : x, y, z \in \{0, 1, \dots, n\},\ x+y+z > 0\}$. Find the smallest number of planes that jointly contain all $(n+1)^3-1$ points of $S$ but none of which passes through the origin. [IMO 2007 P6]

**Problem 15.445.** ★ Initially all positive integers are coloured black. We repeatedly perform the following procedure: in the $n$-th step we recolour the smallest black number $c_n$ blue and simultaneously recolour the number $c_n + n$ red. Find an exact formula for the sequence $(c_n) = (1, 3, 4, 6, 8, 9, \dots)$ of all blue numbers.

**Problem 15.446.** The sequence $(a_n)$ is defined by $a_1 = a_2 = 1$ and $a_{n+1} = a_n + a_{\lfloor n/2 \rfloor}$ for $n \geq 2$. Prove that this sequence contains infinitely many terms divisible by 7.

# Chapter 16

# Combinatorics

**Sources for this chapter:** Camp handouts by Nikola Petrović (April and June 2020; November 2021; February 2022), Dušan Djukić (June 2020; May–June 2021), Łukasz Bożyk (KAUST–Mawhiba website threads; January and June 2022), Arsenii "Senya" (November 2022), Regis Barbosa (November 2022), Emil Kolev (KAUST, February 2024), Bojan Bašić (June 2024), and the January 2025 camp leaflets.

The methods of Chapter 11 return here in the shape they take at the international level. Graphs arrive with named theorems attached; counting arrives with generating functions evaluated at roots of unity; and existence proofs are obtained by averaging rather than by construction. The boxes below record only the statements the trainers put on the board — the proofs, like the solutions, are left to you.

## 16.1 Graph theory

*Revisits the language of graphs (see Chapter 11); the problems below push it further.*

Almost every hard combinatorics problem can be redrawn as a graph, and once it is drawn the named theorems of this section do the work. Each subsection states the results the camps actually used, and then the problems that need them.

### 16.1.1 Degrees, trees and cycles

**Theorem 16.1 (Trees and bipartite graphs).** *Let $G$ be a graph in which there are no closed paths and in which any vertex can be reached from any other along the edges, that is, a tree. Then the vertices of $G$ can be coloured in two colours so that no two vertices of the same colour are connected; such a graph is called* bipartite*. More generally, a graph is bipartite if and only if it contains no cycle of odd length.*

**Problem 16.1.** In a simple graph on $n$ vertices, $n-1$ of its vertices have degrees $1, 2, \ldots, n-1$. What is the degree of the $n$-th vertex?

**Problem 16.2.** In a group of people with at least one pair of acquaintances, every two persons with the same number of acquaintances have no acquaintances in common. Prove that there is a person knowing exactly one other person.

**Problem 16.3.** Every vertex of a simple graph has degree 3. Prove that this graph has an even cycle (i.e. of even length).

**Problem 16.4.** In a connected simple graph on $n$ vertices, every vertex has degree at least $d$. Prove that there is a path of length at least $\min\{2d, n-1\}$.

**Problem 16.5.** Given a tree with 100 leaves, prove that one can add 50 edges so that the so obtained graph has no bridges. (A bridge is an edge whose removal disconnects the graph.)

**Problem 16.6.** In a group of $2n+1$ persons, for every group of $n$ persons there is a person (other than themselves) knowing them all. Prove that there is a person knowing all other $2n$ persons.

**Problem 16.7.** There are 30 teams in NBA and every team plays 82 games in the year. Bosses of NBA want to divide all teams into Western and Eastern Conferences (not necessarily equally), such that the number of games between teams from different conferences is half of the number of all games. Can they do it?

**Problem 16.8.** In a badminton tournament $n$ players took part, where $n$ is an odd positive integer. Each pair of participants played exactly two games with each other, and no game ended in a draw. It turned out that everyone won and lost equally many times. Prove that you can cancel precisely half of all the games and keep this property satisfied.

**Problem 16.9.** In a simple directed graph with 102 vertices, every vertex has the indegree and the outdegree both equal to 40. Prove that from every vertex one can reach every other vertex using at most three edges.

**Problem 16.10.** A graph has $n$ vertices and its maximum vertex degree is $d$.

(a) Prove that there exists an independent set of size at least $\dfrac{n}{d+1}$.

(b) Suppose $k \leq d$. Prove that there exists a set of at least $\dfrac{kn}{d+1}$ vertices containing no $(k+1)$-clique.

(c) The same as in (b), but now $d$ denotes the average degree.

**Problem 16.11.** ★ At a university dinner, there are 2017 mathematicians who each order two distinct entrees, with no two mathematicians ordering the same pair of entrees. The cost of each entree is equal to the number of mathematicians who ordered it, and the university pays for each mathematician's less expensive entree (with ties broken arbitrarily). Over all possible sets of orders, what is the maximum total amount the university could have paid?

**Problem 16.12.** ★ In the simple and connected graph $G$ let $x_i$ be the number of vertices with degree $i$. Let $d > 3$ be the biggest degree in the graph $G$. Prove that if

$$x_d \geq x_{d-1} + 2x_{d-2} + \cdots + (d-1)x_1,$$

then there exists a vertex with degree $d$ such that after removing that vertex the graph $G$ is still connected.

### 16.1.2 Bipartite graphs and matchings

**Definition 16.2 (Matching, system of distinct representatives).** A *matching* in a graph is a set of pairwise disjoint edges; it is *perfect* if it covers every vertex. Given finite sets $A_1, \ldots, A_n$, a *system of distinct representatives* for them is a choice of pairwise distinct elements $a_1 \in A_1, \ldots, a_n \in A_n$; this is the same thing as a matching covering $A_1, \ldots, A_n$ in the bipartite graph that joins each $A_i$ to its elements.

**Theorem 16.3 (Hall's marriage theorem).** *Let $G$ be a finite bipartite graph whose vertex set is split into two parts $A$ and $B$. For a subset $X \subseteq A$ denote by $F(X)$ the set of vertices in $B$ joined to some vertex of $X$ by an edge. Then a matching that covers all of $A$ exists if and only if $|F(X)| \geq |X|$ holds for every $X \subseteq A$.*

**Problem 16.13.** A pack of 52 cards is split into 13 piles, 4 cards each. Prove that it is possible to choose one card from each pile such that the 13 chosen cards are $A, 2, 3, 4, 5, 6, 7, 8, 9, 10, J, Q, K$ in some order.

**Problem 16.14.** There are 42 students in a class. Among any 31 of them there are a girl and a boy who know each other. Prove that there exist 12 disjoint acquainted girl-boy pairs.

**Problem 16.15.** For $0 \leq k \leq 2n+1$, denote by $S_k$ the family of all $k$-element subsets of $\{1, 2, \ldots, 2n+1\}$. Prove that there is a bijection $f : S_n \to S_{n+1}$ with the property that $A \subset f(A)$ for all $A \in S_n$.

**Problem 16.16.** Let $X$ be a finite set and let $\mathcal{F}$ be a family of subsets of $X$ that is closed with respect to taking subsets (that is, if $A \in \mathcal{F}$ and $B \subset A$, then also $B \in \mathcal{F}$). Prove that there is a bijection $f : \mathcal{F} \to \mathcal{F}$ such that $A \cap f(A) = \emptyset$ for all $A \in \mathcal{F}$.

**Problem 16.17.** In a graph with $3n$ vertices, the vertices can be divided into three complete $n$-vertex subgraphs, but the graph contains no complete subgraph on $n+1$ vertices. Prove that the vertices of this graph can be properly coloured in $\lfloor 5n/3 \rfloor$ colours, that is, so that no two adjacent vertices have the same colour.

**Problem 16.18.** A *hollow triangle* is an equilateral triangle $T$ of side $n$ pointing up, divided into $n^2$ unit triangles, from which $n$ upward-pointing unit triangles have been cut out. A *diamond* is a unit rhombus with an angle of $60°$. Prove that a hollow triangle $T$ can be tiled with diamonds if and only if, for each $k = 1, 2, \ldots, n$, every upward-pointing equilateral triangle of side $k$ contained in $T$ has at most $k$ holes.

**Problem 16.19.** On some planet there are three genders: male, female and emale. The inhabitants marry in triples, one of each gender. Suppose there are $3n$ persons, $n$ of each gender, and each of them likes at least $k$ persons of each of the other two genders.

(a) If $k > \frac{3n}{4}$, prove that we can form $n$ happy married triples.

(b) If $k = \frac{n}{2}$, show that it might be impossible to form a single triple.

The next problem is the harder companion of the hat-guessing puzzle of Chapter 6, in which one hundred prisoners wear caps numbered 1 to 100 with repetitions allowed; here the hats are all different and the counting strategy no longer works.

**Problem 16.20.** We are given $n$ dwarves caught by an evil wizard who has a collection of hats, numbered from 1 to $2n-1$. The wizard puts $n$ different hats on the dwarves' heads in such a way that each of them can see the hats of all $n-1$ other dwarves, but cannot see his own hat. Then the wizard asks each dwarf to guess the hat on his own head. If at least one dwarf gives the correct answer, all of them will be set free, otherwise they will remain imprisoned forever. The dwarves can agree on their strategy before, but once the hats are placed on their heads, they lose the possibility of communication (in particular they do not hear the other dwarves' answers). Does there exist a strategy for the dwarves which guarantees freedom?

**Problem 16.21.** Two mathematicians, $A$ and $B$, are performing the following magic trick. Firstly $A$ hands a deck of 124 (distinguishable) cards to somebody from the audience and asks them to choose any 5 cards (the cards are known to $A$ and the audience but remain secret to $B$). Then $A$ shows, one by one, four of the five cards to $B$, and at the climax $B$ guesses the fifth card correctly. How is this trick done?

**Problem 16.22.** The columns and the rows of a $3n \times 3n$ square board are numbered 1, 2, ..., $3n$. Every square $(x, y)$ with $1 \leq x, y \leq 3n$ is colored asparagus, byzantium or citrine according as the modulo 3 remainder of $x + y$ is 0, 1 or 2, respectively. One token colored asparagus, byzantium or citrine is placed on each square, so that there are $3n^2$ tokens of each color. Suppose that one can permute the tokens so that each token is moved to a distance of at most $d$ from its original position, each asparagus token replaces a byzantium token, each byzantium token replaces a citrine token, and each citrine token replaces an asparagus token. Prove that it is possible to permute the tokens so that each token is moved to a distance of at most $d + 2$ from its original position, and each square contains a token with the same color as the square. [IMO Shortlist 2012 C5]

### 16.1.3 Hamiltonicity and tournaments

**Definition 16.4 (Tournament, triangle, king).** A *tournament* is a competition in which every two of the participants play exactly one game and no game ends in a tie; equivalently, it is a complete graph each of whose edges has been given a direction. Three participants form a *triangle* if they can be denoted $A$, $B$, $C$ so that $A$ defeated $B$, $B$ defeated $C$ and $C$ defeated $A$. A participant $A$ is a *king* if for every $B \neq A$ either $A$ defeated $B$ in a direct game, or there is a participant $C$ with $A$ defeating $C$ and $C$ defeating $B$.

**Theorem 16.5 (Ore's theorem).** *Call a pair of vertices of a graph on $n$ vertices* good *if the sum of their degrees is at least $n$. If every pair of non-adjacent vertices is good, then the graph has a Hamiltonian cycle.*

**Problem 16.23.** Given a graph $G$ with $n$ vertices, its "closure" is obtained by repeatedly adding edges between any two non-adjacent vertices $u$ and $v$ with $\deg u + \deg v \geq n$. Prove that $G$ has a Hamiltonian cycle if and only if so does its closure.

**Problem 16.24.** ★

(a) Let a graph $G$ with $n$ vertices have two vertices $a$ and $b$ connected by an edge, such that the number of the remaining edges coming out of these two vertices is at least $n$. Prove that if you can arrange the vertices in a circle so that each two neighbouring vertices are connected, then you can also do so in such a way that $a$ and $b$ are not adjacent to each other in the cycle.

(b) In a room full of $2n$ people each person has at least $n + 1$ friends (friendship is always a mutual relation). Prove that you can arrange the people to be seated around a round table so that each person sits next to two of his friends.

**Problem 16.25.** In a simple graph $G$ with 100 vertices, in every set of four vertices there are at least two edges. Moreover, every two vertices have a non-neighbor in common. Prove that this graph has a Hamiltonian path.

**Problem 16.26.** After an $n$-player tournament ($n \geq 4$) participants took seats at a round table. It turned out that everyone has defeated their direct right-neighbor. Prove that one of the players can leave the table and the remaining players can be relocated so that this property will still be satisfied.

**Problem 16.27.** In a tournament $n$ players took part. Every pair of participants played exactly one game, and there were no ties. Three participants are called a *triangle* if they can be denoted by $A$, $B$, $C$ in such a way that $A$ defeated $B$, $B$ defeated $C$, and $C$ defeated $A$. Suppose that all participants can be placed by a round table in such a way that everyone defeated their closest right neighbor.

(a) Prove that there is at least one triangle.

(b) Prove that there are at least $n/3$ triangles.

(c) Determine, in terms of $n$, the smallest possible number of triangles.

**Problem 16.28.** A *king* of a tournament is every participant $A$ with the following property: for every $B \neq A$ either $A$ defeated $B$ in a direct game, or there exists a participant $C$ such that $A$ defeated $C$ and $C$ defeated $B$. In other words: $A$ is a king if he defeated every other participant either directly or indirectly (with one other player on the way). Find all $n \geq 1$ such that there exists a tournament with $n$ players in which everyone is a king.

**Problem 16.29.** In a table tennis tournament with $2n+1$ contestants, each pair of contestants played each other exactly once and none of the games resulted in a tie. We will call a *triangle* any collection of 3 contestants $A$, $B$ and $C$ such that $A$ defeated $B$, $B$ defeated $C$ and $C$ defeated $A$. Let $T$ be the number of triangles that occurred in the tournament. (Two different triangles may overlap in zero, one or two contestants.)

(a) Find the smallest possible value of $T$.

(b) Find the largest possible value of $T$.

**Problem 16.30.** Exactly $2n + 3$ players took part in a badminton tournament. Every pair of participants played exactly one game and there was only one court, so all the matches were arranged in a sequence. The schedule of the tournament had the following property: each player after taking part in a match was not scheduled to play in any of the $n$ directly following (consecutive) matches. Prove that one of the players of the opening match also played in the closing match.

**Problem 16.31.** 110 teams participated in a volleyball tournament; every two teams played against each other exactly one game (in volleyball, there are no draws). It happened that in every group of 55 teams there exists a team which has lost to at most four out of the other 54 teams in this group. Prove that in the whole tournament there exists a team which has lost to at most four out of the other 109 teams.

**Problem 16.32.** Given is a tournament with $n \geq 2$ participants. Each two of them played against each other exactly once and there were no tie games. Each game was colored either red, or blue. Prove that there exists a monochromatic master, i.e. a participant $M$ with the following property: for every player $A \neq M$ there exist players $A_1, A_2, \ldots, A_k$ with $A_1 = M$ and $A_k = A$ such that $A_i$ defeated $A_{i+1}$ for all $i = 1, \ldots, k-1$, and all of these $k-1$ games have the same color.

**Problem 16.33.** In Puerto Paranoia there are 16 secret agents, each of which follows at least one of the others, but no two follow each other. Assume that every 10 agents can be ordered so as to form a cycle (i.e. the first follows the second, the second follows the third, etc, the tenth follows the first). Prove that every 11 agents can also be so ordered.

### 16.1.4 Extremal graph theory

The first few problems are the classical edge-count bounds, set as exercises in Dušan Djukić's May 2021 graph-theory sheets; the theorem printed after them is the general statement, and the problems following it apply it.

**Problem 16.34.** If a graph on $n$ vertices contains no triangles, prove that it has at most $n^2/4$ edges.

**Problem 16.35.** If a graph on $2n$ vertices has $n^2+1$ edges, prove that it contains at least $n$ triangles.†

**Problem 16.36.** A simple graph has 160 vertices and 500 edges. Prove that it contains a cycle of length at most 8.

**Problem 16.37.** In a graph with 300 vertices, each vertex has degree 3. At most how many cycles of length 4 can this graph contain?

**Problem 16.38.** Find the smallest $n > 4$ for which there is a group of $n$ people in which any two acquaintances have no acquaintances in common, whereas any two non-acquaintances have exactly two acquaintances in common.

**Theorem 16.6 (Turan's theorem).** *If a graph on $n$ vertices contains no complete subgraph on $m$ vertices, then the number of its edges is at most*

$$\frac{n^2}{2}\left(1-\frac{1}{m-1}\right).$$

**Problem 16.39.** ★ A graph has 2017 vertices. Some pairs of vertices are connected with either blue or red edges. In every subset of 4 vertices there is an even number of edges, at least one edge is red, and if there are blue edges then there are more blue edges than red edges. Prove that there are 673 vertices such that each pair of vertices is connected with a red edge.

**Problem 16.40.** ★ Let $n \geq 3$ distinct points be marked on a plane so that no three of them lie on the same line. All points are connected with segments. All segments are painted one of four colours so that if in some triangle with the vertices at the marked points two sides have the same colour, then all its sides have the same colour. In addition, each of the four colours is used to paint at least one segment. What is the largest possible value of $n$?

**Problem 16.41.** Let $G$ be an undirected simple triangle-free $n$-vertex graph of minimum degree greater than $\frac{2n}{5}$. Prove that in $G$ there is no cycle of odd length (i.e. $G$ is bipartite). †

**Problem 16.42.** ★ In a mathematical competition some competitors are friends. Friendship is always mutual. Call a group of competitors a clique if each two of them are friends. (In particular, any group of fewer than two competitors is a clique.) The number of members of a clique is called its size.

Given that, in this competition, the largest size of a clique is even, prove that the competitors can be arranged into two rooms such that the largest size of a clique contained in one room is the same as the largest size of a clique contained in the other room.

**Problem 16.43.** The sides and the diagonals of a regular $n$-gon are colored in $k \geq 3$ colors in such a way that for any color $i$ and any two vertices $X$ and $Y$ there exists a path between $X$ and $Y$ all segments of which are of color $i$. Prove that there exist three vertices $A$, $B$ and $C$ such that the segments $AB$, $BC$ and $AC$ are of different colors.

**Problem 16.44.** ★ Let $n \geq 3$ be an integer. In a country there are $n$ airports and $n$ airlines operating two-way flights. For each airline there is an odd integer $m \geq 3$ and $m$ distinct airports $c_1, \ldots, c_m$ such that the flights offered by that airline are exactly those between the following pairs of airports: $c_1$ and $c_2$; $c_2$ and $c_3$; $\ldots$; $c_{m-1}$ and $c_m$; $c_m$ and $c_1$. Prove that there is a closed route consisting of an odd number of flights where no two flights are operated by the same airline.

**Problem 16.45.** ★ Given any positive real number $\varepsilon$, prove that, for all but finitely many positive integers $v$, any graph on $v$ vertices with at least $(1+\varepsilon)v$ edges has two distinct simple cycles of equal lengths. (A simple cycle does not allow repetition of vertices in a cycle.)

### 16.1.5 Planarity and colouring

**Theorem 16.7 (Euler's polyhedron formula).** *If a connected planar graph is drawn in the plane with $V$ vertices, $E$ edges and $F$ faces (counting the unbounded face), then $V - E + F = 2$. In particular the numbers of vertices, edges and faces of a convex polyhedron satisfy the same relation.*

**Theorem 16.8 (Vizing's theorem).** *To colour the edges of a graph so that no two edges of the same colour meet at the same vertex, one needs either $\Delta$ or $\Delta + 1$ colours, where $\Delta$ is the largest number of edges leaving a single vertex.*

**Problem 16.46.** Prove that every planar graph on $n$ vertices has at most $3n - 6$ edges, and that every bipartite planar graph has at most $2n - 4$ edges.

**Problem 16.47.** Prove that the vertices of a planar graph can be colored with 5 colors so that no edge connects two vertices of the same color.

**Problem 16.48.** Every edge of a convex polyhedron is assigned $+$ or $-$. Prove that there is a vertex having at most two facial angles with differently signed edges.

**Problem 16.49.** ★ Each edge of a polyhedron is given an orientation such that each vertex of the polyhedron has at least one edge entering it and one edge exiting it. Prove that there exists a face of the polyhedron whose edges form an oriented cycle.

**Problem 16.50.** ★ A rectangle $R$ is partitioned into smaller rectangles whose sides are parallel with the sides of $R$. Let $B$ be the set of all boundary points of all the rectangles in the partition, including the boundary of $R$. Let $S$ be the set of all segments whose points belong to $B$. Let a maximal segment be a segment in $S$ which is not a proper subset of any other segment in $S$. Let an intersection point be a point in which 4 rectangles of the partition meet. Let $m$ be the number of maximal segments, $i$ the number of intersection points and $r$ the number of rectangles. Prove that $m + i = r + 3$.

**Problem 16.51.** A polyhedron with at least 5 vertices is given, where exactly three edges meet in each vertex. Prove that it is possible to write a rational number in every vertex of the polyhedron in such a way that:

(a) at least one of those numbers equals 2020;

(b) for every face of the polyhedron, the product of all the numbers in the vertices of that face equals 1.

[Serbian MO 2020]

**Problem 16.52.** ★ In a country called Graphia each city is connected with three other cities via a road. A single road starts and ends in a city without passing through any other city. Prove that it is possible to colour all the roads in four colours so that no two roads of the same colour share a common endpoint city.

## 16.2 Partitions, bijections and Catalan structures

*Revisits partitions, bijections and the Catalan recurrence (see Chapter 11); the problems below push it further.*

Ferrers diagrams turn statements about partitions into statements about pictures, and reflecting a picture proves a theorem.

**Definition 16.9 (Partition and its Ferrers diagram).** A *partition* of a positive integer $n$ is a representation of $n$ as a sum of positive integer summands, called *parts*, where the order does not matter; one lists the parts in non-increasing order. A partition can be represented graphically as an array of dots, by the so-called *Ferrers diagram*: each part contributes one row of that many dots, the rows being left-aligned and listed from the largest part downwards. For example, the partition $13 = 4+4+2+1+1+1$ has the diagram



**Definition 16.10 (Conjugate partition).** Reflect a Ferrers diagram in the line $x = y$. The result is the Ferrers diagram of another partition of $n$, called the *conjugate partition*; equivalently, one reads the original diagram in columns instead of rows. In the example above the conjugate partition is $13 = 6+3+2+2$. Conjugation is obviously a reversible operation, as each partition is conjugate to its conjugate. A partition equal to its own conjugate is called *self-conjugate*.

**Theorem 16.11 (The Catalan generating function).** *The generating function $C(x) = \sum_{n\ge 0} c_n x^n$ of the Catalan numbers satisfies $C(x) = 1 + xC(x)^2$; solving this quadratic equation and choosing the sign that avoids a singularity gives the closed forms*

$$C(x) = \frac{1-\sqrt{1-4x}}{2x}, \qquad c_n = \frac{1}{n+1}\binom{2n}{n}.$$

*Consequently every family counted by the Catalan numbers — rooted ordered trees with $n+1$*

*vertices, rooted ordered binary trees, the strings of $n$ letters $a$ and $n$ letters $b$ in which every prefix has at least as many $a$'s as $b$'s, the triangulations of an $(n+2)$-gon, the partitions of the vertices of an $n$-gon into subsets with pairwise disjoint convex hulls, the dissections of a staircase of rank $n$ into $n$ rectangles, and the non-crossing pairings of the vertices of a regular $2n$-gon — has the same closed count.*

**Problem 16.53.** For any $n, k \in \mathbb{N}$, prove that the number of partitions of $n$ into $k$ parts equals the number of partitions of $n$ whose largest part is equal to $k$.

**Problem 16.54.** We call a partition of a positive integer $n$ *self-conjugate* if it is conjugate to itself. Prove that the number of self-conjugate partitions of $n$ equals the number of partitions of $n$ into distinct odd parts.

**Problem 16.55.** We are given one weight of each of the masses $1, 3, 5, 7, 9, \ldots$. For a positive integer $n$, define $A_n$ to be the number of ways to choose a set of weights with the total mass $n$. Prove that $A_n \leq A_{n+1}$ for $n \geq 1$.

**Problem 16.56.** ★ A partition of $n$ is the division of the set $\{1, 2, \ldots, n\}$ into several nonempty disjoint sets. Prove that the number of partitions of $n$ equals the number of partitions of $n+1$ such that no two consecutive numbers are in the same set. (For example, the number of partitions of 3 is 5: $\{\{1\},\{2\},\{3\}\}$, $\{\{1,2\},\{3\}\}$, $\{\{1,3\},\{2\}\}$, $\{\{2,3\},\{1\}\}$ and $\{\{1,2,3\}\}$; and the number of partitions of 4 such that no two consecutive numbers are in the same set is also 5: $\{\{1\},\{2\},\{3\},\{4\}\}$, $\{\{1,3\},\{2\},\{4\}\}$, $\{\{1,4\},\{2\},\{3\}\}$, $\{\{2,4\},\{1\},\{3\}\}$, $\{\{1,3\},\{2,4\}\}$.)

**Problem 16.57.** Prove that the number of sequences of digits $0, 1, 2, 3$ of length $n$ with an even total number of 0's and 1's is equal to the number of binary sequences (i.e. sequences of digits $0, 1$) of length $2n$ with an even number of 1's.

**Problem 16.58.** Let $n$ be a positive integer. Each point $(x, y)$ in the coordinate plane, where $x$ and $y$ are non-negative integers with $x + y < n$, is colored red or blue, subject to the following condition: if a point $(x, y)$ is red, then so are all points $(x', y')$ with $x' \leq x$ and $y' \leq y$. Let $A$ be the number of ways to choose $n$ blue points with distinct $x$-coordinates, and let $B$ be the number of ways to choose $n$ blue points with distinct $y$-coordinates. Prove that $A = B$.

**Problem 16.59.** Let $n$ be a positive integer. A sequence of $n$ positive integers (not necessarily distinct) is called *full* if it satisfies the following condition: for each positive integer $k \geq 2$, if the number $k$ appears in the sequence then so does the number $k-1$, and moreover the first occurrence of $k-1$ comes before the last occurrence of $k$. For each $n$, how many full sequences are there?

**Problem 16.60.** For a positive integer $n$, denote by $A_n$ the number of partitions of the set $\{1, 2, \ldots, n\}$ into subsets in which every two adjacent numbers in ascending order have different parity, and by $B_n$ the number of partitions of $\{1, 2, \ldots, n\}$ into subsets with all elements of the same parity. Prove that $A_n = B_{n+1}$.

[E.g. for $n = 7$, $\{1,4\} \cup \{2,5,6,7\} \cup \{3\}$ is a partition of the first type, and $\{1,3,7\} \cup \{2,6\} \cup \{4\} \cup \{5\}$ is of the second type.]

**Problem 16.61.** There are 100 towns on Mars, and there are 1000 highways, each connecting two towns, projected in such a way that from any town one can reach any other town. The government intends to close some highways (maybe all) in such a way that at each town there are an even number of highways (0 is even). In how many ways can this be done?

**Problem 16.62.** Let $n$ and $k \geq n$ be positive integers of the same parity. We are given $2n$ lamps labelled $1, 2, \ldots, 2n$, which can be either on or off. Initially all the lamps are off. We consider sequences of $k$ switches (each applied on one lamp):

(a) Denote by $N$ the number of such sequences resulting in the state where lamps $1, \ldots, n$ are all on and lamps $n+1, \ldots, 2n$ are all off.

(b) Denote by $M$ the number of such sequences resulting in the state where lamps $1, \ldots, n$ are all on and lamps $n+1, \ldots, 2n$ are all off, but where none of the lamps $n+1, \ldots, 2n$ is ever switched on.

Determine $N/M$.

## 16.3 Generating functions II and the roots-of-unity filter

*Revisits generating functions and the snake oil method (see Chapter 11); the problems below push it further.*

**Technique 16.12 (The roots-of-unity filter).** Let $A(x) = \sum_{n \geq 0} a_n x^n$ and let $\varepsilon$ be a primitive $m$-th root of unity. For $x = \varepsilon^i$ with $i = 1, 2, \ldots, m-1$ one has $1+x+x^2+\cdots+x^{m-1} = 0$, whereas the same sum equals $m$ at $x = 1$. Consequently

$$\sum_{n \equiv r \pmod m} a_n \;=\; \frac{1}{m} \sum_{i=0}^{m-1} \varepsilon^{-ir} A(\varepsilon^i),$$

so evaluating a generating function at roots of unity extracts exactly the part of a sum lying in one residue class — which is what counting configurations of prescribed weight modulo $m$ requires.

**Problem 16.63.** Let $G(x) = \dfrac{1}{1 - 2x - x^2}$ be the generating function of a sequence $a_n$, $n \geq 0$. Prove that for each $n$,

$$a_n^2 + a_{n+1}^2 = a_{2n+2}.$$

**Problem 16.64.** Let $f_1 = 1$, $f_{2n} = f_n$ and $f_{2n+1} = f_n + f_{n+1}$ for all positive integers $n$. Prove that the generating function $F(x) = \sum_{k=1}^{\infty} f_k x^{k-1}$ satisfies

$$F(x) = (1 + x + x^2) F(x^2).$$

**Problem 16.65.** Find the number of ways of tiling a $2 \times n$ rectangle using:

(a) $1 \times 1$ unit squares and L-trominoes;

(b) $2 \times 1$ dominoes and L-trominoes.

†

**Problem 16.66.** ★ Find the values of the following sums of binomial coefficients:

(a) $\displaystyle\sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} 4^k$;

(b) $\sum_{k=0}^{\lfloor n/3 \rfloor} \binom{n}{3k}$;

(c) $\sum_{k=m}^{n} \binom{n}{k}\binom{k}{m}$;

(d) $\sum_{k=m}^{n} (-1)^k \binom{n}{k}\binom{k}{m}$;

(e) $\sum_{k=0}^{n} \binom{2n}{2k}\binom{2k}{k} 2^{2n-2k}$.

**Problem 16.67.** ★ For any positive integer $n$, prove that the number of natural numbers whose digits are all from the set $\{1, 3, 4\}$ and whose sum of digits is $2n$ is a perfect square. For example, for $n = 3$ there are $9 = 3^2$ such numbers: 111111, 1113, 1131, 1311, 3111, 33, 114, 141, 411.

**Problem 16.68.** ★ Find the sum of the number of non-negative integer solutions of each of the following equations:

$$x + 2y = n, \qquad 2x + 3y = n - 1, \qquad \ldots, \qquad (n+1)x + (n+2)y = 0.$$

†

**Problem 16.69.** A finite sequence of integers $a_1, a_2, \ldots, a_n$ is called $p$-balanced if and only if the sum $a_k + a_{k+p} + a_{k+2p} + \cdots$ is constant for $k = 1, 2, 3, \ldots, p$. Prove that, if a sequence of 50 members is $p$-balanced for $p = 3, 5, 7, 11, 13, 17$, then all the members of that sequence are equal to 0. [Leningrad MO 1991]

**Problem 16.70.** Let $n, a_1, a_2, \ldots, a_m$ be positive integers. Let $f$ be defined as follows: if $k$ is an integer, then $f(k)$ is the number of integer $m$-tuples $(c_1, c_2, \ldots, c_m)$ such that $c_1 + c_2 + \cdots + c_m \equiv k \pmod{n}$ and $1 \leq c_i \leq a_i$ for each $i$. Prove that $f$ is constant if and only if $n$ divides at least one among the numbers $a_1, a_2, \ldots, a_m$. [MOP 1999]

**Problem 16.71.** Let $p > 3$ be a prime number. For a nonempty subset $T$ of $\{0, 1, 2, \ldots, p-1\}$, let $E(T)$ denote the set of all $(p-1)$-tuples $(x_1, x_2, \ldots, x_{p-1})$ such that $x_i \in T$ and $p \mid x_1 + 2x_2 + \cdots + (p-1)x_{p-1}$. Prove that

$$|E(\{0, 1, 3\})| \geq |E(\{0, 1, 2\})|,$$

where the equality is reached if and only if $p = 5$. [IMO Shortlist 1999]

**Problem 16.72.** Let $m, n \geq 2$ be positive integers, and let $a_1, a_2, \ldots, a_n$ be integers such that none of them is divisible by $m^{n-1}$. Prove that there exist integers $e_1, e_2, \ldots, e_n$ that are not all 0 and $|e_i| < m$ for each $i$, such that $e_1 a_1 + e_2 a_2 + \cdots + e_n a_n$ is divisible by $m^n$. [IMO Shortlist 2002]

**Problem 16.73.** A doubly indexed sequence of rational numbers $f(n, i)$, where $n, i \in \mathbb{N}$ and $i \geq n$, is defined recursively as follows:

$$f(1, i) = \frac{1}{i}; \qquad f(n+1, i) = \frac{n+1}{i}\big(f(n, n) + f(n, n+1) + \cdots + f(n, i-1)\big).$$

If $p$ is a prime number, prove that, for $n > 1$, the denominator of the fraction $f(n, p)$ (in its reduced form) is not divisible by $p$. [Putnam 1980]

**Problem 16.74.** A finite set of positive integers $S_0$ is given. Let us define the sequence $S_1, S_2, S_3, \ldots$ as follows: $a \in S_n$ if and only if exactly one number of $a$, $a-1$ is a member of $S_{n-1}$. Prove that there exist infinitely many positive integers $N$ such that $S_N = S_0 \cup \{a + N : a \in S_0\}$. [Putnam 2000]

**Problem 16.75.** For a set $S$ of nonnegative integers, let $r_S(n)$ denote the number of pairs $(s_1, s_2)$ such that $s_1 \in S$, $s_2 \in S$, $s_1 \neq s_2$, and $s_1 + s_2 = n$. Is it possible to partition the set of nonnegative integers into two sets, $A$ and $B$, such that for each $n$ we have $r_A(n) = r_B(n)$? [Putnam 2003]

**Problem 16.76.** For a positive integer $n$ and a real number $c$, let $x_k$ be defined recursively by $x_0 = 0$, $x_1 = 1$ and, for $k \geq 0$,

$$x_{k+2} = \frac{c\,x_{k+1} - (n-k)x_k}{k+1}.$$

Let $n$ be fixed, and let $c$ be the largest possible value such that $x_{n+1} = 0$. Find $x_k$ in terms of $n$ and $k$ for $1 \leq k \leq n$. [Putnam 1997]

**Problem 16.77.** Let $a_{m,n}$ be the coefficient of $x^n$ in the expanded version of $(1 + x + x^2)^m$. Prove that for all nonnegative integers $k$,

$$0 \leq \sum_{i=0}^{\lfloor 2k/3 \rfloor} (-1)^i a_{k-i,\,i} \leq 1.$$

[Putnam 1997]

**Problem 16.78.** Let $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_n$ be two sequences of distinct integers such that $a_i + b_j \neq 0$ for all $i, j$. Let us consider $n^2$ numbers $c_{jk}$ ($j$ and $k$ can be any values from 1 to $n$) such that, for given $i$ and $k$, the following relation holds:

$$\sum_{j=1}^{n} \frac{c_{jk}}{a_i + b_j} = \begin{cases} 1, & \text{if } i = k; \\ 0, & \text{otherwise.} \end{cases}$$

Prove that the sum of all $c_{jk}$ equals $a_1 + a_2 + \cdots + a_n + b_1 + b_2 + \cdots + b_n$. [MOP 1998]

## 16.4 Double counting and designs

*Revisits double counting (see Chapter 11); the problems below push it further.*

**Technique 16.13 (Three faces of double counting).** Double counting most often means counting pairs $(a, b)$ of related objects: the first count is made by initially choosing $a$, the second by starting with $b$. The camps recommend three concrete ways of setting this up. *Flags:* assign a flag to each admissible pair and count the flags. *Bipartite graphs:* put the possible values of $a$ in one part and those of $b$ in the other, joining $a$ to $b$ when the pair is admissible; then count the total degree of one part and afterwards of the other. *Incidence matrices:* form the matrix whose rows are indexed by the values of $a$ and whose columns are indexed by the values of $b$, with entry 1 or 0 according to whether the pair is admissible; count the ones by summing over rows, and then by summing over columns. The strength of the approach is that hardly anything is invented: one reads the problem statement and follows a standard sequence of actions.

**Problem 16.79.** In a parliament with several members, a number of committees are formed, each of the same size. For any pair of parliament members, the number of committees they both belong to does not depend on which pair is chosen. Prove that all members belong to the same number of committees.

**Problem 16.80.** Let $n$ be a positive integer. In a family of finite sets, call an element a *splitting element* if it belongs to at least $n$ of the sets and is omitted by at least $n$ of the sets. Determine the maximum possible size of a family of subsets of $\{1, \ldots, n\}$ for which there is no splitting element.

**Problem 16.81.** A social club has $2k + 1$ members, each of whom is fluent in the same $k$ languages. Any pair of members always talk to each other in only one language. Suppose that there were no three members such that they use only one language among them. Let $A$ be the number of three-member subsets such that the three distinct pairs among them use different languages. Find the maximum possible value of $A$.

**Problem 16.82.** A group of $n$ students attends several classes, in such a way that if two students attend the same two classes, then the numbers of students in these two classes are different from each other. Prove that the largest possible number of classes is $(n-1)^2$. (A class must have at least 2 students.)

**Problem 16.83.** At a party attended by $n$ people each person has exactly $k$ friends, each two people who are friends have exactly $\ell$ (other) friends in common, and each two people who are not friends have exactly $m$ friends in common. Prove that

$$m(n-k) - k(k-\ell) + k - m = 0.$$

**Problem 16.84.** ★ At a bookstore 10 people bought 3 books each, in such a way that each two of these persons have at least one book in common that they bought. Let $M$ be the largest number of people that purchased the same book. Find the smallest possible value of $M$.

**Problem 16.85.** ★ Let $n > 1$ be a positive integer and let a set $P$ of $n^2$ points be given which form a grid of $(n-1) \times (n-1)$ unit squares. Let $S$ be the set of all squares whose vertices are in $P$, and let $a_k$ be the number of unordered pairs of different points of $P$ such that both of them together are vertices of exactly $k$ squares in $S$. Prove that $a_0 = a_2 + 2a_3$.

**Problem 16.86.** There are three disjoint committees of bureaucrats. Any two bureaucrats are either acquainted or unacquainted. For any pair of bureaucrats from two different committees, exactly 10 bureaucrats of the third committee are acquainted with both of them, and exactly 10 bureaucrats of the third committee are unacquainted with both of them. Determine the total number of bureaucrats.

**Problem 16.87.** A rectangular grid with both sides longer than one cell is tiled with dominoes. Let $A$ be the number of $2 \times 2$ squares of the grid that consist of two dominoes, and let $B$ be the number of $2 \times 2$ squares that consist of cells of four different dominoes. Prove that $A > B$.

**Problem 16.88.** At an Olympiad, several students were offered 6 problems. Each pair of problems was solved by more than $2/5$ of all the participants, and nobody solved all 6 problems. Prove that there are at least 2 participants who solved exactly 5 problems each.

**Problem 16.89.** Given a natural number $n > 1$, find the largest possible value of $m$ for which there exist an $mn$-element set $S$ and $2n$ of its $m$-element subsets such that any two of these subsets intersect in at most one element, and each element of $S$ belongs to exactly two of them.

**Problem 16.90.** At a university with 10 001 students, several clubs were formed (a student may join several clubs), and the clubs formed $k$ communities (a club may belong to several communities). It is known that:

(a) for every pair of students there is exactly one club that they both attend;

(b) for every student and every community there is exactly one club of that community which the student attends;

(c) each club has an odd number of students, and a club with $2m + 1$ students belongs to exactly $m$ communities.

Find all possible values of $k$.

**Problem 16.91.** There is a set $S$ consisting of $n$ elements and there are $\left\lfloor \sqrt{2n} \right\rfloor + 2$ subsets of it. It turned out that the union of any three of these subsets is $S$. Prove that the union of some two of these subsets is $S$. (As always, $\lfloor x \rfloor$ stands for the integer part of a real number $x$.)

**Problem 16.92.** Let $a$ be the number of all binary sequences of length 2023 having 7 consecutive 1's. Let $b$ be the number of all binary sequences of length 2024 having 8 consecutive equal elements. Find the ratio $a : b$.

**Problem 16.93.** Consider all $r$-element subsets of $\{1, 2, \ldots, n\}$, where integers $1 \le r \le n$ are fixed. From each of those subsets select the smallest number. Find (in terms of $n$ and $r$) the arithmetic mean of the selected numbers.

**Definition 16.14 (Cycles of a permutation).** A permutation $\sigma$ of $\{1, \ldots, n\}$ is written in *matrix notation* as the two-row array with $i$ in the top row and $\sigma(i)$ below it; it is determined by its bottom row alone, that is, by an $n$-term sequence in which every number from 1 to $n$ occurs exactly once. The *cycle* of $\sigma$ containing $i$ is the maximal sequence of pairwise different terms $i, \sigma(i), \sigma(\sigma(i)), \ldots$ taken together with all its cyclic shifts; a one-element cycle is a *fixed point*. The *cycle notation* of $\sigma$ is its decomposition into disjoint cycles, each written starting from its least element, the cycles being sorted by those least elements.

**Problem 16.94.** Denote by $p_k$ the number of permutations of $\{1, 2, \ldots, n\}$ having exactly $k$ fixed points (a point $x$ is a fixed point of $f$ if $f(x) = x$). Prove that

$$p_1 + 2p_2 + \cdots + np_n = n!.$$

**Problem 16.95.** For positive integers $n$ and $k$ let $c(n, k)$ denote the number of permutations of $\{1, 2, \ldots, n\}$ having exactly $k$ cycles. Prove that

$$\sum_{k=1}^{n} k\, c(n, k) = c(n + 1, 2).$$

**Problem 16.96.** For $n \in \mathbb{N}$, denote by $\{0, 1\}^n$ the set of all 0, 1-sequences of length $n$. Suppose there is a map $f : \{0, 1\}^n \to \{1, 2, \ldots, n\}$ with the following property: whenever two sequences $x$ and $y$ differ in exactly two positions, $f(x)$ and $f(y)$ are different. Prove that $n$ is a power of 2.

**Problem 16.97.** A regular hexagon of side 3 is divided into 54 unit equilateral triangles. These triangles have 37 vertices in total, numbered from 1 to 37 in some order. A unit triangle is called *clocked* if its three numbers, listed in ascending order, are arranged clockwise. Prove that at least 19 triangles are clocked.

**Problem 16.98.** Consider an $n$-vertex simple graph in which every vertex has at least one nonneighbor and every two nonneighbors have at least one common neighbor. Moreover suppose that the sum of squares of degrees of all the vertices equals $n^2 - n$. Find all possible lengths of the shortest cycle in such a graph.

**Problem 16.99.** There are $n$ cities in a country, some pairs of cities linked two-way through a straight flight. For every pair of cities there is exactly one aviaroute (which can have interchanges). The mayor of every city $X$ counted the number of numberings of all cities from 1 to $n$ such that on every aviaroute with the beginning in $X$ the numbers of the cities are in ascending order. Every mayor, except one, noticed that the result of his counting is a multiple of 2016. Prove that the result of the last mayor is a multiple of 2016 too.[†]

**Problem 16.100.** There are $n$ students in a class and their teacher wants to split them into groups, so that each pair of students participate in exactly one group together, every group consists of at least two students and no group contains all students (one student can participate in more than one group). Prove that for these conditions to be satisfied, the teacher needs to split the students into at least $n$ groups.

**Problem 16.101.** Subsets $A_1, A_2, \ldots, A_k$ of $\{1, 2, \ldots, n\}$ are such that $|A_i \cap A_j| = 1$ whenever $i \neq j$. Prove that $k \leq n$.

## 16.5 The probabilistic lens: expectation and averaging

**Definition 16.15 (Probability space, event, random variable, expectation).** A *finite probability space* is a set $\Omega = \{\omega_1, \ldots, \omega_k\}$ in which every outcome $\omega_i$ carries a probability $p(\omega_i) \in [0, 1]$ with $p(\omega_1) + \cdots + p(\omega_k) = 1$. A subset $A \subseteq \Omega$ is an *event*, and its probability is $p(A) = \sum_{\omega_i \in A} p(\omega_i)$. Any function $X : \Omega \to \mathbb{R}$ is a *random variable*, and its *expected value* (or average) is $E(X) = \sum_i X(\omega_i)\, p(\omega_i)$.

**Theorem 16.16 (Linearity of expectation).** *If $X_1, \ldots, X_m$ are random variables on the same probability space, then $E(X_1 + \cdots + X_m) = E(X_1) + \cdots + E(X_m)$.*

**Technique 16.17 (Indicator random variables).** To find the expected number of objects with a prescribed property, write that number as a sum $S = S_1 + \cdots + S_m$ of indicators, where $S_i$ equals 1 if the $i$-th object has the property and 0 otherwise. Linearity then turns the problem into $m$ separate elementary probability computations, and no independence is needed.

**Problem 16.102.** A fair coin is flipped 100 times. Find the expected number of heads.

**Problem 16.103.** The digits $1, 2, 3, 4, 5$ are rearranged at random to form a five-digit number. Find the expected value of the resulting number.

**Problem 16.104.** There are 4 people and 4 name badges, one for each person. Each person receives a random badge. Find the expected number of people who get their own badge.

**Problem 16.105.** Peter lists all $2^{100}$ sequences of 0s and 1s of length 100. On average, how many times does the fragment "00" occur in such a sequence? (Equivalently, find the expected number of occurrences of "00" in a random binary sequence of length 100.)

**Problem 16.106.** Ten boys and ten girls are arranged in a row in a random order. Find the expected number of pairs of adjacent boys.

**Problem 16.107.** Vasya shuffles a deck of 52 cards and draws cards one by one until he gets the third ace. What is the expected number of cards drawn?

**Problem 16.108.** Seven boys and thirteen girls stand in a row in a random order. Find the expected number of pairs of adjacent people consisting of a boy and a girl.

**Problem 16.109.** Fifteen eagles stand in a circle, facing the centre. Each of them pecks either its left or its right neighbour, the choice being made at random and independently, with probability 1/2 each. On average, how many eagles remain unpecked? On average, how many eagles are pecked twice?

**Problem 16.110.** A deck of 54 cards is shuffled. What is the expected number of cards lying between the two jokers?

**Problem 16.111.** In each cell of an $n \times n$ board a quarter-circle is drawn: it is centred at one of the four corners of the cell and joins the midpoints of the two sides meeting at that corner. The corner is chosen uniformly at random, independently for each cell. Find the expected number of complete circles formed on the board.

**Problem 16.112.** Each of the numbers $a_1, \ldots, a_n$ is chosen at random from $\{1, 2, \ldots, n\}$, independently and uniformly. Find the expected size of the set $\{a_1, \ldots, a_n\}$.

**Problem 16.113.** A set of 100 numbers is chosen uniformly at random from the set $\{1, \ldots, 2024\}$. Find the expected value of the smallest chosen number.

**Problem 16.114.** A country has $n$ cities and no roads. The government assigns costs to all possible roads at random, using each of the numbers $1, 2, \ldots, n(n-1)/2$ exactly once. Then each mayor selects and builds the cheapest road available from his own city; some roads are chosen by two mayors and some by only one. The resulting graph has $M$ connected components. Find the expected value of $M$.

**Problem 16.115.** Vasya has cookies labelled by all nonempty subsets of $\{1, 2, \ldots, n\}$. Each day he picks one of the remaining cookies uniformly at random and eats it together with all the cookies whose labels are subsets of its label. Find the expected number of days needed until all the cookies are eaten.

**Problem 16.116.** A team of $n$ schoolchildren takes part in a game: each of them is given a hat of one of $k$ predefined colours. Then, at a whistle, all the schoolchildren simultaneously choose a scarf for themselves. The team earns as many points as there are participants whose hat colour matches the colour of the scarf they chose. (There are enough scarves and hats of each colour; during the game each participant cannot see their own hat but sees the hats of all the others, and they are not allowed to communicate any information after the hats are distributed.)

(a) Find the expected number of points the team will score, and prove that no algorithm can guarantee more than $\lfloor n/k \rfloor$ points.

(b)* Prove that the schoolchildren can guarantee exactly $\lfloor n/k \rfloor$ points.

**Problem 16.117.** Marisa has a collection of the $2^8 - 1 = 255$ distinct nonempty subsets of $\{1, 2, 3, 4, 5, 6, 7, 8\}$. At each step she takes two subsets chosen uniformly at random from the collection and replaces them by either their union or their intersection, chosen at random with equal probability. (The collection is allowed to contain repeated sets.) She repeats this process $2^8 - 2 = 254$ times, until only one set is left in the collection. What is the expected size of this set?

**Problem 16.118.** In a plane there are $n$ seats ($n \geq 2$), each on a ticket of exactly one passenger. The first $n - 1$ passengers to enter the plane take their seats at random (independently of the seat number on their tickets). When the last passenger enters the board: if their seat is free they take it, otherwise they ask the passenger sitting on the taken seat to switch place to the one they have on their ticket. Then the other passenger stands up, goes to their seat and the procedure is repeated (they sit if their assigned seat is free and ask for switching seats otherwise). This goes on until someone sits on the empty place.

(a) What is the probability that the passenger who was the first to board the plane would have to switch seats?

(b) What is the average (expected) number of passengers who would have to switch seats?

**Problem 16.119.** We are given an $(n+1)$-floor building with floors numbered from 0 (ground floor) to $n$. A mathematician is in an elevator on the last floor and pursues the following strategy until he gets to the ground floor: if he is on the $k$-th floor, he chooses a lower floor (from 0 to $k-1$) at random and takes the elevator there.

(a) Find the average number of elevator rides.

(b) Find the average number of the last floor visited before the ground floor.

**Technique 16.18 (Averaging: existence without construction).** Sometimes the existence of a construction is proved indirectly: it is enough to show that the average value of the required quantity, taken over all admissible configurations, is already good enough, since some configuration must be at least as good as the average. Usually the hardest part is deciding to start solving the problem this way.

**Theorem 16.19 (A probabilistic bound for Ramsey numbers).** *Recall that the Ramsey number $R(k, \ell)$ is the smallest integer $n$ such that in any red-blue colouring of the edges of the complete graph on $n$ vertices there is either a complete red subgraph on $k$ vertices or a complete blue subgraph on $\ell$ vertices. If $\binom{n}{k} 2^{1-\binom{k}{2}} < 1$, then $R(k,k) > n$; consequently $R(k,k) > \left\lfloor 2^{k/2} \right\rfloor$ for every $k \geq 3$.*[†]

**Problem 16.120.** On an $8 \times 8$ chessboard, each square contains either a plus or a minus sign, with exactly 32 pluses and 32 minuses. Prove that it is possible to place 8 non-attacking rooks on the board so that at least 4 of them stand on pluses.

**Problem 16.121.** Two circles of radius 1 are each divided into 100 equal arcs. On each circle, 50 of the arcs are coloured blue and 50 are coloured red. Prove that the circles can be superimposed so that at least 50 pairs of overlapping arcs have the same colour.

**Problem 16.122.** Sasha has several light bulbs and several switches. Each switch is connected to some of the bulbs, and each bulb is connected to at least one switch. When a switch is pressed, all the bulbs connected to it change their state: bulbs that are off turn on, and bulbs that are on turn off. The initial state of the bulbs is unknown.

(a) Prove that by pressing some switches in succession one can always turn on at least half of the bulbs.

(b) Solve the same problem by induction on the number of switches.

**Problem 16.123.** In the setting of the previous problem, suppose there are exactly 99 bulbs and 50 switches, and each bulb is connected to exactly 25 of the switches. Prove that Sasha can press 17 switches so that at least 50 bulbs are turned on.

**Problem 16.124.** Engineer Sasha has several bulbs and switches. Each bulb is connected to either one or two switches. Sasha wants to remove some switches and some bulbs so that every remaining bulb is connected to exactly one remaining switch. Prove that Sasha can do this while keeping at least half of all the devices.

**Problem 16.125.** Prove that the numbers $1, 2, \ldots, 2016$ can be coloured with four colours in such a way that no monochromatic arithmetic progression of length 10 exists.

**Problem 16.126.** An $n \times n$ table is filled with the numbers $1, 2, \ldots, n$, each appearing exactly $n$ times. Prove that there is a row or a column containing at least $\sqrt{n}$ distinct numbers.

**Problem 16.127.** There are 2019 light bulbs, some pairs of which are connected by wires. Prove that it is possible to select two bulbs such that among the remaining bulbs at least 1009 are either connected to both of the selected bulbs or connected to neither of them.

**Problem 16.128.** Let $n$ be an even number. The squares of an $n \times n$ board are coloured with $n^2/2$ colours, exactly two squares of each colour. Prove that one can place $n$ rooks on squares of $n$ different colours so that no two rooks threaten each other.

**Problem 16.129.** In a school there are 90 students, each of whom is friends with at least 30 of the others. Prove that they can be divided into three classes of 30 students each so that everyone has at least one friend in his or her own class.

**Problem 16.130.** Let $v_1, v_2, \ldots, v_n \in \mathbb{R}^n$ be unit vectors, that is, $|v_i| = 1$ for all $i$. Prove that there is a choice of signs $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n \in \{-1, 1\}$ such that

$$|\varepsilon_1 v_1 + \varepsilon_2 v_2 + \cdots + \varepsilon_n v_n| \leq \sqrt{n}.$$

**Problem 16.131.** Consider a bipartite graph $G = (V, E)$ with $n$ vertices. Each vertex $v \in V$ is assigned a list $S(v)$ of more than $\log_2 n$ colours. Prove that there exists a proper colouring of $G$ in which each vertex $v$ receives a colour from its own list $S(v)$.

**Problem 16.132.** Baron Munchausen, returning from Kitay, claimed that a round-robin ping-pong tournament had been held there with an extremely balanced line-up: in any group of 1000 players there is at least one who has beaten all the others in that group. Is the baron exaggerating?

**Problem 16.133.** Several cities, regarded as points, are located in the plane. Any two cities at a distance of at most 1 are connected by a bus route. Each route has a positive real cost, and the sum of the costs of all routes is 100. Prove that the cities can be divided into several regions so that:

(a) the distance between any two cities in the same region is at most 1000;

(b) the total cost of the routes joining cities of different regions is at most 1.

**Problem 16.134.** A natural number $n$ is given. Arina has $k$ double-sided cards; each card contains all the numbers from 1 to $n$ exactly once, some of them on one side and the rest on the other side. What is the smallest $k$ for which Arina can always lay the cards out so that all the numbers from 1 to $n$ appear on the visible sides?

**Problem 16.135.** Several tattoo artists have gathered for a convention. Each person has a certain number of tattoos on each of their two arms. There are $n$ types of tattoos, and each of the $n$ types appears on the arms of at least 100 people. What is the largest $n$ for which it is guaranteed that each participant can raise one of their hands in such a way that all $n$ types of tattoos are represented on the raised hands?

**Problem 16.136.** Let $n$ and $t$ be natural numbers. A sports league assigns to each team at most $t$ of $n$ different colours, and each of the $n$ colours is assigned to at least one team. A set $S$ of teams is called *colour-separable* if each team of $S$ can choose one of the colours assigned to it so that this colour does not coincide with any of the colours assigned to any other team of $S$. What is the largest number of teams that is guaranteed to form a colour-separable set?

**Problem 16.137.** There are $n$ points in the plane. A pair of points is called *aligned* if the two points have the same $x$-coordinate or the same $y$-coordinate. Prove that it is possible to colour each of these points in such a way that

(a) all points of the same colour lie on a single straight line, and

(b) there are at most $n^{3/2}$ pairs of points that are aligned but have different colours.

**Problem 16.138.** Let $n \geq 2$ and let $a_1, a_2, \ldots, a_n$ be pairwise distinct integers. Prove that there exists an integer $x$, different from all the $a_i$, such that among the numbers $v_2(x-a_1), v_2(x-a_2), \ldots, v_2(x-a_n)$ at least $n/4$ are even and at least $n/4$ are odd. (Here $v_2(m)$ denotes the exponent of 2 in the prime factorization of $m$.)

## 16.6 Invariants III and processes

*Revisits invariants and monovariants (see Chapter 11); the problems below push it further.*

**Technique 16.20 (The global parameter).** Rather than following a process step by step, attach one number to the whole configuration — a total, a sum of residues, the length of a boundary, the value of a symmetric function — and ask only how that single number can change in one move. When only the set of numbers is known and not their order, Vieta's formulas suggest looking at symmetric polynomials of that set; the plain sum is often already insightful.

**Problem 16.139.** On a $100 \times 100$ field there are 99 squares that have been overrun by weeds. If a square is adjacent to two squares with weeds, it too will grow weeds. Is it possible for all the squares to have weeds? Two squares are considered adjacent if they share a common side.

**Problem 16.140.** A square table $3 \times 3$ is filled with numbers $0, 1, \ldots, 8$ in this order. In every step we can choose two adjacent cells and the numbers in them both decrease by the smaller of them, without changing the others. Can we obtain the zero-table after several steps?

**Problem 16.141.** Several real numbers are written on the board. Whenever there are two numbers $a$ and $b$ with $a - b \geq 100$, we can switch them to $a - 1$ and $b + 1$. Prove that the process will stop after finitely many steps.

**Problem 16.142.** A nonnegative integer is written on each vertex of a regular hexagon, and the sum of all the numbers is odd. Each number can be replaced by the absolute value of the difference of the two numbers written at the neighbouring vertices. Prove that all the numbers can be made equal to 0.

**Problem 16.143.** There is a board $100 \times 100$ and there is either a plus or a minus in each of its cells. In one move you can select any cross (i.e. the union of a row and a column) and change the signs in it to the opposite ones.

(a) With the help of the moves described, how can one get a board with only pluses in its cells?

(b) What is the smallest number of moves needed to achieve that no matter what the initial board is?

**Problem 16.144.** Some rectangle $R$ with odd sides has been divided into smaller rectangles. Prove that among those smaller rectangles there is at least one such that the distances from its sides to the sides of $R$ are of the same parity. [ISL 2017 P1]

**Problem 16.145.** Consider an infinite board with square cells, each containing a light bulb, which is initially switched off. In one move one can change the state (on/off) of all bulbs within a $3 \times 3$ or $4 \times 4$ square. Is it possible that after a finite number of moves there are only four light bulbs which are switched on, and they form a $2 \times 2$ square?

**Problem 16.146.** For which natural numbers $n$ is it possible to find a permutation $a_1, a_2, \ldots, a_n$ of the numbers from 1 to $n$ such that the numbers $a_1 - 1,\ a_2 - 2,\ \ldots,\ a_n - n$ give pairwise distinct remainders upon division by $n$?

**Problem 16.147.** On a $9 \times 9$ chessboard, 9 rooks are placed so that no two of them attack each other. Each rook is then moved in an L-shape, as a knight would move. Prove that after these moves there are at least two rooks attacking each other.

**Problem 16.148.** In a round-robin tournament with 15 teams, prove that there was at least one game before which the two teams playing it had participated in a combined odd number of games.

**Problem 16.149.** Two points $G$ and $H$ lie on opposite sides of the plane of a hexagon $ABCDEF$. Each of these two points is joined to all six vertices of the hexagon. Can the numbers $1, 2, 3, \ldots, 18$ be assigned to the resulting 18 edges, and real numbers be assigned to the eight points $A$, $B$, $C$, $D$, $E$, $F$, $G$, $H$, in such a way that the number on each edge equals the absolute value of the difference of the numbers assigned to its endpoints?

**Problem 16.150.** In the final standings of a round-robin chess tournament, can it happen that for each participant, the total number of points of the players he defeated is less than the total number of points of the players he lost to?

**Problem 16.151.** In a $10 \times 10$ square grid the numbers from 1 to 100 are written in order. Each move allows one to do one of the following:

(a) choose a number, add 2 to it and subtract 1 from two of its neighbours (the two neighbours being of the same type: both vertical, both horizontal, or both diagonal);

(b) choose a number, subtract 2 from it and add 1 to two of its neighbours (again of the same type).

In the end the grid again displays the numbers from 1 to 100. Prove that the final arrangement must coincide with the original one.

**Problem 16.152.** On a rectangular grid some cells are painted black. Then all cells are simultaneously repainted according to the following rule: a cell with an even number of black neighbours becomes white, and a cell with an odd number of black neighbours becomes black. (Neighbours are cells sharing a side.) Let $B_1, B_2, \ldots, B_r$ be a sequence of configurations of black cells in which $B_1$ turns into $B_2$, $B_2$ into $B_3$, and so on, with $B_r$ turning back into $B_1$. Prove that the total number of black cells in $B_1, B_2, \ldots, B_r$ is even.

**Problem 16.153.** On the board are written $2^m$ units. In one step we choose two numbers $a$ and $b$ and change both to their sum. Prove that after $m \cdot 2^{m-1}$ steps the sum of all numbers will be at least $4^m$.

**Problem 16.154.** We are initially given the polynomials $x, x^3, \ldots, x^{2k+1}, \ldots$ on the board. If in some moment we have $f(x)$ and $g(x)$, we may append the list with any of the polynomials $af(x)+b$ $(a, b \in \mathbb{R}, a \geq 0)$, $f(x)+g(x)$ or $f(g(x))$. Can we obtain the polynomial $x^{2011}-20x+11$?

**Problem 16.155.** There are $101^2$ light bulbs arranged in a square lattice $101 \times 101$. Initially, $k$ bulbs are on. In each step we choose four bulbs forming a unit square, and if three of the bulbs are on, we turn the fourth on as well. For which smallest $k$ is there an initial arrangement for which we can finally turn all the bulbs on?

**Problem 16.156.** The numbers $1, 2, \ldots, n$ are arranged on a circle in some order. Whenever $b$ is the right neighbor of $a$ and $b \leq a-2$, we can switch their positions. Prove that after at most $n^3$ switches no more moves will be possible.

**Problem 16.157.** There are 100 pirates sitting at a round table, with the captain holding all $n$ gold coins. At each step, if some pirates hold more than one coin, one of them will give a coin to each of his two neighbors.

(a) If $n \geq 100$, prove that the pirates will never stop.

(b) If $n < 100$, prove that they must stop at some moment.

**Problem 16.158.** On a rectangular board $m \times n$ we play a game with markers having one side black and the other side white. Initially, on each of the $mn$ squares there is a marker with the white side up, except for one marker at a corner with the black side up. A move consists of picking a black marker and turning all adjacent markers on the other side. For which $m$ and $n$ can we remove all markers?†

**Problem 16.159.** There are $2n-1$ light bulbs in a line. Initially the one in the middle is on and the others are off. At each step we can choose two nonadjacent bulbs that are off, under the condition that the bulbs between them must all be on, and change the state of all of them (thus, e.g. the configuration $\bullet\circ\circ\circ\bullet$ changes to $\circ\bullet\bullet\bullet\circ$). At most how many steps can we perform?

**Problem 16.160.** We have $n$ weights with masses less than 1, arranged in a sequence. A block is a set of several consecutive weights. Prove that, for every $k$ ($1 \le k \le n$), it is possible to partition the weights into $k$ (disjoint) blocks so that the total masses of any two blocks differ by at most 1.

**Problem 16.161.** ★ On an infinite one-dimensional board $n(n+1)/2$ coins are distributed among the squares. In one move we may take all the coins from one square and distribute them on the consecutive squares to the right of it, one coin per square, until no coin is left in the hand. Prove that after a finite number of moves the coins will be distributed on $n$ consecutive squares as follows: $n$ coins, $n-1$ coins, ..., 2 coins, 1 coin.

**Problem 16.162.** ★ We have $n$ coins placed on one square of an infinite one-dimensional grid. In each move we select two consecutive squares such that the one on the left has at least 2 coins more than the one on the right, and move a coin from the left square to the right square. The process continues until no more moves can be made. Show that the final configuration of coins is independent of the sequence of moves made.

**Problem 16.163.** ★ On a blackboard $n$ non-negative integers are written. One is allowed to select two numbers $x$ and $y$ from the board such that $x \ge y$ and replace them with $x-y$ and $2y$. Determine for which $n$-tuples of initial numbers it is possible to increase the number of zeros on the blackboard to $n-1$.

**Problem 16.164.** A group of $n$ children arranged in a circle plays a sharing game. In the beginning one child has $m$ candies while the remaining children have none, and in each move a child with at least two candies gives one candy to each of his two neighbours. The process continues until no more moves are possible.

(a) If $m = n$, for which values of $n$ will the game end?

(b) If $m < n$, show that the game ends regardless of the sequence of moves.

**Problem 16.165.** Let $n$ be an even positive integer. On the board $n$ real numbers are written. In a single move we can choose two of these numbers and replace each of them with their product. Prove that for every $n$ initial numbers one can in a finite number of moves obtain $n$ equal numbers on the board.

**Problem 16.166.** An even number of people are seated around a table. After a break, they are again seated around the same table, not necessarily in the same places. Prove that at least two persons have the same number of persons between them as before the break.

**Problem 16.167.** Let $a$, $b$, $c$ be fixed positive integers. There are $a+b+c$ ducks sitting in a circle, one behind the other. Each duck picks either rock, paper, or scissors, with $a$ ducks picking rock, $b$ ducks picking paper, and $c$ ducks picking scissors. A move consists of an operation of one of the following three forms:

- If a duck picking rock sits behind a duck picking scissors, they switch places.
- If a duck picking paper sits behind a duck picking rock, they switch places.

- If a duck picking scissors sits behind a duck picking paper, they switch places.

Determine, in terms of $a$, $b$, and $c$, the maximum number of moves which could take place, over all possible initial configurations.

**Problem 16.168.** ★ Let $n$ be a positive integer. Define a chameleon to be any sequence of $3n$ letters, with exactly $n$ occurrences of each of the letters $a$, $b$, and $c$. Define a swap to be the transposition of two adjacent letters in a chameleon. Prove that for any chameleon $X$, there exists a chameleon $Y$ such that $X$ cannot be changed to $Y$ using fewer than $3n^2/2$ swaps.

**Problem 16.169.** On one square of the infinite chessboard (covering the entire plane) there is a single checker. We can repeatedly perform the following move: if an occupied square $S$ shares sides with two unoccupied squares $S_1$ and $S_2$, then we can remove the checker from $S$ and put two new checkers on $S_1$ and $S_2$. Prove that there exists a finite set $F$ of squares with the following property: regardless of the performed moves there is always at least one checker on some square of $F$.

**Problem 16.170.** A car races along a circular track along which there are several gas stations, each with a fixed (non-replenishable) amount of gas. The total amount of gas at the stations is just large enough to complete one round trip along the track. Prove that there exists a gas station from which the car can start with an empty (and large enough) tank and complete a round trip, i.e. return to the starting station without running out of gas. Assume that the car consumes a fixed amount of gas per unit length.

**Problem 16.171.** ★ On a spherical planet $X$ there are $2n$ gas stations. Each station is paired with the gas station located at the diametrically opposite point of the planet. Each station has a fixed (non-replenishable) amount of gas. The arrangement of the stations and the amounts of gas at each station are such that it is possible for a car with an empty (and large enough) reservoir to start from any station and reach its paired station, filling up its reservoir with any gas it encounters along the way. For which $n$ does it follow that it is possible for the car to start from some gas station and visit all the other gas stations on the planet? Assume that the car consumes a fixed amount of gas per unit length.

## 16.7 Discharging

**Technique 16.21 (Discharging).** Charge each object of the configuration — a cell, a row, a line, a vertex, an edge, a face — with a fixed amount, chosen so that the total initial charge is known in advance. Then let the objects redistribute their charges by a fixed local rule ("every row and every column splits its charge equally among its shaded cells", "every edge gives its charge to the face on its east side"); the total charge does not change during the discharging. Finally bound the terminal charge of every single object locally. Comparing that local bound with the unchanged global total yields the desired inequality, or a contradiction. The whole art lies in choosing the initial charges and the redistribution rule.

**Problem 16.172.** In the KAUST library there were exactly $k$ empty bookshelves yesterday. Today in the morning, some books were rearranged in such a way that every shelf is not empty anymore. A book is called *boring* if the number of books on its current shelf is less than the number of books on its shelf before the rearrangement. Prove that there are at least $k+1$ boring books.

**Problem 16.173.** In a group of people the average number of friends a person has is strictly smaller than 3, but everyone has at least two friends. Prove that there exists a person who has exactly two friends and moreover one of these two friends has at most five friends.

**Problem 16.174.** A square is cut into triangles. Prove that at least two triangles have a whole side in common. (Note that this is not necessarily the case if a triangle is cut into triangles.)

**Problem 16.175.** A set of lines in the plane is in *general position* if no two are parallel and no three pass through the same point. A set of lines in general position cuts the plane into regions, some of which have finite area; we call these its *finite regions*. Prove that for all sufficiently large $n$, in any set of $n$ lines in general position it is possible to color at least $\sqrt{n}$ of the lines blue in such a way that none of its finite regions has a completely blue boundary.

## 16.8 Boards, tilings and domino counting

*Revisits colourings, markings and boards (see Chapter 11); the problems below push it further.*

**Theorem 16.22 (Three basic facts about domino tilings).** *Colour the cells of a region in chessboard fashion.* (Chessboard condition.) *If the region can be tiled with dominoes, then it has equally many black and white cells — but the converse fails.* (Hall condition.) *If the region can be tiled with dominoes, then every set $W$ of white cells has at least $|W|$ black neighbours in total; and when the numbers of black and white cells are equal, this condition is also sufficient, by Hall's theorem.* (Cut condition.) *If a line divides a tiled region into two parts each containing an even (respectively odd) number of cells, then that line cuts an even (respectively odd) number of dominoes in half.*

**Theorem 16.23 (Counting domino tilings).** *Let $A_n$ denote the number of tilings. For a $2 \times n$ rectangle tiled by dominoes, $A_0 = A_1 = 1$ and $A_{n+2} = A_{n+1} + A_n$, so the count is a Fibonacci number. For a $3 \times 2n$ rectangle tiled by dominoes, $A_0 = 1$, $A_1 = 3$ and $A_{n+2} = 4A_{n+1} - A_n$. For a $2 \times 2 \times n$ chimney built from $2 \times 2 \times 1$ bricks, $A_0 = A_1 = 1$ and $A_{n+2} = A_{n+1} + 2A_n$, whence $A_n = (2^{n+1} - (-1)^{n+1})/3$.*

**Problem 16.176.** At most how many squares on a chessboard $8 \times 8$ can we cut along both diagonals so that the chessboard does not fall apart?

**Problem 16.177.** Can you tile a $25 \times 25$ board with $3 \times 3$ and $4 \times 4$ squares?

**Problem 16.178.** There is a cube $4 \times 4 \times 4$. Prove that it is impossible to cover the three faces having a common vertex using $1 \times 3$ rectangles (no overlaps are allowed).

**Problem 16.179.** ★ On each square of a $3 \times n$ board there is a coin, black on one side and white on the other, placed with the black side facing up. In one move we select a square and flip all the coins lying on the squares adjacent to it. Two squares are adjacent if they have at least one common vertex; a square is not adjacent to itself. For which $n$ is it possible to reach a position in which all the coins have the white side facing up?

**Problem 16.180.** There is a board $12 \times 15$. What is the largest possible number of $1 \times 3$ rectangles one can place on the board in such a way that no two rectangles touch each other, even via a corner? [Soviet Union Olympiad]

**Problem 16.181.** Define the distance between two cells of a board as the minimal number of steps a king needs to get from one of them to the other. Several squares of a $100 \times 100$ board are colored so that there are no two cells such that the distance between them is exactly 15. What is the largest number of cells that could be colored?

**Problem 16.182.** A chessboard has been divided into dominoes. Call two dominoes knight-connected if it is possible to jump from one of them to the other with a knight. What is the minimal number of colours needed so that, no matter how the board is divided, one can always colour the dominoes so that no two dominoes of the same colour are knight-connected?

**Problem 16.183.** Let us call a block a staircase with three steps consisting of 12 unit cubes. Find all $n$ such that the cube $n \times n \times n$ can be divided into such staircases.[†]

**Problem 16.184.** There is a $10 \times 11$ board consisting of 110 unit cells. All of its cells are divided into pairs. Prove that among these 55 pairs there is one such that the length of the common border between this pair and any other one is at most 2.

**Problem 16.185.** In each cell of a $10 \times 10$ board one arrow is placed. Each arrow is pointing in one of the four directions $\{\uparrow, \rightarrow, \downarrow, \leftarrow\}$. Find the smallest number $n$ with the following property: it is always (regardless of the initial placement of the arrows) possible to remove at most $n$ arrows from the board in such a way that among the remaining ones no two are pointing at each other. Note: arrows are pointing at each other also if there are some other arrows or empty cells between them.

**Problem 16.186.** Given are integers $n$ and $k$ with $2 \leq k \leq n$. Determine the largest possible number of rooks that can be placed on an $n \times n$ chessboard in such a way that among any $k$ of them there were two sharing a row or a column.

**Problem 16.187.** Given is a $1 \times (2n + 1)$ board with some cells colored black. We call a rectangle consisting of whole cells *odd* if it contains an odd number of black cells. Determine the largest possible number of odd rectangles.

**Problem 16.188.** Consider a three-dimensional chessboard $n \times n \times n$. Let a *liftrook* be a piece moving in all three directions parallel to edges of the chessboard. What is the minimum number of liftrooks to be put on the chessboard to have every cell attacked by at least one liftrook?

**Problem 16.189.** A field has the shape of a checkerboard $41 \times 41$ square. A tank is concealed in one of the cells of the field. By one shot, a fighter airplane fires at one of the cells. If a shot hits the tank, then the tank moves to a neighboring cell of the field, otherwise it stays in its cell (the cells are neighbors if they share a side). The pilot has no information about the tank, and one needs to hit it twice. Find the least number of shots sufficient to destroy the tank for sure.

**Problem 16.190.** The cells of an $8 \times 8$ chessboard are numbered along the diagonals running from the top-left to the bottom-right, starting with 1 in the top-left corner and ending with 64 in the bottom-right corner. Petya placed 8 pieces on the board so that there is exactly one piece in each row and each column. He then rearranged the pieces so that each piece ended up on a cell with a larger number. Could it still happen that there is exactly one piece in each row and each column?

**Problem 16.191.** For which natural numbers $n$ is it possible to mark several cells of an $n \times n$ board so that

(a) each row and each column contains an even number of marked cells, and

(b) each of the $4n-6$ diagonals of length greater than one contains an odd number of marked cells?

**Problem 16.192.** A $(2n+1)\times(2n+1)$ board is coloured in the chessboard fashion. In one move one is allowed to select a rectangle of squares of the board and reverse the colours of all the squares in it. Find the minimum number of moves needed to make the entire board be of the same colour.

**Problem 16.193.** ★ Two unit squares of a board are *adjacent* if they share a common edge. A *snake* is a collection of $k$ unit squares $A_1A_2\ldots A_k$ such that $A_i$ and $A_j$ are adjacent if and only if $|i-j|=1$; the squares $A_1$ and $A_k$ are the *ends* of the snake. A $(2n+1)\times(2n+1)$ board is partitioned into two snakes. Prove that an end of one of the snakes is the central square of the board.

**Problem 16.194.** ★ A coin is placed on every square of a $2020\times 2020$ board. In each move we remove a coin which is adjacent to a positive even number of coins, and moves are made as long as at least one such coin exists. Find $k$, the lowest possible number of coins that can remain on the board, and prove that if we end up with $k$ coins then no two of them are adjacent. Two squares are adjacent if they have at least one common vertex.

**Problem 16.195.** ★ In every square of a $2n\times 2n$ board there is a lightbulb that can be on or off. In each step one can choose a row or a column in which at least $n$ lightbulbs are on and change the state of all the lightbulbs in this row or column.

(a) Prove that there is a starting arrangement of bulbs such that no matter how we play there will always be at least $2n(n-1)$ lightbulbs that are on.

(b) Prove that we can always reach a state with at most $2n(n-1)$ lightbulbs on.

†

**Problem 16.196.** An $n\times n$ toroidal chessboard is an $n\times n$ chessboard whose top and bottom edge are "glued together," as well as the left and the right edge. (For example, a queen that stands on the $(3,2)$ square of such a board can move to any square to which it could move on a regular chessboard, but also to some other squares, such as $(5,n)$, $(6,n-1)$, …, since the sequence $(3,2)$, $(4,1)$, $(5,n)$, $(6,n-1)$, … consists of diagonally neighboring squares.) Find all positive integers $n$ for which it is possible to place $n$ mutually non-attacking queens on a toroidal $n\times n$ chessboard.

**Problem 16.197.** Determine the maximum number of queens that can be placed on a $2017\times 2017$ chessboard so that each queen attacks at most one of the others. [Serbian MO 2017]

**Problem 16.198.** ★ Consider two separate tilings of a $2n\times 2n$ board: one with red dominoes and the other with blue dominoes. We say that two squares are red neighbours if they are covered by the same red domino in the red tiling, and we similarly define blue neighbours.

Suppose we can assign a non-zero integer to each of the squares such that the number on any square equals the difference between the numbers on its red and blue neighbours, i.e. the number on its red neighbour minus the number on its blue neighbour. Show that $n$ is divisible by 3.

**Problem 16.199.** An $n\times n$ square is divided into unit squares. Let a unit side be a side of a unit square. For which $n$ can we cover the board with isosceles right-angled triangles with legs of length $\sqrt{2}$ and hypotenuse of length 2, so that the vertices of all the triangles lie on the vertices of unit squares and all unit sides lie in the interior or on the boundary of exactly one triangle?†

**Problem 16.200.** On an infinite square grid we place finitely many cars, which each occupy a single cell and face in one of the four cardinal directions. Cars may never occupy the same cell. It is given that:

(a) the cell immediately in front of each car is empty, and

(b) no two cars face towards each other (no right-facing car is to the left of a left-facing car within a row, etc.).

In a move, one chooses a car and shifts it one cell forward to a vacant cell. Prove that there exists an infinite sequence of valid moves using each car infinitely many times. Is an infinite sequence of moves where each car is used infinitely many times still possible if we drop the first condition?

**Problem 16.201.** ★ Let $n \geq 2$ be a natural number. An $n \times n$ board is filled in with integers from 1 to $n^2$ such that each number is used exactly once. Two fields are considered adjacent if they have a common edge. It is known that two numbers in adjacent fields differ by at most $n$. Prove that there is a $2 \times 2$ subset of the board such that the pairs of diagonally opposite numbers in the subset have the same sum.

**Problem 16.202.** ★ Given positive integers $n$ and $k$ with $n > k^2 > 4$. In an $n \times n$ grid, a $k$-group is a set of $k$ unit squares lying in different rows and different columns. Determine the maximal possible $N$ such that one can choose $N$ unit squares in the grid and colour them with the following condition holding: in any $k$-group from the coloured $N$ unit squares there are two squares with the same colour, and there are also two squares with different colours.

**Problem 16.203.** ★ A snake of length $k$ is an animal which occupies an ordered $k$-tuple $(s_1, \ldots, s_k)$ of cells in an $n \times n$ grid of square unit cells. These cells must be pairwise distinct, and $s_i$ and $s_{i+1}$ must share a side for $i = 1, \ldots, k-1$. If the snake is currently occupying $(s_1, \ldots, s_k)$ and $s$ is an unoccupied cell sharing a side with $s_1$, the snake can move to occupy $(s, s_1, \ldots, s_{k-1})$ instead. The snake has turned around if it occupied $(s_1, s_2, \ldots, s_k)$ at the beginning, but after a finite number of moves occupies $(s_k, s_{k-1}, \ldots, s_1)$ instead.

Determine whether there exists an integer $n > 1$ such that one can place some snake of length $0.9n^2$ in an $n \times n$ grid which can turn around.

**Problem 16.204.** Let $M$ be a finite set of lattice points and $n$ be a positive integer. A mine-avoiding path is a path of lattice points with length $n$, beginning at $(0, 0)$ and ending at a point on the line $x + y = n$, that does not contain any point in $M$. Prove that if there exists a mine-avoiding path, then there exist at least $2^{n-|M|}$ mine-avoiding paths.

**Problem 16.205.** ★ From an $n \times n$ sheet of stamps, we must tear out blocks of three adjacent stamps in a single row or column. (We can only tear along the perforations separating adjacent stamps, and each block must come out of the sheet in one piece.) Let $b(n)$ be the smallest number of blocks we can tear out and make it impossible to tear out any more blocks. Prove that there are real constants $c$ and $d$ such that

$$\frac{1}{7}n^2 - cn \leq b(n) \leq \frac{1}{5}n^2 + dn$$

for all $n > 0$.

**Problem 16.206.**

(a) Find all integers $n \geq 2$ with the following property: each cell of an $n \times n$ board can be painted black or white in such a way that every cell has the same color as precisely two adjacent cells. (Two cells are adjacent if they share exactly one side.)

(b) Find all integers $n \geq 2$ with the following property: each cell of an $n \times n$ board can be painted black or white in such a way that all border cells are black, each $2 \times 2$ square contains at least one cell of both colors, but no $2 \times 2$ square is colored like a chessboard (i.e. with two opposite corners white and two opposite corners black).

**Problem 16.207.** A *maze* on a board is a closed loop visiting each cell exactly once. There exists a maze on a $6 \times 6$ board with the following property: in order to traverse the maze once, one has to perform equally many steps in each of the four directions. Can you design a maze with this property on an $8 \times 8$ board?

**Problem 16.208.** Calculate the number of ways to build a $2 \times 2 \times n$ chimney from $1 \times 2 \times 2$ bricks.

**Problem 16.209.** Prove that for every $n \geq 1$ the number of ways to build a $2 \times 2 \times 2n$ chimney from $1 \times 1 \times 2$ bricks is a perfect square.

**Problem 16.210.** Prove that for every $n \geq 1$ the number of domino tilings of an $n \times (n+1)$ rectangle is odd.

**Problem 16.211.** Let $n \geq 2018$ be an integer. A board of size $3n \times 3n$ has been tiled with rectangular tiles of size $3 \times 1$ (which may be rotated) so that every cell of the board is covered by exactly one tile. Prove that one may color the cells of the board in three colors so that:

(a) there is the same number of cells of each color;

(b) no two cells of the same color have a common side; and

(c) no tile covers cells of all three colors.

**Problem 16.212.** Let $n$ be a positive integer. Determine the smallest positive integer $k$ with the following property: it is possible to mark $k$ cells on a $2n \times 2n$ board so that there exists a unique partition of the board into $1 \times 2$ and $2 \times 1$ dominoes, none of which contain two marked cells. [ISL 2016 C8]

## 16.9 Arrays, tables and extremal set theory

**Theorem 16.24 (Erdos–Szekeres).** *Let $m$ and $n$ be positive integers. Among any $mn+1$ people of pairwise different heights standing in a row there are $m+1$ whose heights increase from left to right, or $n+1$ whose heights decrease from left to right.*

**Problem 16.213.** ★ There are $n$ boys and $n$ girls in a school class, where $n$ is a positive integer. The heights of all the children in this class are distinct. Every girl determines the number of boys that are taller than her, subtracts the number of girls that are taller than her, and writes the result on a piece of paper. Every boy determines the number of girls that are shorter than him, subtracts the number of boys that are shorter than him, and writes the result on a piece of paper. Prove that the numbers written down by the girls are the same as the numbers written down by the boys (up to a permutation).

**Problem 16.214.** Let the set of integers from 1 to $2n$ be partitioned into two sets of $n$ numbers. Let $a_1 > a_2 > \cdots > a_n$ be the numbers of the first set and $b_1 < b_2 < \cdots < b_n$ be the numbers of the second set. Show that the expression

$$|a_1 - b_1| + |a_2 - b_2| + \cdots + |a_n - b_n|$$

does not depend on the choice of partition, and find the expression as a function of $n$.

**Problem 16.215.** A board $n \times n$ is filled with real numbers such that for each selection of $n$ fields for which no two belong to the same row or column the sum of all elements is the same (as for any other such selection). Let $A$ be the largest among the smallest numbers of each row and let $B$ be the smallest number among the largest numbers of each column. Show that $A = B$.

**Problem 16.216.** Boys and girls, $n^2$ of them in total, are placed in an $n \times n$ array. For each row or column, as well as each of the $2(2n-1)$ diagonals, we know the numbers of girls. For which positions in the array are these information always sufficient to determine whether a boy or a girl occupies it?

**Problem 16.217.** Find all positive integers $n$ with the following property: there exists a permutation $(a_1, a_2, \ldots, a_n)$ of the set $\{1, 2, \ldots, n\}$ such that $a_1 + a_2 + \cdots + a_k$ is divisible by $k$ for $k = 1, 2, \ldots, n$.

**Problem 16.218.** Let $n$ be a positive integer. How many permutations $(a_1, a_2, \ldots, a_n)$ of the set $\{1, 2, \ldots, n\}$ have the following property: for each index $k = 1, 2, \ldots, n$ the number $2(a_1 + a_2 + \cdots + a_k)$ is divisible by $k$?

**Problem 16.219.** Find all positive integers $n$ with the following property: there exists a permutation $(a_0, a_1, \ldots, a_{n-1})$ of the set $\{0, 1, 2, \ldots, n-1\}$ such that the residues of

$$a_0, \quad a_0 + a_1, \quad a_0 + a_1 + a_2, \quad \ldots, \quad a_0 + a_1 + a_2 + \cdots + a_{n-1}$$

modulo $n$ form a permutation of this set as well.

**Problem 16.220.** Let $n$ be a positive integer. Prove that the residues of

$$0, \quad 0+1, \quad 0+1+2, \quad \ldots, \quad 0+1+2+\cdots+(n-1)$$

modulo $n$ form a permutation of the set $\{0, 1, 2, \ldots, n-1\}$ if and only if $n$ is a power of two.

**Problem 16.221.** ★ Let ← denote the left arrow key on a standard keyboard. If one opens a text editor and types the keys "ab←cd←←e←←f", the result is "faecdb". We say that a string $B$ is reachable from a string $A$ if it is possible to insert some amount of ←'s in $A$, such that typing the resulting characters produces $B$. So, our example shows that "faecdb" is reachable from "abcdef".

Prove that for any two strings $A$ and $B$, $A$ is reachable from $B$ if and only if $B$ is reachable from $A$.

**Problem 16.222.** ★ There are $n$ students competing in 6 competitions to enter the National Olympiad team of 6 students. In each competition the students are ordered from the highest to the lowest score and there are no ties. The committee will choose the students by randomly selecting a permutation of competitions and then in order select from each competition the highest rated student that has not yet qualified. Is there a configuration of outcomes such that for each student there is at least one permutation in which the student qualifies for the team, for

(a) $n = 13$,

(b) $n = 14$?

**Problem 16.223.** ★ In a football league, there are $n \geq 6$ teams. Each team has a homecourt jersey and a road jersey with different colour. When two teams play, the home team always wears its homecourt jersey and the road team wears its homecourt jersey if the colour is different from the home team's homecourt jersey, or otherwise the road team shall wear its road jersey. It is required that in any two games with 4 different teams, the 4 teams' jerseys have at least 3 different colours. Find the least number of colours that the $n$ teams' $2n$ jerseys may use.

**Problem 16.224.** ★ Prove that for arbitrary positive integer $n \geq 4$, there exists a permutation of the subsets that contain at least two elements of the set $G_n = \{1, 2, 3, \ldots, n\}$:

$$P_1, P_2, \ldots, P_{2^n-n-1}$$

such that $|P_i \cap P_{i+1}| = 2$ for $i = 1, 2, \ldots, 2^n - n - 2$.

**Problem 16.225.** ★ Let $n \geq 2$ be an integer. Let $2n^2$ contestants participate in a chess competition, where each pair of contestants plays exactly once. A game can end in either a win for one player over the other or a draw. It is known that:

- if $A$ wins $B$ and $B$ wins $C$, then $A$ wins $C$;
- there are at most $\dfrac{n^3}{16}$ draws.

Prove that it is possible to choose $n^2$ contestants and label them $P_{ij}$ $(1 \leq i, j \leq n)$, such that for any $i, j, i', j' \in \{1, 2, \ldots, n\}$, if $i < i'$, then $P_{ij}$ wins $P_{i'j'}$.

**Problem 16.226.** During each of $n$ days every one of seven dwarves either goes into the forest to collect berries, or stays at home, or works in the mine. It is known that for any three dwarves there is a day on which they perform three different activities. Find the least possible value of $n$.†

**Problem 16.227.** Can all positive integers be colored with 5 colors in such a way that in every 5-tuple $(n, 2n, 3n, 4n, 5n)$, where $n$ is a positive integer, every color appears exactly once?

**Problem 16.228.** ★ The positive integer points on a line are coloured with one of two colours. Show that there exists an infinite sequence of natural numbers $a_1 < a_2 < a_3 < \cdots$ such that the points

$$a_1,\ \frac{a_1 + a_2}{2},\ a_2,\ \frac{a_2 + a_3}{2},\ a_3,\ \frac{a_3 + a_4}{2},\ \ldots$$

are all of the same colour.

**Problem 16.229.** ★ Let $n$ be a given positive integer and let $f$ be the function on pairs of non-negative integers determined by

$$f(0, k) = f(k, 0) = 0 \quad (k \in \mathbb{N}_0), \qquad f(1, 1) = n, \qquad f(1, i) = 0 \quad (i \geq 2),$$

$$f(k, i) = \left\lfloor \frac{f(k-1, i)}{2} \right\rfloor + \left\lfloor \frac{f(k-1, i-1)}{2} \right\rfloor \qquad (k > 1,\ i \in \mathbb{N}).$$

Find the number of pairs $(x, y)$ for which $f(x, y)$ is odd.

**Problem 16.230.** ★ Given a set $A$ with $n^2$ elements, where $n \geq 2$, let $F$ be a collection of subsets of $A$, each with $n$ elements, such that each two subsets of the collection share at most one common element. Prove that $|F| \leq n^2 + n$.

**Problem 16.231.** Let a finite set of integers be given, such that each of its elements can be written as a sum of some two elements (not necessarily different) from the same set. For such a set we say that it is of safety of order $n$ if it does not contain a subset of $n$ or less elements whose sum is 0. Prove that there exist sets of arbitrarily large safety order.

**Problem 16.232.** ★ Let $n$ be a positive integer. Suppose we are given $2^n + 1$ distinct sets, each containing finitely many objects. Place each set into one of two categories, the red sets and the blue sets, so that there is at least one set in each category. We define the symmetric difference of two sets as the set of objects belonging to exactly one of the two sets. Prove that there are at least $2^n$ different sets which can be obtained as the symmetric difference of a red set and a blue set.

**Problem 16.233.** ★ Suppose $A_1, A_2, \ldots, A_n \subseteq \{1, 2, \ldots, m\}$ and $|A_i| = 2$, $i = 1, 2, \ldots, n$, satisfying that $A_i + A_j$, $1 \leq i \leq j \leq n$, are distinct from each other, where $A + B = \{a + b \mid a \in A,\ b \in B\}$. Determine the maximal value of $n$.

**Problem 16.234.** ★ Let $A$ be a finite set of positive numbers and let

$$B = \left\{ \frac{a+b}{c+d} \;\middle|\; a, b, c, d \in A \right\}.$$

Show that $|B| \geq 2|A|^2 - 1$, where $|X|$ denotes the number of elements of the finite set $X$.

## 16.10 Combinatorial geometry II

*Revisits combinatorial geometry (see Chapter 11); the problems below push it further.*

**Problem 16.235.** There is a rectangular box that is chock-a-block filled with wooden cubes. Prove that two of the cubes are of the same size.

**Problem 16.236.** A unit square is partitioned into rectangles. From each rectangle we select the smaller of its two sides, or any side if the rectangle is a square. Prove that the sum of the selected sides is at least 1.

**Problem 16.237.** A unit square is partitioned into triangles. Prove that the sum of the perimeters of the triangles is at least $4 + 2\sqrt{2}$. When does equality hold?

**Problem 16.238.** Inside a square, 10 points are marked. Some of these points are joined to the vertices of the square and to each other in such a way that the square is divided into triangles, and all 10 marked points are vertices of the resulting triangles. Find the total number of triangles.

**Problem 16.239.** The plane is divided by vertical and horizontal lines into square cells. A convex polygon $M$ is drawn so that all of its vertices are grid points and none of its sides is vertical or horizontal. Prove that the total length of the vertical grid segments lying inside $M$ equals the total length of the horizontal grid segments lying inside $M$.

**Problem 16.240.** A rectangular sheet of paper has $n$ rectangular holes cut out of it, all of whose sides are parallel to the edges of the sheet. (The holes do not overlap and do not touch

each other.) What is the smallest number of rectangular pieces into which the sheet can always be cut?

**Problem 16.241.** Let $\mathcal{I}$ be a finite collection of intervals on a line whose union has total length $d$. Prove that it is possible to select from $\mathcal{I}$ a subcollection of pairwise disjoint intervals covering a total length of at least

(a) $d/5$;

(b) $d/2$.

**Problem 16.242.** In each vertex of a triangulated $n$-gon an odd number of triangles meet. Prove that $n$ is divisible by 3.

**Problem 16.243.** On a line there are $n$ marked points, whose all mutual distances are integers.

(a) Suppose that $n$ is odd and each marked point has an even sum of distances from the remaining $n-1$ marked points. Prove that every segment with both endpoints marked is of even length.

(b) For each point calculate the sum of distances from the remaining $n-1$ marked points. Find all $n$ for which these sums can be $n$ consecutive integers (in some order).

(c) Suppose that $n$ is even. What is the maximum number of segments of odd length with both endpoints marked?

**Problem 16.244.** Vertices of a regular $2n$-gon are given. How many acute triangles have all three vertices in these points?

**Problem 16.245.** Given are vertices of a regular $2n$-gon. Consider all triangles with vertices in these points. Prove that the total area of all acute triangles is equal to the total area of all obtuse triangles.

**Problem 16.246.** Given is a positive integer $n$. How many squares have both coordinates of all vertices in the set $\{0, 1, 2, \ldots, n\}$?

**Problem 16.247.** The diagonals of a convex 2024-gon are colored in $k$ colors such that no two diagonals of the same color intersect. Find the minimal possible value of $k$.

**Problem 16.248.** Some diagonals of a convex $n$-gon are drawn. A drawn diagonal is called interesting if it intersects (in an inner point) exactly one of the drawn diagonals. Find the maximum possible number of interesting diagonals.

**Problem 16.249.** A square is partitioned into $n^2 \geq 4$ rectangles using $2(n-1)$ lines, $n-1$ of which are parallel to one side of the square and $n-1$ are parallel to the other side. Prove that we can choose $2n$ rectangles of the partition such that, for each two of them, we can place the one inside the other (possibly with rotation).

**Problem 16.250.** Let $n > 3$ straight, infinitely long lines, no two of which are parallel and no three of which intersect in one point, divide the plane into sectors. The *rank* of a sector is the number of the given lines that it touches, and let $m$ be the maximum rank among all sectors. Let $S_k$ be the number of sectors of rank $k$. Prove that

$$S_3 \geq 4 + S_5 + S_6 + S_7 + \cdots + S_m.$$

Can you find examples where equality holds and both sides are larger than 4?

**Problem 16.251.** ★ Show that among any $n$ points in the plane there are at least

$$\sqrt{n-\frac{3}{4}}-\frac{1}{2}$$

different distances between pairs of points.

**Problem 16.252.** In the plane given are $n$ unit circles, no two of them are tangent and no circle is disjoint with all other circles. Every intersection point of at least two of those circles is colored red. Prove that there are at least $n$ red points.

**Problem 16.253.** In the plane given are $3n-1$ points, no three collinear, where $n>1$. Prove that one can choose $2n$ points whose convex hull is not a triangle.

**Problem 16.254.** ★ There are 2015 distinct circles in a plane, with radius 1. Prove that you can either select a subset of 27 circles such that each pair of circles within the subset intersects each other, or that you can select a subset of 27 circles such that no two circles within the subset intersect each other.

**Problem 16.255.** A plane has a special point $O$ called the origin. Let $P$ be a set of 2021 points in the plane such that:

- No three points in $P$ lie on a line, and
- No two points in $P$ lie on a line through the origin.

A triangle with vertices in $P$ is fat if $O$ is strictly inside the triangle. Find the maximum number of fat triangles.

**Problem 16.256.** ★ A collection of $n$ squares on the plane is called tri-connected if the following criteria are satisfied:

- All the squares are congruent.
- If two squares have a point $P$ in common, then $P$ is a vertex of each of the squares.
- Each square touches exactly three other squares.

How many positive integers $n$ are there with $2018 \leq n \leq 3018$, such that there exists a collection of $n$ squares that is tri-connected?

**Problem 16.257.** ★ There is a grid of equilateral triangles with a distance 1 between any two neighbouring grid points. An equilateral triangle with side length $n$ lies on the grid so that all of its vertices are grid points, and all of its sides match the grid. Now, let us decompose this equilateral triangle into $n^2$ smaller triangles (not necessarily equilateral triangles) so that the vertices of all these smaller triangles are all grid points, and all these small triangles have equal areas. Prove that there are at least $n$ equilateral triangles among these smaller triangles.

**Problem 16.258.** ★ Let $n$ be a positive integer. There are given $n$ lines such that no two are parallel and no three meet at a single point.

(a) Prove that there exists a line such that the number of intersection points of these $n$ lines on each of its sides is at least $\left\lfloor \frac{(n-1)(n-2)}{10} \right\rfloor$. Notice that the points on the line are not counted.

(b) Find all $n$ for which there exists a configuration where the equality is achieved.

**Problem 16.259.** Prove that, for every $n \geq 3$, it is possible to choose $n$ points in the plane in such a way that the distance between every two distinct points is irrational, while every three points are vertices of a (nondegenerate) triangle of rational area. [IMO 1987]

**Problem 16.260.** Prove that it is impossible to cut a convex polygon into finitely many nonconvex quadrilaterals.

**Problem 16.261.** Prove that every convex polygon can have at most one triangulation in which all triangles are acute.

**Problem 16.262.** Let $P$ be a regular 1000-gon. A diagonal is called *good* if its endpoints divide the boundary of $P$ into two parts, each composed of an odd number of sides of $P$. The sides of $P$ are also called good. Suppose $P$ has been dissected into triangles by 997 diagonals, no two of which have a common point in the interior of $P$. Find the maximum number of isosceles triangles having two good sides that could appear in such a configuration. [IMO 2006]

**Problem 16.263.** Show that any triangle can, for any integer $n \geq 6$, be divided into $n$ quadrilaterals that are both cyclic and tangential (i.e. have both a circumcircle and an incircle) such that no two of them are mutually congruent.

**Problem 16.264.** ★ Does there exist an $n$-gon $M_0$ such that:

- one can cover the boundary of the $n$-gon with copies of the $n$-gon $M_1, M_2, \ldots, M_k$ such that no two $n$-gons $M_i, M_j$, $i \neq j$, $0 \leq i, j \leq k$, have a common interior point, each $M_i$, $i = 1, \ldots, k$, has at least one common boundary with $M_0$, and $M_0$ is located strictly in the interior of the union of $M_i$, $i = 0, 1, \ldots, k$;
- it is not possible to tile the plane with copies of $M_0$?

**Problem 16.265.** On a circle of length $2^n$ ($n \geq 2$) given are points $P_0, P_1, \ldots, P_{2^n-1}$ which are vertices of a regular $2^n$-gon, denoted in such a way that the length of the arc $P_{k-1}P_k$ (measured clockwise) is equal to $k$ for $k = 1, 2, \ldots, 2^n - 1$. Let

$$E = \{P_k : 2 \mid k\} \qquad \text{and} \qquad F = \{P_k : k < 2^{n-1}\}.$$

Prove that sets $E$ and $F$ are congruent (treated as plane figures).

## 16.11 Games and information

*Revisits games and strategies (see Chapter 11); the problems below push it further.*

At this level the games are rarely about a winning position and almost always about information: how much of it a strategy can transmit, and how little an adversary can be allowed to learn.

**Problem 16.266.** There is a complete graph on 35 vertices. Two players take turns removing the edges of a simple path from it. The player who cannot make a move loses. Which player has a winning strategy?

**Problem 16.267.** A point $X$ is chosen within a convex 100-gon such that $X$ is neither on the sides nor on the diagonals of the polygon. Initially the vertices of the polygon are not marked.

Pete and Bazil alternately mark the vertices in turns. Pete starts and marks 2 vertices on his first turn. Then, on each turn, each player marks a vertex. If, after one of the players moves, the point $X$ appears inside a polygon with all vertices marked, then that player loses the game. Prove that Pete has a winning strategy.

**Problem 16.268.** Mustafa and Zubair play the following game on the $100 \times 100$ chessboard. Mustafa chooses an empty cell and then Zubair places a $1 \times 2$ domino covering this cell and one of the neighboring (free) cells. The game ends when one of the players cannot perform a valid move. Mustafa wins if the entire board is covered by dominoes, otherwise Zubair wins. Who has the winning strategy?

**Problem 16.269.** For any integers $m \geq n \geq 2$ consider the following game. Initially given are two piles consisting of $m$ and $n$ stones, respectively. Two players alternate in moves, each move consisting of taking from one pile a nonzero integral multiple of the number of stones in the other pile. The first player to empty one of the piles wins. Who (in terms of $m$, $n$) has the winning strategy?

**Problem 16.270.** Consider the following game. There are $n$ dominoes in a row, each slightly tilted to the right or to the left, for example $/\,\backslash\,/\,\backslash\,\backslash\,/$. In a move, one domino tile can be pushed in the direction in which it is tilted, which topples this domino together with all the dominoes in the direction of the tilt (independent of the direction in which these dominoes are tilted themselves); for example, pushing the fourth tile of $/\,\backslash\,/\,\backslash\,\backslash\,/$, which is tilted to the left, produces $\backslash\,\backslash\,/$. The first player who cannot make a move loses. For a given $n$, determine the number of positions with $n$ dominoes for which the first player has a winning strategy.

**Problem 16.271.** Let $k$ be a positive integer. Given are two boards $1 \times (k+1)$, each having $k!$ pawns on the first (starting) cell. Alice and Bob play the following game. Alice in each move chooses two pawns, one from each board, and moves each of them by 1 square. Bob in his move destroys one pawn of his choice (from only one board). Alice wins if she can reach the last square with at least one pawn on at least one board, Bob wins if he destroys all pawns. Who has the winning strategy?[†]

**Problem 16.272.** In space given are 2022 points, no three collinear, no four coplanar. Two players alternate in joining pairs of these points with segments. A player loses the game if after their move the following property holds: for every point $v$, any two of the remaining points can be joined with a broken line consisting of drawn segments and not passing through $v$. Who has the winning strategy?

**Problem 16.273.** ★ Let $k$ be a positive integer. Two players $A$ and $B$ play a game on an infinite grid of regular hexagons. Initially all the grid cells are empty. Then the players alternately take turns with $A$ moving first. In his move, $A$ may choose two adjacent hexagons in the grid which are empty and place a counter in both of them. In his move, $B$ may choose any counter on the board and remove it. If at any time there are $k$ consecutive grid cells in a line all of which contain a counter, $A$ wins. Find the minimum value of $k$ for which $A$ cannot win in a finite number of moves, or prove that no such minimum value exists.

**Problem 16.274.** Integers $n$ and $k$ are given, with $n \geq k \geq 2$. You play the following game against an evil wizard.

The wizard has $2n$ cards; for each $i = 1, \ldots, n$, there are two cards labelled $i$. Initially, the wizard places all cards face down in a row, in unknown order.

You may repeatedly make moves of the following form: you point to any $k$ of the cards. The wizard then turns those cards face up. If any two of the cards match, the game is over and you

win. Otherwise, you must look away, while the wizard arbitrarily permutes the $k$ chosen cards and then turns them back face-down. Then, it is your turn again.

We say this game is winnable if there exist some positive integer $m$ and some strategy that is guaranteed to win in at most $m$ moves, no matter how the wizard responds. For which values of $n$ and $k$ is the game winnable?

**Problem 16.275.** ★ Let $n$ be a positive integer. Players $A$ and $B$ are given a circular necklace with $3n$ blue beads and $3n$ green beads, such that no three consecutive beads have the same colour. They play a cooperative game where they alternate turns removing three consecutive beads, subject to the following conditions:

- Player $A$ must remove three consecutive beads which are blue, green and blue, in that order, on each of his turns.
- Player $B$ must remove three consecutive beads which are green, blue and green, in that order, on each of his turns.

They win if all the beads are removed in $2n$ turns. Prove that if they can win with player $A$ going first, they can also win with player $B$ going first.

**Problem 16.276.** Ana and Banana are playing a game. First Ana picks a word, which is defined to be a nonempty sequence of capital English letters. (The word does not need to be a valid English word.) Then Banana picks a nonnegative integer $k$ and challenges Ana to supply a word with exactly $k$ subsequences which are equal to Ana's word. The subsequences of a string of length $n$ are the $2^n$ strings which are formed by deleting some of its characters, possibly all or none, while preserving the order of the remaining characters. Ana wins if she is able to supply such a word, otherwise she loses.

For example, if Ana picks the word "TST", and Banana chooses $k = 4$, then Ana can supply the word "TSTST" which has 4 subsequences which are equal to Ana's word.

Which words can Ana pick so that she wins no matter what value of $k$ Banana chooses?

**Problem 16.277.** A traveler visits a conference of wise men on an island solely inhabited by liars and truth-tellers. Liars always lie, while truth-tellers always tell the truth. At this conference, $n$ wise men are seated at a round table with seats labelled from 1 to $n$ clockwise (from above). The traveler goes around and interviews the wise men. For each $i$, $i = 1, 2, \ldots, n$, the wise man at the $i$-th seat told the traveler that the wise men seated in the next $i$ seats in the clockwise direction were all liars.

(a) For which values of $n$ was it possible for the wise men to give out these statements?

(b) Among the solutions to the previous part, for which values of $n$ will the traveler still be unable to determine with certainty for each wise man whether they are a liar or a truth-teller?

(c) Let $n = 2022$. Identify the seats where truth-tellers sit.

**Problem 16.278.** There is a round table with 30 people sitting, some of which are smart and some are stupid. We only know that at most $G$ of them are stupid. We ask each of them whether their neighbor to the right is smart or stupid. The smart will tell us the truth, but the stupid will give us random answers. For which largest $G$ will we always be able to identify at least one smart person?

**Problem 16.279.**

(a) Two players perform a trick with cards. First player is given five random cards. Upon seeing the cards, he places them on the table in a line so that four cards are face-up and the fifth is face-down. The second player then guesses the hidden card. Prove that they can make a plan so that the trick always succeeds.

(b) The same question, but now the first player shows off four cards and does not place the fifth card at all.

**Problem 16.280.** A bat flies along the real line seeking his house which lies at point $A$. He starts from the origin $O$ and knows he is at least 1 meter away from the house, but he does not know how far it is, and on which direction. Moreover, he is blind, so he won't see it until he bumps into it. Prove that the bat has a strategy that will ensure finding the house flying a total distance not exceeding $9 \cdot OA$. Can the constant 9 be improved?

**Problem 16.281.** Player $A$ chooses $n$ points in a plane such that all points lie in a given circle. Player $B$ wants to find the points by asking questions. He chooses a point $X$ in the plane and $A$ answers by giving the least distance from $X$ to one of his points. A point is considered known if the answer to a question is 0. Prove that $B$ can find all the points with $(n+1)^2$ questions.

**Problem 16.282.** Player $A$ chooses an integer. In one question player $B$ tells $A$ five distinct digits, and $A$ tells $B$ the sum of his number and one of these five digits. Find the minimum number of questions such that $B$ can always find the number.[†]

**Problem 16.283.** Let $n$ and $k < n/2$ be given numbers. We are given $n$ balls labelled 1 to $n$ of which some are made of silver and the remaining from lead. The balls are identical in appearance, except that a lead ball is slightly heavier than a silver ball. (All silver balls have mutually equal weight and so do all the lead balls.) The weight difference is insufficient to be detectable by touch. To identify which balls are made of silver, we use a scale that compares the weight of the balls placed on each side and determines if one side is lighter, heavier or of equal weight in comparison with the other side. We are allowed to select sequences $A$ and $B$ of $k$ consecutive balls each and place on each side the balls belonging to the sets $A \setminus B$ and $B \setminus A$. If we know that the number of silver balls is at least 1 and at most $d$, find the largest value of $d$ for which it will be possible to identify all the silver balls in a finite number of measurements.[†]

**Problem 16.284.** ★ A group of $n$ spies has gathered to share a code for their mission. Each spy must remember a number from 1 to $n$ so that: the numbers the spies have are all different from each other, and no spy has any knowledge of any other spy's number, i.e. he cannot guess it with a probability larger than $\frac{1}{n-1}$. In order to assist them with this task, the spies will lock themselves in $n$ different rooms and use phones to communicate with each other, so that only two spies talk to each other at one time. Before going to their rooms the spies have a chance to discuss their strategy. What strategy will the spies use to achieve their goal? (The spies are in possession of a computer which can perform calculations and randomly generate numbers.)

**Problem 16.285.** ★ A group of 2020 prisoners is gathered by the guard and told that they will receive freedom in the following way. Each will be given a hat of one of five colours and be given a chance to see everyone else's hat except his own. Then they will be arranged randomly in a line and one by one each will be interrogated by the guard. First they will be asked if they know the colour of their hat. If they answer "no", they will be returned to their cell. If they answer "yes" they must whisper the guess in the warden's ear so that the other prisoners do not hear it; if the guess is correct the prisoner is freed, otherwise he is returned to his cell. (Prisoners not yet interrogated can see whether someone is freed or returned to his cell.) As usual, the

prisoners are allowed to discuss their strategy beforehand, but the guard will listen in and exploit any strategy that is not fool-proof. Find the largest possible number of prisoners whose release can be guaranteed, and a strategy which achieves this.

# Chapter 17

# Geometry

**Sources for this chapter:** Camp handouts by Dominik Burek (January 2020, November 2020, May 2025), Mikhail (March 2020, Level 4/4+), Lê Phúc Lũ (April 2021, June 2022), Łukasz Bożyk (November 2021), Regis Barbosa (November 2022, online 2023, February 2025), Emil Kolev (February 2024), Dren Neziri (November 2024).

This is the IMO-track chapter. Everything the earlier levels built — angle chasing, the power of a point, homothety, spiral similarity — is assumed, and the emphasis moves to the transformations and the projective language that let a hard configuration be rewritten into an easy one: isogonal conjugation, harmonic division, poles and polars, and inversion. The final section collects the elite mixed sets used at the January camps.

## 17.1 Advanced angle chasing and classical configurations

The classical named configurations are posed here as problems rather than quoted as theory: at this level the proof of Miquel's theorem or of the Japanese theorem is itself the exercise.

**Problem 17.1.** Prove that the points $A$, $B$, $C$ and $D$ lie on a circle if and only if the bisectors of the angle between $AB$ and $CD$ are parallel to the bisectors of the angle between $AD$ and $BC$.

**Problem 17.2.** Let $ABCD$ be a cyclic quadrilateral and let $E$ be the midpoint of the arc $AD$ which does not contain the points $B$ and $C$. The chords $EB$ and $EC$ intersect $AD$ at the points $P$ and $Q$, respectively. Prove that the quadrilateral $PQCB$ is cyclic.

**Problem 17.3.** The incircle of the triangle $ABC$ touches the sides $AB$ and $AC$ at the points $C_1$ and $B_1$. Prove that the incentre of the triangle $AB_1C_1$ lies on the incircle of $ABC$.

**Problem 17.4.** Let $ABC$ be a triangle and let $A_1$, $B_1$ be points on the sides $BC$, $AC$ respectively such that the quadrilateral $AA_1B_1B$ is cyclic. Let $O$ be the circumcentre of the triangle $CA_1B_1$. Prove that $CO$ is perpendicular to $AB$.

**Problem 17.5.** Let $AA_1$ and $BB_1$ be the altitudes and let $H$ be the orthocentre of the triangle $ABC$. Denote by $M$ the midpoint of $AB$. Prove that the lines $MA_1$ and $MB_1$ are tangent to the circumcircle of the cyclic quadrilateral $CA_1HB_1$.

**Problem 17.6.** The sides $AB$, $BC$, $CD$ and $DA$ of the quadrilateral $ABCD$ touch its incircle at the points $K$, $L$, $M$ and $N$ respectively. The segments $KM$ and $LN$ meet at $S$. Prove that the quadrilateral $SKBL$ is cyclic if and only if the quadrilateral $SNDM$ is cyclic.

**Problem 17.7.** Let $ABC$ be an acute triangle with circumcentre $O$, and let $K$ be a point such that $KA$ is tangent to the circumcircle of $ABC$ and $\angle KCB = 90°$. The point $D$ lies on $BC$ and satisfies $KD \parallel AB$. Prove that the line $DO$ passes through $A$.

**Problem 17.8.** Let the lines $AB$ and $CD$ intersect at the point $P$ and let the lines $AD$ and $BC$ intersect at the point $Q$.

(a) Prove that the circumcircles of the triangles $PBC$, $PAD$, $QAB$ and $QCD$ have a common point $M$ (the *Miquel point* of the complete quadrangle).
(b) Prove that $M$ lies on the line $PQ$ if and only if the quadrilateral $ABCD$ is cyclic.
(c) Prove that $M$ and all four centres of these circles lie on one circle.

**Problem 17.9.** Let $ABCD$ be a cyclic quadrilateral. Denote by $M_1$, $M_2$, $M_3$ and $M_4$ the incentres of the triangles $ABD$, $ABC$, $BCD$ and $ACD$. Prove that $M_1M_2M_3M_4$ is a rectangle. (*Japanese theorem.*)

**Problem 17.10.** Let $A$, $B$, $C$ and $D$ be four points on a circle. The circle passing through $A$, $B$ and the circle passing through $A$, $D$ intersect at $A$ and at a second point $A_1$. Define $B_1$, $C_1$ and $D_1$ similarly (in total there are five circles and eight points). Prove that the points $A_1$, $B_1$, $C_1$ and $D_1$ lie on a circle. (*Miquel's six circles theorem.*)

**Problem 17.11.**

(a) Let $ABC$ be a triangle. Prove that a point $D$ lies on the circumcircle of $ABC$ if and only if the feet of the perpendiculars from $D$ to the sides of $ABC$ are collinear. (The line through these feet is the *Simson line* of $D$.)
(b) Prove that "perpendiculars" may be replaced by "the lines through $D$ forming equal angles with the sides".

**Problem 17.12.** Let $O$ be the circumcentre of the cyclic quadrilateral $ABCD$. The circumcircles of the triangles $ABO$ and $CDO$ meet again at the point $F$. The segments $AC$ and $BD$ intersect at $H$. Prove that the points $A$, $F$, $H$ and $D$ are concyclic. [Tournament of Towns 1999]

**Problem 17.13.** The diagonals $AC$ and $BD$ of the cyclic quadrilateral $ABCD$ intersect at $X$. Let $C_1$, $D_1$ and $M$ be the midpoints of the segments $CX$, $DX$ and $DC$ respectively. Let $Y$ be the intersection point of the lines $AD_1$ and $BC_1$. The line through $Y$ and $M$ intersects the diagonals $AC$ and $BD$ at $E$ and $F$ respectively. Prove that the circumcircle of the triangle $EFX$ is tangent to $XY$. [EGMO 2016]

**Problem 17.14.** The incircle of the triangle $ABC$ touches the sides $AB$, $BC$ and $CA$ at the points $C_1$, $A_1$ and $B_1$ respectively. Let $H$ be the orthocentre of $A_1B_1C_1$ and denote by $P$ and $Q$ the feet of the perpendiculars from $H$ to the lines $AA_1$ and $BC$. Prove that $P$, $Q$ and the midpoint of $B_1C_1$ are collinear. [Bulgaria 2012]

**Problem 17.15.** Let $ABCDE$ be a convex pentagon. Denote by $A_1$, $B_1$, $C_1$, $D_1$ and $E_1$ the intersections of the lines $BC$ and $DE$, $CD$ and $AE$, $DE$ and $AB$, $AE$ and $BC$, $AB$ and $CD$. Let $A_2$ be the second intersection point of the circumcircles of $ABD_1$ and $AEC_1$. Define $B_2$, $C_2$, $D_2$ and $E_2$ similarly. Prove that the five points $A_2$, $B_2$, $C_2$, $D_2$, $E_2$ are concyclic. (*Miquel's pentagon theorem.*)

**Problem 17.16.** The incircle of the circumscribed quadrilateral $ABCD$ touches its sides $AB$, $BC$, $CD$ and $DA$ at the points $K$, $L$, $M$ and $N$ respectively. Denote by $A'$, $B'$, $C'$ and $D'$ the midpoints of the segments $LM$, $MN$, $NK$ and $KL$. Prove that the quadrilateral formed by the lines $AA'$, $BB'$, $CC'$ and $DD'$ is cyclic. [Sharygin Geometry Olympiad 2011]

**Problem 17.17.** Let $BB_1$ and $CC_1$ be the altitudes of the triangle $ABC$. The bisectors of the angles $BB_1C$ and $CC_1B$ intersect the line $BC$ at the points $D$ and $E$ respectively and meet each other at the point $X$. Prove that the intersection points of the circumcircles of the triangles $BEX$ and $CDX$ lie on the line $AX$. [Belarus 2019]

**Problem 17.18.** The internal bisectors of the angles $\angle DAB$ and $\angle BCD$ of a quadrilateral $ABCD$ intersect at the point $X_1$, and the external bisectors of these angles intersect at the point $X_2$. The internal bisectors of the angles $\angle ABC$ and $\angle CDA$ intersect at the point $Y_1$, and the external bisectors of these angles intersect at the point $Y_2$. Prove that the angle between the lines $X_1X_2$ and $Y_1Y_2$ equals the angle between the diagonals $AC$ and $BD$. [Belarus 2019]

**Problem 17.19.** The point $O$ is situated inside the parallelogram $ABCD$ so that $\angle AOB + \angle COD = 180°$. Prove that $\angle OBC = \angle ODC$. [Canada 1997]

**Problem 17.20.** Let $ABCD$ be a quadrilateral with $\angle DAB = \angle ABC = 90°$ and $AB = AD$. Let $E$ be the midpoint of $AD$, and suppose that $CD = BC + AD$. Prove that $\angle ADC = 2\angle ABE$.†

**Problem 17.21.** Let $ABC$ be a triangle with $\angle ABC = 2\angle ACB$. Take points $P$ and $Q$ on the perpendicular bisector of $BC$ such that

$$\angle CAP = \angle PAQ = \angle QAB = \tfrac{1}{3}\angle BAC.$$

Prove that $Q$ is the circumcentre of the triangle $CPB$.

**Problem 17.22.** Let $ABC$ be a triangle with $\angle ACB = 90° + \angle BAC$. Take $D$ on the extension of the side $BC$ beyond $C$ such that $AC = AD$. Take a point $E$ on the opposite side of $BC$ from $A$ such that $\angle EBC = \angle BAC$ and $\angle EDC = \frac{1}{2}\angle BAC$. Prove that $\angle CED = \angle ABC$.†

**Problem 17.23.** Let $ABC$ be a triangle. Take points $E$ and $F$ on $CA$ and $AB$ respectively such that $BE = CF$, and denote by $O$ the intersection of $BE$ and $CF$. Let $X$ be a point inside the triangle $ABC$ such that $\angle ACO = \angle XCB$ and $\angle ABO = \angle XBC$. Prove that $\angle EXF = 90° - \angle BAC$.†

**Problem 17.24.** Let $ABCD$ be a convex quadrilateral and let $X$ be a point inside it such that

$$XA \cdot XC^2 = XB \cdot XD^2 \qquad \text{and} \qquad \angle AXD + \angle BXC = \angle CXD.$$

Prove that $\angle XAD + \angle XCD = \angle XBC + \angle XDC$.†

**Problem 17.25.** Let $ABCD$ be a convex quadrilateral in which the angles at the vertices $B$ and $D$ are 120°. The point $E$ lies on the segment $AD$ and the point $F$ lies on the segment $BC$, and

$$AE \cdot BC = AB \cdot DE, \qquad BF \cdot CD = AD \cdot FC.$$

Prove that the lines $BE$ and $DF$ are parallel.

**Problem 17.26.** Let $ABC$ be a triangle with $AB < AC$ and let $M$ be the midpoint of the side $BC$. A point $P$ lies on the segment $AB$ with $AP > PB$, and a point $Q$ lies on the segment $AC$ with $\angle MPA = \angle AQM$. The perpendicular bisectors of the segments $BC$ and $PQ$ intersect at $S$. Prove that

$$\angle BAC + \angle QSP = \angle QMP.$$

**Problem 17.27.** In a convex pentagon $ABCDE$ the following conditions hold:

$$AB = BC, \qquad \angle ABE + \angle DBC = \angle EBD, \qquad \angle AEB + \angle BDC = 180°.$$

Prove that the orthocentre of the triangle $BDE$ lies on the line $AC$.

**Problem 17.28.** Given a convex pentagon $ABCDE$, let $F$ be a point on the side $AB$ such that the triangles $ADE$, $ECF$ and $DBC$ are similar. Prove that

$$\frac{AF}{BF} = \left(\frac{EF}{CF}\right)^2.$$

**Problem 17.29.** Let $ABC$ be an acute triangle inscribed in a circle with centre $O$. Let $I_B$ and $I_C$ be the incentres of the triangles $AOB$ and $AOC$ respectively, and let $M$ be the midpoint of the arc $BC$ of the circumcircle of $ABC$ that contains the point $A$. Prove that the points $M$, $A$, $I_B$ and $I_C$ lie on one circle.

**Problem 17.30.** Let $ABCD$ be a parallelogram and let $M$ be the midpoint of the arc $ABC$ of the circumcircle of the triangle $ABC$. The point $E$ lies on the segment $AD$ and the point $F$ lies on the segment $CD$, and it is known that $ME = MD = MF$. Prove that the points $B$, $M$, $E$ and $F$ lie on one circle.

**Problem 17.31.** Given a triangle $ABC$ with $AC < BC$, a circle through $A$ and $B$ intersects the segments $AC$ and $BC$ at the points $A_1$ and $B_1$ respectively. The circumcircles of the triangles $ABC$ and $A_1B_1C$ intersect at a point $P \neq C$. The segments $AB_1$ and $A_1B$ intersect at the point $S$. The points $Q$ and $R$ are the reflections of $S$ in the lines $AC$ and $BC$ respectively. Prove that the points $P$, $Q$, $R$ and $C$ lie on one circle.

**Problem 17.32.** Given a triangle $ABC$ with $AB < AC$, a circle $\omega$ is tangent to the line $AB$ at the point $B$ and tangent to the segment $AC$ at the point $D$. Let $E$ be the foot of the perpendicular from $D$ to the line $BC$. The circle $\omega$ intersects the circumcircle of the triangle $ABC$ at two distinct points $B$ and $P$. Prove that $\angle CPE = 2\angle ACB$.

**Problem 17.33.** Let $ABCD$ be a cyclic quadrilateral with $\angle A + \angle D = 90^\circ$, whose diagonals intersect at the point $E$. A line $l$ intersects the segments $AB$, $CD$, $AE$ and $ED$ at the points $X$, $Y$, $Z$ and $T$ respectively. It is known that $AZ = CE$ and $BE = DT$. Prove that the length of the segment $XY$ equals the diameter of the circumcircle of the triangle $ETZ$.

**Problem 17.34.** In the isosceles triangle $ABC$ with $AB = AC$, a point $O$ lies strictly inside the triangle and the circle $\omega$ centred at $O$ and passing through $C$ meets the sides $BC$ and $AC$ (excluding their endpoints) at $D$ and $E$ respectively. Let $\Gamma$ be the circumcircle of the triangle $AEO$, and let it meet $\omega$ again at $F \neq E$. Prove that the circumcentre of the triangle $BDF$ lies on $\Gamma$. [Japan 2022]

**Problem 17.35.** A convex quadrilateral $ABCD$ satisfies $AB \cdot CD = BC \cdot DA$. The point $X$ lies inside $ABCD$ and satisfies

$$\angle XAB = \angle XCD \quad \text{and} \quad \angle XBC = \angle XDA.$$

Prove that $\angle BXA + \angle DXC = 180^\circ$. [IMO 2018]

**Problem 17.36.** Let $ABC$ be an acute, non-isosceles triangle inscribed in a circle $(O)$, and let $BB'$ and $CC'$ be its altitudes. Denote by $E$ and $F$ the second intersections of $BB'$ and $CC'$ with $(O)$, and let $D$, $P$, $Q$ be the projections of $A$ onto $BC$, $CE$ and $BF$ respectively. Prove that the perpendicular bisector of $PQ$ bisects both of the segments $AO$ and $BC$.

**Problem 17.37.** Let $ABC$ be a triangle with incircle $(I)$, tangent to $BC$, $CA$, $AB$ at $D$, $E$, $F$ respectively. On the line $DF$ take points $M$, $P$ such that $CM \parallel AB$ and $AP \parallel BC$. On the line $DE$ take points $N$, $Q$ such that $BN \parallel AC$ and $AQ \parallel BC$. Denote by $X$ the intersection of $PE$ and $QF$, and by $K$ the midpoint of $BC$. Prove that if $AX = IK$, then $\angle BAC \leq 60°$.†

**Problem 17.38.** Let $ABC$ be an acute, non-isosceles triangle inscribed in a circle $(O)$ and let $M$, $N$ be the midpoints of the segments $AB$, $AC$. Denote by $D$ the midpoint of the arc $BAC$ of $(O)$. Take a point $K$ inside the triangle $ABC$ such that $\angle KAB = 2\angle KBA$ and $\angle KAC = 2\angle KCA$. Prove that $KA = KD$.

**Problem 17.39.** Let $ABCD$ be a rectangle with $P$ on the segment $AC$. Denote by $Q$ a point on the minor arc $PB$ of the circle $(PAB)$ such that $QB = QC$, and by $R$ a point on the minor arc $PD$ of the circle $(PAD)$ such that $RC = RD$. The lines $CB$, $CD$ meet the circle $(CQR)$ again at $M$, $N$ respectively. Prove that $BM = DN$.

**Problem 17.40.** Let $ABC$ be a non-isosceles triangle with $M$, $N$, $P$ the midpoints of the segments $BC$, $CA$, $AB$ respectively. Suppose that $I$ is the intersection of the bisectors of the angles $BPM$ and $MNP$, and that $J$ is the intersection of the bisectors of the angles $CNM$ and $MPN$. Let the circle with centre $I$ be tangent to $MP$ at $D$, and the circle with centre $J$ be tangent to $MN$ at $E$. Prove that $DE \parallel BC$ and that the radical axis of $(I)$ and $(J)$ bisects $DE$.

**Problem 17.41.** Let $ABC$ be a triangle with circumcentre $O$ and median $AM$. Let $N$ be the midpoint of the segment $AM$, and take $J$ such that $AJ \parallel BC$ and $NJ \perp ON$. The circle $(AOJ)$ meets $JN$ again at $K$. Prove that $\angle KAB = \angle MAC$.

## 17.2 The orthocentre, the Euler line and the nine-point circle

*Revisits the orthocentre, the Euler line and the nine-point circle (see Chapter 4); the problems below push it further.*

**Problem 17.42.** Let $AA_1$ and $BB_1$ be the altitudes of the triangle $ABC$, let $CC_1$ be the third altitude and let $H$ be the orthocentre. Prove that the second intersection point of the circumcircles of $AB_1C_1$ and $ABC$ is collinear with $H$ and the midpoint of the side $BC$.

**Problem 17.43.** Let $H$ be the orthocentre of an acute triangle $ABC$ and let $M$ be the midpoint of $AB$. The circumcircle of $ABH$ intersects the median $CM$ at an interior point $T$. Prove that $\angle HTC = 90°$.

**Problem 17.44.** Let $H$ be the orthocentre of an acute triangle $ABC$ and let $M$ be the midpoint of $AB$. An arbitrary point $D$ is chosen on the arc $AB$ of the circumcircle of $ABC$ not containing $C$. Denote by $P$ the foot of the perpendicular from $H$ to $CD$. Prove that $MP = MD$.

**Problem 17.45.** Let $H$ be the orthocentre of the triangle $ABC$ and let $l$ be an arbitrary line through $H$. Denote by $l_A$, $l_B$ and $l_C$ the reflections of $l$ in the lines $BC$, $CA$ and $AB$.

(a) Prove that the three lines $l_A$, $l_B$ and $l_C$ pass through a common point $P$ on the circumcircle of $ABC$ (the *anti-Steiner point* of $l$).

(b) Prove that $l$ is the image of the Simson line of $P$ under the homothety with centre $P$ and ratio 2; in particular the Simson line of $P$ bisects the segment $HP$.

**Problem 17.46.** Let $H$ be the orthocentre of a non-isosceles acute triangle $ABC$ and let $l$ be an arbitrary line through $H$. Points $X$ and $Y$ are chosen arbitrarily on $l$. Denote by $X_A$, $X_B$, $X_C$ and $Y_A$, $Y_B$, $Y_C$ the reflections of $X$ and of $Y$ in the lines $BC$, $CA$ and $AB$ respectively. Prove that the circumcircles of the triangles $X_AX_BC$, $Y_BY_CA$ and $CAB$ have a common point. [EGMO 2017]

**Problem 17.47.** Let $O$ and $H$ be the circumcentre and the orthocentre of an acute triangle $ABC$. Prove that among the areas of the triangles $AOH$, $BOH$ and $COH$ one is equal to the sum of the other two. [APMO 2004]

**Problem 17.48.** The nine-point circle of an acute triangle $ABC$ is reflected in the altitude from $A$ to $BC$, and the reflected circle intersects the circumcircle of $ABC$ at two distinct points $X$ and $Y$. Let $H$ be the orthocentre of $ABC$. Prove that $AH$ is the external bisector of $\angle XHY$. [Iran 2019]

**Problem 17.49.** Let $H$ be the orthocentre of an acute non-isosceles triangle $ABC$. Points $D$ and $E$ are chosen on the sides $AB$ and $AC$ respectively such that $AD = AE$ and $H$ lies on the segment $DE$. Let $M$ be the midpoint of the side $BC$. The circumcircle of $ABC$ intersects the half-line $MH$ at $K$. Prove that the points $A$, $D$, $E$ and $K$ are concyclic. [Western China Olympiad 2009]

**Problem 17.50.** Let $H$ be the orthocentre of an acute triangle $ABC$ and let $M$ be the midpoint of the side $AB$. The line $l$ passes through $H$ perpendicular to $HM$. Denote the intersections of $l$ with the sides $AC$ and $BC$ by $E$ and $F$. Prove that $HE = HF$. [Saint Petersburg 1995]

**Problem 17.51.** Let $AA_1$ and $CC_1$ be the altitudes of an acute non-isosceles triangle $ABC$. Denote by $H$ the orthocentre, by $O$ the circumcentre and by $B_0$ the midpoint of the side $AC$. The line $BO$ intersects the side $AC$ at $P$, and the lines $BH$ and $A_1C_1$ meet at $Q$. Prove that the lines $HB_0$ and $PQ$ are parallel. [Russia 2007–2008]

**Problem 17.52.** Let $ABC$ be an acute triangle with circumcircle $\omega$, circumcentre $O$ and orthocentre $H$. The lines $AH$ and $CH$ meet $\omega$ again at the points $A_0$ and $C_0$. The line $A_0C_0$ intersects the sides $AB$ and $BC$ at the points $A_1$ and $C_1$. The lines through $O$ parallel to $BC$ and to $AB$ intersect $AC$ at the points $A_2$ and $C_2$ respectively. The lines $A_1A_2$ and $C_1C_2$ meet at $T$. Prove that $HT \parallel AC$. [Ukraine Jr 2018]

**Problem 17.53.** ★ Let $X$ be a point in the interior of an acute-angled triangle $ABC$, different from the circumcentre. The lines $AX$, $BX$ and $CX$ intersect the circumcircle of $ABC$ again at the points $A_1$, $B_1$ and $C_1$ respectively. Let $A_2$, $B_2$ and $C_2$ be the reflections of $A_1$, $B_1$ and $C_1$ in the midpoints of $BC$, $CA$ and $AB$ respectively. Prove that the circumcircle of $A_2B_2C_2$ passes through the orthocentre of $ABC$; in particular it passes through a point that does not depend on $X$. [China TST 2005]

**Problem 17.54.** ★ Let $H$ be the orthocentre of the triangle $ABC$. Denote by $M$ and $N$ the midpoints of the segments $BC$ and $AH$. The point $D$ lies on the line $MH$ so that $AD \parallel BC$, and the point $K$ lies on the line $AH$ so that $DNMK$ is cyclic. The points $E$ and $F$ lie on the lines $AC$ and $AB$ such that $\angle EHM = \angle C$ and $\angle FHM = \angle B$. Prove that the points $D$, $E$, $F$ and $K$ are concyclic. [Iran 2019]

**Problem 17.55.** ★ Let $H$ be the orthocentre and let $G$ be the centroid of an acute triangle $ABC$ with $AB \neq AC$. The line $AG$ intersects the circumcircle of $ABC$ at $A$ and at $P$. Let $P'$

be the reflection of $P$ in the line $BC$. Prove that $\angle CAB = 60°$ if and only if $HG = GP'$. [EGMO 2017]

**Problem 17.56.** ★ Let $H$ be the orthocentre of an acute triangle $ABC$ with $AB \neq AC$ and let $F$ be the point on the circumcircle of $ABC$ such that $\angle AFH = 90°$. The point $K$ is symmetric to $H$ with respect to $B$, the point $P$ is such that $\angle PHB = \angle PBC = 90°$, and $Q$ is the foot of the perpendicular from $B$ to $CP$. Prove that the line $HQ$ is tangent to the circumcircle of the triangle $FHK$. [Serbia 2018]

**Problem 17.57.** ★ Let $I$ be the incentre of the triangle $ABC$. Prove that the Euler lines of the three triangles $ABI$, $BCI$ and $CAI$ all pass through a point $S$ lying on the Euler line of $ABC$. (The point $S$ is the *Schiffler point*.)

**Problem 17.58.** ★ Let $ABC$ be an acute triangle with $AB > AC$, and let $\Gamma$ be its circumcircle. Let $H$, $M$ and $F$ be the orthocentre of the triangle, the midpoint of $BC$ and the foot of the altitude from $A$, respectively. Let $Q$ and $K$ be the two points on $\Gamma$ that satisfy $\angle AQH = 90°$ and $\angle QKH = 90°$, and assume that $A$, $B$, $C$, $K$, $Q$ are all different and lie on $\Gamma$ in this order. Prove that the circumcircles of the triangles $KQH$ and $KFM$ are tangent to each other. [IMO 2015]

**Problem 17.59.** Let $ABC$ be an acute, non-isosceles triangle with centroid $G$. Take $D$ and $E$ on $AB$ and $AC$ respectively such that $G$ is the orthocentre of the triangle $ADE$. Denote by $M$ the midpoint of the segment $DE$. Prove that $GM$ is perpendicular to $BC$.

**Problem 17.60.** Let $ABC$ be an acute, non-isosceles triangle with altitudes $AD$, $BE$, $CF$ and orthocentre $H$, and let $d$ be the tangent line to the circumcircle of $ABC$ at $A$. The line through $H$ parallel to $EF$ cuts $DE$ and $DF$ at $Q$ and $P$ respectively. Prove that $d$ is tangent to the excircle of the triangle $DPQ$ opposite the vertex $D$.

**Problem 17.61.** Let $ABC$ be an acute, non-isosceles triangle with orthocentre $H$ and circumcentre $O$, and let $AD$ be a diameter of the circumcircle. Suppose that the circle $(AHD)$ meets the lines $AB$ and $AC$ at $F$ and $E$ respectively. Denote by $J$ and $K$ the orthocentre and the nine-point centre of the triangle $AEF$. Prove that $HJ \parallel BC$ and $KO = KH$.

**Problem 17.62.** Let $ABC$ be a triangle with orthocentre $H$ and circumcentre $O$. Denote by $M$ and $N$ the midpoints of $AB$ and $AC$, and let $BD$ and $CE$ be the altitudes of $ABC$. The line $MN$ cuts $DE$ at $K$.

(a) Prove that $AK \perp HO$.

(b) The line $DE$ cuts $BC$ at $T$, and the tangent line from $A$ to the circumcircle of $ABC$ cuts $MN$ at $S$. Prove that $ASTK$ is a parallelogram.

**Problem 17.63.** Let $ABC$ be a triangle inscribed in a circle $(O)$ with $BC$ fixed and $A$ moving on $(O)$. Denote by $H$ its orthocentre and let $BE$, $CF$ be its altitudes. Let $M$, $N$, $I$ be the midpoints of $BC$, $AH$, $EF$ respectively, and let the line $AM$ cut $(O)$ again at $T$. Prove that the power of $I$ with respect to the circle $(ANT)$ does not depend on the position of $A$.

**Problem 17.64.** ★ Let $ABC$ be a triangle with circumcentre $O$, orthocentre $H$, and let $M$ be the midpoint of $AH$. Denote by $(O_1)$ and $(O_2)$ the circles that both pass through $H$ and are tangent to $BC$ at $B$ and at $C$ respectively.

(a) Prove that $O_1$, $O_2$ and $M$ are collinear.

(b) The line $O_1O_2$ cuts $BC$ at $T$. The circle $(MOT)$ cuts the line $HO$ again at $K$. Prove that the centre of the circle $(AOK)$ lies on a midline of the triangle $ABC$.

**Problem 17.65.** Let $ABC$ be an acute-angled triangle with $\angle BAC = 60^\circ$ and $AB > AC$. Let $I$ be the incentre and $H$ the orthocentre of the triangle $ABC$. Prove that $2\angle AHI = 3\angle ABC$. [APMO 2007]

**Problem 17.66.** In an acute-angled triangle $ABC$, $AM$ and $BN$ are altitudes. A point $D$ is chosen on the arc $ACB$ of the circumcircle of the triangle. Let the lines $AM$ and $BD$ meet at $P$ and let the lines $BN$ and $AD$ meet at $Q$. Prove that $MN$ bisects the segment $PQ$. [Russia 2005]

**Problem 17.67.** Let $O$ be the circumcentre and $H$ the orthocentre of an acute triangle $ABC$. Show that there exist points $D$, $E$ and $F$ on the sides $BC$, $CA$ and $AB$ respectively such that

$$OD + DH = OE + EH = OF + FH$$

and the lines $AD$, $BE$, $CF$ are concurrent. [IMO Shortlist 2000]

**Problem 17.68.** Let $ABC$ be an acute triangle with orthocentre $H$ and circumcircle $\Gamma$. A line through $H$ intersects the segments $AB$ and $AC$ at $E$ and $F$ respectively. Let $K$ be the circumcentre of $AEF$, and suppose the line $AK$ intersects $\Gamma$ again at a point $D$. Prove that the line $HK$ and the line through $D$ perpendicular to $BC$ meet on $\Gamma$. [USA TSTST 2019]

**Problem 17.69.** Let $ABC$ be an acute triangle with altitudes $AD$ and $BE$ intersecting at the point $H$, and let $M$ be the midpoint of the segment $AB$. The circumcircles of the triangles $DEM$ and $ABH$ intersect at the points $P$ and $Q$, where $P$ lies on the same side of the line $CH$ as $A$. Prove that the lines $ED$, $PH$, $MQ$ and the circumcircle of the triangle $ABC$ all pass through one common point. [Canada 2016]

**Problem 17.70.** ★ Let $AD$, $BE$ and $CF$ be the altitudes of a scalene triangle $ABC$ with circumcentre $O$. Prove that the circles $(AOD)$, $(BOE)$ and $(COF)$ intersect at a point $X$ other than $O$.

## 17.3 Symmedians and isogonal conjugates

*Revisits the Humpty and Dumpty points (see Chapter 12); the problems below push it further.*

Isogonal conjugation in a triangle is standard by now. The new object of this section is isogonal conjugation inside a *quadrilateral*, where the conjugate of a point exists only under one extra condition.

**Lemma 17.1 (Ratio lemma).** *Let $P$ be the intersection point of the lines $AB$ and $CD$. Then*

$$\frac{PA}{PB} = \frac{[CAP]}{[CBP]} = \frac{[ADP]}{[BDP]} = \frac{[ACD]}{[BCD]} = \frac{CA \cdot AD \cdot \sin\angle CAD}{CB \cdot BD \cdot \sin\angle CBD}.$$

**Theorem 17.2 (Steiner's theorem).** *Let $D$ and $E$ be points on the side $BC$ of the triangle $ABC$. The cevians $AD$ and $AE$ are isogonal in the angle $A$ if and only if*

$$\frac{DB}{DC} \cdot \frac{EB}{EC} = \frac{AB^2}{AC^2},$$

*and this happens if and only if the circle $(ADE)$ is tangent to the circumcircle of $ABC$. In particular, if a line $AD$ with $D$ on the line $BC$ is tangent to the circumcircle of $ABC$, then*

*$\angle DAB = \angle ACB$, the triangles $DAB$ and $DCA$ are similar, and*

$$\frac{DB}{DC} = \frac{AB^2}{AC^2}.$$

**Definition 17.3 (Isogonal conjugates in a quadrilateral).** Points $P$ and $Q$ are called *isogonal conjugates in the quadrilateral $ABCD$* if they are conjugate in each of its four angles: that is, $AP$ and $AQ$ are isogonal in $\angle DAB$, $BP$ and $BQ$ are isogonal in $\angle ABC$, $CP$ and $CQ$ are isogonal in $\angle BCD$, and $DP$ and $DQ$ are isogonal in $\angle CDA$. For four arbitrary distinct points $A$, $B$, $C$, $D$ one takes the symmetries with respect to the bisectors of the directed angles $\angle DAB = \angle(DA, AB)$, $\angle ABC = \angle(AB, BC)$, $\angle BCD = \angle(BC, CD)$, $\angle CDA = \angle(CD, DA)$.

**Theorem 17.4 (Criterion for an isogonal conjugate).** *A point $P$ has an isogonal conjugate in the quadrilateral $ABCD$ if and only if*

$$\angle APB + \angle CPD = 180^\circ.$$

**Remark.** In directed angles the criterion reads $\angle(PA, PB) + \angle(PC, PD) = 0$. Note also that if $P$ and $Q$ are isogonal conjugates in $ABCD$, then $B$ and $D$ are isogonal conjugates in $APCQ$; it is therefore always worth looking for other pairs of points and other quadrilaterals in the same picture.

**Technique 17.5 (Using conjugates in a quadrilateral).** The usual scheme has three steps: (1) prove that some point has a conjugate, using the angle criterion; (2) locate that conjugate using the isogonality in two of the four angles; (3) then exploit the isogonality in the remaining two angles.

**Theorem 17.6 (Known isogonal-conjugate pairs).** *In a triangle $ABC$ the following pairs of points are isogonal conjugates: the orthocentre $H$ and the circumcentre $O$; the centroid $G$ and the symmedian point; the intersection point $T$ of the tangents to the circumcircle at $B$ and $C$ and the point $D$ for which $ABDC$ is a parallelogram. The incentre and the three excentres are each their own isogonal conjugate. If $M$ and $N$ are isogonal conjugates then their six projections onto the sides lie on one circle, whose centre is the midpoint of $MN$; for $(M, N) = (H, O)$ this circle is the nine-point circle.*

**Definition 17.7 (Kosnita point).** Let $O_a$, $O_b$, $O_c$ be the circumcentres of the triangles $OBC$, $OCA$, $OAB$, where $O$ is the circumcentre of $ABC$. The lines $AO_a$, $BO_b$, $CO_c$ are concurrent; their common point is the *Kosnita point* of $ABC$, and it is the isogonal conjugate of the nine-point centre.

**Problem 17.71.** Given is a triangle $ABC$ with $AC = BC$. Points $P$, $Q$ lie inside this triangle and satisfy $\angle PAC = \angle ABQ$ and $\angle PBC = \angle BAQ$. Prove that $C$, $P$, $Q$ are collinear.

**Problem 17.72.** Given is a triangle $ABC$ with $AC = BC$ and let $M$ be the midpoint of the side $AB$. A point $P$ lies inside $ABC$ and satisfies $\angle PAB = \angle PBC$. Prove that $\angle APC + \angle BPM = 180^\circ$.

**Problem 17.73.** The incircle of the triangle $ABC$ is tangent to the side $BC$ at the point $D$.

The line through $D$ perpendicular to $AD$ intersects the bisectors of the angles $ABC$ and $ACB$ at the points $P$ and $Q$ respectively. Prove that $PD = DQ$.

**Problem 17.74.** Let $O$ be the circumcentre and $H$ the orthocentre of an acute triangle $ABC$. The perpendicular bisector of $BH$ intersects the sides $AB$ and $BC$ at $E$ and $F$. Prove that $OB$ bisects the angle $EOF$.

**Problem 17.75.** Let $O$ be the circumcentre of an acute triangle $ABC$. The line $CO$ intersects the altitude from $A$ at $K$. Let $P$ and $M$ be the midpoints of the segments $AK$ and $AC$ respectively. The lines $PO$ and $BC$ meet at $Y$. The circumcircle of the triangle $BCM$ intersects the line $AB$ again at $X$. Prove that the quadrilateral $BXOY$ is cyclic.

**Problem 17.76.** Let $ABCD$ be a convex quadrilateral whose diagonals intersect at $E$. Points $P$, $Q$ lie outside $ABCD$ and satisfy

$$\angle PAB = \angle CAD, \quad \angle PBA = \angle CBD, \quad \angle QCD = \angle ACB, \quad \angle QDC = \angle ADB.$$

Prove that $P$, $E$, $Q$ are collinear.

**Problem 17.77.** Let $ABC$ be a triangle with the median $AD$. Denote by $DM$, $DN$, $AE$ the symmedians of the triangles $ABD$, $ACD$, $ABC$ respectively.

(a) Prove that $MNED$ is an isosceles trapezoid.

(b) If instead $AD$ is the bisector of the angle $BAC$, prove that $AE$, $BN$ and $CM$ are concurrent.

**Problem 17.78.** Let $ABC$ be a non-isosceles triangle with $AB^2 + AC^2 = 2BC^2$. Denote by $AD$ and $AM$ the internal angle bisector and the median from $A$. Take $E$ such that $D$ is the incentre of the triangle $AME$. Prove that $MA = 3\,ME$.

**Problem 17.79.** Let $ABCD$ be a cyclic quadrilateral, let $E = AB \cap CD$ and $F = AD \cap BC$, and let $M$, $N$ be the midpoints of the segments $AC$, $BD$ (so that $MN$ is the Gauss line of the complete quadrilateral). Prove that the bisectors of the angles at $E$ and at $F$ meet on the line $MN$.

**Problem 17.80.** Let $ABC$ be a triangle with $M$, $N$ the midpoints of $AB$, $AC$ and with centroid $G$. The circles $(BMG)$ and $(CNG)$ meet again at $K$.

(a) Prove that $AK$ is the symmedian of the triangle $ABC$.

(b) The line $AK$ cuts $BC$ and the circumcircle $(O)$ at $T$ and $D$ respectively, and $DB$, $DC$ cut the circles $(ABT)$, $(ACT)$ again at $R$, $S$. Prove that $OD \perp RS$ and that $T$ is the midpoint of $RS$.

**Problem 17.81.** Let $ABC$ be an acute, non-isosceles triangle inscribed in a circle $(O)$, and let $I$ be the midpoint of $BC$ and $D$ the midpoint of the minor arc $BC$. Take $K$ on the line $AI$ such that $OK \parallel AD$. The line through $K$ perpendicular to $AO$ cuts $AB$, $AC$ at $E$, $F$ respectively. Take $G$ on $KD$ such that $AG \perp BC$. Prove that $A$ lies on the line joining the midpoints of the segments $EF$ and $DG$.

**Problem 17.82.** Let $ABC$ be an acute, non-isosceles triangle inscribed in a circle $(O)$ and let $M$ be the midpoint of $BC$. The tangent lines to $(O)$ at $B$ and $C$ intersect at $T$. Denote by $H$, $K$ the projections of $T$ onto $AB$, $AC$. Take $A'$ on $(O)$ such that $AA' \parallel BC$, and let $L$ be the projection of $A'$ on $BC$. Prove that $H$, $K$, $L$, $M$ are concyclic.

**Problem 17.83.** Let $ABC$ be a triangle inscribed in a fixed circle $(O)$ with $BC$ fixed and $A$ varying on $(O)$. Denote by $H$ the orthocentre of the triangle $ABC$ and take $D$, $E$ on $AB$, $AC$

respectively such that $H$ is the midpoint of $DE$. Prove that as $A$ moves on $(O)$, the centre of the circle $(ADE)$ lies on a fixed circle.

**Problem 17.84.** Let $B$ and $C$ be two fixed points and let $\Gamma$ be a fixed circle such that the line $BC$ does not intersect $\Gamma$. Take a variable point $A$ on $\Gamma$ with $AB \neq AC$, and let $H$ be the orthocentre of the triangle $ABC$. Let $X \neq H$ be the second point of intersection of the circumcircle of $BHC$ with the circle of diameter $AH$. Find the locus of $X$ as $A$ varies on $\Gamma$.

**Problem 17.85.** Let $ABCD$ be a parallelogram with $\angle CAD = 90°$. Denote by $H$ the projection of $A$ onto $CD$, and let the tangent line to the circle $(ABD)$ at $D$ cut $AC$ at $K$. Prove that $\angle KBA = \angle HBD$.

**Problem 17.86.** Let $ABC$ be a triangle with circumcircle $(O)$. A circle $(O')$ is tangent to $BC$ and internally tangent to $(O)$. The bisector of the angle $BAC$ cuts $(O')$ at $D$ and $E$. Prove that $D$ and $E$ are isogonal conjugates with respect to the triangle $ABC$.†

**Problem 17.87.** Let $ABC$ be a triangle with $\angle ABC = 2\angle ACB$. The perpendicular bisector of $BC$ cuts $AC$ at $D$, and $E$ denotes the reflection of $D$ in $A$. Prove that $BE$ is parallel to the symmedian from the vertex $A$ in the triangle $ABC$.

**Problem 17.88.** Let $ABC$ be a triangle with circumcircle $(O)$, let $AD$ be its angle bisector and let $I$ be the incentre. The circle $(BID)$ cuts $AB$ again at $M$ and the circle $(CID)$ cuts $AC$ again at $N$. Denote $X = IB \cap DM$ and $Y = IC \cap DN$. Prove that the altitude and the median from $I$ in the triangle $IXY$ pass through the midpoint of the arc $BC$ of $(O)$.

**Problem 17.89.** ★ Let $ABC$ be a triangle with $AB \neq AC$. Take $P$ inside the triangle $ABC$ such that $\angle BPC = 180° - \angle BAC$. The lines $PB$, $PC$ cut $AC$, $AB$ at $D$, $E$ respectively. Denote by $I$ and $J$ the $A$-excentres of the triangles $ABD$ and $ACE$. The circle $(ADE)$ cuts $IJ$ again at $T$.†

(a) Prove that $TP$ passes through a fixed point as $P$ moves.
(b) Denoting by $O$ the centre of $(ADE)$, prove that $OI = OJ$.

**Problem 17.90.** ★ Let $ABC$ be a triangle and let a circle through $B$ and $C$ cut $AB$, $AC$ at $F$, $E$ respectively. Denote $T = BE \cap CF$ and $D = AT \cap BC$. Suppose that $T$ is the Lemoine (symmedian) point of the triangle $DEF$. Prove that $ABC$ must be isosceles and that $(DEF)$ is its incircle.

**Problem 17.91.** Let $ABC$ be an acute scalene triangle. Let $X$ and $Y$ be two distinct interior points of the segment $BC$ such that $\angle CAX = \angle YAB$. Let $K$ and $S$ be the feet of the perpendiculars from $B$ to the lines $AX$ and $AY$ respectively, and let $T$ and $L$ be the feet of the perpendiculars from $C$ to the lines $AX$ and $AY$ respectively. Prove that $KL$ and $ST$ intersect on the line $BC$.

**Problem 17.92.** Given is an acute triangle $ABC$. Points $X$, $Y$, $Z$ lie on $BC$, $CA$, $AB$ respectively and $XYZ$ is an equilateral triangle of the smallest possible area inscribed in $ABC$. Prove that the perpendiculars to $YZ$, $ZX$, $XY$ through $A$, $B$, $C$ respectively are concurrent.

**Problem 17.93.** Let $P$ be a point inside an acute triangle $ABC$. The lines $AP$, $BP$, $CP$ intersect the circumcircle of $ABC$ again at the points $D$, $E$, $F$ respectively. The points $K$, $L$, $M$ are symmetric to $D$, $E$, $F$ with respect to the lines $BC$, $CA$, $AB$ respectively. Prove that the circumcircle of $KLM$ passes through the orthocentre of $ABC$.

**Problem 17.94.** Triangle $ABC$ is inscribed in a circle $\omega$. The point $P$ is the midpoint of the arc $BC$ of $\omega$ containing $A$. The circle with diameter $CP$ intersects the bisector of the angle $BAC$ at the points $K$ and $L$ (the points $A$, $K$, $L$ lying in this order on that line). Moreover $M$ is symmetric to $L$ with respect to the line $BC$. Prove that the circumcircle of $BKM$ passes through the midpoint of the segment $BC$.

**Problem 17.95.** Let $ABC$ be an acute triangle with $AB < AC$, and let $D$ and $E$ be points on the side $BC$ such that $BD = CE$ and $D$ lies between $B$ and $E$. Suppose there exists a point $P$ inside $ABC$ such that $PD \parallel AE$ and $\angle PAB = \angle EAC$. Prove that $\angle PBA = \angle PCA$.

**Problem 17.96.** In a convex quadrilateral $ABCD$ the diagonal $BD$ bisects neither the angle $ABC$ nor the angle $CDA$. The point $P$ lies inside $ABCD$ and satisfies $\angle PCB = \angle DBA$ and $\angle PDC = \angle BDA$. Prove that $ABCD$ is a cyclic quadrilateral if and only if $AP = CP$. [IMO 2004 P5]

**Problem 17.97.** Let $ABCD$ be a convex circumscribed quadrilateral with incentre $I$. Points $P$ and $Q$ are chosen on the segments $AI$ and $CI$ respectively such that $\angle PBQ = \frac{1}{2}\angle ABC$. Prove that $\angle PDQ = \frac{1}{2}\angle ADC$.

**Problem 17.98.** Let $ABCD$ be a convex circumscribed quadrilateral. Points $M$ and $N$ are chosen on the segment $BC$ such that $N$ lies between $B$ and $M$. The segments $AM$ and $DN$ meet at $X$. Let $J_1$, $J_2$, $J_3$ and $J_4$ be the incentres of the triangles $MNX$, $ADX$, $ABM$ and $DCN$. Prove that $J_1$, $J_2$, $J_3$ and $J_4$ are concyclic.

**Problem 17.99.** There is given a convex quadrilateral $ABCD$. Prove that there exists a point $P$ inside the quadrilateral such that

$$\angle PAB + \angle PDC = \angle PBC + \angle PAD = \angle PCD + \angle PBA = \angle PDA + \angle PCB = 90^\circ$$

if and only if the diagonals $AC$ and $BD$ are perpendicular. [IMO Shortlist 2008]

## 17.4 Harmonic division, poles and polars, projective tools

**Definition 17.8 (Cross-ratio on a line).** Let $A$, $B$, $C$, $D$ be four points on a line. Their *cross-ratio* is

$$(A, B; C, D) = \frac{\overline{CA}}{\overline{CB}} : \frac{\overline{DA}}{\overline{DB}},$$

where $\overline{XY}$ denotes a directed length.

**Definition 17.9 (Cross-ratio on a circle).** Let $A$, $B$, $C$, $D$ be four points lying on a circle. Their cross-ratio is

$$(A, B; C, D) = \pm \frac{CA}{CB} : \frac{DA}{DB},$$

where the sign is $+$ if the segments $AB$ and $CD$ do not intersect and $-$ otherwise.

**Definition 17.10 (Harmonic bundle; harmonic quadrilateral).** Four points $A$, $B$, $C$, $D$ on a line with $(A, B; C, D) = -1$ form a *harmonic bundle* (also: a harmonic division, or simply: they are harmonic). Four points $A$, $B$, $C$, $D$ lying in this order on a circle form a *harmonic quadrilateral* $ABCD$ if $(A, C; B, D) = -1$; equivalently, a cyclic quadrilateral

$ABCD$ is harmonic if and only if $AB \cdot CD = DA \cdot BC$.

**Theorem 17.11.** *Let $D$, $E$, $F$ be points in the interiors of the sides $BC$, $CA$, $AB$ of the triangle $ABC$, and let $G$ be the intersection of the line $FE$ with the extension of the side $BC$ (with $C$ between $B$ and $G$). Then $(B, C; D, G) = -1$ if and only if $AD$, $BE$ and $CF$ are concurrent.*

**Theorem 17.12 (Projection from a point).** *Let $A$, $B$, $C$, $D$ lie in this order on a line $d$ and let $P$ be a point not on $d$. Take another line $d'$ and let $A'$, $B'$, $C'$, $D'$ be the intersections of the lines $PA$, $PB$, $PC$, $PD$ with $d'$. Then*

$$(A, C; B, D) = (A', C'; B', D').$$

*This configuration is denoted $P(A, C; B, D)$ and is called a* pencil.

**Theorem 17.13 (Projection from a point of a circle).** *Let $A$, $B$, $C$, $D$ lie in this order on a circle and let $P$ be a further point of that circle. If the lines $PA$, $PB$, $PC$, $PD$ meet a line $d$ at $A'$, $B'$, $C'$, $D'$, then $(A, C; B, D) = (A', C'; B', D')$.*

**Theorem 17.14.** *Let $A$, $B$, $C$ be three points on a circle $\omega$. Let the tangents to $\omega$ at $A$ and at $C$ meet at a point $P$, and let the line $PB$ meet $\omega$ again at $D$. Then $ABCD$ is a harmonic quadrilateral.*

**Theorem 17.15.** *Let $A$, $B$, $C$, $D$ lie in this order on a line $d$ and let $X$ be a point not on $d$. If two of the following three statements hold, so does the third:*
*(a) $(A, C; B, D) = -1$;*
*(b) $XB$ is the internal bisector of $\angle AXC$;*
*(c) $XB \perp XD$.*

**Theorem 17.16.** *Let $\omega$ be a circle and $P$ a point outside it, and let $PC$, $PD$ be the two tangents from $P$ to $\omega$ (with $C$, $D$ on $\omega$). A line $l$ through $P$ meets $\omega$ at $A$ and $B$, with $A$ between $P$ and $B$, and the segments $AB$ and $CD$ meet at $Q$. Then $(P, Q; A, B)$ is harmonic.*

**Theorem 17.17 (Newton's and Maclaurin's identities).** *Let $A$, $C$, $B$, $D$ lie on a line in this order and let $M$ be the midpoint of $AB$. Then $(A, B; C, D) = -1$ if and only if*

$$MA^2 = MB^2 = MC \cdot MD \qquad \textit{(Newton's identity)},$$

*and in that case also*

$$DC \cdot DM = DA \cdot DB, \qquad CD \cdot CM = CA \cdot CB \qquad \textit{(Maclaurin's identity)}.$$

*Equivalently, if $A$, $C$, $B$, $D$ lie in this order on a line and $M$ is the midpoint of $CD$, then*

*$(A, B; C, D)$ is harmonic if and only if $AC \cdot AD = AB \cdot AM$.*

**Theorem 17.18 (Harmonic quadrilaterals and symmedians).** *Let $T$ be a point and let $T(A, B; C, D) = -1$ be a harmonic pencil. If a line parallel to $TD$ meets $TA$, $TB$, $TC$ at $A'$, $B'$, $C'$, then $C'$ is the midpoint of $A'B'$; and if $S$ is any point and four lines through $S$ are drawn perpendicular to $TA$, $TB$, $TC$, $TD$, the resulting pencil is again harmonic. If $ACBD$ is a harmonic quadrilateral, then $AB$ is a symmedian of each of the triangles $ACD$ and $BCD$, and likewise $CD$ is a symmedian of the triangles $CAB$ and $DAB$.*

**Theorem 17.19 (Butterfly theorem).** *Let $PQ$ be a chord of a circle and let $M$ be its midpoint. Draw two further chords $AB$ and $CD$ through $M$, and let $X$ and $Y$ be the intersection points of $AD$ and of $CB$ with $PQ$. Then $M$ is the midpoint of $XY$.*

**Definition 17.20 (Pole, polar, conjugate points).** Let $\omega$ be a circle with centre $O$ and radius $r$, and let $A \neq O$. Let $A'$ be the point of the ray $OA$ with $OA \cdot OA' = r^2$. The line $a$ through $A'$ perpendicular to $OA$ is called the *polar* of $A$ with respect to $\omega$, and $A$ is called the *pole* of $a$. Two points are said to be *conjugate* with respect to $\omega$ when each of them lies on the polar of the other.

**Theorem 17.21 (La Hire's theorem).** *If a point $A$ lies on the polar $b$ of a point $B$ with respect to $\omega$, then $B$ lies on the polar $a$ of $A$ with respect to $\omega$.*

**Theorem 17.22.** *Let $\omega$ be a circle and $P$ a point outside it, and let $PC$ and $PD$ be the two tangents from $P$ to $\omega$, with $C$, $D$ on $\omega$. Then $CD$ is the polar of $P$ with respect to $\omega$.*

**Theorem 17.23 (Brokard's theorem).** *Let the quadrilateral $ABCD$ be inscribed in the circle $k$ with centre $O$, and let $E = AB \cap CD$, $F = AD \cap BC$, $G = AC \cap BD$. Then $O$ is the orthocentre of the triangle $EFG$. Moreover the triangle $EFG$ is self-polar with respect to $k$: the lines $EF$, $EG$, $FG$ are the polars of $G$, $F$, $E$ respectively.*

**Theorem 17.24 (Brianchon's theorem).** *Let $ABCDEF$ be a hexagon with an inscribed circle $\omega$. Then the lines $AD$, $BE$ and $CF$ are concurrent.*

**Problem 17.100.** Let $A$, $B$, $C$, $D$ be four points on a line and let $M$ be the midpoint of $AC$. Prove that $(A, C; B, D)$ is a harmonic bundle if and only if $MB \cdot MD = MA^2$.

**Problem 17.101.** Prove that two circles are orthogonal if and only if two diametrically opposite points of one of them are conjugate with respect to the other.

**Problem 17.102.** Prove that $A$ and $B$ are conjugate with respect to the circle $\omega$ if and only if

$$AB^2 = \mathrm{pow}_\omega(A) + \mathrm{pow}_\omega(B).$$

**Problem 17.103.** The points $A$, $A'$ are conjugate with respect to $\omega$, and the points $B$, $B'$ are conjugate with respect to $\omega$. Prove that the points $C = AB \cap A'B'$ and $C' = AB' \cap A'B$ are conjugate with respect to $\omega$ as well.

**Problem 17.104.** In the cyclic quadrilateral $ABCD$ with circumcentre $O$ the lines $AB$ and $CD$ meet at the point $E$ and the lines $AD$ and $BC$ meet at the point $F$. Let $P$ be the intersection point of the diagonals $AC$ and $BD$. Prove that $OP \perp EF$.

**Problem 17.105.** Triangle $ABC$ is inscribed in a circle $\omega$. Points $P$ and $Q$ are on the side $AB$ with $AP < AQ$. The rays $CP$ and $CQ$ meet $\omega$ again at $S$ and $T$ (other than $C$), respectively. If $AP = 4$, $PQ = 3$, $QB = 6$, $BT = 5$ and $AS = 7$, then $ST = m/n$, where $m$ and $n$ are relatively prime positive integers. Find $m + n$. [AIME II 2016]

**Problem 17.106.** Let $ABC$ be a triangle and let $H$ be its orthocentre. The lines $BH$ and $CH$ intersect $AC$ and $AB$ at the points $D$ and $E$ respectively. The circumcircle of $ADE$ intersects the circumcircle of $ABC$ at $F \neq A$. Prove that the bisectors of $\angle BFC$ and $\angle BHC$ meet at a point of $BC$. [Brazil 2011]

**Problem 17.107.** The incircle $\omega$ of the triangle $ABC$ is tangent to $BC$ at $X$. Let $Y \neq X$ be the other intersection point of $AX$ with $\omega$. Points $P$ and $Q$ lie on $AB$ and $AC$ respectively, so that $PQ$ is tangent to $\omega$ at $Y$. Assume that $AP = 3$, $PB = 4$, $AC = 8$ and $AQ = m/n$, where $m$ and $n$ are relatively prime positive integers. Find $m + n$. [AIME II 2018]

**Problem 17.108.** $ABC$ is a triangle with incentre $I$. The foot of the perpendicular from $I$ to $BC$ is $D$, and the foot of the perpendicular from $I$ to $AD$ is $P$. Prove that $\angle BPD = \angle DPC$. [ELMO Shortlist 2012]

**Problem 17.109.** Let $E$ and $F$ be the intersections of the opposite sides of a convex quadrilateral $ABCD$, and let the two diagonals meet at $P$. Let $O$ be the foot of the perpendicular from $P$ to $EF$. Prove that $\angle BOC = \angle AOD$. [China TST 2002]

**Problem 17.110.** Let $ABCD$ be a convex quadrilateral, let $P$ be the intersection of the lines $AB$ and $CD$, let $Q$ be the intersection of the lines $AD$ and $BC$, and let $O$ be the intersection of the diagonals $AC$ and $BD$. Prove that if $\angle POQ = 90°$, then $PO$ is the bisector of $\angle AOD$ and $OQ$ is the bisector of $\angle AOB$. [Brazil 2007]

**Problem 17.111.** Let $ABCD$ be a convex quadrilateral, $O \in AC \cap BD$, $P \in AB \cap CD$, $Q \in BC \cap DA$. If $R$ is the projection of $O$ on the line $PQ$, prove that the orthogonal projections of $R$ on the sidelines of $ABCD$ are concyclic. [Romania TST 2008]

**Problem 17.112.** Let $A$ and $B$ be two fixed points and let $C$ be a variable point such that $\angle ACB = \alpha$, a constant in the range $[0°, 180°]$. The incircle of $ABC$ with incentre $I$ touches the sides $AB$, $BC$, $CA$ at the points $D$, $E$, $F$ respectively, and $AI$, $BI$ intersect $EF$ at $M$, $N$ respectively. Prove that the length of $MN$ is constant and that the circumcircle of $DMN$ passes through a fixed point. [Vietnam 2009]

**Problem 17.113.** In a cyclic quadrilateral $ABCD$, let $E$ be the intersection of $AD$ and $BC$ (so that $C$ is between $B$ and $E$), and let $F$ be the intersection of $AC$ and $BD$. Let $M$ be the midpoint of the side $CD$, and let $N \neq M$ be a point on the circumcircle of the triangle $ABM$ such that $MB/MA = NB/NA$. Show that $E$, $F$, $N$ are collinear. [IMO Shortlist 2004]

**Problem 17.114.** Let $ABC$ be an acute, non-isosceles triangle with altitudes $BD$ and $CE$. Take points $M$, $N$ on $BC$ such that the circles $(ADM)$ and $(ADN)$ are tangent to $BC$. Prove that $M$, $N$, $A$, $E$ are concyclic.

**Problem 17.115.** Let $ABC$ be an acute, non-isosceles triangle with circumcentre $O$. The tangent lines to $(O)$ at $B$ and $C$ meet at $T$. A line through $T$ cuts the segment $AB$ at $D$ and the ray $CA$ at $E$. Take $M$ as the midpoint of $DE$ and suppose that $MA$ cuts $(O)$ again at $K$. Prove that the circle $(MKT)$ is tangent to $(O)$.

**Problem 17.116.** Let $ABC$ be an acute, non-isosceles triangle with circumcentre $O$ and incentre $I$, and let the incircle $(I)$ touch $BC$, $CA$, $AB$ at $D$, $E$, $F$ respectively. Suppose that $EF$ cuts $(O)$ at $P$ and $Q$. Prove that the circle $(PQD)$ bisects the segment $BC$. [IMO Shortlist 2019]

**Problem 17.117.** Let $ABC$ be a triangle with $AB \neq AC$ and with incircle $(I)$ tangent to $BC$ at $D$. Take $K$ on $AD$ such that $CD = CK$. Suppose that $AD$ cuts $(I)$ at $G$ and that $BG$ cuts $CK$ at $L$. Prove that $K$ is the midpoint of $CL$. [Chinese TST 2008]

**Problem 17.118.** Let $ABC$ be a triangle with $AB \neq AC$ inscribed in $(O)$. The tangent line to $(O)$ at $A$ cuts $BC$ at $D$. Take $H$ as the projection of $A$ on $OD$ and let $E$, $F$ be the projections of $H$ on $AB$, $AC$. Suppose that $EF$ cuts $(O)$ at $R$ and $S$. Prove that the circle $(HRS)$ is tangent to $OD$.

**Problem 17.119.** Let $ABC$ be an acute, non-isosceles triangle with altitude $AD$ ($D \in BC$), let $M$ be the midpoint of $AD$ and let $O$ be the circumcentre. The line $AO$ meets $BC$ at $K$, and the circle with centre $K$ and radius $KA$ cuts $AB$, $AC$ at $E$, $F$ respectively. Prove that $AO$ bisects $EF$.

**Problem 17.120.** Let $ABCD$ be a quadrilateral inscribed in a circle $(O)$ and let $E$, $F$, $K$ be the intersections of $AC$ and $BD$, of $AD$ and $BC$, and of $AB$ and $CD$. Denote by $M$ the midpoint of $CD$ and suppose that $EF$ intersects the circumcircle of $MAB$ at $N$ and the lines $AB$, $CD$ at $J$, $I$ respectively. Prove that $I$ belongs to the circumcircle of $ABM$ and that $MA/MB = NA/NB$.†

**Problem 17.121.** Let $ABC$ be a non-isosceles triangle with incircle $(I)$ tangent to $BC$, $CA$, $AB$ at $D$, $E$, $F$ respectively. Suppose that $AD$ meets $(I)$ again at $P$, with $\angle BPC = 90°$. Denote $K = BC \cap EF$ and $Q = PC \cap (I)$. Prove that $QP = QD$ and $EA = AP = PD$.†

**Problem 17.122.** Let $ABC$ be a triangle with a point $M$ inside it. Suppose that $AM$, $BM$, $CM$ intersect $BC$, $CA$, $AB$ at $D$, $E$, $F$ respectively. Take a point $X$ on $BC$ such that $\angle AMX = 90°$, and let $Y$, $Z$ be the reflections of $X$ in the lines $DE$, $DF$. Denote by $T$ the reflection of $M$ in $BC$.

(a) Prove that $XM$ and $XT$ are tangent to the circle $(YZT)$.
(b) Prove that $X$, $Y$, $Z$ are collinear.

**Problem 17.123.** Let $ABC$ be a triangle with altitude $AH$ and let $K$ be the midpoint of the segment $AH$. The incircle $(I)$ of $ABC$ is tangent to $BC$ at $D$. Suppose that $DK$ intersects $(I)$ again at $T$. Prove that the circle $(TBC)$ is tangent to $(I)$.

**Problem 17.124.** Let $ABC$ be a triangle with a point $I$ inside it. Suppose that the rays $IA$, $IB$, $IC$ intersect $BC$, $CA$, $AB$ at $D$, $E$, $F$ respectively. A circle $(O)$ through $D$, $E$, $F$ intersects $AD$, $BE$, $CF$ again at $M$, $N$, $P$. The tangents to $(O)$ at $D$ and $M$ meet at $X$, the tangents at $E$ and $N$ meet at $Y$, and the tangents at $F$ and $P$ meet at $Z$. Prove that $X$, $Y$, $Z$ are collinear and that the line through them is perpendicular to $IO$.†

**Problem 17.125.** Let $ABC$ be an acute triangle with altitudes $AD$, $BE$, $CF$ and orthocentre $H$. Suppose that $DE$ and $CF$ intersect at $M$ and that $DF$ and $BE$ intersect at $N$. Prove that the line through $A$ perpendicular to $MN$ passes through the circumcircle of the triangle $BCH$.†

**Problem 17.126.** Let $ABC$ be a triangle with incircle $(I)$ tangent to $BC$, $CA$, $AB$ at $D$, $E$, $F$. On the segment $BC$ take a point $M$ such that $IM \parallel EF$, and on the segment $AC$ take a point $N$ such that $IN \parallel DF$. Denote by $P$ and $Q$ the projections of $D$ on $EF$ and of $E$ on $DF$ respectively.

(a) Prove that $IP \parallel AM$ and $IQ \parallel BN$.

(b) Prove that $AM$, $BN$ and $IF$ are concurrent.

†

**Problem 17.127.** Let $ABC$ be a triangle with incircle $(I)$, and denote by $D$, $E$, $F$ the tangency points of $(I)$ on $BC$, $CA$, $AB$ respectively. Suppose that $DE$ intersects $AB$ at $P$. A line through $C$ intersects $AB$ and $EF$ at $M$ and $N$ respectively, and the lines $PN$ and $AC$ intersect at $Q$. Prove that $IM$ is perpendicular to $FQ$.

**Problem 17.128.** Let $ABCD$ be a quadrilateral and suppose that $(O)$ is a circle tangent to $AB$, $BC$, $CD$, $DA$ at $M$, $N$, $P$, $Q$ respectively. Denote by $d$ the line through $C$ perpendicular to $OC$, and let $E$, $F$ be the intersections of $NQ$, $MP$ with $d$. Suppose that $AD$, $BC$ and $MP$ are concurrent; prove that $EB$, $FD$ and $OC$ are concurrent.

**Problem 17.129.** Circles $\omega_1$ and $\omega_2$ have centres $O_1$ and $O_2$ respectively, and they intersect at the points $X$ and $Y$. The segment $AB$ is a common tangent line of these two circles, with $A$ on $\omega_1$ and $B$ on $\omega_2$. Let the tangents to $\omega_1$ and to $\omega_2$ at $X$ intersect $O_1O_2$ at the points $K$ and $L$ respectively. Suppose that the line $BL$ intersects $\omega_2$ a second time at $M$ and that the line $AK$ intersects $\omega_1$ a second time at $N$. Prove that the lines $AM$, $BN$ and $O_1O_2$ are concurrent.

**Problem 17.130.** Let $M$ be the midpoint of the side $BC$ of the triangle $ABC$. The circle $\omega$ lies inside $ABC$ and touches the sides $AB$ and $AC$ at the points $E$ and $F$ respectively. The tangents from $M$ to $\omega$ meet this circle at $P$ and $Q$, with $P$ and $B$ on the same side of $AM$. The lines $PM$ and $BF$ intersect at $X$ and the lines $QM$ and $CE$ intersect at $Y$. Suppose that $PM = 2\,BC$. Prove that $XY$ is tangent to $\omega$.†

**Problem 17.131.** Let $ABC$ be a triangle and denote by $M_A$ and $P_A$ the midpoint of $BC$ and the foot of the altitude from $A$; define $M_B$, $P_B$, $M_C$, $P_C$ similarly. The lines $M_BM_C$ and $P_BP_C$ meet at $S_A$, and the tangent to the circumcircle of $ABC$ at $A$ intersects the line $BC$ at $T_A$; define $S_B$, $T_B$, $S_C$, $T_C$ similarly. Prove that the perpendiculars through $A$, $B$, $C$ to the lines $S_AT_A$, $S_BT_B$, $S_CT_C$ respectively are concurrent.

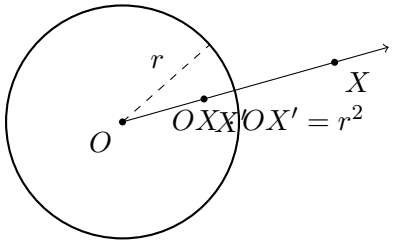
**Problem 17.132.** In a convex quadrilateral $ABCD$ the lines $AB$ and $CD$ meet at the point $E$ and the lines $AD$ and $BC$ meet at the point $F$. Let $P$ be the intersection point of the diagonals $AC$ and $BD$. Denote by $\omega_1$ the circle through $D$ tangent to $AC$ at $P$, and by $\omega_2$ the circle through $C$ tangent to $BD$ at $P$. Let $X$ be the intersection point of $\omega_1$ and $AD$, and $Y$ the intersection point of $\omega_2$ and $BC$. The circles $\omega_1$ and $\omega_2$ intersect a second time at $Q$. Prove that the perpendicular from $P$ to the line $EF$ passes through the circumcentre of the triangle $XQY$.

## 17.5 Inversion

**Definition 17.25 (Inversion).** Let $k(O, r)$ be the circle with centre $O$ and radius $r$. The *inversion* with centre $O$ and radius $r$ (equivalently: the inversion with respect to $k$) is the map $I$ sending each point $X \neq O$ to the point $X'$ of the ray $OX$ with

$$OX \cdot OX' = r^2.$$

The centre $O$ is regarded as being sent to an extra point $\infty$ lying on every line of the plane.




**Theorem 17.26 (Properties of inversion, I).** *Let $I$ be the inversion with centre $O$ and radius $r$.*

*(a) The points of $k$ are fixed: $X \in k \iff X' = X$.*
*(b) $I$ is an involution: $(X')' = X$, so $I(X) = Y$ implies $I(Y) = X$.*
*(c) If $O \notin AB$, then $OA \cdot OA' = OB \cdot OB'$, and the points $A$, $A'$, $B$, $B'$ are concyclic.*
*(d) If $O \notin AB$, then the triangles $OAB$ and $OB'A'$ are similar; in particular $\angle OAB = \angle OB'A'$ and*

$$A'B' = AB \cdot \frac{r^2}{OA \cdot OB}.$$

*(e) Changing the radius of the inversion composes it with a homothety centred at $O$.*

**Theorem 17.27 (Properties of inversion, II: lines and circles).** *Let $I$ be the inversion with centre $O$ and radius $r$.*

*(a) If $s$ is a line through $O$, then $I(s) = s$ as a set of points.*
*(b) If $t$ is a line not passing through $O$, then $I(t)$ is a circle through $O$: taking $P \in t$ with $OP \perp t$ and $P' = I(P)$, the image is the circle with diameter $OP'$.*
*(c) If $k_1$ is a circle through $O$, then $I(k_1)$ is a line not passing through $O$.*
*(d) If $k_1$ is a circle not passing through $O$, then $I(k_1)$ is a circle not passing through $O$: if the line $OO_1$ meets $k_1$ at $A$ and $B$, the image is the circle with diameter $A'B'$. The image of the centre $O_1$ is in general* not *the centre $O_2$ of the image circle, but $O$, $O_1$, $O_2$ are always collinear.*
*(e) Inversion preserves the angle between two curves; in particular it preserves tangency. The number of intersection points of two figures is unchanged, except for the point $O$: if two circles are tangent at $O$, their images are two parallel lines.*
*(f) A circle orthogonal to $k$ is mapped to itself, as is every line through $O$.*

**Problem 17.133.** A line $r$ contains the points $A$, $B$, $C$, $D$ in that order. Let $P$ be a point not on $r$ such that $\angle APB = \angle CPD$. Prove that the bisector of the angle $\angle APD$ intersects the line $r$ at a point $G$ such that

$$\frac{1}{GA} + \frac{1}{GC} = \frac{1}{GB} + \frac{1}{GD}.$$

[IberoAmerican 2015]

**Problem 17.134.** Let $ABCD$ be a convex quadrilateral. Prove that

$$AC \cdot BD \leq AB \cdot CD + AD \cdot BC,$$

with equality if and only if the quadrilateral is cyclic. (*Ptolemy's theorem.*)

**Problem 17.135.** Let $P$ be a point inside the acute angle $\angle ABC$. Show how to draw a line through $P$ cutting the lines $AB$ at $M$ and $BC$ at $N$ so that $\dfrac{1}{MP} + \dfrac{1}{NP}$ is a maximum.

**Problem 17.136.** Circles $\omega_1$ and $\omega_2$ intersect at the points $A$ and $B$. Points $C$ and $D$ are on $\omega_1$ and $\omega_2$ respectively, such that the lines $AC$ and $AD$ are tangent to $\omega_2$ and $\omega_1$ respectively. Let $I_1$ and $I_2$ be the incentres of the triangles $ABC$ and $ABD$ respectively, and let the segments $I_1I_2$ and $AB$ intersect at $E$. Prove that

$$\frac{1}{AE} = \frac{1}{AC} + \frac{1}{AD}.$$

[China TST 2002]

**Problem 17.137.** Let $ABCD$ be a convex quadrilateral and let $P$ and $Q$ be points inside $ABCD$ such that $PQDA$ and $QPBC$ are cyclic quadrilaterals. Suppose that there exists a point $E$ on the line segment $PQ$ such that $\angle PAE = \angle QDE$ and $\angle PBE = \angle QCE$. Show that the quadrilateral $ABCD$ is cyclic. [IMO Shortlist 2008]

**Problem 17.138.** Let $P$ be a point inside a triangle $ABC$ such that

$$\angle APB - \angle ACB = \angle APC - \angle ABC.$$

Let $D$, $E$ be the incentres of the triangles $APB$, $APC$ respectively. Show that the lines $AP$, $BD$, $CE$ meet at a point. [IMO 1996]

**Problem 17.139.** Let $\Gamma_1$, $\Gamma_2$, $\Gamma_3$, $\Gamma_4$ be distinct circles such that $\Gamma_1$, $\Gamma_3$ are externally tangent at $P$, and $\Gamma_2$, $\Gamma_4$ are externally tangent at the same point $P$. Suppose that $\Gamma_1$ and $\Gamma_2$; $\Gamma_2$ and $\Gamma_3$; $\Gamma_3$ and $\Gamma_4$; $\Gamma_4$ and $\Gamma_1$ meet at $A$, $B$, $C$, $D$ respectively, and that all these points are different from $P$. Prove that

$$\frac{AB \cdot BC}{AD \cdot DC} = \frac{PB^2}{PD^2}.$$

[IMO Shortlist 2003]

**Problem 17.140.** Two circles $\omega_1$ and $\omega_2$ of equal radius intersect at different points $X_1$ and $X_2$. Consider a circle $\omega$ externally tangent to $\omega_1$ at a point $T_1$ and internally tangent to $\omega_2$ at a point $T_2$. Prove that the lines $X_1T_1$ and $X_2T_2$ intersect at a point lying on $\omega$. [EGMO 2016]

**Problem 17.141.** Let $\omega$ be the incircle of a triangle $ABC$ and let $I$ be its incentre. Let $\Gamma$ be the circumcircle of the triangle $AIB$. Denote by $X$ and $Y$ the two points of intersection of $\omega$ and $\Gamma$, and by $Z$ the point of intersection of the common tangents to $\omega$ and $\Gamma$. Prove that the circumcircles of the triangles $ABC$ and $XYZ$ are tangent to each other. [Russia 2013]

**Problem 17.142.** Let $ABC$ be a triangle and $O$ its circumcentre. The lines $AB$ and $AC$ intersect the circumcircle of $OBC$ again at points $B_1 \neq B$ and $C_1 \neq C$ respectively. The lines $BA$ and $BC$ intersect the circumcircle of $OAC$ again at points $A_2 \neq A$ and $C_2 \neq C$ respectively. The lines $CA$ and $CB$ intersect the circumcircle of $OAB$ again at points $A_3 \neq A$ and $B_3 \neq B$ respectively. Prove that the lines $A_2A_3$, $B_1B_3$ and $C_1C_2$ have a common point. [Brazil 2009]

**Problem 17.143.** The incircle of the triangle $ABC$ touches the sides $BC$, $AC$ and $AB$ at $D$, $E$ and $F$ respectively. Let $\omega_a$, $\omega_b$, $\omega_c$ be the circumcircles of the triangles $EAF$, $DBF$ and $DCE$ respectively. The lines $DE$ and $DF$ cut $\omega_a$ at $E_a \neq E$ and $F_a \neq F$ respectively, and $r_A$ denotes the line $E_aF_a$; the lines $r_B$ and $r_C$ are defined analogously. Show that the lines $r_A$, $r_B$ and $r_C$ determine a triangle whose vertices lie on the sides of the triangle $ABC$. [Cono Sur 2010]

**Problem 17.144.** Let $\Gamma$ be the circumcircle of a triangle $ABC$. A circle with centre $O$ touches the line segment $BC$ at $P$ and touches the arc $BC$ of $\Gamma$ not containing $A$ at $Q$. If $\angle BAO = \angle CAO$, prove that $\angle PAO = \angle QAO$. [Japan 2009]

**Problem 17.145.** The incircle of $ABC$ touches $BC$, $CA$ and $AB$ at $D$, $E$ and $F$ respectively. $X$ is a point inside $ABC$ such that the incircle of $XBC$ touches $BC$ at $D$ also, and touches $CX$ and $XB$ at $Y$ and $Z$ respectively. Prove that $EFZY$ is a cyclic quadrilateral. [IMO Shortlist 1995]

**Problem 17.146.** Given is a triangle $ABC$ with incircle $\omega$, tangent to $BC$, $CA$, $AB$ at $D$, $E$, $F$. The perpendicular from $B$ to $BC$ meets $EF$ at $M$, and the perpendicular from $C$ to $BC$ meets $EF$ at $N$. Let $DM$ and $DN$ meet $\omega$ at $P$ and $Q$. Prove that $DP = DQ$. [Cono Sur 2022]

**Problem 17.147.** Given a triangle $ABC$ with incentre $I$, let $D$, $E$, $F$ be the tangency points of the incircle with the sides $BC$, $CA$, $AB$ respectively, and let $K$, $L$, $M$ be symmetric to $I$ with respect to $EF$, $FD$, $DE$ respectively. Prove that the circumcentres of $ABC$, $DEF$ and $KLM$ are collinear.

**Problem 17.148.** Let $ABCD$ be a quadrilateral inscribed in a circle $(O)$. The tangent line to $(O)$ at $D$ intersects the lines $AB$ and $AC$ at $M$ and $N$. Prove that

$$\frac{1}{DM} + \frac{1}{DN} = \frac{BC}{DB \cdot DC}.$$

†

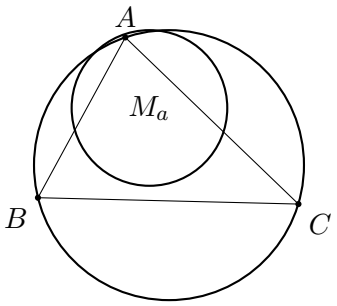
**Problem 17.149.** Let $(O)$ be a fixed circle with a fixed diameter $AB$, and take a fixed point $I$ on the segment $AB$ different from $A$ and $B$. An arbitrary line $d$ (different from $AB$) through $I$ intersects $(O)$ at $P$ and $Q$. The line $m$ is the tangent line to $(O)$ at $B$, and $AP$, $AQ$ intersect $m$ at $M$ and $N$. Prove that the centre of the circle $(AMN)$ lies on a fixed line.

**Problem 17.150.** Let $ABC$ be a triangle with circumcircle $(O)$ and incircle $(I)$. Let $A_0$, $B_0$, $C_0$ be the tangency points of $(I)$ with $BC$, $CA$, $AB$, and let $(O_a)$, $(O_b)$, $(O_c)$ be the circumcircles of the triangles $AB_0C_0$, $BC_0A_0$ and $CA_0B_0$. Suppose that $A_1$, $B_1$, $C_1$ are the second intersection points of $(O_a)$, $(O_b)$, $(O_c)$ with $(O)$ respectively. Prove that $A_0A_1$, $B_0B_1$ and $C_0C_1$ are concurrent at a point $N$ which lies on the Euler line of the triangle $A_0B_0C_0$.

## 17.6 Mixtilinear incircles and Feuerbach's theorem

**Theorem 17.28 (Feuerbach's theorem).** *The nine-point circle of a triangle is tangent to the incircle and to all three excircles.*

**Definition 17.29 (Mixtilinear incircle).** Let $ABC$ be a triangle with circumcircle $(O)$ and incentre $I$. The *$A$-mixtilinear incircle* $(M_a)$ is the circle tangent to the sides $AB$ and $AC$ and internally tangent to $(O)$; the circles $(M_b)$ and $(M_c)$ are defined analogously.




**Theorem 17.30 (Basic facts on the mixtilinear incircle).** *Let $(M_a)$ touch $AB$ and $AC$ at $E$ and $F$ and touch the circumcircle $(O)$ at $T$. Then:*
*(a) the incentre $I$ of $ABC$ is the midpoint of $EF$;*
*(b) the line $TI$ passes through the midpoint of the arc $BAC$ of $(O)$;*
*(c) the lines $AT$ and $AL$ are isogonal in the angle $A$, where $L$ is the point at which the $A$-excircle touches $BC$.*

**Problem 17.151.** Let $X$ be the tangency point of $(M_a)$ and $(O)$. Prove that $AO$ is tangent to the circle $(AIX)$.

**Problem 17.152.** Let $\angle xAy$ be an angle and $(I)$ a fixed circle tangent to the rays $Ax$ and $Ay$. An arbitrary tangent line of $(I)$ intersects the rays $Ax$, $Ay$ at $B$, $C$. Prove that the circle $(ABC)$ is always tangent to a fixed circle.

**Problem 17.153.** Let $A_1$ be the tangency point of $(O)$ and $(M_a)$, and let $A_1A_2$ be a diameter of $(M_a)$. Define $B_2$ and $C_2$ similarly. Prove that $AA_2$, $BB_2$ and $CC_2$ are concurrent.

**Problem 17.154.** Let $X$, $Y$, $Z$ be the tangency points of $(M_a)$ with $(O)$, $AB$, $AC$ respectively. Suppose that $AX$ intersects $YZ$ at $N$ and that $IX$ intersects $BC$ at $M$. Prove that $MN \parallel AI$ and that the circle $(XNI)$ is tangent to $(O)$.

**Problem 17.155.** Let $T$ be the tangency point of $(O)$ and $(M_a)$. Denote by $x$, $y$, $z$ the lengths of the tangent segments from $A$, $B$, $C$ to $(M_a)$. Prove that

$$\frac{x}{TA} = \frac{y}{TB} = \frac{z}{TC}.$$

**Problem 17.156.** Let $X$ be the tangency point of $(O)$ and $(M_a)$. Prove that $XM_b$ and $XM_c$ are symmetric with respect to the bisector of the angle $BXC$.

**Problem 17.157.** ★ Let $ABC$ be a fixed triangle with $\angle A < \angle B < \angle C$ and circumcircle $(O)$. On the minor arc $BC$ of $(O)$ take an arbitrary point $D$. Suppose that $CD$ intersects $AB$ at $E$ and that $BD$ intersects $AC$ at $F$. Denote by $(O_1)$ the circle lying inside the triangle $EBD$, tangent to $EB$, $ED$ and also tangent to $(O)$; denote by $(O_2)$ the circle lying inside the triangle $FCD$, tangent to $FC$, $FD$ and also tangent to $(O)$.

(a) Let $M$ be the tangency point of $(O_1)$ with $BE$ and let $N$ be the tangency point of $(O_2)$ with $CF$. Prove that the circle with diameter $MN$ always passes through a certain fixed point.

(b) The line through $M$ parallel to $CE$ intersects $AC$ at $P$, and the line through $N$ parallel to $BF$ intersects $AB$ at $Q$. Prove that the circumcircles of the triangles $AMP$ and $ANQ$ are both tangent to a fixed circle.

**Problem 17.158.** ★ Let $ABC$ be a triangle with circumcentre $O$, incentre $I$ and $A$-excentre $J$. Denote by $D$ the tangency point of $(I)$ on $BC$, and let the bisector of the angle $A$ cut $BC$ and $(O)$ at $E$ and $F$ respectively. The circle $(DEF)$ meets $(O)$ again at $T$. Prove that $AT$ passes through an intersection point of $(J)$ and $(DEF)$.

**Problem 17.159.** Let $\Omega$ be the circumcircle of an acute triangle $ABC$ and let $D$ be the midpoint of the arc $BC$ of $\Omega$ which does not contain $A$. The circle $\omega$ centred at $D$ is tangent to $BC$ at $E$. The tangents to $\omega$ through $A$ intersect the line $BC$ at points $K$, $L$ in such a way that $B$, $K$, $L$, $C$ is the order of the points on $BC$. The circle $\gamma_1$ is tangent to the segments $AL$, $BL$ and to $\Omega$ at $M$, and the circle $\gamma_2$ is tangent to the segments $AK$, $BK$ and to $\Omega$ at $N$. The lines $KN$ and $LM$ intersect at $P$. Prove that $\angle KAP = \angle EAL$.

**Problem 17.160.** Let $O$ be the midpoint of the hypotenuse $AB$ of a right triangle $ABC$, and let $\Omega$ be the circumcircle of this triangle. The circle $\omega_A$ is tangent to the segment $AO$ at $K$, to the segment $CO$ at $Z$, and to $\Omega$ at $M$. The circle $\omega_B$ is tangent to the segment $BO$ at $L$, to the segment $CO$ at $W$, and to $\Omega$ at $N$. Let $X$ be the intersection of the line $AC$ with the line $MK$, and let $Y$ be the intersection of the line $BC$ with the line $NL$. Prove that the quadrilateral $XWYZ$ is a parallelogram.

**Problem 17.161.** In an acute triangle $ABC$ with $AB < AC$, let $O$ be the circumcentre. Let the tangent to the circumcircle at the point $A$ intersect the line $BC$ at the point $D$. The line $DO$ intersects the segments $AB$ and $AC$ at the points $E$ and $F$ respectively. The point $G$ is constructed so that $AEGF$ is a parallelogram. Let $K$ and $H$ be the points at which the segment $BC$ intersects the segments $EG$ and $FG$ respectively. Prove that the circumcircle of the triangle $GKH$ is tangent to the circumcircle of the triangle $ABC$.

**Problem 17.162.** Given an isosceles triangle $ABC$ inscribed in a circle $\omega$, with $AC = BC$, let $M$ be the midpoint of the segment $AB$. A tangent to the circle with diameter $CM$ passing through the point $B$, different from the line $BC$, intersects the circle $\omega$ again at the point $D$. Prove that there exists a circle tangent to the lines $DC$, $CA$, $AB$ and to the circle $\omega$.

**Problem 17.163.** Triangle $ABC$ is inscribed in a circle $\omega$. Circles $\omega_1$ and $\omega_2$ are tangent to the side $BC$ at the points $E$ and $F$ respectively. Both $\omega_1$ and $\omega_2$ are internally tangent to $\omega$ and lie on the same side of $BC$ as the point $A$. Let $l$ be the common external tangent of $\omega_1$ and $\omega_2$ different from $BC$. Let $\Omega$ be a circle passing through $B$ and $C$ and tangent to the line $l$ at the point $T$, and let $\Omega'$ be a circle tangent to $l$ at $T$ and tangent to $BC$ at the point $S$. Prove that the line $TS$ is the bisector of the angle $ETF$.

## 17.7 Power of a point III: coaxial circles and the shooting lemma

*Revisits the power of a point, the radical axis and the radical centre (see Chapter 12); the problems below push it further.*

Two tools organise this section. The first is the "four-point" form of the radical axis, which turns a perpendicularity into an identity between four squared distances; the second is the shooting lemma, the observation that the midpoint of an arc is the centre of an inversion exchanging the circle with the chord.

**Lemma 17.31 (Four-point theorem).** *For points $X$, $Y$, $Z$, $T$ one has $XY \perp ZT$ if and only if*

$$XZ^2 - XT^2 = YZ^2 - YT^2.$$

*In particular, the locus of the points $X$ with $XZ^2 - XT^2$ constant is a line perpendicular to $ZT$; this is the radical axis of two suitable circles centred at $Z$ and $T$.*

**Lemma 17.32 (Shooting lemma).** *Let $AB$ be a chord of a circle $(O)$ and let $M$ be the midpoint of one of the two arcs $AB$. Then the inversion with centre $M$ and power*

$$k = MA^2 = MB^2$$

*interchanges the circle $(O)$ and the line $AB$. Consequently, for every line through $M$ meeting $AB$ at $C$ and $(O)$ again at $D$ one has $MC \cdot MD = MA^2 = MB^2$.*

**Problem 17.164.** Let $\omega_1$ and $\omega_2$ be two non-intersecting circles with centres $O_1$ and $O_2$. Their common internal tangent touches $\omega_1$ at $A_1$ and $\omega_2$ at $A_2$, and a common external tangent of $\omega_1$ and $\omega_2$ touches $\omega_1$ at $B_1$ and $\omega_2$ at $B_2$. Prove that the lines $A_1A_2$ and $B_1B_2$ intersect at a point of $O_1O_2$.

**Problem 17.165.** Let $H$ be the orthocentre of the triangle $ABC$. Points $K$, $L$ and $M$ are chosen arbitrarily on the sides $BC$, $BA$ and $AC$. Prove that $H$ is the radical centre of the circles with diameters $AK$, $CL$ and $BM$.

**Problem 17.166.** The circle with centre $O$ passes through the points $A$ and $C$ and intersects the sides $AB$ and $BC$ of the triangle $ABC$ a second time at the points $K$ and $N$ respectively. The circumcircles of the triangles $ABC$ and $KBN$ intersect a second time at the point $M$. Prove that $\angle OMB = 90^\circ$. [IMO 1985]

**Problem 17.167.** Let the lines $AB$ and $CD$ intersect at the point $P$ and let the lines $AD$ and $BC$ intersect at the point $Q$. Prove that the orthocentres of the triangles $PBC$, $PAD$, $QAB$ and $QCD$ are collinear. (The line through them is the *Ober line* of the complete quadrilateral.)

**Problem 17.168.** Let $ABCD$ be a cyclic quadrilateral. The line through $A$ parallel to $BD$ intersects the line through $B$ parallel to $AC$ at the point $E$. The circumcircle of the triangle $ABE$ intersects the lines $EC$ and $ED$ a second time at the points $F$ and $G$ respectively. Prove that the lines $AB$, $CD$ and $FG$ are either concurrent or pairwise parallel. [Romania JBMO TST 2018]

**Problem 17.169.** Let $ABCD$ be an isosceles trapezoid with $AB \parallel CD$, and let $E$ be the midpoint of $AC$. Denote by $\omega$ and $\Omega$ the circumcircles of the triangles $ABE$ and $CDE$ respectively. Let $P$ be the intersection point of the tangent to $\omega$ at $A$ and the tangent to $\Omega$ at $D$. Prove that $PE$ is tangent to $\Omega$.†

**Problem 17.170.** Let $ABCD$ be a cyclic quadrilateral. The circles with diameters $AB$ and $CD$ intersect at the points $X_1$ and $Y_1$; the circles with diameters $BC$ and $AD$ intersect at the points $X_2$ and $Y_2$; the circles with diameters $AC$ and $BD$ intersect at the points $X_3$ and $Y_3$. Prove that the lines $X_1Y_1$, $X_2Y_2$ and $X_3Y_3$ are concurrent.

**Problem 17.171.** Two circles $\omega$ and $\Omega$ are inscribed in the angle $\alpha$. A line $l$ intersects the sides of $\alpha$ at $A$ and $F$, intersects $\omega$ at $B$ and $C$, and intersects $\Omega$ at $D$ and $E$; the order of the points on the line is $A$, $B$, $C$, $D$, $E$, $F$. Suppose that $BC = DE$. Prove that $AB = EF$.

**Problem 17.172.** The circle $\omega$ lies inside the circle $\Gamma$. An arbitrary point $C$ is chosen on $\Gamma$ and the tangents from $C$ to $\omega$ intersect $\Gamma$ at the points $A$ and $B$. Find the locus of the incentres of the triangles $ABC$.

**Problem 17.173.** Let $ABC$ be a triangle and let $D$ be a point on the segment $BC$ with $D \neq B$ and $D \neq C$. The circumcircle of $ABD$ meets the segment $AC$ again at an interior point $E$, and the circumcircle of $ACD$ meets the segment $AB$ again at an interior point $F$. Let $A'$ be the reflection of $A$ in the line $BC$. The lines $A'C$ and $DE$ meet at $P$, and the lines $A'B$ and $DF$ meet at $Q$. Prove that the lines $AD$, $BP$ and $CQ$ are concurrent (or all parallel).

**Problem 17.174.** Let $ABC$ be a non-isosceles triangle with incentre $I$, circumcentre $O$, and let $D$ be a point on the segment $BC$ such that the circle $(BID)$ cuts the segment $AB$ at $E \neq B$ and the circle $(CID)$ cuts the segment $AC$ at $F \neq C$. The circle $(DEF)$ cuts the segments $AB$, $AC$ again at $M$, $N$. Let $P = IB \cap DE$ and $Q = IC \cap DF$. Prove that $EN$, $FM$ and $PQ$ are parallel and that the median from the vertex $I$ in the triangle $IPQ$ bisects the arc $BAC$ of $(O)$.

**Problem 17.175.** Let $ABC$ be a triangle inscribed in the circle $(O)$ with incentre $I$ and $A$-excentre $J$. The line through $O$ perpendicular to $AI$ cuts $(O)$ at $M$ (on the same side as $O$ with respect to $AI$). Suppose that $MI$ and $MJ$ cut $(O)$ again at $P$ and $Q$. Take $R$, $S$ on $BC$ such that $PR$ and $QS$ are tangent to $(O)$. Prove that $PQ$, $AI$ and $BC$ are concurrent and that the circle $(ARS)$ is tangent to $(O)$.

**Problem 17.176.** Let $ABC$ be a triangle inscribed in a circle $(O)$ with a diameter $KL$ passing through the midpoint $M$ of $AB$, where $L$ and $C$ lie on different sides of $AB$. A circle through $M$ and $K$ cuts $LC$ at $P$ and $Q$ (with $P$ between $Q$ and $C$). The line $KQ$ cuts the circle $(LMQ)$ again at $R$. Prove that $ARBP$ is cyclic and that $AB$ is the symmedian of the triangle $APR$. [Sharygin Geometry Olympiad 2018]

**Problem 17.177.** Consider circles $(O_1)$, $(O_2)$, $(O_3)$ tangent to a line $d$ at $A$, $B$, $C$, where $(O_2)$ is the largest and is externally tangent to $(O_1)$ and $(O_3)$. Let $BD$ be the diameter of $(O_2)$. The external tangent line of $(O_1)$ and $(O_3)$ different from $d$ cuts $(O_2)$ at $X$ and $Y$, and $K$ is the midpoint of the arc $XBY$ of $(O_2)$.

(a) Prove that $DK$ bisects the segment $AC$.

(b) Prove that the circle with diameter $AC$ touches $DX$ and $DY$.

**Problem 17.178.** Let $ABC$ be a triangle with incircle $(I)$ tangent to $BC$, $CA$, $AB$ at $D$, $E$, $F$. Denote by $K$ the projection of $D$ onto $EF$.

(a) Prove that the powers of $K$ with respect to the circles with diameters $BE$ and $CF$ are equal.

(b) Denoting by $H$ the orthocentre of $ABC$, prove that $KD$ is the bisector of the angle $HKI$. †

**Problem 17.179.** ★ Let $ABC$ be a triangle and let $M$, $N$, $P$ be the midpoints of $BC$, $CA$, $AB$. Denote by $I$ the incentre of the triangle $MNP$. Prove that $I$ is the radical centre of the three excircles of $ABC$.

**Problem 17.180.** ★ Prove that the three Apollonius circles of a triangle are coaxial.

**Problem 17.181.** Let $ABC$ be a triangle. The excircles touch the sides $BC$, $CA$ and $AB$ at the points $U$, $V$ and $W$ respectively. Let $r_U$ be the line through $U$ perpendicular to $BC$, and let $r_V$ and $r_W$ be defined similarly. Prove that $r_U$, $r_V$ and $r_W$ are concurrent. [Brazil 2020]

## 17.8 Master problems

The core of this section is the elite set used at the January 2020 camp in KAUST, in which almost every problem is an application of two or three of the techniques above at once.

**Problem 17.182.** Let $ABCD$ be a cyclic quadrilateral satisfying $AD^2 + BC^2 = AB^2$. The diagonals of $ABCD$ intersect at $E$. Let $P$ be a point on the side $AB$ satisfying $\angle APD = \angle BPC$. Show that the line $PE$ bisects $CD$. [USAMO 2019]

**Problem 17.183.** Let $ABC$ be a triangle and let $M$ and $N$ denote the midpoints of $AB$ and $AC$ respectively. Let $X$ be a point such that $AX$ is tangent to the circumcircle of the triangle $ABC$. Denote by $\omega_B$ the circle through $M$ and $B$ tangent to $MX$, and by $\omega_C$ the circle through $N$ and $C$ tangent to $NX$. Show that $\omega_B$ and $\omega_C$ intersect on the line $BC$. [USA TST 2019]

**Problem 17.184.** Let $ABCD$ be a trapezoid with bases $AB$, $CD$ and with circumscribed circle $\Omega$. Let $M$ be the midpoint of the arc $CD$ of $\Omega$ which does not contain $A$. Consider the circle $\omega$ with centre $M$ tangent to the line $AD$, and let $X$ be an intersection point of $\omega$ and $CD$. Prove that the tangent to $\omega$ at $X$ bisects the segment $AB$. [Polish MO 2020]

**Problem 17.185.** Let $ABC$ be an acute triangle with circumcircle $\omega$, and let $H$ be the foot of the altitude from $A$ to $BC$. Let $P$ and $Q$ be the points on $\omega$ with $PA = PH$ and $QA = QH$. The tangent to $\omega$ at $P$ intersects the lines $AC$ and $AB$ at $E_1$ and $F_1$ respectively; the tangent to $\omega$ at $Q$ intersects the lines $AC$ and $AB$ at $E_2$ and $F_2$ respectively. Show that the circumcircles of $AE_1F_1$ and $AE_2F_2$ are congruent, and that the line through their centres is parallel to the tangent to $\omega$ at $A$. [USA TST 2018]

**Problem 17.186.** Let $ABC$ be an acute triangle with incentre $I$, circumcentre $O$ and circumcircle $\Gamma$. Let $M$ be the midpoint of $AB$, and let the ray $AI$ meet $BC$ at $D$. Denote by $\omega$ and $\gamma$ the circumcircles of $BIC$ and $BAD$ respectively. The line $MO$ meets $\omega$ at $X$ and $Y$, while the line $CO$ meets $\omega$ at $C$ and $Q$. Assume that $Q$ lies inside $ABC$ and $\angle AQM = \angle ACB$. Consider the tangents to $\omega$ at $X$ and $Y$ and the tangents to $\gamma$ at $A$ and $D$. Given that $\angle BAC \neq 60°$, prove that these four lines are concurrent on $\Gamma$. [USA TSTST 2018]

**Problem 17.187.** Let $ABCD$ be a cyclic quadrilateral and let $P$ be a point on the side $AB$. The diagonal $AC$ meets the segment $DP$ at $Q$. The line through $P$ parallel to $CD$ meets the extension of the side $CB$ beyond $B$ at $K$. The line through $Q$ parallel to $BD$ meets the extension of the side $CB$ beyond $B$ at $L$. Prove that the circumcircles of the triangles $BKP$ and $CLQ$ are tangent. [RMM 2018]

**Problem 17.188.** Two circles $\omega_1$ and $\omega_2$, centred at $O_1$ and $O_2$ respectively, meet at the points $A$ and $B$. A line through $B$ meets $\omega_1$ again at $C$, and $\omega_2$ again at $D$. The tangents to $\omega_1$ and $\omega_2$ at $C$ and $D$ respectively meet at $E$, and the line $AE$ meets the circle $\omega$ through $A$, $O_1$, $O_2$ again at $F$. Prove that the length of the segment $EF$ is equal to the diameter of $\omega$. [RMM 2016]

**Problem 17.189.** Let $ABCD$ be a quadrilateral inscribed in a circle $\omega$ with $AC \perp BD$. Let $E$ and $F$ be the reflections of $D$ in the lines $BA$ and $BC$ respectively, and let $P$ be the intersection of the lines $BD$ and $EF$. Suppose that the circumcircle of $EPD$ meets $\omega$ at $D$ and $Q$, and that the circumcircle of $FPD$ meets $\omega$ at $D$ and $R$. Show that $EQ = FR$. [USAJMO 2018]

**Problem 17.190.** Let $ABC$ be a triangle with incentre $I$. Points $K$ and $L$ are chosen on the segment $BC$ such that the incircles of $ABK$ and $ABL$ are tangent at $P$, and the incircles of $ACK$ and $ACL$ are tangent at $Q$. Prove that $IP = IQ$. [USA TSTST 2019]

**Problem 17.191.** In the convex cyclic quadrilateral $ABCD$ the lines $AC$ and $BD$ intersect at $E$, the lines $AB$ and $CD$ intersect at $F$, and the lines $BC$ and $DA$ intersect at $G$. Suppose that the circumcircle of $ABE$ intersects the line $CB$ at $B$ and $P$, and that the circumcircle of $ADE$ intersects the line $CD$ at $D$ and $Q$, where $C$, $B$, $P$, $G$ and $C$, $Q$, $D$, $F$ are collinear in that order. Prove that if the lines $FP$ and $GQ$ intersect at $M$, then $\angle MAC = 90°$. [USAMO 2018]

**Problem 17.192.** The trapezoid $ABCD$ is inscribed in the circle $\omega$, with $AD \parallel BC$. The circles inscribed in the triangles $ABC$ and $ABD$ touch the bases $BC$ and $AD$ of the trapezoid at the points $P$ and $Q$ respectively. The points $X$ and $Y$ are the midpoints of the arcs $BC$ and $AD$ of the circle $\omega$ that do not contain the points $A$ and $B$ respectively. Prove that the lines $XP$ and $YQ$ intersect on the circle $\omega$. [Oral Moscow Geometry Olympiad 2013]

**Problem 17.193.** Let $ABC$ be a triangle with $AB = AC$, and let $M$ be the midpoint of $BC$. Let $P$ be a point such that $PB < PC$ and $PA$ is parallel to $BC$. Let $X$ and $Y$ be points on the lines $PB$ and $PC$ respectively, so that $B$ lies on the segment $PX$, $C$ lies on the segment $PY$, and $\angle PXM = \angle PYM$. Prove that the quadrilateral $APXY$ is cyclic. [IMO Shortlist 2018]

**Problem 17.194.** Let $ABCD$ be a convex quadrilateral with non-parallel sides $BC$ and $AD$. Assume that there is a point $E$ on the side $BC$ such that the quadrilaterals $ABED$ and $AECD$ are circumscribed. Prove that there is a point $F$ on the side $AD$ such that the quadrilaterals $ABCF$ and $BCDF$ are circumscribed if and only if $AB$ is parallel to $CD$. [IMO Shortlist 2002]

**Problem 17.195.** Let $ABC$ be a triangle with $\angle C = 90°$, and let $H$ be the foot of the altitude from $C$. A point $D$ is chosen inside the triangle $CBH$ so that $CH$ bisects $AD$. Let $P$ be the intersection point of the lines $BD$ and $CH$. Let $\omega$ be the semicircle with diameter $BD$ that meets the segment $CB$ at an interior point. A line through $P$ is tangent to $\omega$ at $Q$. Prove that the lines $CQ$ and $AD$ meet on $\omega$. [IMO Shortlist 2015]

**Problem 17.196.** In the triangle $ABC$, let $\omega$ be the excircle opposite to $A$. Let $D$, $E$ and $F$ be the points where $\omega$ is tangent to $BC$, $CA$ and $AB$ respectively. The circle $AEF$ intersects the line $BC$ at $P$ and $Q$. Let $M$ be the midpoint of $AD$. Prove that the circle $MPQ$ is tangent to $\omega$. [IMO Shortlist 2017]

**Problem 17.197.** An equilateral pentagon $AMNPQ$ is inscribed in the triangle $ABC$ such that $M \in AB$, $Q \in AC$ and $N, P \in BC$. Let $S$ be the intersection of $MN$ and $PQ$, and denote by $l$ the bisector of the angle $\angle MSQ$. Prove that $OI$ is parallel to $l$, where $O$ is the circumcentre and $I$ the incentre of the triangle $ABC$. [USAMO 2016]

**Problem 17.198.** Let the excircle of a triangle $ABC$ opposite the vertex $A$ be tangent to the side $BC$ at the point $A_1$. Define the points $B_1$ on $CA$ and $C_1$ on $AB$ analogously, using the excircles opposite $B$ and $C$ respectively. Denote by $\gamma$ the circumcircle of the triangle $A_1B_1C_1$ and assume that $\gamma$ passes through the vertex $A$.

(a) Show that $AA_1$ is a diameter of $\gamma$.

(b) Show that the incentre of the triangle $ABC$ lies on the line $B_1C_1$.

[USA TST for EGMO 2019]

**Problem 17.199.** Let $ABC$ be a triangle with altitude $AE$. The $A$-excircle touches $BC$ at $D$ and intersects the circumcircle at two points $F$ and $G$. Prove that one can select points $V$ and $N$ on the lines $DG$ and $DF$ such that the quadrilateral $EVAN$ is a rhombus. [USA January TST for IMO 2017]

**Problem 17.200.** In the triangle $ABC$ with $AB \neq AC$, let the incircle be tangent to the sides $BC$, $CA$ and $AB$ at $D$, $E$ and $F$ respectively. The internal bisector of $\angle BAC$ intersects the lines

$DE$ and $DF$ at $X$ and $Y$ respectively. Let $S$ and $T$ be distinct points on the side $BC$ such that $\angle XSY = \angle XTY = 90°$. Finally, let $\gamma$ be the circumcircle of $AST$. Prove that $\gamma$ is tangent to the circumcircle and to the incircle of $ABC$. [ELMO 2016]

**Problem 17.201.** Let $\omega$ be the circumcircle of the isosceles triangle $ABC$ with $AB = AC$. The points $P$ and $Q$ lie on $\omega$ and $BC$ respectively, and $AP = AQ$. Suppose that $AP$ and $BC$ intersect at $R$. Prove that the tangents from $B$ and $C$ to the incircle of the triangle $AQR$ (other than $BC$) are concurrent on $\omega$. [Iranian TST 2018]

**Problem 17.202.** Let $P$ be a point in the interior of the quadrilateral $ABCD$ such that

$$\angle BPC = 2\angle BAC, \qquad \angle PCA = \angle PAD, \qquad \angle PDA = \angle PAC.$$

Prove that $\angle PBD = \angle BCA - \angle PCA$. [Iranian TST 2017]

**Problem 17.203.** Let $ABC$ be a triangle and let $K$ be the midpoint of the arc $CAB$ of its circumcircle. Let $M$ be the midpoint of the segment $AB$. The perpendicular bisector of $AC$ intersects the bisector of the angle $CAB$ at the point $P$. Prove that $A$, $P$, $M$ and $K$ lie on one circle.

**Problem 17.204.** Let $I$ be the incentre of the triangle $ABC$. A circle through $A$ and $I$ intersects $AB$ and $AC$ at $X$ and $Y$ respectively. Prove that $BX + CY = BC$.

**Problem 17.205.** Let $D$ be a point on the side $BC$ of the triangle $ABC$, and let $X$ be the tangency point of the incircle of $ABC$ with the side $BC$. Denote by $I_1$, $I_2$ the incentres of the triangles $ABD$ and $ADC$ respectively. Prove that $\angle I_1XI_2 = 90°$.

**Problem 17.206.** Let $I$ be the incentre of a triangle $ABC$ and let $D$ be an arbitrary point of the side $BC$. The perpendicular bisector of $AD$ meets $BI$ and $CI$ at the points $F$ and $E$ respectively. Prove that the orthocentre of the triangle $EIF$ lies on $BC$.

**Problem 17.207.** Let $D$ be a point on the segment $AB$ of the triangle $ABC$. Suppose that the incircle of the triangle $ABC$ is tangent to $AB$ and $AC$ at the points $E$ and $F$. The bisector of the angle $ACD$ intersects the line $EF$ at $M$. Let $I$ be the incentre of the triangle $CDB$. Prove that $\angle CMI = 90°$.

**Problem 17.208.** Let $\Omega$ be the circumcircle of the cyclic quadrilateral $ABCD$. Consider the pairs of points $P$, $Q$ of the diagonal $AC$ for which the rays $BP$ and $BQ$ are symmetric with respect to the bisector of the angle $B$. Find the locus of the circumcentres of the triangles $PDQ$.

**Problem 17.209.** The line touching the incircle of the triangle $ABC$ and parallel to $BC$ meets the external bisector of the angle $A$ at the point $X$. Let $Y$ be the midpoint of the arc $BAC$ of the circumcircle. Prove that the angle $XIY$ is right, where $I$ is the incentre.

**Problem 17.210.** Let $ABC$ be an acute-angled triangle with circumcentre $O$. The circumcircle of the triangle $BOC$ meets the lines $AB$ and $AC$ at the points $A_1$ and $A_2$ respectively. Let $\omega_A$ be the circumcircle of the triangle $AA_1A_2$, and define $\omega_B$ and $\omega_C$ analogously. Prove that the circles $\omega_A$, $\omega_B$, $\omega_C$ meet on the circumcircle of the triangle $ABC$.

**Problem 17.211.** The triangle $ABC$, in which $AB < BC$, is inscribed in a circle $\omega$ and circumscribed about a circle $\gamma$ with centre $I$. The line $l$, parallel to $AC$, touches the circle $\gamma$ and intersects the arcs $BAC$ and $BCA$ at the points $P$ and $Q$ respectively. It is known that $PQ = 2\,BI$. Prove that $AP + 2\,PB = CP$.

**Problem 17.212.** A convex polygon $P$ lies on a flat wooden table. You are allowed to drive some nails into the table. The nails must not go through $P$, but they may touch its boundary. We say that a set of nails *blocks* $P$ if moving $P$ is impossible without lifting it off the table. What is the minimum number of nails that suffices to block any convex polygon $P$?

**Problem 17.213.** Let $A$, $B$, $C$, $D$ be four points in the plane, with $C$ and $D$ on the same side of the line $AB$, such that

$$AC \cdot BD = AD \cdot BC \quad \text{and} \quad \angle ADB = 90^\circ + \angle ACB.$$

Find the ratio $\dfrac{AB \cdot CD}{AC \cdot BD}$, and prove that the circumcircles of the triangles $ACD$ and $BCD$ are orthogonal.

**Problem 17.214.** The triangle $ABC$ is isosceles with $AB = AC > BC$. Let $D$ be a point in its interior such that $DA = DB + DC$. Suppose that the perpendicular bisector of the segment $AB$ meets the external bisector of the angle $ADB$ at $P$, and let $Q$ be the intersection of the perpendicular bisector of the segment $AC$ with the external bisector of the angle $ADC$. Prove that the points $B$, $C$, $P$, $Q$ are concyclic.

**Problem 17.215.** Let $AH_1$, $BH_2$, $CH_3$ be the altitudes of an acute-angled triangle $ABC$, and let its incircle touch the sides $BC$, $AC$ and $AB$ at $T_1$, $T_2$ and $T_3$ respectively. Consider the reflections of the lines $H_1H_2$, $H_2H_3$ and $H_3H_1$ in the lines $T_1T_2$, $T_2T_3$ and $T_3T_1$ respectively. Prove that these images form a triangle whose vertices lie on the incircle of $ABC$.

**Problem 17.216.** ★ A point $E$ is marked on the side $BC$ of the parallelogram $ABCD$, and a point $F$ on the side $AD$, so that the circumscribed circle of $ABE$ is tangent to the line segment $CF$. Prove that the circumcircle of the triangle $CDF$ is tangent to the line $AE$.

**Problem 17.217.** ★ Given a trapezoid $ABCD$ with parallel sides $AB$ and $CD$, assume that there exist points $E$ on the line $BC$ outside the segment $BC$, and $F$ inside the segment $AD$, such that $\angle DAE = \angle CBF$. Denote by $I$ the point of intersection of $CD$ and $EF$, and by $J$ the point of intersection of $AB$ and $EF$. Let $K$ be the midpoint of the segment $EF$, and assume it does not lie on the line $AB$. Prove that $I$ belongs to the circumcircle of $ABK$ if and only if $K$ belongs to the circumcircle of $CDJ$.

**Problem 17.218.** ★ Let $ABC$ be a triangle with circumcentre $O$. The points $P$ and $Q$ are interior points of the sides $CA$ and $AB$ respectively. Let $K$, $L$ and $M$ be the midpoints of the segments $BP$, $CQ$ and $PQ$ respectively, and let $\Gamma$ be the circle passing through $K$, $L$ and $M$. Suppose that the line $PQ$ is tangent to the circle $\Gamma$. Prove that $OP = OQ$.

**Problem 17.219.** ★ Let $\Omega$ and $O$ be the circumcircle and the circumcentre of an acute-angled triangle $ABC$ with $AB > BC$. The angle bisector of $\angle ABC$ intersects $\Omega$ at $M \neq B$. Let $\Gamma$ be the circle with diameter $BM$. The angle bisectors of $\angle AOB$ and $\angle BOC$ intersect $\Gamma$ at the points $P$ and $Q$ respectively. The point $R$ is chosen on the line $PQ$ so that $BR = MR$. Prove that $BR \parallel AC$. (Here an angle bisector is always assumed to be a ray.)

**Problem 17.220.** ★ Let $ABC$ be a triangle with $AB \neq AC$ and circumcentre $O$. The bisector of $\angle BAC$ intersects $BC$ at $D$. Let $E$ be the reflection of $D$ in the midpoint of $BC$. The lines through $D$ and $E$ perpendicular to $BC$ intersect the lines $AO$ and $AD$ at $X$ and $Y$ respectively. Prove that the quadrilateral $BXCY$ is cyclic.

**Problem 17.221.** ★ Let $ABC$ be a triangle for which $AB \neq AC$. The points $K$, $L$, $M$ are the midpoints of the sides $BC$, $CA$, $AB$. The incircle of $ABC$ with centre $I$ is tangent to $BC$ at $D$.

A line passing through the midpoint of $ID$ perpendicular to $IK$ meets the line $LM$ at $P$. Prove that $\angle PIA = 90°$.

**Problem 17.222.** ★ The tangent at $C$ to $\Omega$, the circumcircle of the scalene triangle $ABC$, intersects $AB$ at $D$. Through the point $D$ a line is drawn that intersects the segments $AC$ and $BC$ at $K$ and $L$ respectively. On the segment $AB$ points $M$ and $N$ are marked such that $AC \parallel NL$ and $BC \parallel KM$. The lines $NL$ and $KM$ intersect at a point $P$ lying inside the triangle $ABC$. Let $\omega$ be the circumcircle of $MNP$, and suppose that $CP$ intersects $\omega$ again at $Q$. Show that $DQ$ is tangent to $\omega$.

**Problem 17.223.** Given a trapezoid $ABCD$ in which $AD \parallel BC$, the rays $AB$ and $DC$ intersect at the point $G$. The common external tangents to the circles $(ABC)$ and $(ACD)$ intersect at the point $E$, and the common external tangents to the circles $(ABD)$ and $(CBD)$ meet at $F$. Prove that the points $E$, $F$ and $G$ are collinear. [Russia 2023]

**Problem 17.224.** Consider a fixed circle $\Gamma$ with three fixed points $A$, $B$ and $C$ on it, and fix a real number $\lambda \in (0,1)$. For a variable point $P \notin \{A, B, C\}$ on $\Gamma$, let $M$ be the point on the segment $CP$ such that $CM = \lambda \cdot CP$. Let $Q$ be the second point of intersection of the circumcircles of the triangles $AMP$ and $BMC$. Prove that as $P$ varies, the point $Q$ lies on a fixed circle. [IMO Shortlist 2014]

**Problem 17.225.** Let $ABCD$ be a parallelogram. A variable line $g$ through the vertex $A$ intersects the rays $BC$ and $DC$ at the points $X$ and $Y$ respectively. Let $K$ and $L$ be the $A$-excentres of the triangles $ABX$ and $ADY$. Show that the angle $\angle KCL$ is independent of the line $g$. [IMO Shortlist 2005]

**Problem 17.226.** Let $ABC$ be a triangle and let $I$ and $O$ denote its incentre and circumcentre respectively. Let $\omega_A$ be the circle through $B$ and $C$ which is tangent to the incircle of the triangle $ABC$; the circles $\omega_B$ and $\omega_C$ are defined similarly. The circles $\omega_B$ and $\omega_C$ meet at a point $A_0$ distinct from $A$; the points $B_0$ and $C_0$ are defined similarly. Prove that the lines $AA_0$, $BB_0$ and $CC_0$ are concurrent at a point on the line $IO$. [RMM 2012]

**Problem 17.227.** Let $ABC$ be a given triangle and let $K$ be a point on the internal bisector of $\angle BAC$. The line $CK$ intersects the circumcircle $\omega$ of the triangle $ABC$ at $M \neq C$. The circle $\Omega$ passes through $A$, touches $CM$ at $K$ and intersects the segment $AB$ at $P \neq A$ and $\omega$ at $Q \neq A$. Prove that $P$, $Q$ and $M$ are collinear. [Russia 2010]

**Problem 17.228.** A triangle $ABC$ is inscribed in a circle $\omega$. A variable line $\ell$ chosen parallel to $BC$ meets the segments $AB$, $AC$ at the points $D$, $E$ respectively, and meets $\omega$ at the points $K$, $L$ (where $D$ lies between $K$ and $E$). The circle $\gamma_1$ is tangent to the segments $KD$ and $BD$ and also tangent to $\omega$, while the circle $\gamma_2$ is tangent to the segments $LE$ and $CE$ and also tangent to $\omega$. Determine the locus, as $\ell$ varies, of the meeting point of the common internal tangents to $\gamma_1$ and $\gamma_2$. [RMM 2011]

**Problem 17.229.** Circles $\Omega$ and $\omega$ are tangent at a point $P$, with $\omega$ inside $\Omega$. A chord $AB$ of $\Omega$ is tangent to $\omega$ at $C$, and the line $PC$ meets $\Omega$ again at $Q$. Chords $QR$ and $QS$ of $\Omega$ are tangent to $\omega$. Let $I$, $X$ and $Y$ be the incentres of the triangles $APB$, $ARB$ and $ASB$ respectively. Prove that $\angle PXI + \angle PYI = 90°$. [Romania TST 2013]

**Problem 17.230.** The medians $AM_A$, $BM_B$ and $CM_C$ of the triangle $ABC$ intersect at $M$. Let $\Omega_A$ be the circle passing through the midpoint of $AM$ and through $M_B$ and $M_C$; define $\Omega_B$ and $\Omega_C$ analogously. Prove that $\Omega_A$, $\Omega_B$ and $\Omega_C$ have a common point. [Russia 2016]

**Problem 17.231.** Let $D$ be a point on the side $AB$ of an equilateral triangle $ABC$, and let $E$ and $F$ be the reflections of $D$ in the lines $AC$ and $BC$ respectively. Denote by $P$ the intersection point of $AF$ and $BE$. Prove that the points $D$, $P$, $C$ are collinear.

**Problem 17.232.** Given is a triangle $ABC$ with $AB > AC$. Circles $o_B$, $o_C$ are inscribed in the angle $BAC$, with $o_B$ tangent to $AB$ at $B$ and $o_C$ tangent to $AC$ at $C$. The tangent to $o_B$ from $C$ different from $AC$ intersects $AB$ at $K$, and the tangent to $o_C$ from $B$ different from $AB$ intersects $AC$ at $L$. The line $KL$ and the angle bisector of $BAC$ intersect $BC$ at the points $P$ and $M$ respectively. Prove that $BP = CM$.

**Problem 17.233.** In a big ballroom there is a chandelier in the shape of a perfect cube with vertices $P_1$, $P_2$, ..., $P_8$. It hangs under the ceiling in such a way that the vertex $P_i$ is at distance exactly $i$ from the plane of the ceiling, for each $i = 1, 2, \ldots, 8$. What is the edge length of the chandelier?

**Problem 17.234.** On a circle $2n + 1$ points are given. Determine the maximum number of acute triangles with vertices among these points.

**Problem 17.235.** Determine the product of the lengths of all sides and all diagonals of a regular $n$-gon inscribed in a circle of radius 1.

**Problem 17.236.** The triangle $ABC$ is isosceles with $\angle ACB = 120°$. Let $M$ be the midpoint of $AB$ and let $P$ be a point on the circumcircle of $ABC$. Let $Q$ be the point on the segment $CP$ with $CQ : QP = 1 : 2$, and let $N$ be the point of intersection of the line through $P$ perpendicular to $AB$ with the line $MQ$. Find the locus of the point $N$.

**Problem 17.237.** Let $O$ be the circumcentre of an acute triangle $ABC$, let $M$ and $N$ be the midpoints of $AB$ and $BC$, and let $K$ be the intersection point of $CO$ and $MN$. Find the least possible value of $\angle BAC$ for which $MK = NK$.†

**Problem 17.238.** The quadrilateral $ABCD$ is such that $AC \perp BD$. The perpendicular bisectors of $AD$ and $BC$ intersect at $O$. Prove that $ABCD$ is cyclic if and only if the triangles $OAD$ and $OBC$ have equal areas.

**Problem 17.239.** Consider a triangle $ABC$ and a point $D$ on the segment $AB$. A line intersects $AC$, $DC$ and $BC$ at the points $M$, $N$ and $P$ respectively. Prove that

$$DB \cdot \frac{MA}{MC} + AD \cdot \frac{PB}{PC} = AB \cdot \frac{ND}{NC}.$$

†

**Problem 17.240.** In the triangle $ABC$ let $AS$ (with $S \in BC$) be the bisector of the angle $A$, and let $M \in AB$ and $N \in AC$ be the feet of the perpendiculars from $S$ to $AB$ and $AC$. Let $P$ be the intersection point of $BN$ and $CM$. Prove that $AP \perp BC$.

**Problem 17.241.** Let $ABC$ be an isosceles right triangle with $\angle ACB = 90°$. A point $M$ inside the triangle satisfies $MA = 5$, $MB = 7$ and $MC = 4\sqrt{2}$. Find $\angle AMC$.

**Problem 17.242.** In a triangle $ABC$ the lengths of the sides $BC$, $CA$ and $AB$ are denoted by $a$, $b$, $c$, and $2b = a + c$. Let $O$ and $J$ be the circumcentre and the incentre of $ABC$ respectively. Prove that the lines $OJ$ and $BJ$ are perpendicular.

**Problem 17.243.** Let $ABC$ be an acute triangle with circumcentre $O$. A point $T$ on the altitude from $C$ is such that $\angle TBA = \angle ACB$. The line $CO$ intersects $AB$ at $K$. Prove that the perpendicular bisector of $AB$, the altitude from $A$ and the line $KT$ are concurrent.[†]

**Problem 17.244.** A circle $k$ is tangent to the side $AB$ of the triangle $ABC$ at its midpoint and is internally tangent to the circumcircle of $ABC$ at a point of the arc $AB$ that does not contain $C$. If $t$ is the length of the tangent from $C$ to $k$, prove that

$$t = \frac{BC + AC}{2}.$$

**Problem 17.245.** Let $ABC$ be a triangle and let $D$ be a point on the side $BC$ such that the incircles of the triangles $ABD$ and $ACD$ are congruent. Let $\Omega_B$ be the circle with diameter $AB$ and let $\Omega_C$ be the circle with diameter $AC$. Prove that the line $AD$ is perpendicular to one of the common tangents to the circles $\Omega_B$ and $\Omega_C$.

**Problem 17.246.** On the boundary of the triangle $ABC$, points $D_1$, $D_2$, $E_1$, $E_2$, $F_1$, $F_2$ are chosen so that they appear in the following order: $A$, $F_1$, $F_2$, $B$, $D_1$, $D_2$, $C$, $E_1$, $E_2$. It turns out that

$$AD_1 = AD_2 = BE_1 = BE_2 = CF_1 = CF_2.$$

Prove that the triangles formed by the triples of lines $AD_1$, $BE_1$, $CF_1$ and $AD_2$, $BE_2$, $CF_2$ have equal perimeters.

**Problem 17.247.** Let $ABC$ be a triangle. Points $D$ and $D'$ lie on the side $BC$, points $E$ and $E'$ lie on the side $AC$, and points $F$ and $F'$ lie on the side $AB$, such that

$$AD = AD' = BE = BE' = CF = CF'.$$

Prove that if the lines $AD$, $BE$ and $CF$ are concurrent, then the lines $AD'$, $BE'$ and $CF'$ are also concurrent.

**Problem 17.248.** Let $\Omega_1$ and $\Omega_2$ be circles with diameters $d_1$ and $d_2$ respectively. The circles are externally tangent to each other and tangent to the sides of the convex quadrilateral $ABCD$ as follows: $\Omega_1$ is tangent to the sides $DA$, $AB$ and $BC$, and $\Omega_2$ is tangent to the sides $BC$, $CD$ and $DA$. Let $E$ and $F$ be the points at which $\Omega_1$ and $\Omega_2$ are tangent to the sides $AB$ and $CD$, and let $O$ be the intersection point of the diagonals $AC$ and $BD$. Prove that

$$OE + OF \leq d_1 + d_2.$$

**Problem 17.249.** Given an acute triangle $ABC$ with $AB < BC$, let $M$ and $N$ be the midpoints of the sides $AB$ and $AC$ respectively, and let $H$ be the foot of the altitude from the vertex $B$. The incircle of the triangle $ABC$ touches the side $AC$ at the point $K$. Let the line through $K$ parallel to $MH$ intersect the segment $MN$ at $P$. Prove that the quadrilateral $AMPK$ is circumscribed.

**Problem 17.250.** Inside a parallelogram $ABCD$ a point $P$ is chosen such that $\angle PDA = \angle PBA$. Let $\omega_1$ be the $A$-excircle of the triangle $PAB$ (the excircle opposite the vertex $A$), and let $\omega_2$ be the incircle of the triangle $PCD$. Prove that one of the common tangents to $\omega_1$ and $\omega_2$ is parallel to the side $AD$.

**Problem 17.251.** Let $\omega_1$ and $\omega_2$ be two congruent circles with centres $O_1$ and $O_2$ respectively. Points $X$ and $Y$ lie on the segment $O_1O_2$ such that $O_1Y = O_2X$. Points $A$ and $B$ lie on $\omega_1$ and the line $AB$ passes through the point $X$; similarly, points $C$ and $D$ lie on $\omega_2$ and the line $CD$

passes through the point $Y$. Prove that there exists a circle tangent to the lines $AO_1$, $BO_1$, $CO_2$ and $DO_2$.

**Problem 17.252.** Given an isosceles triangle $ABC$ with $AB = AC$, let $I$ be the incentre of the triangle. Let $\omega$ be a circle that passes through the point $C$ and is tangent to the line $AI$ at the point $I$. Denote by $Q$ and $D$ the points where the circle $\omega$ intersects the line $AC$ and the circumcircle of the triangle $ABC$ respectively. Let $M$ and $N$ be the midpoints of the segments $AB$ and $CQ$ respectively. Prove that the lines $AD$, $MN$ and $BC$ are concurrent.

**Problem 17.253.** Let $AA_1$, $BB_1$ and $CC_1$ be the angle bisectors of the triangle $ABC$. The segments $BB_1$ and $A_1C_1$ intersect at the point $D$. Let $E$ be the foot of the perpendicular from $D$ to $AC$. Points $P$ and $Q$ lie on the sides $AB$ and $BC$ respectively, such that $EP = PD$ and $EQ = QD$. Prove that $\angle PDB_1 = \angle EDQ$.

**Problem 17.254.** Let the incircle $\omega$ of the triangle $ABC$ be tangent to the sides $BC$, $CA$ and $AB$ at the points $D$, $E$ and $F$ respectively. On a line $MN$ choose points $M$ and $N$ such that $MB \parallel AC$ and $NC \parallel AB$. Points $P$ and $Q$ lie on the segments $MD$ and $ND$ respectively, such that $BP \parallel CQ$. Prove that the lines $PF$ and $EQ$ intersect on the incircle $\omega$.

**Problem 17.255.** Let the point $X$ lie inside the circumcircle of the triangle $ABC$. Denote by $I_B$ and $I_C$ the centres of the excircles opposite the vertices $B$ and $C$ respectively. Prove that

$$XI_B \cdot XI_C > XB \cdot XC.$$

**Problem 17.256.** Two triangles intersect in a hexagon which cuts off six small triangles from them. The inradii of these six small triangles are equal. Prove that the inradii of the two original triangles are also equal.

**Problem 17.257.** Let $ABC$ be an acute triangle with $AB < AC$ and let $\Omega$ be its circumcircle. Let $M$ and $N$ be the midpoints of the longer and of the shorter arc $BC$ of $\Omega$ respectively. Points $X \neq M$ and $Y \neq M$ lie on the line $AM$ and satisfy

$$BX = BM = CM = CY.$$

Let $E$ be the foot of the perpendicular from the point $B$ to the line $AC$, and let $F$ be the foot of the perpendicular from the point $C$ to the line $AB$. Prove that $\angle FNX = \angle YNE$.

**Problem 17.258.** Let $ABC$ be an acute scalene triangle inscribed in a circle $\Omega$. Let $M$ be the midpoint of the longer arc $BC$ of $\Omega$, and let $N$ be the midpoint of the shorter arc $BC$ of $\Omega$. The perpendicular bisector of the segment $AN$ intersects the segments $AB$ and $AC$ at the points $X$ and $Y$ respectively. Let $Z$ be the point, different from $N$, where $\Omega$ intersects the circumcircle of the triangle $NXY$. Prove that the line $MZ$ passes through the orthocentre of the triangle $ABC$.

**Problem 17.259.** Let $ABCD$ and $PQRS$ be convex quadrilaterals with equal areas, and suppose that

$$AB = PQ, \quad BC = QR, \quad CD = RS, \quad DA = SP.$$

Prove that there exist points $P'$, $Q'$, $R'$, $S'$ lying in the same plane as the quadrilateral $ABCD$ such that

$$AP' = BQ' = CR' = DS'$$

and the quadrilaterals $P'Q'R'S'$ and $PQRS$ are congruent.

**Problem 17.260.** Let $\omega$ be the circumcircle of the triangle $ABC$, where $AB < AC$. The angle bisectors of the triangle $ABC$ intersect at the point $I$. From the midpoint $M$ of the side $BC$ a perpendicular $MH$ is dropped to the line $AI$. The lines $MH$, $BI$ and $AB$ bound a triangle $T_b$, and the lines $MH$, $CI$ and $AC$ bound a triangle $T_c$. The circumcircles of the triangles $T_b$ and $T_c$ intersect the circle $\omega$ again at the points $B'$ and $C'$ respectively. Prove that the point $H$ lies on the line $B'C'$.

# Chapter 18

# Number Theory

**Sources for this chapter:** Camp handouts by Dušan Djukić (March and June 2020, June 2022, September 2022–January 2023, May 2024, January 2025, November 2025), Miroslav Marinov (November 2024, May 2025, September–October 2025), Nikola Petrović (August 2020), Dmytro Nomirovskii (January 2020), Dima (March–April 2021), Bojan Bašić (February 2024), Marko Radovanović (2024), Regis Barbosa (2023), Arsenii and colleagues (November 2022), Łukasz Bożyk (June 2022 and IMO-team classes).

This is the deepest number-theory chapter of the book. It assumes everything from Chapters 8 and 13: divisibility and the Euclidean algorithm, congruences, the Chinese remainder theorem, arithmetic functions, multiplicative order, Vieta jumping, and the lifting-the-exponent lemma. Built on top of these come the two great structural theories of elementary number theory — quadratic reciprocity and primitive roots — together with the arithmetic of binomial coefficients, Diophantine approximation and Pell's equation, cyclotomic polynomials, and the quadratic integers that make Fermat's two-square theorem and Fermat's last theorem for $n = 3$ accessible.

## 18.1 Orders II and Zsigmondy-type arguments

*Revisits the multiplicative order and its basic properties (see Chapter 13); the problems below push it further.*

**Lemma 18.1 (Coprime orders multiply).** *Let $a_1, a_2, m$ be positive integers with $\gcd(a_1, m) = \gcd(a_2, m) = 1$, and let $n_1$ and $n_2$ be the orders of $a_1$ and $a_2$ modulo $m$. If $\gcd(n_1, n_2) = 1$, then the order of $a_1 a_2$ modulo $m$ equals $n_1 n_2$.*

**Remark.** There is no analogous statement when $\gcd(n_1, n_2) > 1$: the order of $a_1 a_2$ need not be $\operatorname{lcm}(n_1, n_2)$, and counterexamples are plentiful.

**Lemma 18.2 (Exponents and greatest common divisors).** *Let $a > 1$ and let $m, n$ be positive integers. Then*

*(a) $a^m - 1 \mid a^n - 1$ if and only if $m \mid n$;*
*(b) $a^m + 1 \mid a^n + 1$ if and only if $m \mid n$ and $n/m$ is odd;*
*(c) $\gcd(a^m - 1,\, a^n - 1) = a^{\gcd(m,n)} - 1$.*

**Remark.** The companion question for $a^m + 1$ and $a^n + 1$ behaves differently: if $v_2(m) \neq v_2(n)$, then $\gcd(a^m + 1, a^n + 1)$ equals 1 when $a$ is even and 2 when $a$ is odd.

**Lemma 18.3 (Prime divisors of $(a^p-1)/(a-1)$).** *Let $p$ be a prime, let $a \geq 2$ be an integer and let $q$ be a prime divisor of $\dfrac{a^p-1}{a-1}$. Then $q = p$ or $q \equiv 1 \pmod{p}$.*

**Lemma 18.4.** *For every positive integer $n$ one has $\gcd(n, 2^{2^n}+1) = 1$.*

**Problem 18.1.** Let $p$ be a prime and let $n$ be a positive integer which divides $2^p - 1$. Prove that $p$ divides $n - 1$. [Bulgaria, Borislav Boyanov Tournament 2011]

**Problem 18.2.** Determine all pairs $(a, n)$ of positive integers such that $n$ divides $(a+1)^n - a^n$. [China IMO TST 2006]

**Problem 18.3.** Determine all pairs $(p, q)$ of primes such that $p + q$ divides $(2p^2 - 1)^q + 1$ and $(2q^2 - 1)^p + 1$. [Turkey EGMO TST 2017]

**Problem 18.4.** Find all positive integers $n$ for which there exist a positive integer $m$ and primes $p < q$ such that $q - p$ divides $m$ and both $p$ and $q$ divide $n^m + 1$. [Bulgaria Spring Mathematical Competition 2023]

**Problem 18.5.** Let $n \geq 2$ be a positive integer such that $6^n + 11^n$ is divisible by $n$. Prove that $n^{100} + 6^n + 11^n$ is divisible by $17n$, but not divisible by $289n$. [Switzerland IMO TST 2022]

**Problem 18.6.** Let $a, m, n$ be positive integers with $a$ even and $m < n$. Prove that one of the numbers $a^m + 1,\ a^{m+1} + 1,\ \ldots,\ a^n + 1$ is relatively prime to all the others. [Bulgaria Winter Mathematical Competition 2012]

**Problem 18.7.** Solve the equation $\dfrac{x^7 - 1}{x - 1} = y^5 - 1$ in integers. [IMO Shortlist 2006 N5]

**Problem 18.8.** Determine all polynomials $P(x)$ with integer coefficients such that for all $n > 2024$ we have $P(n) > 0$ and the numbers $n^n + 2^n$ and $2P(n) + 1$ are relatively prime. [Bulgaria RMM TST 2024]

**Problem 18.9.** Find all pairs $(p, q)$ of prime numbers with $p > q$ for which

$$\frac{(p+q)^{p+q}(p-q)^{p-q} - 1}{(p+q)^{p-q}(p-q)^{p+q} - 1}$$

is an integer. [IMO Shortlist 2017 N5]

**Problem 18.10.** Determine all positive integers $n$ such that $\tau(n)$ divides $2^{\sigma(n)} - 1$. As usual, $\tau$ denotes the number of divisors and $\sigma$ the sum of divisors. [Iran 2020]

**Problem 18.11.** Prove that $\tau(n)$ does not divide $n^2 + 1$ for any even integer $n$. [Krishna Pothapragada]

**Problem 18.12.** Determine all positive integers $b$ for which there exist infinitely many positive integers $n$ such that $n^2$ divides $b^n + 1$. [USA TSTST 2018]

**Problem 18.13.** ★ Find all triples $(a, m, n)$ of positive integers such that $a^m + 1$ divides $(a+1)^n$.

**Problem 18.14.** Let $m, n, k$ be positive integers such that $5^n - 2$ and $2^k - 5$ both divide $5^m - 2^m$. Prove that $m$ and $n$ are coprime.

**Problem 18.15.** ★ Show that there are infinitely many pairs of distinct primes $p, q$ such that $p \mid 2^{q-1} - 1$ and $q \mid 2^{p-1} - 1$.

**Problem 18.16.** ★ Let $b, m, n$ be positive integers with $b > 1$ and $m \neq n$. Prove that if $b^m - 1$ and $b^n - 1$ have the same prime divisors, then $b + 1$ is a power of 2.

**Problem 18.17.** ★ Prove that for every positive integer $n$ there exists an integer $m$ such that $2^m + m$ is divisible by $n$.

**Problem 18.18.** Prove that $2^{n^2+n-5}$ divides $\varphi(2^{2^n} - 1)$ for all positive integers $n$.

**Problem 18.19.** Let $a$ be a positive integer and let $p$ be any prime divisor of $a^3 - 3a + 1$. If $p \neq 3$, prove that $p \equiv \pm 1 \pmod 9$.

**Problem 18.20.** Let $a > 3$ be an integer. Prove that there exist infinitely many positive integers $n$ such that $a^n - 1$ is divisible by $n$.

**Problem 18.21.** Let $a > 4$ be an even integer. Prove that there exist infinitely many squarefree positive integers $n$ such that $a^n - 1$ is divisible by $n^2$.

**Problem 18.22.** Let $a > 5$ be an odd integer. Prove that there exist infinitely many squarefree positive integers $n$ such that $a^n - 1$ is divisible by $n^2$.

**Problem 18.23.** Let $n > 2$ be a positive integer with $n \mid 3^n - 1$. Prove that $n$ is divisible by 4.

**Problem 18.24.** Let $a > 1$ be a fixed integer and let $n$ be an integer with $n \mid a^n - 1$. Suppose that $n = p_1 p_2 \cdots p_k$, where the $p_i$ are primes (not necessarily distinct) and $k \leq 100$. Prove that $n$ can take only finitely many values.

**Problem 18.25.** Let $a > 2$ be an even integer. Prove that there exists a positive integer $k$ such that $n^k \mid a^n - 1$ holds only for $n = 1$.

**Problem 18.26.** Let $a > 2$ be an odd integer. Prove that there exists a positive integer $k$ such that $n^k \mid a^n - 1$ holds only for $n = 1$.

**Problem 18.27.** Let $a > 1$ be even and let $a - 1$ be squarefree. Prove that $n^3 \mid a^n - 1$ holds only for $n = 1$.

**Problem 18.28.** Let $a > 1$ be an integer such that $n^3$ divides $a^n - 1$ only when $n = 1$. Prove that $a$ is even and $a - 1$ is squarefree.

**Problem 18.29.** Find a squarefree integer $k > 100$ such that $10^k - 1$ is divisible by $k^2$.

**Problem 18.30.** Let $n$ be the least positive integer for which $149^n - 2^n$ is divisible by $3^3 \cdot 5^5 \cdot 7^7$. Find the value of $n$. [American Mathematics Exam 2020]

**Problem 18.31.** Find all positive integers $n$ such that $n^3$ divides $5^n - 1$.

**Problem 18.32.** Let $a$ and $b$ be positive integers with $a - b > 1$. Prove that there exist infinitely many positive integers $n$ such that $n^2$ divides $a^n - b^n$.

**Problem 18.33.** Find all positive integers $n$ for which $(2^n - 1)(3^n - 1)$ is a perfect square.

**Problem 18.34.** Given any integers $a > 1$ and $T$, prove that there exist two different positive integers $m, n$ such that $\gcd(2^{2^m} + a,\ 2^{2^n} + a) > T$.

**Problem 18.35.** Let $a$ and $b$ be coprime odd positive integers. Find all possible values of $\gcd\left(2^a + 2^{(a+1)/2} + 1,\ 2^b + 2^{(b+1)/2} + 1\right)$.

**Problem 18.36.** Prove that there are infinitely many composite numbers $n$ such that $n$ divides $2^{(n-1)/2} + 1$.

**Problem 18.37.** Given a positive integer $n$, prove that for every odd integer $x$ there is an integer $y$ such that $y^y \equiv x \pmod{2^n}$.

**Problem 18.38.** For positive integers $a > b > 1$, define $x_n = \dfrac{a^n - 1}{b^n - 1}$. Find the least $d$ such that, for any $a$ and $b$, the sequence $(x_n)$ does not contain $d$ consecutive prime numbers. [Russia 2011]

**Problem 18.39.** Does there exist an infinite sequence of positive integers $a_1, a_2, a_3, \ldots$ such that $a_m$ and $a_n$ are coprime if and only if $|m - n| = 1$? [RMM 2015]

**Problem 18.40.** Define $a_0 = 2$ and $a_{n+1} = a_n^2 + a_n - 1$ for $n \geq 0$. Prove that $a_n$ is coprime to $2n + 1$.†

## 18.2 Quadratic residues and the reciprocity law

The theory below is specific to prime moduli: most of it breaks down for composite moduli, and composite moduli are in general very hard to handle.

**Definition 18.5 (Quadratic residue, Legendre symbol).** Let $p$ be a prime and let $a$ be an integer not divisible by $p$. We say that $a$ is a *quadratic residue* modulo $p$ if $x^2 \equiv a \pmod{p}$ for some integer $x$, and a *quadratic non-residue* otherwise. The *Legendre symbol* is defined by

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a quadratic residue mod } p, \\ -1 & \text{if } a \text{ is a quadratic non-residue mod } p, \\ 0 & \text{if } p \mid a. \end{cases}$$

**Theorem 18.6 (Counting square roots).** *If $p > 2$ is a prime, then among $1, 2, \ldots, p-1$ there are exactly $\frac{p-1}{2}$ quadratic residues and as many quadratic non-residues. The congruence $x^2 \equiv a \pmod{p}$ has exactly $\left(\frac{a}{p}\right) + 1$ solutions modulo $p$.*

**Theorem 18.7 (Euler's criterion).** *If $p > 2$ is a prime and $a$ is an integer, then $a^{(p-1)/2} \equiv \left(\frac{a}{p}\right) \pmod{p}$.*

**Corollary 18.8.** *The number $-1$ is a quadratic residue modulo $p$ if and only if $p = 2$ or $p \equiv 1 \pmod 4$.*

**Lemma 18.9 (Multiplicativity).** *For any integers $a, b$ and any prime $p$ one has $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$.*

**Lemma 18.10 (Gauss's lemma).** *If $p > 2$ is a prime and $p \nmid a$, then $\left(\frac{a}{p}\right) = (-1)^S$, where $S = \sum_{k=1}^{(p-1)/2} \left\lfloor \frac{2ka}{p} \right\rfloor$.*

**Corollary 18.11 (The rule of 2).** *For a prime $p \geq 3$ one has $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}$; thus 2 is a quadratic residue modulo $p$ if and only if $p \equiv \pm 1 \pmod 8$. Moreover $\left(\frac{-2}{p}\right) = 1$ if $p \equiv 1, 3 \pmod 8$ and $\left(\frac{-2}{p}\right) = -1$ if $p \equiv 5, 7 \pmod 8$.*

**Theorem 18.12 (Law of quadratic reciprocity).** *If $p$ and $q$ are distinct odd primes, then*

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}.$$

**Definition 18.13 (Jacobi symbol).** Given an odd integer $n = p_1 p_2 \cdots p_k$, where the $p_i$ are odd primes (not necessarily distinct), the Jacobi symbol is defined by $\left(\frac{a}{n}\right) = \left(\frac{a}{p_1}\right)\left(\frac{a}{p_2}\right)\cdots\left(\frac{a}{p_k}\right)$.

**Theorem 18.14 (Properties of the Jacobi symbol).** *For odd $m, n$ and integers $a, b$,*

$$\left(\frac{ab}{n}\right) = \left(\frac{a}{n}\right)\left(\frac{b}{n}\right), \qquad \left(\frac{-1}{n}\right) = (-1)^{\frac{n-1}{2}}, \qquad \left(\frac{2}{n}\right) = (-1)^{\frac{n^2-1}{8}}, \qquad \left(\frac{m}{n}\right)\left(\frac{n}{m}\right) = (-1)^{\frac{m-1}{2}\cdot\frac{n-1}{2}}.$$

*The one exception is that $\left(\frac{a}{n}\right) = 1$ does* not *imply that $a$ is a quadratic residue modulo $n$.*

**Lemma 18.15 (Primes with prescribed symbols).** *Let $q_1, \ldots, q_k$ be distinct primes and let each of $\varepsilon_1, \ldots, \varepsilon_k$ be equal to $1$ or $-1$. Then there are infinitely many primes $p$ such that $\left(\frac{q_i}{p}\right) = \varepsilon_i$ for all $i = 1, 2, \ldots, k$.*

**Corollary 18.16.** *Let $a$ be an integer which is not a perfect square. Then there exist infinitely many primes $p$ such that $a$ is not a quadratic residue modulo $p$.*

**Problem 18.41.** Compute the Legendre symbols $\left(\frac{600}{953}\right)$, $\left(\frac{20203}{953}\right)$ and $\left(\frac{-7000}{757}\right)$.

**Problem 18.42.** Compute the Legendre symbols $\left(\frac{-12000}{821}\right)$ and $\left(\frac{22019}{953}\right)$.

**Problem 18.43.** Let $p$ be an odd prime. Prove the following four criteria:
(a) $-2$ is a quadratic residue modulo $p$ if and only if $p \equiv 1, 3 \pmod 8$;
(b) $2$ is a quadratic residue modulo $p$ if and only if $p \equiv \pm 1 \pmod 8$;
(c) $-3$ is a quadratic residue modulo $p$ if and only if $p \equiv 1 \pmod 3$;
(d) $3$ is a quadratic residue modulo $p$ if and only if $p \equiv \pm 1 \pmod{12}$.

**Problem 18.44.** Prove that 5 is a quadratic residue modulo a prime $p \neq 2, 5$ if and only if $p \equiv \pm 1 \pmod{10}$.

**Problem 18.45.** Let $p \equiv 1 \pmod 4$ be a prime and write $p = a^2 + b^2$ with $a$ odd. Prove that $a$ is a quadratic residue modulo $p$.

**Problem 18.46.** Let $p$ be an odd prime. Compute the sum $\left(\frac{1}{p}\right) + \left(\frac{2}{p}\right) + \cdots + \left(\frac{p-1}{p}\right)$.

**Problem 18.47.** Let $p \equiv 1 \pmod 4$ be a prime. Prove that the sum of all quadratic residues in $\{0, 1, \ldots, p-1\}$ equals $\dfrac{p(p-1)}{4}$.

**Problem 18.48.** Given a prime number $p$, prove that there exists $x$ with $p \mid x^2 - x + 3$ if and only if there exists $y$ with $p \mid y^2 - y + 25$.

**Problem 18.49.** Let $p = 4k + 3$ be a prime. Is $3k + 2$ a quadratic residue modulo $p$?

**Problem 18.50.** Let $p = 4k - 1$ be a prime. If the congruence $x^2 \equiv a \pmod p$ has solutions, prove that these solutions are $x = \pm a^k$.

**Problem 18.51.** Prove that for every prime $p > 2$ there exists a quadratic non-residue $a < \sqrt{p}+1$ modulo $p$.

**Problem 18.52.** For any prime $p \geq 3$ prove that the smallest quadratic non-residue modulo $p$ is less than $1 + \sqrt{p}$.

**Problem 18.53.** Evaluate $\left\lfloor \dfrac{1}{101} \right\rfloor + \left\lfloor \dfrac{2}{101} \right\rfloor + \left\lfloor \dfrac{4}{101} \right\rfloor + \cdots + \left\lfloor \dfrac{2^{99}}{101} \right\rfloor$.

**Problem 18.54.** Prove that every prime $p$ has a multiple of the form $x^2 + y^2 + 1$.

**Problem 18.55.** For any given prime $p$ prove that the congruence $x^2 + y^2 \equiv -1 \pmod p$ has an integer solution.

**Problem 18.56.** Prove that (a) every prime divisor of $n^4 - n^2 + 1$ is of the form $12k + 1$; (b) every prime divisor of $n^8 - n^4 + 1$ is of the form $24k + 1$.

**Problem 18.57.** Prove that neither (a) $4xy - x - y$ nor (b) $4xyz - x - y$ can be a perfect square if $x, y, z$ are positive integers.

**Problem 18.58.** Find all positive integers $n$ such that the set $\{n, n+1, \ldots, n+101\}$ can be partitioned into several subsets with equal products of elements.

**Problem 18.59.** Let $P(x) = x^3 + 14x^2 - 2x + 1$. Prove that there exists a positive integer $n$ such that, for every integer $x$, applying $P$ exactly $n$ times to $x$ gives a number congruent to $x$ modulo 101.

**Problem 18.60.** Prove that the number $N = 2^{2^n} + 1$ is prime if and only if $3^{(N-1)/2} \equiv -1 \pmod{N}$.

**Problem 18.61.** Let $k = 2^{2^n} + 1$ for some $n \in \mathbb{N}$. Prove that $k$ is prime if and only if $k \mid 3^{(k-1)/2} + 1$.

**Problem 18.62.** Prove that the sum of all quadratic non-residues modulo a prime $p \neq 3$ is a multiple of $p$.

**Problem 18.63.** Let $p$ be an odd prime and let $a, b, c$ be integers with $p \nmid b^2 - ac$. Prove that

$$\sum_{x=0}^{p-1} \left(\frac{ax^2+bx+c}{p}\right) = -\left(\frac{a}{p}\right).$$

**Problem 18.64.** If $p$ is a prime and $p \nmid a$, prove that the congruence $x^2 + y^2 \equiv a \pmod{p}$ has exactly $p - \left(\frac{-1}{p}\right)$ solutions $(x, y)$ modulo $p$.

**Problem 18.65.** ★ If $a$ is an integer, prove that the congruence $x^2 + y^2 + z^2 \equiv 2axyz \pmod{p}$ has exactly $p^2 + 1 + 3p(-1)^{p'}$ solutions $(x, y, z)$, where $p' = \frac{p-1}{2}$.

**Problem 18.66.** Prove that there are no positive integers $a, b, c$ for which $a^2+b^2+c^2$ is divisible by $3(ab + bc + ca)$.

**Problem 18.67.** Find all positive integers $x$ for which $x^3 + 2x + 1$ is a power of 2.

**Problem 18.68.** Let $p > 2$ be a prime and let $M$ be the set of all numbers $1 \leq x \leq p - 1$ such that both $x$ and $4 - x$ are quadratic non-residues modulo $p$. Compute the product of all elements of $M$ modulo $p$.

**Problem 18.69.** Suppose that $m$ and $n$ are positive integers such that $\varphi(5^m - 1) = 5^n - 1$. Prove that $\gcd(m, n) > 1$.

**Problem 18.70.** Let $p > 2$ be a prime and let $x$ and $y$ be positive integers with $1 \leq x, y \leq \frac{p-1}{2}$. If $(px - x^2)(py - y^2)$ is a perfect square, prove that $x = y$.

**Problem 18.71.** Show that if $p = 6k+5$ is prime, then $p$ does not divide $b^2+3$ for any positive integer $b$.

**Problem 18.72.** Let $p > 3$ be a prime with $p = 3k + 2$. Prove that if $p \mid 3m^2 + n^2$, then $p^2 \mid 3m^2 + n^2$.

**Problem 18.73.** Let $x, y$ be positive integers such that $x^2 + xy + 7y^2$ is divisible by 23. Find the minimum value of $(x + y)^2 - 96$.

**Problem 18.74.** For any prime $p$ with $p \equiv 3 \pmod{4}$, prove that the congruence $x^2 + \frac{p+1}{4} \equiv 0 \pmod{p}$ is not solvable.

**Problem 18.75.** Show that there exist infinitely many primes of the form (a) $4k+1$; (b) $10k+9$.

**Problem 18.76.** ★ Show that every odd divisor of $5x^2 + 1$, where $x$ is a positive integer, has an even tens digit.

**Problem 18.77.** ★ Let $a, b, m, n$ be positive integers such that $a^n \equiv -1 \pmod{m}$ and $b^n \equiv -1 \pmod{m}$. Prove that there exists a positive integer $c$ with $ab \equiv c^2 \pmod{m}$.

**Problem 18.78.** ★ Let $p = 4t+1$ be a prime and let $(a_1, b_1), (a_2, b_2), \ldots, (a_k, b_k)$ be all pairs of positive integers with $a_i < b_i \leq \frac{p-1}{2}$ and $p \mid a_i^2 + b_i^2$. Prove that $2(a_1+a_2+\cdots+a_k) = b_1+b_2+\cdots+b_k$.

**Problem 18.79.** ★ Let $p > 5$ be a prime and let $A = \{b_1, b_2, \ldots, b_{(p-1)/2}\}$ be the set of all nonzero quadratic residues modulo $p$. Prove that there do not exist natural numbers $a, c$ with $\gcd(ac, p) = 1$ such that the sets $B = \{ab_1 + c, ab_2 + c, \ldots, ab_{(p-1)/2} + c\}$ and $A$ are disjoint.

**Problem 18.80.** Determine whether there are infinitely many positive integers $n$ such that $19 \mid 1 + 2^n + 3^n + 4^n$.

**Problem 18.81.** Determine all positive integers $n$ for which there exists an integer $m$ such that $2^n - 1$ is a divisor of $m^2 + 9$.

**Problem 18.82.** Let $a, b, c$ be positive integers such that $|a-b|$ is a prime number and $\gcd(a, b)+\operatorname{lcm}(a, b) = 2021^c$. Prove that $(a + b)^2 + 4$ is composite. [Balkan MO 2021 P3]

**Problem 18.83.** For a positive integer $a$ define the sequence $x_1, x_2, \ldots$ by $x_1 = a$ and $x_{n+1} = 2x_n + 1$ for all $n \geq 1$. Determine the largest $k$ for which there exists $a$ such that $2^{x_1} - 1$, $2^{x_2} - 1$, $\ldots$, $2^{x_k} - 1$ are all prime numbers. [RMM 2013 P1]

**Problem 18.84.** Determine all primes $p$ such that there are exactly $p$ pairs $(x, y)$ of integers with $0 \leq x, y \leq p - 1$ and $y^2 \equiv x^3 + 4x \pmod{p}$. [HMMT 2016 Team Round]

**Problem 18.85.** Prove that $2^n + 1$ does not divide $5^n - 1$ for any positive integer $n$. [Bulgaria RMM TST 2019]

**Problem 18.86.** ★ Let $a, b$ be integers greater than 1 such that $ab$ is not a perfect square. Prove that there exist infinitely many positive integers $n$ such that $(a^n-1)(b^n-1)$ is not a perfect square. [IMO Shortlist 2009 N7, original proposal]

**Problem 18.87.** ★ Prove that there are infinitely many primes, each of which divides an integer of the form $3^n - 2$, but none of which divides an integer of the form $2^m - 3$, where $m$ and $n$ are positive integers. [Saudi Arabia IMO TST 2025]

**Problem 18.88.** Prove that $2^m - 1$ does not divide $3^n - 1$ for any odd integers $m \geq 3$ and $n \geq 3$. [Romania IMO TST 2008]

**Problem 18.89.** ★ For any positive integer $N$ prove that there exist positive integers $a$ and $b$ with $b > a > N$ such that $a - 1$ divides $b - 1$, but $a^n - 1$ does not divide $b^n - 1$ for any integer $n \geq 2$. [Bulgaria RMM TST 2024]

**Problem 18.90.** Let $p$ be a prime. Determine the number of possible remainders when $(x^2+y^2)^2$ is divided by $p$, where $x, y$ are integers not divisible by $p$. [Bulgaria Balkan MO TST 2007, general version]

**Problem 18.91.** Determine all positive integers $n$ for which $2^n + 1$ can be represented as a sum of two squares of integers. [Bulgaria Balkan MO TST 2015]

## 18.3 Primitive roots

*Revisits the definition of a primitive root (see Chapter 13); the problems below push it further.*

**Theorem 18.17 (Existence of primitive roots modulo a prime).** *Let $p$ be a prime number. Then there exists a primitive root modulo $p$, that is, an integer $g$ whose order modulo $p$ equals $p-1$; equivalently, $g, g^2, \dots, g^{p-1}$ run through all of $1, 2, \dots, p-1$ modulo $p$.*

**Remark.** Finding a primitive root efficiently for a given prime $p$ is hard, but knowing one immediately produces all the others: if $g$ is a primitive root, the primitive roots are precisely the numbers $g^k$ with $\gcd(k, p-1) = 1$.

**Definition 18.18 (Primitive root modulo $n$).** We say that $g$ is a primitive root modulo $n$ if the set of all distinct powers of $g$ modulo $n$ coincides with the set of all remainders modulo $n$ that are coprime to $n$.

**Theorem 18.19 (Classification of moduli with a primitive root).** *There exists a primitive root modulo $n$ if and only if $n = 2$, $n = 4$, $n = p^k$ or $n = 2p^k$ for some odd prime $p$ and some positive integer $k$.*

**Lemma 18.20 (Structure of the odd residues modulo $2^n$).** *For $n \geq 2$ the integers $(-1)^a 3^s$, where $a \in \{0, 1\}$ and $s = 1, 2, \dots, 2^{n-2}$, run through all odd remainders modulo $2^n$. The same holds with $3$ replaced by $5$.*

**Lemma 18.21 (Solution counts for $x^k \equiv a$).** *Let $p$ be a prime, $k$ a positive integer and $a$ an integer not divisible by $p$. Then the number of solutions of $x^k \equiv 1 \pmod p$ is $\gcd(k, p-1)$, and the number of solutions of $x^k \equiv a \pmod p$ is either $0$ or $\gcd(k, p-1)$.*

**Lemma 18.22 (Counting $k$-th powers).** *Let $p$ be a prime and $k$ a positive integer. Then the number of distinct $k$-th powers modulo $p$ is $1 + \dfrac{p-1}{\gcd(p-1, k)}$.*

**Corollary 18.23.** *If $\gcd(p-1, k) = 1$, then every remainder modulo $p$ is a $k$-th power.*

**Lemma 18.24 (Power sums modulo $p$).** *Let $p$ be a prime and $k$ a positive integer. Then $1^k + 2^k + \cdots + (p-1)^k$ is congruent to $0$ modulo $p$ if $p-1$ does not divide $k$, and is congruent to $p-1$ modulo $p$ if $p-1$ divides $k$.* *[Folklore; see Hungary–Israel Binational 2009]*

**Problem 18.92.** Let $p \geq 5$ be a prime. Prove that we can arrange the integers from $1$ to $p-1$ on a circle so that for any three consecutive numbers $a, b, c$ on the circle the number $b^2 - ac$ is divisible by $p$. [AoPS]

**Problem 18.93.** Find all primes $p$ for which $x^4 \equiv -1 \pmod p$ has a solution. [Bulgarian Festival

of Young Mathematicians 2011, Senior Group]

**Problem 18.94.** For every prime $p \geq 3$ compute the fractional part of $\frac{1}{p}\sum_{m=1}^{(p-1)/2} m^{120}$. [China IMO TST 1993]

**Problem 18.95.** Let $p \geq 3$ be a prime and let $f(x) = a_0 + a_1 x + \cdots + a_{p-1}x^{p-1}$ be a polynomial with integer coefficients such that $p$ does not divide $a_{p-1}$. Prove that there are integers $a$ and $b$ such that $a - b$ is not divisible by $p$ but $f(a) - f(b)$ is. [Folklore]

**Problem 18.96.** (Roots of unity filter.) Let $R$ be $\mathbb{C}$ or $\mathbb{F}_p$ and suppose $a \in R$ and the positive integer $n$ are such that $a^n = 1$ and $a^l \neq 1$ for $0 < l < n$. Let $P$ be a polynomial with coefficients in $R$. Prove that the sum of all coefficients of $P$ at powers divisible by $n$ equals $\dfrac{P(1) + P(a) + P(a^2) + \cdots + P(a^{n-1})}{n}$.

**Problem 18.97.** For a prime $p \neq 3$ prove that the product of all primitive roots modulo $p$ is congruent to 1 modulo $p$. [PEN; Bulgarian Festival of Young Mathematicians 2015, Senior Group]

**Problem 18.98.** Let $p \equiv 3 \pmod 4$ be a prime. If $g$ is a primitive root modulo $p$ such that $g^2 \equiv g + 1 \pmod p$, prove that the numbers $g - 1$ and $g - 2$ are also primitive roots modulo $p$. [PEN]

**Problem 18.99.** Let $p \geq 3$ be a prime which is not of the form $2^{2^n} + 1$ for an integer $n$. Given a positive integer $a_0$, define the sequence of integers $a_0, a_1, \ldots$ by $a_n = a_{n-1}^2$ for $n \geq 1$. Prove that it is possible to choose $a_0$ so that the subsequence $a_N, a_{N+1}, a_{N+2}, \ldots$ is not constant modulo $p$ for any positive integer $N$. [Baltic Way 2016, stronger version]

**Problem 18.100.** (a) Determine all integers $n \geq 3$ such that every quadratic non-residue modulo $n$ is a primitive root modulo $n$. (b) Determine all integers $n \geq 3$ such that every quadratic non-residue modulo $n$ which is relatively prime to $n$ is a primitive root modulo $n$. [RMM Shortlist 2016; refined version by Ilya Bogdanov]

**Problem 18.101.** Find all primes $p \geq 3$ for which the sets $\{k^2 + 1 : 1 \leq k \leq \frac{p-1}{2}\}$ and $\{g^m : 1 \leq m \leq \frac{p-1}{2}\}$, viewed modulo $p$, are the same for some primitive root $g$. [Mongolia, Olympiad for Teachers 2007]

**Problem 18.102.** Let $k$ be a positive integer such that $p = 2^{2^k} + 1$ is prime. Prove that 3 is a primitive root modulo $p$. [Folklore]

**Problem 18.103.** Let $n$ be a squarefree positive integer. Prove that for all integers $a$ there are integers $x, y, z$ such that $x^3 + y^3 + z^3 \equiv a \pmod n$. [Oxford Number Theory 2019, stronger version]

**Problem 18.104.** Determine the number of primes between 100 and 200 for which there are integers $x$ and $y$ such that $x^{11} + y^{16} \equiv 2013 \pmod p$. [PUMaC 2013]

**Problem 18.105.** The prime $p$ is such that $2^{(p-1)/4} - 1$ is an integer multiple of $p$. Prove that $p \equiv 1 \pmod 8$ and that there exist integers $a$ and $b$ such that $a^4 + 1$ and $b^4 + 2$ are divisible by $p$. [Bulgaria 2012]

**Problem 18.106.** Let $p$ be a prime. Is it true that for any permutation $a_1, \ldots, a_{p-1}$ of $1, \ldots, p-1$ there exist $1 \leq m < n \leq p - 1$ with the same parity such that $a_m + a_n = 2a_{(m+n)/2}$? [Bulgaria IMO TST 2003]

**Problem 18.107.** Determine all positive integers $n$ such that $x^n \equiv x \pmod{n}$ for all integers $x$ and the number of divisors of $n$ is not a multiple of 8. [Taiwan 2023 National Round]

**Problem 18.108.** Is it true that for every prime $p > 2023$ there exist distinct positive integers $a, b, c < p$, none of which is a primitive root modulo $p$, but such that $abc$ is a primitive root modulo $p$? [Thailand TSTST 2024]

**Problem 18.109.** (a) Let $n$ be a positive integer and let $p$ be a prime such that $\frac{p}{2} < n < p$. Determine all positive integers $m$ for which there exist distinct positive integers $a_1, \ldots, a_n$, all less than $p$, such that $a_1^m, a_2^m, \ldots, a_n^m$ give the same remainder upon division by $p$. (b) The same problem for $\frac{p}{3} < n \leq \frac{p-1}{2}$. [(b) Bulgaria, PPMG Burgas Challenge 2023]

**Problem 18.110.** Determine all integers $j$ such that $\sum_{\substack{1 \leq n \leq 2021 \\ \gcd(n, 2021) = 1}} n^j$ is divisible by 2021. [Putnam 2021 A5]

**Problem 18.111.** Let $p \geq 3$ be a prime. Prove that there is a permutation $(a_1, a_2, \ldots, a_{p-1})$ of the set $\{1, 2, \ldots, p-1\}$ such that $\sum_{i=1}^{p-1} i\, a_i$ is divisible by $p$. [Mongolia 2014; Bosnia and Herzegovina 2021]

**Problem 18.112.** Prove that $1^n + 2^n + \cdots + (n-1)^n$ is divisible by $n^2$ for every odd $n$, and divisible by $n^3$ for infinitely many $n$. [Bulgaria National Spring Competition 2009]

**Problem 18.113.** Find all pairs $(p, q)$ of primes such that $3pq$ divides $n^{3pq} - n$ for all positive integers $n$. [Romania IMO TST 1996; Iran National Round 1996]

**Problem 18.114.** Let $x$ and $y$ be integers and let $p$ be a prime. Suppose that there exist coprime positive integers $m$ and $n$ such that $x^m \equiv y^n \pmod{p}$. Prove that there exists a unique integer $z$ modulo $p$ such that $x \equiv z^n$ and $y \equiv z^m$ modulo $p$. [Iran National Round 2017]

**Problem 18.115.** (a) Let $p$ be a prime and $f(x)$ a polynomial with integer coefficients such that $f(0) = 0$, $f(1) = 1$ and $f(n)$ is congruent to 0 or 1 modulo $p$ for every integer $n$. Prove that $\deg f \geq p - 1$. (b) Let $p$ be a prime and let $f(x)$ be a polynomial with integer coefficients for which there is a permutation $\{a_1, \ldots, a_p\}$ of $\{1, \ldots, p\}$ such that $f(a_i) \equiv 2^{p-i} \pmod{p}$ for $i = 1, \ldots, p$. Prove that $\deg f \geq p - 1$. [(a) IMO Shortlist 1997; (b) Bulgaria IMO TST 2017]

**Problem 18.116.** Determine all primes of the form $5n + 1$ for which the remainders of $1^5, 2^5, \ldots, n^5$ modulo $5n + 1$ are pairwise distinct. [Bulgaria Balkan MO TST 2017]

**Problem 18.117.** Let $p < q$ be primes. Denote $A_k = k^{p-1} + k^{p-2} + \cdots + k + 1$ for $k \in \{1, 2, \ldots, q-1\}$. Determine all possible remainders when $A_1 A_2 \cdots A_{q-1}$ is divided by $q$. [Bulgaria RMM TST 2017]

**Problem 18.118.** Let $d$ be a positive integer and let $p$ be a prime. Prove that there is a primitive root $x$ modulo $p$ such that $d^2 x$ is also a primitive root modulo $p$. [Iran National Round 2011]

**Problem 18.119.** Let $n > 1$ be a squarefree positive integer. Prove that there is a prime divisor $q$ of $n$ and a positive integer $m$ such that $n$ divides $q(m^q + q)$. [Turkey EGMO TST 2016]

**Problem 18.120.** Let $n$ be a positive integer such that $125n + 22$ is a power of 3. Prove that $125n + 29$ has a prime divisor greater than 100. [Brazil EGMO TST 2021]

**Problem 18.121.** (a) Let $p \geq 5$ be a prime. Does there exist a sequence of $p-1$ integers $a_1, \ldots, a_{p-1}$, all distinct modulo $p$, such that $a_i a_{i+2} - a_{i+1}^2$ is not divisible by $p$ for $i = 1, \ldots, p-1$ (with $a_p = a_1$ and $a_{p+1} = a_2$), and such that the $p-1$ sums $a_1a_2$, $a_1a_2 + a_2a_3$, $\ldots$, $a_1a_2 + a_2a_3 + \cdots + a_{p-1}a_1$ are all distinct modulo $p$? (b) Let $p \geq 3$ be a prime. Prove that there is a permutation $(x_1, \ldots, x_{p-1})$ of $(1, 2, \ldots, p-1)$ such that $x_1x_2 + x_2x_3 + \cdots + x_{p-2}x_{p-1} \equiv 2 \pmod{p}$. [(a) Imperial Mathematics Competition 2021; (b) IMC 2022]

**Problem 18.122.** Find all positive integers $n$ such that $n$ divides $a^{n+1} - a$ for any integer $a$. [Don Zagier]

**Problem 18.123.** Let $a$ and $b$ be positive integers. Prove that for all positive integers $n$ there exists a positive integer $m$ such that at least one of $a^m b^2 - 1$ and $b^m a^2 - 1$ is divisible by $2^n$. [Bulgaria IMO TST 2020]

**Problem 18.124.** ★ Let $p$ be a prime and let $A$ be a subset of the nonzero remainders $\{1, 2, \ldots, p-1\}$ modulo $p$ with the following properties: (1) $A$ is non-empty and the number of its elements is a multiple of 6; (2) if $x, y \in A$ (not necessarily distinct), then $xy \bmod p$ also lies in $A$; (3) if $x \in A$, then $x^{-1} \in A$. Prove that there are elements $x_1, x_2, x_3 \in A$ with $x_1 + x_2 \equiv x_3 \pmod{p}$. [Vojtech Jarnik 2014, Category II]

**Problem 18.125.** ★ (a) Let $p$ be a prime and let $k$ be a positive integer. The set $S$ consists of all positive integers $a$ less than $p$ such that $x^k \equiv a \pmod{p}$ has a solution. Suppose that $3 \leq |S| \leq p-2$. Prove that the elements of $S$, when arranged in increasing order, do not form an arithmetic progression. (b) For which $p$ and $k$ does the condition $3 \leq |S| \leq p-2$ hold? [(a) China IMO 2018 TST]

**Problem 18.126.** Let $p \geq 5$ be a prime. Prove that there exist positive integers $m$ and $n$ such that $m + n \leq \frac{p+1}{2}$ and $2^n 3^m - 1$ is divisible by $p$. [Mongolian Teachers Olympiad 2017; Moldova IMO TST 2019]

**Problem 18.127.** In terms of the prime $p$, find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that $f(m) = f(n)$ for all $m, n \in \mathbb{Z}$ with $m \equiv n \pmod{p}$, and $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{Z}$. [Mathematical Excalibur 2010]

**Problem 18.128.** ★ 2010 MOPpers are assigned the numbers 1 through 2010. Each one is given a red slip and a blue slip of paper. Two positive integers $A$ and $B$, each at most 2010, are chosen. On the red slip each MOPper writes the remainder of $A$ times his or her number modulo 2011; on the blue slip, the remainder of $B$ times his or her number modulo 2011. They may then repeatedly perform either of the following operations: each gives his or her red slip to the person whose number is written on his or her blue slip; or each gives his or her blue slip to the person whose number is written on his or her red slip. Show that it is always possible to perform some number of these operations so that each MOPper is holding a red slip with his or her own number written on it. [ELMO 2010 P5]

**Problem 18.129.** Let $n$ be a positive integer. Is it possible to fill in the cells of an $n \times n$ table with the integers from 1 to $n^2$ (each appearing exactly once) so that all products of $n$ integers, no two of which lie in the same row or column, give the same remainder when divided by $n^2 + 1$, if (a) $n = 8$; (b) $n = 10$? [Serbia National Round 2010; Baltic Way 2023]

**Problem 18.130.** ★ Let $p$ be a prime and let $n \geq 2$ be an integer. Prove that there exist at least two positive integers $t$ strictly between 1 and $p^n$ such that the product of all positive

integers not exceeding $t$ and congruent to $t$ modulo $p$ is congruent to 1 modulo $p^n$. [Serbia IMO TST 2023]

**Problem 18.131.** Let $p \geq 3$ be a prime and let $m$ be a positive integer. Determine the smallest positive integer $n$ with the following property: for every positive integer $a$ not divisible by $p$, the sum of the positive divisors of $a^n$ that are greater than 1 is divisible by $p^m$. [Bulgarian Festival of Young Mathematicians 2024, Senior Group]

**Problem 18.132.** Determine all pairs $(a, b)$ of positive integers for which there exists an integer $N$ such that $\gcd(a^n + b,\ b^n + a)$ is the same for all $n \geq N$. [IMO 2024 P2]

**Problem 18.133.** ★ Determine all primes $p \geq 7$ for which there exist integers $a$ and $r$ satisfying $1 \leq r \leq p-1$ with the following property: the sequence $1, a, a^2, \ldots, a^{p-5}$ can be rearranged to form a sequence $b_0, b_1, \ldots, b_{p-5}$ such that $b_n - b_{n-1} - r$ is divisible by $p$ for $1 \leq n \leq p-5$. [Putnam 2024 A4]

**Problem 18.134.** Determine all primes $p$ and positive integers $r$ which satisfy $p^p + (p+1)^p + \cdots + (p+r)^p = (p+r+1)^p$. [Taiwan IMOC 2022]

**Problem 18.135.** ★ (a) Let $p \equiv 3 \pmod 4$ be a prime. Prove that $\prod_{1 \leq x < y \leq (p-1)/2} (x^2 + y^2) \equiv (-1)^{\lfloor (p+1)/8 \rfloor} \pmod p$. (b) Let $p = 8k+5$ be a prime and let the integers $r_1, r_2, \ldots, r_{2k+1}$ be such that the numbers $0, r_1^4, r_2^4, \ldots, r_{2k+1}^4$ give different remainders modulo $p$. Prove that $\prod_{1 \leq i < j \leq 2k+1} (r_i^4 + r_j^4) \equiv (-1)^{\lfloor k(k+1)/2 \rfloor} \pmod p$. [(b) Olympiad of Metropolises 2018 P3]

**Problem 18.136.** ★ (a) Is there a prime $p = 6k+1$ with $\binom{3k}{k} \equiv 1 \pmod p$? (b) Let $p = 8k+1$ be a prime and let $r$ be the remainder when $\binom{4k}{k}$ is divided by $p$. Prove that $r$ is not a perfect square. [(a) USA IMO TST 2010; (b) Taiwan IMO TST 2019]

**Problem 18.137.** ★ Let $p \geq 3$ be a prime. Determine the greatest integer $n$ for which there exists a polynomial $Q(x)$ with integer coefficients such that the coefficient at $x^s$ of $(x-1)^n Q(x)$ is congruent modulo $p$ to the coefficient at $x^s$ of $\sum_{k=1}^{p-1} k^{(p-1)/2} x^k$, for all non-negative integers $s$. [Putnam 2019 A5]

**Problem 18.138.** Let $p$ be a prime and let $k$ be a positive integer. Suppose that the numbers $a_i = i^k + i$ for $i = 0, 1, \ldots, p-1$ are pairwise distinct modulo $p$. Determine all possible remainders when $a_2$ is divided by $p$, as $k$ varies. [IMC 2023]

**Problem 18.139.** ★ Let $p \geq 23$ be a prime of the form $2q+1$ where $q$ is also a prime. Find the number of pairs $(m, n)$ of integers with $0 \leq m < n < p-1$ such that $3^m + (-12)^m \equiv 3^n + (-12)^n \pmod p$. [ELMO 2011 P5]

**Problem 18.140.** ★ Let $p$ be a prime of the form $2^{2^k} + 1$, where $k \geq 0$ is an integer. Find the number of pairs $(m, n)$ of integers with $0 \leq m < n < p-1$ such that $2^m + 3^m \equiv 2^n + 3^n \pmod p$. [ELMO 2011, original formulation]

**Problem 18.141.** ★ Let $p \geq 5$ be a prime such that there are at least $\frac{p+1}{3}$ positive integers not exceeding $p-1$ and coprime with $p-1$. Prove that there exists a primitive root $g$ modulo $p$ such that $g+1$ is also a primitive root. [Emanuel Vegh, Pairs of Consecutive Primitive Roots Modulo a Prime]

**Problem 18.142.** Let $\mathcal{P}$ be the set of all primes and let $M$ be a non-empty subset of $\mathcal{P}$. Suppose that for any non-empty subset $\{p_1, p_2, \ldots, p_k\}$ of $M$ all prime factors of $p_1 p_2 \cdots p_k + 1$ are also in $M$. Prove that $M = \mathcal{P}$. [USA TSTST 2015; Romania IMO TST 2003]

**Problem 18.143.** ★ Let $p \geq 3$ be a prime and let $a_1, a_2, \ldots, a_p$ be integers. Prove that there exists a polynomial $P(x)$ of degree at most $\frac{p-1}{2}$ such that $P(i) \equiv a_i \pmod{p}$ for all $1 \leq i \leq p$ if and only if for every positive integer $d \leq \frac{p-1}{2}$ we have $\sum_{i=1}^{p} (a_{i+d} - a_i)^2 \equiv 0 \pmod{p}$, where the indices are taken modulo $p$. [China 2016 National Round]

**Problem 18.144.** ★ Let $p$ be a prime and let $f(x)$ be a polynomial of degree $d$ with integer coefficients. Let $k$ be the number of distinct remainders modulo $p$ among $f(1), f(2), \ldots, f(p)$. If $1 < k < p$, prove that $\dfrac{p-1}{d} \leq k - 1 \leq (p-1)\left(1 - \dfrac{1}{d}\right)$. [International Olympiad of Metropolises 2019 P6]

**Problem 18.145.** ★ Let $p$ be a prime and let $k$ be a positive integer. Assume that $P(x)$ is a non-constant polynomial with coefficients in $\{0, 1, \ldots, p-1\}$, of minimal degree, with the following property: the numbers $1, 2, \ldots, p-1$ can be placed around a circle so that for any $k$ consecutive numbers $a_1, a_2, \ldots, a_k$ the sum $P(a_1) + P(a_2) + \cdots + P(a_k)$ is divisible by $p$. Prove that there exist integers $a, b, d$ such that $P(x) = ax^d + b$. [Iran National Round 2024]

**Problem 18.146.** ★ Let $\mathbb{Z}/n\mathbb{Z}$ denote the set of integers modulo $n$. Determine all positive integers $n$ for which there exists a bijective function $g : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ such that the 101 functions $g(x),\ g(x) + x,\ g(x) + 2x,\ \ldots,\ g(x) + 100x$ are all bijections on $\mathbb{Z}/n\mathbb{Z}$. [USA IMO TST 2019]

**Problem 18.147.** ★ Determine all positive integers $n$ with the following property: for all integers $m$ relatively prime to $n$, there exists a permutation $\pi : \{1, 2, \ldots, n\} \to \{1, 2, \ldots, n\}$ such that $\pi(\pi(k)) \equiv mk \pmod{n}$ for all $k \in \{1, 2, \ldots, n\}$. [Putnam 2023 B5]

**Problem 18.148.** ★ Let $p > 10^9$ be a prime such that $4p+1$ is a prime. Prove that the decimal expansion of $\dfrac{1}{4p+1}$ contains each digit $0, 1, \ldots, 9$ at least once. [Bulgaria IMO TST 2014; Croatia IMO TST 2016]

**Problem 18.149.** ★ For a prime $p$, let $\mathbb{F}_p$ denote the integers modulo $p$ and let $\mathbb{F}_p[x]$ be the set of polynomials with coefficients in $\mathbb{F}_p$. Find all $p$ for which there exists a polynomial $P(x) \in \mathbb{F}_p[x]$ of degree 4 such that for all integers $k$ there exists an integer $l$ with $P(l) \equiv k \pmod{p}$. [ELMO 2024 P6]

**Problem 18.150.** Do there exist 100 positive integers not exceeding $10^{100}$ such that all possible sums of two of them (not necessarily distinct) are different? [IMO Shortlist 2001 N6, stronger version]

**Problem 18.151.** We call a positive integer $m$ an *anagram* of a positive integer $n$ if every digit $a$ appears as many times in the decimal representation of $m$ as it does in the decimal representation of $n$. Do there exist four different positive integers such that each of them is an anagram of the sum of the other three? [Balkan Mathematical Olympiad 2015 Shortlist]

**Problem 18.152.** ★ Let $n = 2^k + 1$, where $k$ is a positive integer. Prove that $n$ is prime if and only if there is a permutation $a_1, \ldots, a_{n-1}$ of $1, 2, \ldots, n-1$ and a sequence of integers $g_1, \ldots, g_{n-1}$ such that $n$ divides $g_i^{a_i} - a_{i+1}$ for every $i \in \{1, \ldots, n-1\}$, where $a_n = a_1$. [IMO Shortlist 2011 N8]

**Problem 18.153.** ★ Prove that for every prime $p \neq 11$ and every integer $r$ there exist integers $a$ and $b$ such that $a^2 + b^5 - r$ is divisible by $p$. [IMO Shortlist 2012 N8, full version]

### 18.3.1 Sidon sets

**Definition 18.25 (Sidon set).** A subset $A = \{a_1 < a_2 < \cdots < a_k\}$ of $\{1, 2, \ldots, n\}$ is called a *Sidon set* if for each $x$ there is at most one solution of $x = a_i + a_j$ with $a_i, a_j \in A$, $i \le j$. Equivalently, all differences $a_k - a_l$ with $k > l$ are distinct. We write $Q(n)$ for the maximal possible size of a Sidon subset of $\{1, 2, \ldots, n\}$.

**Theorem 18.26 (Erdős).** *For every positive integer $n$ one has $Q(n) \ge \lfloor \sqrt[3]{n} \rfloor$.*

**Theorem 18.27 (Ruzsa).** *For every prime $p$ one has $Q(p^2 - p) \ge p - 1$; a witnessing set is $\{\, pi + (p-1)g^i \bmod (p^2 - p) \;:\; 1 \le i \le p-1 \,\}$, where $g$ is a primitive root modulo $p$.*

**Corollary 18.28.** *For all sufficiently large $n$ one has $Q(n) \ge n^{1/2} - n^{21/40}$.*

**Theorem 18.29 (Lindström-type upper bound).** *For every $n \in \mathbb{N}$ and every integer $m$ with $1 \le m \le Q(n) - 1$,*

$$Q(n) \;\le\; \frac{m+1}{2} + \frac{\sqrt{-8m^3 + 12m^2 n - 12m^2 + 12mn - 4m + 3} - \sqrt{3}}{6m}.$$

**Remark.** These bounds give $49 \le Q(2024) \le 51$. The best known upper bound for large $n$ is $Q(n) \le n^{1/2} + 0.99703\, n^{1/4}$ (O'Bryant, 2022). It has been known for decades that $Q(n) \sim n^{1/2}$, but whether $Q(n) \le n^{1/2} + O(n^{\varepsilon})$ for every $\varepsilon > 0$ is a well-known open problem of Erdős.

## 18.4 Lifting the exponent: mastery

*Revisits the lifting-the-exponent lemma (see Chapter 13); the problems below push it further.*

**Definition 18.30 ($p$-adic valuation on the rationals).** For a prime $p$ and a nonzero integer $n$, $v_p(n)$ denotes the largest integer $k$ with $p^k \mid n$. This extends to a function $v_p : \mathbb{Q} \setminus \{0\} \to \mathbb{Z}$ by $v_p(a/b) = v_p(a) - v_p(b)$; by convention $v_p(0) = \infty$. For example $v_3(2025) = 4$, $v_2(8/5) = 3$ and $v_5(7/1000) = -3$.

**Problem 18.154.** For any integers $x, y$ and any positive integers $n$ and prime $p > 2$ such that $n$ is odd, $p \nmid x$, $p \nmid y$ and $p \mid x + y$, prove that $v_p(x^n + y^n) = v_p(x + y) + v_p(n)$.

**Problem 18.155.** For any odd positive integers $x, y$ such that $x - y$ is divisible by 4, prove that $v_2(x^n - y^n) = v_2(x - y) + v_2(n)$ for all positive integers $n$.

**Problem 18.156.** Let $(a_n)_{n \ge 1}$ be the sequence with $a_1 = 1$ and $a_{n+1} = a_n + 2^{a_n}$ for all $n$. Prove that $a_1, a_2, \ldots, a_{3^{2013}}$ leave distinct remainders when divided by $3^{2013}$. [USA TSTST 2013]

**Problem 18.157.** Let $P(x)$ be a polynomial with integer coefficients whose leading coefficient is positive. Prove that there are finitely many positive integers $n$ such that $n! + 1$ is a power of $P(n)$. [Bulgaria Spring Mathematical Competition 2023, Grade 12]

**Problem 18.158.** Let $a$ be a positive integer such that $4(a^n+1)$ is a perfect cube for every positive integer $n$. Prove that $a=1$. [Iran Second Round 2008]

**Problem 18.159.** Determine all positive integers $n$ such that $n(2^n-1)$ is a perfect square. [IMOC 2024; see also RMM 2025 P2]

**Problem 18.160.** ★ We call an infinite set $S$ of positive integers *good* if for all pairwise different integers $a,b,c\in S$ all positive divisors of $\dfrac{a^c-b^c}{a-b}$ are also in $S$. For any positive integer $n>1$ prove that there exists a good set $S$ such that $n$ does not belong to $S$. [Iran IMO TST 2022]

**Problem 18.161.** Prove that for any positive integer $n$ there exists a positive integer $m$ such that $7^n$ divides $3^m+5^m-1$. [Balkan MO Shortlist 2017; Iran National Round 2017]

**Problem 18.162.** Let $a,b$ and $n$ be positive integers with $a>b$ such that $a^{2021}$ divides $n$, $b^{2021}$ divides $n$ and 2022 divides $a-b$. Prove that some of the positive divisors of $n$ have a sum which is divisible by 2022, but not divisible by $2022^2$. [Balkan MO 2022 P2]

**Problem 18.163.** ★ Call a rational number *short* if it has finitely many digits in its decimal expansion. For a positive integer $m$, we say that a positive integer $t$ is *$m$-tastic* if there exists a number $c\in\{1,2,\ldots,2017\}$ such that $\dfrac{10^t-1}{c\cdot m}$ is short and $\dfrac{10^k-1}{c\cdot m}$ is not short for any $1\le k<t$. Let $S(m)$ be the set of $m$-tastic numbers. What is the maximum number of elements of $S(m)$, as $m$ varies over the positive integers? [IMO Shortlist 2017 N4]

**Problem 18.164.** Let $a$ and $b$ be odd positive integers smaller than $2^{100}$. Find all pairs $(a,b)$ for which both $a^b+b$ and $b^a+a$ are divisible by $2^{100}$.

**Problem 18.165.** Show that for $n>1$ the sum $S_n=1^1+3^3+5^5+\cdots+(2^n-1)^{2^n-1}$ is divisible by $2^n$ but not by $2^{n+1}$.

**Problem 18.166.** Let $n$ be a squarefree integer. Find all pairs of coprime positive integers $(x,y)$ such that $x^n+y^n$ is divisible by $(x+y)^3$.

**Problem 18.167.** Prove that for any positive integer $k>1$ there exists a power of 2 such that at least half of its last $k$ digits are equal to 9. For example, $2^{12}=\ldots96$ and $2^{53}=\ldots992$. [Russia 1994]

## 18.5 Wolstenholme, power sums and factorial valuations

**Theorem 18.31 (Wolstenholme-type congruences).** *If $p>5$ is a prime, then*

$$p\mid \frac{1}{1^2}+\frac{1}{2^2}+\cdots+\frac{1}{(p-1)^2},\qquad p^2\mid \frac{1}{1}+\frac{1}{2}+\cdots+\frac{1}{p-1},\qquad p\mid \sum_{1\le i<j<p}\frac{1}{ij},$$

*where the divisibility statements are understood for the numerators of the reduced fractions.*

**Theorem 18.32 (Bounds for $v_p(n!)$).** *For every $n>1$ and every prime $p$, $\dfrac{n-1}{p}<$*

$v_p(n!) < \dfrac{n}{p-1}$.

**Theorem 18.33 (Growth of the primorial).** *Denote by $n\#$ the product of all primes not exceeding $n$. Then $n\# < 4^n$ for all positive integers $n$; more precisely, $m\# < 2^{2m-3}$ for all $m > 3$.*

**Theorem 18.34 (Bertrand's postulate).** *For each $n > 1$ there is a prime $p$ with $n < p < 2n$.*

**Problem 18.168.** Let $n > 1$ be an odd integer. Find the remainder when $\left((n-2)!!\right)^2$ is divided by $n$.

**Problem 18.169.** Let $p$ be a prime and put $N = \dfrac{(p-1)p}{2}$. Find the remainder when $(p-1)!$ is divided by $N$.

**Problem 18.170.** Let $p$ and $p+2$ be twin primes. Find the remainder when $4(p-1)!$ is divided by $p(p+2)$.

**Problem 18.171.** For integers $n$ and $q$ with $n \geq 5$ and $n \geq q \geq 2$, prove that $\lfloor (n-1)!/q \rfloor$ is divisible by $q-1$.

**Problem 18.172.** Let $p > 2$ be a prime and write $\dfrac{m}{n} = \dfrac{1}{1} + \dfrac{1}{2} + \cdots + \dfrac{1}{p-1}$ in lowest terms. Prove that $p \mid m$.

**Problem 18.173.** Let $p > 3$ be a prime and let $m, n$ be relatively prime positive integers with $\dfrac{m}{n} = \dfrac{1}{1^2} + \dfrac{1}{2^2} + \cdots + \dfrac{1}{(p-1)^2}$. Prove that $p \mid m$.

**Problem 18.174.** Let $p > 3$ be a prime and let $S = (p-1)!\left(\dfrac{1}{1} + \dfrac{1}{2} + \cdots + \dfrac{1}{p-1}\right)$. Prove that $S$ is divisible (a) by $p$; (b) by $p^2$.

**Problem 18.175.** Let $p > 2$ be a prime and let $A_k = 1^k + 2^k + \cdots + (p-1)^k$ for $1 \leq k \leq p-2$. Prove that $A_k$ is divisible by $p$.

**Problem 18.176.** Let $p > 2$ be a prime and let $A_k = 1^k + 2^k + \cdots + (p-1)^k$ for $1 \leq k \leq p$. Prove that $A_3, A_5, \ldots, A_{p-2}, A_p$ are divisible by $p^2$.

**Problem 18.177.** Let $p$ be a prime and let $A_k = 1^k + 2^k + \cdots + (p-1)^k$. Prove that $p^2$ divides $A_{pk}$ for $1 \leq k < \frac{p-1}{2}$.

**Problem 18.178.** Let $p > 3$ be a prime and let $A_k = 1^k + 2^k + \cdots + (p-1)^k$. Prove that $p^5$ divides $p^2 A_{p-1} - 2A_p$.

**Problem 18.179.** Let $p > 5$ be a prime, let $k$ be an odd integer with $3 < k < p$, and let $A_k = 1^k + 2^k + \cdots + (p-1)^k$. Prove that $p^4$ divides $kpA_{k-1} - 2A_k$.

**Problem 18.180.** Let $p > 3$ be a prime. Prove that the sum $\frac{(p-1)!^p}{1} + \frac{(p-1)!^p}{2} + \cdots + \frac{(p-1)!^p}{p-1}$ is divisible by $p^3$.

**Problem 18.181.** Let $p$ be an odd prime. Prove that $1 + \frac{1}{3} + \frac{1}{5} + \cdots + \frac{1}{p-2} \equiv \frac{2^{p-1}-1}{p} \pmod{p}$.

**Problem 18.182.** Let $p$ be an odd prime. For every integer $a$ define $S_a = \frac{a}{1} + \frac{a^2}{2} + \cdots + \frac{a^{p-1}}{p-1}$. Prove that $S_3 + S_4 \equiv 3S_2 \pmod{p}$.

**Problem 18.183.** If $p$ is an odd prime, prove that $1^{p-2} + 2^{p-2} + \cdots + (p-1)^{p-2} \equiv \frac{2-2^p}{p} \pmod{p}$.

**Problem 18.184.** Let $p > 3$ be a prime and let $k$ be a positive integer with $k \leq p-2$. Consider all $k$-element subsets of $\{1, 2, \ldots, p-1\}$, multiply the elements of each, and add the results; denote this sum by $S_k$. Prove that $S_k$ is divisible by $p$.

**Problem 18.185.** With the notation of the previous problem, prove that $S_3, S_5, \ldots, S_{p-2}$ are divisible by $p^2$.

**Problem 18.186.** Let $p > 5$ be a prime and let $S_k$ be as above. Prove that $pS_{p-3} - S_{p-2}$ is divisible by $p^4$.

**Problem 18.187.** Let $p > 3$ be a prime and let $k \leq p-2$ be a positive integer. Consider all $k$-element subsets of $\{1, 2, \ldots, p-1\}$, multiply the elements of each, raise the results to the power $p$, and add them all together. Prove that the sum is divisible by $p^2$, and that it is divisible by $p^3$ if $k$ is odd and greater than 1.

**Problem 18.188.** Let $p > 2$ be a prime. Find the parity of the number $\frac{(p-1)!}{p \cdot (p-2)}$.†

**Problem 18.189.** Let $p$ be a prime. Prove that $1^{p-1} + 2^{p-1} + \cdots + (p-1)^{p-1} \equiv p + (p-1)! \pmod{p^2}$.

**Problem 18.190.** Let $p$ be a prime. Determine the smallest positive integer $n$ such that for any integer $a$ there exists a positive integer $k$ with $1^k + 2^k + \cdots + n^k \equiv a \pmod{2015}$. †[Vietnam IMO TST 2015]

**Problem 18.191.** Suppose there exists a permutation $a_1, a_2, \ldots, a_n$ of $0, 1, 2, \ldots, n-1$ such that the remainders of $a_1,\ a_1a_2,\ a_1a_2a_3,\ \ldots,\ a_1a_2\cdots a_n$ upon division by $n$ are distinct. What may the values of $n$ be? Answer the same question with the partial sums $a_1,\ a_1+a_2,\ \ldots,\ a_1+a_2+\cdots+a_n$ in place of the partial products.

**Problem 18.192.** Determine whether it is possible to divide all divisors of 100! into (a) two groups; (b) 100 groups, so that all groups have the same number of elements and the same product of elements.

**Problem 18.193.** Prove that, for each integer $n \geq 2$, the number $n!$ can be represented as a sum of $n$ distinct divisors of itself.

**Problem 18.194.** For a prime $p$ and a positive integer $n$ let $\nu_p(n)$ denote the exponent of $p$ in the prime factorisation of $n!$. Given $d \in \mathbb{N}$ and a set $\{p_1, p_2, \ldots, p_k\}$ of primes, show that there are infinitely many positive integers $n$ such that $d \mid \nu_{p_i}(n)$ for all $1 \le i \le k$.

**Problem 18.195.** For integers $n \ge 2$ and $k \ge 2$ let $\nu_k(n!)$ denote the largest non-negative integer $r$ such that $k^r$ divides $n!$. Prove that the difference $\nu_{10}(n!) - \nu_9(n!)$ is greater than 1000 for infinitely many $n$ and less than $-1000$ for infinitely many $n$. [IMO Shortlist 2023 N3, better version]

**Problem 18.196.** Prove that there is a positive integer $n$ such that $1! + 2! + \cdots + n!$ has a prime factor greater than $10^{2021}$.

**Problem 18.197.** Find all pairs $(a, b)$ of positive integers such that $a! + b! = a^b + b^a$.

**Problem 18.198.** Find all positive integers $n$ and $k$ such that $1^n + 2^n + \cdots + n^n = k!$.

**Problem 18.199.** Prove that there is a prime number between $n$ and $n!$ for every positive integer $n$, without appealing to Bertrand's postulate.

**Problem 18.200.** Let $p_n$ be the $n$-th prime number. Find an explicit function $f(n)$, depending on $n$ alone, such that $p_n < f(n)$ for all $n$.

**Problem 18.201.** Prove that there is no non-constant polynomial $P$ with integer coefficients such that all of $P(1), P(2), P(3), \ldots$ are prime.

**Remark.** There does exist an explicit formula producing the $n$-th prime for every positive integer $n$ — Willans' formula — but it is far too ugly to be of any practical use.

**Problem 18.202.** Let $p$ be a prime and suppose $p^a$ divides $\binom{n}{k}$. Prove that $p^a \le n$.

**Problem 18.203.** Let $p$ be the $k$-th prime number and write $p = 2m+1$. Prove that $\binom{2m+1}{m} < 4^m$ and that $\binom{2m+1}{m}$ is divisible by every prime $q$ with $m+1 < q \le 2m+1$; conclude that $p_{k+1} > 2p_k$ and that $p_k < 4k \log_2 k$.

**Problem 18.204.** Recall that the product of all primes between 1 and $N$ is at most $4^N$. Using this, show that $p_k > \frac{1}{10}\, k \log_2 k$, where $p_k$ denotes the $k$-th prime.

**Problem 18.205.** Prove that for any $n$ it is not possible to divide the numbers $1, 2, \ldots, n$ into two groups so that the product of the numbers in the first group equals the product of the numbers in the second group.

**Problem 18.206.** Prove that in fact there are at least $\sqrt{n}$ primes between $n$ and $2n$ for every $n > 5$.

**Problem 18.207.** Prove that there are only finitely many positive integers $n$ such that $n! + 1$ divides $(100n)!$.

**Problem 18.208.** Let $p$ and $q$ be primes with $p > q$. Prove that $\gcd(p! - 1, q! - 1) \le p^{p/3}$.

**Problem 18.209.** Find all positive integers $n$ and $k$ such that $(n-1)! + 1 = n^k$.

**Problem 18.210.** Let $a$ and $n$ be positive integers with $a > (n-1)!$. Prove that there exist distinct primes $p_1, \ldots, p_n$ such that $p_i \mid a + i$ for each $i = 1, 2, \ldots, n$.

**Problem 18.211.** Let $a_1, a_2, \ldots, a_9$ be distinct positive integers. Prove that, for all sufficiently large $x \in \mathbb{N}$, the number $(x+a_1)(x+a_2)\cdots(x+a_9)$ has at least one prime divisor greater than 20.

**Problem 18.212.** Given $p \geq 3$, show that for all sufficiently large integers $x$ at least one of $x+1,\ x+2,\ \ldots,\ x+\frac{p+3}{2}$ has a prime divisor greater than $p$.

**Problem 18.213.** Let $p_1, \ldots, p_n$ be distinct primes greater than 3. Show that $2^{p_1 p_2 \cdots p_n}+1$ has at least $4^n$ divisors.

**Problem 18.214.** Show that there are infinitely many integers $n$ such that $\pi(n) \mid n$, where $\pi(n)$ denotes the number of primes not exceeding $n$.

## 18.6 The arithmetic of binomial coefficients

**Theorem 18.35 (Legendre's formula).** *For a positive integer $n$ and a prime $p$, $v_p(n!) = \sum_{i \geq 1} \left\lfloor \frac{n}{p^i} \right\rfloor$.*

**Theorem 18.36 (Kummer's theorem).** *For integers $n \geq m \geq 0$ and a prime $p$, $v_p\left(\binom{n}{m}\right)$ equals the number of carries that occur when $m$ and $n-m$ are added in base $p$.*

**Theorem 18.37 (Lucas' theorem).** *Let $p$ be a prime and let $m, n$ be nonnegative integers with base-$p$ representations $m = (m_k m_{k-1} \ldots m_0)_p$ and $n = (n_k n_{k-1} \ldots n_0)_p$. Then $\binom{m}{n} \equiv \prod_{i=0}^{k} \binom{m_i}{n_i} \pmod{p}$.*

**Lemma 18.38 (Binomial coefficients modulo $p$).** *If $p$ is a prime and $0 \leq i, j < p$ are integers, then*

*(a)* $p \mid \binom{p}{i}$ *for* $0 < i < p$;

*(b)* $\binom{p-1}{i} \equiv (-1)^i \pmod{p}$;

*(c)* $\frac{1}{p}\binom{p}{i} \equiv \frac{(-1)^{i-1}}{i} \pmod{p}$ *for* $i > 0$;

*(d)* $\binom{i+j}{j} \equiv (-1)^j \binom{p-1-i}{j} \equiv (-1)^i \binom{p-1-j}{i} \pmod{p}$.

**Problem 18.215.** Let $n$ be an integer and let $m = \gcd\left\{\binom{n}{1}, \binom{n}{2}, \ldots, \binom{n}{n-1}\right\}$. Prove that if $m > 1$, then $m$ is a prime and $n$ is a power of $m$.

**Problem 18.216.** If $k < m < n$ are positive integers, prove that $\binom{n}{k}$ and $\binom{n}{m}$ are not coprime.

**Problem 18.217.** Let $k$ and $m$ be positive integers. (a) Prove that there exists $n$ such that the $k$ numbers $\binom{n}{k}, \binom{n+1}{k}, \ldots, \binom{n+k-1}{k}$ are all divisible by $m$. (b) Prove that the $k+1$ numbers $\binom{n}{k}, \binom{n+1}{k}, \ldots, \binom{n+k}{k}$ are coprime for every $n \in \mathbb{N}$.

**Problem 18.218.** ★ Given a positive integer $r$, denote by $N_r$ the smallest positive integer such that $\frac{N_r}{n+r}\binom{2n}{n}$ is an integer for all $n \in \mathbb{N}$. Prove that $N_r = \frac{r}{2}\binom{2r}{r}$.

**Problem 18.219.** Let $n$ be a positive integer. If $\binom{n-1}{i} \equiv (-1)^i \pmod{n}$ for $i = 0, 1, \ldots, n-1$, prove that $n$ is prime.

**Problem 18.220.** Prove that $\binom{0}{0} - 2\binom{2}{1} + 2^2\binom{4}{2} - \cdots + (-2)^{(p-1)/2}\binom{p-1}{(p-1)/2} \equiv 1 \pmod{p}$ for every prime $p > 3$.

**Problem 18.221.** Denote by $C_n = \dfrac{1}{n+1}\binom{2n}{n}$ the Catalan numbers. Prove that $\sum_{k=1}^{n} C_k \equiv 1 \pmod{3}$ if and only if $n+1$ has a digit 2 in base 3.

**Problem 18.222.** If $n$ is a positive integer, prove that the number of odd terms in the $(n+1)$-st row of Pascal's triangle (that is, among $\binom{n}{i}$ for $0 \le i \le n$) is a power of 2.

**Problem 18.223.** Prove that the number of odd terms in the first $2^m$ rows of Pascal's triangle (that is, among $\binom{n}{k}$ with $0 \le k \le n < 2^m$) equals $3^m$.

**Problem 18.224.** Given a positive integer $n$, prove that the number of odd numbers among $\binom{n-i}{i}$, $0 \le i \le \frac{n}{2}$, is even if and only if $n \equiv 2 \pmod{3}$.

**Problem 18.225.** Let $p$ be a prime and let $n$ be a positive integer. If $1 \le j, k \le p-1$ are integers and $n \equiv k \pmod{p-1}$, prove that $\binom{n}{j} + \binom{n}{j+(p-1)} + \binom{n}{j+2(p-1)} + \cdots \equiv \binom{k}{j} \pmod{p}$.

**Problem 18.226.** Prove that $\binom{2p}{p} \equiv 2 \pmod{p^3}$ for every prime $p > 5$.

**Problem 18.227.** ★ For a positive integer $n$ consider $S_n = \left\{\binom{n}{n}, \binom{2n}{n}, \binom{3n}{n}, \ldots, \binom{n^2}{n}\right\}$. Prove that there are infinitely many composite numbers $n$ such that $S_n$ (a) is not, (b) is, a complete residue system modulo $n$.

**Problem 18.228.** Let $n$ be a positive integer. Prove that the sum $\sum_{k=0}^{3^n-1}\binom{2k}{k}$ is (a) divisible by 3; (b) divisible by $3^n$.

**Problem 18.229.** ★ If $n$ is a positive integer, prove that the numbers $\binom{2^n-1}{0}, \binom{2^n-1}{1}, \ldots, \binom{2^n-1}{2^{n-1}-1}$ leave, upon division by $2^n$, the remainders $1, 3, 5, \ldots, 2^n-1$ in some order.

**Problem 18.230.** ★ If $k \ge 2$ is a positive integer, prove that $\binom{2^{k+1}}{2^k} - \binom{2^k}{2^{k-1}}$ is divisible by $2^{3k}$, but not by $2^{3k+1}$.

**Problem 18.231.** Let $n$ be a positive integer and let $p \ge 5$ be a prime. Prove that

$$\binom{np}{0} + \binom{np}{p} + \binom{np}{2p} + \cdots + \binom{np}{np} \equiv 2^n \pmod{p^3}.$$

**Problem 18.232.** Let $p$ be a prime and let $p^r$ divide $\binom{2n}{n}$. Show that $p^r \le 2n$.

**Problem 18.233.** Show that $\dfrac{4^n}{2n} \le \binom{2n}{n} < 4^n$ for every $n \in \mathbb{N}$.

**Problem 18.234.** Find all positive integers $n$ such that $\binom{2n}{n}$ is not divisible by 4.

**Problem 18.235.** If $n > 3$, $p$ is a prime and $\frac{2n}{3} < p \le n$, prove that $\binom{2n}{n}$ is not divisible by $p$.

**Problem 18.236.** Let $n$ and $k$ be positive integers with $n \ge k + \operatorname{lcm}(1, 2, \ldots, k)$. Prove that $\binom{n}{k}$ has at least $k$ distinct prime factors.

**Problem 18.237.** Prove that in the sequence $\binom{2}{1}, \binom{4}{2}, \binom{8}{4}, \ldots, \binom{2^{n+1}}{2^n}, \ldots$ one may cross out several terms so that the difference of any two remaining terms is divisible by $2^{2020}$.

**Problem 18.238.** Let $p > 2$ be a prime and let $1 \le k < p$. Prove that (a) $\binom{p-1}{k} \equiv (-1)^k \pmod p$; (b) $\binom{p-2}{k} \equiv (-1)^k(k+1) \pmod p$.

**Problem 18.239.** Let $p > 2$ be a prime and let $n > p$. Prove that $\binom{n}{p} \equiv \left\lfloor \frac{n}{p} \right\rfloor \pmod p$.†

**Problem 18.240.** Let $n$ be a given positive integer. Count the number of polynomials $P$ whose coefficients belong to $\{0, 1, 2, 3\}$ and for which $P(2) = n$.

## 18.7 Pell's equation and Diophantine approximation

*Revisits the existence of solutions of Pell's equation (see Chapter 13); the problems below push it further.*

**Theorem 18.39 (Structure of the solutions of Pell's equation).** *Let $D$ be a positive integer that is not a perfect square and consider the equation $x^2 - Dy^2 = 1$. If $(x_1, y_1)$ is its smallest solution in positive integers, then all its solutions in positive integers are given by $x + y\sqrt{D} = \left(x_1 + y_1\sqrt{D}\right)^n$, $n \in \mathbb{N}$.*

**Lemma 18.40 (Dirichlet's approximation lemma).** *Given $n \in \mathbb{N}$ and an irrational number $\alpha$, there exist integers $p, q$ with $0 < q \le n$ such that $\left|\frac{p}{q} - \alpha\right| < \frac{1}{(n+1)q}$. Consequently there are infinitely many fractions $\frac{p}{q}$ with $\left|\frac{p}{q} - \alpha\right| < \frac{1}{q^2}$.*

**Theorem 18.41.** *Every Pell equation $x^2 - Dy^2 = 1$ has a nontrivial solution, that is, one with $y \neq 0$.*

**Theorem 18.42 (The negative Pell equation).** *Closely related is the equation $x^2 - Dy^2 = -1$. If it has solutions in positive integers, the smallest of which is $(x', y')$, then $\left(x' + y'\sqrt{D}\right)^2 = x_1 + y_1\sqrt{D}$, where $(x_1, y_1)$ is the smallest positive solution of $x^2 - Dy^2 = 1$.*

**Definition 18.43 (Farey sequence and mediant).** The *Farey sequence* of order $m$ is the increasing sequence of all rational numbers whose denominator is at most $m$. Given fractions $\frac{a}{b}$ and $\frac{c}{d}$, their *mediant* $\frac{a+c}{b+d}$ always lies between them. Whenever $\frac{a}{b}$ and $\frac{c}{d}$ are consecutive in some Farey sequence one has $bc - ad = 1$, and the fraction with the smallest denominator between them is exactly their mediant; the mediant is consecutive to both terms in the Farey sequence of order $b + d$.

**Theorem 18.44 (Hurwitz).** *Given an irrational number $\alpha$, there exist infinitely many rational numbers $\frac{p}{q}$ such that $\left|\alpha - \frac{p}{q}\right| < \frac{1}{q^2\sqrt{5}}$. The constant $\sqrt{5}$ cannot be improved.*

**Definition 18.45 (Approximability).** A real number $\alpha$ is *approximable with degree $k$* if for some constant $C > 0$ there exist infinitely many rational numbers $\frac{p}{q}$ satisfying $\left|\alpha - \frac{p}{q}\right| < \frac{C}{q^k}$. Rational numbers are not approximable with degree greater than 1, while by Dirichlet's lemma every irrational number is approximable with degree 2.

**Theorem 18.46 (Liouville).** *If $\alpha$ is a zero of a polynomial of degree $k$ with integer coefficients, then $\alpha$ is not approximable with degree greater than $k$. Consequently, if $\alpha$ is approximable with arbitrarily large degrees, then $\alpha$ is transcendental.*

**Problem 18.241.** Prove that the number $e = \sum_{k\geq 0} \frac{1}{k!}$ is irrational.

**Problem 18.242.** Denote by $\varphi = \frac{1+\sqrt{5}}{2}$ the golden section. Prove that for every positive integer $n$ one has $\lfloor \varphi^2 n \rfloor = \lfloor \varphi \lfloor \varphi n \rfloor \rfloor + 1$.

**Problem 18.243.** Suppose that positive integers $x$ and $y$ satisfy $\left\lfloor (4+2\sqrt{3})x \right\rfloor = \left\lfloor (4-2\sqrt{3})y \right\rfloor$. Prove that $x$ and $y$ have opposite parities.

**Problem 18.244.** Let $\alpha$ and $\beta$ be two positive irrational numbers with $\frac{1}{\alpha} + \frac{1}{\beta} = 1$. Prove that in the union of the sequences $\lfloor \alpha \rfloor, \lfloor 2\alpha \rfloor, \lfloor 3\alpha \rfloor, \ldots$ and $\lfloor \beta \rfloor, \lfloor 2\beta \rfloor, \lfloor 3\beta \rfloor, \ldots$ every positive integer occurs exactly once. (Beatty's theorem)

**Problem 18.245.** Find the smallest possible value of $\left|\frac{p}{q} - \sqrt{10}\right|$ over all positive integers $p, q$.

**Problem 18.246.** Is there an irrational number $\alpha$ such that $\lfloor \alpha^n \rfloor$ is divisible by 3 for every positive integer $n$?

**Problem 18.247.** Find all positive integers $n$ for which $n^2$ is divisible by $\left\lfloor n\sqrt{8} \right\rfloor$.

**Problem 18.248.** Let $a, b, c, d$ be integers with $b, d > 0$ and $bc - ad = 1$. What is the rational number strictly between $\frac{a}{b}$ and $\frac{c}{d}$ whose denominator is smallest possible?

**Problem 18.249.** Start with the sequence $\left(\frac{0}{1}, \frac{1}{1}\right)$. Whenever you spot two adjacent fractions $\frac{a}{b}$ and $\frac{c}{d}$ whose sum of denominators is less than $n$, insert $\frac{a+c}{b+d}$ between them. Prove that the final sequence contains every rational number with denominator at most $n$.

**Problem 18.250.** Find the smallest solution of the Pell equation $x^2 - 29y^2 = 1$.

**Problem 18.251.** Suppose that both $x$ and $y = 2 + 2\sqrt{28x^2 + 1}$ are integers. Prove that $y$ is a perfect square.

**Problem 18.252.** Suppose $n$ is a positive integer such that both $3n + 1$ and $4n + 1$ are perfect squares. Prove that $n$ is divisible by 56.

**Problem 18.253.** Let $x, y, n$ be integers with $x^2 - (n^2-1)y^2 = a$. If $0 < a \leq 2n+1$, prove that $a$ is a perfect square.

**Problem 18.254.** Find all positive integers $d$ for which there is a solution $(x, y)$ of Pell's equation $x^2 - dy^2 = 1$ satisfying $x - y = d$.

**Problem 18.255.** If $p \equiv 1 \pmod 4$ is a prime, prove that the equation $x^2 - py^2 = -1$ has a solution in integers.

**Problem 18.256.** Prove that $\left\{n\sqrt{2}\right\} > \dfrac{1}{2n\sqrt{2}}$ for each $n \in \mathbb{N}$. Can the constant $2\sqrt{2}$ be improved?

**Problem 18.257.** Prove that among any 20 consecutive positive integers there is one, say $d$, such that $n\sqrt{d}\left\{n\sqrt{d}\right\} > \dfrac{\sqrt{5}}{2}$ for all $n$.

**Problem 18.258.** Does there exist an infinite bounded sequence $x_1, x_2, \ldots$ such that $|x_m - x_n| \geq \dfrac{1}{|m-n|}$ for any distinct $m, n \in \mathbb{N}$?

**Problem 18.259.** Suppose that $a_1, a_2, \ldots$ is an infinite sequence with $0 \leq a_i \leq c$ for all $i$ and $|a_i - a_j| \geq \dfrac{1}{i+j}$ for all $i \neq j$. (a) Prove that $c \geq 1$. (b) Prove that such a sequence exists if $c = 2$.

**Problem 18.260.** Let $p$ be an odd prime. Find positive integers $x$ and $y$ for which $\sqrt{2p} - \sqrt{x} - \sqrt{y}$ is positive but as small as possible.

**Problem 18.261.** For a positive integer $n$, let $X_n$ be the set of all pairs $(a, b)$ of positive integers with $\gcd(a, b) = 1$ and $0 < a < b \leq n < a + b$. Prove that $\displaystyle\sum_{(a,b)\in X_n} \frac{1}{ab} = 1$.

**Problem 18.262.** Prove that for any irrational number $\alpha$ there are infinitely many positive integers $n$ satisfying $\{n\alpha\} < \dfrac{1}{n}$.

**Problem 18.263.** Define $x * y = xy + \lfloor \alpha x \rfloor \lfloor \alpha y \rfloor$, where $\alpha$ is the positive root of $x^2 = 2024x + 1$. Prove that the operation $*$ is associative on positive integers.

**Problem 18.264.** Do there exist integers $a, b, c > 2021$ such that $a + b\sqrt{c}$ has decimal expansion of the form $\ldots 2022.2023 \ldots$?

**Problem 18.265.** Given $n \in \mathbb{N}$, at most how many numbers from $\{1, 2, 3, \ldots, 2^n\}$ can we choose if for any two distinct chosen numbers $x, y$ the number $v_2(x - y)$ must be even?

## 18.8 Cyclotomic polynomials

**Definition 18.47 (Cyclotomic polynomial).** For a positive integer $n$, the $n$-th cyclotomic polynomial $\Phi_n(x)$ is the monic polynomial whose roots are exactly the primitive $n$-th roots of unity.

**Definition 18.48 (Möbius function).** The Möbius function is defined by $\mu(1) = 1$, $\mu(n) = (-1)^k$ if $n = p_1 p_2 \cdots p_k$ is a product of $k$ distinct primes, and $\mu(n) = 0$ if $n$ is divisible by the square of a prime.

**Lemma 18.49 (Key lemma on prime divisors of $\Phi_n$).** *Let $p$ be a prime, $n$ a positive integer and $a$ an integer with $\Phi_n(a) \equiv 0 \pmod{p}$. Then either $p$ divides $n$, or $n$ divides $p-1$; in the latter case $n$ is exactly the order of $a$ modulo $p$.*

**Problem 18.266.** Polynomials $A(x), B(x), C(x), D(x)$ satisfy $A(x^5) + xB(x^5) + x^2C(x^5) = (1 + x + x^2 + x^3 + x^4)D(x)$ for all $x \in \mathbb{R}$. Find all possible values of $A(1)$.

**Problem 18.267.** A sequence $a_1, a_2, \ldots, a_n$ is called *$k$-balanced* if $a_1 + a_{k+1} + \cdots = a_2 + a_{k+2} + \cdots = \cdots = a_k + a_{2k} + \cdots$. Suppose that the sequence $a_1, a_2, \ldots, a_{50}$ is $k$-balanced for $k = 3, 5, 7, 11, 13, 17$. Prove that all $a_i$ are zero.

**Problem 18.268.** Find all primitive $n$-th roots of unity for (a) $n = 1$; (b) $n = 2$; (c) $n = 3$; (d) $n = 4$; (e) $n = 6$.

**Problem 18.269.** Prove that the number of primitive $n$-th roots of unity is $\varphi(n)$.

**Problem 18.270.** Find the cyclotomic polynomials $\Phi_1(x), \Phi_2(x), \Phi_3(x), \Phi_4(x), \Phi_6(x)$.

**Problem 18.271.** Prove that the degree of $\Phi_n(x)$ is even for any $n > 2$.

**Problem 18.272.** Find the degree of $\Phi_n(x)$.

**Problem 18.273.** Find $\Phi_p(x)$ if $p$ is a prime number.

**Problem 18.274.** Prove that $\prod_{d|n} \Phi_d(x) = x^n - 1$ for any positive integer $n$.

**Problem 18.275.** Prove that $\Phi_n(x)$ is a polynomial with integer coefficients.

**Problem 18.276.** Let $\Phi_n(x) = \sum_{i=0}^{\varphi(n)} a_i x^i$ for $n \geq 2$. Prove that $a_i = a_{\varphi(n)-i}$ for all $0 \leq i \leq \varphi(n)$.

**Problem 18.277.** Find $\Phi_{81}(x)$.

**Problem 18.278.** Let $n > 1$ be an odd integer. Prove that $\Phi_{2n}(x) = \Phi_n(-x)$.

**Problem 18.279.** It is a fact that $\Phi_n(x)$ is irreducible over $\mathbb{Q}$. Prove that $\Phi_p(x)$ is irreducible over $\mathbb{Z}$ if $p$ is a prime number.

**Problem 18.280.** Determine whether there exists a polynomial $P(x)$ with integer coefficients which is irreducible over $\mathbb{Z}$ but reducible over $\mathbb{Z}_p$ for every prime $p$.

**Problem 18.281.** Prove that $\Phi_n(x) = \prod_{d|n} \left(x^d - 1\right)^{\mu(n/d)}$, where $\mu$ is the Möbius function.

**Problem 18.282.** For a positive integer $n$ let $\tau(n)$ be the number of positive divisors of $n$. Prove that $\sum_{d|n} \tau(n/d)\mu(d) = 1$.

**Problem 18.283.** For a positive integer $n$, find the sum of all primitive $n$-th roots of unity.

**Problem 18.284.** Let $x$ be an integer and let $p > 3$ be a prime such that $p \mid x^2 - x + 1$. Prove that $p \equiv 1 \pmod 6$.

**Problem 18.285.** Let $a$ and $b$ be integers such that $a^4 + b^4$ is divisible by 1001. Prove that both $a$ and $b$ are divisible by 1001.

**Problem 18.286.** Let $m$ and $n$ be distinct positive integers and let $p$ be a prime with $p \nmid mn$. Prove that $\Phi_m(a)$ and $\Phi_n(a)$ cannot both be divisible by $p$ for an integer $a$.

**Problem 18.287.** Let $m$ and $n$ be distinct positive integers and let $p$ be a prime with $p \nmid mn$. Prove that $\gcd(\Phi_m(x), \Phi_n(x)) = 1$ over $\mathbb{Z}_p[x]$.

**Problem 18.288.** Prove that for any positive integer $n$ there exist infinitely many primes congruent to 1 modulo $n$.

**Problem 18.289.** Let $p$ be a prime number. Prove that there exists a prime number $q$ such that for every integer $n$ the number $n^p - p$ is not divisible by $q$. [IMO 2003]

## 18.9 Sums of two squares, Thue's lemma and quadratic integers

**Lemma 18.50.** *If $p$ is an odd prime, $a$ and $b$ are integers and $p \mid a^2 + b^2$, then either $p \equiv 1 \pmod 4$, or $p$ divides both $a$ and $b$.*

**Lemma 18.51 (Thue's lemma).** *Given a prime $p$ and an integer $a$, there exist positive integers $x, y \le \lceil \sqrt{p} \rceil$ such that $x \equiv \pm ay \pmod p$.*

**Definition 18.52 (Quadratic integers, norm, units).** Let $d$ be a squarefree integer and consider $\mathbb{Q}[\sqrt{d}] = \{a + b\sqrt{d} : a, b \in \mathbb{Q}\}$. If $d \equiv 2, 3 \pmod 4$, the "integers" of $\mathbb{Q}[\sqrt{d}]$ are the numbers $a + b\sqrt{d}$ with $a, b \in \mathbb{Z}$; if $d \equiv 1 \pmod 4$, they are the numbers $a + b \cdot \frac{-1+\sqrt{d}}{2}$ with $a, b \in \mathbb{Z}$. For $z = a + b\sqrt{d}$ one defines the *conjugate* $\bar{z} = a - b\sqrt{d}$ and the *norm* $N(z) = z\bar{z} = a^2 - db^2$; both are multiplicative. An integer $z$ is a *unit* if $z \mid 1$, which happens precisely when $N(z) = \pm 1$; two integers $x, y$ are *associate* if $x/y$ is a unit; an integer is *prime* if its only divisors are the units and its associates.

**Remark.** For $d = -1$ one obtains the Gaussian integers $a + bi$ with norm $a^2 + b^2$ and the four units $\pm 1, \pm i$. For $d = -3$ the base integers are 1 and $\omega = \frac{-1+\sqrt{-3}}{2}$, so that $\omega^2 = -\omega - 1$ and $\omega^3 = 1$; these are the Eisenstein integers $a + b\omega$, of norm $a^2 - ab + b^2$, and there are six units $\pm 1, \pm\omega, \pm(1 + \omega)$. For $d = 2$ the units are $\pm(1 + \sqrt{2})^n$, $n \in \mathbb{Z}$ — exactly the solutions of the Pell and negative Pell equations.

**Theorem 18.53 (Unique factorisation in some quadratic rings).** *In the Gaussian integers ($d = -1$) factorisation into primes is unique up to order and associates. The same is true for $d \in \{-11, -7, -3, -2, -1, 2, 3, 5, 13\}$; in particular the Eisenstein integers also have the unique factorisation property. For $d = -5$ uniqueness fails: $9 = 3 \cdot 3 = (2 + \sqrt{-5})(2 - \sqrt{-5})$.*

**Theorem 18.54 (Fermat's last theorem for $n = 3$).** *The equation $x^3 + y^3 + z^3 = 0$ has no nontrivial solutions in integers; in fact it has no nonzero solutions in the Eisenstein integers.*

**Theorem 18.55 (No four squares in arithmetic progression).** *No four distinct terms in arithmetic progression can all be squares of rational numbers.*

**Problem 18.290.** If $p$ is a prime, find the remainder when $N = \prod_{k=1}^{p-1}(k^2+1)$ is divided by $p$.

**Problem 18.291.** Prove that the equation $x^3 - 5 = y^2$ has no integer solutions.

**Problem 18.292.** Prove that the equation $x^7 + 7 = y^2$ has no integer solutions.

**Problem 18.293.** Solve the equation $x^3 - 4 = y^2$ in integers.

**Problem 18.294.** Solve the equation $x^2 + 2 = y^3$ in integers.

**Problem 18.295.** Let $p$ be a prime. (a) If $p \equiv 1 \pmod 4$, prove that $p = x^2 + y^2$ for some integers $x, y$. (b) If $p \equiv 1, 3 \pmod 8$, prove that $p = x^2 + 2y^2$ for some integers $x, y$. (c) If $p \equiv 1 \pmod 6$, prove that $p = x^2 + xy + y^2$ for some integers $x, y$.

**Problem 18.296.** Denote $\rho = 1 - \omega$, where $\omega$ is a primitive cube root of unity. If $x$ is an Eisenstein integer and $\rho \nmid x$, prove that $x^3 \equiv \pm 1 \pmod{\rho^4}$.

**Problem 18.297.** Determine whether there exist numbers $n > 15$ and $m > 15$ such that $n^2 + m^2 = 29 \cdot 41$.

**Problem 18.298.** Prove that there are infinitely many integers of the form $4m+1$ that are not a sum of two perfect squares.

**Problem 18.299.** Determine whether there exist integers $m$ and $n > 1$ such that $\dfrac{m^2+1}{n^2-1}$ is an integer.

**Problem 18.300.** Find a number that can be expressed as a sum of two squares in at least four different ways.

**Problem 18.301.** Prove that if a number can be expressed as a sum of two squares in two different ways, then that number is composite.

**Problem 18.302.** Find all integers $x, y, z$ such that $x^2y^2 = x^2 + y^2 + z^2$.

**Problem 18.303.** Factorise the number $1000009 = 235^2 + 972^2$.

**Problem 18.304.** Let $p_1 = 4m_1 + 1$, $p_2 = 4m_2 + 1$ and $p_3 = 4m_3 + 3$ be primes. Prove that the number $N = p_1(p_2p_3)^4$ can be expressed as a sum of two squares in at least five different ways.

**Problem 18.305.** Positive integers $x, y$ are such that $x^2 + 2y + 1$ and $y^2 + 2x + 1$ are both perfect squares. Prove that $x = y$.

**Problem 18.306.** Positive integers $x, y$ are such that $2xy - x^2$ and $2xy - y^2$ are both perfect squares. Prove that $x = y$.

**Problem 18.307.** Does there exist a positive integer $n \geq 3$ such that in the sum $1+2+3+\cdots+(n-1)+n$ two of the plus signs can be changed to equality signs so as to obtain a true system of equalities?

**Problem 18.308.** Determine all positive integers $n \geq 2$ such that in the sum $1+2+3+\cdots+(n-1)+n$ one of the plus signs can be changed to an equality sign so as to obtain a true equality.

**Problem 18.309.** Prove that the equations (a) $x^4+y^4=z^2$ and (b) $x^4+y^2=z^4$ have no solutions in nonzero integers.

## 18.10 Polynomials in number theory

**Lemma 18.56.** *Let $a, b, d$ be integers and let $P(x)$ be a polynomial with integer coefficients. If $a \equiv b \pmod{d}$, then $P(a) \equiv P(b) \pmod{d}$.*

**Lemma 18.57 (Schur).** *Let $P(x)$ be a non-constant polynomial with integer coefficients. Then there exist infinitely many primes $p$ such that $p$ divides $P(n)$ for some integer $n$.*

**Problem 18.310.** A non-constant function $f : \mathbb{N} \to \mathbb{N}$ is such that $a-b$ divides $f(a)-f(b)$ whenever $a \neq b$. Prove that there are infinitely many primes $p$ such that $p \mid f(x)$ for some $x$.

**Problem 18.311.** Determine all polynomials $P(x)$ with integer coefficients such that $P(P(n)+n)$ is a (positive or negative) prime number for infinitely many integers $n$. [Canada 2016]

**Problem 18.312.** Is it true that for any polynomial $P(x)$ with real coefficients of degree 2023 there exists a positive integer $n$ such that the equation $P(x) = n^{-100}$ has no rational root? [Bulgarian Festival of Young Mathematicians 2023, Senior Group]

**Problem 18.313.** A set $S$ of two or more positive integers is called *almost closed under addition* if the sum of any two distinct elements of $S$ also belongs to $S$. Let $P(x)$ be a polynomial with integer coefficients for which there exists an almost closed under addition set $S$ such that for any two distinct $a, b \in S$ the integers $P(a)$ and $P(b)$ are coprime. Prove that $P$ is constant. [Iran 2016 National Round]

**Problem 18.314.** Let $P(x)$ be a polynomial with integer coefficients. Suppose that for every integer $N$ there exists a non-negative integer $k$ such that the $k$-th iterate $P^k(1) = P(P(\cdots P(1)\cdots))$ is divisible by $N$. Prove that $P(x) = x+1$. [Iran IMO 2004 Team Selection Test]

**Problem 18.315.** Determine all polynomials $P(x)$ with integer coefficients such that there exists an integer $N$ with the property that for all integers $n \geq N$ every prime divisor of $n+2^{\omega(n)}$ is also a prime divisor of $P(n)$. Here $\omega(n)$ denotes the number of distinct prime factors of $n$. [Bozhidar Dimitrov]

**Problem 18.316.** Determine all polynomials $P(x)$ with integer coefficients such that there exists an integer $N$ with the property that for all integers $n \geq N$ every prime divisor of $n+2^{\lfloor\sqrt{n}\rfloor}$ is also a prime divisor of $P(n)$. [Bulgaria Autumn Mathematical Tournament 2024, Grade 10]

**Problem 18.317.** Determine all polynomials $P(x)$ with integer coefficients for which there exists an integer $N$ such that $P(n)$ divides $n!$ for all integers $n \geq N$. [Autumn Mathematical Tournament 2012, Grades 11–12]

**Problem 18.318.** ★ Determine all polynomials $P(x)$ with integer coefficients such that $P(0) \neq 0$ and $P^n(m) \cdot P^m(n)$ is a perfect square of an integer for all nonnegative integers $m, n$. Here $P^m(x) = P(P^{m-1}(x))$ for $m \geq 2$. [European Mathematical Cup 2017, Senior Group]

**Problem 18.319.** Suppose that $a_1, \ldots, a_k$ are nonzero integers and $P(x)$ is a polynomial with integer coefficients such that $P(n)$ divides $a_1^n + \cdots + a_k^n$ for every $n \in \mathbb{N}$. Prove that $P(x)$ is a constant polynomial.

**Problem 18.320.** We call a prime $p$ a *nondivisor* of a polynomial $P(x)$ with integer coefficients if there is no integer $x$ with $p \mid P(x)$. Is there a polynomial $P(x)$ with integer coefficients that has exactly one nondivisor?

**Problem 18.321.** Find all polynomials $f$ with integer coefficients such that $f(p)$ divides $2^p - 2$ for every prime $p > 2$.

**Problem 18.322.** Determine all nonconstant polynomials $f(x)$ with integer coefficients such that $f(n)$ divides $f(2n) - 2f(n)$ for all sufficiently large positive integers $n$.

**Problem 18.323.** Let $P(x)$ be a nonconstant polynomial with integer coefficients. For a positive integer $a$, define $a_0 = a$ and $a_k = P(a_{k-1})$ for $k = 1, 2, \ldots$. Assume that for every positive integer $m$ the sequence contains a perfect $m$-th power greater than 1. Prove that $P(x)$ is a linear polynomial.

**Problem 18.324.** Let $a$ and $b$ be positive integers with $a > 2b$ and $b > 1$. Prove that there is a non-constant polynomial $P(x)$ with all coefficients in $\{0, 1, \ldots, b-1\}$ such that $P(a)$ is divisible by $P(b)$.

**Problem 18.325.** Let $P(x)$ be a non-constant monic polynomial with integer coefficients and let $a_1, a_2, \ldots$ be an infinite sequence of positive integers. Prove that there are infinitely many primes each of which divides at least one term of the sequence $P(n)^{a_n} + 1$. [Bulgaria National Round 2025 P3]

**Problem 18.326.** ★ Let $n$ be a positive integer and let $P_1, P_2, \ldots, P_n$ be polynomials with integer coefficients, each of degree at least 2. Let $S$ be the set of all positive integers $N$ for which there exist a positive integer $a$ and an index $1 \leq i \leq n$ with $P_i(a) = N$. Prove that there are infinitely many primes that do not belong to $S$. [Bulgarian Festival of Young Mathematicians 2022, Senior Level]

**Problem 18.327.** ★ Determine all polynomials $P(x)$ with integer coefficients for which there exist non-constant polynomials $Q(x)$ and $R(x)$ with non-negative integer coefficients and a sequence $(a_n)_{n \geq 1}$ of integers greater than 1 such that $a_n^2 \leq n$ and $R(n)$ divides $P\left(a_n^{Q(n)}\right)$ for every integer $n \geq 2025$.† [Bulgaria RMM TST 2025]

**Problem 18.328.** Let $p$ be a prime and let $P(x)$ be an irreducible polynomial with integer coefficients. Suppose that there is no integer $d > 1$ such that $d \mid P(x)$ for every integer $x$. Prove that there is a positive integer $x$ such that $P(x)$ and $P(x+1)$ are coprime.

## 18.11 Advanced Diophantine equations

**Remark.** Several problems in this section are most cleanly finished with Mihailescu's theorem (formerly Catalan's conjecture): if $x, y, a, b > 1$ are integers with $x^a - y^b = 1$, then necessarily $x = b = 3$ and $a = y = 2$.

**Problem 18.329.** A positive integer is said to have property $P$ if it can be represented as $\dfrac{p^a - p^b}{p^c - p^d}$ for some prime $p$ and positive integers $a, b, c, d$. (a) Prove that there are infinitely many positive integers which have property $P$. (b) Prove that there are infinitely many positive integers which do not have property $P$. [Kvant M2355; Bulgarian Festival of Young Mathematicians 2025]

**Problem 18.330.** Determine all pairs $(a, n)$ of positive integers such that $a$ is even and $\dfrac{a^{\varphi(n)} - 1}{n}$ is a perfect square. [Bulgarian Festival of Young Mathematicians 2025, Senior Division]

**Problem 18.331.** Prove that $2x^{2^k} + 1$ is not a perfect square for any integers $x \neq 0$ and $k \geq 2$. [Lemma from Bulgaria National Round 2015]

**Problem 18.332.** Determine the smallest positive integer $n$ such that no multiple of $n$ can be written as a sum of exactly 100 (not necessarily distinct) powers of 2; here $1 = 2^0$ also counts as a power of 2. [APMO 2016 P2]

**Problem 18.333.** Let $a, b$ be (not necessarily positive) integers and let $k \geq 2$ be an integer. Prove that there are infinitely many positive integers which cannot be written as $ax^k + by^k$ for integers $x, y$. [IMC 2010]

**Problem 18.334.** Let $x, y, z, t$ be positive integers such that $xy - zt = x + y = z + t$. Prove that the product $xyzt$ cannot be a perfect square. [IMO Shortlist 2018 N5, generalization]

**Problem 18.335.** Let $P(x)$ be a polynomial with integer coefficients. Prove that there are only finitely many pairs $(m, n)$ of positive integers such that $2^n + P(n) = m!$. [Bulgarian Festival of Young Mathematicians 2024, Senior Division]

**Problem 18.336.** Determine all triples $(a, b, p)$ of positive integers with $p$ prime and $a^p = b! + p$. [IMO 2022 P5]

**Problem 18.337.** Find all positive integers $n$ and $d$ with $d \mid n$ such that $n^2 + d^2$ is divisible by $nd + 1$.

**Problem 18.338.** Suppose that each of the positive integers $a, b, c, d$ is divisible by $ad - bc$. Find all possible values of $ad - bc$.

**Problem 18.339.** Let $n > 1$ be an integer and let $d > \sqrt{n}$ be a divisor of $n^2 + 1$. Prove that $d \geq n + \sqrt{n}$.

**Problem 18.340.** Suppose that $a$ and $b$ are positive integers such that $a^2 + b^2 + a$ is divisible by $ab$. Prove that $a$ is a perfect square.

**Problem 18.341.** If $x, y, z$ are rational numbers such that $xyz = 1$ and both $x + y + z$ and $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ are integers, prove that $|x| = |y| = |z| = 1$.

**Problem 18.342.** Find all pairs of different rational numbers $x$ and $y$ such that $x^y = y^x$.

**Problem 18.343.** Determine all primes $p > 2$ such that both $\frac{p+1}{2}$ and $\frac{p^2+1}{2}$ are perfect squares.

**Problem 18.344.** Find all pairs $(m, n)$ of positive integers such that $mn - 1$ divides $n^3 + 1$.

**Problem 18.345.** Find all positive integers $n$ that cannot be written as $n = \operatorname{lcm}(a, b) + \operatorname{lcm}(b, c) + \operatorname{lcm}(c, a)$ for some $a, b, c \in \mathbb{N}$.

**Problem 18.346.** Find all integers $x$ and $y$ that satisfy $x^2 + x = y^3 + y^2 + y$.

**Problem 18.347.** Find all primes $p$ such that $\dfrac{p^2 - p - 2}{2}$ is a perfect cube.

**Problem 18.348.** Find all positive integers $a, b, c$ such that $2^a + 3^b = 5^c$.

**Problem 18.349.** Find all pairs of nonnegative integers $(x, y)$ satisfying $2^x = 5^y + 3$.

**Problem 18.350.** Solve the equation $4^a + 5^b + 6^c = 7^d$ in nonnegative integers.

**Problem 18.351.** Find all pairs of primes $(p, q)$ such that $pq \mid p^q + q^p + 7$.

**Problem 18.352.** Find all pairs $(k, n)$ of positive integers such that $7^k - 3^n$ divides $k^4 + n^2$.

**Problem 18.353.** Find all primes $p$ such that there exist positive integers $n, x, y$ with $p^n = x^3 + y^3$.

**Problem 18.354.** Find all positive integers $a$ and $b$ such that $(a - b)^{ab} = a^b b^a$.

**Problem 18.355.** Find all natural numbers $a$ and $b$ such that $a \mid b^2$, $b \mid a^2$ and $a + 1 \mid b^2 + 1$.

**Problem 18.356.** Prove that for each positive integer $n$ there are pairwise relatively prime integers $k_0, k_1, \ldots, k_n$, all greater than 1, such that $k_0 k_1 \cdots k_n - 1$ is the product of two consecutive integers.

**Problem 18.357.** Let $k$ be a fixed positive integer. Let $F_k(n)$ be the smallest positive integer greater than $kn$ such that $F_k(n)\, n$ is a perfect square. Prove that $F_k(n) = F_k(m)$ implies $m = n$.

**Problem 18.358.** ★ Find all positive integers $m, n \geq 2$ such that (a) $m + 1$ is a prime of the form $4k - 1$, and (b) there exist a prime $p$ and a nonnegative integer $a$ with $\dfrac{m^{2n-1} - 1}{m - 1} = m^n + p^a$.

**Problem 18.359.** ★ Find all pairs of nonnegative integers $(x, y)$ such that $(2^{2015} + 1)^x + 2^{2015} = 2^y + 1$.

**Problem 18.360.** Find all nonnegative integer solutions $(p, a, b, m)$ of $p^2 + 4^a 9^b = m^2$, where $p$ is a prime.

**Problem 18.361.** ★ Find all natural numbers $n$ for which there exists an infinite subset $A$ of $\mathbb{N}$ such that for each $a \in A$, $a^n + a^{n-1} + \cdots + a + 1$ divides $a^{n!} + a^{(n-1)!} + \cdots + a^{1!} + 1$.

**Problem 18.362.** Do there exist positive integers $a, b, c$ and $n \geq 2$ such that $a^n - 2b^n = 1$ and $b^n - 2c^n = 1$?

**Problem 18.363.** Do there exist positive integers $a, b, c$, all greater than $10^{10}$, such that $abc$ is divisible by each of $a + 2022$, $b + 2022$ and $c + 2022$?

**Problem 18.364.** Are there integers $a, b, c$, all greater than 2022, that satisfy $a^3 + 2b^3 + 4c^3 = 6abc + 1$?

**Problem 18.365.** Find all triples of positive integers $a, b, c$ such that $ab+bc+ca = 4\,\mathrm{lcm}(a, b, c)$.

**Problem 18.366.** Find all positive integers $n$ for which one can find several (at least two) positive rational numbers $a_1, a_2, \ldots, a_k$ such that $a_1 + a_2 + \cdots + a_k = a_1 a_2 \cdots a_k = n$.

**Problem 18.367.** Find all pairs of positive rational numbers $x, y$ satisfying $y^{xy} = y + 1$.

**Problem 18.368.** Positive integers $a, b, c, d$ and $n$ are such that $a+c < n$ and $\frac{a}{b} + \frac{c}{d} < 1$. Prove that $\frac{a}{b} + \frac{c}{d} < 1 - \frac{1}{n^3}$.

**Problem 18.369.** Find all real numbers $\alpha$ with the following property: there exist a real number $r > \alpha$ and an irrational number $x$ such that both $x^2 - rx$ and $x^3 - rx$ are rational.

**Problem 18.370.** Given a positive integer $n$, prove that there are at most two pairs of positive integers $(a, b)$ such that $a + b$ is a power of 2 and $a^2 + b = n$.

**Problem 18.371.** Find all triples $(x, y, z)$ of nonnegative integers such that $x^3 + 2y^3 = 4z^3$.

**Problem 18.372.** Find all pairs $(x, y)$ of positive integers such that $y + x^2 + \gcd(x, y)^3 = xy \gcd(x, y)$. [IMO Shortlist 1995 N4; Russia IMO TST 1997]

**Problem 18.373.** Determine all positive integers $a$ for which there exists a positive integer $b$ such that $b$ divides $a - 1$ and $a^2 + 2ab$ is a perfect square. [Easier version of a problem by Kiran Kedlaya]

**Problem 18.374.** The positive integers $a, b, c, d$ are such that $(a+c)(b+d) = (ab-cd)^2$. Prove that $4ad + 1$ and $4bc + 1$ are perfect squares of positive integers. [AoPS]

**Problem 18.375.** For a positive integer $n$ denote by $T_n$ the set of all primes $p$ such that there exist positive integers $a$ and $b$ with $n = a^p + b^p$. Is there a finite set of primes which is distinct from every $T_n$? [All-Russian Olympiad 2008, Grade 11]

**Problem 18.376.** Given a finite set $P$ of prime numbers, prove that there exists a positive integer $x$ which can be written in the form $a^p + b^p$ (with $a, b$ positive integers) for each $p \in P$, and cannot be written in that form for any $p$ not in $P$. [Russia 2008]

**Problem 18.377.** Find all positive integers $a, b, c$ such that $2^{a!} + 2^{b!} + 2^{c!}$ is a perfect cube.

**Problem 18.378.** Do there exist integers $x, y, t$ such that $(x^2 + y)(y^2 + t)(t^2 + x) = 2015^{2015}$?

**Problem 18.379.** Find all integers $a, b, c, d$ such that $a^5 + b^5 + c^5 + 5 = d^5$.

**Problem 18.380.** Let $a$ and $b$ be two distinct positive integers such that $ab(a + b)$ is divisible by $a^2 + ab + b^2$. Prove that $|a - b|^3 > ab$.

**Problem 18.381.** For a given positive integer $n$, find all pairwise different positive integers $x_1, x_2, \ldots, x_n$ such that $x_1^3 + \cdots + x_n^3 = (x_1 + \cdots + x_n)^2$.

**Problem 18.382.** Solve $a^b + 1 = (a + 1)^c$ in positive integers.

**Problem 18.383.** Find all nonnegative integers $x$ and $y$ such that $(xy-7)^2 = x^2+y^2$.

**Problem 18.384.** Solve the equation $(x+2)^4 - x^4 = y^3$ in integers.

**Problem 18.385.** Positive integers $x$ and $y$ are such that $2x^2 - 1 = y^{15}$. Prove that if $x > 1$ then $x$ is divisible by 5.

**Problem 18.386.** Find all triples of primes $p, q, r$ such that $p^2 + pq + q^2 = r^2$.

**Problem 18.387.** Find all primes $p$ and $q$ such that $p^3 - q^5 = (p+q)^2$.

**Problem 18.388.** Find all positive integers $m, n, k$ such that $m + n = \gcd(m,n)^2$, $m + k = \gcd(m,k)^2$ and $n + k = \gcd(n,k)^2$.

**Problem 18.389.** Numbers $m$ and $n$ satisfy $\operatorname{lcm}(m,n) + \gcd(m,n) = m + n$. Prove that one of the numbers $m$, $n$ is divisible by the other.

**Problem 18.390.** Show that for any $n$ the number $2^{2n} - 1$ has at least $n - 1$ distinct positive divisors.

**Problem 18.391.** There is a sequence of positive integers $a_i$ such that $\gcd(a_i, a_j) = \gcd(i,j)$ for all $i \neq j$. Does it follow that $a_i = i$ for all $i$?

**Problem 18.392.** Let $a$ and $b$ be integers greater than 2. Prove that there exist a positive integer $k$ and a finite sequence $n_1, n_2, \ldots, n_k$ of positive integers such that $n_1 = a$, $n_k = b$ and $n_i n_{i+1}$ is divisible by $n_i + n_{i+1}$ for each $i$ with $1 \leq i < k$.

**Problem 18.393.** Let $n$ be a positive integer. Prove that there are at most $\dfrac{n(n+1)}{2}$ different remainders of the numbers $x^n + y^n$, where $x$ and $y$ are positive integers, upon division by $n^2$.

**Problem 18.394.** Determine whether there exists a positive integer $a$ such that for every positive integer $k$ at least one of the numbers $a^k + 1$ and $a^k - 1$ is of the form $x^y$ with $x, y$ positive integers and $y > 1$.

## 18.12 Master problem sets

The remaining banks are grouped by theme rather than by technique; each of them mixes several of the tools developed above.

### 18.12.1 Functional equations with divisibility

**Problem 18.395.** Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that, whenever $a_1 + a_2 + \cdots + a_n$ is a square (for some $n$), $f(a_1) + f(a_2) + \cdots + f(a_n)$ is also a square.

**Problem 18.396.** Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that $\gcd(f(x), f(y)) = \gcd(x,y)$ holds whenever $x$ and $y$ are different positive integers.

**Problem 18.397.** Does there exist a bijective map $f : \mathbb{N} \to \mathbb{N}$ such that for any two distinct positive integers $x$ and $y$ it holds that (a) $100 \mid xf(y) + yf(x)$; (b) $100 \mid xf(x) + yf(y)$?

**Problem 18.398.** A function $f : \mathbb{Z} \to \mathbb{N}$ is such that $f(m) - f(n)$ is divisible by $f(m-n)$ for all integers $m$ and $n$. Prove that $f(n)$ is divisible by $f(m)$ whenever $f(m) \leq f(n)$.

**Problem 18.399.** A function $f : \mathbb{N} \to \mathbb{N}$ is such that $(m^2+n)^2$ is divisible by $f(m)^2+f(n)$ for all $m, n \in \mathbb{N}$. Prove that $f(n) = n$ for all $n \in \mathbb{N}$.

**Problem 18.400.** Find all surjective functions $f : \mathbb{N} \to \mathbb{N}$ such that for every $m, n \in \mathbb{N}$ the numbers $f(m)+f(n)$ and $f(m+n)$ have the same set of prime divisors.

**Problem 18.401.** ★ Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that (i) $f(x)$ has exactly $x$ divisors for all $x \in \mathbb{N}$, and (ii) $f(xy)$ divides $(x-1)y^{xy-1}f(x)$ for all $x, y \in \mathbb{N}$.

**Problem 18.402.** ★ Find all surjective functions $f : \mathbb{N} \to \mathbb{N}$ such that for every $m, n \in \mathbb{N}$ and every prime $p$ we have $p \mid f(m+n)$ if and only if $p \mid f(m)+f(n)$. [IMO Shortlist 2007]

**Problem 18.403.** Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that for every $m, n \in \mathbb{N}$ and every prime $p$ we have $p \mid m+n$ if and only if $p \mid f(m)+f(n)$.

**Problem 18.404.** ★ Let $\sigma(x)$ denote the sum of divisors of $x$, including 1 and $x$. For every $n \in \mathbb{N}$ define $f(n)$ as the number of positive integers $m \le n$ for which $\sigma(m)$ is odd. Prove that there are infinitely many $n$ with $f(n) \mid n$.

**Problem 18.405.** ★ Determine all functions $f$ from the set of nonnegative integers to itself such that $f(a+b) = f(a)+f(b)+f(c)+f(d)$ whenever $a, b, c, d$ are nonnegative integers satisfying $2ab = c^2+d^2$.

**Problem 18.406.** Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that $f(n)+m! \mid n+f(m)!$ for all $m, n \in \mathbb{N}$.

**Problem 18.407.** ★ Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that $f(f(m)^2+2f(n)^2) = m^2+2n^2$ for all $m, n \in \mathbb{N}$.

**Problem 18.408.** Find all $f : \mathbb{N} \to \mathbb{N}$ such that $f(f(n)) < f(n+1)$ for every $n \in \mathbb{N}$. [IMO 1977 P6]

**Problem 18.409.** Find all $f : \mathbb{N} \to \mathbb{N}$ such that $f(n+1) > \dfrac{f(n)+f(f(n))}{2}$ for every $n \in \mathbb{N}$. †

**Problem 18.410.** Find all $f : \mathbb{N} \to \mathbb{N}$ such that, for every $x, y \in \mathbb{N}$, the numbers $x$, $f(y)$ and $f(y+f(x)-1)$ are the side lengths of some triangle. [IMO 2009 P5]

**Problem 18.411.** Find all surjective functions $f : \mathbb{N} \to \mathbb{N}$ such that $f(n) \ge n+(-1)^n$ for every $n \in \mathbb{N}$.

**Problem 18.412.** Find all functions $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ such that (i) $f(x, y) = f(y, x)$ for every $x, y \in \mathbb{N}$; (ii) $f(x, x) = x$ for every $x \in \mathbb{N}$; (iii) $(y-x)f(x, y) = yf(x, y-x)$ for every $x, y \in \mathbb{N}$ with $y > x$.

**Problem 18.413.** Find all $f : \mathbb{N} \to \mathbb{N}$ such that $f(f(f(n))) = f(n+1)+1$ for every $n \in \mathbb{N}$. [IMO Shortlist 2013]

**Problem 18.414.** Find all increasing functions $f : \mathbb{N} \to \mathbb{R}$ such that $f(mn) = f(m)f(n)$ whenever $\gcd(m, n) = 1$.

**Problem 18.415.** Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that $\operatorname{lcm}(m, f(m+f(n))) = \operatorname{lcm}(f(m), f(m)+n)$.

**Problem 18.416.** Find all functions $f : \mathbb{N} \to \mathbb{N}$ such that for all positive integers $m, n$ the number $f(m) + n - m$ is divisible by $f(n)$.

**Problem 18.417.** Find all nondecreasing functions $f : \mathbb{N} \to \mathbb{N}$ such that, for every $n$, both $f(n) + n + 1$ and $f(f(n)) - f(n)$ are perfect squares.

**Problem 18.418.** A function $f : \mathbb{N} \to \mathbb{N}$ is such that $f(ab) \mid \max\{f(a), b\}$ for all $a$ and $b$. Is it necessarily true that $f(x) = 1$ holds for infinitely many $x$?

**Problem 18.419.** ★ Denote $x!_2 = (x!)!$. Find all functions $f : \mathbb{N} \to \mathbb{N}$ that satisfy $x!_2 + y!_2 \mid f(x)!_2 + f(y)!_2$ for all $x, y$.

**Problem 18.420.** Determine all functions $f$ defined on the natural numbers and taking natural values for which $(f(n))^p \equiv n \pmod{f(p)}$ for all $n \in \mathbb{N}$ and all primes $p$. [Canada 2008]

**Problem 18.421.** Determine all functions $f : \mathbb{Z}_{\geq 10^{100}} \to \mathbb{Z}_{>0}$ such that $mn + 1$ divides $f(m)f(n) + 1$ for any integers $m, n \geq 10^{100}$ and such that there exists a polynomial $P$ with integer coefficients with $f(n) \leq P(n)$ for all $n \geq 10^{100}$. [Bulgaria Winter Mathematical Competition 2025, Grade 12]

**Problem 18.422.** Let $f : \mathbb{Z}_{>0} \times \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ be a function such that $f(a, b) + f(b, c) = f(ac, b^2) + 1$ for any positive integers $a, b, c$. Assume also that there exists a positive integer $n$ such that $f(n, m) \leq f(n, m + 1)$ for all positive integers $m$. Determine all possible values of $f(2025, 2025)$. [Bulgaria Winter Mathematical Competition 2025, Grade 10]

**Problem 18.423.** ★ Determine all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that for any integers $a, b, c$ the sets of distinct prime divisors of $ab + bc + ca$ and of $f(a)f(b) + f(b)f(c) + f(c)f(a)$ coincide. [Malaysia IMO TST 2022]

**Problem 18.424.** ★ Fix a prime $p$ and let $f : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ satisfy: $f(mn) = f(m)f(n)$ for any coprime positive integers $m, n$; and there exists a constant $c \neq 0$ such that $\mathrm{rad}_p(n)$ divides $f(n + 1) - c$ for all positive integers $n$, where $\mathrm{rad}_p(n)$ is the product of all distinct prime factors of $n$ other than $p$. Prove that there exists a nonnegative integer $r$ such that $f(n) = n^r$ for all positive integers $n$. [Bulgaria RMM TST 2023, Navid Safaei]

### 18.12.2 Digits and representations

**Problem 18.425.** Prove that there are only finitely many powers of 2 whose sum of digits does not exceed 2020.

**Problem 18.426.** Denote by $S(n)$ the sum of decimal digits of $n$. Prove that there are infinitely many positive integers $k$ such that $S(2^k) \geq S(2^{k+1})$.

**Problem 18.427.** Prove that the number of integers $n = \overline{a_k \ldots a_1 a_0}$ with $k \geq 10$ that satisfy $n = \overline{a_k \ldots a_{10}}^2 + \overline{a_9 \ldots a_0}^2$ is even.

**Problem 18.428.** A 22-digit number contains no decimal digit 9. Prove that one can choose two equal digits and increase them both by 1 so that the obtained number is not divisible by 7.

**Problem 18.429.** Can the number obtained by writing the numbers $1, 2, \ldots, n$ one after another be a palindrome for some $n > 1$?

**Problem 18.430.** Prove that every positive integer $n > 1$ has a multiple less than $n^4$ whose decimal expansion contains at most four distinct digits.

**Problem 18.431.** We call a positive integer *alternate* if its decimal digits are alternately odd and even. Find all positive integers $n$ that have an alternate multiple.

**Problem 18.432.** Denote by $S(x)$ the sum of decimal digits of a positive integer $x$. Prove that there exist 50 distinct positive integers $n_1, n_2, \ldots, n_{50}$ such that $n_1 + S(n_1) = n_2 + S(n_2) = \cdots = n_{50} + S(n_{50})$.

**Problem 18.433.** Prove that for every $n \in \mathbb{N}$ there exists an $n$-digit number all of whose digits are 1, 2 or 3 and which is divisible by its sum of digits.

**Problem 18.434.** Denote by $o(x)$ the number of odd digits in the decimal representation of $x$. Evaluate the series $\sum_{n\geq 1} \dfrac{o(2^n)}{2^n}$.

**Problem 18.435.** Prove that there are only finitely many powers of 2 whose sum of digits is 2024. †

**Problem 18.436.** ★ Prove that, in the decimal representation of $\sqrt[3]{3}$, there is a digit different from 2 on some position between the 1 000 000-th and the 3 141 592-nd place after the decimal point.

**Problem 18.437.** Let $m$ and $n$ be perfect 17th powers with 15 062 017 and 15 062 017 000 digits, respectively. Can the number obtained by writing $m$ and $n$ one after the other be a perfect 17th power?

**Problem 18.438.** Let positive integers $K$ and $d$ be given. Prove that there exists a positive integer $n$ and a sequence of $K$ positive integers $b_1, b_2, \ldots, b_K$ such that $n$ is a $d$-digit palindrome in all number bases $b_1, b_2, \ldots, b_K$. [BMO Shortlist 2016]

**Problem 18.439.** ★ Let the function rot map the digits $0, 1, 2, 5, 6, 8, 9$ to $0, 1, 2, 5, 9, 8, 6$, respectively. A positive integer $n = \overline{t_k t_{k-1} \ldots t_1 t_0}$ is called *rotatable* if all of its digits belong to $\{0, 1, 2, 5, 6, 8, 9\}$ with $t_0 \neq 0$, and we set $\mathrm{rot}(n) = \overline{\mathrm{rot}(t_0)\mathrm{rot}(t_1)\ldots\mathrm{rot}(t_k)}$ (rotating a calculator display by 180°). Prove that there exist infinitely many positive integers $n$ such that $n$ is rotatable with $\mathrm{rot}(n) = n$, $n^2$ is rotatable with $\mathrm{rot}(n^2) = n^2$, and $41 \mid n$.

**Problem 18.440.** ★ Let the function mirror map the digits $0, 1, 2, 5, 8$ to $0, 1, 5, 2, 8$, respectively. A positive integer $n = \overline{t_k t_{k-1} \ldots t_1 t_0}$ is called *mirrorable* if all of its digits belong to $\{0, 1, 2, 5, 8\}$ with $t_0 \neq 0$, and we set $\mathrm{mirror}(n) = \overline{\mathrm{mirror}(t_0)\mathrm{mirror}(t_1)\ldots\mathrm{mirror}(t_k)}$. Find all positive integers $n$ such that $n$ is mirrorable with $\mathrm{mirror}(n) = n$ and $n^2$ is mirrorable with $\mathrm{mirror}(n^2) = n^2$.

**Problem 18.441.** Let $f(x)$ denote the sum of digits of a positive integer $x$. Prove that for each positive integer $n$ there exists a positive integer $p$ such that the equation $f(n^p x) = f(x)$ has a solution $x$ that does not contain the digit 9 among its digits.

**Problem 18.442.** Initially a positive integer $n$ is written on a blackboard. We repeatedly alter the number in one of two ways: multiply it by a positive integer, or reorder its digits. (For example $43 \to 301 \to 013 = 13 \to 52 \to 25 \to 100 \to 001 = 1$.) Find all numbers $n$ for which it is possible to obtain the number 1 after finitely many such moves.

**Problem 18.443.** How many positive integers $x$ are there for which each of the fractional parts $\{x/10^k\}$, $k = 1, 2, \ldots, 9$, is strictly larger than $x/10^{10}$?

**Problem 18.444.** Let $a_1, a_2, \ldots$ be an infinite sequence of nonzero digits and define $b_n = \overline{a_1 a_2 \ldots a_n}$. Prove that there are infinitely many primes each of which divides at least one $b_n$. [Saint Petersburg 2007, Grade 11]

**Problem 18.445.** ★ Does there exist an infinite sequence of nonzero digits $a_1, a_2, a_3, \ldots$ and an integer $N$ such that for every integer $k > N$ the number $\overline{a_k a_{k-1} \ldots a_1}$ is a perfect square? [IMO Shortlist 2013 N4]

**Problem 18.446.** Prove that every integer can be uniquely written in the form $a_0 + a_1(-4) + a_2(-4)^2 + \cdots + a_k(-4)^k$ for some integer $k \geq 0$ and $a_0, a_1, \ldots, a_k \in \{0, 1, 2, 3\}$.

### 18.12.3 Sequences

**Problem 18.447.** Find all pairs $(a_1, a_2)$ of positive integers such that the sequence $(a_n)_{n \geq 1}$ with $a_{n+2} = |3a_{n+1} - 2a_n|$ for all $n \geq 1$ has only finitely many terms that are not primes. [Bulgaria National Round 2002]

**Problem 18.448.** Let $p$ and $q$ be primes and let $\{a_n\}$ be the sequence of integers defined by $a_0 = 0$, $a_1 = 1$ and $a_{n+2} = pa_{n+1} - qa_n$ for all $n \geq 0$. Find all possible values of $p$ and $q$ for which there exists an integer $k$ with $a_{3k} = -3$. [Bulgaria Autumn Mathematical Tournament 2007, Grade 12]

**Problem 18.449.** Let $c$ be a positive integer. The sequence $a_1, a_2, \ldots$ is defined by $a_1 = c$ and $a_{n+1} = a_n^2 + a_n + c^3$ for all positive integers $n$. Determine all values of $c$ for which there are integers $k \geq 1$ and $m \geq 2$ such that $a_k^2 + c^3$ is the $m$-th power of an integer. [Balkan Mathematical Olympiad 2008 P4]

**Problem 18.450.** ★ Let $a_1 < a_2 < a_3 < \cdots$ be a sequence of positive integers such that $a_{k+1}$ divides $2(a_1 + a_2 + \cdots + a_k)$ for every positive integer $k$. Suppose that for infinitely many primes $p$ there exists $k$ such that $p$ divides $a_k$. Prove that for every positive integer $n$ there exists $k$ such that $n$ divides $a_k$. [IMO Shortlist 2023 N5]

**Problem 18.451.** ★ A sequence of integers $a_0, a_1, \ldots$ is called *kawaii* if $a_0 = 0$, $a_1 = 1$ and $(a_{n+2} - 3a_{n+1} + 2a_n)(a_{n+2} - 4a_{n+1} + 3a_n) = 0$ for all integers $n \geq 0$. An integer is called kawaii if it belongs to some kawaii sequence. Suppose that two consecutive integers $m$ and $m + 1$ are both kawaii (not necessarily belonging to the same kawaii sequence). Prove that $m$ is divisible by 3 and that the integer $m/3$ is also kawaii. [IMO Shortlist 2023 N6]

**Problem 18.452.** For $x \in \mathbb{N}$ denote by $p(x)$ the smallest prime not dividing $x$, and by $q(x)$ the product of all primes less than $p(x)$ (with $q(x) = 1$ if $p(x) = 2$). Define $x_0 = 1$ and $x_{n+1} = \dfrac{x_n p(x_n)}{q(x_n)}$ for $n \geq 0$. Find all $n$ for which $x_n = 1111$.

**Problem 18.453.** The sequence $(x_n)$ is defined by $x_0 = 1$ and $x_{i+1} = x_i + y_i + 1$ for $i \geq 0$, where $y_i$ is the smallest positive integer not dividing $x_i$. Does the number 20! occur in this sequence?

**Problem 18.454.** Given an integer $n \geq 2$, find the smallest value of $m$ for which there exist positive integers $a_1 < a_2 < \cdots < a_n = m$ such that $\frac{1}{2}(a_i^2 + a_{i+1}^2)$ is a perfect square for each $i = 1, 2, \ldots, n-1$.

**Problem 18.455.** ★ The sequence $a_1 = 1, a_2, a_3, \ldots$ is defined as follows: for $n \geq 2$, $a_n$ is the smallest positive integer not occurring among $a_1, \ldots, a_{n-1}$ such that $n \mid a_1 + a_2 + \cdots + a_n$. Prove that $a_{a_n} = n$.

**Problem 18.456.** For a positive integer $k$, define the sequence $(x_n)$ by $x_0 = -1$, $x_1 = 1$ and $x_{n+2} = kx_{n+1} - x_n$. Prove that there are infinitely many values of $k$ for which this sequence contains no prime.

**Problem 18.457.** Let $n$ be a nonnegative integer. Set $a_0 = n$ and define $a_k$ $(k \geq 1)$ to be the smallest integer greater than $a_{k-1}$ for which $a_{k-1} + a_k$ is a square. Prove that there are exactly $\left\lfloor \sqrt{2n} \right\rfloor$ positive integers that cannot be written as a difference of two terms of the sequence $(a_i)_{i \geq 0}$.

**Problem 18.458.** By $\operatorname{rad}(x)$ we denote the product of all distinct prime factors of a positive integer $x$. A sequence $(a_n)$ is defined by $a_0 = a \in \mathbb{N}$ and $a_{n+1} = a_n + \operatorname{rad}(a_n)$. Prove that there exists an index $n$ for which $\dfrac{a_n}{\operatorname{rad}(a_n)} = 2022$.

**Problem 18.459.** Given a rational number $x_1 > 1$, a sequence is defined inductively by $x_{n+1} = \dfrac{nx_n}{\lfloor nx_n \rfloor}$. Show that there exists $k$ for which $x_k = 1$.

**Problem 18.460.** Let $a_1 > 1$ be a rational number. For $n \geq 1$ define $a_{n+1} = a_n + \dfrac{1}{\lfloor a_n \rfloor}$. Prove that $a_n$ is an integer for some $n \in \mathbb{N}$.

**Problem 18.461.** Define $x_1 = 2$, $x_2 = 7$ and $x_{n+1} = 4x_n - x_{n-1}$. Find all $n$ for which $x_n$ is a perfect square.

**Problem 18.462.** Find all real constants $c$ for which there is a function $f : \mathbb{N} \to \mathbb{N}$ satisfying $|f(x) - cx| < 1000$ for all $x$, and such that $f(x) - f(y)$ is divisible by $x - y$ if and only if $|x - y| = 1$.

**Problem 18.463.** Let $a$ and $b$ be given positive integers. Given the initial values $x_1$ and $x_2$, the sequence $(x_n)$ is defined by $x_{n+1} = a \gcd(x_n, x_{n-1}) + b$ for $n \geq 2$. Prove that this sequence is eventually periodic.

**Problem 18.464.** A sequence $(a_n)$ of positive integers is such that every prime greater than 2024 divides at least one of its terms. Define $b_n = a_1 a_2 \cdots a_n - 1$. Prove that there exist infinitely many primes that divide at least one term of the sequence $(b_n)$.

**Problem 18.465.** ★ The sequence $(a_n)$ is defined by $a_1 = 1$, $a_2 = 2$ and $a_{n+2} = a_n(a_{n+1} + 1)$ for all $n \geq 1$. Prove that $a_{a_n}$ is divisible by $a_n$ for every $n \geq 100$.

**Problem 18.466.** Two rational numbers $\frac{m}{n}$ and $\frac{n}{m}$ are written on a blackboard, where $m$ and $n$ are coprime positive integers. At each step, you may pick two of the numbers $x$ and $y$ written on the board and write either their arithmetic mean $\frac{x+y}{2}$ or their harmonic mean $\frac{2xy}{x+y}$ on the board as well. Find all pairs $(m, n)$ for which you can write 1 on the board in finitely many steps.

**Problem 18.467.** Denote by $p(x)$ the smallest prime divisor of a nonzero integer $x$. Given an integer $a_1 > 2$, define a sequence $(a_n)$ by $a_{n+1} = a_n^{a_n} - 1$ for each $n \geq 1$. For which values of $a_1$ is the sequence $p(a_n)$ bounded?†

**Problem 18.468.** Let $x_1 = x_2 = x_3 = 1$ and $x_{n+3} = x_n + x_{n+1}x_{n+2}$ for all positive integers $n$. Prove that for any positive integer $m$ there is a positive integer $k$ such that $m$ divides $x_k$.

**Problem 18.469.** Is there an infinite sequence of positive integers such that for any positive integer $k$ the sum of any $k$ consecutive terms is divisible by $k + 1$? [All-Russian Olympiad 2015, Grade 10]

**Problem 18.470.** Let $a_1 < a_2 < a_3 < \cdots$ be an infinite strictly increasing sequence of positive integers such that for some integer $k$ and all integers $n > k$ the term $a_n$ is a sum of two members of the sequence. Prove that the sequence contains infinitely many composite numbers. [Bulgaria EGMO TST 2015]

**Problem 18.471.** Let $k \geq 3$ be an integer. The sequence $(a_n)_{n\geq 1}$ is defined by $a_1 = 1$, $a_2 = k$ and $a_{n+2} = ka_{n+1} + a_n$ for any positive integer $n$. Prove that there are only finitely many pairs $(m, l)$ of positive integers such that $a_m = l!$. [Bulgaria Balkan MO TST 2025]

**Problem 18.472.** Let $(a_n)_{n\geq 1}$ be the sequence defined by $a_1 = 2$ and $a_{n+1} = a_n^{n+1} - 1$ for each integer $n \geq 1$. Suppose that $p > 2$ is a prime and $k$ is a positive integer. Prove that for some $n$ the number $a_n$ is divisible by $p^k$.† [USAJMO 2024 P3]

**Problem 18.473.** ★ Does there exist an infinite sequence of positive integers $a_1, a_2, \ldots$ such that every positive integer appears exactly once as a member of the sequence and $a_1 + a_2 + \cdots + a_i$ divides $a_1 + a_2 + \cdots + a_j$ if and only if $i = j$? [Bulgaria Team Contest, March 2025, Vlad Spataru]

**Problem 18.474.** The sequence $a_1, a_2, \ldots$ is defined by $a_1 = 2$, $a_2 = 12$ and $a_{n+1} = 6a_n - a_{n-1}$ for every $n \geq 2$. Prove that no term of this sequence is a perfect power of a positive integer. [Bulgaria National Round 2015 P3]

**Problem 18.475.** ★ Let $(x_n)_{n\geq 1}$ be a strictly increasing sequence of nonnegative integers such that for all $n \geq 2$ each term can be uniquely written as the arithmetic mean of two distinct terms of the sequence. (a) Show that there exists a real number $a$ such that $x_n \geq 2n + a$ for infinitely many $n$. (b) Show that there exists a sequence with the required property such that $x_n \leq 4n^2$ for all $n \geq 1$. [Bulgaria RMM TST 2025]

**Problem 18.476.** Find all sequences of positive integers $a_1, a_2, \ldots$ such that $\left\lfloor \frac{a_m}{a_n} \right\rfloor = \left\lfloor \frac{m}{n} \right\rfloor$ holds for all $m$ and $n$.

**Problem 18.477.** Define $a_1 = 1$ and $a_{n+1} = a_n + 2^{a_n}$ for $n \geq 1$. Prove that $a_1, a_2, \ldots, a_{3^{100}}$ give all possible remainders upon division by $3^{100}$.

### 18.12.4 Divisors and arithmetic functions

**Problem 18.478.** Let $d_1, d_2, \ldots, d_s$ be all the positive divisors of $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$. What is the value of $\frac{1}{d_1} + \cdots + \frac{1}{d_s}$?

**Problem 18.479.** There is a house with 100 flats numbered $1, 2, \ldots, 100$; initially the light works in all of them. On day $k$ the administrator changes the state of the light in every flat whose number is divisible by $k$. After 100 days, in which flats does the light still work?

**Problem 18.480.** Prove that a positive integer less than a million has at most 2000 divisors.

**Problem 18.481.** The largest proper divisor of $n$ equals $d$. Is it possible that the largest proper divisor of $n + 2$ equals $d + 2$? (A proper divisor of a number is a divisor other than 1 and the number itself.)

**Problem 18.482.** There is a number 2022 written on the board. Ahmad and Khalid play a game: if at some moment the number $n$ is on the board, a player may replace it by $n - d$, where

$d$ is a divisor of $n$. They take turns, Ahmad going first. The player who obtains 0 loses. Who has a winning strategy?

**Problem 18.483.** Given a positive integer $n$, define a sequence $(a_k)$ by $a_0 = n$ and $a_{k+1} = \tau(a_k)$, where $\tau(x)$ denotes the number of positive divisors of $x$. Find all $n$ for which no term $a_k$ is a perfect square.

**Problem 18.484.** For $n \in \mathbb{N}$ denote by $f(n)$ the smallest positive integer having exactly $n$ divisors; for instance $f(5) = 16$ and $f(6) = 12$. Prove that $f(2^k)$ divides $f(2^{k+1})$ for every $k \in \mathbb{N}$.

**Problem 18.485.** Let $n > 2$ be a positive integer and let $a_1 < a_2 < \cdots < a_k$ be all positive integers less than $n$ and coprime to $n$. Find all $n$ for which none of the sums $a_1 + a_2,\ a_2 + a_3,\ \ldots,\ a_{k-1} + a_k$ is divisible by 3.

**Problem 18.486.** Let $b_1, b_2, b_3, \ldots$ be a sequence of positive integers such that, for each positive integer $n$, $b_{n+1}$ is the square of the number of positive divisors of $b_n$. Prove that if $b_1 > 1$, then the sequence contains a term equal to 9.

**Problem 18.487.** Let $1 = d_1 < d_2 < \cdots < d_k = n$ be all the divisors of a positive integer $n$, and let $a_i$ be the number of positive divisors of $d_i$. Prove that $a_1^3 + \cdots + a_k^3 = (a_1 + \cdots + a_k)^2$.

**Problem 18.488.** Let $\tau(n)$ denote the number of positive divisors of $n$. Find all positive integers $n$ such that $\tau(n)^3 = 4n$.

**Problem 18.489.** ★ Prove that there exist infinitely many positive integers $a$ such that the equation $\tau(an) = n$ has no positive integer solution $n$.

**Problem 18.490.** ★ A number $n$ is *interesting* if 2018 divides $\tau(n)$. Determine all positive integers $k$ such that there exists an infinite arithmetic progression with common difference $k$ all of whose terms are interesting.

**Problem 18.491.** ★ Let $k$ be a positive integer and let $n$ be the smallest number with exactly $k$ divisors. Assuming $n$ is a perfect cube, is it possible that $k$ is divisible by a prime factor of the form $3j + 2$ with $j \in \mathbb{N}$?

**Problem 18.492.** Given arbitrary positive integers $c$ and $m$, prove that there are infinitely many positive integers $n$ such that the number of divisors of $n \cdot c^n + 1$ is divisible by $m$.

**Problem 18.493.** Let $\tau(n)$ denote the number of positive divisors of $n$. Determine all positive integers $m$ for which there exists a positive integer $n$ such that $\dfrac{\tau(n^2)}{\tau(n)} = m$.

**Problem 18.494.** Let $p$ be an odd prime. For $k = 1, 2, \ldots, p-1$, denote by $a_k$ the number of divisors of $kp + 1$ lying in the interval $(k, p)$. Evaluate $a_1 + a_2 + \cdots + a_{p-1}$.

**Problem 18.495.** Determine all positive integers $n$ such that $n$ divides $\varphi(n)^{\tau(n)} + 1$ but $\tau(n)^5$ does not divide $n^{\varphi(n)} - 1$.

**Problem 18.496.** Prove that for every $n$ there exist positive integers $a_1 < a_2 < \cdots < a_n$ such that $\varphi(a_1) > \varphi(a_2) > \cdots > \varphi(a_n)$.

**Problem 18.497.** Let $A$ be the set of positive integers $x$ for which $\lfloor 10^{10}/x \rfloor$ is odd. Evaluate $\sum_{x \in A} \varphi(x)$.

**Problem 18.498.** Given a positive integer $n$, denote by $f(n)$ the number of ordered pairs $(a, b)$ of positive integers for which $\dfrac{ab}{a+b}$ is a divisor of $n$. Prove that $f(n)$ is a perfect square.

**Problem 18.499.** For an integer $x > 1$, denote by $f(x)$ its largest divisor that is less than $x$. Is there a positive integer $a$ for which the equation $x - f(x) = a$ has exactly 1001 solutions $x \in \mathbb{N}$?

**Problem 18.500.** Let $a$ and $b$ be positive integers. A set $M$ of positive integers has the following properties: (1) the numbers $1, a, a^2, a^3, \ldots$ are in $M$; (2) for any $x \in M$, the number $ax + b$ and all divisors of $x$ are also in $M$; (3) for any coprime elements $x, y \in M$, also $xy \in M$. Prove that $M$ contains all numbers coprime to $b$.

**Problem 18.501.** Denote by $f(n)$ the number of ways of factoring a positive integer $n$ into a product of several (at least one) integers greater than 1; the order does not matter, so $f(12) = 4$. If $p$ is any prime divisor of $n$, prove that $f(n) \le \dfrac{n}{p}$.

**Problem 18.502.** Two positive integers $a$ and $b$ are *prime-related* if $a/b$ or $b/a$ is a prime. Find all composite positive integers $n$ such that all divisors of $n$ can be arranged around a circle so that every two adjacent divisors are prime-related.

**Problem 18.503.** Find all positive integers $n$ with the following property: there is a unique positive integer $a \le n!$ for which $n!$ divides $a^n + 1$.

**Problem 18.504.** Let $A$ be a set of $n$ positive integers. For any nonempty subset $B$ of $A$ consider the greatest common divisor of all elements of $B$. Suppose that all the $2^n - 1$ values thus obtained are distinct. At least how many distinct prime divisors must the elements of $A$ have?

**Problem 18.505.** Find all positive integers $n$ for which $2^n - 1$ has exactly $n$ divisors.

**Problem 18.506.** ★ On the board is written a positive integer $n$. In one move it is allowed to choose a number on the board, delete it, and write all of its positive divisors except itself; equal numbers may be present on the board. For which $n$ is it possible that at some point there are exactly $n^2$ (not necessarily distinct) numbers on the board? [All-Russian Olympiad 2017, Grade 11]

**Problem 18.507.** For a positive integer $N$ let $f(N)$ denote the sum of all positive integers not exceeding $N$ and having a common divisor greater than 1 with $N$. Prove that for any integer $n \ge 2$ and any prime $p$ we have $f(n + p) \ne f(n)$. [Balkan Mathematical Olympiad 2010 P4]

**Problem 18.508.** ★ Prove that there does not exist a positive integer $n$ for which the set of its positive divisors can be divided into two groups, each with at least three numbers, so that the numbers in one group form an arithmetic progression while the numbers in the other group form a geometric progression. [China National Round 2017]

**Problem 18.509.** All positive divisors of an integer $N$ are written in a row: $1 = d_1 < d_2 < \cdots < d_s = N$. It turns out that $\gcd(d_1, d_2) + \gcd(d_2, d_3) + \cdots + \gcd(d_{s-1}, d_s) = N - 2$. What values can $N$ take?

**Problem 18.510.** There are 100 distinct positive integers on the board. We simultaneously increase each number by the greatest common divisor of the other 99 numbers. At least how many of the obtained numbers can be distinct?

**Problem 18.511.** Define $f(1) = 1$, $f(2n) = f(n)$ and $f(2n + 1) = f(n) + f(n + 1)$ for $n \ge 1$. Given $m \in \mathbb{N}$, how many odd positive integers $x$ are there for which $f(x) = m$?

**Problem 18.512.** Prove that for positive integers $m$ and $n$ one has $\gcd(m,n) = m+n-mn+2\sum_{k=0}^{m-1}\left\lfloor \frac{kn}{m} \right\rfloor$.

**Problem 18.513.** Let $n > 1$ be an integer. Prove that neither $1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{n}$ nor $1+\frac{1}{3}+\frac{1}{5}+\cdots+\frac{1}{2n-1}$ is an integer.

**Problem 18.514.** ★ Let an $n \times n$ table be filled with integers such that each entry is congruent to 1 modulo $n$, and the sum of the entries of each row, as well as of each column, is congruent to $n$ modulo $n^2$. Let $R_i$ denote the product of the entries of the $i$-th row and $C_j$ the product of the entries of the $j$-th column. Prove that $R_1 + R_2 + \cdots + R_n \equiv C_1 + C_2 + \cdots + C_n \pmod{n^4}$.

### 18.12.5 Combinatorial number theory

**Problem 18.515.** Let $a_1 < a_2 < \cdots < a_n \leq 2n-2$ be positive integers, where $n > 1$. Prove that there exist $i$ and $j$ such that $a_i = a_j + a_1$.

**Problem 18.516.** Prove that among any $n+1$ numbers from the set $\{1, 2, \ldots, 2n\}$ there exist two, one of which divides the other.

**Problem 18.517.** Suppose that we have $k$ positive integers not exceeding $n$, none of which divides the product of the others. Prove that $k \leq \pi(n)$, where $\pi(n)$ denotes the number of primes not exceeding $n$.

**Problem 18.518.** Let $n$ be a positive integer. Prove that among any $2n-1$ integers there exist $n$ whose sum is divisible by $n$.

**Problem 18.519.** Prove that we cannot choose more than $2\lfloor\sqrt{n}\rfloor$ distinct positive integers such that the least common multiple of every two of them is at most $n$.

**Problem 18.520.** Prove that, if $n$ is large enough, we can choose more than $1.01\sqrt{n}$ distinct positive integers such that the least common multiple of every two of them is at most $n$.

**Problem 18.521.** Let $n$ be a positive integer and let $a_1, a_2, \ldots, a_k \leq n$ be positive integers such that the least common multiple of every two of them is greater than $n$. Prove that $\frac{1}{a_1}+\frac{1}{a_2}+\cdots+\frac{1}{a_k} < \frac{3}{2}$.

**Problem 18.522.** Positive integers $a_1, \ldots, a_n$ do not exceed $2n-1$, but $\operatorname{lcm}(a_i, a_j) \geq 2n$ for all $i \neq j$. Prove that $a_i \geq \frac{2n}{3}$ for each $i$.

**Problem 18.523.** (a) An infinite sequence $a_1 < a_2 < a_3 < \cdots$ of positive integers is such that $a_{2n} = 2a_n$ for all $n$. If $p > a_1$ is a prime, prove that there is a term of the sequence divisible by $p$. (b) Given an odd prime $p$, prove that there is a sequence $(a_n)$ with the above properties none of whose terms is divisible by $p$.

**Problem 18.524.** We select 50 numbers from the set $S = \{1, 2, \ldots, 100\}$. Prove that there are two selected numbers whose sum is a perfect square.

**Problem 18.525.** Let $A_1, A_2, \ldots, A_k$ be subsets of $\{1, 2, \ldots, n\}$ such that $A_i \cap A_j \neq \emptyset$ for all $1 \leq i, j \leq k$. Prove that there are $n$ distinct positive integers $x_1, x_2, \ldots, x_n$ such that for each $1 \leq j \leq k$ we have $\operatorname{lcm}_{i \in A_j}\{x_i\} > \operatorname{lcm}_{i \notin A_j}\{x_i\}$.

**Problem 18.526.** Hesam chose 10 distinct positive integers and gave all pairwise greatest common divisors and all pairwise least common multiples (a total of 90 numbers) to Masoud. Can Masoud always recover the original 10 numbers from these 90 numbers?

**Problem 18.527.** Determine whether there exists an increasing sequence $a_1, a_2, \ldots$ of positive integers with the following property: for any integer $k$, only finitely many of the numbers $a_1 + k, a_2 + k, \ldots$ are prime.

**Problem 18.528.** Is there an infinite set of positive integers such that no matter how we choose some of its elements, their sum is not a perfect power?

**Problem 18.529.** The set of all positive integers is split into finitely many subsets. Prove that one of these subsets has the following property: for any positive integer $m$ there are infinitely many numbers in this subset that are divisible by $m$.

**Problem 18.530.** ★ A deck of $n > 1$ cards is given. A positive integer is written on each card. The deck has the property that the arithmetic mean of the numbers on each pair of cards is also the geometric mean of the numbers on some collection of one or more cards. For which $n$ does it follow that the numbers on the cards are all equal? [IMO 2020 P5]

**Problem 18.531.** There are $n$ pairwise coprime numbers, each between 1 and $(2n-1)^2$. Prove that one of them must be prime.

**Problem 18.532.** ★ Determine all integers $n \geq 2$ with the following property: every $n$ pairwise distinct integers whose sum is not divisible by $n$ can be arranged in some order $a_1, a_2, \ldots, a_n$ so that $n$ divides $1 \cdot a_1 + 2 \cdot a_2 + \cdots + n \cdot a_n$. [IMO Shortlist 2021 N7]

**Problem 18.533.** Let $n > 6$ be an integer with the following property: all the integers in $\{1, 2, \ldots, n-1\}$ that are relatively prime to $n$ form an arithmetic progression. Prove that $n$ is either prime or a power of 2. [IMO 1991]

**Problem 18.534.** We say that an ordered triple $(a, b, c)$ of positive integers is *$n$-powerful* if $a \leq b \leq c$, $\gcd(a, b, c) = 1$, and $a^n + b^n + c^n$ is divisible by $a + b + c$. (a) Determine all ordered triples (if any) which are $n$-powerful for all $n \geq 1$. (b) Determine all ordered triples (if any) which are 2004-powerful and 2005-powerful, but not 2007-powerful. [Canada 2005]

**Problem 18.535.** Prove that, for each integer $n > 2$, there is a set $S$ of $n$ integers such that $ab$ is divisible by $(a-b)^2$ for all distinct $a, b \in S$. [USAMO 1998]

**Problem 18.536.** Find all pairs of distinct nonzero integers $(a, b)$ such that there exists a set $S$ of integers with the following property: for any integer $n$, exactly one of $n$, $n + a$, $n + b$ is in $S$. [China 2009]

**Problem 18.537.** We say that a subset $C$ of the natural numbers is *good* if for each integer $k$ there exist $a, b \in C$, $a \neq b$, such that $a + k$ and $b + k$ are not relatively prime. Prove that if the sum of the elements of $C$ equals 2003, then for some $c \in C$ the set $C \setminus \{c\}$ is also good. [Bulgaria 2003]

**Problem 18.538.** Let $d$ be a positive integer. Show that for every integer $S$ there exist an integer $n > 0$ and a sequence $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$ with $\varepsilon_k \in \{1, -1\}$ for every $k$, such that $S = \varepsilon_1(1+d)^2 + \varepsilon_2(1+2d)^2 + \cdots + \varepsilon_n(1+nd)^2$. [Canada 2011]

**Problem 18.539.** ★ Let $p$ be a fixed odd prime. A $p$-tuple $(a_1, a_2, \ldots, a_p)$ of integers is said to be *good* if (i) $0 \le a_i \le p-1$ for all $i$; (ii) $a_1 + a_2 + \cdots + a_p$ is not divisible by $p$; and (iii) $a_1a_2 + a_2a_3 + \cdots + a_pa_1$ is divisible by $p$. Determine the number of good $p$-tuples. [Canada 2014]

**Problem 18.540.** Prove that any positive integer larger than 6 is a sum of distinct primes.

**Problem 18.541.** Find all pairs of sets $(A, B)$ satisfying the following conditions: (i) $A \cup B = \mathbb{Z}$; (ii) if $x \in A$, then $x - 1 \in B$; (iii) if $x \in B$ and $y \in B$, then $x + y \in A$.

**Problem 18.542.** Suppose that $a$ and $b$ are positive real numbers such that $a - b$, $a^2 - b^2$, $a^3 - b^3$, ... are all positive integers. Show that $a$ and $b$ must be positive integers.

**Problem 18.543.** Let $a_0 < a_1 < \cdots < a_n$ be real numbers such that $A_k = a_0^k + a_1^k + \cdots + a_n^k = 0$ for all odd positive $k > 2021$. Prove that $a_i = -a_{n-i}$ for all $i$.

**Problem 18.544.** Let $n$ be a positive integer. A set $A$ of positive integers satisfies the following condition: if $m \le n$, then there exists an element of $A$ divisible by $m$. What is the smallest possible sum of all elements of $A$?

**Problem 18.545.** Determine whether there exist five consecutive positive integers whose least common multiple is a perfect square.

**Problem 18.546.** Find all positive integers $n > 2$ with the following property: for any two positive integers $a, b$ coprime to $n$ and satisfying $a, b < n < a + b$, the product $(a - 1)(b - 1)$ is not divisible by $n$.

**Problem 18.547.** ★ Let $n$ be a positive integer. Suppose that $A$ and $B$ are subsets of $\{1, 2, \ldots, n\}$ with the property that for every $a \in A$ and $b \in B$ the number $ab + 1$ is a square. Prove that at least one of the sets $A$ and $B$ has fewer than $\log_2 n$ elements.

**Problem 18.548.** If $a_1, a_2, \ldots, a_n \in \mathbb{N}$ are pairwise distinct, prove that $\displaystyle\sum_{k=1}^{n} \frac{1}{\operatorname{lcm}(a_1, \ldots, a_k)} < \frac{7}{4}$; prove also that the upper bound cannot be improved to $\frac{5}{4}$.†

**Problem 18.549.** Let $n$ be a positive integer. Prove that there exist distinct integers $n^2 < a, b, c < n^2 + n + 3\sqrt{n}$ such that $c$ divides $ab$.†

**Problem 18.550.** Suppose that $p$ and $\frac{p-1}{2}$ are primes, and $a, b, c$ are integers not divisible by $p$. Prove that there are at most $\lceil \sqrt{2p} \rceil$ exponents $n$ with $1 \le n \le p - 1$ for which $p \mid a^n + b^n + c^n$.†

**Problem 18.551.** Let $p$ and $q$ be coprime positive integers. We are given $pq$ candies and need to place them in $n$ bags so that both of the following are satisfied: the bags can be packed into $p$ white boxes so that each box contains exactly $q$ candies; and the bags can be packed into $q$ black boxes so that each box contains exactly $p$ candies. What is the smallest $n$ for which this is possible?

**Problem 18.552.** Let $n$ be a fixed positive integer. For distinct positive real numbers $x_1, x_2, \ldots, x_n$, consider all sums $x_i + x_j + x_k$ with $1 \le i < j < k \le n$. At most how many distinct powers of 3 can occur among these sums?

**Problem 18.553.** Prove that for any two Pythagorean triples $P$ and $Q$ there is a sequence of Pythagorean triples $P = T_0, T_1, T_2, \ldots, T_n = Q$ such that $T_{i-1}$ and $T_i$ have a common element for each $i = 1, \ldots, n$.

**Problem 18.554.** Positive integers $a, b$ are such that $a+1 \mid b^2+1$ and $b+1 \mid a^2+1$. Prove that $a$ and $b$ are both odd.

**Problem 18.555.** If the difference of two consecutive cubes is $n^2$, where $n$ is a positive integer, prove that $2n-1$ is a perfect square.

**Problem 18.556.** A function $f : \mathbb{Z} \to \mathbb{N}$ is such that $f(x) < 2^{1000}$ and $\gcd(f(x), f(y)) = \gcd(f(x), x-y)$ for any two integers $x, y$. Prove that there exist positive integers $a, b$ such that $f(x) = \gcd(x+a, b)$ for all $x$.

**Problem 18.557.** Let $n$ be a positive integer. Suppose that $x^2+x+n$ is prime for $x = 0, 1, \ldots, \lfloor \sqrt{n/3} \rfloor$. Prove that $x^2+x+n$ is prime for $x = 0, 1, \ldots, n-2$.

**Problem 18.558.** Find all positive integers $n > 1$ for which there exists a permutation $a_1, a_2, \ldots, a_n$ of $1, 2, \ldots, n$ such that $a_{i+1} \equiv a_i^2 + 1$ or $a_{i+1} \equiv a_i^2 - 1 \pmod{n}$ for $i = 1, 2, \ldots, n$, where $a_{n+1} := a_1$.

**Problem 18.559.** Find all positive integers $n > 1$ for which there exists a set $A$ of $n$ distinct positive integers such that $\dfrac{x+y}{\gcd(x,y)} \leq 2n-2$ for any $x, y \in A$ with $x \neq y$.

**Problem 18.560.** Prove that there are infinitely many positive integers $x$ for which the number of distinct odd prime factors of $x(x+3)$ is a multiple of 3.

**Problem 18.561.** Are there 20 positive integers $a_1, \ldots, a_{10}, b_1, \ldots, b_{10}$ with the following property: for every subset $S \subseteq \{1, \ldots, 10\}$ of indices, the sum $\sum_{i \in S} a_i$ divides $12 + \sum_{i \in S} b_i$?

**Problem 18.562.** Given a positive integer $n$, define the two-dimensional array $f_{i,j}$ $(i, j \geq 0)$ by $f_{i,0} = f_{0,j} = 0$, $f_{1,1} = n$ and $f_{i,j} = \left\lfloor \frac{f_{i-1,j}}{2} \right\rfloor + \left\lfloor \frac{f_{i,j-1}}{2} \right\rfloor$ for all $i, j \in \mathbb{N}$ except $(i, j) = (1, 1)$. How many terms in this array are odd?

**Problem 18.563.** For which positive integers $n$ and $k$ can one find integers $a_1, a_2, \ldots, a_n$ and $b_1, b_2, \ldots, b_k$ such that the products $a_i b_j$ for $1 \leq i \leq n$ and $1 \leq j \leq k$ are all distinct modulo $nk$?

**Problem 18.564.** Initially there are two distinct positive integers on the board. At each step one can pick two numbers $a, b$ on the board and additionally write the number $\operatorname{lcm}(a, b) + \gcd(a, b)$ (nothing is ever erased). Given $n \in \mathbb{N}$, prove that one can make a number of steps to obtain a multiple of $n$.

**Problem 18.565.** Let $p_1, \ldots, p_k$ be distinct primes and let $a_1, \ldots, a_n$ be the first $n$ positive integers coprime to $p_1 p_2 \cdots p_k$. Prove that $\sum_{i=1}^{n} \frac{1}{a_i}$ is not an integer.

**Problem 18.566.** Prove that every large enough positive integer $n$ can be written in the form $n = a_1 + \cdots + a_{1000}$, where $a_1 < \cdots < a_{1000}$ are positive integers satisfying $a_i \mid a_{i+1}$ for all $i$.

**Problem 18.567.** Given a positive integer $n$, find the largest $m$ for which there exist positive integers $a_1 < \cdots < a_n = m$ such that $\frac{1}{2}(a_i^2 + a_{i+1}^2)$ is a square for each $i$.

**Problem 18.568.** Let $x > y > 1$ be positive integers such that $xy + x + y$ is a perfect square. Prove that there exists a positive integer $z$ such that $2x + 4y < z < 4x + 2y$ and $\gcd(z, x^2 + xy + y^2) = 1$.

**Problem 18.569.** ★ Prove that there are constants $a > 0$ and $b > \frac{1}{2}$ such that for every $n$ there is a subset of $\{1, 2, \ldots, n\}$ of cardinality at least $an^b$ in which no pairwise difference is a perfect square.

**Problem 18.570.** Find all integers $a$ for which there is a set $X$ of six integers with the property that $\{x + ay : x, y \in X\} = \{1, 2, \ldots, 36\} \pmod{37}$.

**Problem 18.571.** Let $a, b, c, d$ be elements of the set $\{0, 1, \ldots, 99\}$ such that $a + b \equiv c + d \pmod{100}$ and $2^a + 2^b \equiv 2^c + 2^d \pmod{101}$. Prove that $\{a, b\} = \{c, d\}$.

**Problem 18.572.** ★ Is it true that if $n$ is sufficiently large and $a_1, a_2, \ldots, a_n$ is an arbitrary permutation of $(1, 2, \ldots, n)$, then we can find integers $i, d$ such that $1 \le i < i + d < i + 2d \le n$ and $a_i, a_{i+d}, a_{i+2d}$ form an arithmetic progression?

**Problem 18.573.** ★ Is there a positive integer $n$ such that every nonzero digit appears the same number of times in the decimal representation of each of the numbers $n, 2n, 3n, \ldots, 2021n$?

**Problem 18.574.** Prove that there exists a bijective function $f : \mathbb{N}_0 \to \mathbb{N}_0$ such that $f(3xy + x + y) = 4f(x)f(y) + f(x) + f(y)$ for all $x, y$.

**Problem 18.575.** For a positive integer $n$, define $\lambda(n) = (-1)^k$ if $n$ is a product of $k \ge 0$ primes (not necessarily distinct). Prove that there are infinitely many $n$ with $\lambda(n) = \lambda(n + 1)$ and infinitely many with $\lambda(n) \ne \lambda(n + 1)$.

**Problem 18.576.** Denote by $P(n)$ the largest prime divisor of $n \in \mathbb{N}$. Prove that there are infinitely many positive integers $n$ such that $P(n) < P(n + 1) < P(n + 2)$.

**Problem 18.577.** Denote by $P(a)$ the largest prime divisor of $a^2 + 1$. Prove that there are infinitely many triples $a, b, c \in \mathbb{N}$ for which $P(a) = P(b) = P(c)$. [Russia 2011]

**Problem 18.578.** ★ For each integer $n \ge 2$ let $F(n)$ denote the greatest prime factor of $n$. A pair of distinct primes $(p, q)$ is called *strange* if there is no integer $n \ge 2$ with $F(n)F(n+1) = pq$. Prove that there exist infinitely many strange pairs. [RMM 2020 P6]

**Problem 18.579.** Suppose that a nonempty set of primes $M$ has the property that, for any distinct primes $p_1, p_2, \ldots, p_k \in M$, all prime factors of $p_1p_2 \cdots p_k + 1$ also lie in $M$. Prove that $M$ is the set of all primes.

**Problem 18.580.** Let $n \ge 3$ be an odd positive integer. Show that $n$ is prime if and only if, however we choose $n + 1$ pairwise distinct positive integers, we can find two of them, $a$ and $b$, such that $\dfrac{a + b}{\gcd(a, b)} \ge n$.

**Problem 18.581.** Prove that a positive integer $n$ can be written in the form $n = \dfrac{a^2 + b^2 + c^2 + d^2}{abcd + 1}$ for some $a, b, c, d \in \mathbb{N}$ if and only if $n$ is a sum of three squares.

**Problem 18.582.** Can all integers be divided into three disjoint sets such that for every $n \in \mathbb{Z}$ the numbers $n$, $n - 26$ and $n + 36$ all lie in different sets?

**Problem 18.583.** Find all $n \ge 1$ such that both $1\underbrace{77\ldots7}_{n}$ and $3\underbrace{77\ldots7}_{n}$ are prime.

**Problem 18.584.** Solve $2^x + 17 = y^4$ over the positive integers.

**Problem 18.585.** Let $a$ be a positive integer. Prove that there exists a positive integer $b$ with the following property: for each positive integer $n$ there exists a prime $p$ (possibly depending on $a, b, n$) such that $a^n + b$ is divisible by $p$ but not by $p^2$. [Vojtech Jarnik IMC 2025, Category I]

**Problem 18.586.** Let $n$ be a positive integer. Prove that there exists a positive integer $a$ such that exactly $\lfloor \frac{n}{4} \rfloor$ of the integers $a+1, a+2, \ldots, a+n$ are squarefree. [Bulgaria National Round 2025 P5]

**Problem 18.587.** Prove that there are infinitely many pairs $(a, b)$ of positive integers such that $a$ divides $b^n + 1$ and $b$ divides $a^n + 1$, for any fixed integer $n \geq 2$. [Folklore for $n = 2$]

**Problem 18.588.** ★ For two sets of integers $X$ and $Y$ define $X \cdot Y$ as the set of all products of an element of $X$ and an element of $Y$. We call a set $S$ of positive integers *good* if there do not exist sets $A, B$ of positive integers, each with at least two elements, such that $A \cdot B$ and $S$ are the same. Prove that the set of all perfect powers greater than or equal to 2025 is good. [Al-Khwarizmi International Junior Olympiad 2025]

**Problem 18.589.** Prove that there exist pairwise coprime integers $a, b, c > 10^{10}$ such that the sets of prime divisors of $a + b + c$ and of $ab + bc + ca$ are the same. [Iran National Round 2024]

**Problem 18.590.** ★ Determine the smallest positive integer $k$ for which there exists a sequence $(a_n)_{n \geq 1}$ of positive integers such that $\gcd(a_m + n,\ a_n + m) \leq k$ for any distinct positive integers $m$ and $n$. [IMO Shortlist 2015 N7]

**Problem 18.591.** Let $N > 25000$ be an integer. Prove that if $a_1 < \cdots < a_k < 100$ are $k$ distinct positive integers, then the product $\prod_{i=1}^{k}(N^{a_i} + a_i)$ has at least $k$ distinct prime factors. [Taiwan IMO TST 2020]

**Problem 18.592.** ★ Determine all pairs $(a, b)$ of odd positive integers for which there exists a positive integer $c$ such that $2^n a + b$ divides $c^n + 1$ for all positive integers $n$. [Ukraine IMO TST 2010; see also IMO Shortlist 2012 N6]

**Problem 18.593.** Find all positive integers $n$ for which $3^n - 2^n - 1$ is a perfect square.

**Problem 18.594.** Suppose that $a, b, c, d, e$ are distinct positive integers satisfying $a^4 + b^4 = c^4 + d^4 = e^5$. Show that $ac + bd$ is a composite number.

**Problem 18.595.** Find all positive integers $n$ for which there exists an even positive integer $a$ such that the product $(a-1)(a^2-1)\cdots(a^n-1)$ is a perfect square. [All-Russian Olympiad 2025, Grades 9 and 10]

### 18.12.6 A closing miscellany

**Problem 18.596.** Positive integers $a, b, c, d$ are such that $ab = cd$. Is it possible that $a+b+c+d$ is a prime?

**Problem 18.597.** Is it true that the quadratic polynomial $n^2 + n + 41$ takes only prime values for positive integers $n$?

**Problem 18.598.** There are 101 integers written in a circle. Is it possible that the ratio of any two neighbours (the larger divided by the smaller) is a prime number?

**Problem 18.599.** One number consists of $n$ ones in its decimal representation, another one consists of $m$ ones. What is their greatest common divisor?

**Problem 18.600.** Prove that for every natural number $m$ there exists a natural number $n$ such that $n^2 + 7$ is divisible by $2^m$.

**Problem 18.601.** Find all pairs of primes $p, q$ satisfying $2^p = 2^{q-2} + q!$.

**Problem 18.602.** Prove that for every prime $p$ there exists a positive integer $n$ such that $1^n + 2^{n-1} + \cdots + n^1 \equiv 2022 \pmod{p}$.

**Problem 18.603.** Find all composite positive integers $n$ with the property that, whenever $xy = n$ with $x, y \in \mathbb{N}$, the sum $x + y$ is a power of two.

**Problem 18.604.** Find all pairs $(a, b)$ of positive integers with the property that $\sqrt[n+1]{a^n + b^n}$ is an integer for every $n = 1, 2, \ldots$.

**Problem 18.605.** Let $n$ be an odd integer greater than 1 and let $c_1, c_2, \ldots, c_n \in \mathbb{Z}$. For a permutation $a = (a_1, a_2, \ldots, a_n)$ of $\{1, \ldots, n\}$ define $S(a) = \sum_{i=1}^{n} c_i a_i$. Prove that there are permutations $a \neq b$ such that $n!$ divides $S(a) - S(b)$.

**Problem 18.606.** ★ Define a *step* to be the map that sends a pair of positive integers $(x, y)$ with exactly one even coordinate to $(\frac{x}{2},\ y + \frac{x}{2})$ if $2 \mid x$, and to $(x + \frac{y}{2},\ \frac{y}{2})$ if $2 \mid y$. Prove that for each odd positive integer $n > 1$ there exists an even positive integer $b < n$ such that by iteratively applying finitely many steps to the pair $(n, b)$ we reach the pair $(b, n)$.

# Chapter 19

# Mixed Practice

**Sources for this chapter:** Saudi e-training weekly problem sheets, Level 4, three seasons (2019–2020, 2021–2022, 2022–2023); KSA e-training weekly problem sets (2016–2017).

This closing chapter gathers the hardest surviving weekly-sheet problems from three seasons of Level 4 e-training, then closes with the full run of the 2016–2017 KSA weekly problem sets, whose difficulty sits at IMO-shortlist level throughout.

## Practice set 1

**Problem 19.1.** Let $n$ be a positive integer. Prove that at least one coefficient of the polynomial

$$P(x) = (x^4 + x^3 - 3x^2 + x + 2)^n$$

is negative.

**Problem 19.2.** Let $a, b, c, d$ be positive integers such that the sum of the squares of any two of them is divisible by the product of the other two (for instance, $cd \mid a^2 + b^2$). Prove that at least three of the four numbers are equal.

**Problem 19.3.** The speed of a robber's car is 90% of the speed of a policeman's car. The robber and the policeman are on the same line, and the policeman does not know in which direction the robber is traveling. Prove that the policeman can always catch the robber.

**Problem 19.4.** In acute triangle $ABC$, angle $A$ equals $60^\circ$. Prove that the bisector of one of the angles formed by the altitudes from $B$ and $C$ passes through the center of the circumcircle.

**Problem 19.5.** The bisectors of angles $A$, $B$, $C$ of triangle $ABC$ meet the circumcircle $c_1$ of triangle $ABC$ again at $A_2, B_2, C_2$ respectively. The tangents to $c_1$ at $A_2, B_2, C_2$ pairwise intersect at $A_3, B_3, C_3$ (where $A_3$ lies on the same side of $BC$ as $A$, $B_3$ lies on the same side of $CA$ as $B$, and $C_3$ lies on the same side of $AB$ as $C$). The incircle of triangle $ABC$ touches $BC, CA, AB$ at $A_1, B_1, C_1$ respectively. Prove that the lines $A_1A_2$, $B_1B_2$, $C_1C_2$, $AA_3$, $BB_3$, $CC_3$ are all concurrent.

## Practice set 2

**Problem 19.6.** Let $n$ be a positive integer. Prove that

$$\frac{n^2 - n}{2} \le \sum_{k=1}^{n^2} \{\sqrt{k}\} \le \frac{n^2 - 1}{2},$$

where $\{x\}$ denotes the fractional part of $x$.

**Problem 19.7.** Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(xf(y) + 2y) = f(xy) + xf(y) + f(f(y))$$

for all real $x, y$.

**Problem 19.8.** Determine the smallest positive integer $n$ for which the following statement holds: from any $n$ consecutive integers, one can select a nonempty set of consecutive integers whose sum is divisible by 2019.

**Problem 19.9.** Let $n \geq 3$ be an integer. We say that a vertex $A_i$ $(1 \leq i \leq n)$ of a convex polygon $A_1A_2 \ldots A_n$ is Bohemian if its reflection with respect to the midpoint of the segment $A_{i-1}A_{i+1}$ (with $A_0 = A_n$ and $A_1 = A_{n+1}$) lies inside or on the boundary of the polygon $A_1A_2 \ldots A_n$. Determine the smallest possible number of Bohemian vertices a convex $n$-gon can have (as a function of $n$).

**Problem 19.10.** Let $M$ be the midpoint of $AC$ and let $H$ be the foot of the altitude from vertex $B$ of triangle $ABC$. Let $P$ and $Q$ be the orthogonal projections of $A$ and $C$ onto the bisector of angle $B$. Prove that the four points $M$, $H$, $P$, $Q$ lie on the same circle.

**Problem 19.11.** Let $ABCD$ be a trapezoid with $AD \parallel BC$. Let $P$ be the point on line $AB$ for which $\angle CPD$ is maximal, and let $Q$ be the point on line $CD$ for which $\angle BQA$ is maximal. Given that $P$ lies on segment $AB$, prove that $\angle CPD = \angle BQA$.

## Practice set 3

**Problem 19.12.** Let $(a_n)$ be a sequence of positive integers such that the first $k$ terms $a_1, a_2, \ldots, a_k$ are distinct positive integers, and for each $n > k$, the number $a_n$ is the smallest positive integer that cannot be represented as a sum of several (possibly just one) of the numbers $a_1, a_2, \ldots, a_{n-1}$. Prove that $a_n = 2a_{n-1}$ for all sufficiently large $n$.

**Problem 19.13.** Let $p$ be a prime and let $f(x)$ be a polynomial of degree $d$ with integer coefficients. Assume that the numbers $f(1), f(2), \ldots, f(p)$ leave exactly $k$ distinct remainders when divided by $p$, and $1 < k < p$. Prove that

$$\frac{p-1}{d} \leq k - 1 \leq (p-1)\left(1 - \frac{1}{d}\right).$$

**Problem 19.14.** There are 100 students taking an exam. The professor calls them one by one and asks each student the same question in private: "How many of the 100 students will have a passing mark by the end of the exam?" The student's answer must be an integer. Upon receiving the answer, the professor immediately and publicly announces that student's mark, either "pass" or "fail." After all students have received marks, an inspector checks whether any student who gave the correct answer was marked "fail." If at least one such student exists, the professor is suspended and all marks are changed to "pass"; otherwise no changes are made. Can the students agree on a strategy beforehand that guarantees a "pass" mark to every one of them?

**Problem 19.15.** In a social network with a fixed, finite set of users, each user has a fixed set of followers among the other users. Each user has an initial positive integer rating (not necessarily

the same for all users). Every midnight, the rating of every user increases by the sum of the ratings that user's followers had just before midnight. Let $m$ be a positive integer. A hacker, who is not a user of the network, wants all users to have ratings divisible by $m$. Every day the hacker may either choose a user and increase that user's rating by 1, or do nothing. Prove that the hacker can achieve this goal after some number of days.

**Problem 19.16.** A circle with center $O$ is inscribed in an angle. Let $A$ be the reflection of $O$ across one side of the angle. Tangents to the circle from $A$ intersect the other side of the angle at points $B$ and $C$. Prove that the circumcenter of triangle $ABC$ lies on the bisector of the original angle.

**Problem 19.17.** Segment $AB$ meets two congruent circles in 4 points and is parallel to the line joining their centers. From point $A$, draw the two tangents to the circle nearer to $A$; from point $B$, draw the two tangents to the circle nearer to $B$. These four tangent lines form a quadrilateral containing both circles. Prove that a circle can be inscribed in this quadrilateral, tangent to all four sides.

## Practice set 4

**Problem 19.18.** Find the largest $k$ such that, for all positive reals $a, b, c$, the inequality $kabc > a^3 + b^3 + c^3$ implies that $a, b, c$ are the side lengths of a triangle.

**Problem 19.19.** Let $n, p > 1$ be integers with $p$ prime. Given that $n \mid p - 1$ and $p \mid n^3 - 1$, prove that $4p - 3$ is a perfect square.

**Problem 19.20.** Does there exist a 100-term sequence of rational numbers $a_1$, $a_2$, $a_3$, ..., $a_{100}$ such that all terms are of the form $\frac{1}{n}$ for some positive integer $n$, and for all $3 \le k \le 100$, we have $a_k = a_{k-2} - a_{k-1}$?

**Problem 19.21.** For a partition $\pi$ of $\{1, 2, 3, \ldots, 9\}$, let $\pi(x)$ be the number of elements in the part containing $x$. For example, for the partition $\pi$ given by $\{1, 4, 6, 7\} \cup \{2, 8\} \cup \{3\} \cup \{5, 9\}$, we have $\pi(5) = 2$, $\pi(6) = 4$, and $\pi(3) = 1$. Prove that for any two partitions $\pi$ and $\pi'$, there are two distinct numbers $x$ and $y$ in $\{1, 2, 3, \ldots, 9\}$ such that $\pi(x) = \pi(y)$ and $\pi'(x) = \pi'(y)$.

**Problem 19.22.** Let $AD$, $BE$, $CF$ be the altitudes of triangle $ABC$ (with $D, E, F$ on $BC, CA, AB$ respectively). Let $K$, $M$, $N$ be the orthocenters of triangles $AEF$, $BFD$, $CDE$ respectively. Prove that triangles $KMN$ and $DEF$ are congruent.

**Problem 19.23.** In triangle $ABC$, let $X$ be a point on $AB$ and $Y$ a point on $BC$. Segments $AY$ and $CX$ meet at $Z$. Given that $AY = YC$ and $AB = ZC$, prove that $B, X, Y, Z$ lie on a circle.

## Practice set 5

**Problem 19.24.** Show that for positive reals $a, b, c$ we have $abc = 1$ if and only if

$$\frac{1}{1 + a + ab} + \frac{1}{1 + b + bc} + \frac{1}{1 + c + ca} = 1.$$

**Problem 19.25.** Let $p > 3$ be a prime with $p \equiv 3 \pmod 4$. Given a positive integer $a_0$, define the sequence $a_0, a_1, \ldots$ of integers by $a_n = a_{n-1}^{2^n}$ for $n = 1, 2, \ldots$. Prove that $a_0$ can be chosen so that, for every positive integer $N$, the sequence $a_N, a_{N+1}, a_{N+2}, \ldots$ is not constant modulo $p$.

**Problem 19.26.** Let $a, b, c$ be pairwise distinct positive integers with $\gcd(a, b, c) = 1$ satisfying

$$a \mid (b-c)^2, \qquad b \mid (c-a)^2, \qquad c \mid (a-b)^2.$$

Prove that there is no non-degenerate triangle with side lengths $a, b, c$.

**Problem 19.27.** There are $n$ integers, each equal to 1, written on a blackboard. At each step, erase any two numbers $a$ and $b$ and replace them with $\frac{a+b}{4}$. After $n-1$ steps, only one number remains on the blackboard. Prove that this number is at least $\frac{1}{n}$.

**Problem 19.28.** Is it true that in any convex $n$-gon with $n > 3$, there exists a vertex and a diagonal through this vertex such that the angles this diagonal makes with both sides adjacent to the vertex are acute?

**Problem 19.29.** Circles $\omega_1$ and $\omega_2$ have centers $O_1$ and $O_2$ respectively, and intersect at points $X$ and $Y$. $AB$ is a common tangent line of these two circles, with $A$ on $\omega_1$ and $B$ on $\omega_2$. Let the tangents to $\omega_1$ and $\omega_2$ at $X$ intersect $O_1O_2$ at points $K$ and $L$, respectively. Suppose line $BL$ meets $\omega_2$ a second time at $M$, and line $AK$ meets $\omega_1$ a second time at $N$. Prove that lines $AM$, $BN$, and $O_1O_2$ are concurrent.

**Problem 19.30.** Let points $A, B, C$ lie on a parabola $\Delta$ such that the orthocenter $H$ of triangle $ABC$ coincides with the focus of $\Delta$. Prove that as $A, B, C$ vary on $\Delta$ subject to the orthocenter remaining at $H$, the inradius of triangle $ABC$ stays unchanged.

## Practice set 6

**Problem 19.31.** Let $a_1, a_2, \ldots, a_{2n}$ be a permutation of $1, 2, \ldots, 2n$ such that

$$|a_1 - a_2| + |a_2 - a_3| + \ldots + |a_{2n-1} - a_{2n}| + |a_{2n} - a_1| = 2n^2.$$

Prove that

$$|a_1 - a_2| + |a_3 - a_4| + \ldots + |a_{2n-1} - a_{2n}| = n^2.$$

**Problem 19.32.** Find the number of odd coefficients of the polynomial $(x^2 + x + 1)^{31}$.

**Problem 19.33.** Let the sequence of integers $\{a_n\}$, $n \in \mathbb{N}$, be defined by $a_0 = 1$ and $a_n = a_{n-1} + a_{\lfloor n/3 \rfloor}$ for $n \geq 1$. Show that there are infinitely many integers $k$ such that $5 \mid a_k$.

**Problem 19.34.** Forty-nine students each solve a set of 3 problems. The score on each problem is a whole number of points from 0 to 7. Prove that there exist two students $A$ and $B$ such that, on every problem, $A$ scores at least as many points as $B$.

**Problem 19.35.** A club has 42 members. It is known that among any 31 members of the club, there is a boy and a girl who know each other. Show that we can choose 12 pairs of acquainted boys and girls (from among the club's members).

**Problem 19.36.** The chords $AC$ and $BD$ of a circle with center $O$ intersect at the point $K$. The circumcenters of triangles $AKB$ and $CKD$ are $M$ and $N$ respectively. Prove that $OM = KN$.

**Problem 19.37.** A convex quadrilateral $ABCD$ is inscribed in a circle with center $O$. The angles $AOB$, $BOC$, $COD$, $DOA$, taken in some order, are equal to the angles of quadrilateral $ABCD$. Prove that $ABCD$ is a square.

## Practice set 7

**Problem 19.38.** Find all positive integers $n < 10^{100}$ for which simultaneously $n$ divides $2^n$, $n-1$ divides $2^n - 1$, and $n-2$ divides $2^n - 2$.

**Problem 19.39.** For any positive integer $n$, let $d(n)$ denote the number of positive divisors of $n$ (including 1 and $n$ itself). Determine all positive integers $k$ such that $\frac{d(n^2)}{d(n)} = k$ for some $n$.

**Problem 19.40.** Given a real number $a$, define a sequence by $x_0 = 1$, $x_1 = x_2 = a$, and

$$x_{n+1} = 2x_n x_{n-1} - x_{n-2}$$

for $n \geq 2$. Prove that if $x_n = 0$ for some $n$, then the sequence is periodic.

**Problem 19.41.** Determine the greatest possible value of $\sum_{i=1}^{10} \cos(3x_i)$ for real numbers $x_1, x_2, \ldots, x_{10}$ satisfying $\sum_{i=1}^{10} \cos(x_i) = 0$.

**Problem 19.42.** Suppose $A, B, C, D$ are distinct points in the plane, no three of which are collinear. Show that if the squares of the lengths of segments $AB, AC, AD, BC, BD, CD$ are all rational numbers, then the ratio of the areas of triangles $ABC$ and $ABD$, that is $\dfrac{S_{ABC}}{S_{ABD}}$, is rational.

**Problem 19.43.** Suppose a polygon $P$ has vertices $(a_1, b_1), (a_2, b_2), \ldots, (a_n, b_n)$, listed in clockwise order. Prove that the area of $P$ is

$$\frac{1}{2} \left| (a_1 b_2 + a_2 b_3 + \cdots + a_n b_1) - (b_1 a_2 + b_2 a_3 + \cdots + b_n a_1) \right|.$$

†

**Problem 19.44.** The inscribed circle of triangle $A_1A_2A_3$ touches the sides $A_2A_3$, $A_3A_1$, $A_1A_2$ at points $S_1$, $S_2$, $S_3$ respectively. Let $O_1$, $O_2$, $O_3$ be the centers of the inscribed circles of triangles $A_1S_2S_3$, $A_2S_3S_1$, $A_3S_1S_2$ respectively. Prove that the lines $O_1S_1$, $O_2S_2$, $O_3S_3$ are concurrent.

## Practice set 8

**Problem 19.45.** Let $a_0 = 1$, $a_1 = 2$, and $a_n = 4a_{n-1} - a_{n-2}$ for $n \geq 2$. Find an odd prime factor of $a_{2019}$.

**Problem 19.46.** Find all functions $f : \mathbb{N}_+ \to \mathbb{N}_+$ satisfying

$$3f(n) - 2f(f(n)) = n$$

for all $n \in \mathbb{N}_+$, where $\mathbb{N}_+$ denotes the set of positive integers.

**Problem 19.47.** Let $A$ be an $m \times n$ table with a rational number written in each cell, such that among the absolute values of the entries of $A$ there are at least $m + n$ distinct prime numbers. Denote by $a_{ij}$ the entry in row $i$ and column $j$. Prove that there exist two rows $i, k$ and two columns $j, s$ such that $a_{ij} \cdot a_{ks} \neq a_{is} \cdot a_{kj}$.

**Problem 19.48.** Let $n > 1$ be an odd integer and let $c_1, c_2, \ldots, c_n$ be arbitrary integers. For a permutation $a = (a_1, a_2, \ldots, a_n)$ of $1, 2, \ldots, n$, denote

$$S(a) = a_1c_1 + a_2c_2 + \ldots + a_nc_n.$$

Prove that there exist two different permutations $a$ and $b$ such that $S(a) - S(b)$ is divisible by $n!$.

**Problem 19.49.** Five distinct points $A$, $B$, $C$, $D$, $E$ lie on a line in that order such that

$$AB = BC = CD = DE.$$

The point $F$ lies outside the line. Let $G$ be the circumcenter of triangle $ADF$ and $H$ be the circumcenter of triangle $BEF$. Show that lines $GH$ and $FC$ are perpendicular.

**Problem 19.50.** In acute triangle $ABC$, the bisectors of angles $A$, $B$, $C$ meet the circumcircle again at $A_1$, $B_1$, $C_1$ respectively. Let $M$ be the intersection of $AB$ and $B_1C_1$, and let $N$ be the intersection of $BC$ and $A_1B_1$. Prove that $MN$ passes through the incenter of triangle $ABC$.

## Practice set 9

**Problem 19.51.** Let $d(n)$ denote the sum of the decimal digits of $n$. Prove that there exist infinitely many positive integers $n$ such that

$$d(2^n) > d(2^{n+1}).$$

**Problem 19.52.** Let the sequence $(a_i)$ be defined by $a_1 = m \in \mathbb{Z}_+$ and, inductively, $a_{i+1} = a_i + \lfloor \sqrt{a_i} \rfloor$. Prove that the sequence $(a_i)$ contains infinitely many perfect squares.

**Problem 19.53.** Does there exist an infinite sequence $p_1, p_2, p_3, \ldots$ of prime numbers such that, for every positive integer $n$,

$$|p_{n+1} - 2p_n| = 1?$$

**Problem 19.54.** Prove that for any increasing sequence of positive integers $0 < a_0 < a_1 < \ldots < a_n$, the following inequality holds:

$$\frac{1}{\operatorname{lcm}(a_0, a_1)} + \frac{1}{\operatorname{lcm}(a_1, a_2)} + \ldots + \frac{1}{\operatorname{lcm}(a_{n-1}, a_n)} \leq 1 - \frac{1}{2^n}.$$

**Problem 19.55.** Let $f : \mathbb{N} \to \mathbb{N}$ be a strictly increasing function (that is, $f(n+1) > f(n)$ for all $n$) such that

$$f(f(n)) = 3n.$$

Evaluate $f(2017)$.

**Problem 19.56.** Let $S$ be the set of all collections of 3 (not necessarily distinct) positive irrational numbers with sum 1, where exactly one of the numbers is greater than $\frac{1}{2}$. If $\{x, y, z\} \in S$ with $x > \frac{1}{2}$, define $f(\{x, y, z\}) = \{2x - 1, 2y, 2z\}$. Is it possible to choose $\{x, y, z\} \in S$ such that for every positive integer $n$,

$$f^{(n)}(\{x, y, z\}) = \underbrace{f(f(\cdots f}_{n}(\{x, y, z\})\cdots)) \in S\,?$$

(Note that $f$ is defined on three-element sets, not on ordered triples.)

**Problem 19.57.** There are $n$ points in the plane, no three of which are collinear. Prove that the number of triangles with vertices among these $n$ points having area exactly 1 is at most $\frac{2(n^2-n)}{3}$.

## Practice set 10

**Problem 19.58.** Does there exist a polynomial $P(x)$ of degree 2 with integer coefficients such that the leading coefficient is not divisible by 2022, and the numbers $P(1), P(2), \ldots, P(2022)$ give pairwise different residues modulo 2022?

**Problem 19.59.** Prove that for any two positive integers $m$ and $n$ with $(m, n) \neq (1, 1)$, the number

$$\frac{1}{m} + \frac{1}{m+1} + \cdots + \frac{1}{m+n-1}$$

is not an integer.

**Problem 19.60.** Let $ABCDE$ be a pentagon such that $BC \parallel AE$, $AB = BC + AE$, and $\angle B = \angle D$. Let $M$ be the midpoint of $CE$ and let $O$ be the circumcenter of triangle $BCD$. Show that if $OM \perp MD$, then $\angle CDE = 2\angle ADB$.

**Problem 19.61.** Let the incircle of triangle $ABC$ have center $I$ and touch sides $BC$, $CA$, $AB$ at $D$, $E$, $F$ respectively. Let $J_1$, $J_2$, $J_3$ be the excenters opposite $A$, $B$, $C$ respectively. Let $J_2F$ and $J_3E$ meet at $P$, let $J_3D$ and $J_1F$ meet at $Q$, and let $J_1E$ and $J_2D$ meet at $R$. Show that $I$ is the circumcenter of triangle $PQR$.

**Problem 19.62.** Do there exist positive integers $m$ and $n$ such that the decimal representation of $5^m$ starts with the digits of $2^n$, and the decimal representation of $2^m$ starts with the digits of $5^n$?

**Problem 19.63.** On each cell of an $8 \times 8$ chessboard sits a rook. At each step one may remove from the board a rook that attacks an odd number of the remaining rooks (for example, in the initial configuration the top-left rook attacks 2 other rooks). Find the maximal possible number of rooks that can be removed from the board.

**Problem 19.64.** Let triangle $ABC$ satisfy $AB < AC$, and let $M$ be the midpoint of $BC$. The circle $O$ passes through $A$ and is tangent to $BC$ at $B$; it meets lines $AM$ and $AC$ again at $D$ and $E$ respectively. Let $F$ be the point where the line through $C$ parallel to $BE$ meets line $BD$ extended, and let $G$ be the intersection of line $BC$ with line $EF$. Show that $AG = DG$.

## Practice set 11

**Problem 19.65.** Let $x_1, x_2, \ldots, x_n$ be positive numbers such that

$$\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} \frac{1}{x_i}.$$

Prove that

$$\sum_{i=1}^{n} \frac{1}{n-1+x_i} \leq 1.$$

**Problem 19.66.** Find all integers $m$ and $n$ such that

$$(5 + 3\sqrt{2})^m = (3 + 5\sqrt{2})^n.$$

**Problem 19.67.** Twenty children are queueing for ice cream that is sold at SR5 per cone. Ten of the children have a SR5 coin, and the other ten want to pay with a SR10 bill. At the beginning the ice-cream vendor has no change. How many arrangements of the twenty children in the queue are there so that the vendor never runs out of change?

**Problem 19.68.** Let $D$ be a point on side $BC$ of triangle $ABC$, and let $P$ be a point on segment $AD$. A line through $D$ meets $AB$ and $PB$ at $M$ and $E$ respectively, and meets line $AC$ extended and line $PC$ extended at $F$ and $N$ respectively. Given that $DE = DF$, prove that $DM = DN$.

**Problem 19.69.** Let $A$ be one of the two points of intersection of two unequal circles in the same plane. Two particles start from $A$ at the same instant and move, each along one of the circles, in a clockwise direction with (their own) uniform velocities, returning to $A$ at the same instant. Prove that there is always a point in the plane that is equidistant from the two particles at every moment while they are in motion.

**Problem 19.70.** Let $AD$ be the altitude from $A$ in triangle $ABC$, and let $J$, $K$ be the incenters of triangles $ABD$, $ACD$ respectively. Let line $JK$ meet $AB$, $AC$ at $E$, $F$ respectively. Prove that $AE = AF$ if and only if $AB = AC$ or $\angle A = 90^\circ$.

**Problem 19.71.** A quadrilateral $ABCD$ is inscribed in a circle with $AD \perp CD$. Let $BE \perp AC$ at $E$ and $BF \perp AD$ at $F$. Show that line $EF$ passes through the midpoint of segment $BD$.

**Problem 19.72.** Given a quadrilateral $ABCD$, the external bisectors of $\angle CAD$ and $\angle CBD$ meet at $P$. Show that if $AD + AC = BC + BD$, then $\angle APD = \angle BPC$.

## Practice set 12

**Problem 19.73.** Solve the equation

$$27^x - 8^x = 3(18^x - 12^x).$$

**Problem 19.74.** Prove that 1280000401 is composite, without using a calculator.

**Problem 19.75.** There are 2017 coins on a table, each showing heads or tails. For $i = 1, 2, \ldots, 2017$, in succession, one must turn over exactly $i$ of the coins (chosen freely at each step). Prove that, starting from any initial arrangement, it is always possible to choose the turns so that all coins end up showing heads, or always possible to choose the turns so that all coins end up showing tails, but never both.

**Problem 19.76.** In triangle $ABC$, $AH$ is an altitude with $H$ on $BC$, and $BE$ is an angle bisector with $E$ on $AC$. Given that $\angle BEA = 45^\circ$, prove that $\angle EHC = 45^\circ$.

**Problem 19.77.** The bisector of angle $BAD$ in parallelogram $ABCD$ (which is not a rhombus) meets lines $BC$ and $CD$ at points $K$ and $L$, respectively. Prove that the center of the circle through $C$, $K$, $L$ lies on the circle through $B$, $C$, $D$.

## Practice set 13

**Problem 19.78.** Solve the system of equations

$$\begin{cases}(x-1)(y-1)(z-1) = xyz - 1\\ (x-2)(y-2)(z-2) = xyz - 2.\end{cases}$$

**Problem 19.79.** Prove that there are infinitely many positive integers $m$ such that the number of distinct odd prime factors of $m(m+3)$ is a multiple of 3.

**Problem 19.80.** Let $k$ be a positive integer and let $1 = d_0 < d_1 < \cdots < d_m = 4k$ be all the positive divisors of $4k$. Prove that there exists an index $i$, $1 \le i \le m$, such that $d_i - d_{i-1} = 2$.

**Problem 19.81.** Given a positive integer $n$, find a finite set $S$ of integers larger than $n$ with the following property: for every $k \ge n$, the $k \times k$ square can be tiled by squares whose side lengths all belong to $S$.

**Problem 19.82.** Ann and Max play a game on a $100 \times 100$ board. First, Ann writes an integer from 1 to 10000 in each square of the board, using each number exactly once. Max then places a token on some square of the leftmost column and moves it, step by step, to the rightmost column; each move shifts the token to a square adjacent by side or by corner. For every square visited (including the starting one), Max pays Ann a number of coins equal to the number written on that square. Max wants to pay as little as possible, while Ann wants to have written the numbers so as to maximize the amount she receives. If both play optimally (Ann writing the numbers first, then Max choosing his path), how much does Max pay Ann?

## Practice set 14

**Problem 19.83.** Let $x$ be a positive real number and $n$ a positive integer. Prove that

$$\lfloor nx \rfloor \ge \frac{\lfloor x \rfloor}{1} + \frac{\lfloor 2x \rfloor}{2} + \frac{\lfloor 3x \rfloor}{3} + \cdots + \frac{\lfloor nx \rfloor}{n}.$$

**Problem 19.84.** The sequence defined by $x_0 = a$, $x_1 = b$, and

$$x_{n+1} = \frac{1}{2}\left(x_{n-1} + \frac{1}{x_n}\right)$$

is periodic. Prove that $ab = 1$.

**Problem 19.85.** Prove the inequality

$$\frac{1}{2\sqrt{1}} + \frac{1}{3\sqrt{2}} + \cdots + \frac{1}{(n+1)\sqrt{n}} < 2.$$

**Problem 19.86.** Find a triple of rational numbers $(a, b, c)$ such that

$$\sqrt[3]{\sqrt[3]{2} - 1} = \sqrt[3]{a} + \sqrt[3]{b} + \sqrt[3]{c}.$$

**Problem 19.87.** The positive integers $a$ and $b$ are such that $15a + 16b$ and $16a - 15b$ are both perfect squares of positive integers. Find the least possible value of the smaller of these two squares.

**Problem 19.88.** Let $x_1, x_2, \ldots, x_n$ be real numbers in the interval $\left(\frac{1}{4}, 1\right)$. Find the minimum value of

$$\log_{x_1}\left(x_2 - \frac{1}{4}\right) + \log_{x_2}\left(x_3 - \frac{1}{4}\right) + \cdots + \log_{x_n}\left(x_1 - \frac{1}{4}\right).$$

## Practice set 15

**Problem 19.89.** Let $a_1 = t$ and $a_{n+1} = 4a_n(1 - a_n)$ for $n \geq 1$. For how many distinct values of $t$ does $a_{2018} = 0$?

**Problem 19.90.** Prove that $\binom{2n}{n}$ divides $\operatorname{lcm}(1, 2, \ldots, 2n)$ for every positive integer $n$.

**Problem 19.91.** Does there exist an integer greater than $2018^{2018}$ that cannot be written as $x^2 + y^3 + z^6$ for positive integers $x, y, z$?

**Problem 19.92.** A circle is divided into $n$ equal sectors, and each sector is colored with one of $m$ colors so that neighboring sectors receive different colors. Find the total number of such colorings (two colorings that differ by a rotation are counted as different).

## Practice set 16

**Problem 19.93.** Prove that for all natural $n \geq 2$, the number

$$\frac{n^{1003} + n^{1002} + n^{1001} + 1}{n + 1}$$

is composite.

**Problem 19.94.** There are four numbers on a board: $1, 3, 6, 10$. Each time, we may erase two numbers $a, b$ written on the board and write the numbers $a + b$, $ab$ in their place. Can we obtain the four numbers: a) $2015, 2016, 2017, 2018$; b) $2016, 2017, 2019, 2022$, after several moves?

**Problem 19.95.** Let $a, b, c$ be positive real numbers such that $a + b + c = 3$. Prove that

$$\frac{ab}{2a + b + 2c} + \frac{bc}{2b + c + 2a} + \frac{ca}{2c + a + 2b} \leq \frac{3}{5}.$$

**Problem 19.96.** The internal bisector of angle $BAC$ of an acute-angled triangle $ABC$ (with $AB \neq AC$) meets its circumcircle a second time at the point $W$. Let $O$ be the circumcenter of $\triangle ABC$. Line $AW$ meets the circumcircles of triangles $OWB$ and $OWC$ a second time at points $N$ and $M$ respectively. Prove that $BN + MC = AW$.

**Problem 19.97.** The numbers 1 through 19 are written on a board. At any time, three numbers $a, b, c$ on the board may be replaced by

$$\frac{2b + 2c - a}{3}, \quad \frac{2c + 2a - b}{3}, \quad \frac{2a + 2b - c}{3}.$$

Can a number greater than 32 appear on the board?

**Problem 19.98.** For each natural number $n$, find the number of permutations $(a_1, a_2, \ldots, a_n)$ of $(1, 2, \ldots, n)$ satisfying $k \mid (a_1 + a_2 + \cdots + a_k)$ for every $1 \leq k \leq n$.

## Practice set 17

**Problem 19.99.** In triangle $ABC$, the points $M, N, K$ are the midpoints of $BC, CA, AB$ respectively. Let $\omega_B$ and $\omega_C$ be the semicircles with diameters $AC$ and $AB$, drawn outside the triangle. Suppose $MK$ and $MN$ meet $\omega_C$ and $\omega_B$ at $X$ and $Y$ respectively, and let $Z$ be the intersection of the tangents to $\omega_C$ at $X$ and to $\omega_B$ at $Y$. Prove that $AZ \perp BC$.

**Problem 19.100.** Find all positive integers $m, n$ such that $6^m + 2^n + 2$ is a perfect square.

**Problem 19.101.** Let $a, b, c$ be positive real numbers such that $(a+b)(b+c)(c+a) = 8$. Prove that

$$\frac{\sqrt{a}}{a^2+3} + \frac{\sqrt{b}}{b^2+3} + \frac{\sqrt{c}}{c^2+3} \leq \frac{3}{4}.$$

†

**Problem 19.102.** In parallelogram $ABCD$, the perpendicular from $B$ to $CD$ and the perpendicular from $D$ to $BC$ meet at $P$. The point $T$ lies on $PC$ such that $\angle CAT = 90°$. A line $d$ through $C$ meets $PD$ and $PB$ at $X$ and $Y$ respectively, such that $CX = CY$. Prove that the circumcircle of triangle $PXY$ is tangent to the circumcircle of triangle $PAT$.

**Problem 19.103.** A $100 \times 100$ board is divided into unit squares, colored with 4 colors so that every row and every column has exactly 25 squares of each color. Prove that there exist 2 rows and 2 columns whose four intersection squares are painted with four different colors.

**Problem 19.104.** Let $k$ be a positive integer, and let $s(n)$ denote the sum of the decimal digits of $n$. Among the positive integers with $k$ digits, show that there are as many numbers $n$ satisfying $s(n) < s(2n)$ as there are numbers $n$ satisfying $s(n) > s(2n)$.†

## Practice set 18

**Problem 19.105.** Prove that $n^n - n$ is divisible by 24 for every odd positive integer $n$.

**Problem 19.106.** Let $x \neq 0, 1$ and let a positive integer $n$ be given. Define $a_0 = x$, $a_1 = 1 - x$, and $a_k = 1 - a_{k-1}(1 - a_{k-1})$ for $k = 2, 3, \ldots, n$. Prove that

$$\frac{1}{a_0} + \frac{1}{a_1} + \cdots + \frac{1}{a_n} = \frac{1}{a_0 a_1 \cdots a_n}.$$

**Problem 19.107.** Let $ABC$ be an isosceles triangle with $AB = AC$. Let $D, E, F$ be points on segments $BC, CA, AB$ respectively, such that $BF = BE$ and $ED$ is the exterior bisector of $\angle BEC$. Prove that $BD = EF$ if and only if $AF = EC$.

**Problem 19.108.** There are 8 white pawns on the squares of one edge of an $8 \times 8$ chessboard and 8 black pawns on the squares of the opposite edge. On each move, a player shifts one of their

pawns by one or more squares forward (toward the opponent's edge) or backward, but moving a pawn onto or over a square occupied by an opponent's pawn is prohibited. The players move alternately, White starting. The player who cannot move loses. Which player has a winning strategy?

**Problem 19.109.** A positive integer is called charming if it equals 2 or has the form $3^i 5^j$ for nonnegative integers $i, j$. Prove that every positive integer can be written as a sum of distinct charming integers.

## Practice set 19

**Problem 19.110.** Find all pairs of nonnegative integers $m, n$ satisfying $3 \cdot 2^m + 1 = n^2$.

**Problem 19.111.** Consider the following system of 10 equations in real variables $v_1, \ldots, v_{10}$:

$$v_i = \frac{6v_i^2}{1 + v_1^2 + v_2^2 + \cdots + v_{10}^2} \qquad (i = 1, 2, \ldots, 10).$$

Find all 10-tuples $(v_1, \ldots, v_{10})$ satisfying this system.

**Problem 19.112.** Let $ABCD$ be a quadrilateral. The circumcircle of triangle $ABC$ meets sides $CD$ and $DA$ again at points $P$ and $Q$, respectively, while the circumcircle of triangle $CDA$ meets sides $AB$ and $BC$ again at points $R$ and $S$. Lines $BP$ and $BQ$ meet line $RS$ at points $M$ and $N$ respectively. Prove that $M, N, P, Q$ lie on a common circle.

**Problem 19.113.** Let $a, b, c, d$ be real numbers with $abcd > 0$. Prove that there exists a permutation $x, y, z, t$ of $a, b, c, d$ such that

$$2(xy + zt)^2 > (x^2 + y^2)(z^2 + t^2).$$

**Problem 19.114.** One or more powers of 2 are written on each of $n$ sheets of paper, so that the sum of the numbers on each sheet is the same. If each number value appears at most 5 times in total among all the sheets, what is the largest possible value of $n$?

## Practice set 20

**Problem 19.115.** In trapezoid $ABCD$ with $AB \parallel CD$, the diagonals are perpendicular and meet at $O$; the base angles $\angle DAB$ and $\angle CBA$ are both acute. A point $M$ on segment $OA$ satisfies $\angle BMD = 90^\circ$, and a point $N$ on segment $OB$ satisfies $\angle ANC = 90^\circ$. Prove that triangles $OMN$ and $OBA$ are similar.

**Problem 19.116.** A doubles tennis tournament is to be arranged so that the total number of matches equals the total number of players, and every pair of players plays as either teammates or opponents in at least one match (each match has exactly 4 players). Find the largest possible number of players in such a tournament.

**Problem 19.117.** Let $a, b$ be positive integers such that $c = \dfrac{a + ab - 1}{b}$ is an integer. Prove that $c$ is a perfect square.

**Problem 19.118.** Let $a, b, c$ be real numbers with $a \geq 2$, $b \geq 3$, $c \geq 4$, and $a^2 + b^2 + c^2 = 38$. Find the minimum possible value of $a + b + c$.

**Problem 19.119.** Let $ABC$ be a triangle with $\angle A = 90^\circ$. Points $D$ and $E$ lie on $AC$ and $AB$ respectively, such that $\angle ABD = \angle DBC$ and $\angle ACE = \angle ECB$. Segments $BD$ and $CE$ meet at $I$. Determine whether it is possible for the six lengths $AB, AC, BI, ID, CI, IE$ to all be integers.

**Problem 19.120.** Two players alternate moves in a game on a $1 \times 100$ strip of unit squares. On each of the first player's moves, they color four consecutive squares; on each of the second player's moves, they color three consecutive squares. No square may be colored twice. The player who cannot move loses. Which player has a winning strategy?

## Practice set 21

**Problem 19.121.** Find all positive integers $n$ such that $2^n + 153$ is a perfect square.

**Problem 19.122.** In an acute triangle $ABC$, let $M$ be the midpoint of $BC$, and let $D, E, F$ be the feet of the altitudes from $A, B, C$ respectively. Let $H$ be the orthocenter, $S$ the midpoint of $AH$, and $G$ the intersection of lines $FE$ and $HA$. If $N$ is the second intersection of segment $AM$ with the circumcircle of triangle $BCH$, prove that $\angle HMA = \angle GNS$.

**Problem 19.123.** On a $1000 \times 1000$ board, dominoes are placed so that each domino covers exactly two squares, and no two dominoes are adjacent (though they may touch at a single vertex). Determine the maximum number of dominoes that can be placed on the board in this way.

**Problem 19.124.** Find all triples of nonnegative integers $(x, y, z)$ with $x \leq y$ such that

$$x^2 + y^2 = 3 \cdot 2016^z + 77.$$

**Problem 19.125.** Quadrilateral $APBQ$ is inscribed in circle $\omega$ with $\angle P = \angle Q = 90^\circ$ and $AP = AQ < BP$. Let $X$ be a variable point on segment $PQ$. Line $AX$ meets $\omega$ again at $S$, and $T$ is the point on arc $AQB$ of $\omega$ such that $XT \perp AX$. Let $M$ be the midpoint of chord $ST$. Show that as $X$ varies on segment $PQ$, the point $M$ moves along a circle.

**Problem 19.126.** Let $a, b, c$ be nonnegative real numbers such that $a + b + c = 2$. Prove that

$$\sqrt[4]{a^3b + b^3c + c^3a} + \sqrt[4]{ab^3 + bc^3 + ca^3} \leq 2.$$

## Practice set 22

**Problem 19.127.** Through the incenter $I$ of an acute triangle $ABC$, a line is drawn meeting $AB$ and $AC$ at points $M$ and $N$ respectively, so that triangle $AMN$ is acute. Points $K, L$ are chosen on side $BC$ such that $\angle ILB = \angle IMA$ and $\angle IKC = \angle INA$. Prove that $BC = BM + KL + CN$.

**Problem 19.128.** Let 100 pairwise distinct real numbers be arranged around a circle. Prove that there exist four consecutive numbers (along the circle) such that the sum of the middle two is strictly less than the sum of the other two.

**Problem 19.129.** Prove that for any three distinct positive real numbers $a, b, c$,

$$\frac{(a^2-b^2)^3+(b^2-c^2)^3+(c^2-a^2)^3}{(a-b)^3+(b-c)^3+(c-a)^3} > 8abc.$$

**Problem 19.130.** On the sides $AB$, $BC$, $CA$ of triangle $ABC$, points $L, M, N$ are chosen respectively, such that lines $CL, AM, BN$ meet at a common interior point $O$, and quadrilaterals $ALON$, $BMOL$, $CNOM$ all admit inscribed circles. Prove that

$$\frac{1}{AL\cdot BM}+\frac{1}{BM\cdot CN}+\frac{1}{CN\cdot AL}=\frac{1}{AN\cdot BL}+\frac{1}{BL\cdot CM}+\frac{1}{CM\cdot AN}.$$

**Problem 19.131.** Two players $A$ and $B$ alternately write, from left to right, the digits of a six-digit number; $A$ writes the first digit (which must be nonzero), and no digit may be repeated. Player $A$ wins if the resulting number is divisible by 2, 3, or 5; otherwise $B$ wins. Who has a winning strategy?

**Problem 19.132.** Let $a > b > c > d$ be positive integers such that $ac+bd = (b+d+a-c)(b+d-a+c)$. Prove that $ab+cd$ is not prime.

## Practice set 23

**Problem 19.133.** Let $A$ be a 101-element subset of $S = \{1, 2, \ldots, 1000000\}$. Prove that there exist numbers $t_1, t_2, \ldots, t_{100}$ in $S$ such that the sets $A_j = \{x + t_j : x \in A\}$, $j = 1, \ldots, 100$, are pairwise disjoint.

**Problem 19.134.** Let $ABCD$ be a cyclic quadrilateral, and let $P, Q, R$ be the feet of the perpendiculars from $D$ to lines $BC, CA, AB$ respectively. Show that $PQ = QR$ if and only if the bisectors of $\angle ABC$ and $\angle ADC$ are concurrent with $AC$.

**Problem 19.135.** Let $a, b, c$ be positive real numbers such that $abc \geq 1$. Prove that

$$\frac{a^5-a^2}{a^5+b^2+c^2}+\frac{b^5-b^2}{b^5+c^2+a^2}+\frac{c^5-c^2}{c^5+a^2+b^2}\geq 0.$$

**Problem 19.136.** Let $S$ be a set with $n^2+n-1$ elements, for a positive integer $n$. Suppose the $n$-element subsets of $S$ are partitioned into two classes. Prove that some class contains $n$ pairwise disjoint subsets.

## Practice set 24

**Problem 19.137.** Points $P, Q$ are chosen on side $BC$ of an acute triangle $ABC$ so that $\angle PAB = \angle ACB$ and $\angle QAC = \angle CBA$. Points $M, N$ are taken on rays $AP, AQ$ respectively so that $AP = PM$ and $AQ = QN$. Prove that lines $BM$ and $CN$ meet on the circumcircle of triangle $ABC$.

**Problem 19.138.** Determine whether there exists a positive integer $n < 10^9$ that can be expressed as a sum of three squares of positive integers in more than 1000 distinct ways.

**Problem 19.139.** Let $a, b, c$ be positive real numbers such that $a^2 + b^2 + c^2 \geq 3$. Prove that

$$\frac{(a+1)(b+2)}{(b+1)(b+5)} + \frac{(b+1)(c+2)}{(c+1)(c+5)} + \frac{(c+1)(a+2)}{(a+1)(a+5)} \geq \frac{3}{2}.$$

**Problem 19.140.** Let $S$ be a set with 2016 elements, and let $N$ be an integer with $0 \leq N \leq 2^{2016}$. Prove that every subset of $S$ can be colored red or blue so that: (i) the union of any two red subsets is red; (ii) the union of any two blue subsets is blue; (iii) exactly $N$ subsets are red.[†]

**Problem 19.141.** Six points $A_1, A_2$ on $BC$, $B_1, B_2$ on $CA$, and $C_1, C_2$ on $AB$ are chosen on the sides of an equilateral triangle $ABC$ so that they are the vertices of a convex equilateral hexagon $A_1A_2B_1B_2C_1C_2$. Prove that lines $A_1B_2$, $B_1C_2$, $C_1A_2$ are concurrent.

**Problem 19.142.** Find all triples of positive integers $(a, b, c)$ such that $a^3 + b^3 + c^3$ is divisible by $a^2b$, by $b^2c$, and by $c^2a$.

## Practice set 25

**Problem 19.143.** Do there exist positive integers $a, b, c$ such that all three of $a^2+b+c$, $b^2+c+a$, $c^2 + a + b$ are perfect squares?

**Problem 19.144.** Triangle $ABC$ is inscribed in circle $\omega$. A circle through chord $BC$ meets segments $AB$ and $AC$ again at $S$ and $R$ respectively. Segments $BR$ and $CS$ meet at $L$, and rays $LR$, $LS$ meet $\omega$ again at $D$, $E$ respectively. The internal bisector of $\angle BDE$ meets line $ER$ at $K$. Prove that if $BE = BR$, then $\angle ELK = \frac{1}{2}\angle BCD$.

**Problem 19.145.** Let $a, b, c$ be positive real numbers with $a + b + c = 3$. Prove that

$$\frac{1}{a^2 + b^2 + 2} + \frac{1}{b^2 + c^2 + 2} + \frac{1}{c^2 + a^2 + 2} \leq \frac{3}{4}.$$

**Problem 19.146.** Suppose each point of the plane is colored either black or white. Show that there exist three points of the same color that form the vertices of an equilateral triangle.

**Problem 19.147.** Three distinct points $A, B, C$ are fixed on a line, in this order. Let $\Gamma$ be a circle through $A$ and $C$ whose center does not lie on line $AC$. Let $P$ be the intersection of the tangents to $\Gamma$ at $A$ and $C$. Suppose $\Gamma$ meets segment $PB$ at $Q$. Prove that the intersection of the bisector of $\angle AQC$ with line $AC$ does not depend on the choice of $\Gamma$.

**Problem 19.148.** Determine all pairs $(x, y)$ of positive integers such that $\sqrt[3]{7x^2 - 13xy + 7y^2} = |x - y| + 1$.

## Practice set 26

**Problem 19.149.** Let $ABC$ be a triangle with $AC \neq AB$, and let $B_1$ be a point on ray $AC$ such that $AB = AB_1$. Let $\omega$ be the circle through $C$, $B_1$, and the foot of the internal bisector of angle $CAB$. Let $\omega$ meet the circumcircle of triangle $ABC$ again at $Q$. Prove that $AC$ is parallel to the tangent to $\omega$ at $Q$.

**Problem 19.150.** Is it possible to find 100 positive integers not exceeding 25000 such that all pairwise sums of them are distinct?

**Problem 19.151.** Does there exist a polynomial with real coefficients $P(x)$ of degree 2017 such that $P(x^2 - 2017)$ is divisible by $P(x)$?

**Problem 19.152.** Find all primes $p$ for which there exist positive integers $x$, $y$, $n$ such that

$$p^n = x^3 + y^3.$$

**Problem 19.153.** Let $ABC$ be an acute-angled triangle, and let $P$ and $Q$ be two points on side $BC$. Construct point $C_1$ so that the convex quadrilateral $APBC_1$ is cyclic, $QC_1 \parallel CA$, and $C_1$ and $Q$ lie on opposite sides of line $AB$. Construct point $B_1$ so that the convex quadrilateral $APCB_1$ is cyclic, $QB_1 \parallel BA$, and $B_1$ and $Q$ lie on opposite sides of line $AC$. Prove that points $B_1$, $C_1$, $P$, $Q$ lie on a circle.

**Problem 19.154.** Three nonnegative real numbers $r_1, r_2, r_3$ are written on a blackboard. These numbers have the property that there exist integers $a_1, a_2, a_3$, not all zero, satisfying $a_1 r_1 + a_2 r_2 + a_3 r_3 = 0$. We are permitted to perform the following operation: find two numbers $x, y$ on the blackboard with $x \le y$, then erase $y$ and write $y - x$ in its place. Prove that after a finite number of such operations, we can end up with at least one 0 on the blackboard.

## Practice set 27

**Problem 19.155.** In a triangle $ABC$ satisfying $AB + BC = 3AC$, the incircle has center $I$ and touches sides $AB$ and $BC$ at $D$ and $E$, respectively. Let $K$ and $L$ be the reflections of $D$ and $E$ in $I$. Prove that the quadrilateral $ACKL$ is cyclic.

**Problem 19.156.** Let $m$ be a fixed integer greater than 1. The sequence $x_0, x_1, x_2, \ldots$ is defined by

$$x_i = 2^i \text{ if } 0 \le i \le m-1, \qquad x_i = \sum_{j=1}^{m} x_{i-j} \text{ if } i \ge m.$$

Find the greatest $k$ for which the sequence contains $k$ consecutive terms divisible by $m$.

**Problem 19.157.** Find all polynomials with real coefficients $P(x)$ such that

$$P(P(x) + x) = P(x)P(x+1)$$

for all $x \in \mathbb{R}$.

**Problem 19.158.** 100 queens are placed on a $100 \times 100$ chessboard so that no two attack each other. Prove that each of the four $50 \times 50$ corners of the board contains at least one queen.

**Problem 19.159.** Let $a, b, c, d, e, f$ be positive integers. Suppose that the sum $S = a + b + c + d + e + f$ divides both $abc + def$ and $ab + bc + ca - de - ef - fd$. Prove that $S$ is composite.

**Problem 19.160.** The median $AM$ of a triangle $ABC$ intersects its incircle $\omega$ at $K$ and $L$. The lines through $K$ and $L$ parallel to $BC$ intersect $\omega$ again at $X$ and $Y$. The lines $AX$ and $AY$ intersect $BC$ at $P$ and $Q$. Prove that $BP = CQ$.

## Practice set 28

**Problem 19.161.** Let $f(x)$ be a monic polynomial (the coefficient of the highest-degree term equals 1) of degree four such that $f(1) = 2017$, $f(2) = 4034$, $f(3) = 6051$. Find $f(5) + f(-1)$.

**Problem 19.162.** Let $a_1, a_2, \ldots$ be a sequence of integers with infinitely many positive and infinitely many negative terms. Suppose that for every positive integer $M$ the numbers $a_1, a_2, \ldots, a_M$ leave distinct remainders upon division by $M$. Prove that every integer occurs exactly once in the sequence $a_1, a_2, \ldots$.

**Problem 19.163.** Color the vertices of a convex $2n$-gon alternately with two colors, and draw the diagonals with differently colored endpoints. What is the maximum number of intersection points determined by these diagonals?

**Problem 19.164.** Find all monic integer polynomials $p(x)$ (the coefficient of the highest-degree term equals 1) of degree two for which there exists an integer polynomial $q(x)$ such that $p(x)q(x)$ is a polynomial having all coefficients $\pm 1$.

**Problem 19.165.** Let $ABC$ be an acute-angled triangle with $AB \neq AC$, let $H$ be its orthocenter and $M$ the midpoint of $BC$. Points $D$ on $AB$ and $E$ on $AC$ are such that $AE = AD$ and $D, H, E$ are collinear. Prove that $HM$ is perpendicular to the common chord of the circumcircles of triangles $ABC$ and $ADE$.

## Practice set 29

**Problem 19.166.** A point $X$ is marked on the base $BC$ of an isosceles triangle $ABC$, and points $P$ and $Q$ are marked on sides $AB$ and $AC$ so that $APXQ$ is a parallelogram. Prove that the point $Y$ symmetric to $X$ with respect to line $PQ$ lies on the circumcircle of triangle $ABC$.

**Problem 19.167.** Find all polynomials with real coefficients $P(x)$ such that $P(3) = 2016$ and

$$P(x^2) = x^2(x^2 + 1)P(x)$$

for all real $x$.

**Problem 19.168.** Find all triples of positive rational numbers $(a, b, c)$ such that all three of the numbers $a + \dfrac{1}{b}$, $b + \dfrac{1}{c}$, $c + \dfrac{1}{a}$ are integers.

**Problem 19.169.** A $2 \times n$ table (with $n > 2$) is filled with numbers so that the sums in all the columns are different. Is it always possible to permute the numbers in the table so that the sums in the columns remain different and the sums in the rows also become different?

**Problem 19.170.** There are 2017 positive integers arranged in a circle. The difference between any two adjacent numbers equals their greatest common divisor. Determine the maximal value of $N$ which divides the product of the numbers, regardless of their choice.

**Problem 19.171.** Consider a polynomial $P(x)$ such that $P(0) = 1$ and

$$(P(x))^2 = 1 + x + x^{100}Q(x)$$

for some polynomial $Q(x)$. Prove that in the polynomial $(P(x) + 1)^{100}$ the coefficient of $x^{99}$ is zero.

## Practice set 30

**Problem 19.172.** Define the sequence $a_1, a_2, a_3, \ldots$ by $a_1 = 1$, $a_2 = 1$, $a_3 = 2$, and

$$a_{n+3} = \frac{a_{n+2} \cdot a_{n+1} + 7}{a_n}$$

for $n \geq 1$. Prove that all terms of the sequence are positive integers.

**Problem 19.173.** Find all pairs of primes $(p, q)$ such that $p^2 - pq - q^3 = 27$.

**Problem 19.174.** Find all polynomials $f$ with real coefficients such that for all real numbers $a, b, c$ with $ab + bc + ca = 0$, we have

$$f(a-b) + f(b-c) + f(c-a) = 2f(a+b+c).$$

**Problem 19.175.** Triangle $ABC$ is inscribed in circle $\omega$. The tangent lines to $\omega$ at $B$ and $C$ meet at $T$. Point $S$ lies on ray $BC$ such that $AS \perp AT$. Points $B_1$ and $C_1$ lie on ray $ST$ (with $C_1$ between $B_1$ and $S$) such that $B_1T = BT = C_1T$. Prove that triangles $ABC$ and $AB_1C_1$ are similar.

**Problem 19.176.** On a table there are 11 piles of ten stones each. Alzubair and Omar play the following game: in turns they take 1, 2, or 3 stones at a time, with Alzubair taking stones from any single pile while Omar takes stones from different piles, no more than one from each pile. Alzubair moves first. The player who cannot move loses. Which of the players, Alzubair or Omar, can guarantee a victory regardless of the opponent's play?

## Practice set 31

**Problem 19.177.** Among $2n + 1$ integers there is exactly one 0, while each of the numbers $1, 2, \ldots, n$ appears exactly twice. For which $n$ can one arrange these numbers so that for every $m = 1, \ldots, n$ there are exactly $m$ numbers between the two copies of $m$?

**Problem 19.178.** A $100 \times 100$ table is filled with numbers such that the sums in all the columns are different. Is it always possible to permute the numbers in the table so that the sums in the columns remain different and the sums in the rows also become different?

**Problem 19.179.** Given a polynomial $P(x) = x^n + 5x^{n-1} + 3$, where $n$ is a positive integer, prove that $P(x)$ cannot be factored as a product of two positive-degree polynomials with integer coefficients.

**Problem 19.180.** Define the sequence $a_1, a_2, \ldots$ by $a_1 = 3$, $a_2 = 7$, and $a_{n+2} = 3a_{n+1} - a_n$ for $n \geq 1$. Prove that $a_n - 2$ is a perfect square for all odd positive integers $n$.

**Problem 19.181.** There is a scalene triangle. Alzubair and Omar play the following game: on each turn Alzubair chooses a point in the plane, and Omar responds by coloring it red or blue. Alzubair wins if some triangle similar to the given one has all vertices the same color. Find the minimal number of moves Alzubair needs to guarantee a win, no matter how Omar plays (independently of the shape of the given triangle).

**Problem 19.182.** Let $l$ be a tangent to the incircle of triangle $ABC$. Let $l_a, l_b, l_c$ be the images of $l$ under reflection across the external bisectors of $\angle A$, $\angle B$, $\angle C$, respectively. Prove that the triangle formed by these three lines is congruent to $ABC$.

## Practice set 32

**Problem 19.183.** Find all pairs $(a, b)$ of non-negative integers such that $2017^a = b^6 - 32b + 1$.

**Problem 19.184.** Determine all polynomials $P(x) \in \mathbb{R}[x]$ satisfying both: (a) $P(2017) = 2016$, and (b) $(P(x) + 1)^2 = P(x^2 + 1)$ for all real $x$.

**Problem 19.185.** The nonnegative integers 2000, 17, and $n$ are written on a blackboard. Alice and Bob play the following game: Alice begins, then they play in turns. A move consists of replacing one of the three numbers by the absolute difference of the other two. No move is allowed that leaves all three numbers unchanged. A player who cannot make a legal move loses the game.

a) Prove that the game ends for every value of $n$.

b) Who wins the game when $n = 2017$?

**Problem 19.186.** Let $ABCDE$ be a regular pentagon with center $M$. A point $P \neq M$ is chosen on segment $MD$. The circumcircle of $ABP$ meets segment $AE$ again at $Q$, and meets the line through $P$ perpendicular to $CD$ again at $R$. Prove that $AR$ and $QR$ have the same length.

**Problem 19.187.** Find all functions $f : (0, +\infty) \to (0, +\infty)$ such that

$$f(x + f(xy)) = xf(1 + f(y))$$

for all positive $x, y$.

**Problem 19.188.** Let $AA_1, BB_1, CC_1$ be the altitudes of the acute triangle $ABC$. Let $J_a, J_b, J_c$ be the incenters of triangles $AC_1B_1$, $BA_1C_1$, $CB_1A_1$, respectively. Prove that the circumradius of triangle $J_aJ_bJ_c$ equals the circumradius of triangle $ABC$.

**Problem 19.189.** The central square of a town has the form of an $n \times m$ rectangle formed by $1 \times 1$ tiles. To illuminate the area, one-lamp lampposts are used; they are placed at corners of tiles, including on the boundary of the area. The lamp on a lamppost illuminates all tiles having a corner at that lamppost's position, and only those tiles. Find the smallest number of lampposts required to illuminate the whole area even if any one of the lamps burns out.

## Practice set 33

**Problem 19.190.** A sequence of integers $\{a_n\}$ satisfies, for every positive integer $n$,

$$a_n + a_{n+1} = 2a_{n+2}a_{n+3} + 2016.$$

Find all possible values of $a_1$ and $a_2$.

**Problem 19.191.** Each unit square of a $2016 \times 2016$ grid contains a positive integer. You play a game on the grid using the following two types of moves: (i) choose a row and multiply every number in it by 2; (ii) choose a positive integer and a column, and subtract the positive integer from every number in the column. You win if all numbers in the grid become 0. Is it always possible to win after a finite number of moves?

**Problem 19.192.** In triangle $ABC$, let $AD$ be the bisector of $\angle BAC$ with $D$ on $BC$. The perpendicular from $B$ to $AD$ meets the circumcircle of $ABD$ again at $E$. Prove that $E$, $A$, and the circumcenter $O$ of $ABC$ are collinear.

**Problem 19.193.** Show that in any sequence of six consecutive integers there is at least one integer $x$ such that $(x^2+1)(x^4+1)(x^6-1)$ is a multiple of 2016.

**Problem 19.194.** Triangle $ABC$ is right-angled at $A$ and satisfies $AB > AC$. The tangent line to the circumcircle of $ABC$ at $A$ meets line $BC$ at $M$. Let $D$ be the point such that $M$ is the midpoint of $BD$. The line through $D$ parallel to $AM$ meets the circumcircle of $ACD$ again at $E$. Prove that $A$ is the incenter of triangle $EBM$.

**Problem 19.195.** Let $a, b, c$ be positive integers such that $a^3 + b^3 = 2^c$. Prove that $a = b$.

**Problem 19.196.** Consider the sequence $a_1, a_2, a_3, \ldots$ defined by $a_1 = 1$ and $a_n = n - \lfloor\sqrt{a_{n-1}}\rfloor$ for $n \geq 2$. Determine the value of $a_{800}$.

**Problem 19.197.** A binary sequence is a sequence each of whose terms equals 0 or 1. Call a binary sequence superb if each term is adjacent to at least one term equal to 1 (for example, $0, 1, 1, 0, 0, 1, 1, 1$ is superb). Let $B_n$ denote the number of superb binary sequences with $n$ terms. Determine the smallest integer $n \geq 2$ such that $B_n$ is divisible by 20.

## Practice set 34

**Problem 19.198.** Does there exist a two-variable polynomial $P(x, y)$ with real coefficients such that $P(x, y)$ is positive exactly when $x$ and $y$ are both positive?

**Problem 19.199.** A group of students entered a mathematics competition consisting of five problems. Each student solved at least two problems, and no student solved all five. For each pair of problems, exactly two students solved both of them. Determine the minimum possible number of students in the group.

**Problem 19.200.** Let $ABCD$ be a rectangle with $AB > BC$. Let $E$ be the point on diagonal $AC$ such that $BE \perp AC$. Let the circle through $A$ and $E$ whose center lies on line $AD$ meet side $CD$ at $F$. Prove that $BF$ bisects $\angle AFC$.

**Problem 19.201.** Find all pairs $(a, b)$ of positive integers such that $a^{2017} + b$ is a multiple of $ab$.

**Problem 19.202.** Let $ABC$ be an acute triangle. The altitudes $BE$ and $CF$ meet at the orthocenter $H$, and $O$ denotes the circumcenter. Point $P$ is chosen so that $\angle APH = \angle OPE = 90°$, and point $Q$ is chosen so that $\angle AQH = \angle OQF = 90°$. Lines $EP$ and $FQ$ meet at point $T$. Prove that $A$, $T$, $O$ are collinear.

**Problem 19.203.** Start by writing the integers $1, 2, 4, 6$ on a blackboard. At each step, write the smallest positive integer $n$ satisfying both of the following properties: (i) $n$ is larger than every integer currently on the board; (ii) $n$ cannot be written as the sum of 2 distinct integers on the board.

Find the 100th integer that gets written on the board (there are already 4 numbers on the board at the start).

## Practice set 35

**Problem 19.204.** Consider all ordered pairs of integers $(a, b)$ such that $1 \leq a \leq b \leq 100$ and

$$\frac{(a+b)(a+b+1)}{ab}$$

is an integer. Find all such pairs $(a, b)$.†

**Problem 19.205.** Find all positive integers $n$ such that the product of all divisors of $n$ equals $10^9$.

**Problem 19.206.** The area of square $ABCD$ is 196 cm$^2$. Point $E$ is inside the square, equidistant from $D$ and $C$, and such that $\angle DEC = 150^\circ$. Find the perimeter of triangle $ABE$.

**Problem 19.207.** Two three-dimensional objects are said to have the same coloring if one object can be oriented (by moving or turning it) so that it is indistinguishable from the other. For example, if we have two unit cubes sitting on a table, one with all faces black except a red top face, and the other with all faces black except a red bottom face, then these two cubes have the same coloring. In how many different ways can the edges of a regular tetrahedron be colored using two red edges, two black edges, and two green edges?

**Problem 19.208.** Compute the number of functions $f : \mathbb{N} \to \{0, 1, \ldots, 16\}$ such that

$$f(x+17) = f(x) \quad \text{and} \quad f(x^2) \equiv f(x)^2 + 5 \pmod{17}$$

for all integers $x \geq 1$.†

**Problem 19.209.** Call a number $T$ persistent if the following holds: whenever $a, b, c, d$ are real numbers different from 0 and 1 such that

$$a+b+c+d = T, \qquad \frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d} = T, \qquad \frac{1}{1-a}+\frac{1}{1-b}+\frac{1}{1-c}+\frac{1}{1-d} = T,$$

a) Prove that if $T$ is persistent, then $T$ must equal 2.
b) Prove that 2 is persistent.

## Practice set 36

**Problem 19.210.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying $f(x)f(y) = f(x-y)$ for all real $x, y$.

**Problem 19.211.** In an isosceles triangle $ABC$ with $AB = BC$, points $K$ and $M$ are the midpoints of sides $AB$ and $AC$, respectively. The circumcircle of triangle $CKB$ meets line $BM$ again at a point $N \neq M$. The line through $N$ parallel to $AC$ meets the circumcircle of $ABC$ at points $A_1$ and $C_1$. Prove that triangle $A_1BC_1$ is equilateral.

**Problem 19.212.** Consider a shortest lattice path along the edges of a $7 \times 7$ square grid from its bottom-left vertex to its top-right vertex. How many such paths have no edge strictly above the main diagonal joining these two vertices?

**Problem 19.213.** A positive integer is called nice if it equals the sum of the squares of three distinct divisors of itself.
a) Prove that every nice number is divisible by 3.

b) Are there infinitely many nice numbers?

**Problem 19.214.** Determine the largest real number $c$ such that for any 2017 real numbers $x_1, x_2, \ldots, x_{2017}$, the inequality

$$\sum_{i=1}^{2016} x_i(x_i + x_{i+1}) \geq c \cdot x_{2017}^2$$

holds.

**Problem 19.215.** Consider the $n \times n$ multiplication table formed as follows: write $1, 2, \ldots, n$ from left to right along the first row and $1, 2, \ldots, n$ from top to bottom along the first column; each remaining entry is the product of the corresponding first-row and first-column numbers. A path runs from the upper-left square to the lower-right square, at each step moving one square either right or down. Over all such paths:

(a) What is the smallest possible sum of the numbers along the path? Express the answer in terms of $n$, and prove it is correct.

(b) What is the largest possible sum of the numbers along the path? Express the answer in terms of $n$, and prove it is correct.

# Source Materials

This book was compiled exclusively from the training materials of the Saudi Arabian Mathematical Olympiad Program. The table below lists the material by season; each chapter's opening box names the specific handouts it draws from.

- **2016–2017 e-training** — weekly problem sets (KSA 2016 sets 4–21; Group-4 sets, 2017).
- **2019–2020** — November 2019 NT course (Mikhail, L2–L4); November 2019 algebra (Dmytro Nomirovskii); January 2020 camp (Dominik Burek geometry L3/L4; Majid Almarhoumi C/NT; Dmytro cyclotomic polynomials); March–April 2020 online camps (Dominik Burek NT L2/L3; Marko Radovanović polynomials and sequences; Mikhail geometry L4/L4+; Alexander Antropov combinatorics L2; Dušan Djukić NT L4); June–August 2020 online camps (Nikola Petrović A/C/NT; Dušan Djukić NT+C; Mikhail L3); weekly e-training, weeks 1–16, L2–L4.
- **2020–2021** — November 2020 – January 2021 (Dominik Burek A/NT/G); winter camp 2021 (Asaad Mohammedsaleh, algebra L1/L2); March–April 2021 (Dima, NT L4 problem lists; Lê Phúc Lũ, geometry L4; Shaden, JBMO algebra and geometry L3); April–June 2021 daily sheets (Dušan Djukić, NT L2 and combinatorics L2/L4); April–June 2021 (Marko Radovanović, functional equations and inequalities; Nikola Petrović, JBMO).
- **2021–2022** — September–October 2021 e-training (weeks 1–8, L2–L4); November 2021 camp (Dušan Djukić NT L2–L4; Łukasz Bożyk geometry; Nikola Petrović combinatorics L2–L4); January 2022 (Łukasz Bożyk, combinatorics); February–March 2022 camp (Marko Radovanović algebra; Nikola Petrović combinatorics; Dušan Djukić NT; Bojan Bašić NT; Łukasz Bożyk algebra); June 2022 camp (Dušan Djukić NT L2/L4/L4+; Dominik Burek G/NT; Lê Phúc Lũ geometry and algebra; Łukasz Bożyk).
- **2022–2023** — September–October 2022 acceleration (Bojan Bašić, Dušan Djukić, Marko Radovanović); weekly e-training weeks 1–14, L1–L4 (the program's only continuous Level-1 record); November 2022 camp (Arsenii/Senya and Eyad; Regis Barbosa geometry L3/L4; Lu algebra L3/L4); February–March 2023 camp (Dušan Djukić algebra; Bojan Bašić combinatorics); online training March–May 2023 (Marko Radovanović; Łukasz Bożyk; Rafael Kazuhiro Miyazaki; Regis Barbosa); May 2023 (Łukasz Bożyk L2–L4).
- **2023–2024** — remote lessons (Arsenii/Senya; Nikola Petrović; Marko Radovanović); February 2024 camp (Bojan Bašić NT L2–L4; Emil Kolev combinatorics and geometry); May–June 2024 camp (Dušan Djukić NT L2/L4; Bojan Bašić combinatorics L4).
- **2024–2025** — November 2024 camp (Navid Safaei algebra L2–L4; Miroslav Marinov NT L2–L4; Dren Neziri geometry); January–February 2025 camp (combinatorics method leaflets L2–L4; Regis Barbosa geometry L2–L4; Dušan Djukić NT L2–L4; Dren Neziri, Level-1 triangle centers); May–June 2025 (David Zmiaikou algebra; Miroslav Marinov NT; Dominik geometry L2–L4).

- **2025–2026** — September–October 2025 online NT (Miroslav Marinov, L2 and L4, including the primitive-roots course); November 2025 elite camp (Marko Radovanović algebra L2–L4; Dušan Djukić NT L2–L4).
- **Łukasz Bożyk's training website** — combinatorics threads (Level 2, Level 3, IMO Team).

## Materials not included

Third-party published books stored alongside the program materials (S. Seraj, *Rigorous Elementary Mathematics*, Vol. 1–3; T. Andreescu et al., *Contests Around the World* and the *106/107 Geometry Problems* chapter excerpts) are not part of the program's own material and are not reproduced here. Camp tests, team selection tests, and the national-olympiad item banks are likewise outside the scope of this volume. A small number of handouts survive only as handwritten scans (two April 2021 handouts by Shaden; whiteboard notes from the 2023–2024 remote groups; one January 2020 combinatorics scan) and could not be transcribed; the weekly e-training sets also contain geometry problems whose statements exist only as embedded images. These are the only omissions.

## Statements pending verification

Problems marked with † in the text were reconstructed from damaged source files; the list below records them so they can be checked against the original PDFs.

- Chapter 1 (Level 1 Numbers): Problem 1.31.
- Chapter 2 (Level 1 Algebra): Problem 2.17, Problem 2.23.
- Chapter 3 (Level 1 Counting): Problem 3.24.
- Chapter 4 (Level 1 Geometry): Problem 4.23.
- Chapter 5 (Level 2 Algebra): Problem 5.8, Problem 5.22, Problem 5.96.
- Chapter 6 (Level 2 Combinatorics): Problem 6.2, Problem 6.36.
- Chapter 7 (Level 2 Geometry): Problem 7.8, Problem 7.33, Problem 7.85, Problem 7.172.
- Chapter 8 (Level 2 Number Theory): Problem 8.200, Problem 8.315, Problem 8.431.
- Chapter 10 (Level 3 Algebra): Problem 10.20, Problem 10.33, Problem 10.109, Problem 10.127, Problem 10.138, Problem 10.147, Problem 10.148, Problem 10.206, Problem 10.213, Problem 10.214, Problem 10.215, Problem 10.216, Problem 10.217, Problem 10.218, Problem 10.219, Problem 10.220, Problem 10.221, Problem 10.222, Problem 10.224, Problem 10.225, Problem 10.227, Problem 10.299, Problem 10.303, Problem 10.305, Problem 10.309, Problem 10.310, Problem 10.313, Problem 10.319, Problem 10.322.
- Chapter 11 (Level 3 Combinatorics): Problem 11.176, Problem 11.191.
- Chapter 13 (Level 3 Number Theory): Problem 13.27, Problem 13.28, Problem 13.40, Problem 13.60, Problem 13.108, Problem 13.206.
- Chapter 14 (Level 3 Mixed Practice): Problem 14.19.

- Chapter 15 (Level 4 Algebra): Problem 15.49, Problem 15.51, Problem 15.53, Problem 15.54, Problem 15.55, Problem 15.56, Problem 15.115, Problem 15.122, Problem 15.124, Problem 15.137, Problem 15.138, Problem 15.139, Problem 15.141, Problem 15.142, Problem 15.143, Problem 15.145, Problem 15.154, Problem 15.168, Problem 15.171, Problem 15.172, Problem 15.174, Problem 15.175, Problem 15.186, Problem 15.206, Problem 15.207, Problem 15.208, Problem 15.209, Problem 15.211, Problem 15.212, Problem 15.213, Problem 15.217, Problem 15.218, Problem 15.225, Problem 15.226, Problem 15.227, Problem 15.228, Problem 15.272, Problem 15.273, Problem 15.282, Problem 15.283, Problem 15.285, Problem 15.287, Problem 15.320, Problem 15.324, Problem 15.338, Problem 15.339, Problem 15.340, Problem 15.342, Problem 15.343, Problem 15.346, Problem 15.347, Problem 15.352, Problem 15.373, Problem 15.383, Problem 15.384, Problem 15.385, Problem 15.412, Problem 15.413, Problem 15.420, Problem 15.431, Problem 15.433.
- Chapter 16 (Level 4 Combinatorics): Problem 16.35, Problem 16.41, Problem 16.65, Problem 16.68, Problem 16.99, Problem 16.158, Problem 16.183, Problem 16.195, Problem 16.199, Problem 16.226, Problem 16.271, Problem 16.282, Problem 16.283; and the probabilistic Ramsey-bound theory box.
- Chapter 17 (Level 4 Geometry): Problem 17.20, Problem 17.22, Problem 17.23, Problem 17.24, Problem 17.37, Problem 17.86, Problem 17.89, Problem 17.120, Problem 17.121, Problem 17.124, Problem 17.125, Problem 17.126, Problem 17.130, Problem 17.148, Problem 17.169, Problem 17.178, Problem 17.237, Problem 17.239, Problem 17.243.
- Chapter 18 (Level 4 Number Theory): Problem 18.40, Problem 18.188, Problem 18.190, Problem 18.239, Problem 18.327, Problem 18.409, Problem 18.435, Problem 18.467, Problem 18.472, Problem 18.548, Problem 18.549, Problem 18.550.
- Chapter 19 (Level 4 Mixed Practice): Problem 19.43, Problem 19.101, Problem 19.104, Problem 19.140, Problem 19.204, Problem 19.208.

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